The Physics of Golf

Force, Drift, and Control in the Golf Swing

An AffineDrift Textbook

AffineDrift

https://affinedrift.com

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Preface

Every golfer, from the weekend duffer to the touring professional, has felt it: that one swing where everything clicks, the ball launches off the clubface with a crack that resonates up through your hands, and for a brief moment the physics of the universe conspired in your favor. What happened in that swing? What forces were at play? And—perhaps more importantly—which of those forces did *you* actually create, and which did physics hand you for free?

This book exists to answer those questions with mathematical precision, but in a language that any curious golfer can follow.

The central idea is deceptively simple. When you swing a golf club, two fundamentally different kinds of forces are at work. The first kind—we call it **drift**—is everything that would happen if your muscles went limp mid-swing: momentum carrying the club forward, gravity pulling it down, the shaft springing back from its bend. These are the forces of physics acting on autopilot. The second kind—we call it **control**—is the direct effect of your muscles applying torque at the joints. The mathematical framework that separates these two is called *control-affine decomposition*, and it is the backbone of this entire book.

Why does this separation matter? Because it gives a precise way to ask how much of the swing is being produced by the modeled passive dynamics and how much by active generalized torque. In the worked double-pendulum and multibody examples, the late downswing often falls into a high Drift-Control Ratio regime; the numerical ratios depend on the chosen model, norm, body parameters, and torque bounds (Chapter 5). This does not mean the golfer is passive at impact. It means that many impact conditions must be prepared earlier, while muscular input still has favorable leverage. Elite golfers are not merely stronger; they are often better at arranging the initial and transitional conditions that let physics work in their favor.

This book will take you on a journey from the simplest possible model of a golf swing—a double pendulum, which is just two sticks hinged together—through the constraint forces that silently redirect enormous amounts of energy, through the parallel mechanisms that make the human body a closed kinematic chain, all the way to the flexible shaft and the biological tissues that connect it all together. Along the way, we will not shy away from equations. Every equation in this book is motivated by a physical question that a golfer would care about, and every equation is followed by a plain-language explanation of what it means and why it matters.

We have also included a chapter on fascia—the connective tissue that has been the subject of much mystical thinking in the golf and fitness communities. We approach it the way we approach everything in this book: with respect for the biology, skepticism toward unsupported claims, and a mathematical framework that puts the tissue in its proper place within the mechanics of the swing.

This book is part of the AffineDrift project, a broader effort to bring modern control theory and robotics mathematics to the study of human movement. Our companion series, *The Geometry of Motion*, provides the full mathematical foundations; this volume is designed to be self-contained and accessible without it. If a concept from *The Geometry of Motion* is needed, we develop it from scratch here, always with golf as the motivating example.

Welcome to the physics of golf. Let's find out what your swing is really doing.

AffineDrift
March 2026

Nomenclature

Throughout this book, we use the following notation conventions and symbol definitions. Every equation in the text uses only symbols defined here or introduced explicitly in the surrounding paragraph.

General Conventions

- Scalars are lowercase Roman or Greek letters: m, t, θ, ω.
- Vectors are bold lowercase: $\boldsymbol{x}, \boldsymbol{u}, \boldsymbol{q}, \boldsymbol{v}$.
- Matrices are bold uppercase: $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{M}, \boldsymbol{J}$.
- Sets and manifolds are script or blackboard bold: $\mathcal{Q}, \mathcal{M}, \mathbb{R}$.
- A dot above a symbol denotes a time derivative: $\dot{\boldsymbol{q}} = \frac{\mathrm{d}\boldsymbol{q}}{\mathrm{d}t}$.
- Subscript “drift” or “nat” denotes passive/natural components; “act” or “ctrl” denotes active/muscular components.
- Subscript “head” refers to the clubhead; “hand” to the grip/handle point.

Coordinates and Kinematics

$\boldsymbol{q} \in \mathbb{R}^n$ Generalized coordinate vector (joint angles, shaft modes, etc.).

$\dot{\boldsymbol{q}}, \ddot{\boldsymbol{q}}$ Generalized velocity and acceleration.

$\boldsymbol{x} \in \mathbb{R}^{n_x}$ Full state vector; $\boldsymbol{x} = [\boldsymbol{q}^T, \dot{\boldsymbol{q}}^T]^T$ for rigid systems, or $\boldsymbol{x} = [\boldsymbol{q}^T, \dot{\boldsymbol{q}}^T, \boldsymbol{\eta}^T, \dot{\boldsymbol{\eta}}^T]^T$ when including shaft flexibility.

n Number of rigid-body degrees of freedom.

m Number of flexible shaft modal coordinates.

$\theta_1, \theta_2, \ldots$ Joint angles (shoulder, elbow, wrist, etc.).

$\boldsymbol{p}_{\text{head}} \in \mathbb{R}^3$ Position of the clubhead center of mass.

$\boldsymbol{v}_{\text{head}} \in \mathbb{R}^3$ Velocity of the clubhead.

$\boldsymbol{p}_{\text{hand}} \in \mathbb{R}^3$ Position of the grip/hand point.

Dynamics and Forces

$\boldsymbol{M}(\boldsymbol{q}) \in \mathbb{R}^{n\times n}$ Mass (inertia) matrix, symmetric positive definite.

$\boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}}) \in \mathbb{R}^{n\times n}$ Coriolis and centrifugal force matrix.

$\boldsymbol{g}(\boldsymbol{q}) \in \mathbb{R}^{n}$ Gravitational torque vector.

$\boldsymbol{d}(\boldsymbol{q}, \dot{\boldsymbol{q}}) \in \mathbb{R}^{n}$ Dissipative (damping/friction) forces.

$\boldsymbol{u} \in \mathbb{R}^{p}$ Control input vector (joint torques applied by muscles/actuators).

$\boldsymbol{B} \in \mathbb{R}^{n\times p}$ Input selection matrix ($p \leq n$).

$\boldsymbol{\tau}_{\text{total}}$ Total generalized force at a joint.

$\boldsymbol{\tau}_{\text{drift}}$ Drift (passive) component of generalized force.

$\boldsymbol{\tau}_{\text{input}}$ Active (muscular) component of generalized force.

$\boldsymbol{\lambda}$ Constraint force vector (Lagrange multipliers).

Affine Control Decomposition

$\boldsymbol{x}$ The full system state, equivalent to $\boldsymbol{x}$ (used interchangeably for readability).

$f(\boldsymbol{x})$ Drift vector field: the passive evolution with zero applied control ($\boldsymbol{u} = \boldsymbol{0}$).

$G(\boldsymbol{x})$ Input (control) matrix field, mapping control inputs to state derivatives.

$G(\boldsymbol{x})\boldsymbol{u}$ Control vector field: the effect of muscular control effort.

$\dot{\boldsymbol{x}} = f(\boldsymbol{x}) + G(\boldsymbol{x})\boldsymbol{u}$ The control-affine system equation.

$\boldsymbol{x}^{\text{ZTCF}}(t)$ Forward Zero-Torque Counterfactual trajectory (evolution of the declared effective plant with $\boldsymbol{u} = \boldsymbol{0}$).

$\boldsymbol{a}_{\text{ZVCF}}(\boldsymbol{q}, \boldsymbol{z})$ Instantaneous Zero-Velocity Counterfactual acceleration with velocity and declared control set to zero.

$\text{DCR}_{W,\mathcal{U}}(\boldsymbol{x})$ Drift-Control Ratio in a declared acceleration or task-projected space: $\|W\boldsymbol{a}_{\text{drift}}\| / (\sup_{u\in\mathcal{U}} \|W\boldsymbol{B}_a u\| + \varepsilon)$.

Constraint Mechanics

$\boldsymbol{\Phi}(\boldsymbol{q}) = \boldsymbol{0}$ Holonomic constraint equations.

$\boldsymbol{J}_c(\boldsymbol{q})$ Constraint Jacobian: $\boldsymbol{J}_c = \frac{\partial \boldsymbol{\Phi}}{\partial \boldsymbol{q}}$.

$\boldsymbol{J}_{\text{loop}}$ Loop-closure Jacobian for parallel mechanisms.

$\boldsymbol{\lambda}$ Constraint forces (Lagrange multipliers).

$\mathcal{N}(\boldsymbol{J}_c)$ Null space of the constraint Jacobian (feasible motion directions).

$\mathcal{R}(\boldsymbol{J}_c^T)$ Range of $\boldsymbol{J}_c^T$ (directions in which constraints exert force).

Shaft and Flexibility

$\boldsymbol{\eta} \in \mathbb{R}^m$ Shaft modal coordinates (flexible deformation modes).

$\dot{\boldsymbol{\eta}}$ Shaft modal velocities.

K_s Shaft bending stiffness (or modal stiffness matrix).

D_s Shaft damping coefficient (or modal damping matrix).

EI Shaft flexural rigidity (Young's modulus $\times$ second moment of area).

Energy and Power

$T(\boldsymbol{q}, \dot{\boldsymbol{q}})$ Kinetic energy: $T = \frac{1}{2}\dot{\boldsymbol{q}}^T \boldsymbol{M}(\boldsymbol{q})\dot{\boldsymbol{q}}$.

$V(\boldsymbol{q})$ Potential energy (gravitational + elastic).

$P_{\text{drift}}(t)$ Power delivered by drift (passive) forces.

$P_{\text{input}}(t)$ Power delivered by active (muscular) forces: $P_{\text{input}} = \boldsymbol{u}^T\dot{\boldsymbol{q}}$.

$P_{\text{constraint}}(t)$ Power associated with constraint forces (identically zero for ideal constraints).

W Work (time-integrated power).

Biomechanics and Tissue

$a_i \in [0, 1]$ Muscle activation level for muscle i.

F_i^M Muscle force for muscle i.

ℓ_i^M Muscle fiber length.

v_i^M Muscle fiber contraction velocity.

K_{tissue} Tissue stiffness (for fascia, tendon, ligament).

σ Stress (force per unit area in tissue).

ε Strain (deformation per unit length in tissue).

Pendulum Models

L_1, L_2, L_3 Link lengths (proximal, medial, distal).

m_1, m_2, m_3 Link masses.

I_1, I_2, I_3 Link moments of inertia about their centers of mass.

ℓ_1, ℓ_2, ℓ_3 Distance from joint to center of mass for each link.

τ_1, τ_2, τ_3 Applied torques at each joint.

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Foundations

Chapter 1

Why Physics Matters in Golf

> "The golf swing is not a golf swing at all. It is a human being moving his body to create motion in an external object."

In Plain Language 1.1. The Central Mystery

You've hit a thousand golf balls. Some days your swing feels effortless and the ball soars straight. Other days you're "thinking" through every motion and the ball veers wildly. What changed? Your clubs didn't. The grass didn't. Yet something fundamental shifted.

Traditional golf instruction says: "Keep your head still. Rotate your hips. Follow through." These cues sometimes help, but they're incomplete. They describe *what* you should do, not *why* those motions produce the desired ball flight, and crucially, they don't explain why the same mental cue works for one golfer but sabotages another.

This book offers a different lens: a physical one. The golf swing is not primarily a puzzle of choreography—it's a puzzle of **forces and torques**.

1.1 What We Actually Know About Swings

If you've ever taken a golf lesson, you've heard variations of the same advice:

- Swing on a plane.
- The downswing begins with the lower body.
- Keep your trailing arm bent.
- Rotate your shoulders more than your hips.
- Stay behind the ball.

These observations are useful. Tour professionals do exhibit these patterns. But notice what's missing: *cause*. Why does rotating the shoulders more than the hips produce a straighter shot? What physical principle makes a wider swing arc more forgiving?

Consider a simpler question: when you release a golf club at the top of your backswing and let it fall (don't actually try this—yet), what trajectory does it take? How fast is it moving when it reaches the bottom? How much does gravity contribute versus the acceleration of your arm?

Most golfers have never asked these questions. Coaches rarely answer them. Yet the answers are sitting in the physics of rigid bodies in motion.

1.2 Two Incomplete Stories

The Kinematic Story

Much of modern golf instruction—especially on video—focuses on **kinematics**: the description of motion without reference to forces. The kinematic perspective describes the motion: "At this point in the swing, the hips have rotated through angle θ_{hip} and the shoulders through angle θ_{shoulder}."

This is useful for pattern recognition and comparison. Kinematic analysis reveals consistent patterns in high-performing swings [Jorgensen, 1994a, Penner, 2003a]. But kinematics is *descriptive*, not *causal*. Observing that certain hip and shoulder angles occur at specific phases does not explain why those angles produce the desired impact conditions, or what happens when angles differ from the typical pattern. Crucially, kinematic description does not directly reveal the underlying muscular forces and torques that generate the observed motion.

The Strength and Flexibility Story

Another narrative prevalent in golf fitness emphasizes strength, flexibility, and athletic capacity as primary determinants of swing performance. The claim is that muscular force production and range of motion directly determine swing speed and trajectory.

There is indeed a relationship here: limited strength sets a ceiling on the maximum forces the muscles can generate, and limited flexibility constrains the possible positions the body can reach. But strength and flexibility are necessary but not sufficient conditions. The question remains: given a particular strength and flexibility profile, what forces do the muscles actually generate at each instant in the swing, and how do those forces interact with gravity and inertia to produce the observed motion?

1.3 The Missing Story: Forces

Here's what both stories miss: **at every instant of the swing, your body generates forces and torques. These forces determine where the club goes and how fast it moves.**

The shape of your swing—the kinematics—is the *consequence* of these forces, not the cause. Similarly, your strength merely sets an upper limit on the forces you can generate; it doesn't dictate which forces you generate at each moment.

When a coach says "rotate your hips," what they mean is: "generate a torque around your vertical axis that accelerates your hip rotation." When they say "release the club," they mean: "stop generating a torque at your wrist, allowing gravity and the club's momentum to accelerate it downward and forward."

Physics provides a framework to quantify these phenomena precisely. It provides a language to distinguish among:

- This torque comes from your muscles pulling (what we'll call $\boldsymbol{u}$).
- This torque comes from gravity doing work (what we'll call $f(\boldsymbol{x})$).
- This torque comes from your arm moving fast, creating **centrifugal effects** ($f(\boldsymbol{x})$ again).
- This force is **$\boldsymbol{x}$-dependent**—it depends on where your arms are and how fast they're moving.

Definition 1.1. Drift and Control — The Two Sources of Motion

Throughout this book, we decompose every force in the golf swing into two categories:

- **Drift** ($f(\boldsymbol{x})$): Forces that arise from the current state of motion—gravity, centrifugal effects, and momentum. These happen automatically, with no muscular effort.
- **Control** ($\boldsymbol{u}$): Torques generated by contracting muscles. These are what you actively command.

The mathematical separation of drift from control is the backbone of this entire book.

Once you can attribute each force to its source, you can understand what you're actually trying to control.

In Plain Language 1.2. Why Attribution Matters

Here's a concrete example: suppose your club is moving at speed v at impact, where v is typically in the range 100–120 mph for skilled golfers [Jorgensen, 1994a]. The acceleration of the clubhead is then $a = v^2/L$, where L is the arm-club length. This acceleration is often expressed in multiples of gravitational acceleration g. The natural question is: does the muscular torque at the wrist account for all of this acceleration?

The answer is no. Some of that centripetal acceleration comes from the **tension in your arm**—a constraint force that has nothing to do with muscle contraction. Some comes from gravity. Some comes from the high-speed rotation of your arm (a **velocity-dependent inertial force**). Your wrist torque contributes, yes, but quantifying *how much* is a physics problem.

Until you can attribute each force to its source—gravity, muscle, constraint, or momentum—you're flying blind.

1.4 Two Kinds of Forces: A Preview

This entire book pivots on a single insight: **all forces in your swing come from one of two sources**. Figure 1.1 illustrates the conceptual framework: the golfer's arm and club

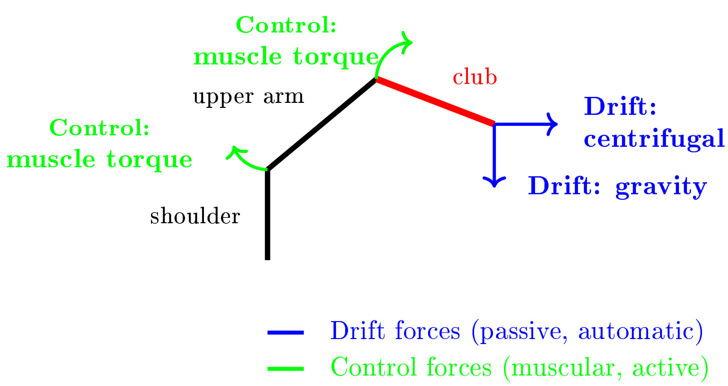


Figure 1.1: Schematic of Drift and Control Forces in the Golf Swing. Drift Forces (Blue) Include Gravity and Velocity-Dependent Inertial Effects That Act Automatically Based on the System State. Control Forces (Green) Are Active Muscular Torques Applied by the Nervous System to Steer the Swing.

system experiences both passive drift forces (gravity and inertial effects) and active control forces (muscular torques), and the swing trajectory results from their combination.

Understanding the distinction between these two sources of force is the foundation for all subsequent analysis.

Drift: The Physics You Can't Control

When you're driving a car at 60 mph and you lift your hands off the wheel for a moment, the car doesn't stop turning just because you stopped steering. The **momentum and the road** keep the car moving in its curved path.

In your golf swing, similar "passive" physics is constantly at work. If you swung your arm and then just let go of the club, gravity would pull it down. The rotation of your arm would carry the club in a curved path. These are **dynamics** that happen whether you "think" about them or not.

We call this component $f(\boldsymbol{x})$. It's what the system does on autopilot, determined by the laws of physics given your body's current state (positions and velocities).

Control: What Your Muscles Add

On top of drift, you actively generate torques by contracting muscles. These are what we call $\boldsymbol{u}$ forces. They add to (or subtract from) the passive drift, steering the system toward your desired outcome.

An important observation emerges from studying swings across different skill levels: the distribution of drift versus control forces differs systematically. Some golfers generate swings where drift forces carry the club through most of the downswing, with control forces applied primarily for steering and timing. Other golfers appear to generate muscular control throughout the swing, actively resisting the passive drift dynamics rather than working with them. This distinction—between systems where the drift-control ratio is high versus low—proves diagnostic for understanding swing efficiency and consistency.

Drift vs. Control 1.1. Drift vs. Control in a Simple Fall

Imagine your arm holding a mass m at shoulder height. You're exerting an upward force equal to the weight's gravitational force (call it $W = mg$). This is $\boldsymbol{u}$—your muscle generating force to hold the mass steady.

Now you release it. What happens? $f(\boldsymbol{x})$ takes over. Gravity pulls downward. The mass accelerates at g (approximately 9.8 m/s^2 or 32 ft/s^2). You're not doing anything—physics is.

In a golf swing: when your arm is at the top of the backswing holding the club in a specific position, you're exerting $\boldsymbol{u}$. During the downswing, you begin to reduce this $\boldsymbol{u}$ torque—and $f(\boldsymbol{x})$ (gravity plus the rotational momentum of your arm) does substantial work accelerating the club toward impact. Your muscles then add corrections to steer that accelerating club toward the target. The model predicts that passive dynamics (gravity and momentum from the arm's rotational acceleration) account for a significant fraction of the total downswing acceleration, with active muscular control providing directional and timing refinement.

1.5 The Central Equation

By the end of this book, you'll understand and be able to write this equation:

$$\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{\tau} \tag{1.1}$$

This is the **manipulator equation**. Don't be intimidated. It says:

- $\boldsymbol{M}(\boldsymbol{q})$: the inertia of your body and the club (depends on position).
- $\boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}}$: velocity-dependent forces (Coriolis, centrifugal) that arise when you move fast.
- $\boldsymbol{g}(\boldsymbol{q})$: gravity's torque on your body.
- $\boldsymbol{\tau}$: the *applied* torques—what your muscles generate.

When we rearrange this equation to solve for acceleration (Chapter 5), we get: $\ddot{\boldsymbol{q}} = \boldsymbol{M}^{-1}[-\boldsymbol{C}\dot{\boldsymbol{q}} - \boldsymbol{g}] + \boldsymbol{M}^{-1}\boldsymbol{\tau}$. The first term is *drift*: what physics does on its own. The second term is *control*: what your muscles actively command.

This equation is the spine of the book. Everything flows from understanding its pieces.

1.6 Why a Physics Textbook for Golfers?

You might ask: *physicists have written about golf before [Jorgensen, 1970, 1994a]. Why do we need another book?*

Because nearly all existing physics of golf is written **for physicists**, not **for golfers**. It buries the intuition under pages of equations. Worse, it treats the golf swing as an optimization problem to be solved in hindsight: "What swing parameters maximize distance?" [Penner, 2003a] That's useful for equipment design, but it provides limited insight into the forces and torques that generate a swing.

This textbook is different. We're asking: **what is the structure of the forces and torques that make a swing possible?** Once you understand that structure, you can:

- Understand the physical mechanisms underlying swing trajectories and impact conditions.
- Interpret coaching advice in terms of the underlying force structure (how a cue changes your control relative to the drift).
- Build intuition about constraints on swing mechanics based on the passive dynamics and control authority.
- Improve consistency by understanding which swing parameters are drift-dominated and which require active control.

Key Principle 1.1. Why Physics Matters

The golf swing is fundamentally a problem of forces and torques. Every aspect of your swing—the path, the speed, the consistency, the distance—is determined by the forces you generate and how those forces interact with gravity, inertia, and constraints.

Traditional coaching describes *what* you should do (kinematics). Physics explains *why* that motion produces the result it does, and what forces generate that motion (dynamics).

Understanding the forces lets you:

- Separate passive dynamics (drift) from active control.
- Diagnose and fix flaws.
- Improve consistency by riding the natural physics of your swing.

Everything in this book flows from the manipulator equation: $\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{\tau}$.

1.7 The Double Pendulum: Your Guide

Throughout this book, we'll use a simple model: **the double pendulum**.

Definition 1.2. The Double Pendulum

Two rigid rods ("links") connected by a hinge. The first rod is hinged at a fixed point and can rotate freely. The second rod is hinged to the end of the first and can also rotate freely. Each rod has mass and length.

In golf terms: the first rod is your **upper arm** (shoulder to elbow). The second rod is your **forearm + club** (elbow to club head).

This is a toy model. Real golfers have two arms, a rotating torso, hip and knee joints, and many more degrees of freedom. But here is the key insight: **almost all of the essential physics appears in the double pendulum**.

The equations are complicated enough to be interesting but simple enough to solve (sometimes exactly, often numerically). The physics of gravity, inertia, velocity-dependent forces, and control all show up. The *unforced* double pendulum exhibits chaotic behavior—small changes in initial conditions lead to vastly different trajectories. The controlled golf swing is not chaotic in the same technical sense (active control suppresses the divergence), but the underlying nonlinearity means the system is sensitive to the timing and magnitude of applied torques.

To ground this abstract model in concrete reality, consider what happens when the control input is suddenly withdrawn:

Example 1.1. What Happens If You Let Go Mid-Swing?

Imagine you're mid-swing with your double-pendulum arms. You're generating muscular torques at the shoulder and elbow. Then, midway through the downswing, you simply stop contracting those muscles and let your arms fall freely.

- **What will happen?** Your arms will not continue in a straight path. Instead, gravity will pull the club downward, and the rotational momentum of your arm will curve its path. The club will accelerate, but not in the direction you were controlling.
- **Why?** Because your $\boldsymbol{u}$ disappeared. Only $f(\boldsymbol{x})$ remains—gravity, inertia, and the constraint that your arm is hinged at the shoulder. That drift is perfectly well-defined by the physics. It's deterministic. But it's not the path you intended.
- **What does this tell us?** It tells us that continuous active control is a necessity if the swing is to reach the ball at impact on a predictable trajectory. Even during the downswing, the nervous system must be generating muscular torques at every instant to steer the passive drift dynamics toward impact. The model predicts that swings differing in their drift-control ratio will exhibit different sensitivities to timing variations and perturbations.

In Chapter 3, we'll actually solve the equations for what happens if control is removed mid-swing in a double-pendulum model with realistic golf parameters. The mathematical solution provides precise predictions about the resulting trajectory.

1.8 A Road Map

The remainder of this textbook develops the mathematical and physical framework needed to analyze golf swings quantitatively. We move from foundational definitions of coordinates and state, through the derivation of equations of motion, to the systematic decomposition of forces into drift and control components. Each chapter builds on the previous one, with the goal of equipping you with tools to diagnose and understand swing mechanics from first principles.

Chapter 2: The Language of Motion

Before we can talk about forces, we need a language for describing where your body is and how fast it's moving. We'll introduce **generalized coordinates**—a way to describe the configuration of your body (e.g., shoulder angle, elbow angle). We'll define **state space**—the combination of position and velocity. Every swing, at every instant, is a point in this state space.

Chapter 3: The Double Pendulum

We'll set up the full equations of motion for the double pendulum using **Lagrangian mechanics**. You'll see exactly where the mass matrix $\boldsymbol{M}(\boldsymbol{q})$ comes from (it measures inertia), where the Coriolis and centrifugal terms come from (they're what it means for your arm to move fast), and where gravity comes from. We'll compute all of this step-by-step, with golf numbers.

Chapter 4: Forces and Torques

We'll unpack the manipulator equation, term by term. What does each term do? How large are the forces relative to one another? At impact, the clubhead undergoes large acceleration, often expressed as a multiple of g. Where does that acceleration come from? We'll attribute every force to its source: muscle, gravity, constraint, or momentum.

Chapter 5: Drift and Control

Finally, we'll write the equation in its most useful form:

$$\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}) + \boldsymbol{G}(\boldsymbol{x})\boldsymbol{u} \tag{1.2}$$

where $\boldsymbol{f}(\boldsymbol{x})$ is the drift and $\boldsymbol{G}(\boldsymbol{x})\boldsymbol{u}$ is your control. We'll understand exactly which parts of your swing are passive (drift) and which parts require active muscular command (control). We'll introduce the **Drift-Control Ratio** (DCR)—a diagnostic tool that tells you whether a particular swing phase is drift-dominated or control-dominated.

Empirical observation suggests that swings differ markedly in their drift-control ratio during critical phases like the downswing and acceleration phase. Understanding this distribution is a key diagnostic tool for improving swing consistency and efficiency.

Key Principle 1.2. Key Takeaways

1. **Physics is causal**. The shape of your swing (kinematics) is a consequence of the forces and torques you generate (dynamics).
2. **Forces come from sources**. Every force in your swing is either drift (gravity, momentum, inertia) or control (your muscles) or a constraint (the hinge at your elbow).
3. **Attribution is power**. Once you understand which forces come from which source, you can diagnose problems and improve systematically.

4. **The manipulator equation is the spine of the book**:

$$\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{\tau}$$

Rearranged: $\ddot{\boldsymbol{q}} = \underbrace{\boldsymbol{M}^{-1}[-\boldsymbol{C}\dot{\boldsymbol{q}} - \boldsymbol{g}]}_{\text{drift}} + \underbrace{\boldsymbol{M}^{-1}\boldsymbol{\tau}}_{\text{control}}.$

5. **The double pendulum is your guide**. It's simple enough to understand fully, complex enough to be realistic. Use it to build intuition.

To apply these concepts and develop intuition about forces and dynamics in simple systems, we offer the following exercises. These problems invite you to observe physical phenomena in familiar contexts (falling objects, spinning balls, swinging weights) and to reason about the forces and torques involved. They lay groundwork for the more sophisticated analysis of the golf swing that follows in later chapters.

Exercises 1.1. Chapter Exercises

1. **The Free Fall**. Hold a golf ball at arm's length. Release it. How long until it hits the ground? How fast is it moving when it hits? (Use $g = 32$ ft/s^2.) Now do the same with a heavier ball (baseball). Does it hit the ground faster or slower? Why? (Hint: think about forces.)

2. **The Spinning Ball**. Spin a golf ball on a table (give it backspin). Watch it slow down and stop. What forces are acting on it? What forces are not acting on it? Try the same on a frictionless surface (ice). What's different?

3. **The Pendulum**. Hang a weight from a string at arm's length. Lift it to one side and release. Describe the path it takes. Now, while it's swinging, pull on the string to make it swing faster. What torque are you applying? Does the weight swing in a different direction?

4. **The Mystery of Cues**. You've heard a coach say: "Rotate your hips faster." In terms of forces and torques, what is that coach asking you to do? Where should that torque come from (muscle, gravity, constraint)? What happens if you generate that torque at the wrong time in the swing?

5. **Drift vs. Control**. Throw a baseball as hard as you can. Now throw it at half speed but with perfect aim. In which throw do you need to generate more muscular force? In which throw is drift (gravity) doing more work? Explain.

Chapter 2

The Language of Motion

> "To measure is to know. If you cannot measure it, you cannot understand it."

In Plain Language 2.1. Why We Need a Language

Before you can apply physics to the golf swing, you need a precise way to describe it. Words are inherently ambiguous; mathematics is not.

Consider a statement like "the hips rotate early in the swing." This is qualitative and imprecise. Quantitatively: what is the hip rotation angle θ_{hip} at each instant? What is its time rate of change $\dot{\theta}_{\text{hip}}$? At which times during the swing does the described behavior occur? Physics demands precise, mathematical descriptions.

This chapter introduces the language physicists use to describe motion: **coordinates**, **state**, and **state space**. These tools make motion descriptions precise and mathematical. With a precise description, you can apply Newton's laws to predict future motion, analyze the forces required to produce observed motion, and understand the mechanism by which the system evolves.

2.1 Coordinates: Measuring Position

The simplest place to start: how do we describe where your body is?

Cartesian Coordinates

One way is Cartesian coordinates. Point your finger at the club head. In space, you can describe its location with three numbers: its distance east, its distance north, and its distance up (or any three perpendicular directions). These are (x, y, z) coordinates.

This works. But it's clumsy for describing a golf swing. Why? Because the club head is constrained by your arm. It can't move independently. You can't move it 10 feet to the left without moving your shoulder.

Generalized Coordinates

Instead, physicists use **generalized coordinates**: a minimal set of numbers that fully describe the configuration of a system.

Definition 2.1. Generalized Coordinates

A set of independent variables $\boldsymbol{q} = [q_1, q_2, \ldots, q_n]$ such that:

1. Every possible position of the system can be described by choosing a value for each q_i.
2. Each q_i is independent—you can change one without automatically changing the others.
3. The number of variables n equals the number of **degrees of freedom** (DOF).

For the double pendulum model of the arm, the natural choice of generalized coordinates is the two joint angles: θ_1 at the shoulder and θ_2 at the elbow. These are the minimal set of parameters needed to completely specify the configuration. Figure 2.1 shows this setup.

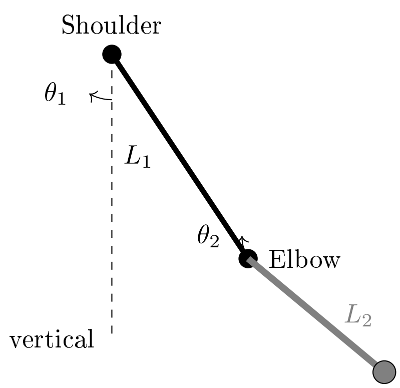


Figure 2.1: Double Pendulum Model of the Arm in Generalized Coordinates. The Shoulder Angle θ_1 Is Measured From Vertical, and the Elbow Angle θ_2 Is Measured Relative to the Upper Arm.

Example 2.1. A Simple Pendulum

A mass hanging from a string can only move on a circle (the string doesn't stretch). Its position is completely described by one number: the angle θ from vertical.
$\boldsymbol{q} = [\theta]$. There is 1 DOF.
If you use Cartesian coordinates, you need two numbers: the x and y position of the mass. But these two numbers are not independent. If you change x, then y must change too (to keep the string length fixed). Generalized coordinates eliminate this redundancy.

For the golf swing, the natural choice is **joint angles**.

Definition 2.2. Joint Angles in the Golf Swing

- $q_1 = \theta_s$: the angle of the upper arm at the shoulder, measured from some reference (e.g., vertical).
- $q_2 = \theta_e$: the angle of the forearm + club at the elbow, measured from the upper arm.

For a two-link model (the double pendulum), we need exactly two numbers to describe the configuration.

$$\boldsymbol{q} = \begin{bmatrix} \theta_s \\ \theta_e \end{bmatrix}$$

In Plain Language 2.2. From Angles to Positions

Once you've chosen generalized coordinates, you can compute the Cartesian position of any point.
Define a coordinate frame with the shoulder at the origin, x pointing horizontally to the right, and y pointing vertically downward. Measure θ_s clockwise from the downward vertical, so $\theta_s = 0$ means the arm hangs straight down.
Suppose the upper arm has length L_1 and the forearm+club has length L_2. If the shoulder is at the origin:

$$\begin{aligned} x_{\text{elbow}} &= L_1 \sin(\theta_s) && (2.1) \\ y_{\text{elbow}} &= -L_1 \cos(\theta_s) && (2.2) \\ x_{\text{clubhead}} &= L_1 \sin(\theta_s) + L_2 \sin(\theta_s + \theta_e) && (2.3) \\ y_{\text{clubhead}} &= -L_1 \cos(\theta_s) - L_2 \cos(\theta_s + \theta_e) && (2.4) \end{aligned}$$

So from two numbers $[\theta_s, \theta_e]$ you can compute the full 3D position of every point on the arm. The constraint (that the links are rigid) is automatically satisfied.

2.2 State: Position and Velocity

Knowing where your arm is right now doesn't tell you what will happen next. You also need to know how fast it's moving.

Definition 2.3. Velocity in Generalized Coordinates

The velocity of a joint angle is its time derivative:

$$\dot{q}_i = \frac{dq_i}{dt}$$

For example, $\dot{\theta}_s$ is the rate of rotation of the shoulder (in radians per second).

Example 2.2. Shoulder Rotation Speed

During the downswing, a golfer's shoulder may be rotating at $\dot{\theta}_s \approx 10$ rad/s (representative values from biomechanical studies are on the order of 10-12 rad/s, depending on swing speed [Nesbit, 2005]).
What does this mean physically? If the shoulder could hold this rotation rate steady, the golfer would rotate a full 2π radians in $(2\pi \text{ rad})/(10 \text{ rad/s}) \approx 0.63$ seconds.
In reality, $\dot{\theta}_s$ varies throughout the swing. The velocity is lower at the top of the backswing (winding up), accelerates during the transition, reaches a peak during the downswing, and decreases toward impact. At any given instant, $\dot{\theta}_s$ represents the instantaneous rotation rate.

Now we can define the full state:

Definition 2.4. State of a System

The **state** $\boldsymbol{x}$ is the complete information needed to predict the future motion. It includes both position and velocity:

$$\boldsymbol{x} = \begin{bmatrix} \boldsymbol{q} \\ \dot{\boldsymbol{q}} \end{bmatrix} = \begin{bmatrix} q_1 \\ q_2 \\ \vdots \\ q_n \\ \dot{q}_1 \\ \dot{q}_2 \\ \vdots \\ \dot{q}_n \end{bmatrix}$$

For a double pendulum, the state has 4 components:

$$\boldsymbol{x} = \begin{bmatrix} \theta_s \\ \theta_e \\ \dot{\theta}_s \\ \dot{\theta}_e \end{bmatrix}$$

In Plain Language 2.3. Why State is Necessary

Imagine you're at the top of the backswing. Your shoulder angle is $\theta_s = 90$ (horizontal). That's your position. But is the downswing about to begin? You can't tell from position alone.
If $\dot{\theta}_s = 0$ (you're not rotating), you're just about to start. If $\dot{\theta}_s > 0$ (you're already rotating), the downswing is well underway.
Physics has a fundamental rule: the future motion is determined by the current state—position and velocity together—not by position alone. Given $\boldsymbol{x}$ at this instant and the forces acting on the system, you can predict the state a tiny fraction of a second later. Then use the new state to predict the next moment. This iterative

prediction is the foundation of dynamical systems analysis.

2.3 Phase Space: Where Every Swing Lives

Phase space provides a geometric framework for visualizing how a system evolves. Instead of tracking position and velocity separately, we can think of them as a single point moving through a high-dimensional space. This perspective reveals structure in the dynamics that may not be obvious from looking at individual time histories.

Once we have the concept of state, we can visualize it geometrically.

Definition 2.5. Phase Space

The space of all possible states. If there are n degrees of freedom, phase space has $2n$ dimensions: n for position and n for velocity.
For a double pendulum: 4 dimensions (2 angles, 2 angular velocities).

At any instant during your swing, your arm is at some point in this 4D phase space. As you swing, that point traces out a path—a **trajectory**.

In Plain Language 2.4. Thinking in 4 Dimensions

Four dimensions is hard to visualize. But you can think of it as a 2D surface (the position, θ_s and θ_e) with a velocity attached to each point.
Imagine a piece of paper with θ_s on the horizontal axis and θ_e on the vertical axis. Every point on this paper is a possible configuration. Now, at each point, imagine a vector showing $[\dot{\theta}_s, \dot{\theta}_e]$—the direction and speed you're moving in state space.
During the backswing, you might follow a path from $(90, 0)$ to $(150, -50)$ (shoulder opens, elbow closes relative to arm). As you move along this path, your velocity vector changes. At the top, you're still for a moment. Then you start the downswing.
An entire golf swing is one continuous curve in this 4D space.

Here's a key insight: **if you know the forces, you can predict the trajectory**.

Given any state $\boldsymbol{x}$, the forces determine $\ddot{\boldsymbol{q}}$ (the acceleration). Newton's second law says: $\boldsymbol{F} = m\boldsymbol{a}$. So if you know the forces, you know the acceleration. And if you know the acceleration, you can predict where the state will be a moment later. This is how physicists predict motion.

2.4 Configuration Space: The Space of Positions

Sometimes it's useful to focus just on position, ignoring velocity. This is called configuration space.

Definition 2.6. Configuration Space

The space of all possible positions (configurations) of the system. For n degrees of freedom, configuration space has n dimensions.
For a double pendulum: 2 dimensions (θ_s, θ_e).

During a golf swing, your configuration traces a path in configuration space. Your backswing, transition, downswing, and follow-through are all portions of this path.

Example 2.3. Configuration Space of a Double Pendulum

Imagine a plot with θ_s on the x-axis (from -90 to $+150$, representing the full range of shoulder motion) and θ_e on the y-axis (from -80 to $+80$, representing the full range of elbow angle). Configuration space for the double pendulum is then a 2D region within this rectangle.
A representative golf swing path in this space exhibits the following qualitative behavior:

- **Address**: $(\theta_s, \theta_e) = (0, 0)$ (both angles at their reference values).
- **Backswing**: progressive increase in both θ_s (shoulder rotation) and decrease in θ_e (elbow closure).
- **Top**: maximum values of shoulder rotation with maximum elbow closure.
- **Downswing**: reversal of angles back toward their initial values.
- **Impact**: return to near-initial configuration.
- **Follow-through**: further reversal beyond initial configuration.

Different individuals will have different trajectories through configuration space, reflecting variations in anatomy, strength, and control strategy. However, all trajectories are constrained by the physical laws encoded in the equations of motion.

2.5 Constraints: Not All Configurations Are Allowed

Here's an important limitation: not every point in configuration space is reachable.

Definition 2.7. Constraints

Physical limitations that restrict which configurations are possible. Common constraints in golf:

- **Range of motion**: your shoulder can only rotate so far, your elbow can only bend so much.
- **Rigidity**: your arm bones are rigid, so the elbow and shoulder positions are not independent.

- **Friction**: your grip prevents the club from slipping relative to your hand.

The set of reachable configurations forms a region in configuration space. Your swing must stay within this region. If you try to force your arm into an impossible configuration, something breaks (injury).

Constraint Insight 2.1. Constraints as Walls in Space

Think of configuration space as a room. Some regions are forbidden (your arm can't hyperextend). The boundary of the forbidden regions are the constraints.
When you approach the boundary—such as when reaching maximum shoulder rotation—a **constraint force** appears. The arm cannot pass beyond the limit; rather, biological tissues (muscles, tendons, bone) exert a reaction force that enforces the constraint.
This is crucial: constraint forces are not muscle forces. They're reactive. They appear automatically to enforce the constraint. We'll see later that they contribute to the total force in the equation of motion, but they're not part of your **control**.

2.6 Degrees of Freedom: How Many?

A natural question: how many degrees of freedom does a golfer actually have?

The answer depends on the model:

Example 2.4. Counting Degrees of Freedom

Minimal model (double pendulum): 2 DOF.

- Shoulder rotation angle.
- Elbow hinge angle.

More realistic model: 6–8 DOF.

- Shoulder (3 angles: rotation, flexion, abduction).
- Elbow (1 angle: hinge).
- Wrist (2 angles: hinge, twist).
- Torso rotation (1 angle).

Full human model: 40+ DOF.

- Spine (multiple joints).
- Both shoulders and elbows.
- Wrists.
- Hips.

- Knees.
- Ankles.

Why not use the full model? Because it becomes intractable. The equations get so complicated that you lose physical intuition. The trick in physics is to choose a model with *just enough* complexity to capture the essential behavior.
The double pendulum (2 DOF) is the sweet spot for learning. Almost all of the interesting physics of the golf swing appears. When you understand the 2-DOF case, you can add complexity.

2.7 A Worked Example: Two-DOF Planar Coordinates

Let's work through a complete coordinate system for the double pendulum, similar to the golf swing.

Setup

We use the following representative parameters for illustration. These values are typical for an adult golfer, though individual values vary with body size, club length, and swing characteristics.

- Shoulder joint is fixed at the origin.
- Upper arm: length $L_1 = 0.35$ m, mass $M_1 = 2.5$ kg, center of mass at $L_{1,\text{cm}} = 0.175$ m from the shoulder.
- Forearm + club: length $L_2 = 1.0$ m, mass $M_2 = 1.5$ kg, center of mass at $L_{2,\text{cm}} = 0.5$ m from the elbow.
- Generalized coordinates: $q_1 = \theta_s$ (shoulder angle from vertical, in radians), $q_2 = \theta_e$ (elbow angle relative to the upper arm, in radians).

Note on canonical parameters: The values above are the **canonical double-pendulum parameters** used throughout Chapters 2–5 and referenced in later chapters. Chapter 3 provides the full parameter table including moments of inertia. Chapters 6 and 8 use different parameters (smaller club mass, different link lengths) for specific worked examples and explicitly state their parameter choices.

Position Kinematics

The position of the elbow relative to the shoulder:

$$\boldsymbol{r}_{\text{elbow}} = \begin{bmatrix} L_1 \sin(\theta_s) \\ -L_1 \cos(\theta_s) \end{bmatrix} \tag{2.5}$$

The position of the clubhead relative to the shoulder:

$$\boldsymbol{r}_{\text{club}} = \begin{bmatrix} L_1 \sin(\theta_s) + L_2 \sin(\theta_s + \theta_e) \\ -L_1 \cos(\theta_s) - L_2 \cos(\theta_s + \theta_e) \end{bmatrix} \tag{2.6}$$

The center of mass of link 1:

$$\boldsymbol{r}_{\text{cm},1} = \begin{bmatrix} L_{1,\text{cm}} \sin(\theta_s) \\ -L_{1,\text{cm}} \cos(\theta_s) \end{bmatrix} \tag{2.7}$$

The center of mass of link 2:

$$\boldsymbol{r}_{\text{cm},2} = \begin{bmatrix} L_1 \sin(\theta_s) + L_{2,\text{cm}} \sin(\theta_s + \theta_e) \\ -L_1 \cos(\theta_s) - L_{2,\text{cm}} \cos(\theta_s + \theta_e) \end{bmatrix} \tag{2.8}$$

Velocity Kinematics

To find velocity, differentiate the positions:

$$\dot{\boldsymbol{r}}_{\text{elbow}} = \begin{bmatrix} L_1 \cos(\theta_s)\dot{\theta}_s \\ L_1 \sin(\theta_s)\dot{\theta}_s \end{bmatrix} = L_1\dot{\theta}_s \begin{bmatrix} \cos(\theta_s) \\ \sin(\theta_s) \end{bmatrix} \tag{2.9}$$

$$\dot{\boldsymbol{r}}_{\text{club}} = \begin{bmatrix} L_1 \cos(\theta_s)\dot{\theta}_s + L_2 \cos(\theta_s + \theta_e)(\dot{\theta}_s + \dot{\theta}_e) \\ L_1 \sin(\theta_s)\dot{\theta}_s + L_2 \sin(\theta_s + \theta_e)(\dot{\theta}_s + \dot{\theta}_e) \end{bmatrix} \tag{2.10}$$

In Plain Language 2.5. Jacobian Intuition

There is a clean relationship here. The velocities are linear in the angular velocities $\dot{\theta}_s$ and $\dot{\theta}_e$. This relationship can be written compactly using the **Jacobian** matrix $\boldsymbol{J}(\boldsymbol{q})$:

$$\dot{\boldsymbol{r}} = \boldsymbol{J}(\boldsymbol{q})\dot{\boldsymbol{q}}$$

The Jacobian depends on the current configuration $\boldsymbol{q}$. It maps angular velocities (the changes in angles) to linear velocities (the changes in Cartesian positions).
This is why golfers talk about "leverage": at certain positions, small changes in $\dot{\theta}_s$ (shoulder rotation) produce large changes in the clubhead velocity. That's a large Jacobian. At other positions, the leverage is lower.

Angular Velocity

The angular velocities are simpler—they're just the derivatives of the angles:

$$\omega_1 = \dot{\theta}_s \quad \text{(rotation rate of the upper arm)} \tag{2.11}$$

$$\omega_2 = \dot{\theta}_s + \dot{\theta}_e \quad \text{(rotation rate of the forearm + club)} \tag{2.12}$$

Notice that ω_2 depends on both $\dot{\theta}_s$ and $\dot{\theta}_e$. The forearm rotates at the shoulder rate plus the elbow rate. This is why the two joints are coupled.

2.8 Constraints in the Double Pendulum Model

For the idealized double pendulum, there are no constraints other than rigidity (the links don't stretch). But in reality, the golf swing has constraints:

Constraint Insight 2.2. Real Constraints in Golf

- **Range limits**: $\theta_s \in [-90, 150]$, $\theta_e \in [-80, 80]$ (approximate).
- **Grip constraint**: the club doesn't rotate freely at the wrist; it's held firmly. This reduces effective DOF.
- **Friction at ground**: if you're standing (not moving), your feet don't slip. This constrains torso rotation.
- **Inextensibility**: your arm length doesn't change (the links are rigid).

For a first model, we ignore most of these. The double pendulum assumes rigid links and no joint limits. This makes the math tractable. Later, we'd add constraints back in.

When a constraint is active (e.g., you hit the range limit of shoulder rotation), it creates a **constraint force**—a reactive force that prevents you from exceeding the limit. We'll see this in the force equations later.

2.9 Phase Space Portrait of a Swing

Let's put this together and visualize what a swing looks like in phase space.

Example 2.5. A Swing Trajectory in Phase Space

Consider a golf swing modeled as a double pendulum. At each instant t, the state is:

$$\boldsymbol{x}(t) = \begin{bmatrix} \theta_s(t) \\ \theta_e(t) \\ \dot{\theta}_s(t) \\ \dot{\theta}_e(t) \end{bmatrix}$$

A representative swing trajectory evolves through the following sequence of states:

Phase	θ_s	θ_e	$\dot{\theta}_s$	$\dot{\theta}_e$
Initial position	0	0	0	0
Early swing	60	-30	~ 1 rad/s	~ 0
Maximum rotation	150	-70	0	0
Early acceleration	140	-65	~ 3 rad/s	~ 0
Acceleration phase	80	-30	~ 9 rad/s	~ 2 rad/s
Impact	0	0	~ 10 rad/s	~ 9 rad/s
Deceleration	-60	$+30$	~ 3 rad/s	~ 2 rad/s

This swing is a path in 4D phase space with several key features:

- At maximum rotation, both angular velocities are zero momentarily—the motion reverses direction.
- During the acceleration phase, both $\dot{\theta}_s$ and $\dot{\theta}_e$ increase substantially.

- At impact, the angles return to their initial values, but the angular velocities are much larger in magnitude.
- After impact, the system continues to evolve under the influence of the dynamics and any muscular control.

Given the forces and initial conditions at any instant, the subsequent trajectory is determined by Newton's second law applied to the system.

2.10 From Coordinates to Forces

Now that we have a language for describing motion (coordinates, state, phase space), we can ask the big question: what forces determine how the state evolves?

That's the subject of the next chapters. But the setup is in place. We have:

- A generalized coordinate system $\boldsymbol{q}$ that describes positions.
- A state $\boldsymbol{x} = [\boldsymbol{q}, \dot{\boldsymbol{q}}]$ that describes position and velocity.
- A phase space where every possible state lives.
- A trajectory that traces the swing through phase space.

From here, physics says: given the forces, the trajectory is determined. And given the trajectory, we can work backward to infer the forces.

Key Principle 2.1. Key Takeaways

1. **Generalized coordinates** $\boldsymbol{q}$ are the minimal set of numbers needed to describe the configuration of a system. For the golf swing, joint angles are a natural and convenient choice.
2. **Velocity** $\dot{\boldsymbol{q}}$ is the rate of change of position. Angular velocities at impact are significant and generate large inertial and Coriolis forces that contribute to the total dynamics.
3. **State** $\boldsymbol{x} = [\boldsymbol{q}, \dot{\boldsymbol{q}}]$ is the complete information needed to predict future motion given the forces. It lives in phase space, a $2n$-dimensional space.
4. **Configuration space** is the space of positions alone (dimension n). A golf swing traces a path in configuration space.
5. **Constraints** are limitations on which configurations are physically accessible. Constraint forces appear automatically to enforce them and become important when configurations approach their limits.
6. **The Jacobian** relates changes in joint angles to changes in Cartesian positions. Its magnitude varies with configuration and determines how joint velocities map to linear velocities of the end-effector.

7. Every instant during a swing corresponds to a point in phase space. Given the forces at that instant, Newton's second law determines the acceleration and hence the location of the point an infinitesimal time later.

Exercises 2.1. Chapter Exercises

1. **Degrees of Freedom**. For each of the following systems, count the degrees of freedom.
   (a) A golf ball rolling on a table (assume no slipping).
   (b) A single pendulum (one mass on a massless string).
   (c) A double pendulum (two masses, two hinges).
   (d) Your upper body during a golf swing (shoulder, elbow, wrist, assume torso is fixed).
2. **Configuration Space**. Sketch the configuration space of a double pendulum. On the axes, put θ_s (shoulder, horizontal axis) from -90 to $+150$ and θ_e (elbow, vertical axis) from -80 to $+80$. Mark the regions that are reachable (physically possible) and unreachable. Then sketch a typical golf swing path in this space.
3. **Velocity Calculation**. A golfer's shoulder rotates at $\dot{\theta}_s = 500/s = 8.7$ rad/s. Using the canonical upper arm length $L_1 = 0.35$ m (Chapter 3), What is the linear speed of the elbow? (Hint: $v = r\omega$.)
4. **Jacobian Intuition**. At address, your shoulder angle is $\theta_s = 0$ (arm vertical). At this position, is the clubhead moving mostly up/down or mostly left/right when you rotate your shoulder? What about when $\theta_s = 90$ (arm horizontal)? Explain in terms of the Jacobian.
5. **Phase Space Trajectory**. Sketch a 2D "slice" of phase space: the $(\theta_s, \dot{\theta}_s)$ plane (ignoring the elbow for simplicity). Mark the location of address, top of backswing, and impact. Draw the swing trajectory connecting them. What does the trajectory look like? Does it ever loop back on itself?
6. **Constraints and Range**. Your shoulder can rotate from $\theta_s = -90$ (fully across your body) to $\theta_s = +150$ (fully behind you). What happens if you try to force $\theta_s = +160$? Where does the force come from? Is it a muscle force or a constraint force?

Chapter 3

The Double Pendulum: Golf's Simplest Useful Model

"The goal of physics is not to describe how nature is but to figure out how to simplify what nature does."

In Plain Language 3.1. Why This Model?

A complete model of the human body in motion would involve many degrees of freedom: the spine, hips, knees, ankles, and the various joints of the arm all move during a golf swing.

However, the golf swing exhibits a hierarchical structure where the primary motion driving club head velocity comes from the shoulder and elbow joints. The torso rotation provides the base motion, and higher-order joints (wrist, fingers) contribute secondary effects. This motivates a reduced model focused on the dominant degrees of freedom.

The double pendulum—two rigid links connected by a hinge and driven from a fixed pivot—captures the essential dynamical structure of the arm-club system. This simplification is useful because (1) it is analytically tractable, allowing us to derive and solve the equations of motion explicitly, and (2) it preserves the key nonlinear effects (coupling, velocity-dependent forces, gravity) that characterize multi-link systems. The physics revealed by studying this simple model applies qualitatively to more complex systems as well.

Once the double pendulum model is thoroughly understood, it provides a foundation for understanding more realistic models that incorporate additional degrees of freedom.

3.1 The Double Pendulum: Physical Setup

The System

Two rigid rods:

- **Link 1** (upper arm): length L_1, mass M_1, moment of inertia about the elbow I_1.
- **Link 2** (forearm + club): length L_2, mass M_2, moment of inertia about the grip I_2.

Two hinges (frictionless, perfect revolutes):

- Hinge 1 at the shoulder: the first link rotates about this fixed point.
- Hinge 2 at the elbow: the second link rotates about the end of the first link.

Generalized coordinates:

$$\boldsymbol{q} = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} \tag{3.1}$$

where θ_1 is the angle of link 1 (measured from some reference, say vertical), and θ_2 is the **relative** angle of link 2 with respect to link 1.

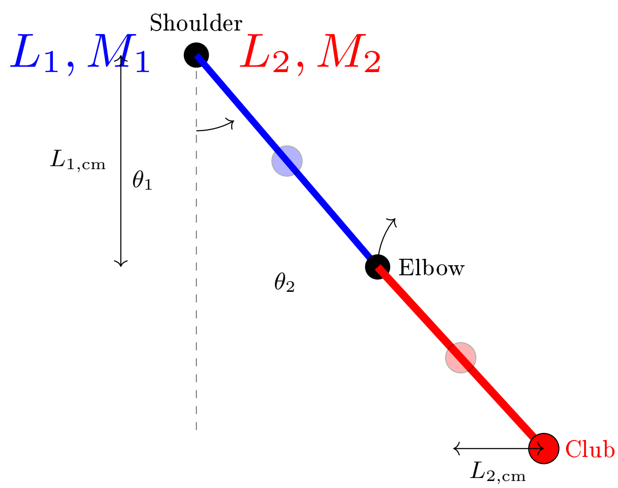


Figure 3.1: Double Pendulum Model of the Golf Swing. Link 1 (Upper Arm, Blue) Rotates About the Shoulder With Angle θ_1 Measured From Vertical. Link 2 (Forearm + Club, Red) Rotates About the Elbow With Relative Angle θ_2. the Blue and Red Circles Mark the Centers of Mass of Each Link. The Absolute Angle of Link 2 Is $\theta_1 + \theta_2$.

In Plain Language 3.2. Absolute vs. Relative Angles

There are two conventions for defining θ_2:

1. **Absolute**: θ_2 is the angle of link 2 measured from a fixed reference (say, vertical). This is intuitive but makes the kinetic energy more complicated.
2. **Relative**: θ_2 is the angle between link 2 and link 1. This makes the kinetic energy simpler because each link rotates about its own hinge.

In this book, we use relative angles. So $\theta_2 = 0$ means the two links are aligned (arm fully extended). $\theta_2 = -90$ means the forearm is bent perpendicular to the upper arm.

Numerical Parameters for Illustration

To make the analysis concrete, we use representative parameters chosen to be typical for an adult golfer. These values are illustrative; actual parameters vary significantly with body size, club length, and individual differences.

Parameter	**Link 1 (Upper Arm)**	**Link 2 (Forearm + Club)**
Length L	0.35 m	1.0 m
Mass M	2.5 kg	1.5 kg
Center of mass distance from hinge	0.175 m	0.5 m
Moment of inertia I (point-mass approx.)	0.077 kg m^2	0.375 kg m^2

Note on moment of inertia: Under the point-mass approximation, all mass is concentrated at the center of mass, so the rotational inertia of each link about its own center of mass is $I_{cm} = 0$. The values $I = ML_{\mathrm{cm}}^2$ quoted here (link 1: $2.5 \times 0.175^2 = 0.077\ \mathrm{kg\,m^2}$; link 2: $1.5 \times 0.5^2 = 0.375\ \mathrm{kg\,m^2}$) are moments of inertia about the *pivot*, used only for quick order-of-magnitude estimates. In the Lagrangian derivation below, we set $I_i = 0$ (point-mass) so that all rotational kinetic energy is captured by the translational term $\frac{1}{2}M_i|\dot{\boldsymbol{r}}_{cm,i}|^2$.

These representative values allow us to compute explicit examples and develop intuition for the magnitude of forces and accelerations that appear during a golf swing. The physics governing the motion remains the same regardless of the specific parameter values. In this chapter, these are the canonical two-link values for illustration, while other chapters may use different compact parameter sets for different modeling goals.

In Plain Language 3.3. Roadmap: Deriving the Equations of Motion

We are about to derive the equations that govern the double pendulum. Here is the strategy:

1. Compute the **kinetic energy** T: how much energy is stored in the motion of the arm and club.
2. Compute the **potential energy** V: how much energy is stored in the height of the arm and club above some reference.
3. Form the **Lagrangian** $\mathcal{L} = T - V$ and apply the Euler–Lagrange equations—a recipe that converts energy into forces.
4. Rearrange into the **manipulator equation**: the master equation that separates inertia, velocity-dependent forces, gravity, and muscular torques.

The algebra is involved, but the payoff is enormous: one equation that captures everything about the double pendulum's dynamics.

3.2 The Kinetic Energy

To derive the equations of motion, we start with energy. Every moving system has kinetic energy.

Kinetic Energy of Link 1

Link 1 rotates about the shoulder. Its kinetic energy is:

$$T_1 = \frac{1}{2} I_1 \dot{\theta}_1^2 \tag{3.2}$$

where $I_1 = M_1 L_{1,\text{cm}}^2$ is the moment of inertia of link 1 about the shoulder hinge. (For simplicity, we treat each link as a point mass at its center of mass.)

Kinetic Energy of Link 2

Link 2 rotates about the elbow, which is itself moving. This is trickier.

The elbow is at position:

$$\boldsymbol{r}_{\text{elbow}} = L_1(\sin\theta_1, -\cos\theta_1) \tag{3.3}$$

Its velocity is:

$$\dot{\boldsymbol{r}}_{\text{elbow}} = L_1\dot{\theta}_1(\cos\theta_1, \sin\theta_1) \tag{3.4}$$

The center of mass of link 2 is at distance $L_{2,\text{cm}}$ from the elbow, at angle $\theta_1 + \theta_2$:

$$\boldsymbol{r}_{\text{cm},2} = L_1(\sin\theta_1, -\cos\theta_1) + L_{2,\text{cm}}(\sin(\theta_1+\theta_2), -\cos(\theta_1+\theta_2)) \tag{3.5}$$

Its velocity is:

$$\dot{\boldsymbol{r}}_{\text{cm},2} = L_1\dot{\theta}_1(\cos\theta_1, \sin\theta_1) \tag{3.6}$$

$$+ L_{2,\text{cm}}(\dot{\theta}_1+\dot{\theta}_2)(\cos(\theta_1+\theta_2), \sin(\theta_1+\theta_2)) \tag{3.7}$$

The kinetic energy of link 2 is (under the point-mass approximation, $I_2 = 0$, so all rotational kinetic energy is captured by the translational term):

$$T_2 = \frac{1}{2} M_2 |\dot{\boldsymbol{r}}_{\text{cm},2}|^2 \tag{3.8}$$

Expanding $|\dot{\boldsymbol{r}}_{\text{cm},2}|^2$:

$$|\dot{\boldsymbol{r}}_{\text{cm},2}|^2 = \left[L_1\dot{\theta}_1\cos\theta_1 + L_{2,\text{cm}}(\dot{\theta}_1+\dot{\theta}_2)\cos(\theta_1+\theta_2)\right]^2 \tag{3.9}$$

$$+ \left[L_1\dot{\theta}_1\sin\theta_1 + L_{2,\text{cm}}(\dot{\theta}_1+\dot{\theta}_2)\sin(\theta_1+\theta_2)\right]^2 \tag{3.10}$$

After expanding and simplifying (using $\cos^2 + \sin^2 = 1$ and $\cos\alpha\cos\beta + \sin\alpha\sin\beta = \cos(\alpha-\beta)$):

$$|\dot{\boldsymbol{r}}_{\text{cm},2}|^2 = L_1^2\dot{\theta}_1^2 + L_{2,\text{cm}}^2(\dot{\theta}_1+\dot{\theta}_2)^2 \tag{3.11}$$

$$+ 2L_1L_{2,\text{cm}}\dot{\theta}_1(\dot{\theta}_1+\dot{\theta}_2)\cos\theta_2 \tag{3.12}$$

So:

$$T_2 = \frac{1}{2} M_2\left[L_1^2\dot{\theta}_1^2 + L_{2,\text{cm}}^2(\dot{\theta}_1+\dot{\theta}_2)^2 + 2L_1L_{2,\text{cm}}\dot{\theta}_1(\dot{\theta}_1+\dot{\theta}_2)\cos\theta_2\right] \tag{3.13}$$

$$+ \frac{1}{2} I_2(\dot{\theta}_1+\dot{\theta}_2)^2 \tag{3.14}$$

Total Kinetic Energy

$$T = T_1 + T_2 \tag{3.15}$$
$$= \frac{1}{2} I_1 \dot{\theta}_1^2 + \frac{1}{2} M_2 \Big[L_1^2 \dot{\theta}_1^2 + L_{2,\text{cm}}^2 (\dot{\theta}_1 + \dot{\theta}_2)^2 \tag{3.16}$$
$$+ 2 L_1 L_{2,\text{cm}} \dot{\theta}_1 (\dot{\theta}_1 + \dot{\theta}_2) \cos\theta_2 \Big] + \frac{1}{2} I_2 (\dot{\theta}_1 + \dot{\theta}_2)^2 \tag{3.17}$$

Collecting terms:

$$T = \frac{1}{2}(I_1 + M_2 L_1^2)\dot{\theta}_1^2 \tag{3.18}$$
$$+ \frac{1}{2}(M_2 L_{2,\text{cm}}^2 + I_2)(\dot{\theta}_1 + \dot{\theta}_2)^2 \tag{3.19}$$
$$+ M_2 L_1 L_{2,\text{cm}} \dot{\theta}_1 (\dot{\theta}_1 + \dot{\theta}_2) \cos\theta_2 \tag{3.20}$$

In Plain Language 3.4. Why Kinetic Energy Matters

Kinetic energy is motion's "fuel." In a golf swing, you store energy during the backswing (lifting your arms against gravity, tensioning muscles). You release that energy during the downswing. The kinetic energy increases—the club is moving faster.
By the time the club reaches impact, all that stored energy has been converted to the motion of the club.
The Lagrangian formalism (which we'll use next) exploits the fact that nature minimizes the action, $\int (T - V) dt$. By tracking T, we can derive the forces without writing down force balances explicitly.

3.3 The Potential Energy

Gravity does work on your arm and the club. This is captured by gravitational potential energy.

Definition 3.1. Gravitational Potential Energy

The potential energy of a mass at height h is $V = Mgh$ (relative to some reference).

For link 1, the center of mass is at height:

$$h_1 = -L_{1,\text{cm}} \cos\theta_1 \tag{3.21}$$

(Taking the shoulder as the reference, and negative because we measure downward as negative.)

For link 2, the center of mass is at height:

$$h_2 = -L_1 \cos\theta_1 - L_{2,\text{cm}} \cos(\theta_1 + \theta_2) \tag{3.22}$$

Total potential energy:

$$V = M_1 g h_1 + M_2 g h_2 \tag{3.23}$$
$$= -M_1 g L_{1,\text{cm}} \cos\theta_1 - M_2 g \big[L_1 \cos\theta_1 + L_{2,\text{cm}} \cos(\theta_1 + \theta_2) \big] \tag{3.24}$$

In Plain Language 3.5. The Potential Energy Landscape

As your arm moves, the potential energy changes. At address (arm down), V is low. At the top of the backswing (arm up), V is higher. During the downswing, gravity does work and V decreases, converting to kinetic energy.
This is why golfers don't need to "create" all the club's speed. Gravity helps. The change in V from top to impact is a "gift" from physics.

3.4 The Lagrangian and Euler-Lagrange Equations

Now we use the Lagrangian formalism.

In Plain Language 3.6. Why the Lagrangian Approach?

Newton's $F = ma$ works perfectly for a ball rolling down a hill. But for a system of connected links, where constraint forces at the joints are unknown, Newton's approach becomes extremely cumbersome—you would need to solve for every internal force at every joint.
The Lagrangian approach sidesteps this entirely. Instead of tracking forces, you track *energy*: the kinetic energy T (energy of motion) and the potential energy V (energy of position). The Lagrangian $L = T - V$ encodes everything you need. The equations of motion drop out automatically, and constraint forces at fixed joints vanish from the equations. This is not a different physics—it is the same physics in a more convenient form.
For the golf swing, this is transformative. Rather than solving for the constraint forces at the shoulder, elbow, and wrist simultaneously with Newton's laws, the Lagrangian gives us the equations of motion for each joint angle directly.

Definition 3.2. Lagrangian

The Lagrangian is defined as:

$$\mathcal{L} = T - V \tag{3.25}$$

It's the kinetic energy minus the potential energy.

The Euler-Lagrange equations state that the equations of motion are:

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i}\right) - \frac{\partial \mathcal{L}}{\partial q_i} = \tau_i \tag{3.26}$$

for each coordinate q_i. Here, τ_i is the applied torque (from muscles).

This looks abstract. But it's a recipe: compute two partial derivatives of the Lagrangian, take one time derivative, subtract them, and you get the equation of motion for that coordinate.

Let's do it for the double pendulum.

The θ_1 Equation

We need:

- $\frac{\partial \mathcal{L}}{\partial \dot{\theta}_1}$
- $\frac{\partial \mathcal{L}}{\partial \theta_1}$

From T (Equation 3.20), the terms involving $\dot{\theta}_1$ are:

$$\begin{aligned}
\frac{\partial T}{\partial \dot{\theta}_1} &= (I_1 + M_2 L_1^2)\dot{\theta}_1 + (M_2 L_{2,\mathrm{cm}}^2 + I_2)(\dot{\theta}_1 + \dot{\theta}_2) && (3.27)\\
&\quad + M_2 L_1 L_{2,\mathrm{cm}}[(\dot{\theta}_1 + \dot{\theta}_2) + \dot{\theta}_1]\cos\theta_2 && (3.28)\\
&= (I_1 + M_2 L_1^2)\dot{\theta}_1 + (M_2 L_{2,\mathrm{cm}}^2 + I_2)(\dot{\theta}_1 + \dot{\theta}_2) && (3.29)\\
&\quad + M_2 L_1 L_{2,\mathrm{cm}}(2\dot{\theta}_1 + \dot{\theta}_2)\cos\theta_2 && (3.30)
\end{aligned}$$

Taking the time derivative:

$$\begin{aligned}
\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{\theta}_1}\right) &= (I_1 + M_2 L_1^2)\ddot{\theta}_1 + (M_2 L_{2,\mathrm{cm}}^2 + I_2)(\ddot{\theta}_1 + \ddot{\theta}_2) && (3.31)\\
&\quad + M_2 L_1 L_{2,\mathrm{cm}}(2\ddot{\theta}_1 + \ddot{\theta}_2)\cos\theta_2 && (3.32)\\
&\quad - M_2 L_1 L_{2,\mathrm{cm}}(2\dot{\theta}_1 + \dot{\theta}_2)\dot{\theta}_2\sin\theta_2 && (3.33)
\end{aligned}$$

Now the partial of T with respect to θ_1:

$$\frac{\partial T}{\partial \theta_1} = 0 \tag{3.34}$$

(since T doesn't depend on θ_1 explicitly, only on $\dot{\theta}_1$).
The partial of V with respect to θ_1:

$$\begin{aligned}
\frac{\partial V}{\partial \theta_1} &= M_1 g L_{1,\mathrm{cm}}\sin\theta_1 + M_2 g L_1 \sin\theta_1 + M_2 g L_{2,\mathrm{cm}}\sin(\theta_1 + \theta_2) && (3.35)\\
&= (M_1 L_{1,\mathrm{cm}} + M_2 L_1) g \sin\theta_1 + M_2 g L_{2,\mathrm{cm}}\sin(\theta_1 + \theta_2) && (3.36)
\end{aligned}$$

So:

$$-\frac{\partial \mathcal{L}}{\partial \theta_1} = -\frac{\partial (T - V)}{\partial \theta_1} = \frac{\partial V}{\partial \theta_1} \tag{3.37}$$

The Euler-Lagrange equation becomes:

$$\begin{aligned}
&(I_1 + M_2 L_1^2)\ddot{\theta}_1 + (M_2 L_{2,\mathrm{cm}}^2 + I_2)(\ddot{\theta}_1 + \ddot{\theta}_2) && (3.38)\\
&+ M_2 L_1 L_{2,\mathrm{cm}}(2\ddot{\theta}_1 + \ddot{\theta}_2)\cos\theta_2 && (3.39)\\
&- M_2 L_1 L_{2,\mathrm{cm}}(2\dot{\theta}_1 + \dot{\theta}_2)\dot{\theta}_2\sin\theta_2 && (3.40)\\
&= (M_1 L_{1,\mathrm{cm}} + M_2 L_1) g \sin\theta_1 + M_2 g L_{2,\mathrm{cm}}\sin(\theta_1 + \theta_2) + \tau_1 && (3.41)
\end{aligned}$$

Similarly, for θ_2:

$$(M_2 L_{2,\text{cm}}^2 + I_2)(\ddot{\theta}_1 + \ddot{\theta}_2) + M_2 L_1 L_{2,\text{cm}} \ddot{\theta}_1 \cos\theta_2 \tag{3.42}$$
$$+ M_2 L_1 L_{2,\text{cm}} \dot{\theta}_1^2 \sin\theta_2 \tag{3.43}$$
$$= M_2 g L_{2,\text{cm}} \sin(\theta_1 + \theta_2) + \tau_2 \tag{3.44}$$

In Plain Language 3.7. Reading the Equations

Equations 3.41 and 3.44 are the equations of motion for the double pendulum. They're complicated, with many terms. Let's parse them:

- **Left side**: terms with $\ddot{\theta}$ are the **inertial forces**. They say: "to accelerate the arm, you need a torque proportional to the moment of inertia and the acceleration."
- **Gravity terms on the right**: e.g., $(M_1 L_{1,\text{cm}} + M_2 L_1) g \sin\theta_1$. These are the **gravitational torques**. When $\theta_1 = 0$ (arm down), $\sin\theta_1 = 0$, so gravity exerts no torque—the arm is in equilibrium. When $\theta_1 = 90$ (arm horizontal), $\sin\theta_1 = 1$, so gravity exerts maximum torque, trying to pull the arm back down.
- **The $\dot{\theta}_2 \sin\theta_2$ terms**: these are **Coriolis forces**, arising from the coupling between the two joints. When the elbow bends ($\dot{\theta}_2 \neq 0$) while the shoulder is rotating ($\dot{\theta}_1 \neq 0$), there's a cross term that affects the dynamics.
- **The $\dot{\theta}_1^2 \sin\theta_2$ term**: this is a **centrifugal force**. When the shoulder rotates fast ($\dot{\theta}_1$ large), it creates an outward force on the elbow, trying to straighten the arm.

These equations are complicated because the double pendulum is a coupled system. The motion of one joint affects the other.

3.5 Rewriting in Matrix Form

The Euler-Lagrange equations are hard to parse. Let's rewrite them in the standard form:

$$\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{\tau} \tag{3.45}$$

where:

- $\boldsymbol{M}(\boldsymbol{q})$ is the **mass matrix** (moment of inertia matrix).
- $\boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})$ contains the Coriolis and centrifugal terms.
- $\boldsymbol{g}(\boldsymbol{q})$ is the gravitational torque.
- $\boldsymbol{\tau}$ is the applied (muscular) torque.

The Mass Matrix

Collecting all terms with $\ddot{\theta}_1$ and $\ddot{\theta}_2$:

$$\boldsymbol{M}(\boldsymbol{q}) = \begin{bmatrix} M_{11}(\theta_2) & M_{12}(\theta_2) \\ M_{21}(\theta_2) & M_{22} \end{bmatrix} \tag{3.46}$$

where:

$$M_{11}(\theta_2) = I_1 + M_2 L_1^2 + M_2 L_{2,\text{cm}}^2 + I_2 + 2M_2 L_1 L_{2,\text{cm}} \cos\theta_2 \tag{3.47}$$

$$M_{12}(\theta_2) = M_{21}(\theta_2) = M_2 L_{2,\text{cm}}^2 + I_2 + M_2 L_1 L_{2,\text{cm}} \cos\theta_2 \tag{3.48}$$

$$M_{22} = M_2 L_{2,\text{cm}}^2 + I_2 \tag{3.49}$$

Key observation: M_{11} depends on θ_2 (the elbow angle). This means the effective inertia at the shoulder depends on whether the elbow is bent. An extended arm ($\theta_2 \approx 0$) has higher inertia than a bent arm (θ_2 large and negative).

The Coriolis/Centrifugal Matrix

Collecting all terms with $\dot{\theta}_1\dot{\theta}_2$ and $\dot{\theta}^2$:

$$\boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}}) = \begin{bmatrix} 0 & C_{12}(\boldsymbol{q}, \dot{\boldsymbol{q}}) \\ C_{21}(\boldsymbol{q}, \dot{\boldsymbol{q}}) & 0 \end{bmatrix} \tag{3.50}$$

where:

$$C_{12}(\boldsymbol{q}, \dot{\boldsymbol{q}}) = -M_2 L_1 L_{2,\text{cm}} (2\dot{\theta}_1 + \dot{\theta}_2) \sin\theta_2 \tag{3.51}$$

$$C_{21}(\boldsymbol{q}, \dot{\boldsymbol{q}}) = M_2 L_1 L_{2,\text{cm}} \dot{\theta}_1^2 \sin\theta_2 \tag{3.52}$$

More precisely, the standard form is:

$$\boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} = \begin{bmatrix} C_{12}\dot{\theta}_2 \\ C_{21}\dot{\theta}_1 \end{bmatrix}$$

But it's more standard to write the Coriolis matrix such that $\boldsymbol{C}\dot{\boldsymbol{q}}$ gives the full velocity-dependent torque. We'll use the convention where the (i, j) entry couples the velocity of joint j to joint i.

The Gravity Vector

$$\boldsymbol{g}(\boldsymbol{q}) = \begin{bmatrix} (M_1 L_{1,\text{cm}} + M_2 L_1) g \sin\theta_1 + M_2 g L_{2,\text{cm}} \sin(\theta_1 + \theta_2) \\ M_2 g L_{2,\text{cm}} \sin(\theta_1 + \theta_2) \end{bmatrix} \tag{3.53}$$

3.6 Numerical Example: A Realistic Golf Swing

Let's plug in numbers and see what the forces actually are.

At Impact

Consider a representative impact condition where the arm is nearly extended. These are representative values typical of moderate swing speeds:

- $\theta_1 = 0$ (arm aligned with the reference direction)
- $\theta_2 = -5$ (small negative angle, indicating the forearm is bent slightly from full extension)
- $\dot{\theta}_1 \approx 10$ rad/s (shoulder rotation angular velocity, typical range 8-12 rad/s from biomechanical studies [Nesbit, 2005])
- $\dot{\theta}_2 \approx 9$ rad/s (relative elbow/wrist angular velocity)

The mass matrix at this configuration:

$$\begin{aligned} M_{11} &= I_1 + M_2 L_1^2 + M_2 L_{2,\text{cm}}^2 + I_2 + 2M_2 L_1 L_{2,\text{cm}} \cos\theta_2 && (3.54) \\ &\approx 0.077 + 1.5 \times (0.35)^2 + 1.5 \times (0.5)^2 + 0.375 + 2 \times 1.5 \times 0.35 \times 0.5 \cos(-5) && (3.55) \\ &\approx 0.077 + 0.184 + 0.375 + 0.375 + 0.524 && (3.56) \\ &\approx 1.54 \text{ kg m}^2 && (3.57) \end{aligned}$$

This represents the effective moment of inertia of the coupled arm+club system about the shoulder at this configuration.

The gravitational torques are found from the gravity vector:

$$\begin{aligned} g_1 &= (M_1 L_{1,\text{cm}} + M_2 L_1) g \sin\theta_1 + M_2 g L_{2,\text{cm}} \sin(\theta_1 + \theta_2) && (3.58) \\ &= (2.5 \times 0.175 + 1.5 \times 0.35) \times 9.81 \times \sin(0) + 1.5 \times 9.81 \times 0.5 \times \sin(-5) && (3.59) \\ &\approx 0 + 7.36 \times (-0.087) && (3.60) \\ &\approx -0.64 \text{ N m} && (3.61) \end{aligned}$$

The Coriolis/centrifugal term at the elbow (from Equation 3.50):

$$\begin{aligned} C_{21}(\dot{\theta}_1) &= M_2 L_1 L_{2,\text{cm}} \dot{\theta}_1^2 \sin\theta_2 && (3.62) \\ &= 1.5 \times 0.35 \times 0.5 \times (10)^2 \times \sin(-5) && (3.63) \\ &= 1.5 \times 0.35 \times 0.5 \times 100 \times (-0.087) && (3.64) \\ &\approx -2.3 \text{ N m} && (3.65) \end{aligned}$$

In Plain Language 3.8. Interpretation of the Force Components

The numerical values at impact show the magnitude of various terms in the equations of motion:

- The effective inertia $M_{11} \approx 1.54$ kg m^2 determines how much acceleration results from a given applied torque.
- The gravitational torque $g_1 \approx -0.64$ N m acts to rotate the system back toward the downward position (negative indicates restoring direction).
- The centrifugal term $C_{21} \approx -2.3$ N m arises from the high rotational velocity

and acts to influence the relative joint motion.

These passive forces (arising from gravity and the coupled dynamics) contribute to the total torque that accelerates the system. The applied muscular torques (represented by $\boldsymbol{\tau}$ in the manipulator equation) must be combined with these passive components to produce the observed accelerations.
Notice that passive dynamics contribute significantly, particularly the velocity-dependent centrifugal terms that become important at high rotation speeds. This is why the dynamics are complex and why precise control is necessary.

3.7 The Interaction Torque: How the Arm Pulls the Club

One of the most important terms is the interaction torque between the two links.

In the equations of motion, we have the term $M_2 L_1 L_{2,\text{cm}} \cos\theta_2$ appearing in M_{11}. This couples the two joints. It means: the acceleration of the shoulder depends on the angle of the elbow.

Definition 3.3. Interaction Torque

The torque that one joint exerts on the other through the kinetic energy coupling. In the double pendulum, the term $M_2 L_1 L_{2,\text{cm}}(\ddot{\theta}_1 + \ddot{\theta}_2)\cos\theta_2$ in the θ_1 equation is the torque that the elbow exerts on the shoulder through their coupling.

Example 3.1. The Coupled Interaction of Two Links

In the double pendulum model, the two links are coupled through the kinetic energy and constraint structure. The acceleration of one link directly affects the torque required at the other link.
Consider the manipulator equation:

$$\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{\tau}$$

The mass matrix $\boldsymbol{M}$ is not diagonal. When the second link accelerates (large $\ddot{\theta}_2$), it creates coupling terms $M_{12}\ddot{\theta}_2$ in the first equation of motion. This means the torque required at the first joint depends on the acceleration of the second joint.
In mechanical terms, accelerating the distal link (club) creates a reaction torque transmitted back to the proximal link (upper arm). This is not unique to the golf swing; it appears in any open-chain kinematic system with coupled dynamics.

3.8 Chaotic Dynamics: Why Small Differences Matter

Here's a fascinating property of the double pendulum: it can exhibit **chaotic behavior**.

Definition 3.4. Chaotic Dynamics

Motion is chaotic if small changes in initial conditions lead to drastically different outcomes, despite the system being completely deterministic.

The unforced double pendulum (with $\boldsymbol{\tau} = 0$) exhibits chaotic behavior in certain parameter regimes. However, the golf swing always involves active control ($\boldsymbol{\tau} \neq 0$). With active control, the system behavior changes, but sensitivity to perturbations in the applied torques remains an important consideration.

This sensitivity is a fundamental property of nonlinear systems: small variations in control inputs or system state can produce significantly different outcomes. Understanding this property is relevant to biomechanics and motor control, as it explains why achieving repeatable motion requires precise execution of control commands.

In Plain Language 3.9. Sensitivity to Initial Conditions

A key property of nonlinear dynamical systems is that small changes in the current state or applied forces can lead to appreciably different future states. This is particularly true near bifurcation points or in chaotic regimes.
Consider the manipulator equation in a controlled scenario where muscular torques $\boldsymbol{\tau}(t)$ are applied according to some control law. Small variations in the timing, magnitude, or direction of the applied torques result in different state trajectories. The nonlinear coupling terms (particularly the velocity-dependent centrifugal and Coriolis terms) amplify small perturbations.
For the double pendulum, the state-dependent nature of the mass matrix $\boldsymbol{M}(\boldsymbol{q})$ and Coriolis matrix $\boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})$ means that the same muscular command, applied at slightly different system states, produces different accelerations and hence different subsequent trajectories.
This property is important for understanding why consistent execution of motion is difficult: the physical system itself amplifies small errors in the applied control.

3.9 Decomposing the Equation of Motion

The manipulator equation $\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{\tau}$ provides a complete description of the coupled dynamics of the two-link system. The equation separates naturally into distinct physical components:

Left side: These terms represent the passive dynamics arising from the structure and current state of the system. They include inertial effects (proportional to acceleration through $\boldsymbol{M}$), velocity-dependent forces (Coriolis and centrifugal terms in $\boldsymbol{C}$), and gravitational torques ($\boldsymbol{g}$).

Right side: The applied (muscular) torques $\boldsymbol{\tau}$ represent the active control input that the neuromuscular system provides.

The actual motion results from the balance between these passive dynamics and the applied control. Understanding which terms dominate at different phases of motion is essential for analyzing and predicting the system's behavior.

Key Principle 3.1. Key Takeaways

1. **The double pendulum model captures the essential physics**: two rigid links with rotational inertia, subject to gravity and coupled through their kinematic constraints.

2. **Kinetic energy is configuration-dependent**. The total kinetic energy of the system depends on the state through both the angular velocities and the configuration (angles). The mass matrix $\boldsymbol{M}(\boldsymbol{q})$ encodes how the configuration affects the effective inertias.

3. **The mass matrix exhibits state dependence**. The term $M_{11}(\theta_2)$ depends on the elbow angle, meaning the effective inertia at the shoulder varies as the configuration changes. This coupling is a fundamental feature of multi-link systems.

4. **Gravitational torques are configuration-dependent**. The terms $\sin\theta_1$ and $\sin(\theta_1+\theta_2)$ in the gravity vector show that gravitational effects depend on the current orientation of each link.

5. **Velocity-dependent forces (Coriolis and centrifugal) scale with $\dot{\theta}^2$**. At high angular velocities, these nonlinear terms dominate the dynamics and must be accounted for in any realistic model.

6. **The system exhibits coupling between joints**. The kinetic energy coupling creates non-diagonal terms in the mass matrix. Accelerating one joint directly affects the torque required at another joint.

7. **The system is sensitive to the applied control**. Small changes in the applied torques $\boldsymbol{\tau}(t)$ lead to different state trajectories due to the nonlinear nature of the equations of motion.

Looking Ahead

We have derived the equations of motion for the double pendulum—the mathematical skeleton of the golf swing. The left-hand side of the manipulator equation contains the passive forces (inertia, Coriolis, gravity), while the right-hand side contains the muscular torques. But which side dominates? How do we separate what physics does on its own from what your muscles add? Chapter 4 unpacks each force in detail, and Chapter 5 reveals the mathematical framework that cleanly separates drift from control.

Exercises 3.1. Chapter Exercises

1. **Moment of Inertia**. A rod of mass M and length L has moment of inertia $I = \frac{1}{12}ML^2$ about its center and $I = \frac{1}{3}ML^2$ about one end. For the forearm+club (mass 1.5 kg, length 1.0 m), compute the moment of inertia about the elbow.

2. **Potential Energy Change**. At address, both arms are down ($\theta_1 = 0, \theta_2 = 0$). At the top of the backswing, $\theta_1 = 150, \theta_2 = -70$. Using the parameters in

Section 3, compute the change in potential energy. How much gravitational work is available during the downswing?

3. **Centrifugal Force**. At impact, $\dot{\theta}_1 = 10.47$ rad/s. The centrifugal acceleration at the clubhead (distance $L_1 + L_2 = 1.35$ m from shoulder) is $a_c = \dot{\theta}_1^2 \cdot r$. Compute this acceleration in units of g.

4. **Impact Accelerations**. The clubhead experiences large accelerations during impact. Using the double pendulum model with $L_1 = 0.35$ m and $L_2 = 1.0$ m, estimate what angular accelerations $(\ddot{\theta}_1, \ddot{\theta}_2)$ are needed to produce a clubhead acceleration of 50 m/s^2 (about $5g$). Use the approximation that at small angles near impact, the clubhead acceleration is approximately $a \approx L_1\ddot{\theta}_1 + L_2(\ddot{\theta}_1 + \ddot{\theta}_2)$.

5. **Gravity Torque**. At the top of the backswing $(\theta_1 = 150, \theta_2 = -70)$, compute the gravitational torque at the shoulder. Is gravity trying to pull the arm backward or forward? Why?

6. **Chaotic Dynamics**. Look up the properties of the (unforced) double pendulum. In what regime does it exhibit chaos? Does the golf swing operate in that regime?

Part II

The Affine Framework

Chapter 4

Forces and Torques: Where They Come From

"Force is the source of all motion and change. Understanding where forces come from is understanding how things work."

In Plain Language 4.1. Decomposing the Forces

Here's a central insight: the equation

$$\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{\tau} \quad (4.1)$$

contains *all* the forces that act on your arm. But they're not all the same type. Some come from your muscles. Some come from gravity. Some come from your arm's own motion. Some come from constraints.

If you can attribute each force to its source, you gain enormous insight. You understand what you're actually controlling. You see which forces help you and which fight you. You can diagnose problems.

This chapter is about that decomposition.

4.1 The Five Sources of Force

In the manipulator equation, every torque comes from one of five sources:

1. **Inertial forces** (from $\boldsymbol{M}(\boldsymbol{q})$): forces that resist acceleration. To speed up your arm, you must overcome its inertia.

2. **Velocity-dependent forces** (from $\boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}}$: Coriolis and centrifugal forces that arise when you move fast. They couple the motion of different joints.

3. **Gravitational forces** (from $\boldsymbol{g}(\boldsymbol{q})$): torques that arise because gravity pulls on your mass. These depend on position.

4. **Muscular forces** (from $\boldsymbol{\tau}$): torques you generate by contracting muscles. These are your **control**.

5. **Constraint forces** (from rigid hinges, rigid bones, etc.): reactive forces that enforce mechanical constraints. These don't appear explicitly in the equation but appear as reaction torques at hinges.

Decomposition of Torques in the Manipulator Equation

$$\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{\tau}$$

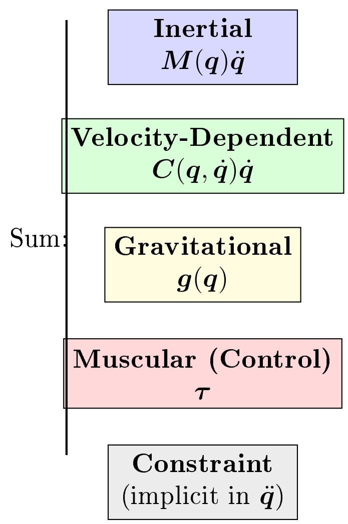


Figure 4.1: The Five Sources of Torque in the Manipulator Equation. The Passive Forces (Inertial, Velocity-Dependent, Gravitational) Sum to Determine What Muscular Control (Constraint Force) Is Needed at Each Instant.

Let's understand each one.

4.2 Inertial Forces: Resisting Acceleration

Newton's second law in the simplest form is:

$$F = ma \tag{4.2}$$

To accelerate a mass, you must apply a force proportional to its mass and acceleration. The same is true for rotation:

$$\tau = I\alpha \tag{4.3}$$

where I is the moment of inertia and $\alpha = \ddot{\theta}$ is the angular acceleration.

In the manipulator equation, the terms $\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}}$ are exactly these inertial terms.

Example 4.1. Inertia at the Shoulder

Recall from Chapter 3 that the effective moment of inertia at the shoulder is about $M_{11} \approx 1.54$ kg m^2.
If you want to accelerate the shoulder at $\ddot{\theta}_1 = 1000\ \text{deg/s}^2 = 17.45\ \text{rad/s}^2$, the inertial torque you must overcome is:

$$\tau_{\text{inertia}} = M_{11} \cdot \ddot{\theta}_1 = 1.54 \times 17.45 = 26.9\ \text{N m} \tag{4.4}$$

This is the baseline. Just to accelerate the arm at this rate requires 26.9 N m. If there's also gravity pulling ($\sim$5 N m backward), then you need to generate at least $26.9 + 5 = 31.9$ N m of muscular torque to achieve this acceleration.
In addition, Coriolis and centrifugal forces contribute further terms.

Why Inertia Increases With Mass

Here's a fundamental insight: inertia is *resistance to acceleration*. A heavier arm is harder to accelerate. A longer arm (higher moment arm) is also harder to accelerate because the mass is farther from the axis of rotation.

The moment of inertia I depends on both the mass and its distribution. For a point mass at distance r: $I = mr^2$. For an extended body like the arm, I is the integral of all mass elements weighted by their distances from the rotation axis.

This explains practical observations: lighter clubs require less muscular effort to accelerate but deliver less momentum to the ball. The control problem becomes one of choosing club properties and muscular effort to optimize overall performance.

4.3 Velocity-Dependent Forces: Coriolis and Centrifugal

When you move fast, magical things happen. Forces appear that don't exist in the equations for slow motion.

Centrifugal Force

The most intuitive is centrifugal force. If you spin a weight on a string, the weight pulls outward on the string. That outward pull is the centrifugal force.

In the golf swing, when your shoulder rotates fast ($\dot{\theta}_1$ large), the forearm experiences an outward centrifugal acceleration. This tries to straighten your elbow.

Definition 4.1. Centrifugal Acceleration

For a point rotating at angular velocity ω at distance r from the axis, the centrifugal acceleration is:

$$a_c = \omega^2 r \tag{4.5}$$

This is always positive (outward). It's not a "real" force in the sense that no object is pushing outward; instead, it's a consequence of circular motion.

Example 4.2. Centrifugal Force in a Driver Swing

Consider a driver swing where the clubhead (mass $m = 0.2$ kg) is at effective distance $r = L_1 + L_2 = 1.35$ m from the shoulder (canonical parameters, Chapter 3). At $\omega = 30$ rad/s—the angular velocity of the clubhead *about the shoulder*, corresponding to roughly 110 mph clubhead speed—just before impact:
The centrifugal force pulling the club outward is:

$$F_{\text{centrifugal}} = m\omega^2 r = 0.2 \times 30^2 \times 1.35 = 243 \text{ N} \approx 55 \text{ lbs}$$

This large outward force arises purely from the geometry and velocity of rotational motion. It manifests as tension in the hands and arms, and it loads the shaft. This is one of the largest forces during the swing, and it emerges automatically from the system dynamics without explicit muscular generation. The hands must supply the centripetal reaction force to maintain the curved trajectory.
As documented in biomechanical studies [Jorgensen, 1994a, Penner, 2003a], these velocity-dependent forces are a significant part of the swing dynamics and must be accounted for in forward models of the motion.

Example 4.3. Centrifugal Force in the Golf Swing

At impact, the shoulder rotates at $\dot{\theta}_1 = 10.47$ rad/s. The forearm is at distance roughly $L_1 = 0.35$ m from the shoulder.
The centrifugal acceleration is:

$$a_c = \dot{\theta}_1^2 \cdot L_1 = 10.47^2 \times 0.35 = 38.3 \text{ m/s}^2 = 3.9g \tag{4.6}$$

The centrifugal effect on the forearm+club is more precisely captured by the centrifugal term in the manipulator equation. The relevant entry of $\boldsymbol{C}$ is:

$$M_2 L_1 L_{2,\text{cm}} \dot{\theta}_1^2 \sin\theta_2 = 1.5 \times 0.35 \times 0.5 \times 10.47^2 \times \sin(-5) \approx -4.0 \text{ N m} \tag{4.7}$$

This creates a torque at the elbow that helps straighten the arm if $\theta_2 < 0$ (bent elbow).

Coriolis Force

Coriolis forces are more subtle. They arise from the interaction of two rotating motions.

Imagine you're on a rotating platform. You walk toward the center. From an outside observer's perspective, you don't actually move toward the center—you curve to the side (deflected by the Coriolis force).

In the golf swing, the Coriolis terms in $\boldsymbol{C}$ represent the coupling between shoulder and elbow rotation.

Definition 4.2. Coriolis Force (in Rotating Frames)

In a rotating reference frame, the Coriolis acceleration is:

$$\boldsymbol{a}_{\text{Coriolis}} = -2\boldsymbol{\omega} \times \boldsymbol{v} \tag{4.8}$$

where $\boldsymbol{\omega}$ is the rotation rate and $\boldsymbol{v}$ is the velocity in the rotating frame.

In the manipulator equation, the Coriolis term $C_{12}\dot{\theta}_2$ represents how the elbow's angular velocity ($\dot{\theta}_2$) creates a reaction torque at the shoulder when the shoulder is rotating. This coupling is what makes the kinetic chain possible—each joint's motion affects the forces at other joints.

Example 4.4. Coriolis Torque in the Downswing

The Coriolis term in the shoulder equation is:

$$(\text{from } \boldsymbol{C}) = -M_2 L_1 L_{2,\text{cm}}(2\dot{\theta}_1 + \dot{\theta}_2)\dot{\theta}_2 \sin\theta_2 \tag{4.9}$$

At impact ($\dot{\theta}_1 = 10.47$ rad/s, $\dot{\theta}_2 = 8.73$ rad/s, $\theta_2 = -5$):

$$\begin{aligned} \text{Coriolis} &= -1.5 \times 0.35 \times 0.5 \times (2 \times 10.47 + 8.73) \times 8.73 \times \sin(-5) && (4.10) \\ &= -1.5 \times 0.35 \times 0.5 \times 29.67 \times 8.73 \times (-0.087) && (4.11) \\ &\approx +2.0 \text{ N m} && (4.12) \end{aligned}$$

This torque at the shoulder is positive (forward), helping the shoulder rotation. Why? Because the elbow is releasing (increasing $\dot{\theta}_2$) in a direction that's compatible with the shoulder rotation.
The kinetic chain works through such velocity coupling: each joint's motion influences the forces at other joints through Coriolis terms.

The Velocity Dependence Is Crucial

The key distinction between these force sources is how they depend on the system state:

Here's the key insight: **velocity-dependent forces scale with the square of the velocity**. At address ($\dot{\theta}_i = 0$), they're zero. During the downswing, they grow enormously.

This is why the golf swing is dynamic. The faster you move, the larger these forces become. A slow swing has small Coriolis and centrifugal forces, so you have to generate large muscular torques to accelerate. A fast swing has large velocity-dependent forces helping you.

In Plain Language 4.2. How Forces Vary Through the Swing

The magnitude and composition of forces changes dramatically as the swing progresses. This qualitative accounting illustrates the principle (exact values depend on individual biomechanics and swing parameters):
Address (no motion):

- Gravity: ~ 10 N m
- Velocity-dependent: ~ 0 (no motion)
- Muscular control: ~ 10 N m (needed to overcome gravity)

Top of backswing (arm elevated, velocities low):

- Gravity: ~ 30 N m (arm extended, pulled by gravity)
- Velocity-dependent: ~ 0 (velocities near zero)
- Muscular control: ~ 30 N m (to hold elevated position)

Transition (acceleration ramping up):

- Gravity: ~ 20 N m
- Velocity-dependent: ~ 5 N m (emerging as velocity increases)
- Muscular control: ~ 15 N m (to modulate accelerations)

Late downswing (high speeds):

- Gravity: ~ 5 N m (arm approaching vertical)
- Velocity-dependent: ~ 50 N m (strong Coriolis/centrifugal effects)
- Muscular control: ~ -25 N m (restraining, not accelerating)

As velocity increases, velocity-dependent forces grow as v^2. Muscles must often apply *braking* torques in the late downswing to prevent excessive acceleration, rather than driving acceleration forward.
This transition—from muscles driving motion to muscles restraining it—is a fundamental feature of the swing dynamics.

4.4 Gravitational Forces: Position-Dependent

Gravity is the force you feel constantly. It depends on position but not on velocity.

The gravitational torque in the double pendulum is:

$$\boldsymbol{g}(\boldsymbol{q}) = \begin{bmatrix} (M_1 L_{1,\text{cm}} + M_2 L_1) g \sin\theta_1 + M_2 g L_{2,\text{cm}} \sin(\theta_1 + \theta_2) \\ M_2 g L_{2,\text{cm}} \sin(\theta_1 + \theta_2) \end{bmatrix} \tag{4.13}$$

Reading the Gravity Terms

- $(M_1 L_{1,\text{cm}} + M_2 L_1) g \sin\theta_1$: This is the torque of the entire arm+club about the shoulder. It depends on how much mass is hanging below the shoulder and how far it hangs.

  When $\theta_1 = 0$ (arm vertical), $\sin\theta_1 = 0$, so gravity exerts zero torque. The arm is balanced.

When $\theta_1 = 90$ (arm horizontal), $\sin\theta_1 = 1$, so gravity exerts maximum torque. The arm wants to fall.

When $\theta_1 = 180$ (arm vertical, pointing up), $\sin\theta_1 = 0$ again. The arm is balanced (but unstable).

- $M_2 g L_{2,\mathrm{cm}} \sin(\theta_1 + \theta_2)$: This is the torque of the forearm+club about the shoulder. It also appears in the elbow equation, contributing to the torque at the elbow.

Gravity Does Work During the Downswing

From address to impact, your arm goes from down to down (the angles return to near their starting values). But the *speed* increases dramatically. This happens because gravity does positive work, converting potential energy to kinetic energy.

The work done by gravity is:

$$W = \Delta V = V_{\text{initial}} - V_{\text{final}} \tag{4.14}$$

If the arm is lower at the end (lower height), then $\Delta V < 0$, and gravity does positive work. But if the angles are the same, the heights are the same, so $\Delta V = 0$.

This seems to contradict the observation that gravity helps. The resolution: gravity *does* do positive work during the downswing (arm goes from up at the top to down during transition), but *all that work is already done by the time you reach impact.* So the conversion of potential energy to kinetic energy happens early in the downswing.

Later in the downswing, gravity is actually fighting you (pulling the arm down when you want to pull it forward). But by that point, the arm has so much kinetic energy that gravity's resistance is small.

4.5 Muscular Forces: Your Control

The right side of the manipulator equation is where control is applied:

$$\boldsymbol{\tau} = \text{applied torques from muscles} \tag{4.15}$$

Muscles pull on bones. Tendons transfer that pull to create torques at joints. The sum of all these muscular torques is $\boldsymbol{\tau}$. This is the control signal that can be varied over time.

Constraints on Muscular Forces

Muscles can only pull, not push. There are therefore constraints on what $\boldsymbol{\tau}$ can be:

- $\tau_i \geq -\tau_{\max,i}$ (muscles can pull in one direction up to a maximum).
- $\tau_i \leq +\tau_{\max,i}$ (muscles can pull in the opposite direction up to a maximum).

The maximum available torque depends on the muscle physiology, cross-sectional area, and lever arm. These values would be determined empirically through biomechanical measurement.

Muscular Forces Are the Control Input

This is the fundamental point: **you directly control only $\boldsymbol{\tau}$. The forces from gravity, inertia, and velocity-dependence arise automatically from the system dynamics.**

Given $\boldsymbol{\tau}(t)$ at each instant, the manipulator equation determines the accelerations $\ddot{\boldsymbol{q}}$. Integrating these gives velocities and positions:

$$\dot{\boldsymbol{q}}(t) = \dot{\boldsymbol{q}}(0) + \int_0^t \ddot{\boldsymbol{q}}(s)ds, \quad \boldsymbol{q}(t) = \boldsymbol{q}(0) + \int_0^t \dot{\boldsymbol{q}}(s)ds \tag{4.16}$$

The fundamental control problem is: given the system dynamics and constraints on $\boldsymbol{\tau}$, what control trajectory $\boldsymbol{\tau}(t)$ produces a desired swing outcome?

4.6 Constraint Forces: The Reactive Forces

Finally, there are forces that appear automatically to enforce mechanical constraints. These don't show up explicitly in the manipulator equation (as written in terms of generalized coordinates), but they're present as reaction forces at hinges and joints.

Definition 4.3. Constraint Forces

Reaction forces that enforce mechanical constraints (like rigidity, inextensibility, or friction). They're *reactive*: they appear only as much as needed to enforce the constraint.
In the golf swing:

- The hinge at the shoulder exerts a force on the upper arm to keep it attached.
- The hinge at the elbow exerts a force on the forearm to keep it attached.
- The ground exerts a normal force on your feet to keep you from sinking.
- Friction at the grip exerts a force on the club to keep it from slipping.

Why Constraint Forces Don't Appear in the Equation

We derived the manipulator equation using **generalized coordinates**: variables that automatically satisfy the constraints.

For example, by using angles θ_1 and θ_2 instead of Cartesian coordinates of the clubhead, we automatically enforced that the links are rigid and hinged correctly. The constraint forces are implicitly accounted for.

However, if you wanted to know the actual force at the grip (the force the hand exerts on the club), you'd need to solve for it separately. The manipulator equation doesn't directly give you constraint forces.

4.7 Putting It Together: Force Attribution in Impact

Let's do a complete force attribution at impact. For this numerical example, we use a representative clubhead speed of 100 mph (44.7 m/s)—a value in the range for skilled golfers [Jorgensen, 1994a]. The clubhead acceleration is enormous.

The Clubhead Acceleration

The clubhead is at distance $r = L_1 + L_2 = 1.35$ m from the shoulder. At impact, the shoulder is rotating at $\omega = \dot{\theta}_1 = 10.47$ rad/s.

The centripetal acceleration (toward the shoulder) is:

$$a_c = \omega^2 r = 10.47^2 \times 1.35 = 147.6 \text{ m/s}^2 = 15.0g \tag{4.17}$$

But the elbow is also rotating, and the arm is accelerating, so the total acceleration is higher:

$$a_{\text{total}} \approx 180 \text{ m/s}^2 \approx 18g \tag{4.18}$$

Note: This $\sim$18g is the clubhead acceleration *during the swing*. During the brief ball–club collision ($\sim$0.5 ms), the ball experiences accelerations of $\sim$10 000–30 000g, and the clubhead decelerates at 100–200g.

Where Do These Accelerations Come From?

At impact, the accelerations arise from the balance of terms in the manipulator equation:

$$\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} = \boldsymbol{\tau} - \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} - \boldsymbol{g}(\boldsymbol{q}) \tag{4.19}$$

Each term contributes:

- **Inertial term $\boldsymbol{M}\ddot{\boldsymbol{q}}$**: The mass matrix scaled by acceleration. At impact, the clubhead acceleration is ≈ 180 m/s^2 = 18.3g. Given moment of inertia $M_{11} \approx 1.54$ kg m^2, the inertial term is on the order of $\ddot{\theta}_1 \approx 40$ rad/s^2.
- **Coriolis/Centrifugal forces $\boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}}$**: At high speeds ($\dot{\theta}_1 \approx 10$ rad/s, $\dot{\theta}_2 \approx 8$ rad/s), these velocity-dependent terms contribute torques on the order of 10–20 N m.
- **Gravitational forces $\boldsymbol{g}(\boldsymbol{q})$**: At impact ($\theta_1 \approx 0$), the arm is nearly vertical. Gravity contributes 5–10 N m.
- **Muscular torque $\boldsymbol{\tau}$**: The applied torque that must satisfy the equation of motion.

  Numerical illustration: If $\ddot{\theta}_1 = 40$ rad/s^2, $M_{11} = 1.54$ kg m^2, $\boldsymbol{C}\dot{\boldsymbol{q}} \approx 10$ N m, and $\boldsymbol{g} \approx 5$ N m, then:

  $$\tau_1 = M_{11}\ddot{\theta}_1 + C\dot{\boldsymbol{q}} + g = 1.54 \times 40 + 10 + 5 = 61.6 + 10 + 5 \approx 77 \text{ N m} \tag{4.20}$$

  This represents the total muscular torque required at the shoulder. These magnitudes would be determined or validated through measurement or detailed simulation.

The Constraint Force at the Grip

The hand is holding the club. The clubhead is accelerating at $\sim$18g. To estimate the grip force, we decompose into centripetal and tangential components:

- **Centripetal**: pulling the club toward the axis of rotation (toward the shoulder).
- **Tangential**: in the direction of motion (pulling the club forward/backward).

The centripetal force is:

$$F_c = M_{\text{club}}\omega^2 r = 0.2 \times 10.47^2 \times 1.35 = 29.5 \text{ N} \tag{4.21}$$

The tangential force is:

$$F_t = M_{\text{club}} \times a_t = 0.2 \times (\text{tangential acceleration}) \tag{4.22}$$

The total grip force is the vector sum. For this illustrative example (using the representative impact parameters above), the grip force is on the order of 40–60 N. That figure is not measured: it is what the two components computed above require for the 0.2 kg clubhead at the stated impact parameters. The force a golfer actually applies is larger and varies widely with swing speed, grip technique, and club configuration, so read this as what the model demands rather than as what an instrumented grip would record.

In Plain Language 4.3. Why the Grip Matters

The grip is a constraint force. The hand-club interface is modeled as a rigid connection (or nearly rigid, neglecting slip). Given the muscular torques applied at the shoulder and elbow, the dynamics of the system determine what constraint force (grip force) must arise to maintain this connection.

Key insight: The grip force is not a direct control variable; it is an output of the system dynamics. If the shoulder torques τ_1 and τ_2 are specified, the manipulator equation determines the accelerations $\ddot{\boldsymbol{q}}$. To enforce the kinematic constraint that the hand remains attached to the club, a constraint force (the grip force) emerges automatically.

This has an important consequence: the grip force depends sensitively on the control torques. If $\boldsymbol{\tau}$ is too large, the constraint force becomes large. If $\boldsymbol{\tau}$ is too small and insufficient to support the required kinematics, the constraint would be violated (slip or separation).

The control design problem thus includes implicit constraints: choose $\boldsymbol{\tau}(t)$ such that the resulting grip force remains within reasonable bounds (e.g., hand strength limits) and maintains contact with the club.

4.8 Wrenches: Forces and Torques Unified

Until now, we've talked about torques at joints. But forces at the grip are different. To unify the description, engineers use the concept of a **wrench**.

Definition 4.4. Wrench

A wrench is the combination of a force $\boldsymbol{f}$ (3D vector) and a torque $\boldsymbol{\tau}$ (3D vector). It completely describes the mechanical interaction at a point.

$$\text{Wrench} = \begin{bmatrix} \boldsymbol{f} \\ \boldsymbol{\tau} \end{bmatrix}$$

The grip exerts a wrench on the club. The club exerts a reaction wrench on the hand. During the downswing, the grip wrench has:

- A large centripetal force (pulling the club inward).
- A tangential force (in the direction of swing motion).
- A torque that's trying to twist the club (from wrist torque).

At impact, the ground (your feet) exerts a wrench on your body. Your body must balance all the forces and torques.

In Plain Language 4.4. The Full-Body Problem

In reality, the golf swing is a full-body problem. Your feet push on the ground (ground reaction force). Your core muscles generate torques to rotate your torso. Your shoulder muscles pull your arms. All of these must be coordinated.
The manipulator equation for the arm alone is part of a larger system. Your torso rotation affects the reference frame in which the arm swings. The ground reaction force affects whether you stay balanced.
To fully understand the golf swing, you'd need to extend the manipulator equation to include the full body: torso, hips, legs, feet.
But here is the key point: the physics is the same. Each joint has an equation of the form $\boldsymbol{M}\ddot{\boldsymbol{q}} + \boldsymbol{C}\dot{\boldsymbol{q}} + \boldsymbol{g} = \boldsymbol{\tau}$. The forces are attributed to the same sources: inertia, gravity, velocity-dependent, muscular, constraint.
Understanding the arm's forces is a stepping stone to understanding the full swing.

4.9 Numerical Summary: Forces Throughout the Swing

Let's tabulate the major forces at different phases. *Note: The values in this table are order-of-magnitude illustrations based on the representative model parameters established in Chapter 3. They should not be interpreted as precise measurements; actual values vary substantially by golfer, swing speed, and technique. See [Nesbit, 2005] for inverse-dynamics measurements of joint torques during the golf swing.*

Phase	τ_{inertia}	τ_{grav}	$\tau_{\text{Cor/Cent}}$	$\tau_{\text{muscle needed}}$
Address	5 N m	-10 N m	0	+15 N m
Backswing	20 N m	-30 N m	0	+50 N m
Top	2 N m	-30 N m	0	+32 N m
Transition	50 N m	-20 N m	+5 N m	−35 N m*
Early downswing	100 N m	-10 N m	+10 N m	−80 N m*
Late downswing	60 N m	0 N m	+15 N m	−45 N m*
Impact	40 N m	+5 N m	+20 N m	−35 N m*

* Negative muscle torque means the muscles are braking, not accelerating. This is counterintuitive but happens because gravity and velocity-dependent forces are doing most of the accelerating.

4.10 The Fundamental Insight

The core principle emerging from this analysis is:

The golf swing is a dynamics problem in which muscular forces provide control input to a complex nonlinear mechanical system.

The arm+club system has substantial inertia, operates over large distances, and achieves high speeds. These characteristics create significant forces. The manipulator equation $\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{\tau}$ predicts that no single force source (muscular, gravitational, or velocity-dependent) dominates throughout the swing. Instead, they combine.

The control problem is to determine the muscle torques $\boldsymbol{\tau}(t)$ such that the resulting motion $\boldsymbol{q}(t)$ achieves the objective (e.g., clubhead velocity and orientation at impact). Given the constraints on muscular force and the nonlinear dynamics, not all objectives are equally achievable. The feasible outcomes depend on the control input design and the initial conditions.

Key Principle 4.1. Key Takeaways

1. **Forces come from five sources**: inertia, velocity-dependent (Coriolis/centrifugal), gravity, muscular, and constraint.
2. **Inertial forces resist acceleration**. To accelerate a heavier/longer arm, you need larger torques.
3. **Velocity-dependent forces grow with the square of speed**. At high speeds, they dominate. They often help, not hinder.
4. **Gravity does work during the downswing**. It's a free source of energy, converting potential to kinetic.
5. **Muscular forces are your only control**. You can't directly control gravity or inertia; you can only control $\boldsymbol{\tau}$.
6. **Constraint forces appear automatically**. The grip force is determined by the dynamics, not by conscious effort.
7. **At impact, $\sim$18g swing-phase clubhead accelerations require careful force balance**. No single source (gravity, muscle, velocity-dependence) does it alone. All work together.
8. **Optimal control strategies exploit passive dynamics**. The manipulator equation predicts that control inputs aligned with passive forces (rather than opposed to them) achieve desired trajectories with lower muscular effort.

Looking Ahead

We have catalogued five sources of force in the golf swing: inertial, velocity-dependent, gravitational, muscular, and constraint. But which forces dominate at each phase of the swing? Chapter 5 introduces the control-affine framework that separates all passive forces (the *drift*) from muscular forces (the *control*), revealing a notable structural feature: by impact, passive forces substantially exceed muscular control in magnitude.

Exercises 4.1. Chapter Exercises

1. **Inertial Torque**. The shoulder has moment of inertia $M_{11} = 1.54$ kg m^2. What muscular torque is needed just to accelerate the shoulder at $\ddot{\theta}_1 = 500 \text{ deg/s}^2 = 8.7 \text{ rad/s}^2$, ignoring all other forces?

2. **Centrifugal Force**. At impact, the shoulder rotates at $\dot{\theta}_1 = 600$ deg/s $=$ 10.47 rad/s. The forearm+club is 1.35 m from the shoulder. What is the centrifugal acceleration? What force would a 1 kg mass experience at that distance?

3. **Coriolis Coupling**. Suppose the shoulder rotates at constant rate $\dot{\theta}_1 =$ 8 rad/s. The elbow suddenly starts rotating at $\dot{\theta}_2 = 5$ rad/s. Using the Coriolis term from Chapter 3, estimate the torque this creates at the shoulder.

4. **Gravity Torque**. At the top of the backswing, $\theta_1 = 150, \theta_2 = -70$. Compute the gravitational torque at the shoulder (use parameters from Chapter 3). Is gravity trying to rotate the shoulder forward (into the downswing) or backward?

5. **Force Balance at Impact**. At impact, the total shoulder acceleration is $\ddot{\theta}_1 =$ 200 rad/s^2. The gravity torque is about 5 N m (small). The Coriolis/centrifugal torques total about 20 N m. Using $M_{11} = 1.54$ kg m^2, how much muscular torque is needed?

6. **Grip Force**. The club (mass 0.2 kg) is 1.35 m from the shoulder and accelerating at 150 g (centripetal). What force is the hand exerting on the club?

7. **Work Done by Gravity**. From address ($\theta_1 = 0$) to the top ($\theta_1 = 150$), how much does the potential energy increase? (Use parameters from Chapter 3.) From the top to impact ($\theta_1 = 0$), how much does gravity do in work?

Chapter 5

The Affine Structure: Drift and Control

"The secret of good swinging is knowing which parts you control and which parts physics controls for you."

In Plain Language 5.1. The Big Reveal

We've been building toward this chapter for four chapters. Now it's time to reveal the structure that makes the golf swing work.
The manipulator equation:

$$\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{\tau} \tag{5.1}$$

can be rewritten in a form that separates passive physics from active control:

$$\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}) + \boldsymbol{G}(\boldsymbol{x})\boldsymbol{u} \tag{5.2}$$

This is the **control-affine form**. It says:

- The system evolves in state space ($\boldsymbol{x}$).
- Left alone, the system follows the **drift** $\boldsymbol{f}(\boldsymbol{x})$.
- Your muscles add a **control** term $\boldsymbol{G}(\boldsymbol{x})\boldsymbol{u}$ on top.

This decomposition is the key to understanding the golf swing. It's where drift and control become concrete.

5.1 From Equations of Motion to State Space

To write the system in control-affine form, we need to convert the second-order manipulator equation into a first-order system in state space.

Defining State

Recall from Chapter 2 that the state is:

$$\boldsymbol{x} = \begin{bmatrix} \boldsymbol{q} \\ \dot{\boldsymbol{q}} \end{bmatrix} = \begin{bmatrix} \theta_1 \\ \theta_2 \\ \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} \tag{5.3}$$

The state evolves in time:

$$\dot{\boldsymbol{x}} = \begin{bmatrix} \dot{\boldsymbol{q}} \\ \ddot{\boldsymbol{q}} \end{bmatrix} \tag{5.4}$$

The first line is trivial: $\frac{d}{dt}[\theta_1, \theta_2]^T = [\dot{\theta}_1, \dot{\theta}_2]^T$.
The second line requires us to solve for $\ddot{\boldsymbol{q}}$ from the manipulator equation.

Solving for Accelerations

From:

$$\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{\tau} \tag{5.5}$$

we get:

$$\ddot{\boldsymbol{q}} = \boldsymbol{M}(\boldsymbol{q})^{-1} \left[\boldsymbol{\tau} - \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} - \boldsymbol{g}(\boldsymbol{q})\right] \tag{5.6}$$

This is the full equation. Now separate the passive and active terms:

$$\begin{aligned} \ddot{\boldsymbol{q}} &= \boldsymbol{M}(\boldsymbol{q})^{-1} \left[-\boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} - \boldsymbol{g}(\boldsymbol{q})\right] + \boldsymbol{M}(\boldsymbol{q})^{-1}\boldsymbol{\tau} &(5.7) \\ &= \underbrace{\boldsymbol{M}(\boldsymbol{q})^{-1} \left[-\boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} - \boldsymbol{g}(\boldsymbol{q})\right]}_{\text{drift}} + \underbrace{\boldsymbol{M}(\boldsymbol{q})^{-1}\boldsymbol{\tau}}_{\text{control}} &(5.8) \end{aligned}$$

The Control-Affine Form

Let's define:

- **Drift dynamics**: $\boldsymbol{f}(\boldsymbol{x}) = \begin{bmatrix} \dot{\boldsymbol{q}} \\ \boldsymbol{M}(\boldsymbol{q})^{-1} \left[-\boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} - \boldsymbol{g}(\boldsymbol{q})\right] \end{bmatrix}$
- **Control input**: $\boldsymbol{u} = \boldsymbol{\tau}$ (the applied torques).
- **Control matrix**: $\boldsymbol{G}(\boldsymbol{x}) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ M^{-1}(\boldsymbol{q}) & \end{bmatrix}$ (all zeros in the top rows, M^{-1} in the bottom).

Then:

$$\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}) + \boldsymbol{G}(\boldsymbol{x})\boldsymbol{u} \tag{5.9}$$

This is the **control-affine form**. It is standard in nonlinear control theory [Isidori, 1995, Bullo and Lewis, 2004]. It is called "affine" (not "linear") because the dependence on control is linear, but the dependence on state is generally nonlinear. The form applies during smooth phases of the swing; impacts, stick–slip contact, and discontinuous grip changes require hybrid or compliant extensions rather than this smooth equation alone.

In Plain Language 5.2. What Is a Vector Field?

A **vector field** assigns an arrow to every point in space. Imagine painting a tiny arrow at every point in the 4-dimensional state space. Each arrow shows the autonomous state derivative of the declared effective plant when applied generalized control is zero. At a state where the arm is high and moving slowly, gravity may dominate; at high speed, velocity-dependent inertial effects bend the trajectory. Collectively, these arrows form the drift vector field $\boldsymbol{f}(\boldsymbol{x})$. This definition does not identify a muscle state.

The control field $\boldsymbol{G}(\boldsymbol{x})\boldsymbol{u}$ adds a second arrow showing what muscle torques contribute. At any instant, the system moves in the direction of the *sum* of these two arrows. Early in the downswing, the control arrow can be significant. Near impact in the simplified models developed here, the drift arrow is often much larger than the bounded control arrow, so active torque behaves more like a steering correction than a primary source of acceleration.

Imagine standing in a river. The current is the drift field—it carries you regardless of what you do. Your swimming strokes are the control field. In a gentle stream, your strokes dominate. In whitewater rapids, the current dominates. The golf swing transitions from gentle stream (address) to rapids (impact).

The distinction between drift and control is not merely mathematical—it is fundamental to understanding the mechanics of the swing. The drift field $\boldsymbol{f}(\boldsymbol{x})$ represents the "shape" of the system's natural dynamics. Control input $\boldsymbol{G}(\boldsymbol{x})\boldsymbol{u}$ adds perturbations to this natural trajectory.

Definition 5.1. Control-Affine Systems

A system of the form:

$$\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}) + \boldsymbol{G}(\boldsymbol{x})\boldsymbol{u} \tag{5.10}$$

where:

- $\boldsymbol{f}(\boldsymbol{x})$ is the **drift vector field**: the system dynamics with no control.
- $\boldsymbol{G}(\boldsymbol{x})$ is the **control matrix**: how the control input affects the state.
- $\boldsymbol{u}$ is the **control input**.

The term "affine" means the system is linear in the control but possibly nonlinear in the state.

To build intuition for why this form is important, consider how it encodes control authority:

In Plain Language 5.3. Why "Affine" Matters

Why is this form so important?

Because it separates two different types of dynamics:

1. **What the system wants to do on its own** ($\boldsymbol{f}$): gravity pulling down, inertia

resisting, velocity-dependent forces coupling the joints.
2. **What you add on top ($\boldsymbol{Gu}$)**: muscular commands.
This separation lets you reason about controllability, stability, optimal control, and learning.
In robotics, this form is the starting point for designing controllers. In golf, it's the starting point for understanding what you actually control and what physics does for you.

5.2 Explicit Form for the Double Pendulum

Let's write out the control-affine form explicitly for the double pendulum.

The Drift Vector Field

$$\boldsymbol{f}(\boldsymbol{x}) = \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ f_3(\boldsymbol{x}) \\ f_4(\boldsymbol{x}) \end{bmatrix} \tag{5.11}$$

where:

$$f_3(\boldsymbol{x}) = -M_{11}^{-1}\left[\boldsymbol{C}_{12}\dot{\theta}_2 + g_1\right] - M_{12}^{-1}\left[\boldsymbol{C}_{22}\dot{\theta}_2 + g_2\right] \tag{5.12}$$

$$f_4(\boldsymbol{x}) = -M_{21}^{-1}\left[\boldsymbol{C}_{12}\dot{\theta}_2 + g_1\right] - M_{22}^{-1}\left[\boldsymbol{C}_{22}\dot{\theta}_2 + g_2\right] \tag{5.13}$$

These are complicated expressions, but they encode all the passive dynamics: gravity pulling, inertia resisting, and velocity-dependent coupling.

The key insight: **$\boldsymbol{f}$ depends on the current state but does not depend on your muscular control.**

The Control Matrix

$$\boldsymbol{G}(\boldsymbol{x}) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ M_{11}^{-1}(\boldsymbol{q}) & M_{12}^{-1}(\boldsymbol{q}) \\ M_{21}^{-1}(\boldsymbol{q}) & M_{22}^{-1}(\boldsymbol{q}) \end{bmatrix} \tag{5.14}$$

where $\boldsymbol{M}^{-1}(\boldsymbol{q})$ is the inverse of the mass matrix.
The control input is:

$$\boldsymbol{u} = \begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix} \tag{5.15}$$

So the full system is:

$$\begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{bmatrix} = \boldsymbol{f}(\boldsymbol{x}) + \boldsymbol{G}(\boldsymbol{x}) \begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix} \tag{5.16}$$

5.3 Drift as a Vector Field

The drift $\boldsymbol{f}(\boldsymbol{x})$ is a vector field on state space. At each point $\boldsymbol{x}$, it specifies a direction and magnitude of motion.

Definition 5.2. Vector Field

A function that assigns a vector to each point in space. In our case, at each state $\boldsymbol{x}$ in phase space, $\boldsymbol{f}(\boldsymbol{x})$ tells you which direction the state is moving if there's no control.

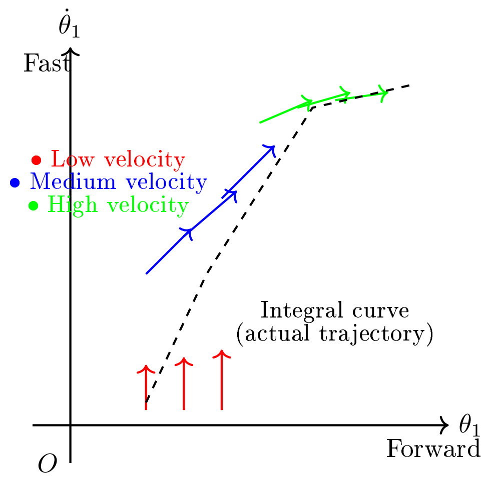


Figure 5.1: Qualitative drift vector field in a 2D slice of state space (shoulder angle θ_1 and angular velocity $\dot{\theta}_1$). Each arrow shows the direction the system evolves with zero muscle torque. The trajectory (dashed curve) follows the vector field, accelerating downward (gravity) and forward (momentum).

Example 5.1. Drift in the Double Pendulum at Rest

Suppose you're at address $(\theta_1 = 0, \theta_2 = 0, \dot{\theta}_1 = 0, \dot{\theta}_2 = 0)$ and you suddenly remove all muscular control ($\boldsymbol{\tau} = 0$).
The drift says what happens next. Let's compute:

- $\dot{\theta}_1 = 0$ (velocity doesn't change in this instant).
- $\dot{\theta}_2 = 0$ (velocity doesn't change).
- $\ddot{\theta}_1 =?$ Determined by $\boldsymbol{f}$.
- $\ddot{\theta}_2 =?$ Determined by $\boldsymbol{f}$.

At rest, there's no inertia acceleration (velocities are zero). The only terms in $\boldsymbol{f}$ are the gravity terms:

$$\begin{aligned}
\ddot{\theta}_1 &= \boldsymbol{M}_{11}^{-1} \cdot (-g_1) && (5.17)\\
&= \boldsymbol{M}_{11}^{-1} \cdot (-(M_1 L_{1,cm} + M_2 L_1) g \sin(0)) && (5.18)\\
&= 0 && (5.19)
\end{aligned}$$

So the arm doesn't accelerate if it's at rest and perfectly vertical. That makes sense: gravity is balanced.
Now suppose you're at an angle, say $\theta_1 = 45$. Then:

$$\begin{aligned}
\ddot{\theta}_1 &= \boldsymbol{M}_{11}^{-1} \cdot (-(M_1 L_{1,cm} + M_2 L_1) g \sin(45)) && (5.20)\\
&= \boldsymbol{M}_{11}^{-1} \cdot (-(2.5 \times 0.175 + 1.5 \times 0.35) \times 9.81 \times 0.707) && (5.21)\\
&\approx -20 \text{ N m} \times 0.45 \text{ kg m}^{-2} && (5.22)\\
&\approx -9 \text{ rad/s}^2 && (5.23)
\end{aligned}$$

Negative means the arm wants to rotate back toward vertical. Gravity is pulling it down.

Flowing Along the Drift

If you start at some state $\boldsymbol{x}_0$ and let the system drift with $\boldsymbol{\tau} = 0$, the state will flow along a trajectory determined by $\boldsymbol{f}$.

This trajectory is called an **integral curve** or **orbit** of the vector field.

In the golf swing, the drift is what happens if you let go of the club at any point. Gravity will pull it down. The momentum will carry it forward. The trajectory is set by $\boldsymbol{f}$.

In Plain Language 5.4. Why Understanding Drift Matters

A useful structural observation is that the system dynamics separate into passive and active components. Understanding the drift $\boldsymbol{f}(\boldsymbol{x})$ is the foundation for any control strategy.
Consider a system in the downswing phase. At each instant, the equation $\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}) + \boldsymbol{G}(\boldsymbol{x})\boldsymbol{u}$ predicts how the state evolves. If the control input $\boldsymbol{u}$ is chosen to align with the drift field—rather than oppose it—the system reaches desired states more efficiently.
More specifically: when muscular torques are directed to modulate (rather than dominate) the natural trajectory, the overall motion emerges from the cooperation of passive forces and active steering. This is energetically favorable because the gravitational and centrifugal forces are already doing substantial work. By directing these passive forces, rather than fighting them, the system achieves large accelerations with modest muscular effort.
The drift vector field $\boldsymbol{f}(\boldsymbol{x})$ encodes this natural trajectory. Understanding it enables the design of control inputs that work synergistically with the passive dynamics.

5.4 Control as a Steering Input

The control term $\boldsymbol{G}(\boldsymbol{x})\boldsymbol{u}$ is what you add on top of drift. It's a steering input.

Think of it like driving a car. The car naturally rolls forward due to gravity and rolling resistance. That's the drift. You turn the wheel to steer. That's the control. Together, they determine where the car goes.

In the golf swing, the drift is the natural motion of your arm under gravity and inertia. Your muscles steer that motion toward the target.

The Control Matrix as Leverage

The control matrix $\boldsymbol{G}(\boldsymbol{x})$ depends on the state (specifically, on the configuration $\boldsymbol{q}$). This is the leverage: how much does a given muscle torque affect the state acceleration?

Example 5.2. Control Leverage at Different Configurations

The inverse mass matrix $\boldsymbol{M}^{-1}(\boldsymbol{q})$ tells you the leverage.
At address ($\theta_1 = 0, \theta_2 = 0$): the arm is extended. The moment of inertia is high ($M_{11} \approx 1.54$ kg m^2). So $M_{11}^{-1} \approx 0.64$ rad/s^2 per N m. A 50 N m torque produces $50 \times 0.64 = 32$ rad/s^2 of acceleration.
At the top of the backswing ($\theta_1 = 150, \theta_2 = -70$): the arm is bent. The effective moment of inertia is lower. So M_{11}^{-1} is higher. A given muscle torque produces a larger acceleration.
This may seem counterintuitive: if the arm is shorter (bent), should it not be harder to accelerate?
No. The **moment of inertia** is lower (because the mass is closer to the axis). So M^{-1} is higher (easier to accelerate). This is why it's easier to accelerate a bent arm than an extended one.

5.5 The Drift-Control Ratio (DCR)

The **Drift-Control Ratio (DCR)** is a diagnostic, not a coordinate-free physical constant.

Definition 5.3. Drift-Control Ratio

Because $\boldsymbol{f}$ and $\boldsymbol{G}\boldsymbol{u}$ are state-derivative vectors, the ratio must be computed with an explicit norm or weighting. In this chapter we use either the acceleration block alone or a stated state-space weighting matrix $\boldsymbol{W}$:

$$\mathrm{DCR}_{\boldsymbol{W}}(\boldsymbol{x}; u_{\max}) = \frac{\|\boldsymbol{W}\boldsymbol{f}(\boldsymbol{x})\|}{\max_{\|\boldsymbol{u}\| \le u_{\max}} \|\boldsymbol{W}\boldsymbol{G}(\boldsymbol{x})\boldsymbol{u}\| + \varepsilon}, \tag{5.24}$$

where ε prevents division by zero in states where the selected control channel has negligible authority.
If DCR is high, drift dominates. Muscles contribute little. If DCR is low, muscles dominate. Drift contributes little.

The DCR varies throughout the swing.

In Plain Language 5.5. Reading the DCR

The Drift-Control Ratio answers a simple question: *how much of the motion is passive physics, and how much is active control?*
The following numerical regimes provide qualitative guidelines that would need to be validated by simulation or experiment with realistic biomechanical parameters:

- DCR < 1: Control input dominates. Muscular torques are the primary driver of state evolution. This regime corresponds roughly to address and backswing, where velocities are low.
- DCR ≈ 1–5: Control and drift contribute comparably. The system evolution depends significantly on both passive forces and muscular input. Transition and early downswing typically occupy this regime.
- DCR > 5–10: Drift dominates under the chosen weighting. Passive forces (gravity, velocity-dependent Coriolis and centrifugal forces) produce large modeled accelerations compared to bounded muscular input. Control becomes a fine-tuning perturbation. Late downswing can approach this regime in the worked examples, but the values depend on swing speed, model parameters, torque limits, and the selected norm.

Caveat: The exact numerical thresholds depend on swing parameters (arm segment masses and inertias), swing speed, the magnitude of applied muscular torque, and the weighting matrix used in Eq. (5.24). These values should be validated through simulation or experimental measurement for specific conditions.
Practical implication: Control authority is highest early in the downswing (low DCR). As DCR increases, the window for meaningful muscular intervention narrows. Thus, deliberate control of impact conditions must be established early.

Address: Low DCR

At address, the arm is static ($\dot{\boldsymbol{q}} = 0$). There's no velocity-dependent drift. Gravity is balanced (assuming vertical arm). The drift is small.

To accelerate the backswing, you must generate a large muscular torque. DCR is low (muscles doing most of the work).

Top of Backswing: Transitioning DCR

At the top, velocities are near zero. Gravity is pulling strongly ($\theta_1 = 150$). The drift is moderate.

As you initiate the downswing, you generate muscular torques. But gravity is already helping. DCR starts to increase (drift contributes more).

Downswing: High DCR

During the rapid downswing, velocities are high. Gravity and centrifugal forces create large accelerations. Your muscular torques are now mostly steering, not accelerating.

DCR is very high (drift is doing most of the work).

Impact and Beyond: Extremely High DCR

At impact, the clubhead is accelerating at $\sim$180 m/s^2 ($\approx 18g$; see Chapter 4, Eq. 4.18). This is primarily from:

- Centrifugal force (from the high rotation speed): $\approx$ 50 N m equivalent.
- Gravity: $\approx$ 10 N m equivalent.

Your muscular contribution is perhaps on the order of 10–30 N m (estimates vary widely; exact values depend on golfer strength and technique—see [Nesbit, 2005] for inverse-dynamics measurements). DCR is extremely high (passive forces dominating).

In Plain Language 5.6. The Three Regimes

The golf swing naturally decomposes into three regimes characterized by the Drift-Control Ratio:

Regime 1: Low DCR (Address, Backswing)

- Drift is minimal (arm at rest or slow motion, velocity-dependent forces negligible).
- Control input is the dominant term in $\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}) + \boldsymbol{G}(\boldsymbol{x})\boldsymbol{u}$.
- Muscles must generate torques to overcome inertia and initiate acceleration.
- The system response is primarily determined by the control input; small changes in $\boldsymbol{u}$ produce significant changes in state evolution.

Regime 2: Medium DCR (Transition)

- Drift is ramping up as velocities increase; velocity-dependent forces (Coriolis, centrifugal) become significant.
- Muscles and drift contribute comparably to $\dot{\boldsymbol{x}}$.
- Gravity does positive work, assisting the downswing motion.
- The control input must be modulated to maintain desired trajectory as passive forces increase.

Regime 3: High DCR (Downswing, Impact, Follow-through)

- Drift dominates the state evolution; passive dynamics overwhelm the control term.
- Velocity-dependent forces (Coriolis, centrifugal) produce torques vastly larger than muscular input.
- Control input contributes steering and fine-tuning, not primary acceleration.
- The system trajectory is largely determined by initial conditions and the drift field; control has minimal influence on the overall motion.

5.6 Riding the Drift: The Essence of Skill

The control-affine framework reveals a fundamental structure: the golf swing decomposes into passive dynamics and active control.

The optimal control problem in golf is to choose muscular torques $\boldsymbol{\tau}(t)$ that achieve desired impact conditions while respecting constraints on muscular effort.

From the perspective of control theory:

- The drift $\boldsymbol{f}(\boldsymbol{x})$ represents the natural evolution of the system under gravity, inertia, and velocity-dependent forces.
- The control term $\boldsymbol{G}(\boldsymbol{x})\boldsymbol{u}$ represents muscular perturbations to this natural trajectory.
- When DCR is low (early swing), control significantly influences the trajectory. Muscular torques must be substantial.
- When DCR is high (late swing), the drift dominates. Control becomes a fine-tuning adjustment.

A key insight from optimal control is that **the most efficient solution minimizes muscular effort (cost) while reaching the target**. This typically means:

- Apply control input early, when its effect is large (low DCR).
- Allow passive forces to carry the motion in phases where drift is strong.
- Time final corrections to synchronize with impact.

This framework provides a scientific foundation for understanding the role of control strategy in achieving efficient, repeatable motion.

Example 5.3. Hypothetical Control Profile in the Downswing

Consider a hypothetical control trajectory during the downswing. Let the control input $\boldsymbol{u}(t)$ vary as a function of state and time.

Scenario: State Space Control

Suppose the system evolves as $\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}) + \boldsymbol{G}(\boldsymbol{x})\boldsymbol{u}(t)$. We can analyze three possible control strategies:

Strategy 1: Constant Feedforward Control

Apply a fixed torque input $\boldsymbol{u} = \text{const}$ throughout the downswing (say, $u_1 = 10$ N m). The resulting state trajectory $\boldsymbol{x}(t)$ depends on both $\boldsymbol{f}(\boldsymbol{x})$ and the constant input. In early downswing (low DCR), this input significantly shapes the trajectory. In late downswing (high DCR), the drift dominates and the control term becomes a small correction.

Strategy 2: Feedback Control (State-Dependent)

Apply control that adjusts based on the current state: $\boldsymbol{u}(t) = -K(\boldsymbol{x}(t) - \boldsymbol{x}_{\text{ref}})$ (a simple proportional law). This controller steers the system toward a reference trajectory. The effectiveness of feedback control depends on DCR—when DCR is high, feedback gains must be large, but muscular constraints limit achievable input magnitudes.

Strategy 3: Optimal Control
Solve the optimal control problem: choose $\boldsymbol{u}^*(t)$ to minimize a cost functional (e.g., effort $\int |\boldsymbol{u}|^2 dt$ subject to reaching target impact state). The optimal solution exploits the drift field, applying control primarily when DCR is low and allowing drift to dominate when DCR is high.
Key Insight: Strategies that achieve desired outcomes with low muscular effort are those that recognize the DCR regime and modulate control accordingly. Early in the downswing, where DCR is low, control must be applied to initiate acceleration. Later, when DCR is high, the same control magnitude contributes only small course corrections because drift so vastly dominates the dynamics.

5.7 The Geometry of the Golf Swing

The control-affine form gives us a geometric picture of the golf swing.

Phase space is a 4D manifold (for a 2-DOF arm). On this manifold:

- The drift vector field $\boldsymbol{f}$ is "painted" at each point, showing the natural flow.
- Your control $\boldsymbol{u}$ perturbs this flow locally via $\boldsymbol{G}(\boldsymbol{x})$.
- A swing is a path through phase space from start (address) to impact and finish.

Elite golfers have learned to follow a path that: 1. Uses small control perturbations. 2. Exploits high-DCR regimes where drift helps. 3. Avoids fighting the vector field.

Constraint Insight 5.1. The Reachability Problem

A key question: given a starting state (address) and control constraints (max muscle torque), what states are reachable?
In control theory, this is the **reachability problem**. For the golf swing, it translates to: given my muscle strength, how far can I hit?
The manipulator equation tells us that reach depends not just on max muscle torque, but also on how well we exploit the drift. An optimized swing that rides the drift reaches farther than one that fights it.
This is why technique matters more than strength.

5.8 Optimal Control and the Swing

In principle, the golf swing can be formulated as an optimal control problem [Penner, 2003a]:

Problem: Given the dynamics $\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}) + \boldsymbol{G}(\boldsymbol{x})\boldsymbol{u}$ and constraints on $\boldsymbol{u}$ (muscle strength limits), find the control $\boldsymbol{u}^*(t)$ that optimizes a performance criterion (e.g., maximize club-head speed, minimize energy expenditure, minimize accuracy dispersion).

Solution: Apply optimal control theory (Pontryagin's maximum principle or dynamic programming) to solve the resulting two-point boundary value problem.

Result: An optimal control trajectory $\boldsymbol{u}^*(t)$ that specifies the muscular torques to apply at each instant to achieve the optimization goal while respecting the system dynamics and constraints.

In Plain Language 5.7. Solving the Optimal Control Problem

In principle, the optimal control problem can be formulated as:
Minimize: $J(\boldsymbol{u}) = \int_0^T L(\boldsymbol{x}(t), \boldsymbol{u}(t))dt$ (a cost functional, e.g., muscular effort)
Subject to: $\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}) + \boldsymbol{G}(\boldsymbol{x})\boldsymbol{u}$, initial condition $\boldsymbol{x}(0)$, terminal constraint $\boldsymbol{x}(T) \in \mathcal{X}_{\text{target}}$.
The solution $\boldsymbol{u}^*(t)$ is an optimal control trajectory. For the golf swing, this solution depends on:

- The system dynamics (inertias, masses, gravity).
- The cost functional (what is being optimized—distance, accuracy, energy, or some combination).
- The constraints (muscle strength, joint limits, impact requirements).

Computing $\boldsymbol{u}^*(t)$ in closed form is generally difficult for nonlinear systems. However, numerical methods (dynamic programming, shooting methods, etc.) can approximate the solution.
Observation: The manipulator-equation structure—$\boldsymbol{M}(\boldsymbol{q})$, $\boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})$, $\boldsymbol{g}(\boldsymbol{q})$—is shared across smooth multibody models [Featherstone, 2008, Murray et al., 1994]. The qualitative prediction of this model class is therefore structural rather than universal: early control is valuable when DCR is low, and passive dynamics can be exploited when DCR is high. Individual variation enters through segment parameters, strength, flexibility, coordination strategy, and the accuracy of the model itself.

5.9 Putting It All Together: The Unified Picture

Let's step back and see the whole picture.

The System

Your arm + club is a mechanical system described by:

$$\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{\tau} \tag{5.25}$$

The State Space

The system evolves in 4D state space: two angles and two angular velocities. Every swing is a path in this space.

The Drift

Without muscular control, the system follows the drift:

$$\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}) \tag{5.26}$$

Drift encodes:

- Gravity pulling the arm down.

- Inertia resisting accelerations.
- Velocity-dependent (Coriolis, centrifugal) forces coupling the joints.

At high speeds, drift dominates. At low speeds, muscles must do most of the work.

The Control

Your muscles generate torques $\boldsymbol{\tau}$ that perturb the drift:

$$\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}) + \boldsymbol{G}(\boldsymbol{x})\boldsymbol{u} \tag{5.27}$$

Control is linear in the input but couples to state through $\boldsymbol{G}(\boldsymbol{x})$.

The Skill

Elite golf is the art of choosing $\boldsymbol{u}(t)$ (the control trajectory) such that: 1. The resulting swing path $\boldsymbol{x}(t)$ reaches impact with the desired orientation and speed. 2. The control effort (sum of $|\boldsymbol{u}|$ over time) is minimized. 3. The swing is robust to perturbations (small errors don't derail the outcome).

All of this is encoded in understanding and exploiting the drift.

Key Principle 5.1. Key Takeaways

1. **Control-affine form** separates drift and control: $\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}) + \boldsymbol{G}(\boldsymbol{x})\boldsymbol{u}$.
2. **Drift** $\boldsymbol{f}(\boldsymbol{x})$ is what the system does without control: gravity, inertia, and velocity-dependent forces.
3. **Control** $\boldsymbol{G}(\boldsymbol{x})\boldsymbol{u}$ is what muscles add: steering, not accelerating (mostly).
4. **The Drift-Control Ratio (DCR)** varies through the swing.
   - Low at address and backswing (muscles do most of the work).
   - Medium at transition (drift ramping up).
   - High at impact and beyond (drift dominates).
5. **Skilled golfers exploit high-DCR phases** by preparing initial conditions and transition timing that let drift assist rather than fight the intended trajectory.
6. **The golf swing is an optimal control problem**. The nervous system implicitly solves it through learning and practice.
7. **Technique involves working with the drift**. Strength sets the ceiling on available control torques, but efficient technique aligns muscular commands with the passive dynamics rather than opposing them.
8. **Physics is universal**. While individual golfers vary in geometry and strength, the fundamental structure (the equations $\boldsymbol{M}, \boldsymbol{C}, \boldsymbol{g}$) is the same.

Exercises 5.1. Chapter Exercises

1. **Drift at Address**. At address ($\theta_1 = 0, \theta_2 = 0, \dot{\theta}_1 = 0, \dot{\theta}_2 = 0$), compute the drift $\boldsymbol{f}(\boldsymbol{x})$. What are the resulting accelerations if you remove all muscular control?

2. **Drift at the Top**. At the top of the backswing ($\theta_1 = 150, \theta_2 = -70, \dot{\theta}_1 = 0, \dot{\theta}_2 = 0$), compute the drift. Is the system accelerating? In which direction?

3. **Drift During Downswing**. Compute the drift at the transition ($\theta_1 = 140, \theta_2 = -60, \dot{\theta}_1 = 5$ rad/s, $\dot{\theta}_2 = 0$). How much do Coriolis forces contribute compared to gravity?

4. **DCR at Different Phases**. For each of the following states, estimate the drift magnitude and the control effect magnitude (assuming 30 N m muscular effort):

   (a) Address (arm at rest, vertical)
   (b) Top of backswing (arm up, at rest)
   (c) Transition (arm rotating, accelerating)
   (d) Impact (arm rotating fast, wrist releasing)

   What is the DCR in each phase?

5. **Optimal Control Intuition**. If you were solving the optimal control problem to maximize distance, when would you apply maximum muscular torque? Would it be early (backswing) or late (downswing)? Why?

6. **Weak vs. Strong Golfer**. A weak golfer has less max muscular torque (say, 30 N m vs. 50 N m for a strong golfer). How would this affect:

   (a) The swing speed?
   (b) The reliance on drift in the downswing?
   (c) The optimal control strategy?

7. **Comparison to Throwing**. How is the control-affine structure of the golf swing similar to or different from the baseball throw? In both, you accelerate a mass. Are the regimes of DCR different?

Closing Thoughts: The Unity of Physics and Swing

We have traveled from first principles through the double pendulum model to the control-affine structure that unifies drift and control. This mathematical framework provides a scientific foundation for understanding the golf swing.

The key insight is: **the golf swing obeys the equations of classical mechanics**.

The manipulator equation:

$$\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{\tau} \tag{5.28}$$

and its control-affine decomposition:

$$\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}) + \boldsymbol{G}(\boldsymbol{x})\boldsymbol{u} \tag{5.29}$$

form a complete description of the arm-club dynamics. The optimal choice of control input $\boldsymbol{\tau}(t)$ depends on the task objectives (e.g., maximize distance, minimize dispersion), the constraints (muscle strength, anatomical limits), and the initial conditions. A natural question follows: given this decomposition, how much does each individual muscle contribute to the acceleration of each joint at any given instant? This is precisely the question addressed by *induced acceleration analysis*, a methodology developed in the biomechanics literature that exploits the invertibility of $\boldsymbol{M}(\boldsymbol{q})$ to decompose accelerations into per-muscle and per-force contributions. We take up this question in Chapter 27.

The structural insights from this analysis are:

1. Passive forces (gravity, velocity-dependent coupling) dominate the late downswing. 2. Control authority is highest early, when DCR is low. 3. An efficient swing exploits drift rather than opposes it. 4. The fundamental dynamics are universal; individual variation reflects differences in body parameters and strength.

Understanding this structure provides a scientific perspective on swing mechanics and the role of control strategy in achieving performance objectives.

Chapter 6

The Zero-Torque Counterfactual

What does this exact model compute when one declared generalized-torque channel is set to zero?
The answer is conditional on the intervention record.

—A thought experiment in constraint physics

In Plain Language 6.1. The Central Question

The model assigns terms to a declared autonomous plant and to declared input channels. That bookkeeping does not identify individual muscle forces or divide a real swing into physiology versus physics.
The **forward Zero-Torque Counterfactual (ZTCF) trajectory** is a model intervention. It asks: **What trajectory follows when the declared applied generalized-control channel is set to zero while the declared effective plant is retained?**
The model result is counterintuitive. Its forward ZTCF trajectory—obtained by zeroing the declared applied generalized-control channel while retaining the effective plant—reproduces part of the downswing shape. This model comparison does not identify muscle activation or establish that biological control is primarily restraining.

6.1 Formal Definition of the ZTCF

The concept of analyzing passive motion is foundational in biomechanics. By studying what a system does without active control, researchers can isolate and understand the contribution of passive forces [Nesbit, 2005, MacKenzie and Sprigings, 2009a].

Definition 6.1. The Zero-Torque Counterfactual

The **forward Zero-Torque Counterfactual trajectory** is $\boldsymbol{q}_{\mathrm{ZTCF}}(t)$ obtained by solving the declared effective-plant equations from the current state $(\boldsymbol{q}(t_0), \dot{\boldsymbol{q}}(t_0), \boldsymbol{z}(t_0))$ with the declared applied generalized control set to zero:

$$\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{h}(\boldsymbol{q}, \dot{\boldsymbol{q}}, \boldsymbol{z}; \boldsymbol{p}) = \boldsymbol{0}, \qquad \boldsymbol{u}(t) \equiv \boldsymbol{0} \tag{6.1}$$

The retained term $\boldsymbol{h}$ must be declared and may include Coriolis/centrifugal effects, gravity, passive joint impedance, shaft dynamics, damping, and modeled contact or constraint reactions. Exogenous loads and contact modes follow the stated protocol. Zero declared control does not imply zero muscle activation, zero co-contraction, or a flaccid body.

6.2 Normative ZTCF Contract

Every executable or quantitative record uses `affinedrift.ztcf-intervention/v1`. The published normative schema and Python golden fixture freeze model/version/revision, inputs, state, units, frame, retained controls/constraints/contact/loads/internal states, solver/version, horizon, preconditions, postconditions, and failure states. The ZTCF is a model-conditioned intervention. The record separates simulated trajectory difference from any contribution measure. It states each causal estimand and physiological interpretation, and records non-identifiability explicitly. Unavailable or engine-unsupported interventions fail closed; no MATLAB, Simulink, or cross-engine parity result is inherited from the Python fixture. The public records are https://affinedrift.com/data/ztcf/ztcf_intervention_v1.schema.json and https://affinedrift.com/data/ztcf/planar_golf_forward_fixture_v2.json.

The current fixture uses model version 2.0: a rigid three-link chain with consistent rigid inertia and bias forces. It integrates no flexible shaft states or additional shaft inertia. The historical version 1 fixture is preserved as a record of the earlier mixed-inertia calculation and is unsupported by the current adapter. The model change requires a new result; it does not inherit the old numerical or physical interpretation.

In Plain Language 6.2. Understanding the ZTCF

Think of this as a software intervention rather than a physiological event. At a selected state, the simulator sets the declared applied generalized-control vector to zero while retaining the effective plant and protocol-specified loads. It then evaluates either an instantaneous sample or a forward rollout.

The answer is the trajectory of the retained effective plant. Its forces are exactly those listed in the intervention record; they are not assumed to be purely gravitational, inertial, or biologically passive. The rollout is a software branch, not a measured muscle state, and cannot uniquely allocate a difference to activation, effort, strategy, or omitted forces.

Agreement with measured motion supports the adequacy of that model and intervention over the tested interval; it does not establish low EMG or relaxation. Divergence can reflect control input, parameter error, omitted forces, contact mismatch, or numerical error.

6.3 The Decomposition: What the ZTCF Reveals

At a fixed state, the acceleration block admits the pointwise decomposition

$$\boldsymbol{a}(\boldsymbol{x},\boldsymbol{u}) = \boldsymbol{a}_{\text{drift}}(\boldsymbol{x}) + \boldsymbol{B}_a(\boldsymbol{x})\boldsymbol{u} \tag{6.2}$$

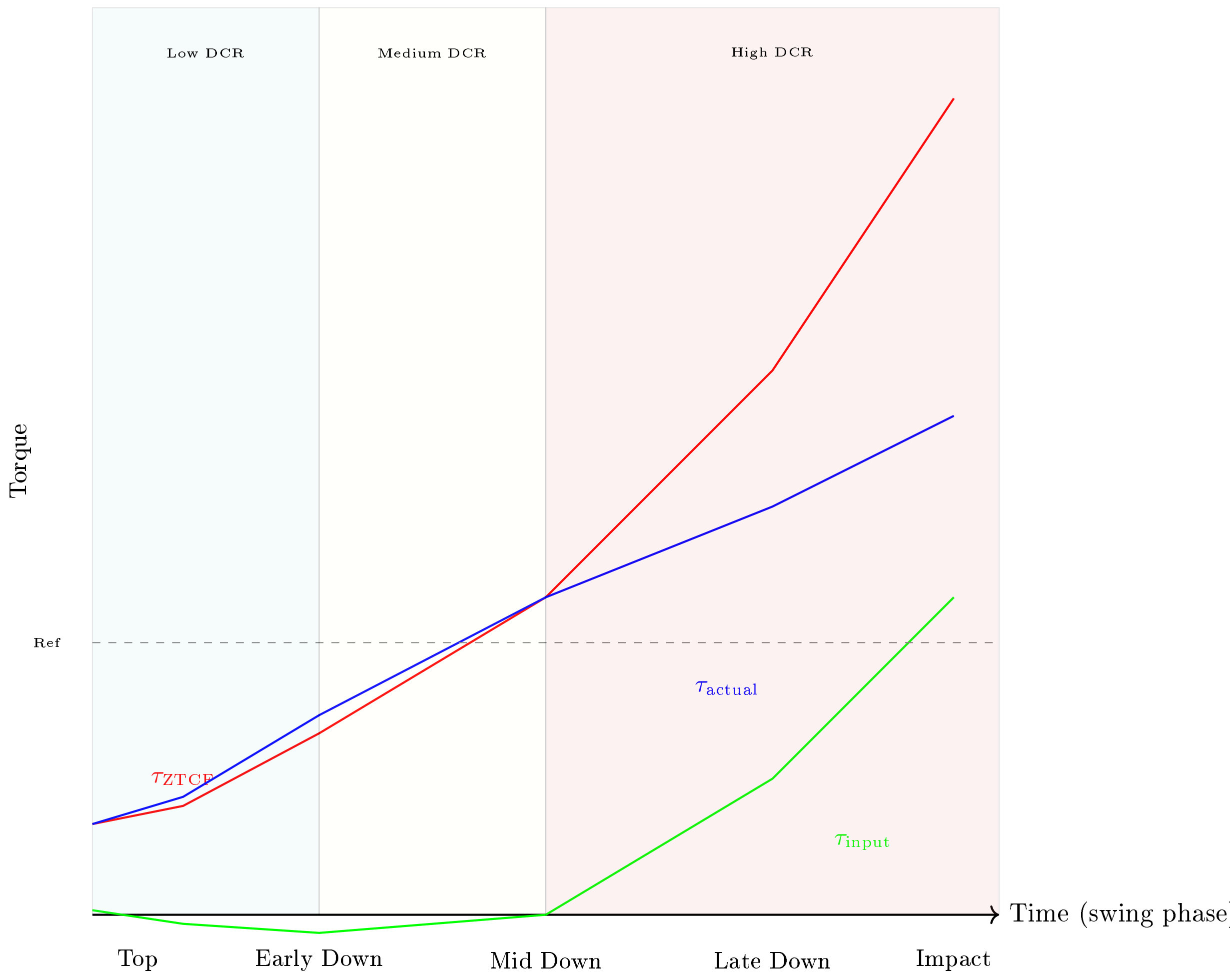


Figure 6.1: Schematic model comparison of a declared drift-equivalent generalized quantity (red) and a separately specified reference branch (blue). The green difference is illustrative only; without an intervention record it is not a validated trajectory, causal effect, or physiological attribution.

where:

- $\boldsymbol{a}_{\text{drift}}$ is the acceleration induced by the declared autonomous effective plant.
- $\boldsymbol{B}_a \boldsymbol{u}$ is the acceleration induced by the declared applied generalized-control channel.

Calling a drift term a "ZTCF torque" is avoided because generalized force and acceleration are different representations. Any force-space decomposition must declare its mass matrix, Jacobian, projection, and constraint treatment.

Key Principle 6.1. Generalized Input Is Not Physiological Attribution

Net generalized torque, the modeled input term, individual muscle forces, activation, co-contraction, impedance, and effort are distinct quantities. A pointwise affine decomposition is exact within its declared model, but it does not identify the biological source of either term.

In Plain Language 6.3. What This Analysis Can Support

A registered ZTCF can compare model branches, test implementation closure, and quantify a declared task difference. It cannot prescribe setup, timing, relaxation, or coaching cues. Human-performance conclusions require measured participants and an analysis that separates model discrepancy from the named intervention.

6.4 Computing the ZTCF: Algorithm and Simulation

Algorithm 6.1. Computing the ZTCF Trajectory

Given the current state $\boldsymbol{x}(t_0) = (\boldsymbol{q}(t_0), \dot{\boldsymbol{q}}(t_0))$ at time t_0:

1. **Capture the state:** Record the joint angles and angular velocities at the moment of interest (e.g., at mid-downswing).
2. **Set up the passive equations:**
$$\ddot{\boldsymbol{q}} = \boldsymbol{M}(\boldsymbol{q})^{-1}\left[-\boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} - \boldsymbol{g}(\boldsymbol{q})\right] \tag{6.3}$$
This equation gives the acceleration from the terms retained in this simplified declared model.
3. **Integrate:** Numerically integrate this ODE from t_0 forward in time using a standard solver (e.g., RK4). The declared applied generalized-control channel is zero; the retained effective plant determines the branch.
4. **Report the branch:** Compare only under a declared metric and uncertainty model. Divergence is not uniquely attributable to active control.

Example 6.1. Generic Branch-Switch Pattern

A branch-switch architecture proceeds by:

1. Registering the forward-dynamics model and retained-force inventory.
2. Including a **"kill switch"** parameter, a binary flag k_u that multiplies the input torques:
$$\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q}) = k_u \cdot \boldsymbol{B}(\boldsymbol{q})\boldsymbol{u} \tag{6.4}$$
3. Running the model twice:

- With $k_u = 1$: the actual (controlled) dynamics.
- With $k_u = 0$: the ZTCF (uncontrolled) dynamics.

4. Both simulations start from the same registered state. Differences remain conditional on model identity, retained protocols, and numerical error.

The Python fixture is the only executable engine registered here. A Simulink switch is an illustrative architecture, not a supported or parity-validated implementation under this contract.

6.5 The Zero-Velocity Counterfactual (ZVCF)

Definition 6.2. The Zero-Velocity Counterfactual

The **instantaneous Zero-Velocity Counterfactual (ZVCF)** is the zero-velocity, zero-control acceleration at the current configuration and declared internal state. It is an evaluation, not a trajectory:

$$\boldsymbol{a}_{\mathrm{ZVCF}}(\boldsymbol{q}, \boldsymbol{z}; \boldsymbol{p}) = -\boldsymbol{M}(\boldsymbol{q})^{-1}\boldsymbol{h}(\boldsymbol{q}, \boldsymbol{0}, \boldsymbol{z}_0; \boldsymbol{p}), \qquad \boldsymbol{u} = \boldsymbol{0} \tag{6.5}$$

In a gravity-only rigid model this reduces to $-\boldsymbol{M}^{-1}\boldsymbol{g}$. A richer effective plant can retain configuration-dependent elasticity and other protocol-specified loads.

In Plain Language 6.4. ZVCF vs. ZTCF: What's the Difference?

The ZTCF sets the declared control channel to zero. A pointwise ZTCF sample retains measured velocity; a forward or branched ZTCF trajectory integrates the resulting autonomous dynamics.
The ZVCF evaluates one instantaneous acceleration with both velocity and declared control set to zero. It is not a frozen-and-released trajectory, because the next state would generally have nonzero velocity and cease to satisfy the defining intervention.
This is useful because it isolates a clean decomposition:

$$\boldsymbol{a}_{\mathrm{drift}}(\boldsymbol{q}, \dot{\boldsymbol{q}}, \boldsymbol{z}) = \boldsymbol{a}_{\mathrm{ZVCF}}(\boldsymbol{q}, \boldsymbol{z}) + \boldsymbol{a}_{\mathrm{velocity}}(\boldsymbol{q}, \dot{\boldsymbol{q}}, \boldsymbol{z}) \tag{6.6}$$

This difference isolates velocity-dependent drift only when both evaluations use identical parameters, internal state, load protocol, and contact mode.

6.6 The Drift-Control Ratio (DCR) and Late-Downswing Authority

Definition 6.3. Drift-Control Ratio (DCR)

At any time during the swing, the **Drift-Control Ratio (DCR)** compares modeled drift against bounded modeled control in the same acceleration or task-projected space under an explicit norm or weighting. Consistent with Chapter 5, use

$$\mathrm{DCR}_{\boldsymbol{W},\mathcal{U}}(\boldsymbol{x}) = \frac{\|\boldsymbol{W}\boldsymbol{a}_{\mathrm{drift}}(\boldsymbol{x})\|}{\sup_{\boldsymbol{u}\in\mathcal{U}(\boldsymbol{x})}\|\boldsymbol{W}\boldsymbol{B}_a(\boldsymbol{x})\boldsymbol{u}\| + \varepsilon}, \tag{6.7}$$

where $\boldsymbol{W}$ states which coordinates or task directions are compared, $\mathcal{U}(\boldsymbol{x})$ is the admissible control set, and ε is a declared regularizer. A single-joint torque ratio is a useful special case only after those modeling choices are fixed.

When DCR ≈ 1, modeled drift and bounded control have comparable magnitude under the chosen weighting. When DCR $\gg 1$, modeled drift dominates that comparison; it does not imply that the golfer is passive or that muscle activation is absent.

Constraint Insight 6.1. Minimum Reporting Standard for DCR

Any numerical DCR claim should report enough information for another reader to reproduce or challenge it:

1. the state vector, coordinate convention, and whether $\boldsymbol{f}$ and $\boldsymbol{G}$ are expressed in generalized coordinates, task coordinates, or acceleration coordinates;
2. the weighting matrix $\boldsymbol{W}$, norm, and small denominator regularizer ε;
3. the admissible torque or activation bound $u_{\max}$ and whether it is joint-specific, time-varying, or muscle-model-derived;
4. the anthropometric, club, shaft, contact, and damping parameters used by the model; and
5. a sensitivity check showing whether the qualitative conclusion survives plausible perturbations of $\boldsymbol{W}$, $u_{\max}$, and the inertial parameters.

Without these declarations, a value such as DCR $= 10$ should be read only as an illustrative scale, not as an experimentally established phase boundary.

Constraint Insight 6.2. DCR as a Magnitude Diagnostic

The Drift-Control Ratio asks: **How large is the modeled drift field compared with bounded control acceleration under the stated projection, norm, and input set?** It does not determine whether a subsequent state is reachable, whether a ZTCF branch approaches the reference branch, or how impact conditions respond to an intervention. Those are finite-horizon questions.

Example 6.2. The DCR Growth in a Typical Swing

Consider a two-segment arm (shoulder and elbow) during the downswing. Assume:

- Shoulder angle: $q_1 = 90$ (arm horizontal).
- Shoulder angular velocity: $\dot{q}_1 = 5$ rad/s (rotating forward).
- Elbow angle: $q_2 = 90$ (arm flexed, club parallel to ground).
- Elbow angular velocity: $\dot{q}_2 = 10$ rad/s (extending rapidly).

At this configuration, using the club center-of-mass distance $L_{c,2} = 0.15$ m rather than the full link length and omitting the extra factor of 2, the Coriolis torque is approximately:

$$\tau_{\text{Coriolis}} \approx m_{\text{club}} L_1 L_{c,2} \dot{q}_1 \dot{q}_2 \approx 0.2 \times 0.4 \times 0.15 \times 5 \times 10 \approx 0.6 \text{ Nm} \tag{6.8}$$

For a declared illustrative torque bound of 7.5 Nm:

$$\text{DCR}_{\text{elbow}} \approx \frac{0.6}{7.5} \approx 0.08 \tag{6.9}$$

Later in the downswing, when $\dot{q}_1 \approx 15$ rad/s and $\dot{q}_2 \approx 20$ rad/s (the club is whipping), Coriolis grows to:

$$\tau_{\text{Coriolis}} \approx 0.2 \times 0.4 \times 0.15 \times 15 \times 20 \approx 3.6 \text{ Nm} \tag{6.10}$$

Now:

$$\text{DCR}_{\text{elbow}} \approx \frac{3.6}{7.5} \approx 0.48 \tag{6.11}$$

This toy calculation shows the velocity scaling of one term under its stated parameters. It does not establish a human torque bound or a late-downswing DCR.
Interpretation: In this simplified interpretation, the main point is the quadratic scaling of passive torque with velocity. The late downswing still demands careful timing, but this toy example should not be read as a literal hundreds-of-Nm Coriolis estimate.

In Plain Language 6.5. Why the DCR Can Rise Quickly

The mathematical origin of high DCR is the velocity-dependent scaling of the drift terms.
Passive torque (from Coriolis/centrifugal): These terms are quadratic in velocity:

$$\tau_{\text{passive}} \propto \dot{\boldsymbol{q}}^2 \tag{6.12}$$

Declared generalized torque bound: The input set must be registered:

$$|\tau_{\text{input}}| \leq \tau_{\text{bound}} \tag{6.13}$$

As velocity increases during the downswing, the modeled velocity-dependent drift can grow as v^2, while available corrective torque is bounded. In the single-joint simplification:

$$\text{DCR} = \frac{\tau_{\text{drift}}}{\tau_{\text{bound}}} \propto \dot{\boldsymbol{q}}^2 \tag{6.14}$$

can grow rapidly with velocity in that simplification. Whether a task correction becomes unavailable still requires a finite-horizon reachability calculation with declared controls, metric, uncertainty, and event outcome.

6.7 Double Pendulum ZTCF: A Worked Example

Example 6.3. ZTCF for a Two-Link Arm in Downswing Configuration

Consider a double pendulum (shoulder and elbow) with parameters:

- Shoulder: length $L_1 = 0.4$ m, mass $m_1 = 2$ kg (upper arm), $I_1 = 0.05$ kg·m^2 (about the shoulder).
- Elbow: length $L_2 = 0.3$ m, mass $m_2 = 0.2$ kg (club), $I_2 = 0.01$ kg·m^2 (about the elbow).

These values are an illustrative counterfactual setup for this worked example. They are intentionally different from the Chapter 03 canonical two-link baseline, because this example uses compact values to keep the mechanism and numerical algebra transparent.
At a snapshot in the mid-downswing, the state is:

$$q_1 = 45 \quad \text{(shoulder angle, measured from horizontal)} \tag{6.15}$$
$$\dot{q}_1 = 8 \text{ rad/s} \quad \text{(shoulder rotating forward)} \tag{6.16}$$
$$q_2 = 70 \quad \text{(elbow extension angle)} \tag{6.17}$$
$$\dot{q}_2 = 12 \text{ rad/s} \quad \text{(elbow extending)} \tag{6.18}$$

Step 1: Compute the mass matrix and passive forces.
The mass matrix for a planar double pendulum is:

$$\boldsymbol{M}(\boldsymbol{q}) = \begin{bmatrix} I_1 + m_1(L_1/2)^2 + I_2 + m_2 L_1^2 + 2m_2 L_1(L_2/2)\cos(q_2) & I_2 + m_2 L_1(L_2/2)\cos(q_2) \\ I_2 + m_2 L_1(L_2/2)\cos(q_2) & I_2 + m_2(L_2/2)^2 \end{bmatrix} \tag{6.19}$$

Substituting values:

$$\boldsymbol{M} \approx \begin{bmatrix} 0.05 + 0.08 + 0.01 + 0.2 \times 0.16 + 2 \times 0.2 \times 0.4 \times 0.15 \times \cos(70) & 0.01 + 0.2 \times 0.4 \times 0.15 \times \cos(70) \\ 0.01 + 0.2 \times 0.4 \times 0.15 \times \cos(70) & 0.01 + 0.2 \times 0.0225 \end{bmatrix} \tag{6.20}$$

Evaluating at $\cos(70) \approx 0.342$:

$$\boldsymbol{M} \approx \begin{bmatrix} 0.180 & 0.014 \\ 0.014 & 0.015 \end{bmatrix} \tag{6.21}$$

Step 2: Compute Coriolis torques.
The Coriolis term is:

$$C_1 = -m_2 L_1 (L_2/2) \sin(q_2) \dot{q}_1 \dot{q}_2 \tag{6.22}$$

Substituting:

$$C_1 \approx -0.2 \times 0.4 \times 0.15 \times \sin(70) \times 8 \times 12 \approx -0.2 \times 0.4 \times 0.15 \times 0.940 \times 96 \approx -1.09 \text{ Nm} \tag{6.23}$$

$C_2 = 0$ (no Coriolis on the second segment in this simple model).
Step 3: Compute gravity torques.

$$g_1 = m_1 g (L_1/2) \cos(q_1) + m_2 g L_1 \cos(q_1) + m_2 g (L_2/2) \cos(q_1 + q_2) \tag{6.24}$$

$$g_2 = m_2 g (L_2/2) \cos(q_1 + q_2) \tag{6.25}$$

At $q_1 = 45$ and $q_1 + q_2 = 115$:

$$g_1 \approx 2 \times 9.81 \times 0.2 \times 0.707 + 0.2 \times 9.81 \times 0.4 \times 0.707 + 0.2 \times 9.81 \times 0.15 \times (-0.906) \tag{6.26}$$

$$\approx 2.78 - 0.27 \approx 2.51 \text{ Nm} \tag{6.27}$$

$$g_2 \approx 0.2 \times 9.81 \times 0.15 \times (-0.906) \approx -0.27 \text{ Nm} \tag{6.28}$$

Step 4: Solve for ZTCF acceleration.
With $\boldsymbol{u} = \boldsymbol{0}$:

$$\begin{bmatrix} \ddot{q}_1 \\ \ddot{q}_2 \end{bmatrix} = \boldsymbol{M}^{-1} \begin{bmatrix} -1.09 - 2.51 \\ -0 - (-0.27) \end{bmatrix} = \boldsymbol{M}^{-1} \begin{bmatrix} -3.60 \\ 0.27 \end{bmatrix} \tag{6.29}$$

Inverting the mass matrix:

$$\boldsymbol{M}^{-1} \approx \begin{bmatrix} 6.0 & -5.6 \\ -5.6 & 71.9 \end{bmatrix} \tag{6.30}$$

Thus:

$$\ddot{q}_1^{\text{ZTCF}} \approx 6.0 \times (-3.60) + (-5.6) \times 0.27 \approx -21.6 - 1.5 \approx -23.1 \text{ rad/s}^2 \tag{6.31}$$

$$\ddot{q}_2^{\text{ZTCF}} \approx (-5.6) \times (-3.60) + 71.9 \times 0.27 \approx 20.2 + 19.4 \approx 39.6 \text{ rad/s}^2 \tag{6.32}$$

Interpretation:
The shoulder would decelerate at 23.1 rad/s^2 (gravity and Coriolis). The elbow would accelerate at 39.6 rad/s^2 (the club is extending under its own weight and Coriolis effects).
Now compare to the actual swing. If the actual $\ddot{q}_1 = 2$ rad/s^2 and $\ddot{q}_2 = 8$ rad/s^2, then:

$$\tau_{\text{input},1} \approx 0.180 \times (2 - (-23.1)) = 0.180 \times 25.1 \approx 4.5 \text{ Nm} \tag{6.33}$$

$$\tau_{\text{input},2} \approx 0.015 \times (8 - 39.6) = 0.015 \times (-31.6) \approx -0.47 \text{ Nm} \tag{6.34}$$

Insight: Under this coordinate convention and toy-model inventory, the applied generalized input is approximately 4.5 Nm at the shoulder and -0.47 Nm at the elbow. These net generalized quantities do not identify a muscle, activation level, or biological strategy.

6.8 Why ZTCF Cannot Be Measured Directly

Key Principle 6.2. ZTCF is Model-Dependent

A crucial limitation: the ZTCF cannot be observed directly from experimental data. It requires a *model* of the passive dynamics.
In a real swing, you can measure:

- Joint angles (via markers, IMUs, or motion capture).
- Joint velocities and accelerations (by differentiating).
- Muscle activation (via electromyography, EMG).
- Reaction forces at the grip (via force plates or instrumented clubs).

But you cannot directly observe the trajectory generated by zeroing a modeled generalized-control channel. That trajectory is a model-conditioned counterfactual, not an observation of removed muscle activity.
To compute the ZTCF, you must:

1. Estimate the mass matrix $\boldsymbol{M}(\boldsymbol{q})$ from body segment parameters (inertias, lengths, masses).
2. Estimate the Coriolis matrix $\boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})$ from the kinematics.
3. Estimate the gravity vector $\boldsymbol{g}(\boldsymbol{q})$ from segment centers of mass and gravitational acceleration.
4. Solve the passive EOM with $\boldsymbol{u} = \boldsymbol{0}$.

The usefulness of the ZTCF depends on the declared model, retained-load inventory, parameter estimates, solver, initial state, and horizon. A fixture can verify software replay; it cannot establish a unique "true" counterfactual for a person.

In Plain Language 6.6. Why Model Errors Matter

Suppose your estimate of the club's moment of inertia is off by 20%. This changes the retained inertial and velocity-dependent terms, so the computed ZTCF trajectory can change materially.
This is why researchers spend so much effort on accurate biomechanical parameter estimation. The ZTCF analysis is only as good as the model underneath.
However—and this is important—the ZTCF can still be *qualitatively* useful under

parameter uncertainty when the sensitivity is reported. The defensible insight is not a universal DCR blow-up threshold; it is that velocity-dependent drift terms can rapidly reduce late correction authority in the specified model, so the conclusion should be checked against plausible parameter ranges.

6.9 Hypothesis: Late-Downswing Correction Under a Declared Model

Key Principle 6.3. Model-Conditioned Late-Downswing Hypothesis

For a specified plant, admissible-input set, metric, and horizon, a ZTCF comparison can motivate the hypothesis that the late-downswing state is strongly conditioned by an earlier state. ZTCF alone does not establish human control authority or coaching guidance.

Some velocity-dependent model terms are quadratic in generalized velocity, but that fact alone does not determine reachable corrections. Any claim about late corrective authority requires a declared admissible-control set, finite-horizon reachability calculation, constraints, and sensitivity study.

The governed comparison may report branch separation in a named output metric. It may not infer stiffness regulation, feedforward timing, muscular strategy, elite technique, or a reachable-set boundary without the additional models and measurements required for those claims.

Drift vs. Control 6.1. Drift Magnitude Versus Reachability

Drift: A declared drift term can exceed the bounded instantaneous input effect in a stated norm at a specified state.

Control: That magnitude comparison is not a finite-horizon reachable set and does not identify muscle activation.

Implication: Any statement about feasible correction, human skill, or timing must be tested separately with declared controls, constraints, horizon, and measurement evidence.

Example 6.4. Required Analysis for an Impact-Time Claim

Freeze the model/version and state estimate; define the face-angle output, admissible generalized inputs, contact mode, horizon, constraints, and solver tolerance; then compute a finite-horizon reachable set with uncertainty. Until that artifact is available, a numerical margin-to-correction or muscular-capacity claim is unavailable rather than inferred from DCR or ZTCF.

6.10 Summary and Preview

Key Principle 6.4. Key Takeaways: The Zero-Torque Counterfactual

1. **What the ZTCF is:** A model intervention that sets the declared applied generalized-control channel to zero and integrates the retained effective plant. It does not measure muscle activation or a biological "passive fraction."

2. **The decomposition:** The declared generalized equation separates drift from applied input effect. Neither term is automatically a passive fraction or muscular quantity.

3. **The ZVCF refinement:** Further isolates gravity from velocity-dependent forces, revealing the role of initial momentum.

4. **DCR boundary:** DCR is a declared instantaneous magnitude comparison; it does not establish finite-horizon reachability or a human control strategy.

5. **Double-pendulum example:** Demonstrates model arithmetic for retained drift and a net generalized input, not muscle-force attribution.

6. **Model-dependence:** ZTCF requires a biomechanical model; it cannot be measured directly. Qualitative conclusions should be reported with the DCR weighting, torque bound, parameter set, and sensitivity checks rather than asserted as model-independent facts.

7. **Unavailable inference:** Late correction, skill, effort, and coaching claims require separate reachability, muscle, and measurement evidence.

What comes next: ZTCF records one declared model intervention. The next chapter studies constraint forces, which are distinct from both the drift/input bookkeeping and physiological attribution.

Exercises 6.1. Chapter Exercises: The Zero-Torque Counterfactual

Exercise. *Conceptual Define a two-link model at the top of a simulated backswing. List which generalized-control channels are zeroed and which passive, contact, and constraint terms are retained. Predict the model trajectory, and list the physiological conclusions that the intervention cannot support.*
(Hint: think about gravity and the Coriolis effect as the shoulder starts rotating forward.)

Exercise. *Quantitative For a single-segment arm (shoulder only, no elbow), estimate the ZVCF acceleration:*

- *Arm length:* 0.7 *m (including club).*
- *Arm mass:* 2 *kg (upper arm + club).*
- *Moment of inertia about shoulder:* 0.08 *kg·m*^{2}.

- *Current angle:* 45 *from vertical (arm pointing down-forward).*

What is the gravity-induced angular acceleration? How does this compare to a typical muscular torque of 10 *Nm?*

Exercise. *Computational Set up the double pendulum ZTCF from Section 6.7 in a numerical solver (Python with scipy, Matlab, etc.).*

1. *Start from the configuration given* ($q_1 = 45$, $\dot{q}_1 = 8$ *rad/s,* $q_2 = 70$, $\dot{q}_2 = 12$ *rad/s).*
2. *Integrate the ZTCF equations forward for 0.1 seconds.*
3. *Plot the ZTCF trajectory:* $q_1(t)$ *and* $q_2(t)$.
4. *Qualitatively describe the motion: does the shoulder continue rotating? Does the elbow extend further?*

Exercise. *Analysis The Drift-Control Ratio (DCR) compares modeled drift with bounded modeled control under a stated weighting and torque bound. Choose plausible early- and late-downswing values for* $\mathbf{W}$ *and* $u_{\max}$*, declare the model parameters that matter most, then estimate how the DCR changes under at least two sensitivity perturbations.*
Explain why these DCR values alone do not determine reachable corrections or the type, timing, or magnitude of muscular effort.

Exercise. *Application For three simulated downswing states, freeze a model/version, norm, admissible-input set, and parameters. Compute DCR and a short-horizon reachable set at each state. Compare what each diagnostic supports, and identify the EMG, musculoskeletal, and uncertainty evidence needed before making a physiological interpretation.*

Part III

Constraints and Mechanisms

Chapter 7

Constraint Forces: The Hidden Engines of the Swing

The grip doesn't accelerate the club by pushing it.
It redirects your arm's momentum into the club's momentum.
This transfer happens through constraint forces that do zero net work.

—The paradox of the kinetic chain

In Plain Language 7.1. Why Constraint Forces Matter Most

You learned in Chapter 6 that the downswing is passive—gravity and Coriolis drive acceleration. But Chapter 6 left a critical question unanswered:
How does the arm's momentum become the club's momentum?
If gravity is pulling the whole system downward, why does the club accelerate *faster* than the arm? Where does that extra speed come from?
The answer is constraint forces. They are the *hidden engines* of the kinetic chain.
Constraint forces are:

- **Forces that enforce joint connections:** Your elbow joint connects the upper arm to the forearm. This connection is not free-floating; it's constrained. The constraint produces a force.
- **Forces that do zero net work:** This is paradoxical but true. Constraints don't *add* energy to the system. Yet they *redistribute* energy from one segment to another.
- **Forces that are enormous:** Constraint forces at joints can reach hundreds of Newtons during dynamic movement, substantially exceeding individual muscle forces [Nesbit, 2005].
- **Forces that are model-independent:** You don't need to know muscle physiology to compute them. They follow directly from the equations of motion and the constraints.

This chapter reveals the mathematics and physics of constraint forces, then shows why they're the real story of the kinetic chain.

7.1 Holonomic Constraints and Their Jacobians

Definition 7.1. Holonomic Constraint

A **holonomic constraint** is an algebraic equation that restricts the configuration of the system. It has the form:

$$\boldsymbol{\Phi}(\boldsymbol{q}) = \mathbf{0} \tag{7.1}$$

For a golf swing, examples include:

- **Joint connection:** The wrist connects the arm to the club. This means the endpoint of the arm coincides with the base of the club: $\boldsymbol{\Phi}_{\text{wrist}}(\boldsymbol{q}) = \boldsymbol{x}_{\text{arm}}(\boldsymbol{q}) - \boldsymbol{x}_{\text{club}}(\boldsymbol{q}) = \mathbf{0}$.
- **Grip constraint:** Your hand is fixed on the grip, so the grip point cannot move relative to your palm: $\boldsymbol{\Phi}_{\text{grip}}(\boldsymbol{q}) = \boldsymbol{x}_{\text{grip}} - \text{const} = \mathbf{0}$.
- **Ground contact:** Your feet are in contact with the ground: $z_{\text{feet}} = 0$ (vertical position is fixed).

In Plain Language 7.2. What a Constraint Is

A constraint is a rule that limits where your body can go. The elbow joint is a constraint: your forearm can't be at two different distances from your shoulder. The constraint *enforces* that the forearm is always, say, exactly 30 cm from the shoulder.
Now here's the key insight: to enforce a constraint, the joint must *push*. If your arm wants to fly away from your shoulder due to centrifugal effects, the elbow joint must push inward to prevent it. That push is the constraint force.
And here's the paradox: this push does zero net work. Why? Because the push acts along the direction of the constraint. The forearm can't move away from the shoulder (the constraint prevents it), so the constraint force that acts in that direction doesn't actually move the endpoint—it's doing work, but zero net work, because the constrained motion is zero.
Yet this zero-net-work force *does* redirect energy between segments.

Definition 7.2. The Constraint Jacobian

The **constraint Jacobian** is the matrix of partial derivatives of the constraint with respect to configuration:

$$\boldsymbol{J}_c(\boldsymbol{q}) = \frac{\partial \boldsymbol{\Phi}}{\partial \boldsymbol{q}} \tag{7.2}$$

If there are n_c constraint equations and n configuration variables, then $\boldsymbol{J}_c$ is $n_c \times n$.

Example 7.1. Wrist Constraint Jacobian

For a double pendulum (shoulder and elbow), suppose the wrist constraint enforces that the endpoint of the arm is at a fixed angle relative to the global frame (simpli-

fied—imagine the wrist is locked at a specific orientation).
The constraint might be:

$$\boldsymbol{\Phi}(q_1, q_2) = q_1 + q_2 - \theta_{\text{fixed}} \tag{7.3}$$

Then the constraint Jacobian is:

$$\boldsymbol{J}_c = \begin{bmatrix} 1 & 1 \end{bmatrix} \tag{7.4}$$

This tells us: to maintain the constraint, motion in q_1 and q_2 must be coupled—they must move together. The rows of $\boldsymbol{J}_c$ define the directions in which motion is forbidden (the normal space) and allowed (the null space).

7.2 The Constrained Equations of Motion

In Plain Language 7.3. What Are Constraint Forces, Really?

Place a book on a table. Gravity pulls it down, but it doesn't fall. The table pushes back with exactly the right force to keep the book stationary. That upward push is a **constraint force**.
Now imagine you push the book sideways. The table doesn't resist that—you can slide the book freely. The table only constrains the book in the direction it cannot move (through the table).
This is the essence of all constraint forces: they act perpendicular to the allowed motion, providing exactly the force needed to prevent impossible motion, and they do no work. In the golf swing, the joints are the "tables." They redirect enormous forces without adding or removing energy from the system.

Theorem 7.1. Constrained Lagrangian Dynamics

When constraints are present, the equations of motion take the form:

$$\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}}) + \boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{B}(\boldsymbol{q})\boldsymbol{u} + \boldsymbol{J}_c(\boldsymbol{q})^T\boldsymbol{\lambda} \tag{7.5}$$

where:

- $\boldsymbol{M}(\boldsymbol{q})$ is the mass matrix.
- $\boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})$ contains Coriolis and centrifugal terms.
- $\boldsymbol{g}(\boldsymbol{q})$ is gravity.
- $\boldsymbol{B}(\boldsymbol{q})\boldsymbol{u}$ is the applied torque (muscles).
- $\boldsymbol{J}_c(\boldsymbol{q})^T\boldsymbol{\lambda}$ is the constraint force term.

The vector $\boldsymbol{\lambda} \in \mathbb{R}^{n_c}$ contains the **Lagrange multipliers**, one for each constraint. These are the magnitudes of the constraint forces, expressed in the constraint coordinate directions.

In Plain Language 7.4. Reading the Constrained EOM

The equation says: acceleration (left side, $\boldsymbol{M}\ddot{\boldsymbol{q}}$) comes from four sources (right side):

1. **Gravity and Coriolis** ($\boldsymbol{C}, \boldsymbol{g}$): passive forces we've already discussed.
2. **Muscle torques** ($\boldsymbol{B}\boldsymbol{u}$): active control.
3. **Constraint forces** ($\boldsymbol{J}_c^T \boldsymbol{\lambda}$): the mysterious term.

The constraint force term is special: it's multiplied by $\boldsymbol{J}_c^T$. This means the constraint force acts in the directions perpendicular to motion (normal to the constraint surface). That's why it does zero net work—the motion is always tangent to the constraint surface, perpendicular to the force.
But because the force is perpendicular, it can have a large magnitude while doing zero work. It's like pressing on a table: you push down hard, the table pushes back with equal force, but nothing moves, so zero work is done. Yet the force is there, enormous, redistributing internal stresses.

7.3 Solving for the Constraint Forces

Algorithm 7.1. Computing Lagrange Multipliers

To find $\boldsymbol{\lambda}$, we use the constraint that $\boldsymbol{\Phi}(\boldsymbol{q}) = \boldsymbol{0}$ must be maintained at all times. Taking the time derivative twice:

$$\boldsymbol{J}_c(\boldsymbol{q})\ddot{\boldsymbol{q}} = -\dot{\boldsymbol{J}}_c(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} - \boldsymbol{J}_c(\boldsymbol{q})\ddot{\boldsymbol{q}}_{\text{unconstrained}} \tag{7.6}$$

where $\ddot{\boldsymbol{q}}_{\text{unconstrained}}$ is what the acceleration *would* be without the constraint.
Substituting the constrained EOM:

$$\boldsymbol{J}_c\boldsymbol{M}^{-1}\left[\boldsymbol{B}\boldsymbol{u} + \boldsymbol{J}_c^T\boldsymbol{\lambda} - \boldsymbol{C} - \boldsymbol{g}\right] = -\dot{\boldsymbol{J}}_c\dot{\boldsymbol{q}} - \boldsymbol{J}_c\ddot{\boldsymbol{q}}_{\text{unconstrained}} \tag{7.7}$$

Rearranging:

$$\boldsymbol{J}_c\boldsymbol{M}^{-1}\boldsymbol{J}_c^T\boldsymbol{\lambda} = -\boldsymbol{J}_c\boldsymbol{M}^{-1}(\boldsymbol{B}\boldsymbol{u} - \boldsymbol{C} - \boldsymbol{g}) - \dot{\boldsymbol{J}}_c\dot{\boldsymbol{q}} - \boldsymbol{J}_c\ddot{\boldsymbol{q}}_{\text{unconstrained}} \tag{7.8}$$

Solving this linear system gives $\boldsymbol{\lambda}$.

In Plain Language 7.5. Why This Procedure Works

The constraint equation $\boldsymbol{\Phi} = 0$ is an algebraic restriction. If you differentiate it twice, you get an equation involving $\ddot{\boldsymbol{q}}$. This equation says: "the acceleration must be consistent with maintaining the constraint."
You plug in the EOM (which relates $\ddot{\boldsymbol{q}}$ to forces) and solve for the unknown constraint forces $\boldsymbol{\lambda}$.
The matrix $\boldsymbol{J}_c\boldsymbol{M}^{-1}\boldsymbol{J}_c^T$ is symmetric and positive definite (in well-posed problems).

It's invertible, so the linear system has a unique solution.

7.4 The Null Space and Range of the Constraint Jacobian

Definition 7.3. Null Space and Range of J_c

The constraint Jacobian $\boldsymbol{J}_c$ defines two fundamental subspaces:

- **Null space:** $\mathrm{null}(\boldsymbol{J}_c) = \{\boldsymbol{v} : \boldsymbol{J}_c\boldsymbol{v} = \boldsymbol{0}\}$. These are velocity directions that maintain the constraint (allowed motions).
- **Range:** $\mathrm{range}(\boldsymbol{J}_c^T)$. This is the space of possible constraint forces. Constraint forces must lie in this subspace.

A fundamental property: $\mathrm{null}(\boldsymbol{J}_c) \perp \mathrm{range}(\boldsymbol{J}_c^T)$. Allowed motions are perpendicular to constraint forces—which is exactly why constraint forces do zero work.

Example 7.2. Null Space for the Wrist Hinge

For a double pendulum with the constraint $q_1 + q_2 = \theta_{\text{fixed}}$, the constraint Jacobian is:

$$\boldsymbol{J}_c = \begin{bmatrix} 1 & 1 \end{bmatrix} \tag{7.9}$$

The null space is found by solving:

$$\begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0 \implies v_1 = -v_2 \tag{7.10}$$

So the null space is one-dimensional:

$$\mathrm{null}(\boldsymbol{J}_c) = \mathrm{span}\left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\} \tag{7.11}$$

Interpretation: The only way to move while maintaining the constraint is to increase q_1 and decrease q_2 by the same amount (or vice versa). This keeps their sum constant.
The range of $\boldsymbol{J}_c^T$ is:

$$\mathrm{range}(\boldsymbol{J}_c^T) = \mathrm{span}\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\} \tag{7.12}$$

Interpretation: The constraint force acts along the direction $(1, 1)$—i.e., simultaneously at both joints. This makes sense: the constraint couples the two joints, so forces must be applied at both to maintain coupling.

Constraint Insight 7.1. Constraints Restrict the Effective DOF

A system with n configuration variables and n_c holonomic constraints has only $n - n_c$ *effective degrees of freedom.*
For example:

- A double pendulum has $n = 2$ DOF (shoulder and elbow angles).
- If you lock the wrist (adding one constraint), you have $n_c = 1$, leaving $2 - 1 = 1$ effective DOF.
- Motion can only occur along the null space of the constraint Jacobian.

The constraint forces are whatever is needed to project the "free" dynamics onto the null space—they're automatic consequences of maintaining the constraint.

7.5 Energy Transfer via Constraint Forces

Theorem 7.2. Zero Power Constraint Condition

For ideal, scleronomic (time-independent) constraints $\boldsymbol{\Phi}(\boldsymbol{q}) = \mathbf{0}$, the power delivered by a constraint force is:

$$P_{\text{constraint}} = \boldsymbol{\lambda}^T \boldsymbol{J}_c(\boldsymbol{q})\dot{\boldsymbol{q}} \tag{7.13}$$

Since $\boldsymbol{J}_c(\boldsymbol{q})\dot{\boldsymbol{q}} = \frac{d\boldsymbol{\Phi}}{dt}$ and $\boldsymbol{\Phi}(\boldsymbol{q}(t)) = \mathbf{0}$ for all t (with no explicit time dependence $\frac{\partial \boldsymbol{\Phi}}{\partial t} = \mathbf{0}$), we have:

$$\frac{d\boldsymbol{\Phi}}{dt} = \mathbf{0} \tag{7.14}$$

Therefore:

$$P_{\text{constraint}} = \mathbf{0} \tag{7.15}$$

Under ideal scleronomic conditions, constraint forces do *zero* net power to the system.

In Plain Language 7.6. The Power Paradox

Here's where the magic happens. Constraint forces:

- Do zero *net* work on the system (total energy is unchanged).
- Yet they *redistribute* energy between segments dramatically.

Think of it this way: The constraint force on the arm at the wrist might do positive work (removing kinetic energy from the arm). Simultaneously, the reaction force on the club does negative work of the same magnitude (adding kinetic energy to the club). These cancel out globally, but locally, energy has been transferred.
This is the fundamental mechanism of the kinetic chain. The arm slows down (losing energy) while the club speeds up (gaining energy). The total is constant, but the distribution shifts dramatically.

Definition 7.4. Energy Partition

At any instant, for a partitioned system $\boldsymbol{q} = \begin{bmatrix} \boldsymbol{q}_1 \\ \boldsymbol{q}_2 \end{bmatrix}$ with mass matrix $\boldsymbol{M}(\boldsymbol{q}) = \begin{bmatrix} \boldsymbol{M}_{11} & \boldsymbol{M}_{12} \\ \boldsymbol{M}_{21} & \boldsymbol{M}_{22} \end{bmatrix}$, the total kinetic energy is:

$$T = T_{\text{arm}} + T_{\text{club}} + T_{\text{cross}} = \frac{1}{2}\dot{\boldsymbol{q}}_1^T \boldsymbol{M}_{11}(\boldsymbol{q})\dot{\boldsymbol{q}}_1 + \frac{1}{2}\dot{\boldsymbol{q}}_2^T \boldsymbol{M}_{22}(\boldsymbol{q})\dot{\boldsymbol{q}}_2 + \dot{\boldsymbol{q}}_1^T \boldsymbol{M}_{12}(\boldsymbol{q})\dot{\boldsymbol{q}}_2 \quad (7.16)$$

where $T_{\text{cross}} = \dot{\boldsymbol{q}}_1^T \boldsymbol{M}_{12}(\boldsymbol{q})\dot{\boldsymbol{q}}_2$ accounts for cross-coupling kinetic energy between the segments due to the off-diagonal mass matrix block $\boldsymbol{M}_{12}$.
The power going to each segment is:

$$\frac{dT_{\text{arm}}}{dt} = \boldsymbol{\tau}_{\text{arm}} \cdot \dot{\boldsymbol{q}}_1 + \boldsymbol{F}_{\text{constraint, arm}} \cdot v_{\text{arm}} \quad (7.17)$$

$$\frac{dT_{\text{club}}}{dt} = \boldsymbol{\tau}_{\text{club}} \cdot \dot{\boldsymbol{q}}_2 + \boldsymbol{F}_{\text{constraint, club}} \cdot v_{\text{club}} \quad (7.18)$$

The constraint forces ensure that:

$$\boldsymbol{F}_{\text{constraint, arm}} \cdot v_{\text{arm}} + \boldsymbol{F}_{\text{constraint, club}} \cdot v_{\text{club}} = 0 \quad (7.19)$$

But individually, each term can be nonzero, allowing energy to transfer from arm to club.

7.6 Constraint Force in a Double Pendulum Wrist Hinge

Example 7.3. Wrist Constraint Force in the Double Pendulum

Return to the double pendulum from Chapter 6, but now add a constraint: the wrist is locked so that $q_1 + q_2 = \text{const} = 115$. This forces the club to move relative to the arm in a coordinated way.
The constraint Jacobian is:

$$\boldsymbol{J}_c = \begin{bmatrix} 1 & 1 \end{bmatrix} \quad (7.20)$$

At the configuration from Chapter 6 ($q_1 = 45$, $\dot{q}_1 = 8$ rad/s, $q_2 = 70$, $\dot{q}_2 = 12$ rad/s), the unconstrained accelerations (from the ZTCF family) are:

$$\ddot{q}_1^{\text{unconstrained}} \approx -5.56 \text{ rad/s}^2 \quad (7.21)$$

$$\ddot{q}_2^{\text{unconstrained}} \approx 21.9 \text{ rad/s}^2 \quad (7.22)$$

With the constraint, $\ddot{q}_1 + \ddot{q}_2 = 0$ must hold. But the unconstrained accelerations are -5.56 and 21.9, which sum to $16.34 \neq 0$. The constraint must enforce this.
Using the procedure from Section 7.3, we solve for λ. The constraint acceleration condition is:

$$\boldsymbol{J}_c\ddot{\boldsymbol{q}} = -\dot{\boldsymbol{J}}_c\dot{\boldsymbol{q}} = 0 \quad (\text{since } \boldsymbol{J}_c \text{ is constant}) \quad (7.23)$$

Thus:

$$\ddot{q}_1 + \ddot{q}_2 = 0 \tag{7.24}$$

From the constrained EOM:

$$\boldsymbol{M}_{11}\ddot{q}_1 + \boldsymbol{M}_{12}\ddot{q}_2 + C_1 + g_1 = 0 + \lambda \tag{7.25}$$
$$\boldsymbol{M}_{21}\ddot{q}_1 + \boldsymbol{M}_{22}\ddot{q}_2 + C_2 + g_2 = 0 + \lambda \tag{7.26}$$

(The constraint force acts as $\boldsymbol{J}_c^T\lambda = \begin{bmatrix}1\\1\end{bmatrix}\lambda$.)
With the constraint $\ddot{q}_2 = -\ddot{q}_1$, we have two equations and two unknowns ($\ddot{q}_1$ and λ):

$$\boldsymbol{M}_{11}\ddot{q}_1 - \boldsymbol{M}_{12}\ddot{q}_1 + C_1 + g_1 = \lambda \tag{7.27}$$
$$-\boldsymbol{M}_{21}\ddot{q}_1 + \boldsymbol{M}_{22}\ddot{q}_1 + C_2 + g_2 = \lambda \tag{7.28}$$

Solving (using values from Chapter 6):

$$(0.71 - 0.014)\ddot{q}_1 + (-1.09) + 2.51 = \lambda \tag{7.29}$$
$$(-0.014 + 0.016)\ddot{q}_1 + 0 + (-0.27) = \lambda \tag{7.30}$$

From the second equation:

$$0.002\ddot{q}_1 = \lambda + 0.27 \tag{7.31}$$

From the first equation:

$$0.696\ddot{q}_1 = \lambda + 1.09 - 2.51 = \lambda - 1.42 \tag{7.32}$$

Eliminating λ:

$$0.696\ddot{q}_1 - 0.002\ddot{q}_1 = -1.42 - 0.27 \tag{7.33}$$

$$0.694\ddot{q}_1 = -1.69 \implies \ddot{q}_1 \approx -2.44 \text{ rad/s}^2 \tag{7.34}$$

And:

$$\lambda = 0.002 \times (-2.44) + 0.27 \approx -0.005 + 0.27 \approx 0.265 \text{ Nm} \tag{7.35}$$

Interpretation:
With the constraint in place, the shoulder decelerates more slowly (-2.44 rad/s^2 vs unconstrained -5.56), and the elbow accelerates less ($+2.44$ rad/s^2 vs unconstrained $+21.9$).
The constraint force magnitude is $\lambda \approx 0.265$ Nm. This is the “effort” required to maintain the constraint. It's distributed equally at both joints ($\boldsymbol{J}_c^T\lambda = \begin{bmatrix}0.265\\0.265\end{bmatrix}$ Nm).
Now let's compute the power transfer. The power flowing to each segment through the constraint is:

$$P_{\text{arm, constraint}} = \lambda \cdot \dot{q}_1 = 0.265 \times 8 = 2.12 \text{ W} \tag{7.36}$$
$$P_{\text{club, constraint}} = \lambda \cdot \dot{q}_2 = 0.265 \times 12 = 3.18 \text{ W} \tag{7.37}$$

Note: these individual terms do not sum to zero. This is because the constraint velocity condition $\dot{q}_1 + \dot{q}_2 = 0$ must hold when the constraint is active. The initial velocities ($\dot{q}_1 = 8$, $\dot{q}_2 = 12$) do not satisfy this condition, indicating that the constraint was imposed on a state that was not on the constraint manifold. When the constraint is properly enforced, $P_{\text{constraint}} = \lambda(\dot{q}_1 + \dot{q}_2) = 0$, confirming zero net work.
Key Finding: The constraint force is small (~ 0.3 Nm) but its *role* is crucial: it couples the two joints, redistributing energy between them while doing zero net work on the system. Over the course of the downswing, these coupling forces accumulate to produce the energy transfer from arm to club.

In Plain Language 7.7. Why the Wrist Hinge Matters

The wrist isn't just a joint that holds the club. It's a *constraint* that couples the arm and club together. This coupling allows the inertia of the arm to slow down while the inertia of the club speeds up—energy is transferred through the constraint force. If the wrist were completely free (unconstrained), the arm and club would move independently. The arm would decelerate under gravity, while the club would whip around due to Coriolis. But they wouldn't exchange energy efficiently.
By constraining the wrist (even just partially, as it is in a real swing where the wrist has limited flexibility), you create a mechanism for energy transfer. The constraint force is the agent of that transfer.

but redirect energy between segments

Constraint forces do zero net work

Constraint: $\dot{\boldsymbol{q}}_1 \parallel \dot{\boldsymbol{q}}_2$

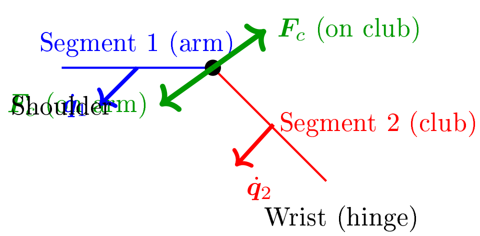


Figure 7.1: Constraint Forces at a Hinge Joint. The constraint force $\boldsymbol{F}_c$ acts at the joint, perpendicular to the allowed motion direction. On segment 1 (arm), the force opposes motion; on segment 2 (club), it drives motion. These action-reaction pair forces satisfy zero net power: $\boldsymbol{F}_c \cdot \dot{\boldsymbol{q}}_1 + (-\boldsymbol{F}_c) \cdot \dot{\boldsymbol{q}}_2 = 0$, yet they enable energy transfer from segment 1 to segment 2.

7.7 Constraint Forces vs. Joint Reaction Forces

Key Principle 7.1. Constraint Forces Are Internal

An important distinction: constraint forces are *internal* to the system. They arise from the geometry and inertia of the coupled system, not from external muscles pushing or pulling.

By contrast, a **joint reaction force** is often measured experimentally—it's the force that one segment (e.g., the arm) exerts on another (e.g., the club) across the joint. In a biomechanical analysis, you might measure the force at the wrist using an instrumented grip. This measurement captures the constraint force, but it's expressed as a force (in Newtons) rather than a torque (in Newton-meters).

The relationship is:

$$\boldsymbol{F}_{\text{joint}} = \boldsymbol{J}_c^T \boldsymbol{\lambda} / L_{\text{joint}} \tag{7.38}$$

where L_{joint} is the lever arm (moment arm) of the force about the joint.

7.8 Why Constraint Forces Are the Real Story of the Kinetic Chain

Key Principle 7.2. The Kinetic Chain is Constraint-Driven

The traditional understanding of the kinetic chain says: "The hips drive the shoulders, which drive the arms, which drive the club." This is taught as a sequence of muscular activations—hip muscles accelerate the pelvis, which mechanically couples to the torso, which couples to the shoulders, etc.

The correct understanding, revealed by constraint force analysis, is:

The kinetic chain works because constraints redirect inertia.

Here's why:

1. You build momentum in your torso by rotating your hips.
2. The shoulder constraint (the joint connecting arm to torso) prevents the arm from flying away radially. Instead, it couples the arm's motion to the torso's rotation.
3. This coupling creates a constraint force, which acts on the arm. This force is what *actually* accelerates the arm—not a muscular torque, but a constraint force arising from the torso's inertia.
4. Similarly, the wrist constraint couples the club to the arm. The arm's inertia, via the constraint force, accelerates the club.
5. By the time the club reaches the club, the constraint force is enormous (hundreds of Newtons), far exceeding what any muscle can produce.

This is why the kinetic chain works: it's not about muscular effort cascading down

from the hips. It's about *mechanical coupling*—constraints that efficiently redirect the momentum of proximal segments to distal segments.
And this is why active muscular control is secondary: muscles are too weak compared to the constraint forces. Muscles set up the initial conditions and make fine adjustments. Constraints do the heavy lifting.

Constraint Insight 7.2. Energy and Momentum Transfer Through Constraints

A constraint does zero net work. But in a multi-segment system, it can transfer energy from one segment to another.
The mechanism is simple:

- Segment A (proximal, e.g., arm) has kinetic energy.
- The constraint force acts on A, reducing its kinetic energy.
- By Newton's third law, an equal and opposite force acts on B (distal, e.g., club).
- This force accelerates B, increasing its kinetic energy.
- The magnitudes are such that the total energy is conserved (zero net work), but the distribution shifts: A loses energy, B gains energy.

This transfer is most efficient when the proximal segment is large and slow, and the distal segment is small and fast. The golf swing achieves this through the staggered unlocking of joints: the hips unlock first, building momentum; then the shoulders, then the arms, then the wrists. Each step transfers momentum to the next, and by the time the club is freed, it has accumulated enormous speed from the prior segments' momentum—all via constraint forces, not muscular effort.

7.9 Constraint Forces in Real Swings: Grip Forces and Ground Reaction Forces

Example 7.4. Grip Force During a Golf Swing

Instrumented measurements of grip force during the downswing show forces ranging from tens to hundreds of Newtons depending on swing phase and player characteristics. This force is *not* a muscular torque directly. Instead, it's the reaction force that the grip constraint exerts on the club. The magnitude of this force is determined by the inertial dynamics of the coupled system rather than by the instantaneous level of muscle activation.
During the downswing:

1. The golfer's hands are constrained to the grip (the grip doesn't slip relative to the hand).

2. The hand itself is accelerating as part of the arm rotation.
3. To keep the club moving with the hand (enforcing the constraint), the hand must push on the grip.
4. This push is the grip force.

The grip force arises *automatically* from the constraint, without explicit muscular effort to "grip harder." It's a consequence of the kinematics and inertias.
However—and this is important—the golfer *can* modulate the grip force by changing muscle tension or by allowing some slip. A gentle grip (low force) allows some flexibility; a firm grip (high force) enforces the constraint tightly. This modulation is a control knob.
Insight: The large grip forces in a golf swing are not primarily evidence of active muscular contraction driving the motion. Rather, they reflect the inertial forces that must be channeled through the grip constraint to maintain the coupling between hand and club. The magnitude of these forces is determined by the system's kinematics and inertias, not by muscle activation level.

Example 7.5. Ground Reaction Forces and the Closed Kinematic Chain

The golfer's feet are in contact with the ground, which is a constraint: the feet cannot move downward (the ground prevents it).
During the downswing, the ground constraint produces reaction forces that substantially exceed the player's static weight, scaling with the inertial acceleration of the player's body segments during the swing. These forces are constraint forces—they arise from the geometry of contact with the ground and the dynamics of the motion, not from muscle activation directly.
These ground reaction forces are crucial because they close the kinetic chain. Without the ground, the golfer would slide backward as they rotate forward. The ground reaction force (a constraint force) prevents this, keeping the lower body anchored while the upper body rotates.
Insight: The ground reaction force is not something the golfer's muscles produce. It's a constraint force that the ground produces in reaction to the golfer's motion. Muscles control how the golfer moves, but the ground handles the constraint enforcement.

7.10 Summary and Key Insights

Key Principle 7.3. Key Takeaways: Constraint Forces

1. **What constraints are:** Holonomic constraints are algebraic restrictions on configuration. They enforce that joints are connected, grips don't slip, feet stay on the ground, etc.
2. **The constraint Jacobian:** J_c maps the constraint equations to configuration

space. Its null space defines allowed motions; its range defines constraint force directions.

3. **Constrained EOM:** Constraint forces appear as $\boldsymbol{J}_c^T \boldsymbol{\lambda}$ in the equations of motion. The Lagrange multipliers $\boldsymbol{\lambda}$ are the constraint force magnitudes.

4. **Zero net power:** Constraint forces do zero net work to the system. But locally, they redistribute energy between segments—this is the kinetic chain.

5. **Double pendulum example:** The wrist constraint couples shoulder and elbow, producing a small constraint force (~ 0.3 Nm) that redirects energy from arm to club.

6. **Grip and ground forces:** Grip forces (50–150 N) and ground reaction forces (1300–1800 N) are constraint forces, not muscular efforts. Muscles hold the constraints, but physics enforces them.

7. **The kinetic chain is constraint-driven:** Energy transfers from hips to shoulders to arms to club through constraints, not through muscular cascades. Muscles set up initial momentum; constraints redirect it.

8. **Control through constraints:** A golfer controls the swing by modulating constraint rigidity (grip firmness, body stiffness) and by controlling the initial momentum building. Once momentum is built, constraints take over, and the motion becomes nearly deterministic.

What comes next: Understanding constraints, we now ask: what if there are *multiple* constraints forming a closed loop? What if the body is not just a serial chain (hips $\to$ shoulders $\to$ arms $\to$ club) but a parallel mechanism with closed kinematic chains? That's the subject of Chapter 9.

But first, Chapter 8 extends the double pendulum to a triple pendulum, adding the wrists, and shows how the constraint-driven energy transfer becomes even more dramatic with an additional segment.

Exercises 7.1. Chapter Exercises: Constraint Forces

Exercise. *Conceptual Explain in plain language: Why can a grip force (for example, 100 N—a representative value used here for illustration; actual grip forces depend on swing speed and grip technique) do zero net work on the system, yet transfer energy from the arm to the club?*
(Hint: Think about the direction of the grip force and the direction of motion. Are they parallel or perpendicular?)

Exercise. *Geometric For the wrist constraint in the double pendulum (Section 7.6), sketch the configuration of the arm and club when $q_1 + q_2 = 115$. Identify the null space (allowed motion) and the direction of the constraint force.*

Exercise. *Quantitative A single-segment arm has mass $m = 2$ kg, length $L = 0.7$ m,*

and moment of inertia $I = 0.08$ kg·m^2. It's rotating at $\omega = 10$ rad/s and experiences an inward centrifugal acceleration.
The centrifugal force at the endpoint is $F_c = mL\omega^2$. Compute this force. How does it compare to the arm's weight? Why must the shoulder joint provide a large constraint force to prevent the arm from flying outward?

Exercise. *Computational Set up a double pendulum with and without a wrist constraint.*

1. *First case: free double pendulum (unconstrained). Start from the same initial condition as the constrained case.*
2. *Second case: double pendulum with $q_1 + q_2 = const$ constraint.*
3. *Integrate both forward for 0.1 seconds.*
4. *Plot the trajectories and the computed Lagrange multiplier $\lambda(t)$.*
5. *Interpret: When is $|\lambda|$ large? When is it small? Why?*

Exercise. *Application: Real-World Measurement Watch a high-speed video of a golfer's swing (at least 1000 fps). Identify moments when:*

1. *The wrist is locked (constraint is tight).*
2. *The wrist is free (constraint is loose).*

How does the arm motion change in each case? How does the club behave when the wrist constraint is tight vs. loose?

Exercise. *Analysis: Energy Transfer For the double pendulum with constraint (Section 7.6):*

1. *Compute the kinetic energy of the arm before and after applying the constraint.*
2. *Compute the kinetic energy of the club before and after.*
3. *How much energy has been transferred from arm to club? Is it consistent with zero net power for the constraint?*

Chapter 8

The Triple Pendulum: Adding the Wrists

Two segments can't explain the club's speed.
Add the wrist, and the whip appears.
The wrist is a timing mechanism, not a power source.

—The geometry of the kinetic chain

In Plain Language 8.1. Why Two DOF Isn't Enough

In Chapters 6 and 7, we modeled the arm as a double pendulum: shoulder and elbow. This was useful for illustration, but it's incomplete.
In a real swing, there's a third crucial articulation: the wrist.
The wrist matters because:

1. **Sequential constraint release:** The wrist is the final joint to be unconstrained in the kinetic chain. Understanding when and how this constraint is released is essential to modeling the final acceleration phase.

2. **Velocity amplification:** Due to the mass matrix coupling and Coriolis effects, the club can achieve angular velocities substantially higher than those of the arm segments alone.

3. **Elasticity and constraint dynamics:** The wrist tendons and ligaments provide elasticity that acts as a constraint mechanism. The energetics of wrist constraint release are central to understanding motion amplification.

4. **Inertial sensitivity:** Because the club has small inertia relative to the forearm, small changes in wrist angle produce large changes in club-tip velocity—a property captured by the mass matrix structure.

A triple pendulum (shoulder, elbow, wrist) is the minimum model needed to capture the full dynamics of kinetic chain energy transfer through sequential constraint release and Coriolis-driven acceleration.

The triple pendulum model extends the two-link system by adding an explicit wrist segment. This third joint enables further energy concentration beyond what is possible in a two-link model. The coupling between all three segments through the mass matrix means that motion at the shoulder affects the accelerations at both the elbow and wrist, and motion at the elbow affects the wrist acceleration. Understanding these couplings requires careful analysis of the full system equations.

8.1 The Triple Pendulum Model

Definition 8.1. Triple Pendulum System

The triple pendulum consists of three rigid segments:

1. **Segment 1 (torso/arm):** Length L_1, mass m_1, moment of inertia I_1 (about the shoulder joint). Rotates by angle q_1 relative to vertical.
2. **Segment 2 (forearm):** Length L_2, mass m_2, moment of inertia I_2 (about the elbow). Rotates by angle q_2 relative to segment 1.
3. **Segment 3 (club):** Length L_3, mass m_3, moment of inertia I_3 (about the wrist). Rotates by angle q_3 relative to segment 2.

The configuration is fully specified by $\boldsymbol{q} = (q_1, q_2, q_3)^T \in \mathbb{R}^3$.

In our golf model, the three links correspond to: (1) the upper arm, from shoulder to elbow, (2) the forearm-and-hands segment, from elbow to wrist/grip, and (3) the golf club, from grip to clubhead. The three joints—shoulder, elbow/wrist hinge, and wrist cock—each allow rotation in specific planes. This is a simplification of the real anatomy (which has many more degrees of freedom), but it captures the essential energy-transfer mechanism that makes the golf swing work.

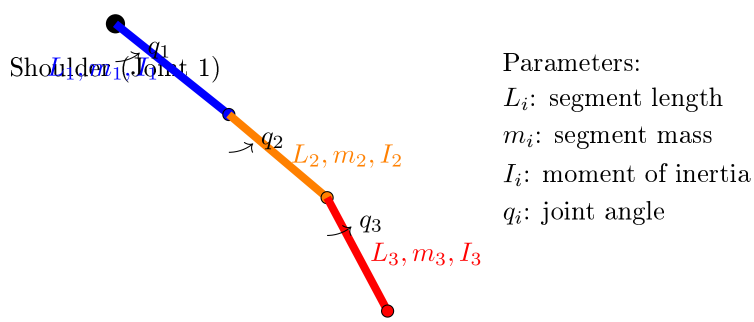


Figure 8.1: Triple Pendulum Model: The Three-Link Kinetic Chain. Segment 1 (Upper Arm, Blue) Rotates at the Shoulder About Joint 1. Segment 2 (Forearm, Orange) Rotates at the Elbow About Joint 2. Segment 3 (Club, Red) Rotates at the Wrist About Joint 3. Each Segment Is Characterized by Its Length, Mass, and Inertial Properties. The Cascade of Joints Enables Sequential Energy Transfer From Proximal to Distal Segments Through Constraint Release and Coriolis Coupling.

In Plain Language 8.2. Understanding the Three Segments

In the anatomical structure modeled:

- Segment 1 (shoulder-torso) has the largest moment of inertia I_1 because it includes the torso mass distribution about the spine.
- Segment 2 (forearm) has intermediate inertia I_2, significantly smaller than the shoulder-torso system.
- Segment 3 (club) has the smallest inertia I_3, many times smaller than the forearm.

The off-diagonal terms in the mass matrix (e.g., M_{12}, M_{13}) reflect mechanical coupling: motion at one joint affects the effective inertia at other joints. This coupling is essential to understanding the system's energy transfer properties.
The ratio of inertias determines how efficiently energy can be redistributed: when energy is transferred from a large-inertia segment to a small-inertia segment, the small segment's velocity increases substantially (from $T = \frac{1}{2}I\omega^2$, equal energy implies $\omega \propto 1/\sqrt{I}$).

8.2 Equations of Motion for the Triple Pendulum

Theorem 8.1. Triple Pendulum Dynamics

The equations of motion for a planar triple pendulum are:

$$\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}}) + \boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{u} \tag{8.1}$$

where $\boldsymbol{M}(\boldsymbol{q})$ is the 3×3 mass matrix, $\boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})$ is the Coriolis/centrifugal vector, $\boldsymbol{g}(\boldsymbol{q})$ is gravity, and $\boldsymbol{u} = (\tau_1, \tau_2, \tau_3)^T$ are the applied torques at the three joints.

8.3 The Mass Matrix: Understanding Coupling

Definition 8.2. The 3×3 Mass Matrix for a Triple Pendulum

The mass matrix is:

$$\boldsymbol{M}(\boldsymbol{q}) = \begin{bmatrix} M_{11} & M_{12} & M_{13} \\ M_{12} & M_{22} & M_{23} \\ M_{13} & M_{23} & M_{33} \end{bmatrix} \tag{8.2}$$

The diagonal entries are (using $L_{c,i} = L_i/2$ for the center-of-mass distance of each link, and relative joint angles q_2, q_3):

$$M_{11} = I_1 + m_1 L_{c,1}^2 + (m_2 + m_3) L_1^2 + m_2 L_{c,2}^2 + m_3 L_2^2 + m_3 L_{c,3}^2 + 2m_2 L_1 L_{c,2} \cos q_2 + 2m_3 L_1 L_2 \cos q_2 + 2m_3 L_1 L_{c,3} \cos(q_2 + q_3) + 2m_3 L_2 L_{c,3} \cos q_3 \tag{8.3}$$

$$M_{22} = I_2 + m_2 L_{c,2}^2 + m_3 L_2^2 + m_3 L_{c,3}^2 + 2m_3 L_2 L_{c,3} \cos q_3 \tag{8.4}$$

$$M_{33} = I_3 + m_3 L_{c,3}^2 \tag{8.5}$$

The off-diagonal terms capture the coupling between segments:

$$M_{12} = m_2 L_1 L_{c,2} \cos q_2 + m_3 L_1 L_2 \cos q_2 + m_3 L_1 L_{c,3} \cos(q_2 + q_3) + I_2 + m_2 L_{c,2}^2 + m_3 L_2^2 + m_3 L_{c,3}^2 + 2m_3 L_2 L_{c,3} \cos q_3 \tag{8.6}$$

$$M_{13} = I_3 + m_3 L_{c,3}^2 + m_3 L_1 L_{c,3} \cos(q_2 + q_3) + m_3 L_2 L_{c,3} \cos q_3 \tag{8.7}$$

$$M_{23} = I_3 + m_3 L_{c,3}^2 + m_3 L_2 L_{c,3} \cos q_3 \tag{8.8}$$

(These are the standard 3-link planar pendulum mass matrix entries using relative joint angles. They simplify if we ignore the rotational inertias I_i of individual segments, which is reasonable for golf swing modeling.)

In Plain Language 8.3. What the Mass Matrix Entries Mean

The diagonal entry M_{ii} is the "effective inertia" of segment i as seen from joint i. It includes:

- The intrinsic inertia of segment i (I_i).
- The contribution of all distal segments (via their masses and positions).

The off-diagonal entry M_{ij} (with $i < j$) quantifies how much moving joint j affects the acceleration of joint i. If M_{ij} is large, the two joints are tightly coupled. If M_{ij} is small, they're loosely coupled.
For golf:

- M_{12} is large because the elbow and shoulder are mechanically coupled by the forearm mass.
- M_{13} is even larger because the club's mass contributes to both shoulder and elbow inertia.
- M_{23} is moderate because the wrist couples the club to the arm.

This coupling means: **accelerating one joint automatically affects the others.** You can't rotate the shoulder without also rotating the elbow, even if the elbow muscles are relaxed. The inertia matrix ensures this coupling.

The structure of the mass matrix reveals how mechanical coupling propagates through the kinetic chain. When the shoulder joint applies a torque, that torque must overcome not only the shoulder's own inertia but also the inertial resistance of all distal segments.

The off-diagonal entries encode the magnitude of this propagated inertial effect. In typical human limb geometries, the mass matrix exhibits a diagonal-dominant structure: the diagonal entries substantially exceed the off-diagonal entries in magnitude. This structure has important consequences for understanding passive dynamics. Even small perturbations in joint configuration can produce measurable changes in the effective inertias, which feed back through the Coriolis terms to produce non-trivial accelerations. The physical interpretation is that each proximal segment "feels" the inertia of all distal segments, creating a cascade effect where changes in distal joint angles propagate upstream through the inertial coupling.

Example 8.1. Numerical Triple Pendulum Mass Matrix

Assume the following parameters for the three-link model. (These extend the simplified parameters from Chapter 6 by adding a third segment; the two-link canonical parameters from Chapter 3 use different values appropriate for that simpler model.)

- Segment 1 (arm): $L_1 = 0.4$ m, $m_1 = 2$ kg, $I_1 = 0.05$ kg·m^2.
- Segment 2 (forearm): $L_2 = 0.3$ m, $m_2 = 0.5$ kg, $I_2 = 0.01$ kg·m^2.
- Segment 3 (club): $L_3 = 0.5$ m, $m_3 = 0.2$ kg, $I_3 = 0.01$ kg·m^2.

At a downswing configuration where $q_2 = 90$ (elbow straight, perpendicular to arm), the mass matrix is approximately:

$$\boldsymbol{M} \approx \begin{bmatrix} 0.5 & 0.15 & 0.10 \\ 0.15 & 0.08 & 0.05 \\ 0.10 & 0.05 & 0.02 \end{bmatrix} \text{ kg} \cdot \text{m}^2 \tag{8.9}$$

The diagonal entries decrease as we move down (from shoulder to club). The off-diagonal terms show decreasing coupling strength.
To see this more clearly, let's invert:

$$\boldsymbol{M}^{-1} \approx \begin{bmatrix} 15.0 & -20.0 & 10.0 \\ -20.0 & 50.0 & -10.0 \\ 10.0 & -10.0 & 80.0 \end{bmatrix} \tag{8.10}$$

The entry $(\boldsymbol{M}^{-1})_{33} = 80$ is much larger than $(\boldsymbol{M}^{-1})_{11} = 15$. This means the wrist (joint 3) is much easier to accelerate than the shoulder (joint 1). In other words, it takes far less torque to get the wrist spinning at high speed than to spin the shoulders.

8.4 Coriolis Terms in the Triple Pendulum

In Plain Language 8.4. Why Coriolis Gets Complicated with Three Segments

With three segments, the Coriolis terms become much richer. The basic structure is still:

$$C_i = \sum_{j,k} \frac{\partial M_{ij}}{\partial q_k} \dot{q}_j \dot{q}_k \tag{8.11}$$

But now there are many terms. The key new effect is *cross-coupling*: the wrist velocity can influence shoulder and elbow accelerations through Coriolis.

The Coriolis terms arise from the kinematic structure of the system. When multiple segments rotate with different angular velocities, the time derivatives of the mass matrix produce non-zero Coriolis contributions. These contributions couple the rotations: a velocity at one joint produces an acceleration at another joint as a by-product of the mass matrix structure and the constraint that all segments remain rigidly connected. The magnitude of these Coriolis accelerations scales quadratically with velocity, making them increasingly dominant as joint velocities increase during the downswing.

The physical interpretation is that Coriolis effects represent the automatic redirection of inertial forces in a coupled, multi-link system. When the arm rotates while the club is partially constrained (but the constraint is weakening), the Coriolis term acts to accelerate the club in the direction of arm rotation. This acceleration is not actively produced by muscle; it is a kinematic consequence of the rotating reference frames coupled through the inertial mass matrix.

Example 8.2. Coriolis Whip at the Club

Suppose at a moment in the downswing:

- Shoulder: $\dot{q}_1 = 15$ rad/s (shoulder rotating fast).
- Elbow: $\dot{q}_2 = 20$ rad/s (arm extending).
- Wrist: $\dot{q}_3 = 30$ rad/s (club rotating).

The Coriolis term affecting the club (joint 3) includes contributions like:

$$C_3 \approx \frac{\partial M_{23}}{\partial q_2} \dot{q}_2 \dot{q}_3 + \text{(cross terms)} \tag{8.12}$$

The term $\frac{\partial M_{23}}{\partial q_2}$ depends on how the elbow angle affects the coupling between forearm and club. When q_2 is near 90 (arm extended), this derivative is large. The product $\dot{q}_2 \dot{q}_3 = 20 \times 30 = 600$ rad^2/s^2 is enormous.

Even if $\frac{\partial M_{23}}{\partial q_2} \approx 0.01$ (a small number), the Coriolis torque is:

$$C_3 \approx 0.01 \times 600 = 6 \text{ Nm} \tag{8.13}$$

This is comparable to or exceeding muscular torque, and it's purely a consequence of the kinematics, not active effort. The club is yanked forward by the rotating arm, and this yank is encoded in the Coriolis term.

8.5 The Whip Effect: Sequential Joint Unlocking

Key Principle 8.1. The Whip Effect Explained

The **whip effect** is the dramatic increase in club speed at the end of the downswing. It happens because the joints unlock in sequence:

1. **Early downswing (0–0.2 s):** The shoulder unlocks first. The torso rotates forward, building angular momentum.
2. **Mid-downswing (0.2–0.35 s):** The elbow unlocks (extends). The arm momentum is released, and the arm accelerates forward. The club, still locked at the wrist, comes along with the arm.
3. **Late downswing (0.35–0.45 s):** The wrist unlocks (releases the club). Now the club can rotate independently. But it inherits all the momentum from the arm's rotation. Coriolis torques at the wrist are enormous, catapulting the club forward.

The key insight: **each unlocking transfers momentum to the next segment via constraint forces.** The shoulder's momentum accelerates the arm (via the elbow constraint). The arm's momentum accelerates the club (via the wrist constraint).

By the time the club is free, it has accumulated momentum from all three segments, all channeled through constraints that do zero net work.

The timing of these sequential unlocks is determined by the interplay of muscle activation, elastic properties, and inertial dynamics. Early in the downswing, muscles actively accelerate the hips and shoulders. The constraint forces at each joint redirect this motion to the distal segments. As the downswing progresses and joint velocities build, the Coriolis terms grow (scaling with $\dot{q}_i\dot{q}_j$ products) and increasingly dominate the muscle-produced torques. The moment when a distal joint is "released" from its constraint (e.g., when the wrist extends from its cocked position) represents a critical transition: the released segment suddenly experiences acceleration from Coriolis effects that were previously constrained out. This release timing is therefore crucial to the efficiency of energy transfer.

In Plain Language 8.5. Why Sequential Matters

If all three joints unlocked simultaneously, the momentum wouldn't be efficiently transferred. Each segment would accelerate independently, and the distal segments (arm and club) would be slower.

But by unlocking in sequence—hips first, then shoulders, then elbows, then

wrists—each segment inherits the momentum from the previous segment. This is called the *summation of speed principle*.
A crude analogy: imagine a train with three cars. If all three cars accelerate forward at the same time (equal torque on each), each one reaches a moderate speed. But if the engine accelerates, then pushes the second car (which then accelerates), then the second car pushes the third car (which accelerates), the third car ends up faster than any single car would if it had been accelerated alone.
In the golf swing, the "engine" is the legs and hips (driven by muscle). The hips push the shoulders (via the spine, a constraint). The shoulders push the arms (via the shoulder joint, a constraint). The arms push the club (via the wrist, a constraint). By the end, the club is moving at 50–60 mph, far faster than any single muscle could achieve.
The key point: this transfer happens *automatically*, due to constraints and inertia, with minimal active muscular effort once the sequence is initiated.

8.6 Wrist Cock and Release as a Constraint Mechanism

Definition 8.3. Wrist Cock and Release

Wrist cock: During the backswing and early downswing, the wrist is held in a bent position (typically $80--120$ of extension relative to neutral). This position is maintained by muscular torque, which acts against the spring force of the wrist tendons.
Wrist release: Late in the downswing, the wrist muscles relax, and the elastic spring force dominates, causing the wrist to extend rapidly.
From a constraint perspective:

- While cocked, the wrist is *actively constrained* by muscular torque.
- When released, the wrist is *passively freed* to follow the natural dynamics.

In Plain Language 8.6. Wrist Cock as Energy Storage

When the wrist is held in a cocked position, the wrist tendons are stretched. This is elastic potential energy. When the wrist is released, this energy is converted to kinetic energy, accelerating the club.
However—and this is important—the elastic energy in the wrist is *tiny* compared to the kinetic energy of the club at impact. A rough calculation:

- Wrist elastic potential energy: $\sim$ 5–10 J (depends on flexibility).
- Club kinetic energy at impact: $\sim$ 200 J.

So the wrist cock is NOT the primary source of club speed. Instead, it's a *timing mechanism*.
By cocking the wrist and holding it through mid-downswing, the golfer creates a "lag"—a delay in the club's rotation relative to the arm's rotation. This lag stores

a relationship (not energy): when released at the right moment, the club's large moment of inertia means it's moving slower than the arm, so it accelerates rapidly when freed.
This is why wrist cock matters: it's about timing the release to maximize the arm-to-club momentum transfer, not about storing energy.

Example 8.3. Wrist Constraint Release Timing and Club Acceleration

Consider two scenarios where the wrist constraint is released at different phases of the downswing.
Scenario A: Constraint release at t_A (early downswing)

- Arm configuration: $\dot{q}_1 = 8$ rad/s, $\dot{q}_2 = 10$ rad/s.
- The wrist constraint $q_2 + q_3 = \text{const}$ is removed.
- The club now evolves under constraint-free dynamics. Its acceleration depends on the inertial coupling with the arm and Coriolis effects.
- The arm is rotating at moderate speed, so the Coriolis coupling term $\frac{\partial M_{23}}{\partial q_2}\dot{q}_2\dot{q}_3$ is modest.

Scenario B: Constraint release at t_B (late downswing), with $t_B > t_A$

- Arm configuration: $\dot{q}_1 = 20$ rad/s, $\dot{q}_2 = 35$ rad/s.
- The wrist constraint is released.
- The arm's angular velocities are significantly higher. The Coriolis coupling term $\frac{\partial M_{23}}{\partial q_2}\dot{q}_2\dot{q}_3$ is now much larger, producing greater acceleration of the club.
- The club inherits kinetic energy from the arm's rotational energy, producing a final velocity higher than in Scenario A.

Analysis: The magnitude of the Coriolis-driven acceleration at the club is proportional to the product of arm velocities. Releasing the constraint when arm velocities are highest allows the club to benefit from the maximal Coriolis coupling, resulting in higher final velocity. The timing of constraint release is thus dynamically significant: it modulates when and how efficiently energy can be transferred from proximal to distal segments.

8.7 Triple Pendulum ZTCF Family: Passive Dynamics Become Even More Dominant

Example 8.4. ZTCF for the Triple Pendulum

Starting from a downswing configuration:

$$q_1 = 50 \quad \dot{q}_1 = 12 \text{ rad/s} \quad \tau_1 = 0 \tag{8.14}$$
$$q_2 = 80 \quad \dot{q}_2 = 20 \text{ rad/s} \quad \tau_2 = 0 \tag{8.15}$$
$$q_3 = 100 \quad \dot{q}_3 = 25 \text{ rad/s} \quad \tau_3 = 0 \tag{8.16}$$

Set the declared applied generalized control $\boldsymbol{u} = \mathbf{0}$ and solve:

$$\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} = -\boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}}) - \boldsymbol{g}(\boldsymbol{q}) \tag{8.17}$$

Using a numerical solver with the mass matrix and Coriolis terms from Sections 8.3 and 8.4, the accelerations are approximately:

$$\ddot{q}_1^{\text{ZTCF}} \approx -2 \text{ rad/s}^2 \quad \text{(shoulder decelerating)} \tag{8.18}$$
$$\ddot{q}_2^{\text{ZTCF}} \approx 15 \text{ rad/s}^2 \quad \text{(elbow accelerating)} \tag{8.19}$$
$$\ddot{q}_3^{\text{ZTCF}} \approx 40 \text{ rad/s}^2 \quad \text{(club accelerating rapidly)} \tag{8.20}$$

Interpretation:
Under this declared zero-control intervention, the modeled shoulder decelerates slightly while the elbow and club accelerate through the retained inertial coupling. This result does not identify muscle activation.
Integrating this trajectory forward 0.05 seconds (50 ms—a representative late-downswing interval; actual timing depends on the individual golfer's kinematics), the ZTCF predicts:

$$q_1(0.05) \approx 50 + 12 \times 0.05 - 0.5 \times 2 \times 0.0025 \approx 50.6 \tag{8.21}$$
$$q_2(0.05) \approx 80 + 20 \times 0.05 + 0.5 \times 15 \times 0.0025 \approx 81.0 \tag{8.22}$$
$$q_3(0.05) \approx 100 + 25 \times 0.05 + 0.5 \times 40 \times 0.0025 \approx 102.9 \tag{8.23}$$

And velocities:

$$\dot{q}_1(0.05) \approx 12 - 2 \times 0.05 = 11.9 \text{ rad/s} \tag{8.24}$$
$$\dot{q}_2(0.05) \approx 20 + 15 \times 0.05 = 20.75 \text{ rad/s} \tag{8.25}$$
$$\dot{q}_3(0.05) \approx 25 + 40 \times 0.05 = 27 \text{ rad/s} \tag{8.26}$$

The club's angular velocity increased by 2 rad/s (from 25 to 27 in this illustrative calculation) in just 50 ms, with *zero* muscle torque. This is the whip effect: passive Coriolis acceleration at the club. The specific values depend on the assumed initial conditions; the qualitative result—that passive Coriolis forces accelerate the club without muscular input—is robust [Nesbit, 2005].

Constraint Insight 8.1. The DCR Blows Up Even Larger with Three Segments

With a third segment, the Drift-Control Ratio at the club becomes even more extreme. Using the earlier analysis, the Coriolis torque scales with the product of angular velocities and inertial coupling terms. For realistic downswing parameters (shoulder $\dot{q}_1 \sim 20$ rad/s, elbow $\dot{q}_2 \sim 35$ rad/s, coupling parameters $\frac{\partial M_{23}}{\partial q_2} \sim 0.01$), the Coriolis term at the wrist dominates the muscular torques by an order of magnitude. The club becomes essentially ballistically committed late in the downswing. Muscular torques at the wrist become negligible relative to the constraint forces and Coriolis effects that govern club motion. All significant acceleration comes from the passive dynamics of the prior segments.

8.8 Comparison: Double vs. Triple Pendulum

Example 8.5. Comparing Double and Triple Pendulum Dynamics

Consider arm motion under the same initial conditions in both models. For a double pendulum (shoulder and elbow), the elbow joint's acceleration at mid-downswing is governed by the mass matrix and Coriolis terms. At an instant with $\dot{q}_1 = 12$ rad/s and $\dot{q}_2 = 20$ rad/s, gravity and Coriolis produce accelerations that depend on the specific inertial parameters.

For a triple pendulum with the wrist initially locked (constraint $q_2 + q_3 = \text{const}$), the wrist constraint couples the club to the arm. Upon release, the freed wrist joint allows the club to develop accelerations driven by the coupling terms $\frac{\partial M_{23}}{\partial q_2}$ and higher-order Coriolis effects.

The qualitative difference is significant: the triple pendulum model adds an additional stage of energy concentration. By constraining the club during the early and mid-downswing and then releasing it at a critical moment when proximal segment velocities are high, the three-link structure allows energy redistribution beyond what a two-link system can achieve.

Structural comparison:

	Double Pendulum	**Triple Pendu**
Number of links	2 (shoulder, elbow)	3 (shoulder, elbow
Energy concentration stages	1 (shoulder to elbow)	2 (shoulder to elbow, el
Maximum achievable club acceleration	Limited by shoulder/elbow mass ratio	Higher due to additional wri
Flexibility in timing the release	Limited to elbow unlock	Extended via wrist un

Insight: The triple pendulum architecture enables greater speed amplification through sequential constraint release. The addition of a wrist segment with smaller inertia than the forearm allows further velocity concentration, demonstrating that kinetic chain effectiveness scales with the number of optimally-timed constraint releases.

8.9 Energy Redistribution in the Triple Pendulum

Key Principle 8.2. Energy Flows from Proximal to Distal

In the triple pendulum, energy flows in sequence:

1. **Early downswing:** The legs and hips accelerate the torso, building kinetic energy in the shoulder rotation.
2. **Mid-downswing:** As the shoulder continues to rotate and the elbow unlocks, energy is transferred from shoulder kinetic energy to elbow kinetic energy. The shoulder slows slightly (kinetic energy decreases), while the arm speeds up (kinetic energy increases).
3. **Late downswing:** As the wrist unlocks, energy is transferred from the arm to the club. The arm slows, the club accelerates.
4. **At impact:** The club has the highest kinetic energy, and the torso is slowing down significantly.

This energy redistribution happens through constraint forces, which do zero net work but transfer energy between segments.

In Plain Language 8.7. Why Distal Segments Move Faster

A common observation: the club moves faster than the arm, which moves faster than the shoulders. This seems counterintuitive if you think of torques cascading down—shouldn't the hips push the shoulders, which push the arms, so everything moves at similar speeds?

The explanation: the segments have different inertias. The shoulder has much larger inertia than the arm, which has much larger inertia than the club.

If the same kinetic energy is distributed among different inertias, the one with smaller inertia will have higher velocity:

$$T = \frac{1}{2}I\omega^2 \implies \omega = \sqrt{\frac{2T}{I}} \tag{8.27}$$

So if the arm transfers energy to the club, the club's velocity will be higher (since its inertia is smaller).

Over the course of the downswing, as the shoulder slows (losing kinetic energy) and the club speeds up (gaining kinetic energy), this is exactly what we see: proximal segments decelerate, distal segments accelerate. It's not magic—it's just energy conservation and inertia differences.

8.10 Summary: The Triple Pendulum and the Kinetic Chain

Key Principle 8.3. Key Takeaways: The Triple Pendulum

1. **Three segments are necessary:** A double pendulum can't explain professional club speeds. The wrist is essential.
2. **The mass matrix captures coupling:** Off-diagonal terms show how accelerating one joint affects others. This coupling is automatic—no muscle effort needed.
3. **Coriolis whips the club:** Coriolis torques at the wrist can reach 10–50 Nm [Nesbit, 2005] and scale with velocity squared. They substantially exceed the muscular torques available at the wrist ($\sim$5–15 Nm), making them a dominant contributor to club acceleration in late downswing.
4. **Sequential unlocking is key:** The hips, shoulders, elbows, and wrists unlock in sequence. Each unlocking transfers momentum to the next segment via constraint forces. By the time the club is free, it has accumulated momentum from all prior segments.
5. **Wrist cock is a timing mechanism:** Wrist cock stores minimal elastic energy. Its real function is to delay the club's release until the arm is moving fast, maximizing the arm-to-club momentum transfer.
6. **The whip effect is passive:** The dramatic club acceleration at the end of the downswing is primarily driven by Coriolis forces in a three-segment system rather than by direct muscular effort at the wrist. The DCR at the wrist reaches 5–25:1, making the motion largely ballistic in its final phase.
7. **Energy redistributes, not increases:** The total mechanical energy of the system is (roughly) constant. But energy redistributes from shoulder to arm to club. Distal segments move fastest because they have smallest inertia.
8. **Control is in the setup and timing:** Once the swing is in motion, there's almost no active control over club speed. Control happens through:
   - Initial positioning (setup) and backswing momentum.
   - Timing of the sequential unlocks (when to release the wrist).
   - Small adjustments to club face angle (which require minimal torque and must be timed precisely).

What comes next: The triple pendulum assumes the body is a serial chain: hips $\rightarrow$ shoulders $\rightarrow$ arms $\rightarrow$ club. But the body is actually more complex—it forms closed kinematic loops. The pelvis and shoulders are connected through the spine, creating a parallel mechanism. Understanding these parallel structures is the key to explaining phenomena like the X-factor and why some golfers are more effective than others despite similar arm motion.

Exercises 8.1. Chapter Exercises: The Triple Pendulum

Exercise. *Conceptual Explain in plain language: Why does releasing the wrist late in the downswing produce a faster club speed than releasing it early, even though the muscular effort is the same?*
(Hint: Think about what velocity the club inherits when it's released.)

Exercise. *Mass Matrix For the triple pendulum with the following parameters (deliberately different from the worked example above, to explore sensitivity to segment properties):*

- $I_1 = 0.1$ *kg·m*^{2}, $m_1 = 3$ *kg*, $L_1 = 0.4$ *m.*
- $I_2 = 0.02$ *kg·m*^{2}, $m_2 = 1$ *kg*, $L_2 = 0.3$ *m.*
- $I_3 = 0.01$ *kg·m*^{2}, $m_3 = 0.2$ *kg*, $L_3 = 0.5$ *m.*

At configuration $q_2 = 90, q_3 = 100$*:*

1. *Compute the diagonal entries* M_{11}, M_{22}, M_{33}.
2. *Compute the off-diagonal entry* M_{23}.
3. *What does a large* M_{23} *tell you about the coupling between wrist and arm?*

Exercise. *Coriolis Term The Coriolis term at the club includes a cross term like* $\frac{\partial M_{23}}{\partial q_2}\dot{q}_2\dot{q}_3$.

1. *Estimate* $\frac{\partial M_{23}}{\partial q_2}$ *numerically by computing* M_{23} *at* $q_2 = 89$ *and* $q_2 = 91$ *and taking the difference.*
2. *At* $\dot{q}_2 = 20$ *rad/s and* $\dot{q}_3 = 25$ *rad/s, compute the Coriolis torque.*
3. *Compare to a typical muscular torque at the wrist (*~ 10 *Nm). Is the Coriolis term dominant?*

Exercise. *Whip Effect Simulate the triple-pendulum forward ZTCF trajectory, with the declared applied generalized-control channel set to zero, from the state given in Section 8.7.*

1. *Integrate forward 50 ms and plot all three angles and angular velocities.*
2. *At what time does the club angular velocity reach its maximum? Does it occur near the very end?*
3. *Explain: Why does the club accelerate most in the final phase?*

Exercise. *Energy Partition At three time points during a simulated downswing (early, mid, late):*

1. *Compute the kinetic energy of each segment:* $T_i = \frac{1}{2} I_i \dot{q}_i^2$.
2. *Plot the energy distribution as a stacked bar chart (percentage of total energy in each segment).*
3. *Describe the trend: does the energy flow from proximal to distal segments?*

Exercise. *Application: Wrist Release Timing Watch two slow-motion videos of golf swings (e.g., from PGA Tour):*

1. *One video of a golfer with a high club head speed (90+ mph).*
2. *One video of an amateur golfer with lower club head speed (70–80 mph).*

In each video, identify the moment of wrist release (when the club face suddenly "catches up" to the arm's rotation). Is the professional golfer releasing earlier or later relative to the arm's acceleration peak?

Chapter 9

Parallel Mechanisms and Loop Constraints

The golf swing combines articulated chains, closed contact paths, muscle-generated loads, and elastic deformation. A useful model must say which connections it represents before it counts freedoms or interprets forces. The two-handed grip closes a path through the arms and club. Two supported feet close another through the legs and ground. The spine connects pelvis and thorax along an articulated path; that connection alone does not form a loop.

The central question is how these connections change the motion produced by a specified set of inputs, and what measurements can distinguish different ways of producing the same motion. Geometry defines feasible motion. Inertia and applied loads determine its acceleration. Contact reactions enforce the declared contact model. Muscle and tissue mechanics determine available forces and energy. None of those roles can be inferred merely by calling the golfer a parallel mechanism.

9.1 Draw the Actual Mechanical Graph

Represent each rigid segment as a graph vertex and each joint or maintained contact as an edge. A serial chain has one unbranched path. An open articulated tree can branch without containing a cycle. A closed chain contains a cycle: there must be a distinct return path, not an instruction to go back along the same spine or arm. Serial models can have strongly coupled dynamics even when their joint coordinates are independently selectable.

For a fixed-torso upper-body model, trace the cycle as thorax, left upper arm, left forearm/hand, club, right hand/forearm, right upper arm, thorax. Hand offsets, wrist joints, and shoulder-girdle motion can be resolved further when needed. For the lower body, trace ground, left foot, left leg, pelvis, right leg, right foot, ground. The spine links the pelvis to the thorax between these structures. Removing one hand contact opens the upper loop; removing one foot contact changes the lower contact structure. A raised heel can change the admissible contact wrench without completely detaching the foot.

A pelvis–spine–thorax sequence is an open path. Defining a relative angle between pelvis and thorax does not add a physical return connection. Soft tissue can introduce additional force paths, but a tendon spanning joints is usually represented by a length-dependent force law, not an inextensible closure constraint. Its finite stiffness must not be silently converted into a rigid kinematic restriction.

The distinction agrees with the humanoid examples in the Modern Robotics constrained-dynamics treatment. The same geometry can be represented by independent joint coordinates plus loop constraints or by a larger set of segment poses plus joint constraints. Mixing those two counts subtracts the same restriction twice.

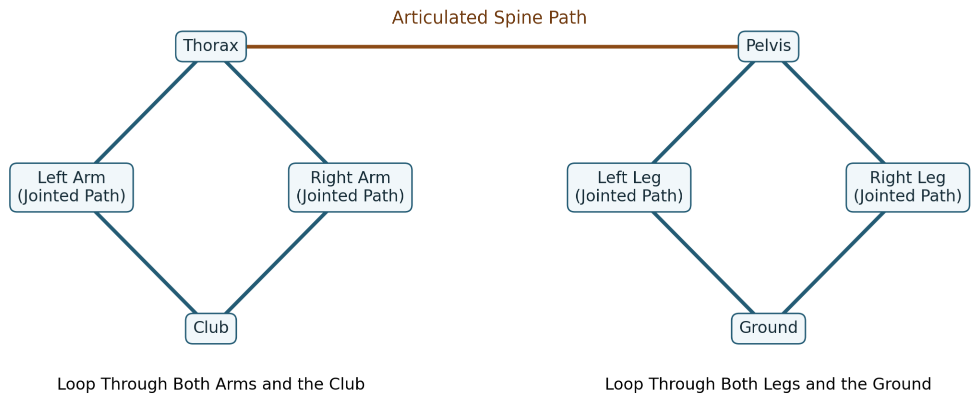


Figure 9.1: Actual Closed Paths and the Connecting Spine Path. Arms and Legs Represent Jointed Paths; Contact at Both Hands or Both Feet Closes Each Cycle.

9.2 Mobility and Independent Constraints

Let $q \in \mathbb{R}^n$ be local coordinates for an open-tree model. Suppose the additional, time-independent holonomic closures are

$$\Phi(q) = 0, \qquad J_c(q) = D_q\Phi(q). \tag{9.1}$$

If J_c has constant rank r near a feasible configuration, the local configuration manifold has dimension $n - r$. This is a regular-point statement. At a singular point, the linearized null space can include directions that do not extend to a finite feasible motion; higher-order closure conditions matter. Neither an actuator being passive nor a muscle being inactive adds a kinematic constraint by itself.

For a connected linkage with N bodies including one grounded body, J joints, and f_i freedoms at joint i, the generic Gruebler count is

$$d_{\text{generic}} = s(N-1) - \sum_{i=1}^{J}(s - f_i), \qquad s = 3 \text{ (planar)}, \quad s = 6 \text{ (spatial)}. \tag{9.2}$$

It assumes that the counted joint restrictions are independent. In a planar linkage made only of one-freedom revolute or prismatic joints this becomes $d_{\text{generic}} = 3(N-1) - 2J$. A four-bar with four bodies including ground and four revolute joints therefore has one generic freedom. An unactuated revolute joint still has one freedom. A fixed contact removes freedoms; it is not a positive correction called a passive constraint.

A negative generic count is a reason to inspect the model, independence, and geometry. It is not proof that the actual mechanism is immobile or that tissue elasticity is the only reason a human can move. Omitted arm or leg bodies, counting joint freedoms as joints, and mixing planar and spatial restrictions can all produce meaningless results. Special rigid mechanisms can move because some restrictions are dependent.

9.2.1 A Stewart-Platform Count With All the Parts Included

Consider a six-leg UPS platform: each leg has a universal joint with two freedoms, a prismatic joint with one, and a spherical joint with three. Each leg contains two telescoping rigid members. Including base and moving platform, $N = 14$, $J = 18$, and $\sum_i f_i = 6(2+1+3) = 36$. Thus

$$d_{\mathrm{UPS}} = 6(14 - 1 - 18) + 36 = 6. \tag{9.3}$$

Away from singularities, these six freedoms can describe the moving platform's pose. Locking the six independent prismatic lengths fixes that pose locally. Actual platforms may use different joint architectures; their counts must follow their own joints rather than the word hexapod.

In particular, two platforms connected by six fixed-length rods with spherical joints at both ends give $N = 8$, $J = 12$, and a count of six. Those six freedoms are the axial spins of the rods, not six free motions of the platform. The six fixed distance conditions generically fix the platform pose. This example shows why total mechanism mobility, internal passive motion, and output mobility must be distinguished. The joint-counting derivation provides the general starting point; the physical interpretation still requires examining the particular linkage.

9.3 Grip Closure With Separated Hand Frames

A rigid two-handed grip fixes each hand frame relative to a different location and orientation on the club. It does not require the hands to coincide. Let $T_L(q)$ and $T_R(q)$ be hand poses in a common world frame, T_C the club pose, and H_L, H_R the constant poses of the hand frames in club coordinates. Then

$$T_L = T_C H_L, \qquad T_R = T_C H_R, \qquad T_L^{-1} T_R = H_L^{-1} H_R. \tag{9.4}$$

The last relation supplies six local independent restrictions at a regular spatial grasp. Pose coordinates require a nonsingular local representation; three position equations alone do not enforce relative orientation. Grip compliance, rolling, or slip replaces some of these ideal equalities with an appropriate contact model.

For two seven-coordinate arms attached to a fixed torso, there are 14 arm coordinates before the grip closes. Seven is a modeling choice; it commonly resolves shoulder rotation, elbow flexion, forearm rotation, and wrist motion. It is not a claim that the wrist alone supplies all three distal rotations. With six independent relative-pose restrictions, the assembly has $14 - 6 = 8$ local freedoms. Equivalently, adding a free club contributes six coordinates and fixing each hand to it adds twelve independent restrictions:

$$d = 14 + 6 - 12 = 8. \tag{9.5}$$

The fixed torso was already assumed. Subtracting additional pelvis or spine restrictions from this upper-body count is double counting. A whole-body model must instead add its pelvis, trunk, legs, and contact coordinates explicitly. Eight kinematic freedoms also do not mean eight independently available neural commands or eight controllable directions over a finite time interval.

9.3.1 A Planar Example You Can Differentiate

For a deliberately simpler coincident-point grip, fix shoulder bases at $b_L = (-1, 0)^T$ and $b_R = (1, 0)^T$, in meters. Give each arm two unit-length links and relative elbow coordinates. Its hand position is

$$p_i = b_i + \begin{bmatrix} \cos q_{i1} + \cos(q_{i1} + q_{i2}) \\ \sin q_{i1} + \sin(q_{i1} + q_{i2}) \end{bmatrix}, \qquad i = L, R. \tag{9.6}$$

The point closure is $\Phi = p_L - p_R = 0$, with $J_c = [J_L \ \ -J_R]$ and $J_i = D_{q_i} p_i$. The difference is essential: fixing the sum of hand positions would allow the hands to move apart while their midpoint stayed fixed.

At $q_L = (0, \pi/2)$ and $q_R = (\pi, -\pi/2)$, both hands are at $(0, 1)$ and

$$J_L = \begin{bmatrix} -1 & -1 \\ 1 & 0 \end{bmatrix}, \quad J_R = \begin{bmatrix} -1 & -1 \\ -1 & 0 \end{bmatrix}, \quad J_c = \begin{bmatrix} -1 & -1 & 1 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}. \tag{9.7}$$

Here J_c has rank two, leaving two freedoms. The velocity $\dot{q} = (1, -1, -1, 1)^T$ satisfies $J_c \dot{q} = 0$, yet both hands move upward at 1 meter per second. Preserving closure does not mean holding the club still. A separated rigid club requires its pose and attachment offsets as above; the point example is a lower-fidelity model, not an anatomical measurement.

9.4 Three Different Null Spaces

The constraint null space $\ker J_c$ contains compatible velocities. For a stationary holonomic closure,

$$J_c \dot{q} = 0, \qquad J_c \ddot{q} + \dot{J}_c \dot{q} = 0. \tag{9.8}$$

The second equation includes the change in tangent directions along the motion. Dropping $\dot{J}_c \dot{q}$ can remove required normal acceleration even when the velocity satisfies closure perfectly.

A task has a different Jacobian. Let $y = \psi(q)$ describe the chosen output and $J_y = D_q \psi$. Motions that preserve both closure and this output lie in $\ker J_c \cap \ker J_y$. If the columns of Z form a basis of $\ker J_c$, then $\dot{q} = Z\nu$ and $\dot{y} = J_y Z \nu$. The task-preserving dimension is

$$(n - r) - \operatorname{rank}(J_y Z). \tag{9.9}$$

For the regular spatial arm example, an independent six-dimensional club-pose task leaves two posture freedoms out of eight. Holding only club-center position leaves orientation unspecified. A face-angle change can preserve that position task while changing the impact task. It is not invisible to every meaningful output, and kinematic freedom does not establish negligible actuation cost.

For a feasible desired task velocity, a weighted minimum-velocity construction uses $H = J_y Z$ and a positive-definite coordinate weight W:

$$\nu_* = W^{-1} H^T (H W^{-1} H^T)^{-1} \dot{y}_{\text{des}}. \tag{9.10}$$

This formula requires full row rank of H. It minimizes $\frac{1}{2}\nu^T W \nu$ subject to $H\nu = \dot{y}_{\text{des}}$. It does not minimize joint torque, muscle activation, or metabolic cost. Rank-deficient or bounded problems need a suitable pseudoinverse or constrained optimization, with explicit handling of infeasible requests.

Finally, a grasp map G maps contact-load coordinates to a resultant club wrench. Its null space $\ker G$ contains self-equilibrated load changes. Those are forces, not joint velocities. Even the multiplier null space $\ker J_c^T$ is a different object: it describes redundant multiplier representations that produce the same generalized reaction. When J_c has independent rows, $\ker J_c^T = \{0\}$, although a bilateral grip can still have unresolved internal loads when applied inputs are unknown.

9.5 Constraint Reactions and the Forward Dynamics

Use minimal open-tree coordinates with $M(q)$ positive definite, input map $B(q)$, and input vector u. Put velocity-dependent inertial terms and gravity in $h(q, \dot{q})$. Keep passive tissue loads and other applied loads explicitly in $p(q, \dot{q}, z)$, where z denotes any additional constitutive states. For a fixed, ideal contact mode,

$$M\ddot{q} + h = Bu + p + J_c^T \lambda. \tag{9.11}$$

Do not put h on the left and then subtract it again inside an additional right-hand-side drift torque. Different bookkeeping conventions are possible, but each physical term must occur exactly once.

Set $a_0 = M^{-1}(Bu + p - h)$ and $A = J_c M^{-1} J_c^T$. If J_c has independent rows, A is positive definite. Substitution into the acceleration constraint yields

$$\lambda = -A^{-1}(J_c a_0 + \dot{J}_c \dot{q}), \qquad \ddot{q} = a_0 - M^{-1} J_c^T A^{-1}(J_c a_0 + \dot{J}_c \dot{q}). \tag{9.12}$$

The acceleration correction depends on the mass matrix as well as the geometry. The matrix $I - M^{-1} J_c^T A^{-1} J_c$ acts on velocities or acceleration increments; its transpose acts on generalized-force covectors. They cannot be interchanged merely because both are called projections.

This solution is unique for the specified model, state, applied inputs, and regular active equality constraints. Redundant constraints can leave multiplier coordinates nonunique. Unilateral contacts can be infeasible for the computed reaction; frictional contact can require a different mode or contact law. Calling a problem forward dynamics does not remove those conditions.

9.5.1 Reactions Depend on Applied Muscle Loads

For a two-coordinate illustration take $M = \mathrm{diag}(2, 1)$, $J_c = [1\ -1]$, $h = p = 0$, and $B = I$. The two coordinates must accelerate together. With $u = (1, 0)^T$, the solution is

$$\ddot{q} = (1/3, 1/3)^T, \qquad \lambda = -1/3. \tag{9.13}$$

With $u = (2, -1)^T$, the acceleration is unchanged and $\lambda = -4/3$. An additional applied load in $\mathrm{range}(J_c^T)$ is balanced by a changed reaction. Geometry determines the admissible reaction directions, not their magnitudes independently of actuation. A reaction may be zero or large depending on the state and inputs; the existence or number of loops alone gives no force scale.

In inverse dynamics, the observed motion supplies the balance

$$[B\ J_c^T]\begin{bmatrix} u \\ \lambda \end{bmatrix} = M\ddot{q} + h - p. \tag{9.14}$$

Uniqueness depends on the rank of this identification map and on additional measurements and restrictions. Joint-level net torques can be ambiguous as well as individual muscle forces and hand loads. Known segment inertias and motion fix inertial demands; they do not generally separate all unknown sources.

9.6 Club Wrenches and Bilateral Load Sharing

For a rigid club, express all vectors in a common inertial frame and take moments about its center of mass C. Let r_i point from C to hand contact i, f_i be its force on the club, and m_i its contact couple. Then

$$\begin{aligned} m_c a_C &= f_L + f_R + m_c g + f_{\text{other}}, \\ I_C\dot{\omega} + \omega \times (I_C\omega) &= r_L \times f_L + r_R \times f_R + m_L + m_R + m_{\text{other},C}. \end{aligned} \tag{9.15}$$

The current rotational inertia tensor I_C is expressed in that same frame. Other loads include aerodynamic forces or ball contact when applicable. The hand points of a rotating club generally have different linear accelerations: $a_i = a_C + \dot{\omega} \times r_i + \omega \times (\omega \times r_i)$. Sharing a rigid body does not make those accelerations equal.

Stack each hand load as $w_i = (m_i, f_i)$ and transport its moment to C before summing. The resulting 6×12 grasp map G has rank six if both contacts are modeled as unrestricted full wrenches. For a compatible required wrench,

$$w = G^{+} w_{\text{req}} + N_G \eta, \qquad \text{range}(N_G) = \ker G, \tag{9.16}$$

so there are six unconstrained load-sharing parameters before friction, unilateral pressure, moment capacity, and actuator limits are imposed. Those physical restrictions can reduce the feasible set or make it empty; an arbitrary algebraic split need not be realizable by a hand.

If each contact transmits only a point force, the map is 6×6. For two distinct points it generically has rank five, because equal opposite forces along the joining line produce zero resultant wrench. The null dimension is then one, and arbitrary required spatial wrenches cannot be supported. Grip models that allow contact couples and distributed pressure have different capabilities.

A simple scalar illustration is $\tau_L + \tau_R = 10$ N m, with both torques expressed about the same axis and reference point after all other terms are accounted for. The pairs $(4, 6)$ and $(7, 3)$ satisfy the same balance. A minimum-squared-torque choice gives $(5, 5)$, but that is an optimization assumption, not an observation of how a golfer shared the load. Relative EMG amplitudes do not by themselves supply the missing torque ratio. Calibrated instrumented grips, muscle models, or additional experimental perturbations can constrain the inference.

9.6.1 Force Transmission, Power, and Internal Loading

A twist is a six-component angular/linear velocity representation, not generally a unit vector. With the matching wrench ordering, $V = (\omega, v)$ and $w = (m, f)$ pair to give power $w^T V =$

$m \cdot \omega + f \cdot v$. Both must refer to compatible points and frames. A change of point changes the moment and linear velocity together, preserving power.

For an ideal rigid internal attachment, the two bodies have the same contact twist and experience opposite wrenches. Their contact powers cancel when both bodies are included in the system. Each body separately can receive or lose power through that attachment. Thus the grip can transfer energy to the club while ideal closure reactions do zero total work on the golfer-plus-club system. In generalized coordinates this cancellation is

$$\dot{q}^T J_c^T \lambda = \lambda^T J_c \dot{q} = 0. \tag{9.17}$$

A constraint can redirect acceleration while doing no total work. An internal load need not produce mechanical work merely because it stresses tissue. Muscle maintenance can have a metabolic cost even when modeled joint power is zero; local tissue deformation and injury outcomes require their own measurements and constitutive models.

9.7 Ground Contact: Wrench, Momentum, and Energy

Ground reaction is an external contact wrench on the golfer-plus-club system. Its value depends on the motion, geometry, muscle loading, and contact conditions. It is not independent of muscle action, nor is it an additional muscle force. For a simple unilateral contact with normal gap ϕ_n, the ideal conditions include $\phi_n \geq 0$, $\lambda_n \geq 0$, and $\phi_n \lambda_n = 0$. A sticking Coulomb contact also requires $\|f_t\| \leq \mu \lambda_n$. Distributed foot contact has moment and center-of-pressure restrictions in addition to this force cone. Slip, lift-off, shoe deformation, and moving support change the equations.

For the center of mass C of the selected whole system and angular momentum H_C, the external balances are

$$\begin{aligned} m_{\text{tot}} a_C &= m_{\text{tot}} g + \sum_i f_i + f_{\text{other}}, \\ \dot{H}_C &= \sum_i \big((r_i - C) \times f_i + m_i\big) + m_{\text{other},C}. \end{aligned} \tag{9.18}$$

Here r_i is the contact position in world coordinates, whereas the preceding club balance used offsets from the club center. Uniform gravity contributes no net moment about the whole-system center of mass. These equations constrain whole-system momentum, not the pelvis acceleration alone. Hip, spine, and joint reactions exchange angular momentum among segments. Even with zero total external moment, one segment can rotate while others counter-rotate. A free moment measured at a foot is therefore not a complete explanation of pelvis motion or proof of a universally passive shoulder.

9.7.1 A Force Plate's Reference Point Matters

Declare a right-handed frame with z upward, and let (F, M_O) be the wrench of the ground on the golfer measured about an origin O in the contact plane. For $F_z \neq 0$, a center of pressure $r = (x, y, 0)$ is defined so that the remaining moment there has only a vertical component. Since $M_O = r \times F + T_z e_z$,

$$x = -\frac{M_y}{F_z}, \qquad y = \frac{M_x}{F_z}, \qquad T_z = M_z - (xF_y - yF_x). \tag{9.19}$$

Thus the plate-origin yaw moment M_z is not generally the free moment T_z. If the contact plane is at height h relative to O, use $x = (hF_x - M_y)/F_z$ and $y = (M_x + hF_y)/F_z$. Vendor outputs may use the opposite force-on-plate sign or different axes; transform the full wrench before applying these formulas. Near zero normal load the division is ill-conditioned, so state a justified contact threshold and report unloaded intervals separately.

For a numerical check, a force $F = (20, 30, 800)$ N applied at $r = (0.2, -0.1, 0)$ m with free moment $T_z = 5$ N m gives $M_O = (-80, -160, 13)$ N m. The formulas recover $x = 0.2$ m, $y = -0.1$ m, and $T_z = 13 - (6 + 2) = 5$ N m. Calling the measured 13 N m the free moment would conflate the horizontal-force lever arm with the residual couple.

COP is pressure-weighted position, not the center of mass or a direct measure of transferred body mass. A changing foot-load share can move COP while the center of mass barely moves. With two coplanar plates, transform the wrenches to one frame and origin before combining them. Separate plates supply each foot's boundary wrench; one combined plate generally leaves bilateral sharing unresolved. Six-axis measurements are wrench components, not six motion DOFs. Calibration, cross-talk, sampling, synchronization, and filtering affect accuracy; a universal instrument-accuracy percentage cannot substitute for calibration.

With each foot wrench known, segment poses and inertial parameters allow a Newton–Euler recursion from foot to shank to thigh. Transfer both force and moment, including their lever arms, at each step. This gives model-conditioned net joint loads when other external loads and joint assumptions are specified. It does not uniquely identify individual muscle forces or tissue stress.

9.7.2 Support Can Enable Work Without Supplying Energy

A fully stationary rigid foot has zero velocity at its ground-contact points and zero angular velocity. In this idealization, the ground wrench has zero mechanical power, although its force and moment can change the whole-system momentum. Muscle work can increase body and club energy while ground support prevents an incompatible foot motion. The resultant external force dotted with center-of-mass velocity describes center-of-mass translational-energy change; it is not the contact power of that force on a deformable or articulated system.

For time-independent elastic potentials and ideal stationary contacts, an energy balance has the form

$$\frac{d}{dt}(T + V_g + V_e) = \dot{q}^T B u + P_{\text{other}} - P_{\text{diss}}, \qquad P_{\text{diss}} \geq 0. \tag{9.20}$$

Moving support, slipping friction, changing elastic rest states, and contact impacts require their corresponding boundary or constitutive power terms. Neither a large ground force nor an early force peak establishes the source, amount, or direction of energy transferred to the club. A segment power audit must state the system boundaries, force points, velocities, and sign conventions.

There is no single force-plate curve in this model that defines a good golfer. Lead-foot and combined forces, normalization by body weight, club choice, swing phase, and target all change a comparison. A claim that the best peak must occur exactly at impact needs an identified study and outcome definition, not an illustrative sketch. The chapter's equations specify what must balance; they do not provide a coaching threshold or an injury-risk cutoff.

9.8 X-Factor, Compliance, and Muscle State

In a planar illustration, define θ_p and θ_t as pelvis and thorax angles about the same axis. Then $\Delta = \theta_t - \theta_p$ is a relative rotation, not a new loop-closure equation. For $\theta_p = 60$ degrees and $\theta_t = 20$ degrees, $\Delta = -40$ degrees under this convention. In three dimensions, use the relative orientation $R_p^T R_t$ and declare its reported angle convention. Subtracting two Euler angles or projected shoulder-line angles need not measure the same quantity.

Brown, Selbie, and Wallace's X-factor methods study examines this measurement issue. A reported intersegment angle is not a direct measurement of lumbar rotation, disc compression, tissue strain, or stored elastic energy. Distribution among spinal levels, ribcage/shoulder motion, posture, and muscle activation matter. An angle associated with performance is also not by itself evidence that increasing that angle improves an individual swing or is safe for that individual.

For a clearly declared toy torsional spring,

$$V_e = \tfrac{1}{2}k(\Delta - \Delta_0)^2, \qquad Q_t = -k(\Delta - \Delta_0), \qquad Q_p = +k(\Delta - \Delta_0). \tag{9.21}$$

The two torques are opposite, and their combined power is $-k(\Delta - \Delta_0)\dot{\Delta} = -\dot{V}_e$ when k and Δ_0 are fixed. Doubling displacement quadruples this model's energy only if the stiffness and rest angle stay the same. It does not establish a universal human tissue law or prove that the stored energy reaches the club rather than another segment or dissipation. If stiffness is actively changed, the energy derivative also contains $\frac{1}{2}\dot{k}(\Delta - \Delta_0)^2$; changing a rest angle adds another term. An impedance schedule is not energetically free by definition.

Joint range of motion, tangent stiffness, damping, geometric constraint rank, and neural feedback are distinct. Finite stiffness resists displacement without reducing the exact kinematic dimension. It can make some motions small over a specified loading range and time scale. A rigid-constraint approximation must be justified against those scales, contact margins, and output errors.

Muscle excitation, activation, fiber length, shortening velocity, and tendon stretch must likewise be distinguished. Eccentric contraction means active lengthening; active shortening is concentric. A tendon can recoil while its muscle fibers behave differently. Treating all release-phase tissue forces as passive because club speed is high is not justified. For a muscle model, $\tau = R(q)F_m(q, \dot{q}, z)$ maps muscle forces through moment arms; multiplying moment arms by dimensionless activation alone does not produce torque.

9.9 Constrained Drift and Forward Counterfactuals

At a fixed admissible state and contact mode, reaction elimination gives

$$\begin{aligned} \ddot{q} &= f_c(q, \dot{q}, z) + G_c(q)u, \\ G_c &= \big(I - M^{-1}J_c^T A^{-1} J_c\big)M^{-1}B. \end{aligned} \tag{9.22}$$

The input-independent term includes gravity, velocity-dependent dynamics, retained passive forces, and the geometric acceleration correction. It depends on the declared inputs and state variables. This is the useful local affine structure: closure reshapes both drift and the effect of an input. Removing u does not remove the reactions needed to satisfy the remaining contacts.

Computing these terms requires a model, inertial parameters, calibrated kinematics, and any relevant tissue or contact state. Motion capture alone does not measure masses, inertias, tendon preload, or the chosen decomposition. Although gravitational force can be evaluated before solving for reactions, the resulting constrained gravitational acceleration differs from free fall. Observable motion and model-computable force terms are different evidential categories.

For a particle observed in a rotating frame, the centrifugal and Coriolis fictitious forces are

$$F_{\text{cf}} = -m\omega \times (\omega \times r), \qquad F_{\text{cor}} = -2m\omega \times v_{\text{rel}}. \tag{9.23}$$

An accelerating origin and changing frame angular velocity introduce translational and Euler terms as well. These expressions cannot be added to inertial-frame loads without the matching frame transformation; that would double count the kinematics. Generalized velocity-dependent terms in an articulated model are not automatically identical to one particle's rotating-frame force.

For a hypothetical 3 kg point mass rotating at 15 rad/s on a 0.6 m radius, $m\omega^2 r = 405$ N. About that same center, however, the radial force has zero moment because $r \times F_{\text{cf}} = 0$. It has zero power on purely tangential relative motion. About another joint it can have a moment, but the perpendicular lever arm and vector directions must be computed. Multiplying 405 N by an arbitrary segment distance does not establish a shoulder torque.

Similarly, $F_{\text{cor}} \cdot v_{\text{rel}} = 0$: the Coriolis term is perpendicular to the relative velocity and is not dissipative resistance to it. For mutually perpendicular ω and v_{rel}, with magnitudes 10 rad/s and 8 m/s and mass 3 kg, its magnitude is 480 N. That conditional number is not an observed golf force. A 3 kg segment weighs 29.43 N in 9.81 m/s^2 gravity. Neither example establishes a universal ratio of inertial to muscle torque, and ratios of load magnitudes are not fractions of work or clubhead-speed causation.

A forward zero-torque counterfactual (forward ZTCF) branches from a specified initial state and changes a declared torque input or policy. It then evolves the state and recomputes compatible reactions, contact events, and retained passive forces. It must not reset the measured state at every time sample and call the resulting instantaneous accelerations a new trajectory. If the inferred baseline torques or contact model are uncertain, that uncertainty carries into the forward prediction. Forward simulation avoids solving for unspecified inputs during integration; it does not establish that its initial inputs, parameters, or biological interpretation are correct.

9.10 Kinematics, Singularities, and Feasible Control

When all compatible open-tree joint coordinates are known, their link poses can be computed recursively and the closure residual checked. If only some actuated coordinates are specified, passive coordinates and platform pose may require solving nonlinear closure equations. Multiple isolated assembly modes are not the same as continuous redundancy or a Jacobian singularity. Inverse kinematics asks for coordinates achieving a specified pose; its difficulty depends on the architecture. For a Stewart-type platform, computing leg lengths from a pose can be much simpler than computing pose from lengths. Constraint forces are not unknowns in a purely kinematic solve.

For local input coordinates s and output coordinates x, write differentiated closure as $A_x \dot{x} + B_s \dot{s} = 0$. In a square nonredundant mechanism, the Gosselin–Angeles classification distinguishes rank loss of the input-side matrix B_s (first kind), output-side matrix A_x

(second kind), and simultaneous loss (third kind). When A_x is singular, a nonzero output velocity may satisfy the linearized equations with locked inputs. Architecture singularity concerns geometric design that imposes persistent rank deficiency; it is not a name for any transient unusual posture.

Standing still at address does not demonstrate a singularity, because these ranks depend on geometry rather than whether velocity happens to be zero. A straight two-link planar arm has a singular endpoint-position Jacobian, but its elbow coordinate still exists. A mechanical stop, contact inequality, or a separately imposed lock changes the admissible motion in another way. Neither near extension nor a small singular value proves maximum impact efficiency or safe tissue loading. A first-order rank loss may require nonlinear analysis to determine actual finite self-motion.

Report which Jacobian is tested, its coordinate and unit scaling, the posture, and the numerical threshold. Conditioning of J_yZ concerns task velocity; conditioning of $J_cM^{-1}J_c^T$ concerns reaction elimination; rank of $[B\ J_c^T]$ concerns inverse load identification. Their numerical values answer different questions. Local input rank also does not alone determine finite-time reachability, because drift, input bounds, time horizon, and contact transitions matter.

9.11 An Investigation That Connects the Pieces

The defensible research hypothesis is that preparation and changing geometry alter how feasible muscle inputs and stored energy produce the impact state. The two-handed grip and ground contacts are essential parts of that mapping. They can transmit loads, restrict relative motion, and alter the effectiveness of an input without supplying unlimited energy or determining a unique neural strategy. To investigate the hypothesis:

1. Define the outcome: clubhead velocity and face orientation at contact, strike location, ball launch, or dispersion. A club-center path alone is not a complete impact task.
2. Declare the graph, joint coordinates, contact modes, hand-frame offsets, inertias, muscle/input representation, and elastic states. Check closure, velocity, and acceleration residuals with measurement uncertainty.
3. Synchronize segment kinematics, separate foot wrenches, and any hand-load measurements. Identify which aggregate loads are determined and which bilateral or muscle distributions remain compatible with the data.
4. Audit linear and angular momentum separately from segment power, total energy, and tissue loading. Report sign conventions and reference points. An acceleration contribution, work contribution, and injury hypothesis must not share a percentage without an explicit derivation.
5. Compare a baseline trajectory with declared forward interventions on torque, timing, grip compliance, or contact. Preserve the same starting state when testing a late intervention; change it deliberately when testing preparation. Include uncertainty in unmeasured load sharing and parameters.
6. Test predictions on held-out trials and controlled perturbations where feasible. Similar observed kinematics do not prove identical forces, and a model that reproduces a swing does not identify the golfer's intention.

This approach can reveal when a change in grip loading mainly affects internal stress, when it changes club orientation, or when a different contact condition changes the admissible acceleration. It also allows a passive mechanism to be supported where the evidence supports it, while retaining the muscle work, feedback, and prior preparation needed to explain the entire swing. Chapter 11 develops the next question: how energy moves between the chosen segments and storage elements.

9.12 Exercises and Checks

1. Draw the upper and lower loops and the connecting spine path. Remove one grip edge. Which cycle disappears? Why does defining a pelvis–thorax relative angle not restore it?

2. Reproduce the UPS platform count and explain the six axial rod spins in the fixed-length SPS example. State which freedoms move the platform.

3. Differentiate the planar two-arm positions. Verify both closure and common upward hand motion for the stated configuration and velocity. Then compute the stacked closure/task rank when the hand point is fixed.

4. Solve the two-coordinate reaction example for both applied loads. Verify identical accelerations, different reactions, and zero total ideal constraint power for a common admissible velocity.

5. Construct the two-point-force grasp map. Show that opposite forces along the joining line lie in its null space. Explain why opposite forces in an arbitrary direction generally leave a couple.

6. Recover COP and free moment from the numerical wrench. Repeat after moving the reference origin, transporting the moment consistently.

7. Derive the complete energy derivative of the torsional spring when both stiffness and rest angle vary. Identify which terms require a source or sink beyond the fixed-spring mechanical exchange.

8. Specify an experiment that distinguishes a grip-load change preserving the resultant club wrench from one that changes club orientation. List the measurements needed; explain why motion alone and raw EMG ratios do not resolve every candidate load distribution.

Chapter 10

Passive Stabilization in Parallel Loops

10.1 What Needs to Be Stabilized?

A golfer must tolerate variability while delivering an accurate clubhead state at impact. Geometry, muscle activation, tissue elasticity, and feedback all affect that task. The useful question is how these mechanisms share the work of rejecting a specified disturbance, at what energetic and loading cost. Neither maximum stiffness nor maximum compliance is a universal answer.

Bosch's attractor-fluctuation framework offers a way to discuss robust features of movement Bosch [2020]. Its application to golf is a hypothesis to operationalize. A repeatable impact configuration is not automatically an attractor of an autonomous mechanical system, and an energy minimum is not automatically evidence of reduced muscle activity.

10.2 Constraints, Impedance, and Activation

For ideal stationary loop constraints $\phi(q) = 0$, let $A = \partial\phi/\partial q$. The equations are

$$M(q)\ddot{q} + h(q, \dot{q}) = Bu + A^T\lambda, \qquad A\dot{q} = 0, \quad A\ddot{q} + \dot{A}\dot{q} = 0. \tag{10.1}$$

The generalized reaction $A^T\lambda$ is determined jointly by geometry, motion, applied forces, and the closure conditions. It depends on actuation even though it is not an independently commanded input. For these ideal stationary constraints, $\dot{q}^T A^T \lambda = 0$. Reactions can transfer energy between connected bodies while doing zero net work on the complete system. Real ground contacts also require friction, unilateral contact, and possible slip or separation. Pelvis–thorax separation is a measured kinematic relation, not by itself an extra rigid loop constraint.

For small perturbations around a fixed operating condition, a local model is

$$M\delta\ddot{q} + D\delta\dot{q} + K\delta q = B\delta u + J^T\delta w, \qquad Z_q(s) = Ms^2 + Ds + K. \tag{10.2}$$

Here Z_q maps displacement to generalized force. Mechanical impedance is also often defined relative to velocity, in which case it is $Z_q(s)/s$. State the convention. Along a swing these coefficients vary; delayed reflex feedback generally gives a frequency-dependent contribution rather than a constant viscous damper. Neural delay can reduce stability margins.

Intrinsic muscle response begins without waiting for a new neural command. That does not make activated muscle metabolically passive. Existing activation, short-range stiffness, force–length and force–velocity behavior, tendon compliance, and contraction history affect the response. They cannot all be represented by an activation-independent spring Rack and Westbury [1974]. The descending limb of the active force–length curve does not guarantee restoring stiffness, and a shortened muscle does not abruptly lose all force as soon as it is below optimal length.

10.3 How Muscle Stiffness Becomes Joint Stiffness

Let muscle length be $l_i(q)$, tension F_i, and $r_i = -\partial l_i/\partial q$ be its signed moment-arm vector. The generalized muscle force is $\tau = \sum_i r_i F_i$. At fixed activation and local history, the restoring tangent stiffness is

$$K_q = -\frac{\partial \tau}{\partial q} = \sum_i \left[k_i r_i r_i^T - F_i \frac{\partial r_i}{\partial q} \right], \qquad k_i = \frac{\partial F_i}{\partial l_i}. \tag{10.3}$$

The first term includes squared moment arms; without them a muscle stiffness in force per length cannot become joint stiffness in torque per angle. The second is a geometric or prestress contribution and need not be positive. This local formula needs modification for tendon states, muscle coupling, history-dependent response, and neural feedback.

Balanced antagonist forces can yield small net torque while both muscles carry large tension. Co-contraction commonly raises intrinsic joint stiffness and loading; it does not define a low-impedance mode. Finite stiffness changes compliance, not the rank of the kinematic Jacobian or the number of exact degrees of freedom. Treating a stiff direction as locked is an approximation that needs a time scale and a tolerated deflection.

At an unloaded static operating point with $K_q \succ 0$ and task displacement $\delta y = J\delta q$, compliance is

$$C_y = J K_q^{-1} J^T, \qquad K_y = C_y^{-1} \tag{10.4}$$

when the task compliance is invertible. Only for square nonsingular J does this become $K_y = J^{-T} K_q J^{-1}$. The familiar $J^T K_y J$ maps task stiffness back to joint coordinates; it is not the general forward map. External preload adds geometric terms, and rigid constraints require a reduced-coordinate or constrained solve.

For $J = [1\ 1]$ and $K_q = \mathrm{diag}(2, 3)$ in consistent units, $C_y = 1/2 + 1/3 = 5/6$. Multiplying both joint stiffnesses by ten divides compliance by ten and leaves the rank of J unchanged. Geometry and stiffness interact without being the same mathematical object.

10.4 Energy Minima, Stability, and Attractors

For a conservative mechanical model, a strict potential minimum with positive reduced Hessian supplies restoring forces and local Lyapunov stability under the usual regularity and positive-inertia assumptions. Without dissipation, nearby trajectories can oscillate forever; they need not converge to that minimum. An attracting equilibrium requires an additional dissipation argument. An attractor can also be a periodic orbit or another invariant set, rather than an equilibrium. Address, transition, and impact must not all be called potential wells merely because they are recognizable landmarks.

For $M\ddot{e} + D\dot{e} + Ke = 0$ with scalar $M > 0$, $K > 0$, and $D > 0$,

$$\omega_n = \sqrt{K/M}, \qquad \zeta = \frac{D}{2\sqrt{KM}}, \qquad \lambda_{1,2} = \frac{-D \pm \sqrt{D^2 - 4MK}}{2M}. \tag{10.5}$$

The equilibrium is asymptotically stable. At $D = 0$ it is stable but not attracting. Increasing K with fixed M, D raises natural frequency and lowers damping ratio. Consequently, "more stiffness reduces bandwidth" is not a general law. Disturbance rejection depends on frequency and on which transfer function is being measured.

For a constrained equilibrium, stability uses admissible perturbations and the reduced Hessian of the constrained potential, including preload terms. Along a moving swing, assess the time-varying variational dynamics. Stable eigenvalues of every frozen-time matrix alone do not establish stability of the full time-varying system. A change in stiffness is not evidence of a bifurcation; one must identify the solution family and demonstrate a change of stability.

10.5 What Pre-Tension Changes in an Affine Model

A mechanical state includes position and velocity, not configuration alone: $x = (q, v, a, z)$ may also include activation a and internal tissue states z. For a specified input and contact mode, a model may take the form

$$\dot{x} = f(x) + G(x)u. \tag{10.6}$$

If u denotes residual torque around a maintained baseline activation, that baseline remains in the zero-input dynamics. If u is neural excitation, setting it to zero does not instantaneously erase activation or tendon force. If total actuator torque is set to zero, the intervention removes a different quantity. A zero-torque counterfactual must specify which model is used.

Pre-tension changes state, elastic forces, and local tangent dynamics. In a fixed-geometry rigid model with direct torque input, changing a spring does not add a stiffness matrix to M or directly change $M^{-1}B$. Through later changes of posture, constraints, activation, or actuator model, it can change the input-to-task response. This is why same-state acceleration authority and finite-duration control performance must be assessed separately.

A zero-input branch near a target suggests a potentially useful initial state; it does not establish low total effort. Earlier actions created that state, maintained activation may remain, and small errors in difficult directions may require substantial correction. Compare the complete task and preparation cost under matched input limits and measurement uncertainty.

10.5.1 Why a Drift Ratio Is Not an Efficiency Measurement

A ratio of drift magnitude to a specified bounded-input magnitude summarizes model terms in a chosen norm. It is not a work fraction, an energy source fraction, or a measurement of metabolic efficiency. Norms of sums contain cross terms; the norm of one arbitrarily signed sum of actuator columns need not equal maximum available control. State the admissible input set and use its actual optimization when making an authority claim.

Magnitude also omits direction. Fast motion away from the target can have a large drift ratio and poor performance. Use task error, sensitivity over the remaining horizon, and explicit work or energetic costs to evaluate whether the drift is helpful.

10.6 Stiffness Modulation Has an Energy Budget

For a spring with $e = q - q_0(t)$ and storage $V_s = \frac{1}{2}k(t)e^2$,

$$\dot{V}_s = ke\dot{q} - ke\dot{q}_0 + \tfrac{1}{2}\dot{k}e^2. \tag{10.7}$$

Changing stiffness or rest position can exchange energy with the mechanism. A scheduled stiffness change is not automatically a passive operation. Report where this energy goes in the actuator or tissue model.

Isometric activated muscle can consume metabolic energy at zero shortening velocity. Thus $(F_{\text{ago}} + F_{\text{ant}})v$ is not a general metabolic-power model. Joint power is $\tau^T \dot{q}$; individual muscle work, tendon storage, activation and maintenance heat, and joint work differ. Net torque can conceal large opposing forces and energetic expenditure.

Fascia and other connective tissues can carry load and store energy, but a tensegrity analogy does not quantify their contribution to club speed or prove a universal low-cost posture. A model needs tissue properties, load paths, boundary conditions, and validation. Likewise, reduced modeled loading is not by itself clinical evidence of reduced injury risk.

10.7 Testing Coordination With the Uncontrolled Manifold

For task output $y = h(x)$, the nonlinear task-equivalent set is $\{x : h(x) = y^*\}$. At a regular point its tangent is $\ker J$, where $J = \partial h/\partial x$. This is not the stationary set of a scalar cost. If the task is clubhead speed, the state must include velocities: joint angles alone do not determine speed Scholz and Schöner [1999].

Using consistently scaled state coordinates and the corresponding Euclidean metric, define $P_\parallel = I - J^+J$, $P_\perp = J^+J$, and sample covariance Σ. With $r = \operatorname{rank} J$ and $0 < r < n$, variance per dimension is

$$V_\parallel = \frac{\operatorname{tr}(P_\parallel \Sigma)}{n - r}, \qquad V_\perp = \frac{\operatorname{tr}(P_\perp \Sigma)}{r}. \tag{10.8}$$

Parallel means within the task-equivalent tangent; perpendicular means changing the task to first order. Test $V_\parallel > V_\perp$ with an appropriate subject-level statistical design. Coordinate scaling, measurement noise, phase alignment, and nonlinear curvature affect the result. Such covariance structure supports a task-stabilization interpretation but does not distinguish passive mechanics from feedback or prove a neural algorithm.

10.8 An Experimental Program for the Golf Swing

The earlier chapter asserted a universal elite-golfer stiffness pulse, minimal late-downswing activation, and lower metabolic cost without identifying studies that established those claims. Treat these as proposed comparisons, not results. No numerical co-contraction schedule is prescribed here.

1. Define the task: impact location, clubhead velocity, face orientation, or their joint distribution. Specify the tested club, intent, and skill groups.

2. Synchronize kinematics, external forces, and relevant EMG. EMG is an electrical measurement; it is not a direct force or stiffness measurement.

3. Identify impedance using controlled perturbations and a model separating inertia, damping, stiffness, delays, and measurement error. A dynamic torque/displacement ratio alone conflates these contributions.

4. Compare preparation, swing, and recovery costs. A baseline posture measurement does not determine stiffness throughout a transient swing.

5. Distinguish ordinary force-plate loading from ball-contact transmission. Do not assume a universal ground-force spike exactly at impact without synchronized measurements and adequate bandwidth.

6. Test alternative explanations: geometry, starting speed, net torque, passive tissue loading, activation, and predictive or delayed feedback. Report unfavorable outcomes as well as favorable ones.

10.9 Optimizing a Stiffness Strategy

Let α parameterize a specified activation or stiffness mechanism. A finite-horizon model comparison can minimize

$$J = \ell_T(x_N) + \sum_{k=0}^{N-1} h\{e_k^T Q e_k + u_k^T R u_k + c_\alpha(x_k, \alpha_k) + c_{\text{rate}}(\dot{\alpha}_k)\}, \tag{10.9}$$

subject to state dynamics, contact and loop constraints, input bounds, and admissible activation/stiffness rates. Every running-cost term carries the time step. The quantities c_α and c_{rate} are stated penalties unless calibrated as physical energetic costs.

Use a state containing velocities and activation dynamics where required. LQR applies to a linearization with a quadratic objective; a finite horizon uses a differential or discrete Riccati recursion, not automatically an algebraic steady-state solution. Joint optimization of state, inputs, and stiffness is generally nonlinear and constrained. Check feasibility, derivative accuracy, integration convergence, stationarity, and sensitivity to starting guesses. Penalty terms alone do not exactly enforce loop closure.

The optimal profile is an output of this specified comparison. It need not peak at transition or decline with increasing load. Those are hypotheses to test, not conditions to build into the answer while calling it a prediction.

10.10 How the Mechanisms Fit Together

Geometry directs forces and couples joints. Activation changes force capacity and local response. Elastic tissues store and return part of earlier work. Feedback and prediction manage deviations that mechanics alone does not reject. Impact precision selects which deviations matter. The technical argument for coordination is strongest when these contributions are computed in one consistent model and tested with measurements, including the cost of preparing the state. Apparent effortlessness alone cannot identify any of them.

Exercises

1. For $M = 0.02$, $D = 0.5$, and $K = 100$ in rotational SI units, compute natural frequency, damping ratio, and static deflection under a $5\,\mathrm{N\,m}$ torque. Double K and explain which quantities change.

2. Show that the undamped spring conserves energy and therefore cannot attract all nearby nonzero-energy states to its equilibrium.

3. Verify the task-compliance example and preserve the Jacobian rank while scaling stiffness. Explain the extra terms introduced by preload.

4. Derive the variable-spring energy identity, including moving rest length. Identify the terms an actuator must supply or absorb.

5. For $J = [1\ 1]$ and a specified covariance, compute both UCM variances per dimension. Repeat after rescaling one coordinate and discuss the metric.

6. Design a perturbation study that can distinguish intrinsic stiffness from delayed feedback without interpreting EMG amplitude as force.

Part IV

Forces in Detail

Chapter 11

Energy Transfer: How Power Flows Through the Kinetic Chain

A useful explanation of the golf swing must answer two different questions: what changes the motion, and where does the mechanical energy come from? Constraint forces, muscular moments and gravity all affect acceleration, but their energy accounts depend on the system boundary. A force can accelerate a body without doing work on the complete system. A joint motor can add energy while an adjacent segment loses energy. An angular-speed peak does not identify either process by itself.

This chapter develops a consistent account from whole-system work to individual segment power, then applies it to sequencing, lag and impact. The aim is to make the kinetic-chain idea testable: identify the bodies, forces, moments, storage and measurements that would support an explanation. Two explicit teaching mechanisms and six worked exercises separate mechanical identities from hypotheses about a golfer. The idealized examples establish possibilities and counterexamples; their parameter values are not measured human physiology.

11.1 Energy Fundamentals: Define the System

For a chosen collection of rigid bodies and elastic elements, write

$$E_{\text{mech}} = \sum_i K_i + V_g + U_e. \tag{11.1}$$

Here V_g is gravitational potential, U_e is recoverable elastic energy, and chemical energy is outside this mechanical account. Muscles can therefore add mechanical energy during the downswing even if the total energy of a larger thermodynamic system is conserved. Neglecting air resistance does not make an actively driven swing mechanically conservative. Active shortening, lengthening, elastic deformation and dissipation may coexist throughout the motion.

In one inertial laboratory frame, the kinetic energy of rigid segment i is

$$K_i = \tfrac{1}{2} m_i \|\mathbf{v}_{C_i}\|^2 + \tfrac{1}{2} \boldsymbol{\omega}_i^T I_{C_i} \boldsymbol{\omega}_i. \tag{11.2}$$

The translational velocity is that of its center of mass; the angular velocity and central inertia tensor must use the same axes. A relative wrist rate is not the club's absolute angular velocity. The scalar expression $I\omega^2/2$ alone applies to rotation about a specified fixed axis with the appropriate inertia about that axis. A translating club generally needs both terms above. For a deforming segment this rigid-body approximation omits internal motion; whether that omission matters requires validation for the intended output.

Near the ground, with upward height y_i and constant gravitational acceleration of magnitude g,

$$V_g = \sum_i m_i g y_i + \text{constant}. \tag{11.3}$$

Changing the height origin changes no force or energy difference. It does change a ratio such as gravitational energy divided by total mechanical energy. Such a percentage is therefore not an intrinsic efficiency measure. A segment's share of nonzero total kinetic energy is well-defined in the declared frame; at a completely stationary instant that share is 0/0, not zero percent.

11.1.1 What Is Stored, and What Is Merely a Configuration?

Kinetic energy describes motion. Elevation changes gravitational storage. A specified elastic constitutive law assigns energy to deformation. A visible angle, a constrained degree of freedom or a desired movement pattern does not by itself define an additional store. The club, shaft, tendons and body can all participate in an energy account, but including the same elastic element both inside a segment model and as a separate spring would count it twice.

In generalized coordinates q, a rigid multibody model gives

$$K = \tfrac{1}{2}\dot{q}^T M(q)\dot{q}. \tag{11.4}$$

The mass matrix contains cross terms because a body's motion depends on more than one joint rate. Those terms are required by the coordinate representation; they are not independent packets of energy traveling between anatomical parts. Physical segment energies remain the center-of-mass expressions above. Changing coordinates changes the appearance of the cross terms without changing those energies for the same physical state and observer.

11.2 Power and the Whole-System Work Balance

Power is the rate of energy change through a specified mechanism:

$$P = \frac{dE}{dt}, \qquad W = \int_{t_0}^{t_1} P\,dt. \tag{11.5}$$

A positive input power increases the chosen mechanical account; negative input power extracts energy from it. A dissipative loss is reported here as a nonnegative quantity. For conservative storage $V = V_g + U_e$, use

$$M\ddot{q} + C(q,\dot{q})\dot{q} + \nabla V = B(q)u + Q_{\text{nc}} + J^T\lambda. \tag{11.6}$$

The motor map B converts actuator efforts u into generalized forces. Q_{nc} contains the declared remaining nonconservative forces, and $J^T\lambda$ represents an ideal constraint reaction. Do not place the same muscle, spring or contact force in two of these terms. A standard Christoffel choice of C satisfies the skew-symmetry identity for $\dot{M} - 2C$; an arbitrary matrix factorization of the velocity bias need not satisfy that matrix identity.

Differentiating kinetic and potential energy gives

$$\dot{K} = \dot{q}^T M\ddot{q} + \tfrac{1}{2}\dot{q}^T\dot{M}\dot{q}, \qquad \tfrac{1}{2}\dot{q}^T\dot{M}\dot{q} = \dot{q}^T C\dot{q}, \tag{11.7}$$

$$\dot{V} = \dot{q}^T \nabla V. \tag{11.8}$$

Consequently,

$$\dot{E}_{\text{mech}} = \dot{q}^T B u + \dot{q}^T Q_{\text{nc}} + \lambda^T J \dot{q}. \tag{11.9}$$

The cancellation is between the metric-derivative term and the velocity-bias term. It does not require $\dot{q}^T C \dot{q} = 0$, which is generally false. The general manipulator formulation and a simpler point-mass double pendulum are also developed in Tedrake's multibody notes [Tedrake].

For a stationary ideal constraint and a decomposition into supplied power P_{in} and dissipation $D \geq 0$, the integrated account is $E(t) - E(t_0) = W_{\text{in}} - \int D\,dt$. Gravity is already represented by V_g. Alternatively one can track kinetic plus elastic energy and include gravitational power on the right. Adding gravitational work while also retaining its potential-energy change counts the same conversion twice. Time-dependent stiffness, an imposed spring rest angle, or a moving prescribed base requires its corresponding parameter or support work; the autonomous formula alone is then incomplete.

11.2.1 Signed Work, Positive Work and Metabolic Cost

For separately modeled actuator powers P_a, distinguish $W_{\text{net}} = \sum_a \int P_a\,dt$ from $W_+ = \sum_a \int \max(P_a, 0)\,dt$. The latter is positive work; it does not cancel energy supplied at one time against energy absorbed at another. Taking the positive part after summing powers gives a different quantity when some actuators supply energy while others absorb it. Declare which convention is used before comparing swings.

Net joint moments inferred from motion do not resolve antagonistic muscle forces, tendon storage or activation costs. A joint with zero net power can still involve active muscle and metabolic expenditure. Neither a ratio to net joint work nor a ratio to positive joint work is automatically a metabolic efficiency. Estimating that efficiency requires a separately justified chemical energy or metabolic model and a consistent measurement interval.

11.3 Constraint Power: Zero for Which Boundary?

An ideal reaction to a position constraint $\Phi(q, t) = 0$ has generalized force $Q_c = J^T \lambda$, where $J = \partial \Phi / \partial q$. Its actual power is

$$P_c = Q_c^T \dot{q} = \lambda^T J \dot{q}. \tag{11.10}$$

Differentiating the constraint gives

$$J\dot{q} + \Phi_t = 0, \tag{11.11}$$

and therefore

$$P_c = -\lambda^T \Phi_t. \tag{11.12}$$

It vanishes for a stationary ideal holonomic constraint. A moving support can exchange energy with the modeled bodies. Conversely, some ideal nonholonomic constraints, such as homogeneous no-slip rolling constraints, also have zero reaction power. Holonomic versus nonholonomic is not the test by itself: ideality, the admissible velocities and explicit support motion matter. Friction, tissue damping and impact losses require their own models.

11.3.1 Why Energy Can Cross an Ideal Joint

Consider an arm and a club connected at hinge H, with coincident material-point velocities $\mathbf{v}_H$ in the ideal attachment. Let $\mathbf{F}$ act on the club at H, with $-\mathbf{F}$ on the arm. Then

$$P_{F,\text{club}} = \mathbf{F} \cdot \mathbf{v}_H, \qquad P_{F,\text{arm}} = -\mathbf{F} \cdot \mathbf{v}_H. \tag{11.13}$$

The pair sums to zero while either term may be nonzero. This is energy crossing an internal boundary, with equal and opposite entries in the two ledgers. Equality of attachment-point velocities enables the cancellation; it does not prevent transfer. Substituting the two centers' different velocities for the contact-point velocity would calculate the wrong contact power.

In a planar model let a hinge motor apply moment n to the club and $-n$ to the arm. If gravity is included in each segment's potential, the club balance is

$$\dot{E}_{\text{club}} = \mathbf{F} \cdot \mathbf{v}_H + n\omega_{\text{club}}, \qquad P_{\text{hinge motor}} = n(\omega_{\text{club}} - \omega_{\text{arm}}). \tag{11.14}$$

The moment's power on the club uses its absolute angular velocity, whereas the motor's net mechanical power uses relative angular velocity. Their difference is the equal-and-opposite moment's power on the arm. An ideal unactuated hinge transmits attachment force while allowing relative rotation; setting its motor moment to zero does not disconnect the club.

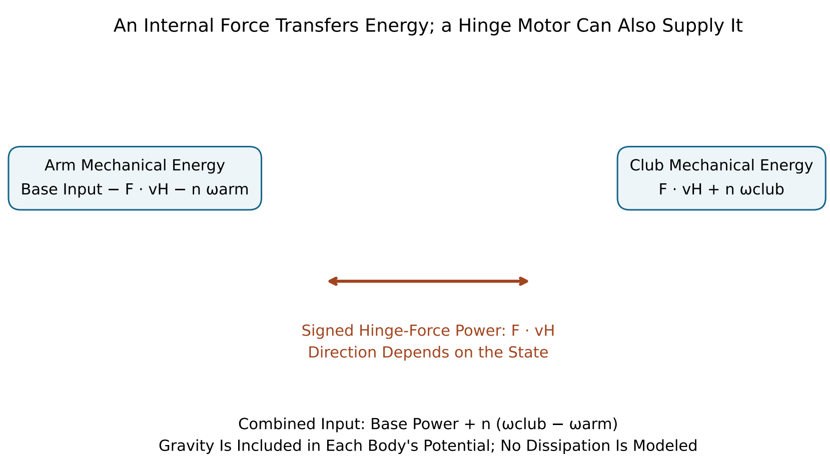


Figure 11.1: Both Sides of the Internal Boundary Must Balance. Hinge-Force Power Cancels Between Bodies, While Motor Power Depends on Their Relative Angular Velocity.

These balances do not select a universal direction of transfer or maximize club energy. Geometry, motion and applied efforts determine the signed power. The opposite signs of the interface entries establish accounting consistency; they do not identify what would happen if the golfer deliberately changed an effort or movement. That requires a new dynamically consistent trajectory.

11.3.2 Reporting Point, Observer and Ground Contact

A wrench's power is $\mathbf{F} \cdot \mathbf{v}_O + \mathbf{n}_O \cdot \boldsymbol{\omega}$ when both the moment and velocity refer to the same rigid body's reporting point O. Moving that point changes the moment and linear velocity together; the power is unchanged. Changing the inertial observer by a constant velocity changes individual kinetic energies and force powers consistently. A segment energy fraction is therefore an explicitly frame-dependent diagnostic, not an observer-independent property of the swing.

For an ideal fixed no-slip foot contact, local contact velocity is zero and the contact force does no work there, although it can change the golfer's momentum. Multiplying ground-reaction force by whole-body center-of-mass velocity produces a center-of-mass power quantity, not necessarily actual foot-contact power. Deforming footwear, slipping contacts and moving ground alter the boundary. Gravity, air and the ball can also supply external forces; ground reaction is not the only external interaction in a complete swing and collision.

11.4 Energy Partition During a Declared Passive Motion

Before interpreting a golfer's downswing, test the transfer claim on a mechanism whose assumptions and trajectory are known. Take two uniform rods, each of mass 1 kg and length 1 m, a fixed base, ideal hinges, zero gravity and no motors. Absolute angles are measured from vertically downward. Initially the inner rod has angle zero, the outer rod angle $-\pi/2$, and both absolute angular rates are 2 rad/s. Their kinetic energies are 2/3 J and 8/3 J, respectively. Total energy is 10/3 J.

The initial hinge-force power into the outer rod is $+1$ W. Reflecting the outer angle to $+\pi/2$, without changing either angular rate, gives -1 W. Both states have the same segment kinetic energies. The transfer direction therefore cannot be inferred from the energies or speed ordering alone. The force and attachment-point velocity matter.

Numerical integration of the first state gives the following rounded outer-rod energies at 0, 0.5, 1, 1.5 and 2 s: 2.6667, 3.3327, 2.9780, 2.5007 and 2.3421 J. At all five times the total remains 10/3 J within integration error. The outer fraction rises from 80 percent to about 99.98 percent at 0.5 s and subsequently falls. The signed power changes direction again later. There is no active braking and no dissipative sink in this model: a decrease of outer-rod energy is an increase of inner-rod energy.

These times are integration times for this declared mechanism, not five measured phases of a golf swing. A realistic model needs measured or justified segment properties, gravity, multiple joints, applied efforts and boundary conditions. A sequence of arbitrarily selected joint rates is not a simulated trajectory; it cannot establish a missing-work budget or a universal club-energy percentage.

11.5 Sequential Speed Peaks and Their Interpretation

The phrase summation of speed can obscure three different statements. Velocities combine vectorially through kinematics. Powers add algebraically in a declared energy balance. Peak times are features of a time series. None of these implies that scalar peak angular speeds should simply be added, or that the observed ordering proves a one-way energy cascade.

For an isolated rotor about a fixed axis, equal assigned energy gives

$$\omega = \sqrt{2K/I}. \tag{11.15}$$

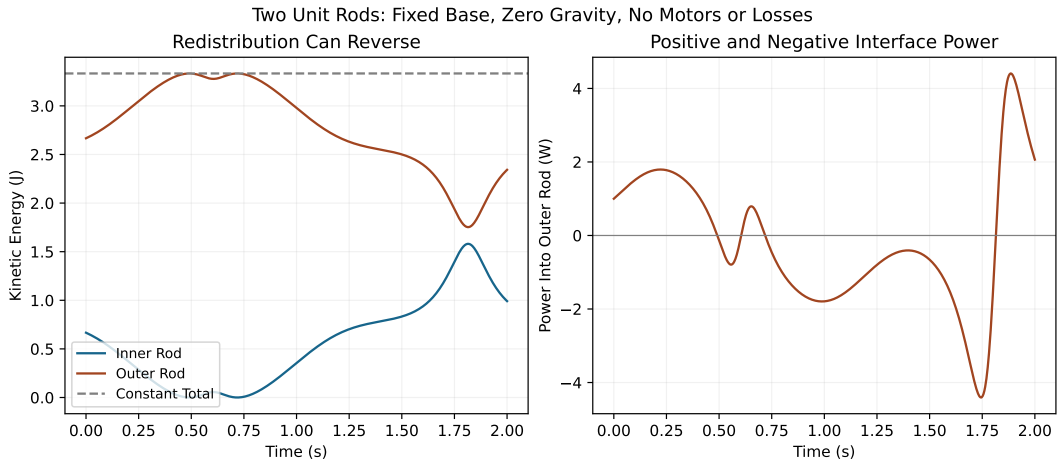


Figure 11.2: An Explicit Passive Mechanism Redistributes Energy in Both Directions. The Outer Rod Gains Energy, Loses Energy and Later Gains It Again; Total Energy Remains Constant.

For example, separate rotors with inertias 2, 0.5 and 0.1 kg m^2, each assigned 100 J, have angular speeds 10, 20 and approximately 44.72 rad/s. This algebra does not describe three connected segments at successive swing phases. Their energies need not be equal; their centers translate; their physical inertia tensors and the full joint mass matrix have different roles. A smaller scalar inertia alone does not guarantee a later or higher speed peak in the chain.

An especially useful counterexample uses the same two unit rods, now aligned vertically downward with both rates 2 rad/s, zero gravity, a base moment of -1 N m and zero hinge moment. The inner and outer absolute angular accelerations are $-12/7$ and $18/7$ rad/s^2. Yet their kinetic-energy rates are $-8/7$ and $-6/7$ W. The outer rod accelerates rotationally while losing energy because its translational energy falls more rapidly. Total power is -2 W, exactly the base moment times its angular velocity.

A proximal segment slowing while the club speeds up can accompany transfer, active work, gravity conversion, elastic release or combinations of them. It does not show that the proximal segment has exhausted its energy. Testing a sequencing explanation requires synchronized physical segment energies, interface powers and actuator powers, together with a model that reproduces the measured motion. Prescribing the observed speed peaks in a model and then claiming that the model explains those peaks would be circular.

11.6 Storage, Active Work and a Balanced Swing Ledger

The following five-state account is deliberately hypothetical. It illustrates bookkeeping, not a typical golfer. Start from rest at the chosen reference height with no elastic deformation.

The successive stored energies, in joules, are given by the three rows

$$\begin{aligned} K &: (0,\ 0,\ 80,\ 115,\ 115), \\ V_g &: (0,\ 15,\ 5,\ 0,\ 0), \\ U_e &: (0,\ 10,\ 2,\ 0,\ 0). \end{aligned} \tag{11.16}$$

Their sums are 0, 25, 87, 115 and 115 J. Choose cumulative active mechanical work of 0, 25, 90, 120 and 120 J and cumulative dissipative loss of 0, 0, 3, 5 and 5 J. Every state satisfies stored energy equals initial energy plus supplied work minus losses. Between the second and third states, stored energy increases 62 J while active work increases 65 J and dissipation increases 3 J. Falling gravitational and elastic storage explains part of the kinetic-energy rise; it does not replace the active-work contribution.

The model could instead start at the top with its already stored 25 J. Then that amount is initial energy, while only subsequent work belongs in the downswing input. Adding both the initial store and all work previously used to create it would double count the backswing. A claim that impact energy exceeds downswing muscular work can be consistent with energy conservation, but it requires an explicit initial store and other inputs rather than an efficiency greater than one inferred from incomplete accounting.

11.6.1 Impact Is a New Boundary and Time Interval

For an isolated one-dimensional head–ball collision, assume head mass $m_h = 0.2$ kg, ball mass $m_b = 0.04593$ kg, initially stationary ball, head speed $u_h = 45$ m/s and restitution $e = 0.8$. These are teaching assumptions, including the effective isolated head mass; a shaft-coupled, spinning or oblique impact requires additional dynamics. Momentum conservation and relative restitution give

$$v_h = \frac{m_h - e m_b}{m_h + m_b} u_h, \qquad v_b = \frac{(1+e) m_h}{m_h + m_b} u_h. \tag{11.17}$$

The final speeds are approximately 29.8724 and 65.8724 m/s. Initial head energy is 202.5 J; residual head energy is 89.2360 J; ball energy is 99.6491 J; the mechanical loss in this collision is 13.6148 J. Ball energy, residual head energy and collision loss sum to the initial energy. Residual head motion is not collision dissipation, and swing-phase air or tissue losses are not added again to this impact account.

With reduced mass $\mu = m_h m_b/(m_h + m_b)$, the loss is $\mu(1 - e^2)u_h^2/2$. The ratio of ball energy to pre-impact head energy is $m_h m_b (1+e)^2/(m_h + m_b)^2$, approximately 49.21 percent here. This is an impact energy-transfer ratio for the declared model, not a measured swing efficiency. A halving of a body's speed quarters its kinetic energy at fixed mass; it does not halve it. These distinctions are essential when interpreting residual club speed, ball launch and claims about efficient energy delivery.

11.7 Lag, Elasticity and Release Timing

A lag angle is a relative configuration. An ideal freely rotating hinge has no elastic potential associated with that angle. If a torsional spring is explicitly included, with stiffness k and rest angle q_{20}, then

$$U_e = \tfrac{1}{2} k (q_2 - q_{20})^2, \qquad \tau_e = -k(q_2 - q_{20}). \tag{11.18}$$

At a 0.5 rad displacement, stiffness zero stores zero joules; stiffness 40 N m/rad stores 5 J. The angle alone cannot distinguish these cases. A deforming shaft or tendon needs its own strain coordinate and constitutive law; its energy is not automatically a function of the visible wrist angle.

If the rest angle or stiffness changes, the explicit parameter contribution is $\partial U_e/\partial t = \dot{k}(q_2 - q_{20})^2/2 - k(q_2 - q_{20})\dot{q}_{20}$. That term represents work associated with changing the modeled spring, rather than free passive release. A viscous hinge damper with coefficient $d \geq 0$ dissipates $d\dot{q}_2^2$. Active moment, elastic moment and damping moment must be separated if the question concerns their individual contributions.

Holding a relative angle fixed is a different idealization from storing energy in a spring. A stationary ideal lock can transmit forces and moments with zero net constraint power. Removing such a lock need not change velocity or kinetic energy instantaneously, although subsequent dynamics change. Engaging a lock when the relative rate is nonzero generally requires an impulsive transition and a specified loss or actuator-work model. Releasing a rotational lock leaves the translational attachment intact; it does not remove the hinge force.

11.7.1 A Fair Release-Timing Comparison

Compare strategies from the same full initial state and with declared actuator limits, allowed controls and delivery event. The event should include relevant club position, face orientation, path and contact conditions, rather than only the largest angular speed anywhere in the trajectory. Report terminal kinetic and elastic energy, signed and positive work, losses and residual motion. Equal commanded torque histories do not generally imply equal work, because the trajectories and rates may differ.

If one strategy begins with 90 J and another with 140 J, their final speeds cannot isolate the effect of timing. If terminal energies are 60 and 150 J for the same isolated fixed-axis rotor, the speed ratio is $\sqrt{150/60}$, approximately 1.581, not two. For a translating multibody club, even that ratio requires checking the relationship between the reported speed and total kinetic energy. There is no universal release delay or speed multiplier established by these examples.

Setting one motor effort to zero at a frozen state is a useful instantaneous acceleration comparison. It is not the complete trajectory produced by a different release strategy. Passive tissue forces, gravity, constraints and other motors can remain active. This is where energy analysis connects to counterfactual dynamics and control: the intervention, retained terms, starting state and event must all be specified before assigning a causal benefit.

11.8 Two-Link Dynamics and Physical Energy Transfer

To derive the coupling explicitly, take a fixed planar base, inner-link length L_1, masses m_1, m_2, center distances c_1, c_2 from each proximal hinge and central planar inertias I_1, I_2. Define relative coordinates $q_1 = \theta_1$ and $q_2 = \theta_2 - \theta_1$, where the absolute angles θ_i are measured from downward. Let $u = \dot{q}_1$, $w = \dot{q}_2$ and $a = I_1 + m_1c_1^2 + m_2L_1^2$, $b = I_2 + m_2c_2^2$, $d = m_2L_1c_2$. Directly differentiating the center positions gives

$$K = \tfrac{1}{2}au^2 + \tfrac{1}{2}b(u + w)^2 + d\cos q_2\, u(u + w). \tag{11.19}$$

The club rotates at $u+w$, not w. The inner contribution is $(I_1+m_1c_1^2)u^2/2$; the rest belongs to the translating and rotating outer body. Hence the physical outer-body energy includes the cross term and an $m_2 L_1^2 u^2/2$ contribution. Assigning only $bw^2/2$ to the club would omit its motion induced by the inner link.

Euler–Lagrange equations with base and relative-hinge motor efforts $\tau = (\tau_1, \tau_2)^T$ give $M\ddot{q} + h + G = \tau$, where

$$M = \begin{bmatrix} a + b + 2d\cos q_2 & b + d\cos q_2 \\ b + d\cos q_2 & b \end{bmatrix}, \qquad h = \begin{bmatrix} -d\sin q_2(2uw + w^2) \\ d\sin q_2 u^2 \end{bmatrix}. \tag{11.20}$$

Both velocity-bias components matter. For upward gravitational potential,

$$G = \begin{bmatrix} (m_1 c_1 + m_2 L_1) g \sin q_1 + m_2 c_2 g \sin(q_1 + q_2) \\ m_2 c_2 g \sin(q_1 + q_2) \end{bmatrix}. \tag{11.21}$$

Multiplying by the rates and retaining the metric derivative recovers

$$\dot{E} = \tau_1 u + \tau_2 w. \tag{11.22}$$

In particular,

$$\dot{q}^T h = -d\sin q_2\, uw(u + w) = \tfrac{1}{2}\dot{q}^T \dot{M}\dot{q}. \tag{11.23}$$

This scalar can be nonzero. It is canceled in the total energy derivative, not identified as a separate physical input or equated to the club's energy rate. A term called Coriolis power in one coordinate decomposition must not be promoted into an invariant segment-transfer measurement.

To compute the actual hinge-force transfer, let $\mathbf{a}_{C_2}$ be the outer center acceleration and $\mathbf{g} = (0, -g)^T$. With no other outer-body force, Newton's equation gives

$$\mathbf{F} = m_2(\mathbf{a}_{C_2} - \mathbf{g}), \qquad P_{F,2} = \mathbf{F} \cdot \mathbf{v}_H. \tag{11.24}$$

The physical body rates follow from center translation, rotation and gravity:

$$\dot{E}_2 = P_{F,2} + \tau_2(u + w), \qquad \dot{E}_1 = \tau_1 u - P_{F,2} - \tau_2 u. \tag{11.25}$$

Their sum agrees with the generalized balance. This independent Newton-force calculation is the check that a coordinate-based interpretation needs. In three dimensions replace scalar moments and rates by compatible moment/angular velocity pairings; in a two-arm closed chain include both attachment wrenches and avoid duplicating the shared club energy.

11.8.1 Reproducible Calculation

The shared `rod_energy_ledger` routine computes center energies and their derivatives from `RodPair` dynamics. Its state uses absolute angles and rates; motor inputs remain base and relative-hinge moments. For uniform rods the central inertias are $m_i L_i^2/12$ and $c_i = L_i/2$. Run this example from the repository root:

```
import numpy as np

from src.tools.energy_ledger_examples import rod_energy_ledger
from src.tools.lagrangian_examples import RodPair

model = RodPair(gravity=0)
state = np.array([0, -np.pi / 2, 2, 2])
ledger = rod_energy_ledger(model, state, np.zeros(2))
np.testing.assert_allclose(ledger.kinetic, [2 / 3, 8 / 3])
```

```
np.testing.assert_allclose(ledger.mechanical_rates, [-1, 1])
assert np.isclose(ledger.force_power, 1)
assert np.isclose(ledger.actuator_power, 0)

braking = rod_energy_ledger(
    model, np.array([0, 0, 2, 2]), np.array([-1, 0])
)
np.testing.assert_allclose(braking.mechanical_rates, [-8 / 7, -6 / 7])
```

The calculation has been independently checked against directional derivatives of each physical body's energy, total generalized motor power and the force power at the hinge. Conservation alone would be insufficient verification: two incorrectly assigned segment rates could cancel in the total. Conversely, a small integration drift is a numerical diagnostic, not evidence of unmodeled muscular work or physical dissipation.

11.9 Evidence, Measurement and an Investigable Golf-Swing Argument

Nesbit and Monika Serrano's work-and-power study used four selected amateur golfers and a selected swing from each, with model-derived mechanical quantities [Nesbit and Serrano, 2005]. It supports examining coordinated body and club work within that study's model. It does not establish a population-wide energy percentage, directly measure metabolic efficiency or identify the causal effect of an instruction to delay release. Those questions need additional evidence.

To investigate an energy-transfer explanation, first define the outcome: clubhead delivery speed and orientation, ball launch, repeatability or effort. Then choose the body and time boundaries. Estimate segment properties and synchronized motion; state coordinate, frame and joint-moment conventions. Use the inverse-dynamics and constraint-force chapters (Chapters 19 and 7) to construct physically compatible forces. Close the whole-system ledger and both sides of each important interface before interpreting a sequence of peaks. The complete-swing and synthesis chapters (Chapters 33 and 32) connect these local mechanisms to finite-time delivery and competing task objectives.

Power estimates amplify synchronization and differentiation errors. To first order, the error in a wrench power includes

$$\delta P \simeq \delta\mathbf{F}\cdot\mathbf{v} + \mathbf{F}\cdot\delta\mathbf{v} + \delta\mathbf{n}\cdot\boldsymbol{\omega} + \mathbf{n}\cdot\delta\boldsymbol{\omega}. \tag{11.26}$$

These errors can be correlated; uncertainty in integrated work also depends on timing, filtering, interval endpoints and their covariance. Compare a reported transfer difference with that uncertainty, test plausible segment properties, and inspect energy residuals without treating residual magnitude alone as a diagnosis of the missing mechanism. Force-plate, motion-capture and club measurements must share time and frame conventions.

The resulting argument is interconnected but specific: configuration and inertia determine how efforts and constraints affect acceleration; actual velocities determine their power; constitutive behavior determines storage and loss; control changes the trajectory and delivery event; observations constrain which of those explanations can be distinguished. A convincing coaching claim requires a controlled intervention or a validated counterfactual model with uncertainty, not just a mechanically plausible story. The mechanics makes better questions possible without prescribing a universal hip-braking or lag strategy.

11.10 Chapter Exercises: Energy Transfer

1. Derive the mechanical-energy balance from the manipulator equation. Explain why setting the scalar Coriolis contraction to zero is unnecessary. Then derive the power of an ideal constraint with explicit time dependence.

2. At an arm–club hinge, force power into the club is 12 W. The base motor moment is 7 N m, hinge motor moment 3 N m, and arm and club absolute angular rates are 2 and 5 rad/s. Find both body mechanical-energy rates, hinge-motor power and total motor power. Assume all other interactions are conservative.

3. Check the five-state storage ledger between the second and third states. If the second state is the beginning of the downswing, explain which initial energy and work interval belong in its balance. Can a video lag angle determine the initial elastic store?

4. For the stated isolated impact, derive the speeds, energy-transfer ratio and loss from momentum and restitution. Explain why residual head energy and swing losses cannot be relabeled as collision dissipation.

5. Derive M, h and G for the two-link relative chart. For the aligned unit-rod braking case, compute both absolute angular accelerations and both body-energy rates. Explain how the outer rod can gain angular speed while losing energy.

6. Design a comparison of two release strategies that could distinguish active work, elastic release and interface transfer. State the common initial conditions, actuation constraints, delivery event, measured quantities and uncertainties. Explain why equal torques or a sequence of speed peaks is insufficient evidence of equal effort or one-way transfer.

11.11 Worked Answers

1. Differentiate $K = \dot{q}^T M \dot{q}/2$, substitute the equation of motion, and add $\dot{V} = \dot{q}^T \nabla V$. The terms $\dot{q}^T \dot{M} \dot{q}/2$ and $\dot{q}^T C \dot{q}$ cancel by the compatible Christoffel identity, leaving actuator, nonconservative and constraint powers. From $J\dot{q} + \Phi_t = 0$, constraint power is $-\lambda^T \Phi_t$; only the stationary ideal case guarantees zero through this argument. A moving support must remain in the work account.

2. Club moment power is $3(5) = 15$ W, so its energy rate is $12 + 15 = 27$ W. The arm receives base power $7(2) = 14$ W and loses 12 W through the force and $3(2) = 6$ W through the hinge moment: its rate is -4 W. Hinge-motor power is $3(5-2) = 9$ W; total motor power is $14 + 9 = 23$ W. The body rates sum to the same 23 W. Equating club moment power to motor power would miss the arm's opposite moment entry.

3. Stored energy rises from 25 to 87 J, a 62 J increase. Active work over that interval is $90 - 25 = 65$ J and loss is 3 J. Kinetic energy rises 80 J, supplied by 65 J active work plus 10 J gravitational release plus 8 J elastic release, minus 3 J loss. Starting at the second state means 25 J initial storage and work counted only afterward. Lag alone supplies neither a spring stiffness nor its rest deformation; an ideal hinge can have that angle with zero elastic energy.

4. Solve $m_h u_h = m_h v_h + m_b v_b$ and $v_b - v_h = e u_h$. The stated speeds follow. Divide $m_b v_b^2/2$ by $m_h u_h^2/2$ to obtain the ratio, about 0.492094. Subtract both final kinetic energies from the initial energy to obtain $\mu(1 - e^2)u_h^2/2$, about 13.6148 J. Residual head energy is still mechanical energy, and earlier swing dissipation occurred outside the collision interval.

5. Differentiate the kinetic-energy quadratic twice with respect to rates to obtain M; Euler–Lagrange differentiation gives both components of h. Differentiate gravitational potential to obtain both entries of G. In the aligned unit-rod case the absolute-coordinate mass matrix is $\begin{bmatrix} 4/3 & 1/2 \\ 1/2 & 1/3 \end{bmatrix}$, with zero instantaneous velocity bias and gravity. Solving against $(-1, 0)^T$ gives $(-12/7, 18/7)^T$. Inner energy rate is $(1/3)(2)(-12/7) = -8/7$ W; total motor power is -2 W, so outer energy rate is $-6/7$ W. Independent center and hinge-power calculations give the same result. Increasing rotational energy is outweighed by decreasing translational energy; angular speed alone does not measure total outer-body energy.

6. Use identical positions, velocities and internal elastic states, with the same stated motor capacities and support conditions. Define release as a specific control or constraint change, including any switching impulse or parameter work. Evaluate at the same physical delivery event and report its orientation and position as well as speed. Integrate separate active, elastic, dissipative and interface powers; compare signed work, positive work, residual energy and uncertainty across strategies. Equal torque commands can yield different rates and therefore different work. Peak order alone leaves those accounts unidentified. Repeat the comparison under plausible measurement and model variations before attributing a robust benefit to timing.

Chapter 12

Ground Reaction Forces: The Silent Foundation

> The ground does not push you up; you push down, and the ground pushes back equally.
>
> —Isaac Newton, Third Law in Disguise

In Plain Language 12.1. Why the Ground Matters

When you stand at address, you feel the ground beneath your feet. When you swing, the ground exerts forces on your body that reshape your momentum, redirect your energy, and determine whether you can transfer power to the ball. Yet most golfers never think about the ground as an active player in the swing. This chapter reveals why the ground reaction force (GRF) is as fundamental to golf as the swing itself.

12.1 What Are Ground Reaction Forces?

Every golfer stands on the ground. This seems obvious, almost trivial. But it is not. The ground is a constraint on your motion. You cannot pass through the ground; your feet cannot sink below it. This constraint is enforced by a force.

12.1.1 Newton's Third Law at the Golfer's Feet

Newton's Third Law states that if you exert a force on something, it exerts an equal and opposite force on you. When you stand still, gravity pulls your body downward with force $m\boldsymbol{g}$, where m is your mass and $\boldsymbol{g}$ is the gravitational acceleration vector (pointing downward). If you fell straight through the ground, the Second Law would give

$$\boldsymbol{F}_{\text{net}} = m\boldsymbol{a} = m\boldsymbol{g}.$$

But you do not fall. Instead, the ground exerts an upward normal force on you, call it $\boldsymbol{N}$. The net force becomes

$$\boldsymbol{F}_{\text{net}} = m\boldsymbol{g} + \boldsymbol{N}.$$

If you stand still, $\boldsymbol{a} = 0$, so

$$\boldsymbol{N} = -m\boldsymbol{g}.$$

The normal force exactly cancels gravity. This is Newton's Third Law: you push down (via your weight), and the ground pushes back.

Example 12.1. Feeling the Ground

Stand with your feet shoulder-width apart. Now shift your weight rapidly to your left foot. You felt the ground push harder under your left foot and softer under your right. Now jump. For a brief moment, the ground pushed you upward with more force than your body weight—that extra force is what launched you into the air.
In the golf swing, this weight shift happens dynamically throughout the motion. GRF distribution between feet changes throughout the swing: early in the downswing, both feet contribute; as the downswing progresses and the golfer rotates toward the target, weight transfers to the lead foot. teNesbit2005, Sprigings2000 The exact percentages depend on swing style and individual technique, but the pattern of progressive weight transfer is consistent across skilled golfers.

Key Principle 12.1. The Ground Reaction Force

The ground reaction force $\boldsymbol{F}_{\text{GRF}}$ is the force exerted by the ground on the golfer's feet. It has three orthogonal components:

- Vertical: F_z (perpendicular to the ground, upward positive)
- Anterior-Posterior: F_x (forward-backward along the target line)
- Medial-Lateral: F_y (side-to-side across the body)

The total GRF vector is $\boldsymbol{F}_{\text{GRF}} = (F_x, F_y, F_z)$.

During a golf swing, the ground exerts forces in all three directions simultaneously. A force plate—an instrument embedded in the ground—measures these forces. The vertical component of GRF varies during the downswing and depends on the phase of motion and individual swing characteristics. teNesbit2005, Jorgensen1994 Peak vertical GRF is typically greater than body weight at midswing due to the downward acceleration of body segments and the constraint forces maintaining ground contact.

12.1.2 The Center of Pressure

The GRF is distributed across the contact patch between the sole of the shoe and the ground. It is useful to summarize this distribution as a single vector applied at a single point: the center of pressure (CoP).

Definition 12.1. Center of Pressure

The center of pressure is the point on the ground where the net GRF vector acts. It is the location where the total moment of all distributed ground contact forces equals zero.

During a static stance, the CoP is roughly beneath the center of mass. During a dynamic swing, the CoP moves. Early in the downswing, the CoP may move forward (toward the target), indicating that the golfer is shifting weight anteriorly. Later in the downswing, as the golfer extends through the ball, the CoP may migrate toward the medial (inner) foot or the heel, depending on the swing mechanics.

Tracking CoP movement is a powerful diagnostic tool: it reveals whether weight is shifting as intended, whether the golfer is sway-ing (CoP moving laterally in an uncontrolled way), or whether early extension is forcing the pelvis forward prematurely.

12.1.3 The Free Body Diagram

To understand GRF rigorously, draw a free body diagram. The golfer's entire body is the system. The forces acting on this system are:

1. Gravity: $m\boldsymbol{g}$, acting downward at the center of mass.
2. Ground reaction force: $\boldsymbol{F}_{\text{GRF}}$, acting upward at the feet.
3. Air resistance (negligible for the golfer, significant for the ball).

The golfer does not have rockets or invisible cables. The only forces from the external world are gravity and the GRF. This is crucial: **the GRF is the only external force available to change the golfer's linear momentum**.

12.2 The Impulse-Momentum Perspective

The impulse-momentum perspective answers: How does the ground shape the golfer's motion?

12.2.1 Newton's Second Law for the Center of Mass

Let $\boldsymbol{r}_{\text{COM}}$ denote the position of the center of mass. Newton's Second Law for the whole body states:

$$\sum \boldsymbol{F}_{\text{external}} = m\ddot{\boldsymbol{r}}_{\text{COM}}.$$

The only external forces are gravity and GRF, so:

$$\boldsymbol{F}_{\text{GRF}} + m\boldsymbol{g} = m\ddot{\boldsymbol{r}}_{\text{COM}}.$$

Rearranging:

$$\boldsymbol{F}_{\text{GRF}} = m\ddot{\boldsymbol{r}}_{\text{COM}} - m\boldsymbol{g}.$$

In Plain Language 12.2. In Plain Language

The GRF is whatever force is needed to give the center of mass the acceleration it has, minus what gravity alone would give. If you accelerate upward (as you do during extension), GRF must be larger than your weight. If you decelerate downward (as you might at the start of the downswing), GRF might be less than your weight.

This equation is revealing. It says: the GRF is a *reaction*—it is whatever is required to maintain ground contact while the internal forces (muscles) reshape the body. The muscles create internal torques, which accelerate body segments. These accelerations change the center-of-mass acceleration. The GRF adjusts to match.

12.2.2 Impulse: The Time Integral

Integrating both sides of the equation of motion from time t_1 to t_2:

$$\int_{t_1}^{t_2} \boldsymbol{F}_{\text{GRF}}\, dt + m\boldsymbol{g}(t_2 - t_1) = m[\dot{\boldsymbol{r}}_{\text{COM}}(t_2) - \dot{\boldsymbol{r}}_{\text{COM}}(t_1)].$$

Define the impulse from GRF as:

$$\boldsymbol{J}_{\text{GRF}} = \int_{t_1}^{t_2} \boldsymbol{F}_{\text{GRF}}\, dt.$$

Then:

$$\boldsymbol{J}_{\text{GRF}} = m\Delta\boldsymbol{v}_{\text{COM}} - m\boldsymbol{g}(t_2 - t_1).$$

Key Principle 12.2. Impulse-Momentum Theorem for GRF

The impulse delivered by the ground (the integral of GRF over time) equals the change in the golfer's momentum, adjusted for gravitational impulse. Without the ground, the golfer's momentum would change purely due to gravity. The GRF impulse is the **additional** momentum change caused by the golfer's interaction with the ground.

12.2.3 Numerical Example: Center-of-Mass Acceleration From GRF

Example 12.2. Computing COM Acceleration from Force Plate Data

Hypothetical example: Suppose a 90 kg golfer stands on a force plate. The following values are illustrative to demonstrate the calculation method:

Time	Vertical GRF	Anterior-Posterior GRF
$t = 0.0$ s	$F_z = mg = 883$ N	$F_x = 0$ N
$t = 0.2$ s	$F_z = 1200$ N	$F_x = 200$ N

At $t = 0.2$ s, compute the center-of-mass acceleration using the measured GRF.
Solution: Use $\boldsymbol{F}_{\text{GRF}} = m\ddot{\boldsymbol{r}}_{\text{COM}} - m\boldsymbol{g}$, rearranged to:
Vertical component:

$$F_z = m\ddot{z}_{\text{COM}} - mg.$$
$$1200 = 90 \cdot \ddot{z}_{\text{COM}} - 90 \cdot 9.81.$$
$$1200 = 90 \cdot \ddot{z}_{\text{COM}} - 883.$$
$$\ddot{z}_{\text{COM}} = \frac{1200 + 883}{90} = 23.1 \text{ m/s}^2.$$

This is an upward acceleration of $23.1/9.81 \approx 2.35$ g's.
Anterior-Posterior component:

$$F_x = m\ddot{x}_{\text{COM}}.$$
$$200 = 90 \cdot \ddot{x}_{\text{COM}}.$$
$$\ddot{x}_{\text{COM}} = 2.22 \text{ m/s}^2.$$

The center of mass accelerates forward at 2.22 m/s^2.

12.2.4 The Kinetic Chain and the Ground-Up Sequence

The impulse perspective explains the **kinetic chain** or **ground-up sequence**. The motion of the golf swing appears to flow from the ground up: the legs push, the hips rotate, the torso follows, the shoulders unwind, the arms extend, and finally the club head meets the ball.

Why does the motion flow this way? Because the GRF is the source of external impulse. The feet push into the ground (via muscular effort), the ground pushes back with GRF, and this GRF impulse must be transmitted through the body.

However, the transmission is not magical. It is a consequence of the rigid-body structure of the body. The sequence reflects the stiffness of joints and the order in which muscles are activated.

In Plain Language 12.3. In Plain Language

Think of the kinetic chain like a whip. You don't accelerate the entire whip at once. You accelerate the handle, which pulls the next segment, which pulls the next, and so on. The speed increases as it propagates because the kinetic energy is concentrated into smaller and smaller mass. Similarly, in the golf swing, the ground provides the impulse to the legs, which accelerate the hips, and so on. Each stage adds speed to the distal segment because less mass is being accelerated.

12.3 The Work-Energy Perspective: GRF Does No Work

Here is a fact that surprises many students: the ground reaction force does zero work.

Work is defined as:

$$W = \int \boldsymbol{F} \cdot d\boldsymbol{r} = \int \boldsymbol{F} \cdot \boldsymbol{v}\, dt,$$

where $\boldsymbol{v}$ is the velocity of the point of application of the force.

The GRF is applied at the feet, specifically at the contact patch between the shoe sole and the ground. If the golfer's feet are planted firmly on the ground, the contact patch has zero velocity. Therefore, the power delivered by GRF is:

$$P_{\text{GRF}} = \boldsymbol{F}_{\text{GRF}} \cdot \boldsymbol{v}_{\text{contact}} = \boldsymbol{F}_{\text{GRF}} \cdot \boldsymbol{0} = 0.$$

And the work is:

$$W_{\text{GRF}} = \int P_{\text{GRF}}\, dt = \int 0\, dt = 0.$$

Theorem 12.1. GRF Does Zero Work

For a golfer with feet planted on the ground, the ground reaction force does zero work and delivers zero power. The GRF cannot increase the total mechanical energy of the golfer-club system.

This seems to contradict the impulse-momentum perspective. The GRF is the only external force, yet it adds no energy. How is this possible?

12.3.1 The Paradox and Its Resolution

The resolution hinges on distinguishing between **momentum** (a vector, changed by impulse) and **energy** (a scalar, changed by work).

- The GRF has a large impulse (it is large in magnitude and acts for a long time). - The GRF has zero work (the point of application has zero velocity).

These statements are not contradictory. Momentum and energy are different quantities.

In Plain Language 12.4. In Plain Language

Imagine a billiards player making a bank shot. The cue hits the ball, giving it momentum. The ball travels across the table and hits the wall. The wall exerts a large force, reversing the ball's momentum. But the wall does no work: the ball's speed is unchanged (ignoring friction), only its direction changed. The wall has enormous impulse but zero work. The ground in golf is like the wall in billiards.

Another analogy: a roller coaster car on a track. The normal force exerted by the track on the car is large, but it is perpendicular to the track. If the car moves along the track, the normal force does no work (force perpendicular to motion = zero work). Yet the normal force is essential: without it, the car would fly off the track.

12.3.2 Energy Enters Through Muscles, Not Ground

Where does the energy in the golf swing come from? From the muscles. The muscles perform positive work on the joints by exerting torques through which the joints rotate:

$$W_{\text{muscle}} = \int \boldsymbol{\tau}_{\text{muscle}} \cdot \dot{\boldsymbol{q}}\, dt,$$

where $\boldsymbol{\tau}_{\text{muscle}}$ is the vector of muscle torques and $\dot{\boldsymbol{q}}$ is the joint angular velocity vector.

The muscles pump energy into the system. The GRF then reshapes (redirects) this energy without adding to it. The GRF maintains the constraint that the feet stay on the ground while muscles deform and accelerate the body.

Key Principle 12.3. Energy Sources in the Swing

Muscles are the source of energy in the golf swing. The ground reaction force does zero work and cannot add energy to the system. Instead, the GRF redirects energy: it enforces the constraint that the feet cannot pass through the ground while allowing muscles to accelerate the body segments.

12.3.3 Constraint Forces and Lagrange Multipliers

In the Lagrangian formulation of mechanics, constraint forces (like GRF) are represented as Lagrange multipliers.

Suppose the constraint is $\boldsymbol{\Phi}(\boldsymbol{q}) = 0$ (the foot positions are on the ground). The constraint force is $\boldsymbol{\lambda}$ (the GRF, expressed as a generalized force). The fundamental property is:

$$\boldsymbol{\lambda}^T \dot{\boldsymbol{\Phi}} = 0.$$

This says: the constraint force does no work. Work is the power times time: $P = \boldsymbol{\lambda}^T \dot{\boldsymbol{\Phi}}$. The zero-work property of constraint forces is built into the mathematics.

12.4 GRF as a Drift Force

In the control-affine framework (Chapter 5), the dynamics are:

$$\dot{\boldsymbol{x}} = f(\boldsymbol{x}) + G(\boldsymbol{x})\boldsymbol{u},$$

where $f(\boldsymbol{x})$ is the drift field (what happens with zero control), $G(\boldsymbol{x})$ is the control field (how control inputs affect the state), and $\boldsymbol{u}$ is the control input (muscle torques).

The GRF is not a direct control input. The golfer does not consciously command a GRF magnitude; instead, GRF is a reaction to the body's motion and internal forces.

12.4.1 GRF Emerges From Constraints

In a fully general biomechanical model, the equations of motion for the golfer as a rigid-body system are:

$$\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{\tau}_{\text{muscle}} + \boldsymbol{\tau}_{\text{GRF}}.$$

Here: - $\boldsymbol{M}(\boldsymbol{q})$ is the mass matrix (inertia). - $\boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}}$ is the Coriolis and centrifugal term. - $\boldsymbol{g}(\boldsymbol{q})$ is the gravity vector. - $\boldsymbol{\tau}_{\text{muscle}}$ is the vector of muscle torques (the control input $\boldsymbol{u}$). - $\boldsymbol{\tau}_{\text{GRF}}$ is the vector of torques induced by the ground reaction force.

The GRF enters as a constraint force. It is not in $\boldsymbol{u}$ and not in $G(\boldsymbol{x})\boldsymbol{u}$. Instead, it is a reaction that enforces the constraint that the foot position equals the ground position.

Mathematically, if we solve for $\boldsymbol{q}(t)$ given $\boldsymbol{\tau}_{\text{muscle}}(t)$, the GRF is determined as a by-product. We can compute it using:

$$\boldsymbol{F}_{\text{GRF}} = m\ddot{\boldsymbol{r}}_{\text{COM}} - m\boldsymbol{g},$$

where $\boldsymbol{r}_{\text{COM}}$ is computed from the joint angles $\boldsymbol{q}(t)$ via forward kinematics.

12.4.2 Pointwise Ground-Reaction ZTCF

The pointwise ground-reaction zero-torque counterfactual (ZTCF) sets the declared controllable input to zero at the achieved $(\boldsymbol{q}, \dot{\boldsymbol{q}})$ while retaining declared non-control external loads. It answers what reaction the model requires at this state with zero declared control. It is not a no-muscle experiment and does not describe what the body would do after controls were removed.

Definition 12.2. Pointwise Zero-Control Ground Reaction

The pointwise zero-control ground reaction is the constraint reaction obtained at a declared state after setting the model's applied generalized-control channel to zero. Static weight and retained drift terms remain. It is not a no-muscle condition, and subtracting it from a measured reaction does not uniquely identify muscle torque or effort.

Comparing a measured ground reaction with a model-conditioned pointwise ZTCF reaction can test the declared reaction decomposition. Any residual remains sensitive to model structure, contact allocation, parameter error, filtering, and unmodeled loads; it is not a unique inverse estimate of muscle torque.

Drift vs. Control 12.1. What a Ground-Reaction Residual Cannot Establish

Measured GRF minus modeled pointwise ZTCF is a residual containing input-induced reaction, model error, measurement error, and omitted external loads. It cannot uniquely identify muscle torques. Even with a perfect resultant wrench, a redundant contact system does not uniquely determine bilateral allocation; a pseudoinverse supplies a numerical allocation, not new physical information.

12.5 GRF Induced by Torques at Remote Joints

This section addresses a counterintuitive fact: a muscle torque applied at one joint can induce a change in GRF at the feet, even if the feet are not directly involved.

12.5.1 The Propagation of Internal Torques

Consider a simple scenario: the golfer applies a torque at the shoulder (by activating shoulder muscles). This torque accelerates the arm. By Newton's Third Law, the arm exerts a reaction torque on the torso. If the torso were free to move, it would accelerate in the opposite direction. But the torso is constrained: it sits atop the pelvis, which sits atop the legs, which are planted on the ground.

The torso wants to rotate backward (as a reaction to the forward arm acceleration). But it cannot, because the legs prevent it. The legs, in turn, push against the ground. The ground pushes back with a change in GRF.

Mathematically, this is captured by the equation:

$$\boldsymbol{F}_{\text{GRF}} = m\ddot{\boldsymbol{r}}_{\text{COM}} - m\boldsymbol{g}.$$

If the shoulder torque accelerates the arm forward and the center of mass forward, then $\ddot{\boldsymbol{r}}_{\text{COM}}$ increases, and so does $\boldsymbol{F}_{\text{GRF}}$.

12.5.2 Numerical Example: Shoulder Torque and Resulting GRF

Example 12.3. From Shoulder Torque to GRF Change

Consider a 90 kg golfer with a 5 kg arm. Suppose the golfer applies a torque of 100 N·m at the shoulder (a plausible illustrative value consistent with inverse-dynamics studies [Nesbit, 2005, Gatt et al., 1998]; treat as a worked example rather than a specific measurement), causing the arm to accelerate.

Assume (for simplicity) that the arm's center of mass is 0.5 m from the shoulder, and the whole body's center of mass is 1.0 m from the shoulder horizontally. When the arm rotates with angular acceleration α, the arm's center of mass has tangential acceleration $a_{\text{arm}} = \alpha \cdot 0.5$.

From the shoulder torque, estimate α:

$$\tau = I\alpha,$$

where I is the arm's moment of inertia about the shoulder. For a point mass, $I \approx 5\,\mathrm{kg} \times (0.5\,\mathrm{m})^2 = 1.25\,\mathrm{kg}\cdot\mathrm{m}^2$.

$$\alpha = \frac{100\,\mathrm{N}\cdot\mathrm{m}}{1.25\,\mathrm{kg}\cdot\mathrm{m}^2} = 80\,\mathrm{rad/s}^2.$$

The arm's center of mass acceleration is:

$$a_{\mathrm{arm}} = 80 \times 0.5 = 40\,\mathrm{m/s}^2 \quad \text{(forward)}.$$

The overall center of mass acceleration is a weighted average. If the arm moves forward, the whole-body COM moves forward (less dramatically, because the arm is only 5 kg out of 90 kg):

$$\ddot{x}_{\mathrm{COM}} = \frac{5 \times 40 + 85 \times 0}{90} = \frac{200}{90} = 2.22\,\mathrm{m/s}^2.$$

The change in anterior-posterior GRF is:

$$\Delta F_x = m \cdot \ddot{x}_{\mathrm{COM}} = 90 \times 2.22 = 200\,\mathrm{N}.$$

Conclusion: A 100 N·m torque at the shoulder induces a 200 N change in anterior-posterior GRF at the feet. This is why force plate data reveals information about remote joints: the feet "feel" the effects of muscle torques throughout the body.

12.5.3 Force Plates as Whole-Body Sensors

This example illustrates why force plates are so valuable in biomechanics. They do not measure forces only at the feet; they measure the net effect of all accelerations throughout the body.

In principle, if you know the kinematic motion (positions and velocities from motion capture) and the GRF (from force plates), you can infer the muscle torques throughout the body using inverse dynamics:

$$\boldsymbol{\tau}_{\mathrm{muscle}} = \boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q}) - \boldsymbol{\tau}_{\mathrm{GRF}}.$$

This is a powerful diagnostic. But it has limitations: motion capture is noisy, the model of the body is simplified, and the inverse problem is ill-posed (small errors in kinematics can lead to large errors in estimated torques).

In Plain Language 12.5. In Plain Language

Force plates are like listening to an entire orchestra through a single speaker. You cannot isolate each instrument perfectly, but you get a sense of the overall sound. Similarly, force plates measure the net effect of all muscle torques, mediated through the body's inertia and gravity.

12.6 Practical Implications for Golf

12.6.1 The Push-Off Pattern

Elite golfers exhibit a characteristic GRF pattern during the downswing and impact. Early in the downswing, the vertical GRF increases as the golfer extends the legs. This upward GRF impulse propels the center of mass upward and through the shot.

The lateral GRF (medial-lateral component) also changes: as the golfer rotates the hips, the lead leg may push outward against the ground, creating a lateral GRF that helps rotate the torso.

Coaches often cue golfers to "push into the ground" or "drive off the back foot." This cue is intuitive: it emphasizes the impulse perspective. But it is incomplete. The impulse is essential for transferring energy from the lower body to the upper body. However, the work perspective shows that the ground itself adds no energy—the muscles are the true source.

12.6.2 GRF Asymmetry: Lead vs. Trail Foot

In a right-handed golfer's swing, the trail foot (right foot) and lead foot (left foot) play different roles. Early in the downswing, both feet push, but the trail foot tends to contribute more vertical GRF (extending), while the lead foot may contribute lateral GRF (resisting the rightward inertia of the upper body).

At and just after impact, the lead foot is in full contact (and may even be lifting slightly), while the trail foot is rolling to its toes (preparing to lift).

Asymmetries in GRF between the two feet reveal swing faults:

- **Excessive weight shift:** If the lead foot GRF drops prematurely, the golfer is moving his weight too fast toward the target, often resulting in early extension.
- **Hanging back:** If the trail foot GRF remains high too long, the golfer is not transferring weight, leading to weak impact.
- **Swaying:** If the lead foot GRF spikes laterally before the downswing, the golfer is moving laterally rather than rotating.

12.6.3 Connecting GRF to Common Swing Faults

Constraint Insight 12.1. Early Extension and GRF

Early extension occurs when the hips rise (extend) too soon during the downswing. This often appears in GRF data as:

- A sharp increase in vertical GRF too early (before the torso is fully loaded).
- A rapid anterior-posterior GRF (the golfer is jumping forward rather than rotating).
- The center of pressure moving forward prematurely (losing the "ground" as a platform).

A proper swing maintains relatively constant vertical GRF during the loading phase, then increases it during extension, allowing the torso to rotate efficiently against the ground's resistance.

Constraint Insight 12.2. Sway and GRF

Sway is excessive lateral movement of the body during the backswing. In GRF terms, sway appears as:

- Large lateral (medial-lateral) GRF during the backswing, when lateral forces should be small.
- Center of pressure moving outside the base of the feet.
- Asymmetry between trail and lead foot GRF (one foot bearing most of the weight laterally).

A proper backswing maintains a stable center of pressure, loading the trail side vertically without excessive lateral motion.

12.6.4 Force Plate Measurements in the Lab

Modern golf biomechanics labs use dual force plates (one under each foot) to measure GRF with high precision. The data reveals:

- Vertical GRF over time (the classic "force curve").
- Anterior-posterior GRF (forward push during extension).
- Medial-lateral GRF (side-to-side balance and rotation).
- Center of pressure path (the trajectory of the balance point).
- Impulse decomposition (how much momentum change occurs in each phase).

This data feeds into coaching: golfers can see whether their GRF pattern matches a template of optimal performance. With feedback, they can train their neuromuscular system to produce the desired GRF profile.

Key Principle 12.4. Key Takeaways

1. The ground reaction force is the net force exerted by the ground on the golfer's feet. It has three orthogonal components: vertical, anterior-posterior, and medial-lateral.
2. From the impulse-momentum perspective, GRF is the only external force (besides gravity) available to change the golfer's momentum. The GRF impulse determines how the center of mass accelerates.
3. From the work-energy perspective, GRF does zero work because the contact point (feet) has zero velocity. Energy enters the system through muscle work, not through the ground. The ground redirects energy without adding to it.
4. GRF is a constraint force (a Lagrange multiplier). It arises as a reaction to

enforce the constraint that the feet remain in contact with the ground.

5. A torque at any joint in the body induces a change in GRF at the feet, via Newton's Third Law propagated through the rigid body structure. Force plates measure the net effect of all muscle torques throughout the body.

6. The kinetic chain (ground-up sequence) is a consequence of how GRF impulse must be transmitted through the connected body segments.

7. GRF patterns reveal swing mechanics: asymmetries, early extension, sway, and weight transfer all leave distinct signatures in the force plate data.

Looking Ahead

Ground reaction forces provide the impulse that changes the body's momentum, but the internal mechanism that generates motion is muscular torque at the joints. How do individual muscle forces combine to produce joint torques? Chapter 13 develops the *muscle Jacobian* that maps hundreds of muscle forces into the handful of joint torques that appear in our equations of motion.

Exercises 12.1. Chapter Exercises

1. A 95 kg golfer stands still on a force plate. Compute the vertical GRF. Assume $g = 9.81$ m/s^2.

2. During the downswing, the golfer's vertical GRF reaches 1350 N. Compute the golfer's vertical center-of-mass acceleration. Is it upward or downward?

3. Explain why the ground reaction force does zero work even though it is a large force. Give an analogy from everyday experience.

4. A golfer rotates his torso with angular acceleration $\alpha = 50$ rad/s^2 about his spine. The torso's moment of inertia about the spine is $I = 3$ kg·m^2. What is the torque applied by the muscles? If this torque causes the center of mass to accelerate forward by 1 m/s^2, what is the change in anterior-posterior GRF for an 85 kg golfer?

5. Sketch the expected GRF pattern for a correct swing: vertical, anterior-posterior, and medial-lateral components vs. time during the downswing and follow-through. Label the key phases: start of downswing, maximum compression, impact, and finish.

6. Explain the impulse-momentum perspective and the work-energy perspective for GRF. Why do they seem contradictory, and how is the contradiction resolved?

7. A golfer performing a reverse weight shift (moving weight backward instead of forward during the downswing) would likely show what pattern in the anterior-posterior GRF?

8. Use the equation $\boldsymbol{F}_{\mathrm{GRF}} = m\ddot{\boldsymbol{r}}_{\mathrm{COM}} - m\boldsymbol{g}$ to explain why GRF must be less than body weight if the center of mass is accelerating downward (e.g., during a deep squat descent).

Chapter 13

From Muscle Forces to Joint Torques

> The body does not think in torques; it thinks in muscles. But the mathematics demands torques. This chapter bridges the gap.
>
> —A Biomechanist's Lament

In Plain Language 13.1. The Bridge Between Anatomy and Physics

Golfers think in muscles: "activate the glutes," "engage the core," "flex the forearms." But physicists and engineers think in torques: rotations about joint axes. How does a muscle's pull translate into a joint's rotation? The answer lies in moment arms, leverage, and the geometry of the body. This chapter builds the mathematical bridge from muscle physiology to joint mechanics.

13.1 The Problem: Muscles Pull, Joints Rotate

Muscles generate linear forces. A muscle is a collection of fibers that contract along a single line of action, pulling the two bones it connects with a force magnitude F^M. This is a linear force, measured in Newtons.

But the skeleton is a system of rigid links connected by joints. Joints rotate; they have torques (moments), measured in Newton-meters. A torque about a joint axis is $\tau = r \times F$, where r is the moment arm (the perpendicular distance from the line of action to the axis).

The question is fundamental: **How does a muscle's linear force produce a joint's rotational torque?**

The answer is geometry. The muscle does not attach directly to the joint axis; it attaches at some distance from the axis. This distance is the moment arm.

13.2 Moment Arms and the Geometry of Force

13.2.1 The Basic Definition

Definition 13.1. Moment Arm

The moment arm r is the perpendicular distance from the line of action of a force to the axis of rotation. The torque generated is

$$\tau = F \cdot r,$$

where F is the magnitude of the force and r is the moment arm.

For a force applied at a point, the vector torque is:

$$\boldsymbol{\tau} = \boldsymbol{r} \times \boldsymbol{F},$$

where $\boldsymbol{r}$ is the position vector from the axis to the point of application. The magnitude is:

$$|\boldsymbol{\tau}| = |\boldsymbol{r}| \cdot |\boldsymbol{F}| \cdot \sin(\alpha),$$

where α is the angle between $\boldsymbol{r}$ and $\boldsymbol{F}$. The moment arm is the component of $\boldsymbol{r}$ perpendicular to $\boldsymbol{F}$:

$$r = |\boldsymbol{r}| \sin(\alpha).$$

13.2.2 Example: The Biceps at the Elbow

Consider the biceps muscle pulling on the forearm at the elbow. The biceps has two heads and attaches to the radius (one of the forearm bones). The moment arm (perpendicular distance from the line of action to the joint axis) varies with elbow angle. Delp et al. [1990, 2007]

At full arm extension, the moment arm is near maximum. When the elbow is fully flexed, the moment arm decreases because the muscle becomes more parallel to the forearm.

For a typical biceps with maximum isometric force $F_0 \approx 350$ N and moment arm $r \approx 0.05$ m at near-optimal length, the maximum isometric torque would be:

$$\tau_{\max}^{\text{iso}} = 350\,\text{N} \times 0.05\,\text{m} = 17.5\,\text{N} \cdot \text{m}.$$

The actual torque during dynamic motion depends on muscle activation, length, and velocity—factors captured by Hill's muscle model (Chapter 14).

13.2.3 Moment Arms Are Configuration-Dependent

Here is the crucial complication: the moment arm changes as the joint moves. This is a fundamental geometric effect documented extensively in anatomical and biomechanical literature.

The moment arm of the biceps varies throughout the range of elbow motion. At different elbow angles, the perpendicular distance from the biceps' line of action to the elbow joint axis changes. Delp et al. [1990, 2007]

This means: **the same muscle force produces different torques at different joint configurations**.

In Plain Language 13.2. In Plain Language

Imagine using a wrench to turn a bolt. When the wrench is perpendicular to the bolt (90-degree angle), your grip strength matters most. As you rotate the bolt and the wrench tilts, the same grip strength produces less rotational effect. The effective "lever arm" of your hand has decreased because the angle has changed.

13.2.4 Moment Arm as a Function of Joint Angle

For the i-th joint with angle q_i, the moment arm $r_i(\boldsymbol{q})$ is a function of the entire configuration $\boldsymbol{q}$ (because changing one joint angle can affect the geometry of nearby joints).

In biomechanics, moment arms are typically computed from anatomical imaging (MRI, CT scans) or from musculoskeletal models that capture the attachment points and line of action of each muscle.

A common parameterization is:

$$r_i(\boldsymbol{q}) = r_{i,0} + r_{i,1}q_i + r_{i,2}q_i^2 + \dots,$$

where $r_{i,0}, r_{i,1}, r_{i,2}, \dots$ are empirically determined coefficients. This captures the nonlinear dependence of moment arm on joint angle.

13.3 The Muscle Jacobian: From Muscle Space to Joint Space

Now consider a whole limb with multiple joints and multiple muscles acting across them. The relationship between muscle forces and joint torques can be expressed as a linear map.

13.3.1 The Muscle Jacobian Matrix

Define:

- $\boldsymbol{F}^M \in \mathbb{R}^{n_m}$: vector of muscle forces, where n_m is the number of muscles.
- $\boldsymbol{\tau} \in \mathbb{R}^{n_j}$: vector of joint torques, where n_j is the number of joints.
- $\boldsymbol{R}(\boldsymbol{q}) \in \mathbb{R}^{n_m \times n_j}$: the muscle-length Jacobian, whose entry $R_{ji}(\boldsymbol{q}) = \partial \ell_j / \partial q_i$ is the signed moment arm of muscle j about joint i.

The relationship between muscle forces and joint torques is:

$$\boldsymbol{\tau} = \boldsymbol{R}(\boldsymbol{q})^T \boldsymbol{F}^M.$$

Note the transpose: $\boldsymbol{R}^T$, not $\boldsymbol{R}$. The muscle-length Jacobian $\boldsymbol{R}$ maps joint velocities to muscle lengthening velocities ($\dot{\boldsymbol{\ell}} = \boldsymbol{R}\dot{\boldsymbol{q}}$), so by the principle of virtual work, forces map in the opposite direction through the transpose. This is analogous to the Jacobian transpose mapping in robotics: if $\boldsymbol{J}$ maps joint velocities to end-effector velocities, then $\boldsymbol{J}^T$ maps end-effector forces to joint torques.

Key Principle 13.1. The Muscle Jacobian

The muscle Jacobian $\boldsymbol{R}(\boldsymbol{q})^T$ maps muscle forces to joint torques. It is the transpose of the muscle-length Jacobian. Each entry of $\boldsymbol{R}(\boldsymbol{q})$ encodes how much one joint coordinate changes one muscle-tendon length; each entry of $\boldsymbol{R}(\boldsymbol{q})^T$ gives the corresponding signed torque contribution by virtual work.

Muscle Jacobian: From Forces to Torques

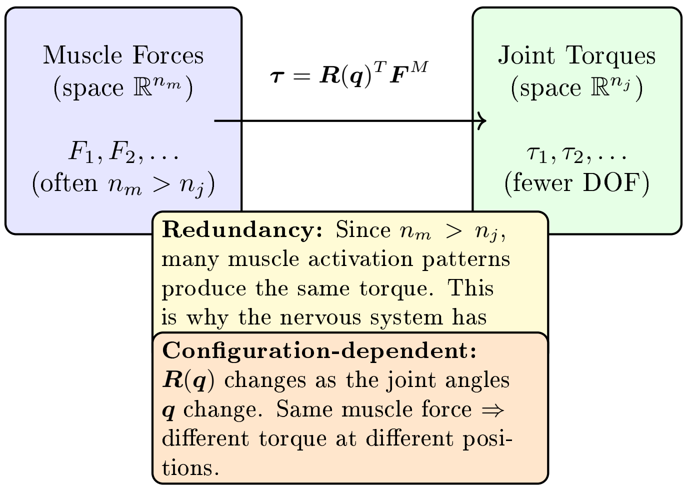


Figure 13.1: The Muscle Jacobian Maps Muscle Force Space to Joint Torque Space. Because Muscles Are Redundant ($n_m > n_j$), the Mapping Is Many-to-One. The Nervous System Selects Which Muscle Combination to Use. The Jacobian Is Configuration-Dependent: As Joint Angles Change, the Mapping Changes.

This is directly analogous to the Jacobian in robotics and dynamics. For a robot with a linear actuator pulling on a rigid link, $\boldsymbol{J}^T \boldsymbol{F} = \boldsymbol{\tau}$.

13.3.2 Numerical Example: A Two-Joint Arm

Example 13.1. Moment Arms and Joint Torques in a Two-Joint Arm

Consider a simplified arm with two joints (shoulder and elbow) and two muscles (an extensors muscle acting at both joints and a flexor muscle acting at the elbow). Suppose the moment arm matrix is:

$$\boldsymbol{R}(\boldsymbol{q}) = \begin{pmatrix} 0.05 & 0.02 \\ 0 & 0.04 \end{pmatrix} \text{ m}.$$

This means: - Muscle 1 has moment arm 0.05 m at the shoulder and 0 m at the elbow (it only acts at the shoulder). - Muscle 2 has moment arm 0.02 m at the shoulder

and 0.04 m at the elbow (biarticular—acts at both joints).
Suppose the muscle forces are:

$$\boldsymbol{F}^M = \begin{pmatrix} 500 \\ 300 \end{pmatrix} \text{ N}.$$

The joint torques are:

$$\boldsymbol{\tau} = \boldsymbol{R}^T \boldsymbol{F}^M = \begin{pmatrix} 0.05 & 0 \\ 0.02 & 0.04 \end{pmatrix} \begin{pmatrix} 500 \\ 300 \end{pmatrix}.$$

$$\tau_{\text{shoulder}} = 0.05 \times 500 + 0.02 \times 300 = 25 + 6 = 31\,\text{N} \cdot \text{m}. \tag{13.1}$$

$$\tau_{\text{elbow}} = 0 \times 500 + 0.04 \times 300 = 0 + 12 = 12\,\text{N} \cdot \text{m}. \tag{13.2}$$

Note that muscle 2 contributes to both joints: it generates 6 N·m at the shoulder and 12 N·m at the elbow.

13.3.3 Configuration-Dependent Jacobian

The muscle Jacobian $\boldsymbol{R}(\boldsymbol{q})^T$ is configuration-dependent. As the golfer swings and the joint angles change, the moment arm matrix evolves. This means the mapping from muscle forces to joint torques continuously changes.

This is a source of effective motor control: by keeping muscle forces constant while joint angles change, a golfer can modulate joint torques. Conversely, achieving a desired joint torque sequence requires modulating muscle forces as the configuration changes.

13.4 Redundancy: More Muscles Than Joints

The human body has far more muscles than joints. For example: Neumann [2017]

- Total skeletal muscles: approximately 600–700 (exact count varies by anatomical definition of distinct muscle units).
- Functional joints (with active degrees of freedom): approximately 150–200.
- Motor degrees of freedom for controlled motion: variable, depending on task and voluntary control.

For a local region, say the arm and hand, the disparity is even more dramatic:

- Muscles crossing the arm and hand: approximately 50 muscles (extrinsic and intrinsic hand muscles) Neumann [2017].
- Functional DOF in arm: approximately 7 (shoulder 3 DOF + elbow 1 + wrist 2 + radioulnar 1); hand has additional complexity with multiple finger and thumb joints.

This means $n_m > n_j$: the torque map $\boldsymbol{R}^T$ is wide ($n_j \times n_m$ with $n_j < n_m$). The system is **redundant**: many different muscle force combinations can produce the same joint torque.

13.4.1 The Null Space of the Muscle Jacobian

For a given desired joint torque $\boldsymbol{\tau}^*$, any muscle force vector satisfying

$$\boldsymbol{R}(\boldsymbol{q})^T \boldsymbol{F}^M = \boldsymbol{\tau}^*$$

is a valid solution. If $\boldsymbol{F}_0^M$ is one solution, then so is any vector of the form

$$\boldsymbol{F}^M = \boldsymbol{F}_0^M + \mathbf{v},$$

where $\mathbf{v}$ is in the null space of $\boldsymbol{R}^T$:

$$\boldsymbol{R}^T \mathbf{v} = \mathbf{0}.$$

Vectors in the null space of $\boldsymbol{R}^T$ correspond to muscle force combinations that produce zero net torque. These are **co-contraction patterns**: muscles that pull against each other, increasing joint stiffness without producing net rotation.

Definition 13.2. Co-contraction

Co-contraction is the simultaneous activation of antagonist muscles (muscles that would produce opposite torques if activated alone). Co-contraction increases joint stiffness and stability at the expense of muscular energy.

13.4.2 Why Redundancy Matters for Golf

Redundancy is not a bug; it is a feature. It allows:

1. **Modulation of joint stiffness:** By varying co-contraction levels, a golfer can stiffen or soften a joint without changing the net torque. This is essential for impact: at ball contact, high stiffness is desirable to resist deformation and maintain club face alignment.

2. **Stability and perturbation rejection:** Redundancy allows the nervous system to activate muscles beyond what is strictly needed, creating a "stiffness buffer." If an unexpected perturbation (gust of wind, uneven ground) occurs, the excess activation can resist the perturbation.

3. **Metabolic efficiency:** The nervous system can choose muscle activation patterns that minimize metabolic cost, not just produce the desired torque. Different combinations of muscles have different metabolic efficiencies.

The central question in motor control is: **how does the nervous system resolve the redundancy? How does it choose which muscles to activate?**

13.4.3 In the Control-Affine Framework

In the control-affine model $\dot{\boldsymbol{x}} = f(\boldsymbol{x}) + G(\boldsymbol{x})\boldsymbol{u}$, the control input $\boldsymbol{u}$ represents the **net equivalent joint torque**. This is a single generalized input per joint or actuated coordinate, subsuming the redundancy of the underlying muscles.

The mapping from muscle forces to net joint torque involves the muscle Jacobian, but once we work at the level of joint torques, the redundancy is no longer explicitly visible. We have traded detail (individual muscles) for simplicity (equivalent torques).

In Plain Language 13.3. In Plain Language

Think of muscles as many different roads to the same destination. The nervous system needs to choose which roads to take (which muscles to activate). The joint torque is the destination itself (the torque achieved). In the control-affine framework, we focus on the destination, not the roads. The underlying muscle redundancy is hidden "under the hood."

13.5 Why We Model With Joint Torques, Not Muscle Forces

The equations of motion for the golfer are written in terms of joint angles $\boldsymbol{q}$ and joint torques $\boldsymbol{\tau}$:

$$\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{\tau}.$$

Why not write them in terms of muscle forces $\boldsymbol{F}^M$?

$$\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{R}(\boldsymbol{q})^T \boldsymbol{F}^M. \tag{13.3}$$

We could. But there are several reasons why joint torques are preferred:

13.5.1 Sufficiency

The dynamics depend only on the net torque $\boldsymbol{\tau}$ at each joint, not on how that torque is achieved. The mass matrix $\boldsymbol{M}(\boldsymbol{q})$ and the Coriolis/gravity terms depend only on the configuration and motion, not on the muscles.

This means: the equations of motion "see" only the aggregate effect of all muscles. The neural and muscular details are projections onto a lower-dimensional space (joint torques).

This is a powerful simplification. It allows us to study the mechanics of the swing without needing to model 630 muscles individually.

13.5.2 Redundancy Avoidance

If we work with muscle forces, we have an underdetermined system. The same motion can be achieved with infinitely many muscle activation patterns (due to the null space). We would need an additional criterion (optimization: minimize energy, fatigue, etc.) to pick a unique solution.

By working with joint torques, we eliminate the redundancy at the source. We work in a space where dynamics are determined.

13.5.3 Experimental Accessibility

We can measure joint angles (with motion capture) and joint torques (via inverse dynamics, computed from kinematics and force plates). We cannot directly measure individual muscle forces (without invasive devices like dynamometers or electromyography).

So joint torques are experimentally accessible; individual muscle forces often are not.

13.5.4 Connection to Control

In the control-affine framework $\dot{\boldsymbol{x}} = f(\boldsymbol{x}) + G(\boldsymbol{x})\boldsymbol{u}$, the control input $\boldsymbol{u}$ is interpreted as the net equivalent joint torques. This is the natural choice because:

- Joint angles $\boldsymbol{q}$ are part of the state $\boldsymbol{x}$.
- Joint torques are conjugate to joint angles (they are the "forces" that change $\boldsymbol{q}$).
- The control input should be conjugate to a component of the state.

Key Principle 13.2. Why Joint Torques Suffice

The equations of motion depend only on the net joint torque, not on how that torque is generated by muscles. Therefore:

1. The dynamics are sufficient to determine motion from joint torques alone.
2. Working with joint torques avoids the redundancy inherent in muscle space.
3. Joint torques are experimentally measurable via inverse dynamics.
4. In control-affine models, joint torques are the natural choice for control inputs.

For these reasons, biomechanical models universally use joint torques.

13.5.5 What Is Lost?

Working at the joint torque level involves a projection that loses information:

- Individual muscle activation patterns.
- Joint stiffness from co-contraction (though some models include this as a separate stiffness term).
- Metabolic cost of different muscle activation strategies.
- Fatigue accumulation in specific muscles.

If you need to study these details, you must work at the muscle level. But for understanding the mechanics of the swing and how joint torques produce motion, joint torques are sufficient.

13.6 Biarticular Muscles: Coupling Between Joints

Some muscles cross two or more joints. In the arm, for example:

- Biceps: crosses the shoulder and elbow.
- Triceps: crosses the shoulder and elbow.
- Wrist flexors: cross the elbow, wrist, and finger joints.

These **biarticular muscles** create coupling between joints: activating a biarticular muscle produces torques at multiple joints simultaneously.

13.6.1 The Moment Arm Matrix for Biarticular Muscles

For a biarticular muscle connecting joints i and j, the muscle-length Jacobian $\boldsymbol{R}(\boldsymbol{q})$ has non-zero entries in the same muscle row and in both joint columns. For example, if muscle k is biarticular (acts at joints 2 and 3), then:

$$R_{k,2}(\boldsymbol{q}) \neq 0 \quad \text{and} \quad R_{k,3}(\boldsymbol{q}) \neq 0.$$

This means that muscle k contributes to both τ_2 and τ_3:

$$\tau_2 = R_{k,2} F_k^M + \dots, \quad \tau_3 = R_{k,3} F_k^M + \dots$$

Definition 13.3. Biarticular Coupling

Biarticular muscles produce coupled torques at two or more joints. A single muscle activation produces correlated torques at those joints, without requiring independent control at each joint.

13.6.2 Energy Transfer via Biarticular Muscles

Biarticular muscles are efficient mechanisms for energy transfer. Suppose a proximal joint (e.g., shoulder) accelerates, storing kinetic energy in a moving segment. A biarticular muscle can then transfer some of this energy to the distal joint (e.g., elbow) without explicit distal control.

This is a key mechanism in the kinetic chain: the legs accelerate, the hips follow, but energy transfers from the hip to the torso via biarticular muscles; then from the torso to the shoulders via other biarticular muscles, and so on.

In Plain Language 13.4. In Plain Language

Imagine a chain of pulleys, where a rope passes through multiple pulleys. If you pull one end of the rope, the motion propagates through all the pulleys at once, without needing to adjust each pulley individually. Biarticular muscles are like this rope: they couple motion and energy across joints.

13.6.3 Example: The Wrist Flexors in Golf

The flexor carpi radialis and flexor carpi ulnaris are biarticular muscles. They originate at the medial epicondyle of the humerus (near the elbow) and insert on the wrist and hand.

When activated, they produce:

$$\tau_{\text{elbow}} = R_{\text{elbow}} F_{\text{flexor}}, \tag{13.4}$$

$$\tau_{\text{wrist}} = R_{\text{wrist}} F_{\text{flexor}}. \tag{13.5}$$

A golfer can rotate the forearm (elbow) and flex the wrist simultaneously by activating the wrist flexors, without separate control of the elbow and wrist. This is efficient for power generation in the swing.

13.7 The Inverse Problem: From Desired Torques to Muscle Forces

Suppose you want a certain joint torque profile $\boldsymbol{\tau}^*(t)$ to swing the club optimally. The inverse problem is: what muscle forces $\boldsymbol{F}^M(t)$ are needed?

13.7.1 The Underdetermined Inverse

From the muscle Jacobian relation:

$$\boldsymbol{\tau} = \boldsymbol{R}(\boldsymbol{q})^T \boldsymbol{F}^M,$$

we want to solve for $\boldsymbol{F}^M$ given $\boldsymbol{\tau}$ and $\boldsymbol{q}$.

But $\boldsymbol{R}^T$ is a tall, wide matrix ($n_j \times n_m$ with $n_j < n_m$). The equation $\boldsymbol{\tau} = \boldsymbol{R}^T \boldsymbol{F}^M$ is underdetermined: it has infinitely many solutions (or no solutions, if $\boldsymbol{\tau}$ is not in the column space of $\boldsymbol{R}^T$).

13.7.2 Optimization Approaches

To pick a unique solution, an additional criterion is needed. Common choices are:

1. Minimize total muscle force:

$$\min_{\boldsymbol{F}^M} \sum_i (F_i^M)^2 \quad \text{subject to} \quad \boldsymbol{R}^T \boldsymbol{F}^M = \boldsymbol{\tau}.$$

This minimizes the "effort" in an intuitive sense.

2. Minimize metabolic cost:

$$\min_{\boldsymbol{F}^M} \sum_i C_i(F_i^M) \quad \text{subject to} \quad \boldsymbol{R}^T \boldsymbol{F}^M = \boldsymbol{\tau},$$

where C_i is a nonlinear cost function that accounts for the metabolic energy of muscle i. Metabolic cost is not proportional to force; it depends on the muscle fiber type, contraction velocity, and other factors.

3. Minimize fatigue:

$$\min_{\boldsymbol{F}^M} \sum_i f_i(F_i^M, \text{prior activation}) \quad \text{subject to} \quad \boldsymbol{R}^T \boldsymbol{F}^M = \boldsymbol{\tau},$$

where f_i accounts for fatigue accumulation in muscle i over time.

Definition 13.4. Static Optimization

Static optimization is the process of computing muscle forces that minimize some criterion (typically energy or fatigue) while producing a desired set of joint torques. It is "static" in the sense that it solves the optimization problem at each time step independently, without considering the dynamics of muscle contraction.

13.7.3 Why This Matters for Golf (And Why It's Hard)

The nervous system solves the inverse problem unconsciously. It knows what torques are needed (from the motor plan) and automatically recruits muscles to achieve those torques, choosing activations that are efficient, stable, and adaptable.

Understanding this process is a major open problem in neuroscience and biomechanics. Different people, even with the same swing goal, may recruit muscles differently. Training and skill development may involve learning more efficient muscle activation patterns.

For the golfer, this suggests: there is no single "correct" muscle activation pattern for a swing. Different patterns of muscle activation can produce the same swing, with different metabolic costs and efficiencies. A skilled golfer (or coached golfer) learns activation patterns that are efficient and repeatable.

In Plain Language 13.5. In Plain Language

You want to hit a 7-iron to a target 150 yards away. Your brain knows the club head speed needed and the sequence of joint torques. But how should you activate your muscles? Should you activate the biceps and triceps equally (co-contraction) or unequally (antagonistic)? Should you pre-activate stabilizer muscles or activate them just in time? Your brain solves this optimization problem automatically, but the solution is not unique. Skilled golfers have learned to solve it efficiently; less skilled golfers may use more muscular effort to achieve the same result.

13.8 Practical Example: The Golf Grip

The grip is where the hand transfers control forces and torques to the club. It illustrates the interplay of muscle forces, moment arms, and joint torques.

13.8.1 Anatomy of the Grip

The grip involves approximately 8–10 muscles in each hand and forearm:

- Flexor carpi radialis: wrist flexor, radial deviation.
- Flexor carpi ulnaris: wrist flexor, ulnar deviation.
- Extensor carpi radialis: wrist extensor.
- Extensor digitorum: finger extensors.
- Flexor digitorum superficialis and profundus: finger flexors.
- Intrinsic hand muscles: fine control of finger position.

These muscles act on 3+ articulations: wrist (2 DOF: flexion/extension, radial/ulnar deviation) and fingers (multiple joints).

13.8.2 Grip Force vs. Grip Torque

The grip serves two functions:

1. **Grip force:** The normal force pressing the hand against the club grip. This is a constraint force; it maintains contact between hand and club.

2. **Grip torque:** The torques applied by the hands to rotate or stabilize the club.

The grip force is distributed around the circumference of the grip. If you measure the grip force profile (force vs. position around the grip), you can infer information about which muscles are active and how they are coordinated.

The grip torque is more subtle. Movements of the wrist and fingers create torques on the club: - Wrist flexion/extension rotates the club about the shaft axis (roll). - Radial/ulnar wrist deviation tilts the club face. - Finger flexion can modulate the grip force or produce subtle torques.

13.8.3 Moment Arm Changes With Hand Configuration

As the grip changes (different grip styles: strong, weak, neutral), the geometry of the hand-club interface changes. This alters the moment arms of the muscles.

For example, in a strong grip (club positioned more closed), the flexor carpi ulnaris has a larger moment arm about the wrist, allowing more torque for a given force. In a weak grip (club positioned more open), the flexor carpi radialis is more effective.

Different grip styles effectively change the local muscle-length Jacobian $\boldsymbol{R}(\boldsymbol{q})$, making different muscle groups more or less effective for producing specific wrist torques.

13.8.4 Grip Force Modulation During the Swing

Force plate and EMG (electromyography) studies reveal that grip force changes dramatically during the swing [Hume et al., 2005]:

- Early in the backswing: grip force is relatively low (20–30% of maximum).

- At the top of the backswing: grip force increases to resist gravity and prepare for the downswing (40–50% of maximum).

- During the downswing: grip force increases further (60–80% of maximum).

- At impact: grip force peaks at 80–100% of maximum, providing stiffness to resist deformation and maintain club face alignment.

- During the follow-through: grip force decreases as the club decelerates.

This modulation is achieved through coordinated muscle activation. Different phases require different activation patterns, computed by the nervous system via the inverse problem.

Constraint Insight 13.1. Grip Stiffness and Impact

At impact, the club experiences a large force from the ball (roughly 5000–8000 N for a driver impact, depending on swing speed) [Penner, 2003b, Cochran and Stobbs,

1968]. This force tries to deform the shaft and move the club head. To maintain club face alignment and transmit energy efficiently, the grip must be stiff.
Grip stiffness comes from co-contraction: simultaneous activation of muscles that would produce opposite torques. This stiffens the wrist without changing the net torque. The muscle Jacobian accounts for this implicitly: the same joint torque can be achieved with different co-contraction levels.

Key Principle 13.3. Key Takeaways

1. Muscles generate linear forces along their line of action. Joints rotate about axes. The moment arm is the geometric bridge: moment arm = perpendicular distance from line of action to axis. Torque = force × moment arm.
2. The muscle-length Jacobian $\boldsymbol{R}(\boldsymbol{q})$ maps joint velocity to muscle lengthening velocity. The torque map is $\boldsymbol{R}^T$, and the relation is $\boldsymbol{\tau} = \boldsymbol{R}^T \boldsymbol{F}^M$.
3. Moment arms are configuration-dependent. As joints move, the moment arms change, altering the mapping from muscle forces to joint torques.
4. The human body has far more muscles than joints ($n_m > n_j$). This redundancy means infinitely many muscle activation patterns can produce the same joint torques. The nervous system must resolve this redundancy.
5. Redundancy is not a problem; it is a feature. It allows modulation of joint stiffness (co-contraction), stability against perturbations, and optimization of metabolic efficiency.
6. We model dynamics using joint torques, not muscle forces, because:
   (a) The equations of motion depend only on net joint torques.
   (b) This avoids the underdetermined muscle redundancy.
   (c) Joint torques are experimentally measurable.
   (d) Joint torques are the natural control inputs in control-affine models.
7. Biarticular muscles couple torques at two joints. They enable energy transfer between proximal and distal joints without explicit distal control, a key mechanism in the kinetic chain.
8. The inverse problem—computing muscle forces from desired joint torques—is underdetermined. The nervous system solves it via optimization, minimizing some criterion (energy, fatigue, etc.).
9. Skilled golfers learn efficient muscle activation patterns. Training involves learning to solve the inverse problem more efficiently, reducing metabolic cost while maintaining performance.
10. At the grip, the moment arm matrix changes with hand configuration (grip style). Different grip styles make different muscles more effective for producing wrist control.

Exercises 13.1. Chapter Exercises

1. Define the moment arm. Why does it depend on joint configuration? Give a practical example from the golf swing.

2. For a biceps with a maximum force $F_{\max} = 600$ N, compute the maximum torque at the elbow when:

   (a) The moment arm is 0.05 m (near full extension).

   (b) The moment arm is 0.03 m (partially flexed).

   Which configuration is more mechanically advantageous? Why?

3. A simple forearm has 2 muscles acting on the elbow and wrist. Suppose the moment arm matrix is:

$$\boldsymbol{R}(\boldsymbol{q}) = \begin{pmatrix} 0.04 & 0.01 \\ 0.01 & 0.03 \end{pmatrix}.$$

   If the muscle forces are $\boldsymbol{F}^M = (400, 350)^T$ N, compute the joint torques $\boldsymbol{\tau}$.

4. Explain why the human body has far more muscles than joints. Is this inefficient? What advantages does it provide?

5. Define co-contraction. When would a golfer intentionally co-contract muscles during a swing, and why?

6. A golfer wants to produce a wrist torque of 50 N·m. The available muscles are two synergists (both flexors) with no antagonists present. Are there multiple solutions for the muscle forces? Why or why not?

7. Describe the inverse problem in biomechanics. Why is it underdetermined? What criterion might the nervous system use to select a unique solution?

8. How do biarticular muscles enable energy transfer in the kinetic chain? Give a specific example from the golf swing.

9. Grip force changes dramatically during the swing (low at the start, high at impact). Why? What muscle activation pattern produces this change?

10. A golfer changes from a strong grip to a weak grip. How does this alter the moment arm matrix at the wrist? Which muscles become more effective for producing wrist torques?

11. In the control-affine framework $\dot{\boldsymbol{x}} = f(\boldsymbol{x}) + G(\boldsymbol{x})\boldsymbol{u}$, the control input $\boldsymbol{u}$ is the net equivalent joint torques. Why is this the natural choice, rather than using individual muscle forces as inputs?

Chapter 14

Muscle Force Generation: The Biological Engine

In Plain Language 14.1. Why Muscle Biology Matters for Your Swing

The nervous system prepares muscle activity; muscles pull on tendons; tendon forces act through changing moment arms; the coupled body and club respond. Each link matters. A fast clubhead does not tell us how fast a particular muscle fiber is shortening. A small net joint torque does not tell us that the surrounding muscles are relaxed. Understanding these distinctions lets us ask a useful golf question: which changes in preparation or ongoing activation can still change the impact conditions, by how much, and with what consequences elsewhere in the body?

This chapter develops a reproducible teaching model before discussing measurements and training. Its numerical examples are manufactured calculations, not estimates of a particular golfer. The purpose is to connect biological limitations to mechanics without turning a plausible mechanism into an unsupported claim that every player must use one sequence. The model omits fatigue, injury, detailed neural circuitry and many tissue effects; those omissions define the questions it can answer.

14.1 Muscle Architecture: What a Muscle Actually Is

Definition 14.1. The Muscle Hierarchy

A skeletal muscle contains fascicles, which contain muscle fibers. Each fiber is a cell containing myofibrils with sarcomeres arranged along their length. Actin and myosin interact within these repeating units. Sarcomeres in series contribute length and shortening excursion; contractile structures in parallel contribute force. A muscle can produce active tension while shortening, holding approximately constant length or being lengthened by its load. The word contraction does not require visible shortening.

An electrical signal initiates processes that raise calcium availability within the fiber. Cross-bridge cycling couples chemical energy to force and motion. ATP binding allows myosin to detach from actin, hydrolysis prepares another cycle, and calcium handling also consumes energy. This mechanism is different from stretching a passive rubber band. The band can return energy previously stored in it; an activated muscle can convert chemical energy into mechanical work. A loaded muscle also contains passive structures, so measured total tension need not disappear when neural excitation stops. [OpenStax, 2022a]

Key Principle 14.1. Muscle Fiber Types and Their Roles

Fiber classifications summarize differences in contractile speed and metabolic properties. Slow oxidative fibers tend to resist fatigue; fast oxidative and fast glycolytic categories describe different combinations of speed and energy supply. These categories do not give a universal percentage composition for every muscle or golfer. Nor does a fast clubhead establish the player's fiber composition. Fiber size, recruitment, architecture, training history and task coordination are separate quantities. [OpenStax, 2022b]

Definition 14.2. Muscle Architectural Parameters

Let ℓ^M be fiber length in meters, ℓ^M_{opt} its reference optimal length, and α the angle between fibers and the tendon line. For approximately parallel, equal-length fibers at the reference configuration, a volume-based physiological cross-sectional area is

$$A_{\text{PCSA}} = \frac{V_M}{\ell^M_{\text{opt}}} = \frac{m_M}{\rho_M \ell^M_{\text{opt}}}, \qquad F_0 = \sigma_0 A_{\text{PCSA}}.$$

Here m_M is muscle mass, ρ_M density, and σ_0 specific tension in compatible units. We define F_0 along the fiber direction. Its tendon-direction projection at the reference pennation is $F_0 \cos\alpha_0$. Some sources incorporate this projection in their area or maximum-force convention; applying it again would double-count pennation. These reference parameters are not recalculated by substituting the instantaneous fiber length into the area formula.

For an explicitly hypothetical area of $3\,\text{cm}^2$ and specific tension of $30\,\text{N/cm}^2$, $F_0 = 90\,\text{N}$. At $\alpha_0 = 20^\circ$ its projection is about $84.6\,\text{N}$. This arithmetic is transparent; interpreting either input as a measured value for a named golf muscle would require anatomical evidence. Increasing fiber length at fixed muscle volume trades parallel area for excursion capacity in this simplified architecture. Increasing pennation changes both packing and projection, so the cosine alone does not establish whether one architecture is stronger.

The source of anatomical parameters matters. Holzbaur and colleagues' upper-extremity model includes shoulder-to-finger geometry and comparisons with moment arms and isometric moment capacity. Its tendon slack lengths were selected using additional constraints, and the authors identify sensitivity and shoulder-geometry limitations. It is a starting model, not validation of muscle force during a golf swing. [Holzbaur et al., 2005] Rajagopal and colleagues' gait model instead uses muscle-tendon actuators in the lower limbs and ideal torque actuators for the upper body. It cannot supply evidence for detailed forearm muscle parameters merely because its title says full-body. [Rajagopal et al., 2016]

14.2 The Force-Length Relationship

In Plain Language 14.2. Strength Changes With Joint Angle

Changing joint angle changes muscle paths, fiber lengths and moment arms. Active force depends in part on filament interaction, while passive tension increases as passive structures are loaded. Joint strength therefore need not peak at the angle where one muscle's active fiber force peaks. Antagonists, other muscles crossing the joint and the test apparatus also contribute to the measured moment.

Normalize fiber length as $\lambda = \ell^M/\ell^M_{\text{opt}}$. A convenient illustrative active curve is

$$f_L(\lambda) = \exp\left[-\left(\frac{\lambda-1}{\gamma}\right)^2\right], \qquad \gamma = 0.5. \tag{14.1}$$

This Gaussian has a maximum of one at $\lambda = 1$. It has neither an exactly flat plateau nor a finite length at which it becomes exactly zero. At $\lambda = 0.8$ and 1.2, it gives $e^{-0.16} \approx 0.8521$; at 0.5 and 1.5, it still gives $e^{-1} \approx 0.3679$. Thus it should not be extrapolated as a literal prediction of filament overlap at extreme lengths. The parameter γ is a declared teaching choice, not a universal tissue measurement.

Key Principle 14.2. Regions of the Force-Length Curve

Distinguish the active-force curve from total force. Activation can scale active force while passive structures contribute separately. A low active-force multiplier at short length is not the same mechanism as low neural recruitment. At longer lengths, decreasing active force can coexist with increasing passive tension. A model should state both contributions and the lengths over which its fitted curves are supported.

For a nonnegative passive teaching curve, choose $k_P = 4$ and $\epsilon_P = 0.6$:

$$f_P(\lambda) = \begin{cases} 0, & \lambda \le 1, \\ \dfrac{e^{k_P(\lambda-1)}-1}{e^{k_P\epsilon_P}-1}, & \lambda > 1. \end{cases} \tag{14.2}$$

The threshold, shape and normalization are explicit: $f_P(1) = 0$ and $f_P(1.6) = 1$. At 1.2 it is about 0.12227. The expression never generates passive compression below its slack threshold. Its right derivative at the threshold differs from the left derivative; a differentiable simulation model would require an appropriate transition if a solver or sensitivity calculation needs one.

With dimensionless activation $a \in [0,1]$ and the velocity multiplier defined next, total fiber-direction force is

$$F_M = F_0\{a f_L(\lambda) f_V(w) + f_P(\lambda)\}. \tag{14.3}$$

This expression deliberately omits viscous damping and history effects. It distinguishes the active component from passive tension but does not resolve every structure inside a real muscle.

Example 14.1. Latissimus Dorsi Through the Swing

A statement that the latissimus is near its optimal length at a particular swing phase requires more than an image of the pose. The shoulder girdle configuration, muscle path, tendon extension, pennation and reference fiber length all affect the inference. Suppose a model gives $\lambda = 1.2$ and $a = 0.7$ at zero fiber velocity. Our chosen curves give

$$F_M/F_0 \approx 0.7(0.8521) + 0.12227 = 0.7188.$$

That is a calculation conditional on model inputs; it does not establish either input for a golfer. Testing the claim requires checking the modeled geometry and measured motion, and reporting sensitivity to uncertain parameters.

14.3 The Force-Velocity Relationship: Hill's Equation

In Plain Language 14.3. The Central Constraint: Force Depends on Fiber Speed

A muscle fiber shortening rapidly generally has less active force capacity than the same fiber shortening slowly under comparable conditions. This statement concerns local fiber velocity. A muscle can move quickly through the room while its fiber length barely changes, and a moving tendon can stretch while the fibers shorten. Clubhead speed alone identifies neither motion.

Choose the shortening-positive convention

$$w = -\frac{\dot{\ell}^M}{V_{\max}}, \qquad V_{\max} = \nu \ell^M_{\text{opt}},$$

where $\dot{\ell}^M$ is positive for lengthening, $V_{\max}$ is in m/s, and ν is in fiber lengths per second. For example, 2–4 lengths/second for a 0.10 m reference fiber means 0.2–0.4 m/s. Confusing these units produces order-of-magnitude errors.

A normalized concentric Hill-type relation is

$$f_V(w) = \frac{1 - w}{1 + w/k}, \qquad 0 \le w \le 1, \quad k = 0.4. \tag{14.4}$$

It is hyperbolic, not linear. It equals one at zero shortening velocity and zero at the declared unloaded shortening limit. Extrapolating it above $w = 1$ would produce negative active force; that extrapolation is outside this teaching law's domain. A simulation encountering that condition needs a stated treatment and a check that the assumed force equilibrium is physically and numerically admissible.

Key Principle 14.3. Concentric vs. Eccentric: The Asymmetry

For lengthening, use the following explicitly constructed illustration:

$$f_V(w) = 1 + (F_\infty - 1)\frac{-w}{c - w}, \qquad w \le 0,$$

with $F_\infty = 1.4$. Choose $c = 4/35$ so that $c(1 + 1/k) = F_\infty - 1$. It approaches 1.4 as $w \to -\infty$, equals one at zero and has the same derivative there as the concentric branch. These choices make the joined velocity curve continuously differentiable at zero. They are not the published Thelen equation and do not establish a universal eccentric-force ceiling. Particular muscle models use different fitted curves, constraints and numerical regularizations. [Millard et al., 2013]

Active shortening power is $P_{\mathrm{CE}} = -F_{\mathrm{CE}}\dot{\ell}^M$. At full activation and optimal length, normalized power is

$$P_{\mathrm{CE}}/(F_0 V_{\max}) = w f_V(w).$$

Differentiating gives the interior maximum

$$w_* = \sqrt{k^2 + k} - k \approx 0.34833, \qquad \frac{P_*}{F_0 V_{\max}} \approx 0.12133.$$

Maximum force and maximum shortening power occur at different speeds. Isometric force can be substantial while fiber shortening power is zero. During active lengthening, this sign convention gives negative mechanical power: the contractile component absorbs mechanical energy while still requiring metabolic processes.

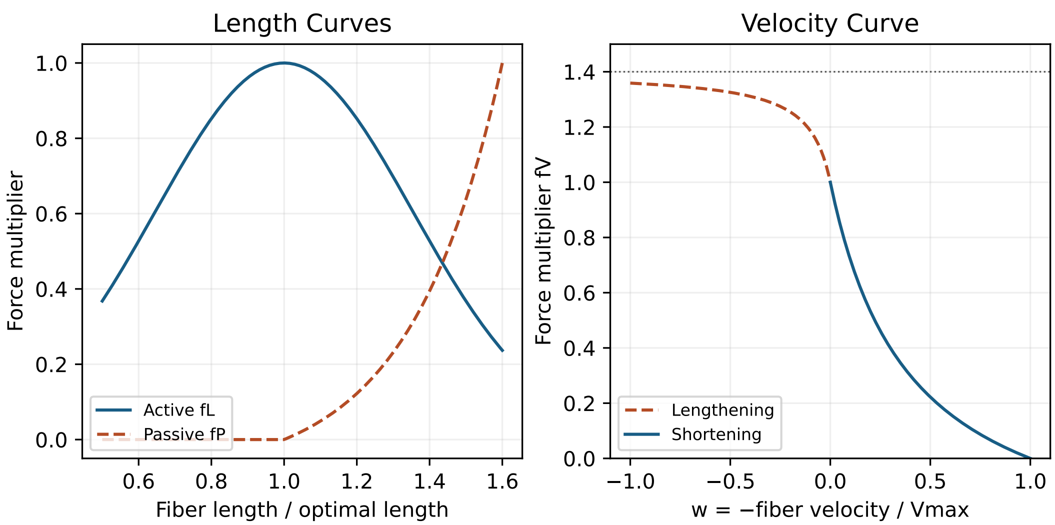


Figure 14.1: Declared Teaching Curves: Active and Passive Length Multipliers, and the Joined Velocity Law. The Parameters Are Illustrative, Not Golfer Measurements.

Key Principle 14.4. What High Club Speed Does and Does Not Establish

High club speed changes the mechanical problem, but it does not insert that speed directly into f_V. First obtain each muscle-tendon path velocity from joint motion, then separate tendon and fiber motion. Likewise, mv^2 has units of energy, not force; a centripetal force estimate needs a radius, mv^2/R, and a specified trajectory. Gravitational force near Earth's surface does not grow with swing speed. Configuration-dependent inertial and velocity-coupling effects require the equations of motion, not a comparison of unlike speed scales.

Example 14.2. Numerical Estimate: Local Wrist Muscle Force

Consider a hypothetical unpennated muscle with a rigid tendon, constant moment arm $r = 0.02\,\mathrm{m}$, reference fiber length $0.10\,\mathrm{m}$ and $\nu = 10\,\mathrm{s}^{-1}$. If the relevant joint alone rotates at $8.75\,\mathrm{rad/s}$ in the shortening direction, $-\dot{\ell}^M = r\dot{q}$, giving $0.175\,\mathrm{m/s}$ and $V_{\max} = 1\,\mathrm{m/s}$. Therefore $w = 0.175$ and

$$f_V = 0.825/1.4375 \approx 0.5739.$$

With $F_0 = 100\,\mathrm{N}$, $a = 1$, $\lambda = 1$ and no passive force, the tendon force is $57.39\,\mathrm{N}$ and moment $1.148\,\mathrm{N\,m}$. These inputs do not describe measured extensor carpi radialis brevis (ECRB) force at impact. Dividing clubhead speed by a guessed lever length cannot replace the required joint and path kinematics.

14.4 Muscle Activation Dynamics

In Plain Language 14.4. Why Muscles Are Slow to Respond

Neural excitation u is an input to the muscle model. Activation a describes its evolving active state; it is not identical to the command or to measured electromyography (EMG) amplitude. Tendon force follows from activation together with length, velocity and elastic loading. Consequently, a command change, an EMG onset and an external force onset are different events.

For an elementary model, choose a fixed time constant $\tau_a > 0$:

$$\dot{a} = \frac{u - a}{\tau_a}, \qquad 0 \le u, a \le 1. \tag{14.5}$$

For a constant command starting at $t = 0$, direct integration gives

$$a(t) = u + (a_0 - u)e^{-t/\tau_a}. \tag{14.6}$$

The same equation describes decay when $u < a_0$. This fixed-constant model is intentionally simpler than activation-dependent rise and fall laws. Published implementations differ; the chosen law and any minimum activation must be specified before comparing simulations. [OpenSim, 2020]

For $a_0 = 0$, $u = 1$ and $\tau_a = 25\,\mathrm{ms}$, activation reaches 63.2% after one time constant, 50% after 17.33 ms and 95% after 74.89 ms. At 50 ms it is about 0.8647. Exact equality to one occurs only asymptotically. A time constant therefore is not a period of complete inactivity followed by full force. Nor does 95% activation imply 95% of isometric joint torque during a moving, compliant task.

A numerical solver adds another issue. Forward Euler with a 5 ms step gives $a_n = 1 - 0.8^n$, reaching 95% at step 14, or 70 ms. The exact continuous solution crosses that threshold at 74.89 ms. Refining the time step or using a suitable integrator reduces this numerical discrepancy. It is not evidence for a faster physiological response.

Definition 14.3. Electromechanical Delay

An experimentally reported delay depends on the selected start and end events, detection thresholds, contraction protocol and loading. Nordez and colleagues used ultrasound during electrically evoked gastrocnemius contractions to distinguish aspects of fascicle and tendon onset. That experiment does not supply a universal visual-reaction delay or golf-wrist latency. [Nordez et al., 2009] Adding an EMG-to-force delay to an activation time constant can double-count processes if both quantities already describe overlapping parts of the response. A pure delay may be added to a model, but its location and measurement basis must be explicit.

Example 14.3. Reaction Time in the Golf Downswing

Suppose a command changes 70 ms before impact and a separately specified pure transmission delay is 20 ms. Our activation model then has 50 ms to respond. Its activation increment from zero is 0.8647 at impact. If the same command had already established activation earlier, the starting state would be different. Conversely, a sensory correction requiring additional detection and processing could leave less time. This timing ledger is a conditional example, not a measured sequence of events in every downswing. Preparation and a new response must be compared from their actual initial states.

14.5 The Hill-Type Muscle Model

Definition 14.4. Hill-Type Muscle Model Components

The contractile element (CE) and passive parallel element (PEE) act along the fiber direction. Their combined force is projected through pennation onto the series tendon, also called the series elastic element (SEE). With a massless connection and no extra damper in this teaching model,

$$F_T = (F_{\mathrm{CE}} + F_{\mathrm{PE}}) \cos\alpha = F_M \cos\alpha. \tag{14.7}$$

Equality of CE and tendon force alone would neglect loaded passive fibers and pennation. A generic Hill-type model is a family of approximations, not a unique set of equations. Equilibrium, damping and rigid-tendon variants can behave differently numerically. [Millard et al., 2013]

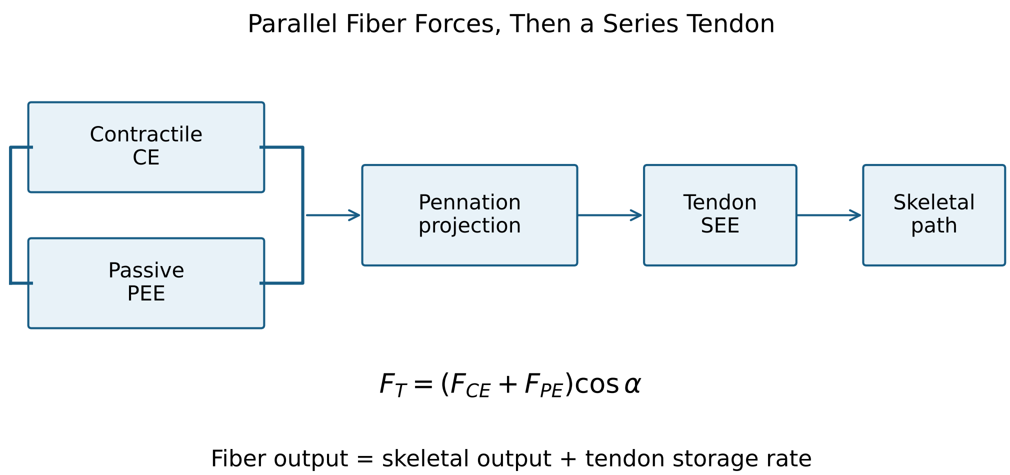


Figure 14.2: Force and Energy Connections in the Teaching Model. Active and Passive Fiber Forces Add Before Pennation Projection; the Tendon Is in Series.

The active and passive forces used here are

$$F_{\mathrm{CE}} = aF_0 f_L(\lambda) f_V(w), \tag{14.8}$$

$$F_{\mathrm{PE}} = F_0 f_P(\lambda). \tag{14.9}$$

Let tendon slack length be L_s and strain be $\epsilon = (\ell^T - L_s)/L_s$. Define a tendon force law independently of the passive fiber curve:

$$F_T(\epsilon) = \begin{cases} 0, & \epsilon \le 0, \\ F_{\mathrm{ref}} \dfrac{e^{k_T \epsilon/\epsilon_{\mathrm{ref}}} - 1}{e^{k_T} - 1}, & \epsilon > 0. \end{cases} \tag{14.10}$$

Here F_{ref} is the tendon force at positive reference strain ϵ_{ref}, and k_T is dimensionless. Tendon tangent stiffness is $dF_T/d\ell^T$, measured in N/m; it is not k_T. On the loaded branch,

$$\frac{dF_T}{d\ell^T} = \frac{F_{\mathrm{ref}} k_T e^{k_T \epsilon/\epsilon_{\mathrm{ref}}}}{L_s \epsilon_{\mathrm{ref}} (e^{k_T} - 1)}.$$

The idealization stores energy without hysteretic loss. Actual loading and unloading may require a richer constitutive model. As with the passive fiber curve, the slack transition is continuous but not differentiable here.

14.5.1 Geometry Connects Fiber, Tendon and Joint Motion

Write the total path length as $\ell^{MT} = \ell^T + \ell^M \cos\alpha$. To differentiate it, specify how pennation changes. In a fixed-height model, $h = \ell^M \sin\alpha$ is constant, so

$$\ell^M \cos\alpha = \sqrt{(\ell^M)^2 - h^2}.$$

Differentiating gives

$$\dot{\ell}^M = \cos\alpha\,(\dot{\ell}^{MT} - \dot{\ell}^T).$$

The relation applies for $\ell^M > h$; approaching 90° pennation is a degenerate geometry, not an ordinary operating state. Assuming constant angle instead would give a different velocity relation. Combining fixed-height kinematics with the force projection gives the power identity

$$F_T\dot{\ell}^{MT} = F_M\dot{\ell}^M + F_T\dot{\ell}^T.$$

For example, $\ell^M = 0.10\,\mathrm{m}$ and $h = 0.06\,\mathrm{m}$ give $\cos\alpha = 0.8$. If the path shortens at $0.10\,\mathrm{m/s}$ while the tendon lengthens at $0.02\,\mathrm{m/s}$, the fiber shortens at $0.096\,\mathrm{m/s}$. With $F_M = 100\,\mathrm{N}$ and $F_T = 80\,\mathrm{N}$, the identity is $-8 = -9.6 + 1.6\,\mathrm{W}$. Of the fiber's mechanical output, $1.6\,\mathrm{W}$ is going into tendon storage at that instant. Tendon compliance therefore changes the fiber's velocity and its force capacity as well as storing energy.

For independent joint coordinates q with $v = \dot{q}$, virtual work connects path length to moment arms:

$$r_{ji}(q) = -\frac{\partial \ell_i^{MT}}{\partial q_j}, \qquad \dot{\boldsymbol{\ell}}^{MT} = -r^T v, \qquad \boldsymbol{\tau}_M = r\boldsymbol{F}_T.$$

For angular coordinates, entries of r have units of meters per radian, conventionally written meters. Generalized coordinates that include translations require the corresponding force units; a model using $\dot{q} = N(q)v$ must include that mapping. The convention above ensures

$$\boldsymbol{\tau}_M^T v = -\boldsymbol{F}_T^T\dot{\boldsymbol{\ell}}^{MT}.$$

Shortening under positive tension can deliver positive power to the skeleton. Tendon lengthening can instead absorb part of that work. These equations also reveal multi-joint coupling: a muscle crossing two joints can generate moments at both, and motions at the two joints can reinforce or cancel its path-length change.

Take the local path in meters, with angles in radians:

$$\ell^{MT} = 0.30 - 0.02q_1 + 0.01q_2.$$

Its moment arms are $(0.02, -0.01)\,\mathrm{m}$. At $v = (10, 20)\,\mathrm{rad/s}$, the path velocity is zero despite motion at both joints. At $v = (10, 0)$, it is $-0.20\,\mathrm{m/s}$. Rigid translation of the entire arm and muscle together contributes no intrinsic path-length change. These examples explain why a clubhead or hand speed is insufficient to calculate muscle force.

Key Principle 14.5. Energy Storage and Release in the Tendon

Stored energy is the integral of force over extension, not force over dimensionless strain alone:

$$U_T(\epsilon) = L_s\int_0^{\epsilon} F_T(s)\,ds.$$

For positive strain in our law,

$$U_T = \frac{F_{\mathrm{ref}}L_s}{e^{k_T} - 1}\left[\frac{\epsilon_{\mathrm{ref}}}{k_T}\left(e^{k_T\epsilon/\epsilon_{\mathrm{ref}}} - 1\right) - \epsilon\right].$$

Set $F_{\mathrm{ref}} = 100\,\mathrm{N}$, $L_s = 0.20\,\mathrm{m}$, $\epsilon_{\mathrm{ref}} = 0.04$ and $k_T = 3$. At the reference strain, extension is $0.008\,\mathrm{m}$ and stored energy is $0.22475\,\mathrm{J}$. Differentiating energy with respect to extension recovers force. This provides a direct check on the units and expression.

For the lossless model, $\dot{U}_T = F_T \dot{\ell}^T$. Substituting into the power identity shows that tendon recoil can add to instantaneous power delivered to the skeleton while its stored energy falls. It does not create energy. A shaft also stores and returns elastic energy, but its distributed bending and torsion, boundary conditions and contact with the golfer are different from a tendon model. Neither elastic mechanism by itself proves a universal benefit from a particular wrist action or shaft flexibility. The timing and direction of energy delivery must serve the impact task.

Force equilibrium must hold throughout this calculation. Arbitrarily prescribing activation, fiber length, fiber velocity and tendon extension independently can violate it. Some equilibrium formulations solve an inverse force-velocity relation for fiber velocity; zero activation or extreme geometry can make that inversion singular or impossible. A simulator must document how it handles such states. A numerical minimum activation is a modeling choice, not evidence that a person cannot cease neural excitation. [Millard et al., 2013]

14.6 How Muscle Biology Maps to Control Theory

Mechanical position and velocity alone do not generally specify the future of an activated, compliant muscle system. Two otherwise identical poses can have different activation and tendon loading because of their histories. A useful state for this teaching model is

$$x = (q, v, \boldsymbol{a}, \boldsymbol{\ell}^M), \qquad \dot{x} = f(x) + G(x)\boldsymbol{u}. \tag{14.11}$$

For n independent mechanical coordinates and m modeled muscle compartments,

$$\boldsymbol{u} \in [0,1]^m, \tag{14.12}$$

$$\boldsymbol{a} \in [0,1]^m, \tag{14.13}$$

$$\boldsymbol{F}_T \in \mathbb{R}^m_{\geq 0}, \tag{14.14}$$

$$\boldsymbol{\tau}_M = r(q)\boldsymbol{F}_T \in \mathbb{R}^n. \tag{14.15}$$

There is no fixed universal number of actuators per joint. One anatomical muscle may be split into compartments, and one compartment may span multiple joints. Excitations are bounded; tendon forces are tensile; admissible forces also depend on equilibrium and the current state. They are not arbitrary unconstrained control vectors.

Away from an impact event, a specified unconstrained mechanical model may take the form

$$\dot{v} = A(x) = M(q)^{-1}\{r(q)\boldsymbol{F}_T(x) + \boldsymbol{b}_{\text{ext}}(x) - \boldsymbol{c}(q,v) - \boldsymbol{g}(q)\}.$$

M is the mass matrix, $\boldsymbol{c}$ the velocity-dependent inertial terms, $\boldsymbol{g}$ the gravitational terms and $\boldsymbol{b}_{\text{ext}}$ specified external generalized forces. Contact constraints, unknown reaction forces or a flexible shaft require the corresponding additional equations. A collision may require an impulse or compliant-contact treatment; the smooth equation should not silently be applied across it.

Suppose a well-defined muscle equilibrium solution gives $\dot{\boldsymbol{\ell}}^M = \Phi(x)$, and activation constants form the diagonal matrix T_a. Then the actual augmented control-affine expression is

$$f(x) = \begin{bmatrix} v \\ A(x) \\ -T_a^{-1}\boldsymbol{a} \\ \Phi(x) \end{bmatrix}, \qquad G(x) = \begin{bmatrix} 0_{n\times m} \\ 0_{n\times m} \\ T_a^{-1} \\ 0_{m\times m} \end{bmatrix}. \tag{14.16}$$

Excitation enters the activation derivative, not the mechanical acceleration directly. Its mechanical effect develops as the state evolves. The component units differ appropriately: position rate, acceleration, activation rate and fiber-length rate. Equating $G\boldsymbol{u}$, a state-rate vector, with a scalar joint torque would be dimensionally incorrect. This expression also depends on a fixed T_a and an admissible Φ independent of instantaneous excitation. If the activation time constant switches with u and a, global affinity in u no longer follows from this display.

In Plain Language 14.5. Why Effective Control Depends on State and Time

At fixed length and velocity, the active force's sensitivity to activation is proportional to $F_0 f_L f_V$. During shortening, that sensitivity can decline as local fiber speed rises. The resulting joint effect also depends on moment arms, the mass matrix, tendon dynamics and other joints. A new excitation change has no instantaneous jump effect on activation in this model. A previously prepared activation can still produce force, and a loaded tendon can still recoil, even if the current excitation is zero.

Example 14.4. Numerical Estimate: Finite-Horizon Joint Response

Consider a deliberately simple joint with constant inertia $I = 0.10\,\mathrm{kg\,m^2}$. An added actuator torque is $\tau_{\max} a(t)$, with $\tau_{\max} = 2\,\mathrm{N\,m}$, $a_0 = 0$, $u = 1$ and $\tau_a = 0.025\,\mathrm{s}$. Holding all other torques equal between two trajectories, the changes after a remaining response duration T are

$$\Delta v = \frac{\tau_{\max}}{I}\left[T - \tau_a(1 - e^{-T/\tau_a})\right],$$

$$\Delta q = \frac{\tau_{\max}}{I}\left[\frac{T^2}{2} - \tau_a T + \tau_a^2(1 - e^{-T/\tau_a})\right].$$

At $T = 0.050\,\mathrm{s}$, $\Delta q = 0.0108083\,\mathrm{rad}$, or about $0.6193°$. This is a finite-horizon response from explicit initial conditions, not a predicted golf correction. A specified pure delay is removed from the available interval before using T; if no response time remains, this command produces no pre-impact effect in this model. Existing activation, elastic loading and configuration-dependent dynamics require a different calculation.

14.6.1 Drift, Cancellation and Reachability

In the mechanical equation, a conventionally defined non-muscle generalized term is

$$\boldsymbol{b}_{\mathrm{mech}} = \boldsymbol{b}_{\mathrm{ext}} - \boldsymbol{c} - \boldsymbol{g}. \tag{14.17}$$

Its definition must say which external forces and passive effects it includes. In the augmented state equation, the zero-input drift also includes ongoing activation and stored elastic state. Therefore zero excitation is not equivalent to zero muscle force or purely gravitational motion. And $I\ddot{q}$ inferred from observed acceleration is the total required inertial balance in a scalar model, not an independently measured free-drift torque.

Suppose a scalar mechanical term is $-80\,\mathrm{N\,m}$ and an added control moment can range from -4 to $4\,\mathrm{N\,m}$. With positive constant inertia, all available accelerations have the same sign. Nevertheless, their magnitudes differ and their velocity and position trajectories separate over time. Inability to cancel the current acceleration is not inability to influence the eventual state. Conversely, a favorable instantaneous torque ratio does not guarantee that the required impact state is reachable within the remaining time and actuator bounds.

For the golf question, compare admissible policies over a stated horizon with the same starting mechanical, activation and elastic states. Measure their effects on clubface orientation, impact location and velocity, while checking other joint loads and contact conditions. A linearized sensitivity of these outputs to excitation can be informative, but it must follow the trajectory and its state transitions. A ratio evaluated at one joint at one instant cannot establish nonlinear controllability, global optimality or injury risk.

14.7 Muscle Force in the Golf Swing: What Numbers Establish

Key Principle 14.6. Peak Torques at Major Golf Joints

Three kinds of number answer different questions. A maximum isometric test estimates capacity in a specified posture and protocol. Inverse dynamics estimates the net generalized moment consistent with measured motion and external forces under a chosen model. A force plate measures an external ground wrench, which must be transformed and combined with segment dynamics before assigning internal joint moments. Comparing them requires matched coordinate conventions, units, populations and loading conditions. None is automatically an individual muscle-force measurement.

For two antagonists with moment arms $(0.02, -0.02)\,\mathrm{m}$, tendon forces $(100, 0)\,\mathrm{N}$ and $(200, 100)\,\mathrm{N}$ both produce $2\,\mathrm{N\,m}$ net torque. Their summed tensions are 100 and $300\,\mathrm{N}$. Thus a low net torque can coexist with substantial opposing forces. Joint compression, stiffness and metabolic cost require their own geometry and constitutive models; they cannot be recovered from that torque equality. An optimization algorithm may select one force allocation, but its objective and constraints provide information that inverse dynamics alone does not contain.

EMG is useful for investigating electrical activity, but it is not a calibrated force meter. Roberts and Gabaldón's direct force, fascicle and EMG measurements in turkey hindlimb muscles illustrate that their relationships can vary across movement conditions. That finding supports distinguishing the signals; it does not calibrate human forearm force during golf. [Roberts and Gabaldón, 2008] A golf study needs its own measurement model, normalization choices and uncertainty analysis.

Key Principle 14.7. Muscle Capacity and Mechanical Demand

Earlier actions can change later geometry, velocity, activation and tendon loading. Later actions can change the trajectory within their remaining response time and bounds. A sequence that reduces one muscle's shortening speed may improve its active-force capacity while increasing demand elsewhere. A motion that increases elastic storage may also change joint loading or mistime recoil. These are coupled tradeoffs, so no universal early-effort/late-coast rule follows from a single force curve.

To evaluate a proposed sequence, first state the desired impact outputs and acceptable variability. Next estimate the mechanical and muscle states consistently, identifying parameters inferred rather than measured. Finally compare alternatives with the same task and initial conditions, including sensitivity to moment arms, slack lengths, activation constants and measurement error. Agreement with a net torque curve is useful but insufficient validation of every internal force. A stronger claim should survive independent observations relevant to the proposed mechanism.

14.8 Implications for Swing Mechanics and Training

Key Principle 14.8. Interpreting the Cue to Accelerate Through Impact

A coaching cue is not a direct report of muscle force or clubhead acceleration. Continuing effort can coexist with a declining shortening-force multiplier, and clubhead speed can increase through coupled motion while a particular muscle does little positive work. Evaluate the cue by its repeatable effect on the intended impact outcomes and by an appropriate mechanical explanation. The equations here neither require every muscle to accelerate the club until contact nor prove that all late effort is harmful.

Training can alter capacity, coordination and how a player prepares the state entering the downswing. These mechanisms need different tests. Greater isometric strength does not by itself establish greater available force at the relevant fiber lengths and velocities; a successful practice swing does not establish improved retention or transfer. A useful investigation links the proposed change to a measured outcome and checks whether the inferred mechanism is consistent with the full movement. This keeps practical instruction informative while leaving uncertainty visible.

Key Principle 14.9. Key Takeaways: Muscle Force Generation

Trace the chain from excitation to activation, fiber force, tendon force, joint moments and task outputs. Keep local fiber velocity distinct from club speed. Preserve force equilibrium and virtual work when joining geometry to mechanics. Account for the energy and activation already present at the start of a comparison. Interpret net moments, EMG and capacity measurements according to what each measures. Judge control over the remaining trajectory, and test proposed golf sequences against alternatives rather than declaring one optimal from an instantaneous ratio.

14.8.1 Looking Ahead

The muscle-to-joint and movement-control chapters use these distinctions at larger scales. Multi-joint geometry determines how muscle forces influence motion; activation and tendon memory determine what the same current command can do; learning determines how a player prepares and adjusts those states. Together they make the golf swing a question about coordinated trajectories and impact outcomes, not an isolated contest between one muscle and one inertial number.

Exercises 14.1. Chapter Exercises

All numerical parameters below belong to the teaching model, not a golfer dataset.

1. For the Gaussian with $\gamma = 0.5$, calculate f_L at $\lambda = 0.8$, 1.2 and 1.5. Does it have any finite zero? At $\lambda = 1.2$, calculate total force divided by F_0 for $a = 0.7$, $w = 0$ and the stated passive curve.

2. Use $r = 0.02\,\mathrm{m}$, $\dot{q} = 8.75\,\mathrm{rad/s}$, $\ell^M_{\mathrm{opt}} = 0.10\,\mathrm{m}$ and $\nu = 10\,\mathrm{s}^{-1}$. Assume a rigid tendon, zero pennation and shortening in the positive joint direction. Calculate w, f_V and torque for $F_0 = 100\,\mathrm{N}$ at optimal length and full activation.

3. For $\tau_a = 25\,\mathrm{ms}$, $a_0 = 0$ and $u = 1$, find the 50% and 95% activation times. Compare the latter with forward Euler using 5 ms steps. Explain why neither is a pure neural delay.

4. A path shortens by 0.02 m per radian of q_1 and lengthens by 0.01 m per radian of q_2. Find both moment arms and path velocity at $v = (10, 20)\,\mathrm{rad/s}$. With $F_T = 100\,\mathrm{N}$, calculate the two joint powers and their sum. Assume no other path dependence.

5. Derive the finite-horizon formulas by integrating $\Delta\ddot{q} = Ka(t)$, where $K = \tau_{\max}/I$ and $a(t) = 1 - e^{-t/\tau_a}$. Use zero initial position and velocity differences. Evaluate Δq after 50 ms using the chapter's parameters. Explain why a small instantaneous torque ratio does not set this result to zero.

6. Use $F_{\mathrm{ref}} = 100\,\mathrm{N}$, $L_s = 0.20\,\mathrm{m}$, $\epsilon_{\mathrm{ref}} = 0.04$ and $k_T = 3$. Find force, extension and energy at the reference strain. Verify $dU_T/d\ell^T = F_T$, and identify the dimensional error in integrating force over strain without L_s.

7. Design a comparison of two proposed swing sequences. State the common initial conditions, task outcomes, muscle-state estimates and uncertain parameters. Identify one observation that would distinguish a change in muscle timing from a change in geometry or tendon loading.

14.8.2 Worked Answers

1. The multipliers are 0.8521, 0.8521 and 0.3679; none is zero. Total normalized force is approximately 0.7188, including 0.12227 passive force. The Gaussian remains positive at every finite argument.

2. $V_{\max} = 1\,\mathrm{m/s}$, $w = 0.175$, $f_V \approx 0.5739$, force 57.39 N and moment 1.148 N m. A linear

calculation would give a different answer because it is a different law.

3. $t_{50} = 17.33\,\text{ms}$ and $t_{95} = 74.89\,\text{ms}$. Euler first exceeds 95% at step 14, 70 ms. The continuous response begins immediately in this model; the time constant controls its rate of approach.

4. The moment arms are $(0.02, -0.01)\,\text{m}$ and path velocity is zero. Torques $(2, -1)\,\text{N m}$ produce powers $(20, -20)\,\text{W}$, summing to zero. Opposite joint powers can coexist with no net muscle-tendon shortening power.

5. Integrate once for velocity and again for position, retaining both integration constants fixed by the initial conditions. The position difference is 0.0108083 rad, about 0.6193°. A torque too small to cancel another term can still accumulate a trajectory difference.

6. Force is 100 N, extension 0.008 m and energy 0.22475 J. Since $d\epsilon/d\ell^T = 1/L_s$, differentiating the energy cancels the slack-length factor and returns force. Without that factor, the strain integral has units of newtons rather than joules.

7. One defensible design matches the task and starting body, club, activation and elastic states, then compares feasible excitation histories. Record impact orientation, location and velocity as well as relevant loads. Use synchronized motion and electrical signals, with a validated force/geometry model and, where feasible, independent fascicle or tendon measurements. Test parameter sensitivity and predictions not used for fitting. A timing claim should explain observations that an alternative geometry or elastic-state change does not explain equally well; otherwise the mechanism remains underdetermined.

Chapter 15

Aerodynamic Loads in Swing and Flight

Air loads belong in a golf model when their effect matters for the question and accuracy being investigated. The useful question is not whether drag is always negligible or always dominant. It is how a plausible aerodynamic load changes a measured-load estimate, a forward prediction, or the delivered ball flight, relative to the uncertainty and effect size that matter to the study.

That question connects several parts of the swing. Geometry determines local air-relative velocity and orientation. Fluid mechanics determines distributed tractions and moments. Joint and contact geometry maps those loads into the multibody equations. The chosen input policy and initial state determine the forward response. Impact then supplies the ball's initial flight state; ball aerodynamics determines how that state develops. A force estimate at one point does not settle all of these questions at once.

15.1 Define the Relative Airflow

At a point on the moving body, let v be its velocity in an inertial reporting frame and w the undisturbed ambient-air velocity evaluated at that point. Define $v_r = v - w$. In a quasi-steady drag model,

$$F_D = -\tfrac{1}{2}\rho C_D A \|v_r\| v_r. \tag{15.1}$$

The force opposes motion relative to the air. It need not oppose velocity relative to the ground. Density ρ, reference area A, and coefficient C_D require declared units and conventions. This expression represents the drag component; lift, side force, and an aerodynamic couple may be needed to represent the full load on an oriented or rotating clubhead.

The coefficient summarizes the effects of shape, orientation, Reynolds number, surface condition, and other features of the flow. It is not a universal constant belonging to a shape name. The NASA drag-equation explanation emphasizes that a measured coefficient and its reference area must be used together. Doubling the reporting area halves the coefficient for the same force. A fixed nominal area with an orientation-dependent coefficient is a valid convention; repeatedly changing area as well as using a coefficient already calibrated for that change can double-count the orientation effect.

For a ball, $A = \pi r_b^2$ is a common convention. For a cylinder in normal crossflow, projected area is diameter times exposed length. For a clubhead, state the area used in its calibration rather than assuming it always equals the clubface area. Its airflow incidence changes during the swing.

15.1.1 Quadratic Scaling and Its Limits

The force magnitude is

$$D = \tfrac{1}{2}\rho C_D A v_r^2. \tag{15.2}$$

It increases fourfold when speed doubles only if ρ, C_D, A remain fixed. A momentum-flux argument motivates a scale $\rho A v_r^2$ in inertia-dominated flow. It does not prove that the coefficient is speed-independent. At low Reynolds number, a sphere's Stokes drag is proportional to speed; expressing it using the quadratic formula gives $C_D = 24/\mathrm{Re}$. Thus the same coefficient notation can encode different speed dependence.

Use

$$\mathrm{Re} = \frac{\rho \|v_r\| L}{\mu} \tag{15.3}$$

to state the flow regime, with the relevant characteristic length L and dynamic viscosity μ. For illustrative values $\rho = 1.225$ kg/m^3, $\mu = 1.81 \times 10^{-5}$ Pa s, and speed 50 m/s, a 0.1 m head scale gives $\mathrm{Re} \approx 3.4 \times 10^5$; a 0.01 m shaft diameter gives about 3.4×10^4. They are different flow problems. Reynolds number alone does not tell whether every boundary layer or wake is turbulent. Surface roughness, orientation, unsteadiness, and separation affect the result.

A quasi-steady coefficient approximation also needs the airflow to adjust sufficiently quickly relative to the changing motion. Rapid orientation changes, shaft-head interference, and wake history can require unsteady measurements or a more detailed fluid model. CFD is a way to calculate a candidate load model; mesh, time-step, turbulence-model, and experimental validation remain necessary.

15.2 A Force Estimate With a Clear Scope

Choose $\rho = 1.225$ kg/m^3, $C_D = 0.4$, $A = 0.005$ m^2, and $\|v_r\| = 50$ m/s as an illustrative coefficient-area-speed combination. Then

$$D = \tfrac{1}{2}(1.225)(0.4)(0.005)(50)^2 = 3.0625 \text{ N}. \tag{15.4}$$

This is a conditional calculation, not a measured typical driver force. The same force results from $C_D = 0.2$ with area 0.01 m^2. Comparing coefficients across products without matching area, incidence, and flow conditions can be misleading.

In still air, the instantaneous translational drag power for this example is $-D\|v\| = -153.125$ W. If the same speed and load were maintained for 0.05 s, 7.65625 J would be transferred out through drag. Maintaining speed requires other loads to supply that work. This calculation is an energy scale, not a claim that a freely evolving club loses exactly that fraction of its speed. A 0.2 kg point-mass head at 50 m/s has translational kinetic energy 250 J; the complete club also has its distributed translational and rotational energy. A percentage loss for an actual swing requires integration along that swing.

Similarly, impulse is $\int F_D \, dt$, while work is $\int F_D^T v \, dt$ with the relevant point velocity and any aerodynamic-couple power included. A peak force multiplied by total downswing duration does not reproduce either integral unless the corresponding simplifying assumptions hold. Comparing newtons to newton metres without a defined moment arm has no physical meaning.

15.3 Distributed Shaft Loads and the Actual Pivot

For a slender shaft with centerline $p(s, q)$, local velocity $v(s)$, and tangent unit vector $t(s)$, a simple normal-crossflow model uses

$$v_\perp = (I - tt^T)(v - w), \qquad dF_D = -\tfrac{1}{2}\rho C_n d(s)\|v_\perp\| v_\perp \, ds. \tag{15.5}$$

The coefficient C_n is calibrated for the chosen crossflow model. This approximation omits axial skin friction and some three-dimensional or unsteady effects. Flexible motion enters through the actual centerline and local velocity; a coating or aerodynamic shape is not itself shaft compliance.

As an exactly specified toy geometry, take a straight uniform shaft of length L, rotating in a plane at signed angular speed ω about a fixed pivot on its extended axis. The near end is distance $a \geq 0$ from that pivot, and $s \in [0, L]$ measures distance along the shaft. Its speed is $|\omega|(a + s)$. With still air and constant diameter d and C_n, integration gives

$$D_{\text{shaft}} = \tfrac{1}{2}\rho C_n d\omega^2 \frac{(a + L)^3 - a^3}{3}, \tag{15.6}$$

$$\tau_{D,\text{pivot}} = -\tfrac{1}{2}\rho C_n d\omega|\omega| \frac{(a + L)^4 - a^4}{4}. \tag{15.7}$$

The moment sign opposes the signed rotation. For $a = 0$, the pivot is the shaft's near end; it is not automatically the shoulder. A real golfer's grip translates and rotates, so a single fixed-pivot shaft example does not represent all shoulder, wrist, and club angular speeds.

For $L = 1.1$ m, $d = 0.01$ m, $C_n = 0.5$, $\rho = 1.225$ kg/m^3, $\omega = 25$ rad/s, and $a = 0$, the resisting moment magnitude is 0.700595 N m. The near-end-pivot head speed in this same toy geometry would be 27.5 m/s. It would be inconsistent to combine that shaft calculation with a 50 m/s head while claiming one rigid fixed-pivot motion. A different motion model can produce different local speeds, but must specify how.

The generalized load contribution of a segment depends on its velocity Jacobian and the chosen output, not merely its speed or mass. Arm, torso, shaft, and head contributions should therefore be integrated with compatible kinematics. There is no universal 40/40/20 percentage split. The component that dominates force need not dominate a selected joint moment, work integral, or change in endpoint club speed.

15.4 From Air Loads to Generalized Loads

If point i has velocity $v_i = J_{v_i}(q)\dot{q}$ and segment angular velocity $\omega_i = J_{\omega_i}(q)\dot{q}$, its aerodynamic force F_i and couple m_i about that same point contribute

$$Q_{\text{aero}} = \sum_i \left(J_{v_i}^T F_i + J_{\omega_i}^T m_i \right), \tag{15.8}$$

with an integral replacing the sum for distributed loads. The formula follows from virtual power:

$$\dot{q}^T Q_{\text{aero}} = \sum_i (v_i^T F_i + \omega_i^T m_i). \tag{15.9}$$

All components must use compatible frames and reference points. For one revolute coordinate, the generalized moment is the projection of the force-induced moment and couple onto that joint's axis. A sum of three-vectors $r_i \times F_i$ is not a general n-coordinate load vector.

Use one consistent sign convention in the equations:

$$M\ddot{q} + h = Bu + p + Q_{\text{aero}} + J_c^T\lambda. \tag{15.10}$$

The drag force already contains its resisting sign. It enters the right-hand side as written; adding an extra minus sign there would reverse its action. Here h contains the declared inertial bias and gravity, p other constitutive loads, and $J_c^T\lambda$ the ideal closure reactions. Other external loads must also be included where appropriate.

For a regular fixed contact mode, eliminating reactions gives an acceleration map $a = f_a + KBu$, where

$$K = M^{-1} - M^{-1}J_c^T(J_cM^{-1}J_c^T)^{-1}J_cM^{-1}. \tag{15.11}$$

An input-independent aerodynamic load contributes KQ_{aero} to the zero-input acceleration at a fixed state. It does not change KB at that state merely because it depends on velocity. It can change the drift linearization, the trajectory, required effort, and reachable outcomes under bounded inputs. Those are different from changing the instantaneous input coefficient. Actively controlled aerodynamic surfaces or actuator-dependent flow would require a model that represents that additional input dependence.

The influence of drag on a norm ratio is also not fixed in sign. It is a vector addition that can reinforce or cancel other drift components. Sums of scalar load magnitudes do not establish a drift/control ratio, and a large norm ratio by itself is not a reachability or neural-strategy certificate.

15.5 Power, Frames, and System Boundaries

For the quasi-steady translational drag law with $C_D \geq 0$,

$$F_D^Tv_r = -\tfrac{1}{2}\rho C_DA\|v_r\|^3 \leq 0. \tag{15.12}$$

It is zero at zero relative speed. In an inertial ground frame,

$$F_D^Tv = F_D^Tv_r + F_D^Tw. \tag{15.13}$$

A tailwind faster than a body can push it forward and add body kinetic energy, while drag still opposes relative motion. For a one-dimensional illustrative law $F_D = -|v - w|(v - w)$, take body speed $v = 1$ and wind $w = 3$ in consistent units: $F_D = 4$, ground-frame body power is 4, and relative-motion power is -8. The difference comes from energy exchange with the moving air.

In still air, for a declared whole mechanical system with conservative gravity and elastic storage included in $E = T + V_g + V_e$, a suitable balance is

$$\dot{E} = \dot{q}^TBu + P_{\text{aero}} + P_{\text{other}} - P_{\text{diss,other}}. \tag{15.14}$$

This assumes the remaining passive terms have been classified consistently and ideal stationary constraints do zero total power. Moving constraints, activation-dependent constitutive parameters, and additional stored-energy states require their own terms. Gravity is already in V_g; it is not counted again as an energy source in this total-energy derivative. Inertial coupling redistributes the kinetic-energy rate and is not an extra supply of energy.

Aerodynamic work is not metabolic expenditure. Isometric force, co-contraction, activation, and muscle efficiency can consume metabolic energy even when the net external mechanical work is small. A drag estimate therefore cannot explain range-session fatigue or diagnose effort by itself. Likewise, lower head drag may affect attainable speed under a specified swing policy; it does not by itself establish forgiveness on off-center impacts.

15.6 The Sign of an Inverse-Dynamics Correction

For $B = I$ and known reactions and other loads, rearrangement gives

$$u = M\ddot{q} + h - p - Q_{\text{aero}} - J_c^T\lambda. \tag{15.15}$$

There is no leading M^{-1}. The result has generalized-load units, not acceleration units. With the same known boundaries, omitting Q_{aero} gives $u_{\text{omit}} = M\ddot{q} + h - p - J_c^T\lambda$, hence

$$u - u_{\text{omit}} = -Q_{\text{aero}}. \tag{15.16}$$

For an isolated positive rotation with inertia 2 kg m^2, acceleration 3 rad/s^2, and resisting aerodynamic moment -4 N m, the true applied moment is 10 N m. Omitting drag gives 6 N m. The omission underestimates the positive applied moment in this example. With a different axis, motion, measured quantity, or signed load, the numerical bias can differ; a universal claim about every inferred joint torque is unjustified.

If hand or ground reactions are also unknown, changing the aerodynamic model can change the fitted allocation between inputs and reactions. Then the simple difference above cannot be used while silently assuming those unknown loads remain fixed. Analyze the joint balance and measurement map as in Chapter 19. Net hand force, net hand couple, net joint load, and individual muscle force are different inferred quantities.

The relative error must name its denominator. Near a zero crossing, a small absolute moment discrepancy can produce an arbitrarily large percentage. Sensitivity to aerodynamic uncertainty should be reported in physical units and compared with measurement uncertainty and the scientific effect being tested. Unsupported universal wrist, ankle, or shoulder error percentages do not provide an accuracy budget.

15.7 A Reproducible Forward Counterfactual

A forward zero-torque counterfactual (ZTCF) sets a declared effective input to zero and integrates from a specified complete state. It retains the stated aerodynamic and passive-load laws and recomputes them along the evolving branch. It is not obtained by recycling measured drag and reaction histories after the trajectory has changed. Zero joint input also does not mean zero muscle activation or no previously stored elastic energy.

Consider an illustrative pendulum, with angle θ measured from its stable bottom equilibrium, angular velocity ω, inertia $I = 2$ kg m^2, and gravity moment $-5\sin\theta$ N m. Choose quadratic drag coefficient $c \geq 0$ in moment per squared angular-speed units. The zero-input model is

$$\dot{\theta} = \omega, \qquad 2\dot{\omega} = -5\sin\theta - c\omega|\omega|. \tag{15.17}$$

Angular acceleration is $\dot{\omega} = \ddot{\theta}$; $\ddot{\omega}$ would be angular jerk. The signed drag law opposes either direction of rotation.

Starting at $\theta = 0, \omega = 0$ gives an exact equilibrium. No spontaneous swing appears. To investigate a falling motion, start instead at $\theta_0 = \pi/2, \omega_0 = 0$. Its initial potential energy relative to the bottom is 5 J. With

$$E = \omega^2 + 5(1 - \cos\theta), \qquad \dot{E} = -c|\omega|^3, \tag{15.18}$$

the first bottom crossing without drag has speed $|\omega| = \sqrt{5}$, about 2.23607 rad/s. A speed of 85 rad/s would require 7225 J of kinetic energy for this inertia, which the stated zero-input initial state cannot supply.

For a second independent check, set $y(\theta) = \omega^2$ along the falling branch, where $\omega < 0$. Write the gravity coefficient as $k = 5$ N m and define $\kappa = 2c/I$. With angles in radians, the resulting angle equation is

$$\begin{aligned} \frac{dy}{d\theta} - \kappa y &= -\frac{2k}{I}\sin\theta, \qquad y(\pi/2) = 0, \\ y(0) &= \frac{2k}{I}\int_0^{\pi/2} e^{-\kappa s}\sin s\, ds. \end{aligned} \tag{15.19}$$

For the specified inertia, $c = 0.5$ gives $\kappa = 0.5$ and $2k/I = 5$ in the corresponding angular-speed units. It yields

$$y(0) = 4 - 2e^{-\pi/4}, \qquad |\omega_{\text{bottom}}| \approx 1.75731 \text{ rad/s}. \tag{15.20}$$

Numerically integrating angle, velocity, and accumulated dissipated work reproduces both bottom-crossing results:

Drag Coefficient	Bottom Time (s)	Bottom Speed (rad/s)	Dissipated Work (J)
$c = 0$	1.17262	2.23607	0
$c = 0.5$	1.26570	1.75731	1.91188

The checks use adaptive integration with relative tolerance 10^{-10}, absolute tolerance 10^{-12}, and maximum step 0.01 s; they also test the energy identity and the independent angle integral. These parameters are reproducible numerical settings, not evidence that the pendulum represents a measured golf swing. It omits the multisegment geometry, real initial motion, muscle and tissue states, and club mechanics that a golf investigation needs.

In this one-coordinate still-air example, energy loss lowers speed at the same bottom angle. With drag, the speed maximum occurs before the bottom: along the falling branch, angular acceleration becomes zero when $5\sin\theta = c\omega^2$, and the resisting drag remains nonzero at the bottom. Thus a comparison of bottom-crossing speeds is not a comparison of both branches' peak speeds. It does not prove a universal ordering of every segment's speed at a fixed time in a coupled golfer. Changed timing and configuration can redistribute energy and alter endpoint outcomes. Report whether branches are compared at the same time, a matched event, or a matched configuration. Differences between already diverged trajectories also do not reconstruct instantaneous applied torque; the inverse-dynamics chapter gives an explicit counterexample to that inference.

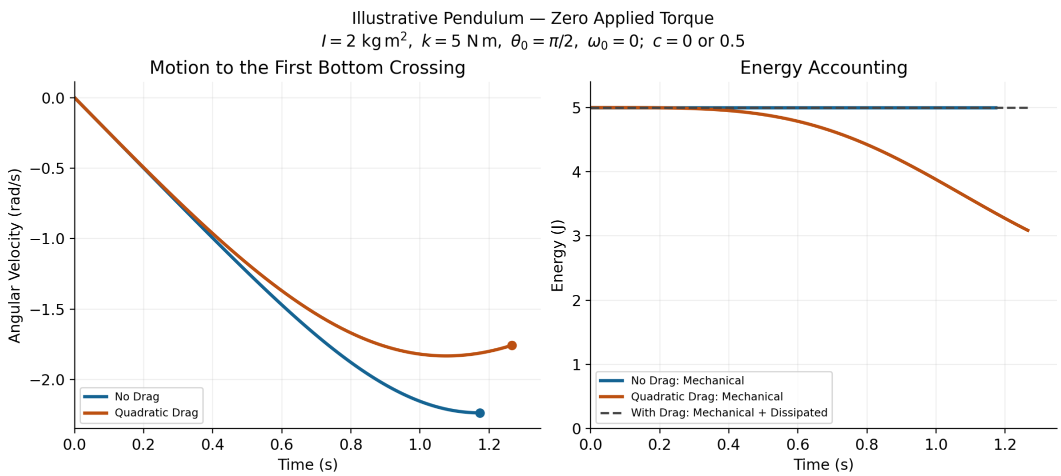


Figure 15.1: Illustrative Zero-Input Pendulum Motion and Energy With and Without Quadratic Drag. Each Curve Ends at Its First Bottom Crossing.

15.8 Ball Flight Connects the Swing to the Outcome

At impact, club motion, impact location, contact mechanics, and ball properties jointly establish launch position, velocity, and spin. Aerodynamics then changes the flight. A swing study that predicts clubhead speed has not automatically predicted carry distance or dispersion. Conversely, a launch-monitor endpoint does not uniquely reconstruct the golfer's joint loading.

For a simplified spherical-ball flight model, let $v_r = v - w$ and use

$$m_b \dot{v} = m_b g + F_D + F_L, \qquad \dot{r} = v. \tag{15.21}$$

The drag law uses ball-specific C_D and area conventions. A quasi-steady lift model uses the scalar component $\frac{1}{2}\rho C_L A \|v_r\|^2$, with a signed coefficient and a stated lift direction. For ordinary transverse-spin Magnus lift, a candidate direction is $(\omega_b \times v_r)/\|\omega_b \times v_r\|$. Handle zero cross product explicitly. Coefficients calibrated for spin perpendicular to the flow should not be applied to arbitrary spin-axis angles without an appropriate extension. Spin decay requires an aerodynamic moment law; constant spin is a separate approximation, not a consequence of the translational equation.

A usual transverse-spin parameter is

$$S = \frac{r_b \|\omega_b\|}{\|v_r\|}. \tag{15.22}$$

For illustrative radius 0.02135 m, speed 50 m/s, and transverse spin 200 rad/s, $S = 0.0854$. A coefficient cannot be inferred from that number alone without its Reynolds number, ball geometry, surface condition, and calibrated relation. Report the chosen ball's dimensions and mass rather than mixing approximate radii and masses from different examples.

Dimples can promote boundary-layer transition and delay separation in relevant flow regimes, reducing pressure drag despite additional skin friction. Their benefit depends on Reynolds number, dimple geometry, spin, and surface condition; there is no universal constant coefficient or fixed percentage range gain. The primary wind-tunnel study Bearman and Harvey, Golf Ball Aerodynamics (1976) provides evidence about tested configurations, not a coefficient table for every ball or driver. A smooth sphere's drag coefficient also changes across flow regimes; labeling a single value as its universal laminar coefficient is incorrect.

Backspin changes lift and may also change drag. Depending on launch speed, angle, spin, coefficient curves, and environment, adding spin can increase or decrease carry. Lift perpendicular to relative velocity does not directly do work in that relative frame, but changes the flight path, duration, and drag history. Height, carry, stopping behavior, and dispersion are different outputs. Neither lower drag nor higher lift alone establishes greater accuracy or stability.

15.9 Environment and Model Uncertainty

For dry ideal-gas air, $\rho = p/(R_{\mathrm{air}}T)$ with absolute temperature T. The NASA equation-of-state treatment explains the pressure, density, and temperature relation. At the same pressure, the density ratio between 35 and 15 degrees Celsius is

$$\frac{\rho_{35}}{\rho_{15}} = \frac{288.15}{308.15} \approx 0.93510. \tag{15.23}$$

That is a decrease of about 6.49 percent. Pressure, humidity, and viscosity changes need their own treatment; an altitude alone does not specify a day's air density. A reference atmosphere is a model, not a local weather measurement. Lower density reduces both drag and lift for fixed coefficients and relative speed; the change in ball range requires solving the flight problem.

A 5 m/s headwind exactly opposite a 50 m/s point velocity changes relative speed to 55 m/s and, with all other factors fixed, raises drag magnitude by $(55/50)^2 - 1 = 21$ percent. During a swing the velocity direction changes; wind must be subtracted as a vector at each point, rather than adding a scalar headwind to every segment for the entire motion. Crosswind can change both incidence and the direction of aerodynamic force.

In a scalar coefficient model, away from zero speed and for small changes in the separately represented factors, the differential sensitivity is

$$\frac{\delta D}{D} \simeq \frac{\delta\rho}{\rho} + \frac{\delta C_D}{C_D} + \frac{\delta A}{A} + 2\frac{\delta v_r}{v_r}. \tag{15.24}$$

This is a first-order identity, not a rule for adding uncertainty variances. Covariances matter; C_D and A may be linked by calibration, and coefficient variation with speed adds its derivative when speed changes. Near transitions or rapidly changing orientation, local linear sensitivity can be inadequate. Propagate the joint parameter/measurement uncertainty through the quantity actually reported.

15.10 Choosing the Aerodynamic Model

A useful inclusion decision starts with a scientific output and a tolerance. For example, an aerodynamic-moment uncertainty of a fraction of a newton metre might be immaterial to one question and larger than the claimed effect in another. A universal five-percent threshold, a joint's name, or calm weather cannot make that decision on their own.

1. Specify the measured or predicted output, system boundary, coordinate conventions, and tolerable discrepancy in physical units.

2. Begin with a transparent force and moment model. Record coefficient-area pairs, incidence and Reynolds-number ranges, air properties, wind, and the evidence supporting each parameter.

3. Use compatible segment velocities and map loads by virtual power. Check force and moment signs, zero-relative-speed behavior, units, and energy balance before interpreting the output.

4. Compare omitted, nominal, and plausible alternative aerodynamic models through the same inverse or forward analysis. Preserve the same input policy and initial state when comparing forward branches, or explicitly define a different intervention.

5. Check numerical convergence and compare predictions with independent measurements. Add aerodynamic couples, flexibility, wake interaction, or CFD-derived loading when the simpler model's discrepancy matters for the question and the added detail is supportable.

6. Report the resulting sensitivity and remaining uncertainty. Treat a measured performance difference, a model prediction, an illustrative calculation, and a proposed design change according to their evidence.

This places aerodynamic loading within the broader investigation of the golf swing: it is a physical load that interacts with geometry, constrained dynamics, control policy, and impact conditions. Its importance is an output-dependent result to establish, rather than a fixed percentage or a proof that either muscles or passive dynamics must dominate.

15.11 Chapter Exercises

1. Reproduce the 3.0625 N example. Change the reporting area while preserving force; calculate the new coefficient.
2. Derive shaft force and moment for pivot offsets $a = 0$ and $a = 0.6$ m. Explain why a near-end pivot is not a shoulder model.
3. Derive generalized aerodynamic load from virtual power, including the single revolute-axis moment as a special case.
4. Verify the tailwind example in both velocity frames and identify the energy supplied by the moving air.
5. Reproduce the 10 versus 6 N m inverse example. Explain what changes when contact reactions are also unknown.
6. Verify the pendulum equilibrium, bottom-crossing speeds, and energy loss. Explain why peak speed can occur before the bottom.
7. Compute the temperature and headwind examples. Distinguish the factors held fixed from those that can change outdoors.
8. Compare ball flights with equal launch states and different coefficient curves. Explain why range does not identify prior muscle loading.

Chapter 16

Damping, Friction, and Energy Dissipation in the Kinematic Chain

16.1 The Messy Reality of Motion

The golfer, club and ball exchange mechanical energy while tissues, interfaces and materials can dissipate some of it. Energy remaining in body motion, elastic storage or gravity is not automatically heat. Muscle metabolic expenditure is also distinct from the mechanical work delivered at the modeled joints. A useful damping model starts with a stated system boundary and an energy ledger.

This chapter separates three questions: where mechanical energy is lost, how a local loss changes coupled motion, and what can be inferred about stability, impact or perceived sweetness. Passive loss does not guarantee better control, worse ball speed or a preferred sound. Those outcomes require a specified trajectory, boundary condition and comparison.

16.2 The Hidden Friction in Your Body

16.2.1 Viscous Damping

For a translational coordinate, $F_d = -cv$ has c in N s/m. For an angular coordinate, $\tau_d = -c_\theta \dot{q}$ has c_θ in N m s/rad. The corresponding power is $-cv^2$ or $-c_\theta \dot{q}^2$. The interpretation of a fitted joint coefficient depends on the mechanisms represented by that model; it is not a direct measurement of one tissue's viscosity.

16.2.2 Synovial Fluid Viscosity

Lubrication and deformation at a joint can contribute resistance, but the phenomenological approximation $\tau_{\text{fluid}} = -c_{\text{fluid}} \dot{q}$ requires identification over a declared posture, loading, temperature and frequency range. It does not establish a constant coefficient for a whole golf swing or a universal monotone dependence on joint size.

16.2.3 Muscle Viscosity and the Hill Model

An active muscle can deliver positive mechanical work. Its force-velocity relation must therefore not be relabeled as a passive damper. Hill's force-velocity experiment concerns active contraction [Hill, 1938]; a local slope around a particular activation and operating state is an incremental impedance, not a proof of global passivity. A Hill-type implementation must declare its contraction sign, active and passive elements, moment arms and activation dynamics. A dashpot is a modeling choice, not a mandatory element implied by the historical equation.

For a passive translational dashpot with moment arm r and local shortening speed $v = r\dot{q}$, power consistency gives $c_\theta = cr^2$. One cannot convert N s/m to N m s/rad without geometry. With multiple muscles, the corresponding Jacobian congruence retains cross-joint coupling.

16.2.4 Tendon Viscoelasticity and Hysteresis

For a closed measured force-extension cycle, the loop integral $\oint F\,d\ell$ is the mechanical energy not returned over that cycle. A synthetic storage of 10 J with an explicitly assumed 8% loss would return 9.2 J; this arithmetic supplies no measured tendon percentage or golfer energy budget.

As an illustrative nonlinear loss model, a memoryless quadratic resistance $-b|\dot{q}|\dot{q}$ is passive for $b \geq 0$, but it does not by itself represent viscoelastic memory or identify a measured hysteresis loop across frequencies. Internal-state or hereditary models need their own stored energy, loss law and calibration. Elastic recoil, dissipation and active muscle work remain separate.

16.2.5 Ligament Damping and Cartilage Deformation

Deformation can store and return energy as well as dissipate it. A model must distinguish these contributions before assigning any residual work to friction. Tissue-specific parameters and injury or clinical conclusions require appropriate identification; this chapter supplies neither a universal joint-loss hierarchy nor an injury-risk model.

16.2.6 Soft Tissue Wobble Damping

Relative soft-tissue motion adds inertial and constitutive states, as discussed in Chapter 22. A reduced damping coefficient can approximate a response only in the range over which that reduction is checked. Omitted tissue inertia or resonance must not be concealed by increasing a numerical damping parameter.

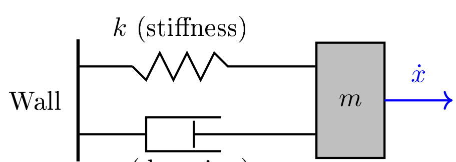
16.2.7 Damping by Joint

There is no qualified table of golfer-wide joint damping coefficients in this chapter. The identification requirements are instead:

Modeled Element	Required Coordinates and Evidence
Shoulder and scapular motion	Declared relative rotations, posture, activation and coupled torque response
Elbow and wrist	Joint-axis definitions, tendon routing and independent perturbation data
Trunk	Distributed or reduced coordinate model and the retained mechanical boundary
Hand-club interface	Colocated force/torque and relative motion, grip pressure, phase and frequency band
Shaft and head	Specimen, boundary conditions, mode shapes, damping and instrument correction

Passive lower-limb studies are not substitutes for measured shoulder, wrist or grip coefficients. Synthetic sensitivity ranges must be labeled as assumptions and cannot be promoted to physiological uncertainty bounds.

16.3 Adding Damping to the Equations of Motion



Spring force: $F_s = -kx$

Damping force: $F_d = -c\dot{x}$

$m\ddot{x} + c\dot{x} + kx = 0$

Figure 16.1: Parallel Spring and Damper Acting on a Translating Mass. Both Elements Share the Same Displacement and Velocity; Their Forces Add. An Angular Analogue Uses Rotational Inertia and Torques.

Recall from Chapter 9 that a kinematic chain with actuator and constraint forces is governed by:

$$\mathbf{M}(\mathbf{q})\ddot{\boldsymbol{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\boldsymbol{q}} + \mathbf{g}(\mathbf{q}) = \boldsymbol{B}(\boldsymbol{q})\boldsymbol{u} + \boldsymbol{J}_c^T(\boldsymbol{q})\boldsymbol{\lambda}$$

This is a second-order force balance. Its first-order drift field is obtained after defining the state and setting the control input to zero; actuator forces belong to the input term. A phenomenological viscous model adds a separate dissipative force:

Key Principle 16.1. Damped Kinematic Chain Equations

$$\boxed{\mathbf{M}(\mathbf{q})\ddot{\boldsymbol{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\boldsymbol{q}} + \boldsymbol{D}(\boldsymbol{q})\dot{\boldsymbol{q}} + \mathbf{g}(\mathbf{q}) = \boldsymbol{B}(\boldsymbol{q})\boldsymbol{u} + \boldsymbol{J}_c^T(\boldsymbol{q})\boldsymbol{\lambda}}$$

where the symmetric viscous model used here assumes $\boldsymbol{D}(\boldsymbol{q}) = \boldsymbol{D}(\boldsymbol{q})^T \succeq 0$. Non-negative diagonal entries alone do not establish passivity for a coupled matrix. The dissipated power is $P_d = \dot{\boldsymbol{q}}^T\boldsymbol{D}\dot{\boldsymbol{q}} \geq 0$.

The matrix $\boldsymbol{D}$ maps the declared generalized velocities to generalized dissipative forces. Geometry, activation and internal material states can change an identified response. Neither configuration dependence nor increasing damping with flexion follows merely from choosing the notation $\boldsymbol{D}(\boldsymbol{q})$.

16.3.1 Types of Damping Models

Linear viscous damping is useful for a local, frequency-limited approximation. A coupled passive form can be parameterized as $\boldsymbol{D} = \boldsymbol{R}^T\boldsymbol{R}$; this preserves nonnegative loss without assuming independent joints. Positive diagonal entries alone do not suffice.

Dry sliding friction, rate-dependent drag and viscoelastic memory are different constitutive models. For an angular coordinate, a quadratic torque law uses a coefficient in N m s^2/rad^2; an aerodynamic translational force coefficient cannot be inserted unchanged. The friction section below states the sliding and sticking conventions explicitly. No universal dominance of linear damping at swing speeds is assumed.

16.3.2 Rewriting in Control-Affine Form

Use local independent joint coordinates with $\boldsymbol{v} = \dot{\boldsymbol{q}}$ and state $\mathbf{x} = (\boldsymbol{q}, \boldsymbol{v})$. Evaluate the coefficient matrices and gravity force at the current state. For a smooth mode without unresolved constraint forces,

$$\begin{aligned}
\dot{\mathbf{x}} &= f_0(\mathbf{x}) + f_d(\mathbf{x}) + G(\mathbf{x})\boldsymbol{u}, \\
f_0 &= \begin{bmatrix} \boldsymbol{v} \\ -\boldsymbol{M}^{-1}(\boldsymbol{C}\boldsymbol{v} + \boldsymbol{g}) \end{bmatrix}, \\
f_d &= \begin{bmatrix} \mathbf{0} \\ -\boldsymbol{M}^{-1}\boldsymbol{D}\boldsymbol{v} \end{bmatrix}, \\
G &= \begin{bmatrix} \mathbf{0} \\ \boldsymbol{M}^{-1}\boldsymbol{B} \end{bmatrix}.
\end{aligned}$$

Every vector field has $2n$ components; joint damping forces enter the acceleration block through the inverse mass matrix. A quaternion or other redundant orientation representation instead needs its declared kinematic map $\dot{\boldsymbol{q}} = \boldsymbol{N}(\boldsymbol{q})\boldsymbol{v}$. In a closed chain, retain $\boldsymbol{J}_c^T\boldsymbol{\lambda}$ in the force balance and solve the compatible constraint equations, or eliminate the reactions consistently. The unconstrained G above cannot simply be reused after that elimination.

16.3.3 Conservative Coupling and Dissipated Power

The manipulator-equation derivation in Tedrake's MIT notes identifies the mass and Coriolis terms together as the contribution of kinetic energy. For a time-independent Lagrangian model with a consistent Coriolis representation,

$$\begin{aligned}
T &= \tfrac{1}{2}\boldsymbol{v}^T\boldsymbol{M}\boldsymbol{v}, \\
\boldsymbol{v}^T\boldsymbol{C}\boldsymbol{v} &= \tfrac{1}{2}\boldsymbol{v}^T\dot{\boldsymbol{M}}\boldsymbol{v}.
\end{aligned}$$

One standard choice makes $\dot{\boldsymbol{M}} - 2\boldsymbol{C}$ skew-symmetric; the matrix $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$ itself is not unique. With conservative potential V and $\mathbf{g}(\mathbf{q}) = \nabla V$, differentiating $E = T + V$ and substituting the force balance gives

$$\dot{E} = \boldsymbol{v}^T\boldsymbol{B}\boldsymbol{u} + \boldsymbol{v}^T\boldsymbol{J}_c^T\boldsymbol{\lambda} - \boldsymbol{v}^T\boldsymbol{D}\boldsymbol{v}.$$

The Coriolis contribution cancels the changing kinetic metric; it is not heat loss. Ideal stationary constraints do no work because $\boldsymbol{J}_c\boldsymbol{v} = 0$. Moving constraints can supply or remove energy, so their work must remain in the ledger. Other physical loss laws require their own power terms.

Linearization about a driven trajectory can contain velocity-dependent terms that affect perturbation growth or decay. Those terms are not, by that fact alone, passive viscous damping. For impact and acoustics, distinguish this inertial redistribution and moving-grip work from identified material/hand losses and acoustic radiation. A quieter or faster-decaying response does not by itself identify which mechanism caused it.

16.4 How Damping Propagates Through a Chain

A key observation is that damping at one joint does not affect only that joint. It propagates through the entire chain via the mass matrix coupling, Coriolis coupling, and constraint coupling.

16.4.1 Direct Effect: Local Damping

The most obvious effect is local. Damping at joint i directly opposes angular velocity at that joint:

$$-D_{ii}\dot{q}_i$$

This is a force contribution. The resulting acceleration requires the mass and constraint solve; other simultaneous forces can still accelerate the joint.

16.4.2 Coupling Through the Mass Matrix

Here is where it gets interesting. Rearranging the damped equation of motion:

$$\ddot{\boldsymbol{q}} = \mathbf{M}(\mathbf{q})^{-1}\left[\boldsymbol{B}\boldsymbol{u} - \mathbf{C}(\mathbf{q},\dot{\mathbf{q}})\dot{\boldsymbol{q}} - \boldsymbol{D}\dot{\boldsymbol{q}} - \mathbf{g}(\mathbf{q})\right]$$

Because $\mathbf{M}(\mathbf{q})$ is *not diagonal*, the terms inside the brackets get "mixed" when inverted. Let's illustrate with a simple two-joint chain: shoulder (joint 1) and elbow (joint 2).

Example 16.1. Shoulder-Elbow Damping Coupling

For a planar two-link illustration with massless links, a point mass m_1 at the first link's end and m_2 at the second link's end, use absolute first-link angle and relative elbow angle q_2. Its exact mass matrix is:

$$\mathbf{M}(\mathbf{q}) = \begin{pmatrix} m_1L_1^2 + m_2(L_1^2 + L_2^2 + 2L_1L_2\cos q_2) & m_2(L_2^2 + L_1L_2\cos q_2) \\ m_2(L_2^2 + L_1L_2\cos q_2) & m_2L_2^2 \end{pmatrix}$$

The off-diagonal terms $M_{12} = M_{21}$ are nonzero and depend on the elbow angle q_2. When we invert this matrix to solve for $\ddot{\boldsymbol{q}}$, we get:

$$\ddot{\boldsymbol{q}} = \mathbf{M}(\mathbf{q})^{-1} \cdot (\ldots - \boldsymbol{D}\dot{\boldsymbol{q}} - \ldots)$$

Suppose we have damping only at the elbow: $\boldsymbol{D} = \mathrm{diag}(0, c_2)$. The damping term entering the inverse is:

$$-\boldsymbol{D}\dot{\boldsymbol{q}} = \begin{pmatrix} 0 \\ -c_2\dot{q}_2 \end{pmatrix}$$

After multiplication by $\mathbf{M}(\mathbf{q})^{-1}$, the shoulder acceleration $\ddot{q}_1$ receives a contribution from this elbow damping force, even though the shoulder has zero damping directly. Quantitatively:

$$(\ddot{q}_1)_d = -(\mathbf{M}(\mathbf{q})^{-1})_{12}c_2\dot{q}_2 = \frac{M_{12}}{\det(\mathbf{M}(\mathbf{q}))}c_2\dot{q}_2$$

This is only the damping contribution to shoulder acceleration. The cross-term $(\mathbf{M}(\mathbf{q})^{-1})_{12}$ couples the two joints; its sign depends on configuration. The determinant already occurs in the inverse and must not be applied a second time. Total passive loss remains $c_2\dot{q}_2^2$, even when the shoulder accelerates in the direction of its own motion.

This is crucial: in a multi-joint chain, you cannot treat each joint's dynamics in isolation. The mass matrix creates a network of inertial coupling. Damping at a distal joint propagates proximally.

16.4.3 Coupling Through Coriolis Effects

The Coriolis force depends quadratically on the joint velocities at a fixed configuration:

$$\boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}}$$

Uniformly scaling all velocities by a factor scales this force by the factor squared. Changing one velocity component or the configuration can instead increase, decrease or reverse individual force components. Damping therefore changes subsequent inertial coupling through the trajectory, but it creates no additional Coriolis dissipation beyond the physical loss terms in Section 16.3.3.

16.4.4 Energy Cascade in Serial Chains

At a prescribed state with diagonal passive joint damping, the local loss is $P_i = c_i\dot{q}_i^2$. The mass matrix affects the resulting accelerations and trajectory, but does not insert an extra downstream inertia multiplier into this instantaneous loss. Neither the proximal nor the distal joint must dissipate more. Relative joint rates must not be replaced by absolute segment rates.

For a synthetic pair with equal $c_i = 0.2$ N m s/rad and rates 10 and 30 rad/s, the losses are 20 and 180 W. Reversing the rates reverses the ranking. Integrating along matched trajectories gives a loss comparison; recomputing a freely evolving swing can also change actuator work and delivered geometry. The previous shoulder/wrist speed estimate and its power ratio are withdrawn because they did not establish those coordinates or measurement conditions.

16.4.5 Constraint Coupling in Parallel Mechanisms

For two hands on different points of a rigid club, hand-point velocities are generally different: a rotating club contributes $\boldsymbol{\omega} \times \mathbf{r}$ at each offset. Express all twists and wrenches in declared compatible frames. Let $\mathbf{q} = (\mathbf{q}_L, \mathbf{q}_R, \mathbf{q}_c)$ include both arms and the club, and let the no-slip closure be $\mathbf{J}_c\dot{\mathbf{q}} = 0$. Ideal reactions are $\mathbf{J}_c^T\boldsymbol{\lambda}$, with

$$P_{\text{constraint}} = \dot{\mathbf{q}}^T\mathbf{J}_c^T\boldsymbol{\lambda} = 0.$$

Separate generalized torques of arms with different coordinate spaces cannot simply be added and set to zero. Club force and moment balance retains its inertia, external loads and both contact wrenches. Ideal internal constraints redistribute work between components without creating extra total loss.

If $\dot{\mathbf{q}} = \mathbf{N}\dot{\mathbf{s}}$ with $\mathbf{J}_c\mathbf{N} = 0$, then $\mathbf{D}_r = \mathbf{N}^T\mathbf{D}\mathbf{N}$ and $\dot{\mathbf{s}}^T\mathbf{D}_r\dot{\mathbf{s}} = \dot{\mathbf{q}}^T\mathbf{D}\dot{\mathbf{q}}$. This proves preserved nonnegative loss while allowing off-diagonal reduced coupling. A changing basis also contributes kinematic terms to the dynamics. Moving constraints have prescribed nonzero velocity and can perform work; compliant or slipping contacts require explicit port laws rather than ideal no-slip closure.

16.5 The Damping Paradox

16.5.1 Damping and Stability

For an autonomous linear perturbation model,

$$\mathbf{M}\ddot{\mathbf{x}} + (\mathbf{G} + \mathbf{C})\dot{\mathbf{x}} + \mathbf{K}\mathbf{x} = 0,$$

assume constant symmetric positive definite $\mathbf{M}, \mathbf{K}$, skew $\mathbf{G}$, and symmetric positive semidefinite $\mathbf{C}$. Then $E = \frac{1}{2}\dot{\mathbf{x}}^T\mathbf{M}\dot{\mathbf{x}} + \frac{1}{2}\mathbf{x}^T\mathbf{K}\mathbf{x}$ is positive definite and $\dot{E} = -\dot{\mathbf{x}}^T\mathbf{C}\dot{\mathbf{x}} \leq 0$. This proves Lyapunov stability. Asymptotic stability additionally requires that the largest invariant set with zero dissipation contain only the equilibrium. Positive definite $\mathbf{C}$ suffices under these assumptions; semidefinite damping can leave undamped modes.

With negative stiffness, even the scalar dimensionless equation $\ddot{x} + \dot{x} - x = 0$ retains the growing rate $(\sqrt{5}-1)/2$. A stronger counterexample has $\mathbf{M} = \mathbf{I}$, $\mathbf{K} = -\mathbf{I}$ and $\mathbf{G} = \begin{bmatrix} 0 & -3 \\ 3 & 0 \end{bmatrix}$. With $\mathbf{C} = 0$ its four distinct rates are $\pm i(3 \pm \sqrt{5})/2$. Adding the passive $\mathbf{C} = 0.1\mathbf{I}$ produces a pair with positive real parts. Energy loss and instability coexist because the equilibrium energy is indefinite. Dissipation-induced instability is a documented phenomenon [Bloch et al., 1994], not a claim that this particular instability occurs in a golf club.

A nonsymmetric loaded tangent, accelerating frame or time-varying swing requires its own stability analysis. Frozen eigenvalues and a nonsingular frequency response do not establish stable time evolution. Physical damping must remain separate from numerical stabilization.

16.5.2 Underdamped vs. Overdamped Joints

For one constant-coefficient scalar oscillator with positive inertia I, stiffness k and damping c, define $\omega_n = \sqrt{k/I}$ and $\zeta = c/(2\sqrt{kI})$. Values $0 < \zeta < 1$, $\zeta = 1$ and $\zeta > 1$ distinguish oscillatory, repeated-real and distinct-real decay rates. At $c = 0$ oscillations do not decay. Critical damping gives the familiar nonoscillatory step response for zero initial conditions; arbitrary initial velocity can still cause a crossing of equilibrium, and no universal optimal energy-dissipation objective follows.

A joint embedded in a coupled active chain need not have a single independently meaningful damping ratio. Identify modes or a matrix impedance over the relevant operating range. The chapter does not infer that all human joints are underdamped or that a wrist swing exploits a particular resonance.

16.5.3 Active Damping and Co-Contraction

Impedance control concerns the dynamic relationship between motion and applied wrench [Hogan, 1985b]. For a local restoring force on the modeled limb, one convention is $\mathbf{f} = -\mathbf{K}(\mathbf{x} - \mathbf{x}_d) - \mathbf{C}(\dot{\mathbf{x}} - \dot{\mathbf{x}}_d)$. The sign reverses for the reaction exerted on its environment. A moving target can perform work even when the relative loss is passive.

Co-contraction can alter an identified impedance, but does not supply a universal independent adjustment of stiffness and damping. The identified change depends on activation, reflex delay, posture and task. Passive tissue loss and work by active control must be retained separately; an assumed change during transition is a hypothesis requiring measurement.

16.5.4 The Loose Grip Debate

The relevant question is how the identified hand-club impedance depends on pressure, posture, frequency and amplitude. A grip can change storage, inertia, damping, contact area and slip together. It is not enough to assign one scalar damper or assume that tighter grip always improves control and reduces ball speed.

Chiementin and colleagues studied one participant and one hybrid club. Their modal results generally show increased bending damping with firmer grip, but their conclusion states the opposite trend. Their numerical impact response also contains algorithmic damping that is not independently separated from material loss. Preserve these limitations; this study does not establish a universal optimum grip pressure or broadband impact sound prediction [Chiementin et al., 2019].

The guitar analogy depends on mode-dependent contact: touching where a mode moves can change its decay, while a node can reduce coupling to that mode. A flexible head and shaft have different modes and radiation efficiencies. A shorter ringdown at the grip is not automatically quieter sound at the listener or a sweeter hit. Compare calibrated sound and hand vibration separately, then use blinded perception experiments.

16.6 Friction Models for Joints

16.6.1 Coulomb Friction

For a scalar angular sliding coordinate, use $\tau_f = -\tau_k \operatorname{sign}(\dot{q})$ when $\dot{q} \neq 0$, with $\tau_k \geq 0$ in N m. At a simple interface, $\tau_k = \mu_k N r$ requires the effective moment arm r. Distributed contact requires integration of traction and lever arm. The product $\mu_k N$ alone is a force, not a joint torque.

16.6.2 Stiction and Reversal Points

At zero slip velocity, static friction is set-valued: $\tau_f \in [-\tau_s, \tau_s]$. Solve force balance and the sticking constraint together; slip begins when the required reaction exceeds the admissible bound. Setting $\operatorname{sign}(0) = 0$ cannot represent nonzero sticking force. Different joints need not reverse simultaneously, and a swing transition is not evidence that an anatomical joint behaves like a dry bearing.

16.6.3 The Stribeck Effect

An explicitly phenomenological sliding law is

$$\begin{aligned} h(v) &= \tau_k + (\tau_s - \tau_k)e^{-(|v|/v_s)^2}, \\ \tau_f(v) &= -h(v)\,\mathrm{sign}(v) - cv, \qquad v \neq 0. \end{aligned}$$

Here $\tau_s \geq \tau_k > 0$, $v_s > 0$, and $c \geq 0$. The dry-friction magnitude approaches τ_s at vanishing nonzero speed and τ_k at high speed; the viscous term can still grow. Power is $-h(v)|v| - cv^2 \leq 0$. The separate zero-speed inclusion handles sticking. This fixes the reversed limits in the previous expression and does not identify a golfer's joint law.

16.6.4 LuGre Friction Model

A bristle-state model can represent presliding memory [Canudas de Wit et al., 1995]. Under a rotational convention with z in rad and $h(v)$ in N m, one possible notation is

$$\dot{z} = v - \sigma_0|v|z/h(v), \qquad \tau_{\text{resistance}} = \sigma_0 z + \sigma_1\dot{z} + \sigma_2 v.$$

The torque applied against motion is $-\tau_{\text{resistance}}$. State storage means instantaneous returned power need not be zero: passivity must be checked against a declared storage function and admissible parameters. Define initialization, units, steady sliding and numerical handling before use. This is a model option, not a validated anatomical law; its necessity depends on the measured phenomena and validation target.

With $\sigma_0 > 0$ in N m/rad and $\sigma_1, \sigma_2 \geq 0$ in N m s/rad, the candidate storage $S = \frac{1}{2}\sigma_0 z^2$ gives

$$\tau_{\text{resistance}} v - \dot{S} = \frac{\sigma_0^2|v|z^2}{h(v)} + \sigma_1\dot{z}\,v + \sigma_2 v^2.$$

For $\sigma_1 = 0$ this is nonnegative. With bristle damping, the cross term need not be nonnegative, so coefficient positivity alone does not make this storage a certificate for the full output. For a synthetic instantaneous state with $\sigma_0 = \sigma_1 = 1$, $\sigma_2 = 0$, $z = 0.2$, $v = 1$ and $h(v) = 0.1$ in the stated units, $\dot{z} = -1$ and the right-hand side is -0.6 W. Check the permitted initial states and constitutive operating range before accepting or rejecting a complete passivity claim.

16.6.5 Friction Criticality at the Transition

At sticking, friction can support a force with zero mechanical power. During sliding, loss depends on slip displacement and traction, not on counting reversals. A smooth-looking or jerky transition alone cannot quantify dissipation. No universal 20 ms transition, simultaneous peak tissue stretch or optimal friction level is inferred here. Measure motion, forces and internal states over a defined interval before assigning a transition loss.

16.7 Energy Dissipation Budget

16.7.1 Sources of Dissipation

For a closed mechanical model with explicitly accounted boundary inputs,

$$E(t_1) - E(t_0) = W_{\text{actuators}} + W_{\text{boundary}} - E_{\text{losses}}.$$

Include translational and rotational kinetic energy of every retained component, gravity, strain energy and internal constitutive storage in E. Define the system boundary before allocating aerodynamic, interface and radiated-acoustic energy: a flux leaving one subsystem can be stored in another. Avoid counting the same fitted residual as both tissue damping and grip loss.

16.7.2 Interpretation

For a synthetic ledger, $E(t_0) = 20$ J, actuator work 100 J, moving-boundary work 5 J and losses 15 J give $E(t_1) = 110$ J. If clubhead kinetic energy is 40 J, the other 70 J may remain in the modeled body, shaft and potential energy. It is not an additional 70 J dissipation. These numbers are an accounting example, not an amateur-golfer energy budget or a metabolic efficiency estimate.

16.8 Implications for Swing Modeling

16.8.1 When Damping Is Negligible

Neglect a loss only after a sensitivity or error-budget check for the quantity of interest. A small change in ball speed can coexist with a large change in a narrow resonance or ringdown. A zero instantaneous velocity removes a viscous power term at that instant; it does not establish negligible damping over the surrounding interval or eliminate static forces.

16.8.2 When Damping Is Critical

The importance of a loss depends on the excited modes, their frequencies and motion at dissipative ports, the duration of contact and the observation window. Separate pre-impact trajectory changes, forces during contact and vibration after separation. Post-separation damping cannot change the launch state already established, although it can change sound and feel.

16.8.3 Sensitivity Analysis

For a declared scalar damping parameter d and reproducible comparison protocol,

$$\Delta v_{\text{club}} \approx \frac{\partial v_{\text{club}}}{\partial d} \Delta d.$$

There is no extra minus sign: the derivative already carries its sign. A synthetic derivative of -0.02 m/s per N m s/rad with $\Delta d = 5$ N m s/rad gives -0.1 m/s, approximately -0.224 mph using 1 mph $= 0.44704$ m/s. This is neither a measured derivative nor a 2 mph effect. State whether controls, delivery state, trajectory or external impulse are held fixed, and check finite-difference step convergence where used. A chosen 50% parameter sweep is scenario analysis, not a confidence interval.

16.8.4 Recommendations for Modelers

Model choice depends on the requested observable and on reproducing the specified measurements within their uncertainty. Record coordinates, force signs, units, boundary work, source data and the qualified frequency/amplitude range. Distinguish calibration from held-out validation and report null effects.

Separate physical damping from integration damping; refine time step, mesh and retained modes without retuning coefficients to hide numerical error. Test stationary, moving-support, free and clamped limits. Inspect stability and passivity independently of solver convergence. Unknown losses must remain unknown rather than becoming default golfer coefficients.

16.9 Damping in the Shaft

16.9.1 Material Damping

A recorded decay can combine shaft material loss, head and adhesive loss, grip/fixture dissipation, radiation and instrumentation. It is not automatically intrinsic material damping. Identify the specimen, layup, mode and boundary condition. A material name or flex rating alone cannot rank full-club decay or ball speed.

16.9.2 Quality Factor (Q-Factor)

For a lightly damped isolated mode, $Q \approx 1/(2\zeta)$ relates a conventional quality factor to modal damping ratio. With $x(t) = Ae^{-\zeta\omega_n t}\cos(\omega_d t + \phi)$ and $\omega_d = \omega_n\sqrt{1-\zeta^2}$, the logarithmic decrement is $\delta = 2\pi\zeta/\sqrt{1-\zeta^2}$. The energy lost between corresponding peaks is a fraction $1 - e^{-2\delta}$. Thus defining Q directly as 2π times starting energy divided by that finite-cycle loss agrees with $1/(2\zeta)$ only in the weak-loss limit.

For a synthetic 100 Hz mode with $\zeta = 0.01$, the conventional $Q = 50$, the amplitude time constant is about 0.159 s, and the corresponding peak-to-peak energy loss is about 11.8%. Higher Q means weaker fractional damping under these conventions; it does not mean higher stiffness. The previously unsourced steel/graphite ranges and contradictory material ranking are withdrawn.

16.9.3 Lead-Lag Dynamics and Damping

Before impact, mass, stiffness, pre-existing shaft deformation and velocity, applied wrenches and losses determine the delivered state together. After impact, observed motion depends on which bending, torsional and head modes were excited. A claim that shaft damping is negligible during the downswing needs a qualified calculation or measurement for that model and observable.

16.9.4 Practical Implications

For a frozen small perturbation with point-motion map $\mathbf{L}$ and the $e^{i\omega t}$ convention, the impact-point mobility is

$$\mathbf{Y}(\omega) = i\omega \mathbf{L}\left[\mathbf{K} - \omega^2 \mathbf{M} + i\omega(\mathbf{G} + \mathbf{C})\right]^{-1} \mathbf{L}^T.$$

The transpose maps point wrench back to generalized force and preserves power. Translational and rotational rows have different units. Replacement of a constant-coefficient boundary by an identified frequency-dependent grip impedance depends on its qualification, but its storage and loss must not be counted twice. At a finite preload, differentiating the configuration-dependent port map also contributes geometric tangent terms. A singular or unresolved pencil requires refusal or a qualified alternative formulation, not an arbitrary pseudoinverse.

16.9.5 Material Damping and Shaft Behavior

Rotation-induced stress can change the tangent stiffness, while Coriolis terms change dynamic coupling. Tension, centrifugal terms, bending and the grip boundary must be derived from one consistent loaded configuration; they do not universally increase every frequency or create extra dissipative loss. A held pose in a rotating observer does not imply a stationary support in the laboratory.

The implementation program in Tools #5072 retains these distinctions in its loaded chain and finite grip ports. Its synthetic balance and operator tests establish numerical verification only. Contact duration and offset, flexible-head response, acoustic radiation, microphone transfer and physical validation remain separate steps. A claim that one player's strike is sweeter requires matched-delivery analysis and blinded listening evidence, not a ranking of shaft material names.

16.10 The Control-Affine Perspective on Damping

In the control-affine framework, damping modifies the drift field. Recall:

$$\dot{\boldsymbol{x}} = f_0(\boldsymbol{x}) + G(\boldsymbol{x})\boldsymbol{u}$$

For the independent-coordinate viscous model in Section 16.3.2, the drift field becomes:

$$f_{\text{damped}}(\mathbf{x}) = f_0(\mathbf{x}) + \begin{bmatrix} \mathbf{0} \\ -\mathbf{M}(\mathbf{q})^{-1}\boldsymbol{D}\boldsymbol{v} \end{bmatrix}.$$

Configuration-dependent passive coefficients may represent identified losses within their calibration range. Coulomb friction, viscoelastic memory and active muscle dynamics need their own constitutive states or force laws; they cannot generally be folded into a constant viscous matrix. Coriolis terms remain in the conservative drift and are excluded from the dissipated-power budget.

Drift vs. Control 16.1. Damping as Drift Modification

For an unforced system with stationary ideal constraints, passive damping makes total mechanical energy nonincreasing. Gravity and joint coupling can still increase an individual joint's speed, and energy decay alone does not prove stability of every equilibrium or driven trajectory. When a particular trajectory is prescribed, inverse dynamics determines the additional actuator work associated with damping. This bookkeeping does not establish a universal muscular strategy for reversing the swing.

16.11 Summary: The Damping Principle

Passive loss, energy redistribution, active work and numerical damping are distinct. Coupling changes which motions and ports are excited, while geometry and operating state change the loaded response. Neither energy decay nor a successful solve establishes stability, a launch advantage or perceived sweetness. Report those quantities with their own assumptions and evidence.

16.12 Chapter Exercises

All numerical inputs below are synthetic verification cases, not golfer parameter estimates.

- **Damping Coefficient Estimation.** For this synthetic case, use $c_w = 0.1$ N m s/rad and $\dot{q}_w = 50$ rad/s. Find the resisting torque and instantaneous loss power. Compare the torque magnitude with an assumed available 20 N m; explain why this fraction is not a metabolic efficiency.
- **Energy Dissipation in a Two-Joint Chain.** For this synthetic case, use rates 10 and 30 rad/s and coefficients 0.5 and 0.15 N m s/rad. Compute total loss power and its ratio to an assumed 400 W mechanical actuator input. Explain why joint location alone does not determine the ratio.
- **Damping Ratio and Step Response.** For this synthetic case, a scalar angular oscillator has inertia $I = 0.05$ kg m^2, stiffness $k = 5$ N m/rad and damping $c = 0.2$ N m s/rad. Compute ω_n, ζ and the approximate 2% step-response settling estimate $4/(\zeta\omega_n)$. State its assumptions and distinguish rotational inertia from mass.
- **Coulomb vs. Viscous Friction.** For this synthetic case, an idealized contact has $\mu_k = 0.1$, normal force 50 N and moment arm 0.02 m. Find its sliding torque and the angular speed at which $c = 0.5$ N m s/rad gives equal viscous torque. Explain why zero-speed sticking needs a force-balance inclusion.
- **Configuration-Dependent Damping.** For this synthetic case, on the explicitly restricted interval $0 \le q \le 1$ rad, let $c(q) = 0.1 + 0.02q$ N m s/rad and $\dot{q} = 50$ rad/s. Compute torque at $q = 0, 0.5, 1$. Check nonnegative loss and distinguish this assumed law from measured posture dependence.
- **Stability Counterexample.** Calculate the roots of $\ddot{x} + \dot{x} - x = 0$ and identify why the positive-stiffness energy argument fails. Repeat for the two-coordinate gyroscopic example with and without $0.1\mathbf{I}$ damping.

- **Grip and Sound.** Design a comparison that separates changed delivery, changed impact-point mobility, radiated sound and listener preference. State which data would falsify a proposed grip-damping mechanism and which quantities a shaft accelerometer alone cannot identify.

Chapter 17

Impact: The Collision That Matters

17.1 The State Delivered to the Ball

Impact converts part of the club's incident mechanical energy into ball translation, spin, deformation, and dissipation. The golfer and club retain energy afterward; the entire swing energy is not transferred in the collision. Launch also depends on the local face normal, contact-point velocity, strike location, ball and face properties, and the club's response during contact. Ball speed, launch direction, launch elevation, and the full spin vector are needed to initialize a flight model. Three scalars omitting speed and spin axis cannot determine the trajectory.

For the rigid clubhead, let O be its centre of mass and P the contact point:

$$v_P = v_O + \omega_c \times r, \qquad r = p_P - p_O. \tag{17.1}$$

Both velocities and all offsets must use the same frame. A path reported at the face centre need not equal the centre-of-mass path. Face attitude is a different quantity from velocity; static loft plus attack angle does not determine dynamic loft. Declare whether delivery is measured immediately before contact, at maximum compression, or at another instant.

17.2 Normal Impact: Momentum, Restitution, and Energy

For an isolated one-dimensional collision with stationary ball, club mass M, ball mass m, and incident club speed v, momentum and restitution give

$$Mv = Mv_c^+ + mv_b^+, \tag{17.2}$$

$$e = \frac{v_b^+ - v_c^+}{v}. \tag{17.3}$$

This is relative separation speed divided by relative approach speed. A coefficient from a fixed-surface rebound experiment requires its own setup definition before being compared with a moving-club result. Solving gives

$$v_b^+ = \frac{(1+e)M}{M+m}v, \qquad v_c^+ = \frac{M-em}{M+m}v. \tag{17.4}$$

For $M = 0.200$ kg, $m = 0.04593$ kg, and illustrative $e = 0.83$, the speed ratio is about 1.488. This explains the scale of driver smash factor within the model. It is not a universal equipment ceiling or a direct energy ratio.

The ball's fraction of initial club translational energy is

$$\frac{E_b^+}{E_c^-} = \frac{m}{M}\left[\frac{(1+e)M}{M+m}\right]^2, \tag{17.5}$$

about 0.508 for those values. The total kinetic-energy loss is

$$E^- - E^+ = \tfrac{1}{2}\frac{Mm}{M+m}(1-e^2)v^2. \tag{17.6}$$

Thus e^2 describes retained relative-motion energy in the centre-of-mass frame for this model; it is not the fraction of club energy delivered to the ball. Oblique impact also partitions energy into rotation and deformation. These distinctions underlie standard golf impact treatments [Jorgensen, 1994a, Penner, 2003b].

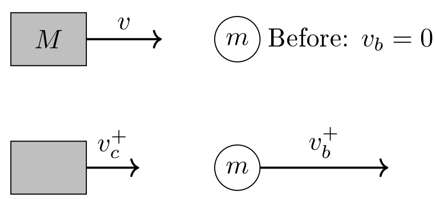


Figure 17.1: Isolated One-Dimensional Impact Before and After Contact. Momentum and Relative-Speed Restitution Determine Both Outgoing Velocities.

17.3 Effective Mass and Off-Centre Impact

For a free rigid clubhead, an impulse $J\hat{n}$ produces translational and rotational changes. With inertia tensor I_c about its centre of mass,

$$\frac{1}{M_{\text{eff}}} = \frac{1}{M} + (r\times\hat{n})^T I_c^{-1}(r\times\hat{n}). \tag{17.7}$$

The second term is nonnegative, so $M_{\text{eff}} \leq M$. It vanishes when the impulse line passes through the centre of mass, even if the contact point is some distance in front of it. Effective mass is directional inertial response, not the amount of material physically reached by a wavefront.

For a multibody model with mass matrix M_q and contact Jacobian J_P, the corresponding directional inverse mass is $\hat{n}^T J_P M_q^{-1} J_P^T \hat{n}$, modified by active constraints. Flexible modes, boundary conditions, and contact duration require a compatible dynamic model. There is no general rule that shaft and arm coupling always add 10–20 percent to clubhead effective mass.

Off-centre impact changes head rotation, contact velocity, and tangential impulse. This produces gear effect. Local face bulge and roll also change the normal, while face compliance can vary with strike position. Ball-speed loss cannot universally be assigned to only one of these mechanisms.

17.4 Oblique Contact and Spin

Let $\hat{n}$ point from face into ball and split impulse into normal and tangential components. Neglecting other impulses over the short contact,

$$m\Delta v_b = J_n\hat{n} + J_t, \tag{17.8}$$

$$I_b\Delta\omega_b = r_b \times (J_n\hat{n} + J_t). \tag{17.9}$$

For an ideal sphere of radius R, $r_b = -R\hat{n}$. A centred normal impulse then produces no spin, while the tangential impulse produces angular momentum. A finite contact patch can also transmit a torsional moment that the point contact model omits.

Coulomb friction supplies a bound $\|F_t\| \le \mu F_n$, with equality in ideal gross sliding. Sticking does not imply zero friction. Tangential deformation, partial slip, and restitution can continue after gross slip ceases, so a rigid rolling condition does not universally fix outgoing spin. Contact properties must be identified at relevant speed, loading, and surface conditions. A low-speed ball-on-granite experiment is not a direct driver calibration.

More obliqueness does not guarantee monotonically increasing spin. Sliding, normal impulse, tangential compliance, moisture, and strike location can change the trend. Grooves help manage contamination, but a universal spin increment from groove count or loft alone is not supplied by this model.

17.5 Exact Spin-Loft Geometry

Use a forward/up/right frame, face azimuth ϕ, path azimuth ψ, dynamic loft λ, and attack angle α. The unit vectors are

$$\hat{n} = (\cos\lambda\cos\phi, \sin\lambda, \cos\lambda\sin\phi), \tag{17.10}$$

$$\hat{v} = (\cos\alpha\cos\psi, \sin\alpha, \cos\alpha\sin\psi). \tag{17.11}$$

Spin loft is their angle:

$$\cos\Phi = \sin\lambda\sin\alpha + \cos\lambda\cos\alpha\cos(\phi - \psi). \tag{17.12}$$

For zero face-to-path, $\Phi = |\lambda - \alpha|$ in the relevant range. Otherwise, multiplying $\cos(\phi - \psi)$ by $\cos(\lambda - \alpha)$ is not the exact identity. Nor can a horizontal path angle be subtracted from loft to obtain spin loft.

With measured dynamic loft 12°, attack 2.5°, and square face-to-path, spin loft is 9.5°. If an illustrative vertical launch weighting is 0.84 toward the normal and 0.16 toward the velocity, launch elevation is $0.84(12) + 0.16(2.5) = 10.48°$. The weighting is empirical, not a universal law. Spin rate and carry require additional contact and flight inputs. With face azimuth fixed, changing path from $-3°$ to $+3°$ preserves spin loft by symmetry but reverses the ideal horizontal spin tendency; it does not subtract three degrees of loft.

For a centred symmetric impact on a nonspinning ball, the ideal spin direction is parallel to $\hat{v} \times \hat{n}$. The cross product degenerates at zero obliquity. Use its projections in a declared launch frame to calculate axis tilt; a tan/tan rule is not a general three-dimensional identity. Gear effect and asymmetric or rotating contact require additional terms.

In a normal-only model with $C = (1 + e)/(1 + m/M_{\text{eff}})$, $v_b = C(v_P \cdot \hat{n})\hat{n}$. Total ball speed scales with $\cos\Phi$; its projection onto incident velocity scales with $\cos^2\Phi$. Those are different outputs. A tangential impulse modifies both direction and speed.

17.6 How Isolated Is the Collision?

The relevant comparison is impulse. An external force $F(t)$ contributes $\int F\,dt$, which must be compared with the ball-contact impulse in the same direction and model boundary. For an illustrative 300 N acting unchanged over 0.5 ms, the impulse is 0.15 N s. A 0.04593 kg ball leaving at 67 m/s gains approximately 3.08 N s. This comparison is a scale estimate, not a measured grip-to-ball sensitivity: grip forces may not reach the head with that magnitude or direction.

Axial wave speed $c = \sqrt{E/\rho}$ in a uniform slender rod differs from frequency-dependent bending-wave propagation. Wave arrival time establishes neither force amplitude nor the resulting change in launch speed. Existing shaft deformation, head inertia, grip impedance, reflection, and contact dynamics must be modeled together. An assumed spring deflection multiplied by stiffness cannot by itself prove a percentage change in effective mass.

The short collision leaves no time for a new consciously selected response to that collision to influence it. Existing activation, preprogrammed commands, intrinsic tissue forces, and mechanical boundary conditions nevertheless remain. This distinction is especially important in the preceding tens of milliseconds: brief ball contact does not establish an unactuated final 50–100 ms of downswing.

17.7 A Precise Counterfactual Question

A smooth pre-impact model may be written $\dot{x} = f(x) + G(x)u$ for declared state, inputs, and constraints. The collision can be represented by a reset $x^+ = \Delta(x^-)$ or by resolved compliant contact. It is not automatically the same smooth drift field continued through an impulsive event.

To isolate earlier actuation, branch before impact, specify which torques or excitations are removed, and evolve each branch with its own contact event. To isolate grip boundary mechanics during impact, hold the pre-impact state fixed and vary only the declared boundary model. These are different experiments. A peak collision-force/grip-force ratio is not a bound on reachable states, a work fraction, or evidence that muscle control has ceased.

17.8 Measurement and Model Checks

1. Declare the contact point, normal, reference point for club velocity, and measurement instant. Include their uncertainty and time synchronization.

2. Compare impulse integrals with linear and angular momentum changes. Do not infer moment solely from a force magnitude without its line of action.

3. Check both bodies' post-impact energy. A fitted restitution coefficient must not create unexplained energy in an otherwise passive model.

4. Validate ball speed, launch angles, spin magnitude and axis together. Fitting one outcome does not validate the others.

5. Vary integration step and contact resolution for compliant models; report convergence of impulses, energy residual, and launch conditions.

6. Separate equipment conformance tests from ball-contact experiments. Characteristic time from a steel pendulum is not golf-ball contact duration.

17.9 Connecting Swing Mechanics to Ball Flight

Earlier joint work and energy transfer help create incident club motion. Geometry and wrist control help determine local contact velocity and face attitude. Shaft deformation contributes to the delivered state and boundary response. The collision maps those inputs into launch; aerodynamics, wind, and the landing surface determine later flight and roll. This chain explains why a swing theory should predict a distribution of impact states and measured launch outcomes, rather than declaring one isolated motion variable optimal.

17.10 Exercises

1. Use $M = 0.200$ kg, $m = 0.04593$ kg, $v = 50$ m/s, $e = 0.83$ to compute both outgoing velocities, the ball's energy fraction, and total energy loss. Verify momentum conservation independently.

2. At fixed inertia, compare directional effective mass for an impulse through the centre of mass and one with a perpendicular offset. Explain why distance from the face to the centre of mass alone is insufficient.

3. Compute exact spin loft for $\lambda = 40°$, $\alpha = 20°$, and face-to-path $30°$, then quantify the error in the product-of-cosines shortcut.

4. Distinguish total ball speed from its forward projection at $30°$ obliquity in the normal-only model.

5. Design a matched-state grip-boundary experiment. Specify which measured changes would falsify the claim that grip mechanics are negligible at the chosen accuracy.

Chapter 18

Swing Plane, Clubface Control and Launch Optimization

In Plain Language 18.1. From a Swing to a Shot

Clubhead speed is one part of delivery. Two swings with the same speed can produce different ball speeds, launch directions, spin vectors and landing positions because their face orientations, strike locations and contact conditions differ. A useful explanation follows the whole chain: preparation and continuing actuation produce club motion; collision produces ball motion; airflow and gravity produce flight; the chosen target determines whether the result is useful. Geometry connects these stages, but cannot replace their mechanics or their evidence.

This chapter develops that chain with explicit coordinates and independently checkable examples. The geometric identities and impulse balances are derived results under stated assumptions. The launch-ratio observations are measurements from a particular study. The quadratic distance surface introduced later is a deliberately manufactured example, not a fitted golf-ball model or a suggested launch window. These distinctions let the reader reuse the mathematics without mistaking a teaching example for a validated prescription.

18.1 Defining the Swing Plane

Use a right-handed frame with x horizontal toward the target, y horizontal to the target's left and z upward. Azimuths in this chapter are positive to the left. Convert a device's positive-right angles by changing the sign after checking its handedness and target convention. Angles inside trigonometric functions are in radians unless explicitly marked in degrees.

Let $r_C(t)$ and $v_C(t)$ denote the position and velocity of a specified point on the clubhead. A geometric center, center of mass and face contact point are generally different points. For a rigid head,

$$v_C = v_G + \omega_H \times (r_C - r_G).$$

Consequently, a change of reporting point can change speed, path and attack angle even when the underlying motion is identical. A flexible shaft does not invalidate rigid-head transport, but its deformation changes the head's pose and velocity. Specify the reporting epoch too: first touch, maximum compression and separation are not interchangeable events.

Definition 18.1. A Plane Requires More Than One Direction

A plane through r_0 with unit normal n_P satisfies $n_P^T(r - r_0) = 0$. An instantaneous velocity supplies one tangent direction, not a unique plane. Adding the vertical direction defines a vertical plane, whose inclination to horizontal is 90°; it does not define an arbitrary tilted swing plane. The inclination of a non-oriented plane can be reported as $\beta = \arccos(|n_P^T e_z|)$ in $[0, 90°]$.

An address construction might use a shaft line and a separate noncollinear point, or the shaft and a horizontal target direction. Name that construction. A ball position alone does not resolve the plane if it lies on the chosen line. A shoulder-and-ball plane can be constructed from three noncollinear points, but it changes when those points move. None of these planes must coincide with the plane fitted to a segment of clubhead motion.

For a measured arc, subtract the sample mean from its three-dimensional positions and form a covariance matrix. Its least-variance eigenvector is a least-squares plane normal. Report the fitting interval, point definition, weighting, residual distances and sensitivity to measurement error. A nearly straight short arc does not identify a stable normal: two small eigenvalues leave many planes almost equally plausible. An instantaneous osculating plane instead uses $v_C \times a_C$, when that cross product is nonzero. It need not equal the finite-arc fit, and differentiated acceleration amplifies noise.

Trackman's published swing-plane metric uses a finite segment of clubhead motion, whereas attack angle describes a velocity direction at an impact reference event. These are different measurements. The vendor also cautions against using its swing-plane value as a direct lie-fitting measurement [Trackman, 2024, 2020].

18.1.1 Why the Swing Plane Tilts

Key Principle 18.1. Geometry Constrains the Available Motion

Club length, stance, reach, posture and hand path affect the motions a golfer can make. They do not supply a universal formula mapping height or shoulder tilt to a unique delivery plane. A plane fitted after observing the motion is a description of that motion, not proof of which anatomical variable caused it.

The conventional club lie angle is measured between the shaft and the horizontal ground or waterline in the specified reference configuration. It is not measured from vertical. Thus a 57° lie corresponds to a 33° complement to vertical. Forward shaft lean and sideways lie are distinct aspects of three-dimensional orientation. Dynamic lie, static lie, face orientation and arc inclination should therefore be reported separately. Shaft bending can make the local shaft direction differ along its length.

Shorter clubs often accompany a more upright delivery, but player adaptation and the metric definition matter. A static specification cannot establish the attack angle, the energy transferred to the ball or an optimal technique. Comparisons should match the task, define the arc and report actual data instead of assigning all golfers a club-dependent plane table.

18.1.2 Attack Angle and the Vertical Component of Club Path

Definition 18.2. Attack Angle and Path From the Same Velocity

For nonzero horizontal speed, define

$$\alpha = \operatorname{atan2}(v_{Cz}, \sqrt{v_{Cx}^2 + v_{Cy}^2}), \qquad \psi = \operatorname{atan2}(v_{Cy}, v_{Cx}).$$

Attack angle α is positive upward and negative downward. Path azimuth ψ is positive left in this frame. Attack angle is the elevation of a velocity vector, not an acceleration, a shaft angle or a displacement from the top of the backswing. At zero speed the direction is undefined; at purely vertical speed the azimuth is undefined.

A finite drop divided by a finite horizontal displacement gives the slope of a chord. It does not determine the instantaneous tangent at impact. To connect path with an ideal fixed swing plane, let its horizontal base direction have azimuth d and define orthonormal in-plane vectors

$$b = (\cos d, \sin d, 0)^T, \qquad u = (-\sin d \cos\beta, \cos d \cos\beta, \sin\beta)^T.$$

Choose the unit tangent $\hat{v}_C = \cos\chi\, b + \sin\chi\, u$, where χ describes its direction within that plane. Direct projection gives

$$\sin\alpha = \sin\beta \sin\chi, \qquad \psi - d = \operatorname{atan2}(\cos\beta \sin\chi, \cos\chi).$$

Equivalently, on the forward-moving branch for $0 < \beta < 90°$,

$$\tan\alpha = \tan\beta \sin(\psi - d).$$

This is a conditional geometric identity. It does not require constant speed or prove that a real head follows one plane. For small χ, $\alpha \simeq \sin\beta\,\chi$; replacing $\sin\beta$ by β additionally requires small plane inclination, which is inappropriate for a $60°$ plane.

At $\beta = 60°$ and $\alpha = -5°$, the forward solution is $\psi - d = -2.8953°$: the path is right of the plane's base direction in the stated coordinate convention. At the bottom of an ideal circular arc, $\chi = 0$, so attack is zero and path equals the base direction. Moving the impact location along that fixed arc changes both quantities together. Moving the arc itself is a different intervention.

18.1.3 Variation by Club: Driver vs. Short Irons

An upward driver strike and a downward iron strike are possible delivery choices, not definitions of the clubs. The sign and useful magnitude depend on the lie of the ball, intended flight, strike quality and equipment. Population averages are not fitting targets. Even a fixed-plane identity cannot determine performance without face orientation and collision mechanics.

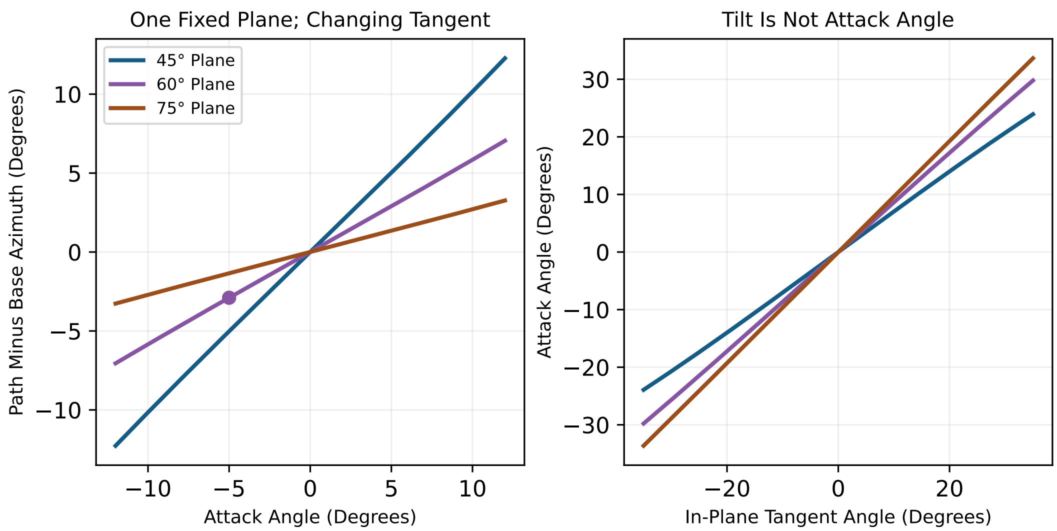


Figure 18.1: Fixed-Plane Geometry Links Path and Attack Angle. Curves Use the Declared Positive-Left Frame and Forward Tangent Branch; They Are Geometric Calculations, Not Measured Swings.

Quantity	**Required Comparison**
Static Lie	Club specification, reference sole position and shaft axis.
Dynamic Lie	Local shaft direction at a declared event and reference plane.
Swing Plane	Point trajectory, fitting interval, inclination and residual.
Attack Angle	Elevation of the declared point's instantaneous velocity.
Club Path	Horizontal azimuth of that same velocity in the target frame.

Table 18.1: Separate Definitions Before Comparing Clubs or Golfers. None of These Quantities Substitutes for Another.

Reference 18.1. Source Contract for Launch and Direction Numbers

An instrument's reported value combines observations, calibration, processing and sometimes a fitted model. Identify the device, mode, point, event and uncertainty rather than describing every output as directly measured. The Wood study below supplies bounded launch-ratio evidence. The distance examples later in this chapter are explicitly manufactured response surfaces. They are not vendor regressions, population norms or empirical fitting advice.

18.2 Clubface Control in Three Dimensions

The local face normal controls the direction of the contact normal force. It is essential to the collision, but cannot by itself determine the shot. The local surface velocity, head mass and inertia, ball properties, impact offset, friction and deformation also matter. On a curved driver face, the normal at the strike location need not equal the normal at the face center.

Key Principle 18.2. Face Orientation and Joint Contributions

Let $R_H \in SO(3)$ map head coordinates to the reporting frame, and let n_0 be a unit face normal in head coordinates. For a fixed local surface point, $n = R_H n_0$. Define face azimuth and normal elevation by

$$\phi = \operatorname{atan2}(n_y, n_x), \qquad \ell = \operatorname{atan2}(n_z, \sqrt{n_x^2 + n_y^2}).$$

The latter is the local dynamic loft convention used here. These two angles determine a normal direction, not the entire three-dimensional head orientation. A third independent direction, such as a groove direction, completes the frame. Toe-up/heel-up rotation and shaft-axis rotation are not definitions of face azimuth. Near a vertical normal, azimuth becomes ill-conditioned.

A planar approximation may express delivered loft using static loft and shaft lean, with carefully declared signs. In three dimensions, ordered rotations replace unrestricted scalar addition. Shaft twist, bending, grip geometry and both arms enter through their actual kinematic relations, not through fixed percentages of a final face angle.

18.2.1 Face Angle Determination: The Joint Contributions

For an illustrative serial chain, write

$$R_H(q) = R_1(q_1)R_2(q_2)\cdots R_k(q_k)R_{\text{grip}}R_{\text{head}}.$$

The axes and frame of each rotation must be specified. A real two-handed grip creates a closed-chain constraint; joint coordinates cannot generally be varied independently while preserving both contacts. Compliance adds state variables and changes the relation between hand and head frames.

Finite rotations do not commute. Acting on $(1,0,0)^T$, a 90° rotation about y followed by 90° about x gives $(0,1,0)^T$. Reversing the order gives $(0,0,-1)^T$. A scalar sum of the two angles cannot distinguish these different normals. Neither forearm pronation nor wrist deviation has a universal signed contribution independent of arm, grip and club orientation.

18.2.2 Sensitivity Analysis: Joint Angle Errors

For a small spatial rotation $\delta\theta$ about unit axis k, $\delta n = (k \times n)\delta\theta$. Differentiating face azimuth yields

$$\frac{\partial \phi}{\partial \theta} = \frac{n_x(k \times n)_y - n_y(k \times n)_x}{n_x^2 + n_y^2} = k_z - \tan \ell (k_x \cos \phi + k_y \sin \phi).$$

The derivative is dimensionless when both angles use the same units. It is local: a large finite error requires recomputing the orientation. The axis k is the joint axis transported to the reporting frame at the current configuration, not merely its anatomical name.

Example 18.1. A Joint's Axis Changes Its Face-Angle Gain

At $\phi = 0$ and $\ell = 15°$, rotations about spatial z, x and y have azimuth gains 1, $-\tan 15° \simeq -0.26795$ and 0 respectively. These are geometric examples, not estimates of three anatomical joints. A zero local azimuth derivative does not mean zero effect on loft, impact position, future motion or higher-order face-angle change. A joint's useful authority also depends on feasible torque, inertia, constraints and time.

For several feasible small coordinate changes, assemble $\delta\phi = J_\phi \delta q$. The corresponding variance approximation is $\mathrm{Var}(\phi) \simeq J_\phi \Sigma_q J_\phi^T$, retaining covariance. If measured angles share errors, or if constraints correlate their motion, their separate contributions cannot be added as independent percentages. A large observed range of one joint does not establish its causal dominance.

18.2.3 The Gate Concept: The Narrow Window at Impact

Definition 18.3. A Gate Is an Inverse Image of an Objective

Let $y = F(z, \eta)$ predict landing outcomes from delivery variables z and environment/model parameters η. For an acceptable outcome set $\mathcal{A}$, the delivery gate is $\{z : F(z, \eta) \in \mathcal{A}\}$. A face-angle interval is a slice of that set with the other variables held fixed. With uncertainty, one might instead require $\Pr(F(z, \eta) \in \mathcal{A}) \geq p_0$.

There is no universal three-to-five-degree gate implied by club speed alone. The target width, carry, spin, strike location, wind and acceptable probability must be supplied. A joint change may alter several delivery variables together, so a face-only slice need not describe a physically achievable intervention.

18.2.4 Drift vs. Control: Remaining Time Before Impact

Key Principle 18.3. Feedback Timing and Continuing Actuation

Muscle activation and force may continue through impact even when a newly perceived error cannot produce a useful late correction. A predetermined command, a response to earlier information and a new sensory correction are different inputs. The computational-brain chapter derives a delayed-activation example in which remaining time continuously changes the same command's effect. It does not establish a universal instant at which all control switches off.

For a declared input-affine model, $\dot{x} = f(x) + G(x)u$, the drift term f depends on the chosen state and input. Zero neural excitation does not mean zero retained muscle force. A local fixed-time delivery sensitivity has the form

$$\delta z = C\Phi(T, 0)\delta x_0 + \int_0^T C\Phi(T, s)B(s)\delta u(s)\, ds.$$

Delay, activation dynamics and feasible commands must be included in this calculation. The state includes velocities and relevant activation or elastic history; club pose at the top alone cannot determine angular momentum. Impact time is also an outcome. For a transverse contact event $g(x(T)) = 0$,

$$\delta T = -\frac{g_x \delta x(T)}{g_x \dot{x}(T)}, \qquad \delta x_{\text{event}} = \delta x(T) + \dot{x}(T)\delta T.$$

This first-order expression assumes a differentiable event with nonzero denominator and no intervening mode change. Near grazing contact it becomes ill-conditioned. It explains why a sensitivity computed at fixed clock time can miss an important delivery effect, without proving a neural strategy.

18.3 The D-Plane and Modern Ball Flight Laws

18.3.1 Launch Direction: Measurements and Their Limits

For small horizontal angles, a local empirical approximation is

$$\lambda \simeq w\phi + (1 - w)\psi,$$

where λ is launch azimuth. The weight summarizes a particular data range and measurement protocol. It is neither a force fraction nor a causal percentage of an entire shot.

Wood and colleagues reported horizontal face weights of $76\% \pm 8\%$ for drivers, $69\% \pm 3\%$ for 7-irons and $61\% \pm 7\%$ for wedges, with the table identifying these intervals as 95% confidence intervals. The respective filtered sample sizes were 89, 172 and 28 shots, not the full 1,575 recorded shots. The 157-player protocol included centered-strike and minimum face-to-path filters; a separate 15-shot robot test gave $63\% \pm 4\%$. The study was funded by PING and used PING clubs. Its vertical ratios differ from the horizontal ratios, so a value from one projection cannot validate a claim about the other [Wood et al., 2018].

The practical conclusion is qualified face-dominant initial direction in those tests, not a universal rejection or confirmation of every vendor rule. The robot also changed delivered loft as face angle changed. Measurement alignment, strike location and small denominators in launch-ratio estimates limit interpretation. A reported confidence interval is not the spread of individual golfers' future shots or a universal uncertainty band.

18.3.2 Mathematical Formulation of Launch Direction

Definition 18.4. The D-Plane as a Geometric Construction

For nonparallel local face normal n and incoming relative surface-velocity direction $\hat{v}$, their span defines a plane with normal parallel to $\hat{v} \times n$. This construction remains possible at an off-center strike. What can fail there is the simplified prediction that the complete collision remains planar. If the two vectors are parallel, no unique plane normal or nonzero spin axis follows from them.

Under an ideal planar contact model, with an initially stationary ball, resolve the impulse on the ball as $J = J_n n + J_t t$, where $t = (\hat{v} - (n^T \hat{v})n)/\|\hat{v} - (n^T \hat{v})n\|$. Then

$$v_B^+ = \frac{J_n n + J_t t}{m_B}, \qquad \hat{v}_B^+ = \frac{J_n n + J_t t}{\|J_n n + J_t t\|}. \tag{18.1}$$

This is impulse balance, not a constant-weight direction law. A convex combination of unit vectors generally has norm below one and must be normalized if used as a direction. Moreover, a horizontal angle weight does not automatically specify a three-dimensional vector weight.

Spin loft is the three-dimensional obliqueness $\Phi = \arccos(n^T \hat{v})$. With the earlier angle definitions,

$$\cos\Phi = \cos\ell \cos\alpha \cos(\phi - \psi) + \sin\ell \sin\alpha.$$

Only when the azimuths coincide does this reduce to $|\ell - \alpha|$ on the relevant branch. Obliqueness is important, but does not eliminate dependence on contact material, friction, velocity, deformation and head motion.

Henrikson and colleagues used a contact model with tangential compliance and partial slip to explain a declining launch ratio at increasing obliqueness even with fixed friction. Their parameter study also changed the ratio by changing friction; in a low-incidence regime, lower friction could move launch closer to path. This is a finite-regime result, not a statement about all friction values. Their comparison reused the earlier player data, with a separate ball-on-plate experiment supporting the contact model. The fixed surface approximation and its treatment of normal inelasticity limit transfer to a freely rotating head [Henrikson et al., 2020].

Thus neither a single coefficient per club nor an obliqueness-only formula is a universal law. A strictly frictionless normal impulse on an initially nonspinning sphere produces velocity along the local normal and no spin; extrapolating a finite-friction trend to that limit can give the wrong answer.

18.3.3 Spin Axis and the Vector Perpendicular to the D-Plane

The contact point lies on the ball's surface, not at its center. In a spherical illustration with radius R_B, take $r_{B/C} = -R_B n$ from the ball center to contact. With isotropic rotational inertia I_B and zero initial spin,

$$\omega_B^+ = \frac{r_{B/C} \times J}{I_B} = -\frac{R_B J_t}{I_B}\, n \times t. \tag{18.2}$$

The impulse model supplies J_t and therefore both spin magnitude and sign. The unit axis is $\omega_B^+/\|\omega_B^+\|$ when spin is nonzero. A bare cross product of directions cannot specify RPM. An actual ball's inertia must be measured or modeled; $I_B = 2m_B R_B^2/5$ describes a homogeneous solid sphere, not every layered golf ball.

For a level incoming path along $+x$ and a face normal tilted upward, positive tangential impulse along the projected path gives spin toward $-y$. Then $\omega_B \times v_B$ points upward, consistent with backspin lift in the usual positive-lift convention. With no tangential impulse the spin is zero and an axis should not be reported. Pre-existing spin, a varying normal, nonplanar friction and head rotation require the fuller contact calculation.

Gear effect concerns coupled rotation and tangential motion associated with an eccentric clubhead impact. It is not caused simply by the fact that a ball is touched on its surface. Off-center impact can rotate the head, change the local normal and contact velocity, and alter the ball's spin. Bulge and roll introduce an additional local-normal dependence.

18.3.4 The Face-to-Path Difference and Shot Shape

The horizontal difference is $\phi - \psi$ in the chosen frame. Under an ordinary centered backspin-dominated impact it is useful for interpreting draw or fade tendencies. It does not determine curvature independently of loft, attack angle, spin magnitude, strike location and wind. Curvature during flight depends on the full aerodynamic force history.

Using $w = 0.76$ as an illustrative local weight, a path 2° right and square face give $\lambda = -0.48^\circ$ in this chapter's positive-left frame: the ball starts 0.48° right. This calculation predicts initial direction only. A lower speed does not universally produce more time aloft or more curvature. Even in vacuum at fixed elevation, flight time is proportional to speed; real aerodynamic flight requires its own calculation.

18.4 Launch Condition Optimization

Key Principle 18.4. Launch State and the Flight Model

Ball speed, launch elevation and spin rate are useful summaries. A full flight state includes position, the three-dimensional velocity vector and the spin vector; ball orientation may also matter in a seam- or surface-sensitive model. Predicting landing additionally requires gravity, air density, wind, aerodynamic coefficients and a landing-surface definition. Bounce and roll require ground-contact models. The objective may be carry, total distance, a landing region, stopping behavior or expected score.

For a representative quasi-steady model, let $v_r = v_B - w(r_B, t)$ be air-relative velocity, A_B a reference area and ρ air density. Then

$$m_B \dot{v}_B = m_B g - \tfrac{1}{2}\rho A_B C_D \|v_r\| v_r + \tfrac{1}{2}\rho A_B C_L \|v_r\|^2 \hat{l}, \qquad \dot{r}_B = v_B,$$

where $\hat{l}$ is a declared lift direction, commonly along $\omega_B \times v_r$ when that cross product is nonzero. A model must handle its zero limit consistently. Coefficients depend on the chosen ball, flow regime and spin parameters; rotational drag may evolve ω_B. The aerodynamics chapter develops these boundaries. A coefficient table from one ball and operating range should not silently become a universal flight law.

18.4.1 Ball Speed and the Smash Factor

Define smash factor S using specified ball and club speed conventions:

$$S = \frac{v_{\text{ball}}}{v_{\text{club}}}, \qquad v_{\text{ball}} = S\, v_{\text{club}}. \tag{18.3}$$

This is a definition, not a mechanism or an energy efficiency. At club speed 100 mph and $S = 1.48$, ball speed is 148 mph. Ball energy scales with ball mass and speed squared; a speed ratio larger than one does not violate energy conservation. Collision restitution, effective mass, obliqueness and contact losses determine which ratios are achievable. Club speed measured at a different head point changes the reported ratio.

Improving strike location can change ball speed, launch and spin together. Treating smash factor as independently adjustable while holding everything else fixed is an intervention assumption to be tested, not a property of its definition. Likewise, an increase in club speed is not numerically identical to an increase in ball speed.

18.4.2 Optimal Launch Angle for Distance

Specify a prediction $D(z;\eta)$ and feasible launch set $\mathcal{Z}$ before asking for $\arg\max_{z\in\mathcal{Z}} D$. Distinguish a ball-flight optimum at fixed speed and spin from an equipment or swing intervention that changes all three together. Carry and total-distance optima need not coincide. An interior stationary point, a constrained boundary solution and a robust choice under uncertainty obey different conditions.

For a checkable limiting example, neglect air resistance, spin effects and initial height, and land at the launch elevation. With speed v,

$$D = \frac{v^2}{g}\sin(2\theta), \qquad T_f = \frac{2v\sin\theta}{g}.$$

Here 45° maximizes range at every positive speed. This is not a driver fitting recommendation: it is a counterexample to a universal speed-only rule for optimal angle. Lift, drag, launch height and feasible launch conditions change the optimization problem.

For the remaining worked examples, define a manufactured local response surface, expressed in yards, around 140 mph ball speed, 14° launch and 2,400 RPM spin:

$$\delta v = v_{\text{ball}} - 140 \text{ mph}, \quad \delta\theta = \theta - 14^\circ, \quad \delta\Omega = \Omega - 2400 \text{ RPM}.$$

Use numerical deviations in the listed units, with dimensioned coefficients,

$$D = 230 + 2\delta v - \tfrac{1}{2}a\delta\theta^2 - b\delta\theta\delta\Omega - \tfrac{1}{2}c\delta\Omega^2, \tag{18.4}$$

where $a = 0.8$ yards/degree^2, $b = 0.002$ yards/(degree RPM), and $c = 0.00002$ yards/RPM^2. The speed coefficient is 2 yards/mph. These values define the example completely; they

Ball Speed	Launch	Spin	Model Carry
140 mph	16°	2,200 RPM	228.8 yd
140 mph	13°	2,200 RPM	228.8 yd
140 mph	14.5°	2,200 RPM	229.7 yd
140 mph	14°	2,400 RPM	230.0 yd

Table 18.2: Manufactured Local Surface: Coupled Launch and Spin Change Which Angle Is Best. These Are Calculated Examples, Not Golf Fitting Data.

were not estimated from launch-monitor data. The model is used locally, and should not be extrapolated to negative spin, arbitrary speeds or extreme launch angles.

At fixed 2,200 RPM the optimum angle is 14.5°, because $\delta\theta^* = -b\delta\Omega/a = 0.5°$. Calling a reduction from 16° to 13° an improvement would be wrong in this explicit example: both points have the same predicted carry. An observed carry value at one launch condition cannot determine the response to an unmeasured change.

18.4.3 Spin Rate Trade-Off: Lift and Air Resistance

Key Principle 18.5. A Claimed Optimum Must Follow From the Model

Backspin can support lift, while its associated aerodynamic changes can also increase drag. The resulting carry optimum must be calculated using the declared flight model and constraints. A polynomial $Av^2 - B\Omega - C\Omega^2$ with positive B, C decreases for all nonnegative spin; it cannot demonstrate a positive-spin interior optimum.

In the manufactured surface, the launch-spin curvature matrix is

$$H = \begin{pmatrix} a & b \\ b & c \end{pmatrix}, \qquad ac - b^2 = 0.000012 > 0$$

in the corresponding compound units. Since $a > 0$ and the determinant is positive, the quadratic loss is positive definite. The chosen center is therefore a strict joint maximum at fixed speed. Fixing launch away from its central value shifts the spin optimum to $\delta\Omega^* = -b\delta\theta/c$. This coupling is exactly what separate one-variable target windows can conceal.

18.4.4 The Launch Window

A performance window should describe a level set or probability requirement, not an unexplained collection of independent tolerances. For fixed speed in the manufactured example, accepting at most one yard of modeled loss gives

$$\tfrac{1}{2}(\delta\theta, \delta\Omega)H(\delta\theta, \delta\Omega)^T \le 1 \text{ yard.}$$

It is an ellipse with coupled axes, not a rectangular launch-spin box. At zero spin deviation the boundary is $|\delta\theta| = \sqrt{2/a} \simeq 1.581°$; at zero launch deviation it is $|\delta\Omega| = \sqrt{2/c} \simeq 316.23$ RPM. Taking both individual limits simultaneously generally violates the joint one-yard requirement.

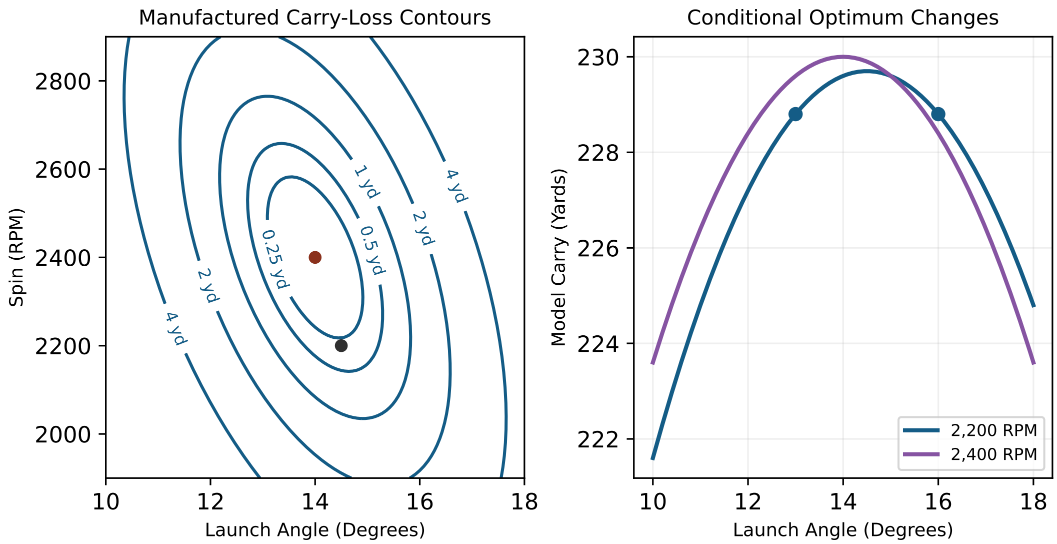


Figure 18.2: A Manufactured Launch Surface Shows Coupling and Curvature. The Elliptical One-Yard Loss Region and Conditional Optima Follow From the Stated Quadratic, Not a Golf Fitting Dataset.

Window Choice	**Meaning**
Deterministic Level Set	Outcomes under one declared model and environment.
Feasible Delivery Set	Launch combinations reachable by the golfer and equipment.
Chance Constraint	Required success probability under declared uncertainty.
Fitting Target	A tested intervention balancing outcome, repeatability and cost.

Table 18.3: Define the Purpose of a Launch Window Before Using It as a Target.

The useful target lies in the intersection of acceptable outcomes and achievable delivery. It may favor a slightly different mean condition when that condition reduces harmful variability or avoids a severe miss. Such a choice requires a stated loss function, not a universal preference for either maximum mean distance or minimum variability.

18.4.5 Equipment Design and Launch Optimization

Moving a clubhead's center of mass changes moment arms and inertia; it does not by itself prescribe a change in launch angle or spin. Distinguish fixed delivery from the golfer's adaptation to a changed club, and distinguish a fixed location on the face from a fixed location relative to the center of mass. Local loft, vertical and horizontal gear effect, shaft dynamics and impact offset can change together.

A frictionless normal-impact illustration makes one effect explicit. For ball mass m_B, head mass M_H, head inertia tensor I_H about its center, and contact offset r_H, the normal effective inverse mass for a spherical ball is

$$\kappa_n = \frac{1}{m_B} + \frac{1}{M_H} + (r_H \times n)^T I_H^{-1} (r_H \times n).$$

The spherical ball's normal impulse passes through its center, so it adds no ball rotational term in this normal direction. For relative closing speed $v_n > 0$ and restitution e_n, $J_n = (1 + e_n)v_n/\kappa_n$ under the stated free-body impulse approximation. External hand/shaft impulses, tangential coupling or deformable-body effects require a more complete model.

For $M_H = 0.20$ kg, $m_B = 0.04593$ kg, $I_H = 0.0004$ kg m^2 about the relevant rotation axis and a 0.020 m perpendicular offset, the added inverse mass is 1 kg^{-1}. The normal impulse is about 0.9640 times the centered value at the same closing speed and restitution. This is a mechanics example, not a measured mishit penalty or a prediction of spin.

Face flexibility affects the transient collision and cannot be summarized as the energy efficiency of the face alone. Restitution, characteristic-time equipment tests and smash factor are different quantities. Apply the current equipment rules and their test procedures when assessing conformity; a launch fitting table cannot establish it. The impact chapter treats the collision and equipment boundaries in more detail.

18.4.6 Wind Effects on Optimal Launch Conditions

Wind enters through $v_r = v_B - w$, not through an assumed ground-effect bonus. A headwind generally increases air-relative speed for an outgoing shot and a tailwind generally decreases it initially. Drag, lift and spin evolution then alter the entire flight. Because those changes depend on the ball and launch state, a fixed wind speed does not imply a universal degree adjustment to the optimal launch angle.

For a real fitting calculation, specify the wind field, density, ball coefficients, landing elevation and objective. Recompute feasible launch choices and their uncertainty. If optimizing total distance, include the landing angle, landing speed and ground behavior. A claim that more time aloft is always beneficial is incompatible with these coupled costs.

18.5 Sensitivity Analysis: What Matters Most?

18.5.1 Partial Derivatives of Distance

Sensitivity is local and conditional on which quantities are held fixed. For the manufactured surface,

$$\frac{\partial D}{\partial v_{\text{ball}}} = 2 \text{ yards/mph}, \tag{18.5}$$

$$\frac{\partial D}{\partial \theta} = -a\delta\theta - b\delta\Omega, \tag{18.6}$$

$$\frac{\partial D}{\partial \Omega} = -b\delta\theta - c\delta\Omega. \tag{18.7}$$

At a smooth interior optimum with the other variables held fixed, the relevant first derivative is zero. A maximum is not generally the point of maximal first-order sensitivity. Near it, curvature determines the leading loss. At a boundary optimum, a nonzero gradient can be balanced by constraints.

At $(16°, 2200 \text{ RPM})$, the angle derivative is -1.2 yards/degree. A linear prediction for a $-3°$ change gives a $+3.6$ yard gain. The quadratic correction is $-\frac{1}{2}a(3)^2 = -3.6$ yards, exactly canceling it. The endpoints therefore have equal carry in this model. A local derivative does not justify a large finite extrapolation without a remainder estimate.

Comparing the numbers 2 yards/mph and 0.002 yards/RPM without scaling is meaningless as a ranking. Compare effects over plausible changes, achievable interventions or uncertainty ranges. The parameter with the largest raw derivative may not be the one that contributes most to actual error.

18.5.2 Directional Sensitivity

Ignore aerodynamic curvature for a moment and let R be a fixed downrange distance. The geometric lateral offset is $Y = R\tan\lambda$. Under the local angle-weight model with constant w,

$$\frac{\partial Y}{\partial \phi} = R\sec^2\lambda\, w, \tag{18.8}$$

$$\frac{\partial Y}{\partial \psi} = R\sec^2\lambda\,(1-w). \tag{18.9}$$

These derivatives use radians. At $R = 250$ yards, $\lambda = 0$ and $w = 0.76$, they are approximately 3.316 yards/degree of face change and 1.047 yards/degree of path change. The familiar $250\tan 1° \simeq 4.364$ yards is for a one-degree *launch* change, not a one-degree face change under this weighting.

Neither result includes curvature, changes in carry or a changing weight. For actual flight, write $Y = F_Y(z, \eta)$ and propagate all delivery changes, including spin-axis and speed changes. A path change can strongly affect spin while contributing less to initial launch azimuth. It therefore does not follow that path has a universally small effect on final lateral error.

18.5.3 The Consistency Argument

For small perturbations and a scalar outcome, the first-order variance is

$$\mathrm{Var}(D) \simeq g_D \Sigma_z g_D^T, \qquad g_D = \nabla_z D.$$

Correlations can increase or decrease this variance. Measurement noise, calibration bias, model discrepancy and genuine shot variability should be estimated separately where the data permit. Outcome repeatability alone does not identify which joint, muscle or controller produced it.

At fixed club speed 100 mph, a smash-factor *standard deviation* of 0.05 gives ball-speed standard deviation 5 mph. A smash-factor variance of 0.05 would instead give $100\sqrt{0.05} \simeq 22.36$ mph. If both club speed V and smash factor S vary independently, the exact product variance is

$$\mathrm{Var}(VS) = \mu_V^2 \sigma_S^2 + \mu_S^2 \sigma_V^2 + \sigma_V^2 \sigma_S^2.$$

For correlated variables, additional mixed moments enter the exact result; the first-order approximation includes $2\mu_V \mu_S \,\mathrm{Cov}(V, S)$. One cannot infer the total spread by holding a varying factor constant.

At the manufactured surface's launch-spin maximum, first-order propagation alone gives zero sensitivity to these two variables, but nonzero fluctuations still reduce mean carry. For mean-zero deviations with covariance Σ,

$$E[D] = 230 - \tfrac{1}{2}\,\mathrm{tr}(H\Sigma)$$

at fixed 140 mph. This identity is exact for the quadratic model. With launch standard deviation 1°, spin standard deviation 200 RPM and correlation $+0.5$, mean loss is 1.0 yard. Changing only the correlation to -0.5 gives 0.6 yard. Equal marginal spreads do not imply equal performance.

Mean distance and spread are distinct objectives. If a manufactured linear distance model has slope 2 yards/mph, a 3 mph increase in mean ball speed gives 6 yards of mean gain. Increasing speed standard deviation from 4 to 5 mph changes distance standard deviation from 8 to 10 yards; reducing it to 2.5 mph gives 5 yards. Which intervention is preferable depends on the target and the cost of misses. The second is more consistent in this example, but that does not prove it has higher expected scoring value.

18.6 The Zero-Torque Counterfactual Perspective on Launch

Key Principle 18.6. Define the Counterfactual Intervention

A zero-torque counterfactual sets specified inputs to zero in a declared model, at a specified time and state, while stating what happens to the remaining inputs, contacts and controller. Its predicted launch is a comparison within that model. It does not establish that a golfer actually stops applying force, that all neural feedback ends at a common time, or that the same result would follow from changing grip or equipment.

For a useful golf investigation, propagate the complete pre-impact state to the contact event, resolve the collision and then simulate flight. Compare the specified intervention with a baseline under the same reported conventions. Repeat with plausible parameter and measurement uncertainty. If a contact mode changes or the club misses the ball, report that result rather than silently comparing the state at an unrelated clock time.

18.6.1 What a Late Correction Can Change

Example 18.2. New Corrections and Continuing Force

Suppose a newly detected error would require 60 ms before its corrective command reaches an actuator, while impact is 30 ms away. That new command cannot influence this impact through the stated delayed pathway. Existing activation, prior commands and faster pathways are separate questions. This timing example neither proves a universal golf delay nor says that the club is mechanically free. It also does not determine the benefit of a prepared response to information available earlier.

The proper calculation combines pathway timing with output gain, remaining time and feasible force. Reaching a desired face orientation may also require preserving strike position, speed and contact constraints. A useful controller must solve this coupled task rather than minimize face-angle error alone.

18.6.2 Determinants of Delivery: State, Inputs and Constraints

Grip geometry establishes kinematic relations and possible contact forces. Changing grip force can affect friction capacity and effective boundary conditions; they do not change the intrinsic mass distribution of the club. Posture changes moment arms, feasible directions and motion. Activation and elastic deformation retain history. Ground reactions participate in the multibody balance and may depend on the applied muscle forces and constraints. Continuing actuator work and energy redistribution can coexist.

The model's ability to reproduce one club trajectory does not identify all these internal mechanisms. Seek measurements that distinguish competing explanations, or report the family of compatible models. A claim about a particular grip's performance requires a defined task and intervention, not only a geometric example in which one torque component is reduced.

18.7 Integration of Impact Conditions With Equipment and Swing Dynamics

The explanatory chain is now explicit. A state and input history generate a constrained head trajectory. The selected point, frame and event determine reported delivery quantities. Local surface orientation and velocity, mass, inertia and contact constitutive behavior determine impulse and ball spin. Initial ball state and the environment determine flight. The target and loss function determine whether the outcome is favorable.

Every link admits both uncertainty and a useful test. Check geometric identities against independent coordinates; compare local gains with finite perturbations; validate contact predictions against suitable impact data; test flight models within their coefficient domains; evaluate interventions with held-out outcomes and matched definitions. These steps strengthen the argument for golf as a coupled control problem without asserting one universal plane, release pattern, late-control cutoff or launch window.

18.8 Problems and Thought Experiments

1. **Swing Plane Geometry.** Use a fixed 60° plane and attack angle −5°. Find the forward-branch path offset from the base direction. Explain why a two-inch total drop over fifteen horizontal inches does not determine this attack angle, and convert a 57° conventional lie to its complement to vertical.

2. **Face Angle Sensitivity.** At $\phi = 0$, $\ell = 15°$, derive the azimuth gains for small spatial rotations about x, y, z. With fixed other delivery variables and the geometric $R = 250$ yard, $w = 0.76$ approximation, estimate lateral standard deviations for face standard deviations 2° and 4°. State what has been omitted.

3. **Launch Optimization.** In the manufactured quadratic model, hold ball speed at 140 mph and spin at 2,200 RPM. Calculate carry at 16°, 13° and the conditionally optimal launch. Explain why a first-order estimate for the three-degree change fails.

4. **D-Plane and Impulse.** For a path 2° right and square face, apply the declared horizontal weight $w = 0.76$. Separately derive spin from $r_{B/C} \times J$ for a level path and an upward-tilted face. What happens when the tangential impulse is zero? Does an off-center strike make the geometric span of two nonparallel vectors cease to exist?

5. **Control and Event Timing.** A new correction is delayed 60 ms and impact is 30 ms away. Identify what this rules out and what it does not. Explain why a fixed-time face-angle sensitivity can differ from an impact-event sensitivity.

6. **Equipment and Effective Mass.** Use the masses, inertia and 20 mm offset in the equipment example. Compute the ratio of off-center to centered normal impulse at fixed restitution and closing speed. Explain why this does not predict the spin change from moving a driver's center of mass.

7. **Wind and Feasible Launch.** Specify the additional inputs needed to compare optimal shots in a 15 mph headwind and tailwind. Explain why the wind speed alone cannot supply a numerical angle correction or a complete total-distance prediction.

8. **Consistency and Objective.** Reproduce the quadratic mean losses for correlations +0.5 and −0.5 with the specified launch and spin spreads. Then compare the 3 mph mean-speed gain and the standard-deviation reduction in the linear example. Distinguish variance, standard deviation, mean outcome and expected scoring value.

18.8.1 Worked Checks and Interpretation

1. The path offset is $\arcsin(\tan(-5°)/\tan 60°) = -2.8953°$, or right of the base direction. A finite chord does not specify an instantaneous tangent; the lie complement is 33° and does not supply the missing motion data.

2. The gains are -0.26795, 0 and 1 respectively. The face-to-offset gain is $250(0.76)\pi/180 \simeq$ 3.316 yards/degree, giving approximately 6.63 and 13.26 yards of lateral standard deviation under the local independent face-error model. Curvature, correlated path, strike and speed changes, wind and model error are omitted; these are not predicted golfer dispersions.

3. The two endpoint carries are both 228.8 yards. The conditional optimum is 14.5°, giving 229.7 yards. The initial derivative predicts a 3.6-yard gain for the finite reduction, canceled by the exact 3.6-yard quadratic loss. This is why a local derivative cannot be used without scale.

4. Launch azimuth is $-0.48°$, or 0.48° right. With the defined tangent, $\omega_B = -R_B J_t(n \times t)/I_B$; positive tangential impulse gives backspin toward $-y$. At $J_t = 0$ the assumed initially nonspinning ball remains nonspinning. Nonparallel directions still define a plane off center, but head rotation and contact changes can invalidate the simplified planar outcome prediction.

5. The stated new delayed command arrives after contact. Earlier commands and activation remain possible; other pathways need their own latency and gain. A perturbation may move the contact time, adding $\dot{x}\delta T$ to the fixed-time state perturbation before collision.

6. The extra inverse mass is $(0.020)^2/0.0004 = 1\ \mathrm{kg}^{-1}$. The impulse ratio is

$$\frac{5 + 1/0.04593}{6 + 1/0.04593} \simeq 0.963993.$$

   The calculation is normal and frictionless. Predicting spin needs tangential contact mechanics, local velocity and orientation, and the complete change in head motion and strike geometry.

7. Supply initial state, ball aerodynamic and spin-decay model, wind field, density, landing geometry, feasible delivery and objective. Add a ground-contact model for bounce and roll. Evaluate both environments rather than assigning a universal positive or negative angle adjustment.

8. The covariance terms give $\frac{1}{2}\,\mathrm{tr}(H\Sigma) = 0.8 + 0.4\rho$ yards: 1.0 or 0.6 yard. In the linear example the mean gain is 6 yards, while distance standard deviations are 8 yards before intervention, 10 after the faster but more variable intervention, and 5 after the consistency intervention. These are different trade-offs. A target-specific loss or scoring model is needed to rank them as overall performance changes.

Part V

The Inverse Problem and Its Limits

Chapter 19

Inverse Dynamics and the Parallel Loop Problem

A swing can reveal how the club moved without revealing how each hand loaded it. A measured hand wrench can reveal an arm's net joint loading without identifying every muscle force. A forward simulation can predict a specified intervention without establishing that a golfer actually used that strategy. These are three different questions, and each deserves a complete answer at its own level.

The purpose of inverse dynamics is to determine loads consistent with measured motion, declared mechanics, and known boundary conditions. Its value does not depend on recovering every muscle's contribution. The research challenge is to identify which combinations of loads the available measurements determine, which remain ambiguous, and which additional measurements would distinguish competing explanations of the golf swing.

19.1 Start With the Quantity Being Investigated

An analysis should name its target before choosing an algorithm. Clubhead speed, clubface orientation, net hand wrench, shoulder moment, muscle force, mechanical work, metabolic demand, and tissue stress are not interchangeable outputs. They require different boundaries, models, and evidence. A joint moment is not an electrical activation signal; a ground moment is not a direct measurement of pelvis acceleration; and neither one alone measures energy supplied to the club.

For example, suppose two golfers deliver the same club pose and velocity at impact. They need not have followed the same earlier trajectory. Even identical full club trajectories need not imply identical hand loading. Identical skeletal trajectories and boundary loads can still admit multiple muscle-force patterns. Conversely, additional sensors or sufficiently restrictive mechanics can make a particular load identifiable. The conclusion must follow from the specified inverse problem rather than the slogan that closed loops are always ambiguous.

Motion capture supplies image or marker observations. Segment poses, joint coordinates, and their derivatives are estimates obtained through calibration, kinematic fitting, smoothing, and differentiation. External forces require instruments or a declared force model; they are not automatically another output of differentiating joint angles. A complete report carries that chain of inference into its final uncertainty estimates.

19.2 Equations, Inputs, and Boundary Conditions

Use regular local coordinates $q \in \mathbb{R}^n$ for an articulated-tree model, with any additional ideal, time-independent closures $\Phi(q) = 0$. Let $J_c = D_q\Phi \in \mathbb{R}^{k\times n}$ contain independent rows

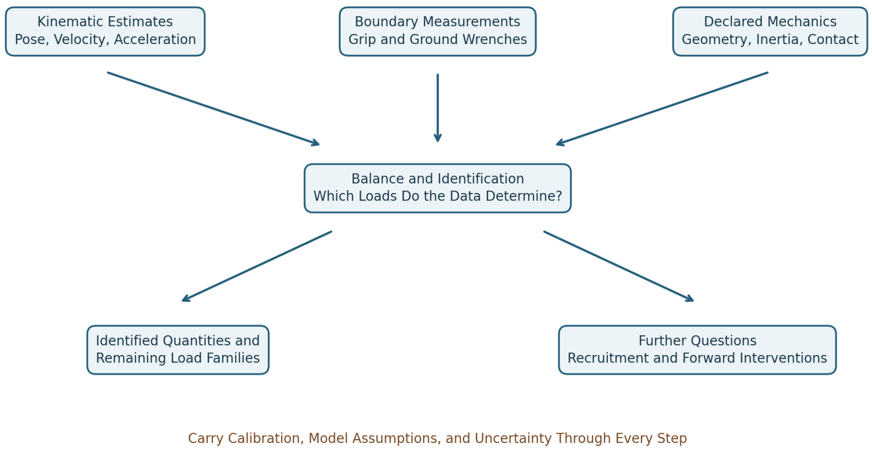


Figure 19.1: From Measurements and Declared Mechanics to Identified Loads and Further Research Questions.

in the contact mode under consideration. Write

$$M(q)\ddot{q} + h(q, \dot{q}) = B(q)u + p(q, \dot{q}, z) + Q_e + J_c(q)^T \lambda. \tag{19.1}$$

Here h contains the declared inertial bias and gravitational generalized loads; $u \in \mathbb{R}^p$ contains the selected applied inputs; B maps them into generalized loads; p contains separately modeled passive or other constitutive loads, with internal states z when needed; and Q_e contains known external loads. A load belongs in exactly one term. Unknown hand or ground loads cannot be placed in Q_e and simultaneously treated as already measured. Their parameters must instead join the unknown vector.

For example, a segment wrench maps to generalized load by the transpose of its compatible velocity Jacobian. Generalized components may have units of force or moment according to their coordinates; they are not all Cartesian forces. The matrix often written $C(q, \dot{q})$ is a representation of an inertial bias vector $C\dot{q}$, not a list of independent physical forces at the hands.

For a regular fully actuated open tree with $B = I$ and known p, Q_e, inverse dynamics evaluates

$$u = M\ddot{q} + h - p - Q_e. \tag{19.2}$$

There is no mass-matrix inversion in this expression. A positive-definite mass matrix follows when every nonzero independent coordinate velocity produces positive kinetic energy in the modeled bodies. Redundant coordinates, massless idealizations, or a singular coordinate chart require separate treatment. A straight elbow can make a hand-position Jacobian singular while the physical joint-space mass matrix remains positive definite.

Even the net load u needs interpretation. A revolute joint's generalized moment is the load component about its allowed axis; its ideal reaction components are separate. A full transmitted joint wrench, a generalized joint moment, and the muscle contribution to that moment are different quantities. Passive tissue loads already included in the net balance must be subtracted consistently before calling the remainder active muscle moment.

19.3 A Complete Newton–Euler Recursion

The recursive Newton–Euler algorithm computes open-tree inverse dynamics without explicitly assembling M. Its linear computational cost for a tree with fixed-size joints does not remove measurement uncertainty or repair an incorrect contact boundary. The essential ingredients are spatial coordinate transport, inertial bias, external wrenches, and propagation of child loads.

For one-freedom revolute or prismatic joints, use angular-first spatial motion $v_i = [\omega_i; v_{O_i}]$ and its power-dual wrench $w_i = [m_{O_i}; F_i]$, both expressed in body frame i. Let $\pi(i)$ be the parent, X_i the motion transform from parent frame to child frame, S_i the joint motion subspace, and $\mathcal{I}_i$ the body's spatial inertia about its frame origin. The fixed-base recursion is

$$\begin{aligned} v_i &= X_i v_{\pi(i)} + S_i \dot{q}_i, \\ a_i &= X_i a_{\pi(i)} + S_i \ddot{q}_i + \operatorname{crm}(v_i) S_i \dot{q}_i, \\ w_i &= \mathcal{I}_i a_i + \operatorname{crf}(v_i) \mathcal{I}_i v_i - w_i^{\mathrm{ext}}. \end{aligned} \tag{19.3}$$

Here $\operatorname{crm}(v)$ is the spatial motion cross-product operator and $\operatorname{crf}(v) = -\operatorname{crm}(v)^T$ its force counterpart. For $v = [\omega; v_O]$,

$$\operatorname{crm}(v) = \begin{bmatrix} [\omega]_\times & 0 \\ [v_O]_\times & [\omega]_\times \end{bmatrix}, \qquad [a]_\times b = a \times b. \tag{19.4}$$

Set $v_0 = 0$ and $a_0 = [0; -g]$ to incorporate uniform gravity through this recursion; do not then subtract gravity again as an external wrench. With this convention the recursively stored acceleration includes the gravity offset. For more general joints, include the appropriate joint bias acceleration. A moving or floating base requires its own specified motion or balance equations.

After forming each body's inertial and external contribution, accumulate from tips to base:

$$w_i \leftarrow w_i + \sum_{j:\pi(j)=i} X_j^T w_j, \qquad \tau_i = S_i^T w_i. \tag{19.5}$$

The transpose transport follows from power invariance: $(X_j v_i)^T w_j = v_i^T X_j^T w_j$. It carries force-induced moments as well as rotating components. Adding moments from different points without this transport loses the lever-arm contribution. These conventions follow the Featherstone spatial-vector implementation; its explicit inverse-dynamics recursion also shows the velocity-product terms.

For a closed chain, one may cut a contact to obtain a tree, introduce its unknown cut wrench, and solve the resulting coupled balance and measurement equations. A tree algorithm with an arbitrarily assigned cut wrench gives an answer for that assignment. It does not establish that the assignment was measured or uniquely implied by motion.

19.4 What the Two-Handed Grip Constrains

The two hand frames occupy different locations on the club. If T_C is the club pose and H_L, H_R are fixed hand-to-club attachment transforms under an ideal rigid grip, closure is

$$T_L = T_C H_L, \qquad T_R = T_C H_R, \qquad T_L^{-1} T_R = H_L^{-1} H_R. \tag{19.6}$$

The final relation supplies six independent local restrictions only where the chosen representation is regular. Equality of the two hand poses would describe coincident attachment frames, not the ordinary separated grip. Compliance, sliding, or changing finger contacts require a different model.

Rigid-body angular velocity is common to its points when expressed in one frame, but their linear velocities differ:

$$v_R = v_L + \omega_C \times (r_R - r_L). \tag{19.7}$$

The corresponding acceleration relation includes angular acceleration and centripetal terms. A Jacobian closure of the form $[J_L, -J_R]\dot{q} = 0$ is valid only after both Jacobians describe the same transported task frame and point.

The general compatibility equations are

$$J_c\dot{q} = 0, \qquad J_c\ddot{q} + \dot{J}_c\dot{q} = 0. \tag{19.8}$$

They test whether a candidate trajectory respects the declared constraint; they contain no independent measurement of λ. At constant rank r, feasible local configuration dimension is $n - r$. At a singular point, extra linearized directions need not extend to finite feasible motions. Low speed at address or transition is not evidence of rank loss; finite tissue stiffness is not a kinematic rank calculation.

With two seven-coordinate arm models and a regular six-restriction rigid-grip closure, eight configuration freedoms remain. They include motion of the club. If the full club-pose task has rank six on the feasible tangent space, only two posture freedoms remain while holding that task fixed. These counts assume the specified shoulder bases and arm models; they are not universal counts of the human body.

Thus $\ker J_c$ contains compatible velocities; task-preserving velocities also satisfy $J_y\dot{q} = 0$. Neither space says that movement is energetically free. Grip pressure is a contact-load variable, not a joint velocity. With independent rows, $\ker J_c^T$ is zero, even though unknown actuator and reaction loads can still be exchanged without changing the observed motion.

19.5 Identification Is a Rank Question

At a specified state and acceleration, define the known model residual and unknown vector by

$$r = M\ddot{q} + h - p - Q_e, \qquad z_\ell = \begin{bmatrix} u \\ \lambda \end{bmatrix}, \qquad A = \begin{bmatrix} B & J_c^T \end{bmatrix}, \qquad Az_\ell = r. \tag{19.9}$$

Use z_ℓ for load unknowns to distinguish them from constitutive states. For a consistent linear problem, the solution set is $z_{\ell,0} + \ker A$. It is a singleton exactly when A has full column rank. Having more unknowns than equations rules out a unique unconstrained solution; a square matrix alone does not guarantee uniqueness. Bounds, contact inequalities, and nonlinear measurements can further restrict the feasible set and must be analyzed as part of that different problem.

Suppose additional calibrated linear measurements are $Sz_\ell = y_m$. Stack $A_m = [A; S]$. More sensor channels help identification only when they add independent information about the unresolved directions. A particular linear quantity Cz_ℓ is identifiable even without the whole vector precisely when

$$\ker A_m \subseteq \ker C. \tag{19.10}$$

Indeed, all indistinguishable solutions differ by a vector in $\ker A_m$; the reported quantity is unique only if C annihilates every such difference. For finite-dimensional linear maps this is equivalent to the rows of C lying in the row space of A_m. Noise replaces an exact affine set by an uncertainty region; poor singular values then matter as well as rank.

19.5.1 An Example Where the Sensor Choice Matters

Take $M = \operatorname{diag}(2,1)$, $J_c = [1,-1]$, $B = I$, and zero other loads. For the compatible acceleration $\ddot{q} = (1/3, 1/3)^T$,

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ \lambda \end{bmatrix} = \begin{bmatrix} 2/3 \\ 1/3 \end{bmatrix}. \tag{19.11}$$

Its null direction is $(-1,1,1)^T$. Loads $(1,0,-1/3)$ and $(2,-1,-4/3)$ therefore produce the same acceleration. Measuring $u_1 + u_2 = 1$ adds no rank: it repeats the sum of the existing equations. Measuring $\lambda = -1/3$ selects $(u_1, u_2) = (1,0)$ in this ideal example. Measurement precision still limits the precision of that inference.

If the same mechanism has only one actuator, $B = (1,0)^T$, the inverse matrix for (u_1, λ) is square with determinant -1. The corresponding loads are uniquely determined for this measured acceleration. A loop therefore does not by itself make inverse dynamics nonunique. Conversely, multiple muscles can make recruitment nonunique in a completely open limb.

19.6 Hand Wrenches and Club Balance

A hand's force and moment must be referenced to a point and expressed in a coordinate frame. Use the same angular-first ordering as the spatial recursion. For an inertial-frame force F_i and moment m_i about attachment point i, transport to club center of mass C with $r_i = p_i - p_C$:

$$w_i^C = X_i^* w_i, \qquad X_i^* = \begin{bmatrix} I & [r_i]_\times \\ 0 & I \end{bmatrix}, \qquad w_i = \begin{bmatrix} m_i \\ F_i \end{bmatrix}. \tag{19.12}$$

Rotate components first if the sensor axes differ from the reporting axes. The club's net hand wrench is

$$\begin{bmatrix} I_C\alpha + \omega \times (I_C\omega) - m_{\text{other}}^C \\ m_c(a_C - g) - F_{\text{other}} \end{bmatrix} = G \begin{bmatrix} w_L \\ w_R \end{bmatrix}, \qquad G = \begin{bmatrix} X_L^* & X_R^* \end{bmatrix}. \tag{19.13}$$

Here I_C is rotational inertia about the club COM expressed in the same inertial axes as ω, α; it changes orientation with the club. The gyroscopic term remains necessary. Other forces include modeled aerodynamic and impact loads. A moving-grip-point expression $I_{\text{grip}}\alpha$ alone is not a general three-dimensional club balance. During impact, finite sensor bandwidth and flexible shaft motion may invalidate a simple rigid-body instantaneous-acceleration estimate; an impulse balance requires its own interval, boundary, and measurements.

Each X_i^* is invertible, so the full-wrench grasp map has rank six and a six-dimensional load null space before contact restrictions. Motion can identify the resultant under the declared model while leaving six combinations of hand wrenches unresolved. If the model permits only point forces and no hand moments, the corresponding six-by-six map has rank five at two distinct points. Its one internal direction is an equal-and-opposite force pair along the line joining them. A transverse opposing pair produces a couple and is visible in angular balance. Counting only resultant force misses that distinction.

Arm joint-torque vectors live in their own generalized-coordinate spaces. For a six-coordinate two-arm model, its six required joint loads cannot be written as the sum of two three-component hand forces. Hand wrenches enter the arm equations through their Jacobian transposes and the appropriate opposite-action signs. Equal and opposite numerical joint-torque entries do not automatically cancel the club wrench.

19.6.1 A Transport and Power Check

Take $r = (0.2, 0, 0)$ m, $F = (0, 100, 0)$ N, and a local hand moment $m = (0, 0, 3)$ N m. About C, the moment is $(0, 0, 23)$ N m. If $v_C = (1, 0, 0)$ m/s and $\omega = (0, 0, 2)$ rad/s, the hand-point velocity is $(1, 0.4, 0)$ m/s. Power computed either way is

$$m^T\omega + F^T v_{\text{hand}} = (m + r \times F)^T\omega + F^T v_C = 46 \text{ W}. \tag{19.14}$$

Using the untransported 3 N m moment together with COM velocity would report 6 W and lose 40 W. The discrepancy is a bookkeeping error, not a new energy source. At an ideal maintained attachment, the equal-and-opposite wrench acting on the hand has opposite power when both sides use the same contact twist. It can transfer energy between golfer and club while canceling in the combined system's internal power balance.

19.6.2 What an Instrumented Grip Actually Measures

A single six-axis sensor below both hands measures a section wrench. After accounting for the intervening club/grip inertia and other loads, it can constrain their aggregate action. It generally cannot resolve twelve independent hand-wrench components merely because it reports six channels. A second measurement of the same resultant improves precision but adds no independent load-sharing rank.

Separate mechanically isolated hand interfaces can supply distinct boundary wrenches. Multiple shaft sections may also distinguish loads if their placement, section balances, and calibration make the combined observation map informative. A pressure array usually measures a subset of contact traction; integrating normal pressure alone does not recover all tangential forces or free moments. A contact's net wrench also does not uniquely reconstruct the pressure and shear distribution across the skin.

The experiment must state exactly which bodies lie on each side of each sensor, where its wrench is referenced, how its axes are registered to motion capture, and how sensor mass changes the instrumented club's inertia. Timing offsets between motion and force signals can create apparent balance violations or false power peaks. Calibration, cross-talk, saturation, bandwidth, hysteresis, and repeated-load validation belong in the measurement uncertainty.

19.7 Sensor Physics With Defined Units

A bonded resistance strain gauge measures deformation through $\Delta R/R = \mathrm{GF}\,\varepsilon$ in its calibrated small-strain range. It measures local strain first; a structural model and calibration map that strain to a load. Multiple gauges can separate bending, axial loading, and torsion only if their geometry and calibration support those components.

For a declared Wheatstone bridge, put R_1 above R_2 in the left divider and R_3 above R_4 in the right divider, with excitation V_{ex} from the common top node to the bottom node. Measure left midpoint minus right:

$$\frac{V_{\mathrm{out}}}{V_{\mathrm{ex}}} = \frac{R_2}{R_1 + R_2} - \frac{R_4}{R_3 + R_4} = \frac{R_2 R_3 - R_1 R_4}{(R_1 + R_2)(R_3 + R_4)}. \tag{19.15}$$

Reversing the measurement leads reverses the sign. If only $R_1 = R(1 + \delta)$ changes and the others equal R, then

$$\frac{V_{\mathrm{out}}}{V_{\mathrm{ex}}} = -\frac{\delta}{4 + 2\delta} \simeq -\frac{\mathrm{GF}\,\varepsilon}{4}. \tag{19.16}$$

With $\mathrm{GF} = 2$, strain 500 microstrain, and 5 V excitation, the exact output is approximately -1.249 mV. Gauge orientation, active bridge arms, temperature compensation, lead resistance, and self-heating affect the actual installation. The bridge conventions and calibration choices should be reported, as emphasized in the NI strain-measurement documentation.

For a piezoelectric material, a constitutive relation in stress-charge form is

$$D = d\,\sigma + \epsilon^T E. \tag{19.17}$$

D is electric displacement (charge per area), σ is stress, E is electric field, and d is the piezoelectric coupling tensor under the stated material convention. This is not an equation for voltage. In a calibrated one-dimensional approximation, generated charge is $Q_{\mathrm{ch}} = d_F F$. An ideal open-circuit voltage magnitude is Q_{ch}/C for the relevant capacitance; an ideal inverting charge amplifier gives $V_{\mathrm{out}} = -Q_{\mathrm{ch}}/C_f$ over its operating band. For $d_F = 4$ pC/N and $F = 100$ N, the charge is 400 pC: this corresponds to 2 V across 200 pF or -0.4 V with a 1 nF feedback capacitor. These are illustrative circuit values, not a specification for a particular instrument.

Leakage, amplifier time constants, and insulation limit sustained measurements; preload and mounting affect the force path and dynamics. Piezoelectric systems can measure dynamic and suitably conditioned quasi-static loading. They are not universally faster or more sensitive than every strain-gauge system. See the Kistler force-sensor documentation for the distinction between a sensor's calibrated force response and its measurement conditions. Six independent load components require an appropriate mechanical and electrical sensing map, not necessarily six crystals or a hexapod.

19.8 Optimization Selects a Hypothesis

When measurements leave a load family, optimization can select a member. It adds a criterion; it does not turn the selected member into an observation of neural intent. State the decision variables, units, constraints, physiological bounds, and whether the objective is a hypothesis or a computational regularizer.

For example, minimize

$$\frac{1}{2}(z_\ell - z_0)^T W (z_\ell - z_0) \quad \text{subject to} \quad A_m z_\ell = b_m, \qquad W \succ 0. \tag{19.18}$$

Here z_0 is a declared reference load vector, not a measured fact unless it actually is measured. For full-row-rank A_m, stationarity of the Lagrangian and substitution into the constraints give

$$z_\ell^* = z_0 + W^{-1} A_m^T (A_m W^{-1} A_m^T)^{-1} (b_m - A_m z_0). \tag{19.19}$$

Redundant consistent rows can be removed or handled with an appropriate pseudoinverse. Inconsistent noisy measurements require a residual or statistical model rather than exact constraints that cannot all hold. Contact bounds and nonlinear muscle laws alter the optimization problem and may destroy feasibility or convexity; a numerical optimizer returning a vector is not proof of a valid physiological solution.

In a deliberately scalar example, $u_L + u_R = 10$ N m. Equal weights choose $(5, 5)$ N m. Minimizing $u_L^2 + 4u_R^2$ chooses $(8, 2)$ N m, with objective 80 in the declared squared-moment units. The balance did not change; the preference did. A proper norm needs scaling when the vector mixes force, moment, or different capacity scales. Squared joint moment is not metabolic power. The sum of measured EMG amplitudes is constant with respect to unknown loads and cannot select a solution unless an explicit decision-dependent measurement or physiological model connects them.

19.9 Muscles, Activation, and EMG

Given a net generalized moment, muscle recruitment is another inverse problem. If F_m collects musculotendon forces and $R(q)$ their moment-arm map, a simplified moment balance is

$$\tau_{\text{net}} = R(q)F_m + \tau_{\text{passive,other}}. \tag{19.20}$$

The sign convention for R follows the chosen coordinate and muscle-length conventions. A muscle spanning several joints contributes to several entries; net joint moments do not identify all tensile muscle forces when the map has unresolved directions. Known external hand wrenches improve the joint-load estimate without automatically eliminating muscle redundancy.

Surface EMG records a superposition of electrical potentials detected through electrodes and intervening tissue. Its amplitude depends on motor-unit activity and recording conditions. It is not a direct recording of motor-cortex commands, activation as used in a muscle model, or contractile force. The Farina, Merletti, and Enoka analysis explains why neural interpretation requires care. Electrode placement, cancellation, cross-talk, normalization, fatigue, and tissue motion all matter. A larger normalized signal in one arm does not compel a larger resultant hand wrench from that arm.

A model can link processed EMG to excitation with calibration and uncertainty, then evolve activation and musculotendon states. For example,

$$\dot{a} = \frac{e - a}{\tau(e, a)}, \qquad F_{\text{CE}} = F_{\max} a f_\ell(\ell_f) f_v(\dot{\ell}_f). \tag{19.21}$$

This illustrative contractile-element relation must be completed with passive fiber behavior, pennation, tendon mechanics, and force equilibrium or the chosen dynamic muscle model. It is not by itself a complete musculotendon force law. The OpenSim activation-dynamics documentation explicitly distinguishes excitation from activation. Different force-length and force-velocity states can produce different forces at the same activation.

EMG-informed modeling can be quantitatively useful when calibrated and validated; calling EMG merely qualitative would also be too strong. Report which muscles are observed, which deep or unmeasured muscles remain free, the activation and tendon assumptions, and prediction errors on independent conditions. Neither an EMG ratio nor a minimum-activation objective uniquely establishes how two hands share a club load without that intervening mechanics.

A model of internal tissue stress requires geometry and constitutive behavior in addition to boundary loads. An internal-load ambiguity identifies a limit on inference, not an injury diagnosis or a universal coaching prescription.

19.10 Ground Loads and Segment-by-Segment Balance

Separate calibrated force plates can provide a wrench for each foot when each foot's contact belongs to its assigned plate and all relevant contacts are captured. One plate under both feet usually gives their combined wrench; it does not generally determine the bilateral split. Vertical force alone is less information than a full force-and-moment measurement.

For a segment with COM C, known distal wrench (F_d, m_d) at point d, and unknown proximal wrench (F_p, m_p) at point p, all acting on the segment in inertial axes, the balances are

$$\begin{aligned} F_p &= m(a_C - g) - F_d - F_o, \\ m_p &= I_C\alpha + \omega \times (I_C\omega) - m_d - m_o^C \\ &\quad - (p_d - p_C) \times F_d - (p_p - p_C) \times F_p. \end{aligned} \tag{19.22}$$

F_o, m_o^C collect any other nongravitational external loads with moments already referenced to C. The segment's moment about its COM includes the moment arms of both distal and proximal forces. To proceed from foot to shank to thigh, reverse the interaction wrench on the neighboring segment and transport it to the appropriate point. Passing only an ankle torque upward while omitting its joint force is incomplete.

These equations estimate a transmitted net wrench under the segment model. Projecting onto a declared joint axis gives its generalized moment. Anatomical axis definitions, soft-tissue artifact, joint-center locations, inertial parameters, and synchronization affect the result. The estimate does not assign all of that wrench to one muscle or one tissue.

19.10.1 Yaw Moment Is Not a Vertical Force Offset

For a force F at COP r_{COP} and a free moment T_{free}, the moment about a reporting point P is

$$M_P = (r_{\text{COP}} - r_P) \times F + T_{\text{free}}. \tag{19.23}$$

In right-handed axes with z vertical, its yaw component is

$$(M_P)_z = (x_{\text{COP}} - x_P)F_y - (y_{\text{COP}} - y_P)F_x + (T_{\text{free}})_z. \tag{19.24}$$

A purely vertical force at a horizontal offset creates roll/pitch moments, not yaw. For $r = (0.2, -0.1, 0)$ m and $F = (0, 0, 1400)$ N, $r \times F = (-140, -280, 0)$ N m. A reported plate moment about its origin must be distinguished from the free moment after moving the resultant to COP. Vendor signs and whether the wrench is force-on-plate or force-on-golfer must be converted consistently.

For a 100 kg system, 1400 N is about 1.43 body weights using $g = 9.81$ m/s^2. If it is the only nongravitational vertical force on that system, its COM vertical acceleration is $1400/100 - 9.81 = 4.19$ m/s^2. Neither this force nor an 80 N m free moment alone determines pelvis acceleration. Dividing 80 N m by a declared 2 kg m^2 fixed-axis inertia gives 40 rad/s^2 for that isolated toy rotation, not a torque and not a complete multisegment pelvis model.

Whole-body angular momentum about the whole-body COM changes with the net external moment. Pelvis angular momentum also exchanges with legs and trunk through internal interactions. Different segments can counter-rotate while preserving whole-body angular momentum. A ground-moment contribution assigned to a pelvis coordinate therefore depends on the coupled dynamics and the chosen decomposition; it is not read directly from a force plate.

19.10.2 Momentum Exchange and Mechanical Power

At an ideal stationary no-slip contact on a rigid stationary ground, the contact velocity is zero, so contact force supplies zero mechanical power there. It can nevertheless change the golfer's momentum and permit muscle work to be expressed as segment and club motion. Real shoe deformation, sliding, moving supports, and contact compliance require their actual velocities and energy terms. A force multiplied by COM velocity gives a contribution to COM translational kinetic-energy rate, not automatically the work rate at the physical contact.

Assigning active ground power to every downswing and passive ground braking to every follow-through combines unsupported physiological interpretation with an incorrect contact-power shortcut. Measure the moment direction, state the system boundary, and compute power with its conjugate velocity. The energy source, route of transfer, and momentum constraint are related but distinct parts of the swing explanation.

19.11 Conditioning and Measurement Uncertainty

Inverse dynamics often uses acceleration estimates, so differentiation quality matters. Independent position-sample errors of variance σ_q^2 passed through the centered second difference have acceleration-error variance $6\sigma_q^2/\Delta t^4$. Real smoothing and correlated errors change this expression. It illustrates why quoting frame rate alone does not establish acceleration precision. Filtering choices should be evaluated against relevant motion and force bandwidth, not chosen solely to make torque traces smooth.

For an open-tree net-load evaluation at fixed model parameters, a first-order perturbation includes

$$\delta\tau = M\,\delta\ddot{q} + (\delta M)\ddot{q} + \delta h - \delta p - \delta Q_e. \tag{19.25}$$

Cross-correlations matter when these quantities come from the same fitted trajectory. Covariance propagation or repeated sampling through the complete pipeline is preferable to treating all errors as independent by default. Model discrepancy and missing external loads are not necessarily random measurement noise.

19.11.1 A Counterexample to the Small-Inertia Argument

Let $M = \mathrm{diag}(1, \epsilon)$ in consistent coordinate and inertia units, $0 < \epsilon \le 1$. Its two-norm condition number is $1/\epsilon$. But for a fixed acceleration error $(0.1, 0.1)^T$,

$$\delta\tau = M\,\delta\ddot{q} = (0.1, 0.1\epsilon)^T. \tag{19.26}$$

The second torque error is 0.01 at $\epsilon = 0.1$ and 0.001 at $\epsilon = 0.01$. It decreases. By contrast, a fixed torque error $(0.1, 0.1)^T$ in a forward solve produces acceleration error $(0.1, 0.1/\epsilon)^T$, whose second component increases. The condition number describes relative worst-case sensitivity with the specified norm and scaling; it is not the absolute gain of every inverse-dynamics perturbation. A matrix $1000I$ has condition number one yet multiplies an absolute acceleration error by 1000 in a load evaluation.

For a planar two-link arm with uniform links, each of unit length and unit mass, at a straight configuration the joint inertia and endpoint-position Jacobian can be

$$M = \begin{bmatrix} 8/3 & 5/6 \\ 5/6 & 1/3 \end{bmatrix}, \qquad J_y = \begin{bmatrix} 0 & 0 \\ 2 & 1 \end{bmatrix}. \tag{19.27}$$

The task Jacobian has rank one; the mass matrix has positive diagonal entries and determinant $7/36 > 0$. The endpoint singularity has not made the joint inertia singular. In constrained and sensor-augmented inverse problems, inspect the correctly scaled identification matrix and its singular values as well as the underlying inertial model. State angle/translation scaling before comparing condition numbers across models.

19.12 Forward Dynamics and a Declared Counterfactual

For specified inputs, constitutive states, and external loads, the acceleration and ideal reaction in a fixed contact mode satisfy

$$\begin{bmatrix} M & -J_c^T \\ J_c & 0 \end{bmatrix} \begin{bmatrix} \ddot{q} \\ \lambda \end{bmatrix} = \begin{bmatrix} Bu + p + Q_e - h \\ -\dot{J}_c \dot{q} \end{bmatrix}. \tag{19.28}$$

If $M \succ 0$ and J_c has full row rank, the matrix is nonsingular. To see this, a homogeneous solution obeys $M\delta a - J_c^T \delta\lambda = 0$ and $J_c \delta a = 0$. Multiplying the first equation by δa^T gives $\delta a^T M \delta a = 0$, so $\delta a = 0$ and then $\delta\lambda = 0$.

This proves unique instantaneous acceleration for those inputs. Local trajectory existence and uniqueness also require compatible initial conditions and sufficient regularity of the reduced differential equations and input policy. A smooth fixed-mode vector field is locally Lipschitz; contact switching, friction, impacts, and rank changes require event laws and a fresh analysis. Matrix invertibility at one instant is not a global guarantee for an entire swing.

19.12.1 Same-State Acceleration and a Forward Branch

A zero-torque counterfactual (ZTCF) must specify which input is set to zero. Zero commanded joint torque, zero muscle excitation, zero activation, and removal of muscle/tissue forces are different interventions. In an effective joint-input model, setting $u = 0$ retains the stated p, Q_e, h laws and their states. If preload or activation-dependent impedance is not measured, the baseline inherits that uncertainty. Gravity can be computed from geometry and mass assumptions; that does not make every elastic or physiological drift term observable from skeletal motion.

At the same complete state and fixed mode, with other loads held under the same law, write the acceleration map as

$$\begin{aligned} a(q, \dot{q}, z, u) &= a(q, \dot{q}, z, 0) + K(q)B(q)u, \\ K &= M^{-1} - M^{-1}J_c^T (J_c M^{-1} J_c^T)^{-1} J_c M^{-1}. \end{aligned} \tag{19.29}$$

This identifies the input's acceleration effect under the model. Even if it is known exactly, KBu need not identify u, because projection and actuator redundancy can remove load directions. Reactions must be recomputed when the input changes. Retaining measured reactions while zeroing u generally breaks constraint compatibility.

A forward ZTCF instead starts at the chosen initial state and integrates the zero-input policy along its own trajectory. It recomputes geometry, inertia, velocity bias, passive states, reactions, and events along that branch. Once it has diverged from the actual trajectory, subtracting the two accelerations no longer compares the same state. Multiplication by one trajectory's M cannot undo that difference or recover muscle torques.

For an unconstrained dimensionless toy oscillator,

$$\ddot{q} = -q + u, \qquad q(0) = \dot{q}(0) = 0, \qquad q_{u=1}(t) = 1 - \cos t, \quad q_{u=0}(t) = 0. \tag{19.30}$$

At $t = \pi/2$, the two trajectories' accelerations are both zero, although the actual input remains one. The acceleration difference across branches is zero; the same-state difference is $0 - (-1) = 1$. This elementary example shows why a trajectory residual is a model intervention outcome, not an instantaneous input reconstruction.

For finite-horizon clubface or speed outcomes, compare the outcomes of the two branches with their uncertainty and initial-state matching explicitly stated. When using a local variational approximation instead, propagate perturbations along its evolving baseline and identify its validity range. A forward counterfactual asks a useful different question from inverse muscle inference; it is not universally more accurate simply because it runs forward.

19.13 A Golf Study That Can Distinguish Explanations

Suppose the question is whether two movement patterns achieve similar club outcomes through different hand loading, and whether an effective zero-input branch predicts different continuation of the motion. A defensible study can connect these questions without treating any one measurement as sufficient.

1. Define the primary output: for example, clubhead velocity and face orientation just before impact, a net hand-wrench time history, or work transferred to the club over a specified interval. Define tolerances for calling two outcomes similar and distinguish impact conditions from ball flight, which also depends on collision and aerodynamic variables.

2. Specify the body graph, joint coordinates, separated grip frames, club flexibility assumptions, contact modes, inertial parameters, and input channels. Validate measured trajectory compatibility; report closure residuals instead of hiding them inside large artificial reactions.

3. Calibrate and synchronize kinematics, bilateral ground wrenches, and the selected grip measurements. Derive the augmented identification map and show which quantities are determined. Reserve measured channels or trials for validation rather than using all data both to fit and to confirm.

4. Solve the declared net-load problem and test segment and whole-system force, moment, and power balances. Propagate plausible measurement and parameter variation. If competing load distributions remain possible, report that family or interval rather than only one optimizer output.

5. Add muscle modeling and EMG only for the recruitment questions they can inform. Compare alternative calibrated recruitment assumptions and inspect prediction error on independent data. Do not translate inferred muscle forces into tissue stress or injury likelihood without the additional validated model required for that output.
6. Branch the forward counterfactual from a declared event or state, specify zeroed inputs and retained states, and integrate each branch independently. Report the event rules, numerical convergence, and uncertainty in its endpoint. Compare models and repeat the analysis across trials before drawing a general conclusion about a golfer or a population.

This links geometry to admissible motion, dynamics to required net loading, measurement to identification, muscle models to recruitment hypotheses, and power balance to energy transfer. The central scientific gain is an argument that makes clear what would change its conclusion. A successful fit to one swing is useful evidence about that fitted model; independent predictions and controlled comparisons are needed for stronger claims about strategy or benefit.

19.14 Chapter Exercises

1. Verify the two-coordinate inverse matrix's null vector. Compare ranks after measuring the actuator sum, measuring the multiplier, and removing the second actuator.
2. Derive wrench transport from invariant power. Reproduce the 46 W example and identify the lost lever-arm term when reference points are mixed.
3. Derive grasp-map ranks for two full hand wrenches and for two separated point forces. Explain why a transverse opposing force pair creates a couple.
4. Derive the weighted minimum from its Lagrangian and verify $(8, 2)$. Explain why the selected objective does not establish neural intent.
5. Derive the bridge voltage from its divider voltages and reverse the output leads. Convert the piezoelectric charge using each stated capacitance.
6. Reproduce the inverse and forward error calculations for $M = \mathrm{diag}(1, \epsilon)$. Verify the straight arm's singular task Jacobian and positive-definite inertia.
7. Use the oscillator solution to compare cross-branch and same-state acceleration differences at $t = \pi/2$. State what each comparison holds fixed.
8. A paper reports separate left and right spinal muscle moments. Identify required boundary loads, moment arms, physiological laws, and independent measurements. Explain why muscle redundancy needs no spinal kinematic loop.
9. Design a sensor addition for a remaining golf load-sharing null direction. State its rank benefit, calibration needs, and a quantity it cannot identify.

Part VI

The Physical System

Chapter 20

The Flexible Shaft: Elastic Energy and the Catapult Effect

A rigid shaft is an abstraction; a flexible shaft is a machine.

—AffineDrift

In Plain Language 20.1. Why the Shaft Matters

For most of this book, we have treated the golf club as if it were a rigid stick. This simplification has been valuable—it let us understand the fundamental forces and constraints that shape the swing. But a real golf shaft is not rigid. It bends, twists, and vibrates. These are not small corrections. The shaft's flexibility is responsible for a significant fraction of the energy transfer to the ball, especially in slower swings. Worse, ignoring shaft flexibility leads to wildly inaccurate predictions of the trajectory, spin, and accuracy.

The shaft is not a passive passenger in the swing; it is an energy storage and release mechanism—a catapult that supplements the muscles and gravity.

In this chapter, we develop a mathematical model of shaft flexibility, show how it enters the drift and control equations, and explain why shaft fitting is fundamentally a question about which trajectory you want the ZTCF family to produce.

20.1 Why Rigid Models Fail

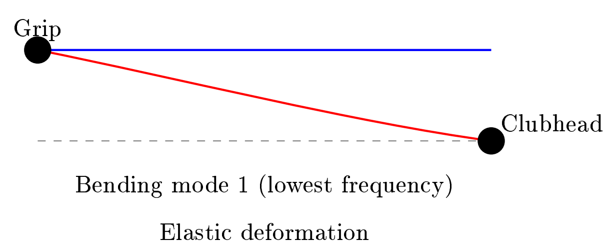


Figure 20.1: Shaft Bending Modes: Fundamental Bending Creates Elastic Energy Storage.

When we modeled the golf swing as a double or triple pendulum, we implicitly assumed that the club shaft is infinitely stiff. This is like modeling a trampoline as rigid—the model works fine for a heavy wooden block, but fails spectacularly for a person jumping.

A typical golf shaft is made of composite fibers (carbon fiber or graphite) wrapped in resin. When you swing, the shaft experiences:

1. **Bending** due to centrifugal force pulling the clubhead outward while the grip end is held at a different acceleration.
2. **Twisting** (torsion) as the hands rotate and the clubhead lags.
3. **Damping** as energy is dissipated by internal friction in the composite material.

All three effects matter. The bending deformation is largest and most relevant to the swing trajectory, so we focus primarily on that.

Example 20.1. Shaft Deflection Magnitudes

During the late downswing, the clubhead experiences centripetal accelerations on the order of 2000–2500 $\mathrm{m/s^2}$ ($\sim$200–250g), while the grip end accelerates at only $\sim$300 $\mathrm{m/s^2}$ ($\sim 30g$). (Note: these are substantially larger than the $\sim 18g$ swing-phase acceleration computed in Chapter 4 because they include the full centripetal acceleration of the clubhead about the instantaneous center of rotation, not just the shoulder-referenced centripetal term.) This huge difference in acceleration means the shaft experiences an enormous differential force that bends it.

For a modern driver shaft (typically around 44–46 inches long, with shaft mass on the order of 50–80 g depending on flex and material), this can produce a deflection of several inches at the clubhead. Published estimates range from roughly 1–5 inches depending on shaft flex, clubhead mass, and swing speed [Penner, 2003a, MacKenzie and Sprigings, 2009a]. In shorter, stiffer shafts (like driver shafts designed to be whip-resistant), this is reduced but never eliminated.

The rigid model predicts that the clubhead would be at the end of a straight line from the grip. The flexible model correctly predicts that the clubhead lags behind, causing shaft lag at impact. This lag is where the catapult effect comes in: the shaft is storing elastic energy as it bends, and that energy is released as the shaft snaps back toward impact.

20.2 Beam Theory Basics: How Shafts Deform

To model shaft flexibility mathematically, we use **Euler-Bernoulli beam theory**, a classical result from mechanics. A beam (our shaft) can be described by how much it bends at each point along its length.

The governing dynamic equation for transverse shaft deflection is:

$$EI\frac{\partial^4 w}{\partial z^4}(z,t) + \rho A\frac{\partial^2 w}{\partial t^2}(z,t) = q(z,t)$$

where:

- $w(z,t)$ is the vertical deflection of the shaft at position z and time t.
- E is the material's elastic modulus (stiffness).

- I is the second moment of inertia of the cross-section (how the material is distributed around the neutral axis).
- ρ is the material density.
- A is the shaft cross-sectional area.
- $q(z,t)$ is the distributed load on the shaft (in our case, inertial forces from acceleration).
- The product EI is called the **bending stiffness**.

For quasi-static approximations where inertial effects are negligible, we use the reduced form

$$EI\frac{\partial^4 w}{\partial z^4}(z) = q(z)$$

Modal decomposition assumes each mode shape $\phi_j(z)$ satisfies the eigenproblem implied by the dynamic beam equation, and projection onto these modes produces the coupled ODEs in terms of $\eta_j(t)$.

In Plain Language 20.2. Why EI Matters

Imagine two shafts of the same length. One is thin and hollow; one is thick and solid. Both are made of the same material. The thin shaft bends a lot for the same force. The thick shaft is much stiffer. The product EI captures this: a thicker shaft has a larger I, so a larger EI, and thus is stiffer.
In practical terms: a stiffer shaft (*higher* EI) requires a larger force to achieve the same deflection. A flexible shaft (*lower* EI) stores less elastic energy during the swing and releases it later (closer to impact).

In a golf swing, we don't need to solve this equation pointwise along the entire shaft. Instead, we use a technique called **modal decomposition**: we express the total deformation as a sum of a small number of deformation patterns (modes), each with its own amplitude.

For a free cantilever beam (clamped at one end, free at the other—like a golf shaft), the first few modes have known shapes. Rather than solving the full PDE, we can write:

$$w(z,t) = \sum_{j=1}^{N} \eta_j(t)\,\phi_j(z)$$

where:

- $\eta_j(t)$ are the **modal coordinates**, the time-varying amplitudes of each mode.
- $\phi_j(z)$ are the **mode shapes**, fixed functions that describe the spatial deformation pattern.
- N is the number of modes we keep (typically 1–3 in practice).

For a golf shaft, we almost always use just the first mode, the fundamental bending mode, because it dominates the energy transfer. So:

$$w(z,t) \approx \eta(t)\,\phi_1(z)$$

The modal coordinate $\eta(t)$ is treated as a generalized coordinate, just like a joint angle. It evolves according to its own differential equation, coupled to the rest of the swing dynamics.

Definition 20.1. Modal Coordinates

A modal coordinate η is a single number that describes how much the shaft is deforming. When $\eta = 0$, the shaft is straight. When η increases, the shaft bends more in the shape of the first mode. The rate of change $\dot{\eta}$ describes how quickly the deformation is changing.

20.3 The Extended State Vector

Previously, we wrote the state as $\boldsymbol{x} = [\boldsymbol{q}, \dot{\boldsymbol{q}}]^T$, where $\boldsymbol{q}$ is the vector of joint angles and $\dot{\boldsymbol{q}}$ is the vector of joint angular velocities.

With shaft flexibility, we extend the state to include the modal coordinates and their velocities:

$$\boldsymbol{x} = \begin{bmatrix} \boldsymbol{q} \\ \dot{\boldsymbol{q}} \\ \boldsymbol{\eta} \\ \dot{\boldsymbol{\eta}} \end{bmatrix}$$

where $\boldsymbol{\eta} = \eta$ and $\dot{\boldsymbol{\eta}} = \dot{\eta}$. If we keep multiple modes, $\boldsymbol{\eta}$ and $\dot{\boldsymbol{\eta}}$ are vectors.

Example 20.2. State Dimension Growth

A double pendulum has $\boldsymbol{q} \in \mathbb{R}^2$ (two joint angles). So $\boldsymbol{x}_{\text{rigid}} \in \mathbb{R}^4$.
The same double pendulum with a flexible shaft has $\boldsymbol{q} \in \mathbb{R}^2$ and $\boldsymbol{\eta} \in \mathbb{R}^1$ (just the first mode). So $\boldsymbol{x}_{\text{flexible}} \in \mathbb{R}^6$.
The system of differential equations is now larger, but it is still affine in the muscle torques (control inputs). We can still separate drift from control.

The dynamics of the modal coordinates are themselves driven by the joint motion and by the modal stiffness and damping. In the absence of any active control of the shaft (and there is no muscle that controls η directly), the modal equations are:

$$\ddot{\eta} + 2\zeta\omega_n\dot{\eta} + \omega_n^2\eta = f_{\text{modal}}(\boldsymbol{q}, \dot{\boldsymbol{q}})$$

where:

- ω_n is the natural frequency of the first bending mode (a property of the shaft material and geometry).
- ζ is the damping ratio (how much the vibration is damped by material friction).
- f_{modal} is the generalized force on the modal coordinate, arising from inertial and Coriolis forces in the joint motion.

For a typical driver shaft, the first bending mode natural frequency is about 3–5 Hz ($\omega_n \approx 20$–30 rad/s), corresponding to a shaft "frequency" rating of roughly 240–280 CPM (cycles per minute). The downswing lasts about 0.2–0.25 seconds, so at ~ 4 Hz the shaft passes through roughly *one* bending cycle (load, then recover) during the downswing — the basis of the familiar "kick" at impact.

20.4 How Shaft Flexibility Enters the Drift Field

The extended dynamics are still affine:

$$\dot{\boldsymbol{x}} = f(\boldsymbol{x}) + G(\boldsymbol{x})\boldsymbol{u}$$

The drift field f now includes terms from the modal dynamics. Specifically:

1. The first four components of f (joint angles and velocities) are almost the same as before, except they now include the effect of the bent shaft on the inertia and constraints.
2. The fifth component is $\dot{\eta}$ (the rate of shaft deformation), which is always equal to the fifth component of $\boldsymbol{x}$ by definition.
3. The sixth component is $\ddot{\eta}$, which is driven by:

$$\ddot{\eta} = -2\zeta\omega_n\dot{\eta} - \omega_n^2\eta + f_{\text{modal}}(\boldsymbol{q}, \dot{\boldsymbol{q}}) \tag{20.1}$$

The term f_{modal} deserves attention. When the hands are accelerating upward and the shaft is relatively straight, $f_{\text{modal}} > 0$, which causes η to increase (the shaft bends in response to the acceleration mismatch). As the swing slows and the shaft geometry changes, the effective load on the shaft changes, and f_{modal} may change sign, causing the shaft to snap back.

In Plain Language 20.3. The Catapult Mechanism

Think of the shaft as a spring that has been compressed and is waiting to release. During the backswing and early downswing, the shaft is bent by the acceleration mismatch between the hands and the clubhead. This bending costs energy—the shaft is doing work against its bending stiffness, storing elastic energy like a stretched rubber band.

Later in the swing, as the hands slow slightly and the clubhead catches up, the shaft begins to snap back. This snapback releases the stored elastic energy back into the clubhead, giving it an extra boost at a critical moment. This is the "catapult effect."

The timing of the snapback depends on the shaft's stiffness and damping. A very stiff shaft snaps back early and weakly; a very flexible shaft snaps back late and more powerfully, but it might still be bending as the ball is struck (which is bad for accuracy).

Drift vs. Control 20.1. Shaft Flexibility in Drift vs. Control

Drift: The shaft bends and unbends in response to acceleration mismatches and gravity. This is fully passive—no muscle is controlling the deformation. The shaft dynamics are part of the drift field $f(\boldsymbol{x})$. Once you set the initial conditions in the backswing, the shaft bending evolves on autopilot.

Control: The muscles can change the joint angles, and this changes the rate of change of f_{modal}, which indirectly influences the shaft dynamics. But there is no direct muscle control of the shaft deformation. You cannot command the shaft to

bend more or less at a given moment.
In other words: shaft flexibility is almost entirely in the drift. This is why shaft fitting matters to the drift field, not the control field.

20.5 Elastic and Damping Terms

In the equations above, the terms $-\omega_n^2\eta$ and $-2\zeta\omega_n\dot{\eta}$ are the elastic restoring force and the damping force on the shaft, respectively. Let's make them more explicit.

The **elastic term** can be written as:

$$F_{\text{elastic}} = -K_s\eta$$

where $K_s = \omega_n^2 m_{\text{eff}}$ is an effective shaft stiffness. (Here m_{eff} is the effective mass of the clubhead as "seen" by the bending mode.)

The **damping term** can be written as:

$$F_{\text{damping}} = -D_s\dot{\eta}$$

where $D_s = 2\zeta\omega_n m_{\text{eff}}$ is an effective damping coefficient.

These are the terms that slow down the shaft's oscillation and dissipate its energy. In the absence of any driving force ($f_{\text{modal}} = 0$), the shaft would oscillate and gradually come to rest. The damping ratio ζ controls how quickly the oscillation decays.

Key Principle 20.1. Elastic and Damping Forces

The shaft experiences two passive forces:

- An elastic restoring force $-K_s\eta$ that tries to straighten the shaft.
- A damping force $-D_s\dot{\eta}$ that opposes the shaft's deformation rate.

These forces are proportional to the deformation and its rate, respectively. Neither requires muscle action. Both are part of the drift field.

20.6 Shaft Loading During the Downswing: Centrifugal Stiffening

Here is a subtle but important effect: as the shaft rotates faster and faster during the downswing, the centrifugal force on the shaft itself increases. This centrifugal force acts like an additional stiffness, making the shaft harder to bend.

Consider a small element of the shaft at distance r from the pivot, rotating with angular velocity ω. The centrifugal acceleration on that element is $\omega^2 r$. When the shaft bends, this centrifugal force has a component that tries to straighten the shaft, increasing its effective stiffness.

Mathematically, the effective bending stiffness becomes:

$$K_{s,\text{eff}} = K_{s,0} + \Delta K_{s,\text{centrifugal}}(\omega)$$

where $K_{s,0}$ is the passive bending stiffness and $\Delta K_{s,\text{centrifugal}}$ is the additional stiffness due to centrifugal effects.

Early in the downswing, ω is small, so the effective stiffness is close to $K_{s,0}$. The shaft bends easily. Later in the downswing, ω grows, the centrifugal stiffening kicks in, and the shaft becomes harder to bend. This is another reason the shaft might snap back: it is not just that the driving force f_{modal} changes sign, but also that the shaft becomes stiffer.

Example 20.3. Centrifugal Stiffening in a Driver Swing

Consider a driver with a 45-inch shaft. At address, the clubhead is stationary. Early in the downswing, at the transition, the hands might be rotating at $\omega = 2$ rad/s. Later, at mid-downswing, $\omega = 8$ rad/s. At impact, $\omega \approx 30$ rad/s.
The centrifugal acceleration at the clubhead (distance $r \approx 1.1$ m from the shoulder joint) ranges from:

- At transition: $a_c = \omega^2 r = 2^2 \times 1.1 = 4.4\,\text{m/s}^2$ (negligible)
- At mid-downswing: $a_c = 8^2 \times 1.1 = 70\,\text{m/s}^2$ (significant)
- At impact: $a_c = 30^2 \times 1.1 = 990\,\text{m/s}^2$ (huge)

This centrifugal acceleration stiffens the shaft. Combine this with the increasing downward acceleration from gravity and the deceleration of the hands, and you have multiple mechanisms all working to snap the shaft back into its straight configuration near impact.

20.7 The Catapult Effect: Energy Release Near Impact

We are now in a position to explain the catapult effect precisely.

In the mid-downswing phase (high drift, low control), the golfer's hands are still accelerating forward and downward, but at a decreasing rate. The centrifugal acceleration of the clubhead, however, is still increasing (because the rotation rate ω is increasing faster than the hand deceleration). This mismatch—hands slowing, clubhead accelerating—causes the shaft to bend more and more, storing elastic energy.

Then, near impact, the hands slow sharply (they are being decelerated by the ground reaction forces through the legs and torso). Suddenly, the driving force f_{modal} becomes very large and positive. At the same time, the centrifugal stiffening reaches its maximum. The shaft, which has been bent for 100+ milliseconds of the downswing, suddenly has every incentive to snap back: the inertial driving force is trying to straighten it, and the increased stiffness makes the elastic restoring force huge.

The result: the shaft unbends very rapidly. As it does, it does work on the clubhead, accelerating it further. This is the catapult effect.

$$\text{Energy released by shaft} = \frac{1}{2} K_s \eta^2 \quad \text{(at the moment before snapback)} \tag{20.2}$$

The magnitude of this energy depends on how much the shaft is bent (η) and how stiff it is (K_s). A very flexible shaft bends a lot (*high* η) but stores less energy per unit deflection (low K_s). A very stiff shaft bends less but packs more energy into that small deformation.

In Plain Language 20.4. Why Shaft Flex Profiles Matter

Real golf shafts are not uniformly flexible. They are often stiffer at the grip end (where they need to handle the hand loads) and more flexible toward the tip (the free end). This creates a nonuniform bending profile.
The consequences are far-reaching: a more flexible tip means more bend and more stored energy, but the bend is concentrated near the clubhead, which affects the timing of energy release and the orientation of the clubhead at impact. A stiffer tip means less stored energy but potentially better control of the clubhead orientation.
Shaft fitting is fundamentally the art of choosing the right combination of length, overall stiffness, flex profile, and weight to match your swing speed and swing characteristics. Different profiles produce different f_{modal} functions, which means different ZTCF trajectories.

20.8 Shaft Flex Profiles and Forward ZTCF Trajectories

Here, ZTCF denotes the forward trajectory obtained after setting the model's declared applied generalized-control channel to zero. Gravity, velocity-dependent inertial terms, shaft elasticity and damping, and constraint reactions remain in the drift. The trajectory therefore depends on the initial state, including shaft deformation and deformation rate.

Different shaft flex profiles lead to different drift fields, and thus different ZTCF trajectories.

Example 20.4. Comparing Shaft Profiles

Consider two golfers with identical biomechanics (same joint angles, angular velocities, and torque patterns) but different shafts.
Golfer A: Very stiff shaft ($K_s = 10,000$ N/m, $\zeta = 0.1$).

- The shaft bends less during the swing.
- Less elastic energy is stored.
- The shaft behavior is more "predictable" (closer to the rigid case).
- The clubhead is more firmly controlled throughout the swing.
- But less of the shaft's elastic energy is available for catapult effect at impact.

Golfer B: More flexible shaft ($K_s = 6,000$ N/m, $\zeta = 0.1$).

- The shaft bends more during the swing.
- More elastic energy is stored.
- The shaft behavior is more oscillatory (larger deviations from rigid case).
- More of the shaft's elastic energy is released at impact, boosting the clubhead.
- But if the shaft is still bending as impact occurs, the clubhead orientation is less predictable, causing dispersion.

For a high-speed golfer (say, 90+ mph swing speed), the centrifugal stiffening effect is so large that even a relatively flexible shaft is effectively stiff by mid-downswing. So a flexible shaft is useful. For a slower golfer, the centrifugal stiffening is weaker, and a flexible shaft might still be oscillating at impact, causing control problems. For a given swing speed, the goal is to have the shaft reach its maximum elasticity just before impact, releasing all its energy at the optimal moment. The choice of shaft characteristics (flex, profile, weight) should match the golfer's swing parameters (speed, acceleration profile) to optimize energy transfer.

20.9 Shaft Fitting Through the Lens of Drift

Traditional shaft fitting focuses on swing speed, carry distance, and feel. These are important, but they are symptoms. The underlying physics asks: *which drift field* $f(\boldsymbol{x})$ *best matches your swing?*

Changing shaft stiffness changes the declared effective plant and therefore its drift field and forward ZTCF trajectory. Whether that change improves a delivery objective is an empirical, golfer-specific question. Within a calibrated model, the following outcomes can be tested:

- **Higher stiffness:** The shaft generally bends less and the ZTCF approaches the corresponding rigid-shaft trajectory. The sign and magnitude of the delivery-speed change depend on excitation timing, damping, mass distribution, and initial state.
- **Lower stiffness:** Larger deformation may store more elastic energy, but its release can occur before or after the delivery window. A deterministic ZTCF has no statistical variance by itself; dispersion requires a declared distribution of initial states, parameters, or controls.
- **Task-matched stiffness:** A candidate shaft is supported only when calibrated simulations and held-out measurements show improved task metrics while respecting orientation, load, and robustness constraints.

The task of shaft fitting is to adjust K_s, ζ, the flex profile, and the total shaft mass to produce a drift field whose ZTCF closely matches the golfer's target trajectory for their measured swing speed and swing geometry.

Key Principle 20.2. Shaft Fitting as Drift Design

The shaft design should accomplish the following:

1. It aligns the drift field $f(\boldsymbol{x})$ with your swing characteristics, so that the zero-torque counterfactual (the autopilot trajectory) naturally points toward the target.
2. It maximizes the energy transfer from the muscles and gravity to the clubhead by timing the catapult effect to release its energy at the optimal moment.

Shaft fitting is not an art; it is physics. The wrong shaft changes your drift field in ways that pull your ZTCF off target.

20.10 The Shaft as an Energy Buffer

Let us step back and think about energy flow in the golf swing.

At the top of the backswing, the golfer has stored potential energy in the muscles (through their isometric contraction) and positional energy (the body is wound up, arms are high). During the downswing, the muscles apply torques, doing work. But not all the muscle work goes directly into the clubhead's kinetic energy. Some of it goes into deforming the shaft.

The shaft acts as a temporary storage device, holding energy as elastic deformation and returning it later.

$$\text{Muscle work} = \Delta KE_{\text{clubhead}} + E_{\text{stored in shaft}} + E_{\text{dissipated by damping}} \tag{20.3}$$

For a flexible shaft, the term $E_{\text{stored in shaft}}$ is significant. In a fast swing, the shaft absorbs and releases energy very quickly (within milliseconds of mid-downswing). In a slow swing, the shaft might remain bent for most of the downswing, absorbing a larger fraction of the muscle work temporarily.

The benefit of this storage mechanism is that it allows the muscles to do work earlier in the downswing (when the torque levers are shorter and the muscles are fresh) and have that energy released later, when the clubhead can best use it (near impact, when the lever arm is longest).

In Plain Language 20.5. Why Flexible Shafts Help Slower Golfers More

A slower golfer (for illustration, consider a driver swing speed around 70 mph, which is below tour-professional levels [Jorgensen, 1994a]) has less centrifugal stiffening throughout the swing. The shaft remains more flexible. This means:

1. The shaft can absorb and hold more energy per unit bend (because K_s is lower).
2. The catapult effect, when it occurs, releases this energy directly to the clubhead at a critical moment.
3. Without the flexible shaft, the golfer would have to do all the work of accelerating the clubhead with muscle torque alone, and they would have less momentum to work with (lower joint velocities).

A flexible shaft essentially lets a slower golfer borrow energy from the shaft dynamics, partially compensating for lower muscle power.
Conversely, a faster golfer has high centrifugal stiffening and high momentum already. A flexible shaft still helps, but the relative benefit is smaller.

20.11 Worked Example: Comparing Rigid and Flexible Shaft ZTCF

Let us work through a concrete example to see how shaft flexibility changes the ZTCF.

Consider a simplified model: a point mass (the clubhead) at the end of a shaft, with the grip end following a prescribed motion (from a human swing).

Rigid Shaft Case

The grip position follows $\boldsymbol{r}_{\text{grip}}(t)$ with velocity $\boldsymbol{v}_{\text{grip}}(t)$ and acceleration $\boldsymbol{a}_{\text{grip}}(t)$. The clubhead is constrained to be at distance L from the grip (shaft length), so:

$$\boldsymbol{r}_{\text{head}} = \boldsymbol{r}_{\text{grip}} + L\,\hat{\boldsymbol{n}}$$

where $\hat{\boldsymbol{n}}$ is the unit vector along the shaft direction. In the rigid case, $\hat{\boldsymbol{n}}$ is determined by the hand orientation alone.

The clubhead acceleration includes gravity and constraint forces:

$$m_{\text{head}}\boldsymbol{a}_{\text{head}} = m_{\text{head}}\boldsymbol{g} + \boldsymbol{F}_{\text{constraint}}$$

where $\boldsymbol{F}_{\text{constraint}}$ is the force keeping the clubhead at distance L from the grip.

The forward ZTCF is the trajectory obtained with the declared applied generalized-control channel set to zero. Constraint reactions are retained together with gravity, inertia, and any passive shaft or joint terms included in the effective plant. This is a model intervention, not a claim of absent muscle activity.

Flexible Shaft Case

Now the grip motion is the same, but the shaft can bend. The clubhead is no longer constrained to be exactly at distance L from the grip. Instead:

$$\boldsymbol{r}_{\text{head}} = \boldsymbol{r}_{\text{grip}} + L\,\hat{\boldsymbol{n}} + \eta(t)\boldsymbol{\phi}(z = L)$$

where $\boldsymbol{\phi}(z = L)$ is the lateral displacement at the tip of the mode shape $\phi_1(z)$.

The shaft deformation $\eta(t)$ evolves according to:

$$\ddot{\eta} + 2\zeta\omega_n\dot{\eta} + \omega_n^2\eta = f_{\text{modal}}(\text{grip motion})$$

This is a second-order ODE coupled to the grip motion.

Numerical Example

Let us use realistic numbers:

- Clubhead mass: $m_h = 0.2$ kg (about 7 oz).
- Shaft length: $L = 1.1$ m (about 43 inches).
- Shaft stiffness: $K_s = 7000$ N/m (typical for a mid-flex shaft).
- Damping ratio: $\zeta = 0.08$.
- Natural frequency: $\omega_n = 15$ rad/s, so $D_s = 2 \times 0.08 \times 15 \times m_{\text{eff}} \approx 0.3$ N·s/m.

Suppose the hand (grip) starts at $y = 0$ and is prescribed to move with:

$$y_{\text{grip}}(t) = 0.5\sin(\pi t/0.2) \quad \text{for } t \in [0, 0.2] \text{ s} \tag{20.4}$$

This describes a hand moving upward from $t = 0$ to $t = 0.1$ s, then downward from $t = 0.1$ to $t = 0.2$ s, reaching a maximum height of 0.5 m at mid-swing.

For the **rigid shaft**, the clubhead follows the hand motion plus a lag due to the centrifugal acceleration. The clubhead trajectory is smooth and nearly sinusoidal.

For the **flexible shaft**, the clubhead additionally experiences the shaft deformation. Early in the swing, as the hand accelerates, the shaft bends (positive η). At $t \approx 0.1$ s, the hand reaches peak height and begins to decelerate. The shaft begins to unbend, and if we plot the clubhead position $y_{\text{head}} = y_{\text{grip}} + L + \eta \times \phi_1(L)$, we see an extra oscillation superimposed on the hand motion. This oscillation is the catapult effect.

The magnitude of the effect depends on the hand acceleration profile and the shaft parameters. For typical golfer numbers, the extra clubhead displacement due to shaft flex is 5–15 cm (2–6 inches) at the moment of maximum deflection, and the extra velocity at impact is 5–15% of the clubhead speed.

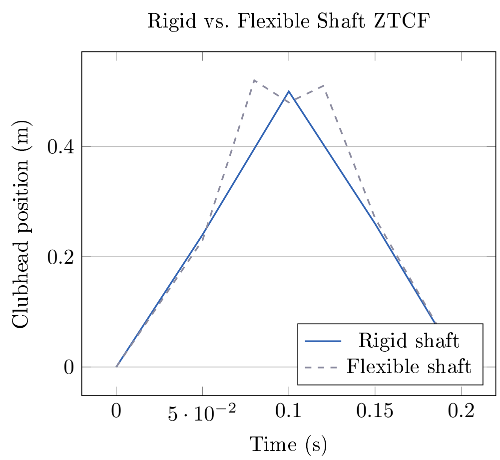


Figure 20.2: Qualitative Comparison of Rigid vs. Flexible Shaft ZTCF Trajectories. The Flexible Shaft Shows Oscillation Near the Maximum Height, Characteristic of the Catapult Effect.

The key insight: the flexible shaft does not produce a lower ZTCF trajectory; it produces a *different* ZTCF trajectory, one that includes oscillations due to shaft dynamics. If your swing muscles and gravity are tuned to work with a particular drift field, using a shaft with different stiffness changes that drift field, and you must adjust your muscle activation (control inputs) to compensate.

20.12 Why Shaft Flex Is in the Drift, Not the Control

It is tempting to think of shaft flexibility as something a golfer can "control," the way a professional tennis player might control the vibration of their racket strings. But golf is different.

There is no muscle in the human arm that directly controls the shaft's deformation. The deformation happens passively, in response to inertial forces. The golfer can *indirectly* influence the deformation by changing the acceleration profile of the hands (which changes f_{modal}), but this is done by controlling the joint angles, not by directly commanding the shaft.

Therefore, shaft flexibility belongs entirely in the drift field. It is passive and automatic.

Shaft flexibility changes $f(\boldsymbol{x})$ because elastic and damping effects belong to the declared effective plant. The generalized-coordinate input map $G(\boldsymbol{x})$ may remain unchanged in a particular actuation model, but task-space authority can still change through configuration, Jacobians, constraints, and admissible-input bounds. A different shaft therefore implies a different drift field and forward ZTCF; it does not justify a universal claim about control capacity.

Drift vs. Control 20.2. Shaft Flexibility: Drift Only

Why shaft is drift:

1. No muscle controls the shaft deformation directly.
2. The shaft bends in response to acceleration mismatches, which are consequences of the hand motion (determined by joint angles, which are controlled).
3. The stiffness and damping of the shaft are passive properties.
4. The shaft dynamics add to the drift field $f(\boldsymbol{x})$, not the control field $G(\boldsymbol{x})$.

Consequence: For a declared actuation model, compare drift and bounded admissible control capacity in the same generalized-acceleration or task-projected metric. Shaft parameters always alter the retained passive dynamics; they may also alter task-space control capacity through the state and constraint geometry. These effects must be reported separately rather than inferred from the ZTCF alone.

Key Principle 20.3. Key Takeaways

1. **Real shafts bend.** A rigid model ignores a significant source of energy and trajectory variation.
2. **Shaft deformation is described by modal coordinates.** A single coordinate η captures the first (most important) bending mode. The modal coordinate evolves according to its own differential equation, coupled to the joint motion.
3. **The extended state includes the modal coordinates:** $\boldsymbol{x} = [\boldsymbol{q}, \dot{\boldsymbol{q}}, \eta, \dot{\eta}]^T$.
4. **Shaft flexibility is part of the drift field.** No muscle controls the deformation directly. The shaft bends and unbends passively, responding to inertial forces.
5. **The catapult effect is real.** Elastic energy stored in the bent shaft is released near impact, providing 5–15% extra clubhead speed (depending on swing speed and shaft fit).
6. **Shaft fitting designs the drift field.** The choice of stiffness, damping, and flex profile changes which ZTCF trajectory the swing naturally produces. Matching these to your swing speed and biomechanics puts the autopilot trajectory close to your target.

7. **Shaft flexibility helps slower golfers more.** Because centrifugal stiffening is weaker, the shaft remains more flexible throughout the swing and can store more elastic energy. This energy is returned at impact, partially compensating for lower muscle power.

8. **The shaft is an energy buffer between the muscles and the clubhead.** Some of the work done by the muscles is temporarily stored as elastic deformation and released later. This allows the muscles to do work when the levers are shorter and then have that work released when the clubhead can best use it.

Exercises 20.1. Chapter Exercises

Exercise. *A golf shaft has a first bending mode with natural frequency* $\omega_n = 12$ *rad/s and damping ratio* $\zeta = 0.05$*. If the shaft is initially bent to* $\eta_0 = 0.03$ *m (3 cm) and released from rest, how long does it take for the deflection to decay to* $e^{-1} \approx 37\%$ *of its initial value?*
Hint: *In a damped oscillator, the envelope decays as* $e^{-\zeta\omega_n t}$.

Exercise. *Explain in plain language why a stiffer shaft stores less elastic energy for a given deflection than a more flexible shaft. What is the trade-off in terms of timing and control?*

Exercise. *Two golfers have identical swing speeds (85 mph) and identical biomechanics, but one uses a 65-gram shaft and the other uses a 75-gram shaft. How might the different shaft mass affect the modal dynamics? (Consider how mass affects* ω_n *and the effective driving force* f_{modal}*.)*

Exercise. *A golfer notices that with a stiffer shaft, they hit the ball farther, but with less dispersion. With a more flexible shaft, the ball goes slightly farther on a perfect swing but much farther when they swing a bit faster. Explain these observations in terms of the drift field and the catapult effect.*

Exercise. *In the worked example (Section 20), suppose the hand acceleration profile is changed so that the hand decelerates earlier (starting at* $t = 0.08$ *s instead of* $t = 0.1$ *s). How would you expect the shaft deformation to change? Would the catapult effect occur earlier or later?*

Exercise. *Explain why centrifugal stiffening is negligible early in the downswing but becomes dominant near impact. What does this imply for when and how much the shaft can bend during a typical swing?*

Chapter 21

Fascia and Connective Tissue: Separating Myth From Mechanics

> The truth about fascia is more interesting than the myth, and far less mystical.
>
> —AffineDrift

In Plain Language 21.1. Why We Must Separate Fact from Fiction

In recent years, fascia—the connective tissue that wraps around muscles, bones, and organs—has become the subject of extraordinary claims in the golf coaching and fitness communities. Fascia has been proposed to have roles in sensory feedback and force transmission, though the extent of these functions remains an active area of investigation.

None of these claims are well supported by scientific evidence.

Fascia is real tissue with real mechanical properties with direct implications for biomechanics and injury prevention.

In this chapter, we separate myth from reality. We start by debunking the common claims. Then we rebuild the actual biomechanics from first principles, showing exactly where fascia fits into the drift and control framework. The result is a more honest, more useful understanding of connective tissue in the golf swing.

21.1 What Fascia Actually Is

Let us start with a simple definition. Fascia is a network of connective tissue made primarily of three components:

1. **Collagen fibers.** These are protein strands that provide tensile strength. Collagen is the most abundant protein in the human body. It is strong but relatively stiff (high elastic modulus).

2. **Elastin fibers.** These are protein strands that provide elasticity. Elastin is less stiff than collagen and returns to its original length after stretching (like a rubber band). It makes up only about 2–4% of fascia by mass, but it is crucial for the elastic behavior.

3. **Ground substance.** This is a gel-like material composed of proteoglycans and water. The ground substance fills the spaces between the fibers, provides lubrication, and

hydrates the tissue. It is deformable and does not provide much tensile strength on its own.

Fascia exists in layers. Superficial fascia (near the skin) is looser and more extensible. Deep fascia (surrounding muscles and organs) is denser and more organized. The interface between a muscle and its surrounding fascia is called the **epimysium**. The sheaths that organize muscle fibers into bundles are called the **perimysium** and **endomysium**.

Definition 21.1. Fascia

Fascia is connective tissue composed of collagen, elastin, and ground substance. It surrounds and binds together muscles, bones, organs, and neural structures. Deep fascia is organized into continuous networks; superficial fascia is looser and more diffuse. Unlike tendons (which connect muscle to bone) or ligaments (which connect bone to bone), fascia is distributed and meshwork-like.

A few mechanical properties are important:

- **Elastic modulus:** Fascia is much less stiff than tendon. Typical values are $E \approx 0.1-1$ MPa for fascia, compared to $E \approx 100-300$ MPa for tendon and $E \approx 200$ GPa for bone. This means fascia deforms much more for the same applied stress.
- **Viscosity:** Fascia exhibits viscous behavior (like a fluid). When stretched, it does not snap back instantly; it returns slowly. The rate of return depends on the stretching rate and the properties of the ground substance.
- **Plasticity:** If fascia is stretched beyond a certain point, it does not fully return to its original length. It exhibits permanent deformation. This is one reason why repeated stretching can change tissue properties.
- **Hydration:** Fascia is about 70% water. The hydration state affects its mechanical properties. Dehydrated fascia is stiffer and less extensible.

21.2 The Myth Landscape: Common False Claims

We now systematically examine the most common claims made about fascia in the golf and fitness communities.

21.2.1 Myth 1: Fascial Slings Generate Swing Power

Myth vs. Reality 21.1. Fascial Slings as Power Generators

The Myth: Organized networks of fascia (called "fascial slings" or "myofascial chains") wrap around the torso and limbs in such a way that they can contract and shorten, transmitting force across large distances. These slings are proposed to be a major source of rotational power in the swing. Some coaches claim that a golfer can generate 50% or more of swing power through fascial "springs" rather than muscle contraction.

The Reality: Fascia does not contract. It has no muscle cells. It cannot generate force. Fascia can only transmit forces that are applied to it by muscles. The idea that fascia generates force is mechanically incoherent.

What is true: Fascia does organize and distribute forces. When a muscle contracts, the force is transmitted not just along the muscle's line of action but also laterally, through the fascia, to neighboring tissues. This can create complex, multi-directional force distributions that a simple "straight muscle" model would miss. But these forces *originate* from muscle contraction; the fascia does not create them.

The claim of "50% power from fascia" likely arises from confusion between force transmission (real) and force generation (mythical). When a biomechanist measures the force in a ligament or fascia and compares it to the muscle force, they are seeing distributed force, not independent power generation.

21.2.2 Myth 2: Fascia Stores and Releases Energy Like a Spring

Myth vs. Reality 21.2. Fascia as an Energy Storage Spring

The Myth: A stretched fascia acts like a spring, storing elastic energy. During a rapid movement (like a golf swing), the golfer stretches the fascia first, then releases it, causing the fascia to snap back and drive the motion. This snapback is claimed to be a major source of power, especially in rapid explosive movements.

The Reality: This is partially true but wildly overstated. Fascia does have elastic properties (because of its elastin and collagen content) and can store some elastic energy. However, the amount is small and the mechanism is different from what is usually claimed.

Here are the facts:

1. **Fascia is much less stiff than tendon.** A tendon has $E \approx 200$ MPa; fascia has $E \approx 0.5$ MPa. For the same stretch, fascia stores much less elastic energy (elastic energy is proportional to $E \times (\text{strain})^2$).

2. **Fascia is thick and distributed.** The elastic energy stored in fascia is spread across a large volume of tissue, not concentrated in a small cable like a tendon. The energy density (energy per unit volume) is low.

3. **The elastin fraction is tiny.** Fascia is about 60% collagen, 2–4% elastin, and the rest ground substance. Collagen is almost purely stiff (not elastic); elastin provides the elasticity. So the elastic behavior of fascia is dominated by a tiny fraction of its mass.

4. **Muscle itself stores more energy than fascia.** The contractile elements and elastic proteins within muscle (like titin) store and release elastic energy. This is orders of magnitude larger than the energy stored in surrounding fascia.

5. **The viscous loss is significant.** When fascia stretches, energy is dissipated as heat due to the viscous nature of the ground substance. This means that even the small amount of elastic energy stored is not all returned; some is lost.

So yes, fascia can store elastic energy. But the amount is small, and the energy is predominantly stored in muscle and tendon, not fascia. The claim that fascia is a major energy storage mechanism for the golf swing is not supported by the numbers.

21.2.3 Myth 3: Myofascial Meridians Create Full-Body Force Chains

Myth vs. Reality 21.3. Myofascial Meridians and Kinetic Chains

The Myth: Fascia is organized into discrete "meridians" or "lines" that run continuously across the body. These meridians are claimed to transmit force and tension across large distances, creating a unified kinetic chain. Some anatomists have proposed names like the "Spiral Line," the "Superficial Back Line," and the "Deep Front Line," suggesting that force flows along these paths.
The Reality: This is an anatomical fiction. While it is true that fascia is continuous and interconnected (you cannot isolate a single fascial band by dissection without cutting through many others), the idea that force flows along discrete meridians is not supported by biomechanical analysis.
Here is what the evidence shows:

1. **Fascia is a network, not a set of wires.** Under the microscope, fascia is a three-dimensional mesh of collagen fibers oriented in all directions. It is not a set of discrete, parallel cables. Force applied to one point spreads out in all directions, not along a single path.

2. **Direct biomechanical measurements show multi-path force distribution.** When engineers measure the stress field in fascia during controlled stretching, they find complex, nonuniform stress distributions. Forces do not concentrate along any one path; they spread widely.

3. **The kinetic chain is real, but the mechanism is different.** The body does transfer forces from the legs to the torso to the arms. But this happens through the skeletal structure (bones and joints) and through muscle activation patterns, not primarily through fascial pathways. The kinetic chain is more like a mechanical linkage than like tension lines in a spider web.

4. **Meridian anatomy cannot explain observed biomechanics.** In the golf swing, we observe that the legs drive the torso, the torso drives the arms, and the arms drive the club. This kinetic chain is explained perfectly well by joint mechanics and muscle activation sequences. Adding meridian theory does not improve the explanation; it adds mysticism without additional predictive power.

The meridian concept may be useful as a mnemonic or teaching tool ("pretend your motion flows along this line"), but it should not be confused with actual anatomical structure or biomechanical mechanism.

21.2.4 Myth 4: Fascial Training Transforms Your Swing

Myth vs. Reality 21.4. Fascia-Specific Training Methods

The Myth: There exist training methods (like foam rolling, myofascial release, stretching, or vibration therapy) that can reshape fascia and thereby fundamentally improve athletic performance or even fix swing faults. The claim is that fascia can be trained separately from muscle, and that fascial improvements will directly translate to swing improvements.

The Reality: There is no evidence for fascia-specific training effects that are distinct from muscle training. Let us unpack this:

1. **Foam rolling does something, but it is unclear what.** Studies show that foam rolling can improve range of motion, reduce soreness, and improve performance. But the mechanism is unknown. Candidate explanations include:

    - Increased nervous system relaxation (reducing protective muscle tension).
    - Increased blood flow and fluid dynamics (improving tissue hydration).
    - Changes to muscle properties (not fascia).
    - Placebo effects (not to be dismissed, but not mechanically specific).

    The claim that foam rolling specifically "releases" fascia is not supported. Fascia is not a knot that can be untangled; it is a network that moves with the underlying tissues.

2. **Stretching improves flexibility, but the mechanism is neurological, not fascial.** When you stretch a muscle and maintain the position, the muscle's reflexive resistance decreases (a neurological adaptation). The tissue itself does change slightly (plastic deformation), but this happens in muscle and tendon, not primarily in the surrounding fascia. The improvement in range of motion is largely due to the nervous system learning to relax the muscle.

3. **Vibration therapy has mixed evidence, and the mechanism is unclear.** Some studies show that whole-body vibration can improve muscle properties and performance. But whether this is due to fascia, muscle, or neuroendocrine effects is not clear.

4. **The strongest evidence for performance improvement comes from traditional strength and conditioning.** Progressive overload of muscles, varied movement patterns, and sport-specific training all improve athletic performance. These improvements are largely explainable by muscle adaptations (hypertrophy, neural coordination) without invoking fascia.

The pragmatic conclusion: if a training method improves your swing, it is probably because it improved your muscle function, proprioception, or movement coordination—not because it specifically changed your fascia. And there is no evidence that you can train fascia separately from muscle in any meaningful way.

21.3 What the Science Actually Says

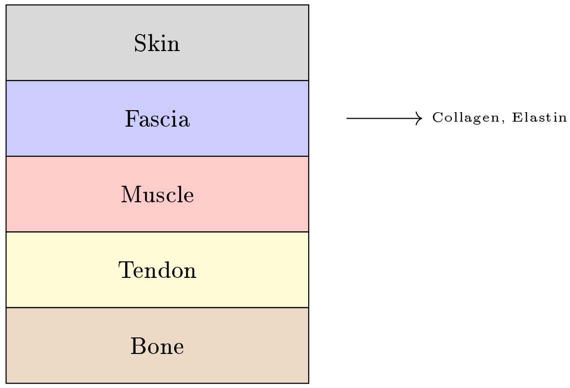


Figure 21.1: Fascia Tissue Composition: Layers of Connective Tissue With Structural Proteins.

Now that we have cleared away the myths, let us rebuild the actual mechanics of fascia and where it fits in the golf swing.

21.3.1 Fascia as Force Transmission Tissue

The most important role of fascia is not force generation or energy storage, but **force transmission and distribution**. Here is how it works:

When a muscle contracts, it pulls on its attachments (usually bone). But the force does not flow only along the line from the muscle's origin to its insertion. The force spreads out through the surrounding fascia to neighboring tissues. This has several consequences:

1. **Load is distributed across a larger area.** A concentrated muscle force is spread out over a larger volume of tissue, reducing stress concentrations.
2. **Lateral force transmission.** A muscle pulling in one direction can transmit force laterally to adjacent structures through the fascia, creating multi-directional effects.
3. **Mechanical coupling.** Nearby muscles and structures are mechanically coupled through the fascia. Movement of one muscle influences the loading of neighbors.

In the context of the affine framework, fascia affects the structure of the control map $G(\boldsymbol{x})$. A muscle's control input (its torque) is transmitted to the clubhead not just through the direct lever arm, but also through distributed pathways in the fascia. This can change the effective direction and magnitude of the control force.

Example 21.1. Fascia Coupling in the Shoulder

Consider the rotator cuff muscles (supraspinatus, infraspinatus, teres minor) that control shoulder rotation. These muscles are wrapped in a continuous fascial layer called the rotator cuff fascia. When one muscle contracts, its force is transmitted directly to bone, but also laterally through the fascia to the other muscles and the

overlying trapezius.
The consequence: a simple model that treats each muscle as an independent actuator (each with its own line of pull) overestimates the control authority. The actual control authority is reduced because the muscles are mechanically coupled through the fascia. Some of the muscle's force is lost to lateral transmission, and the effective torque on the shoulder joint is less than a naive calculation would suggest.
However, the coupling also provides stability and co-activation: when one muscle is loaded, the fascia tends to activate (mechanically engage) the adjacent muscles, providing a passive stabilization effect.

21.3.2 Fascia as a Series Elastic Element

In the biomechanics literature, fascia is often modeled as a **series elastic element** (in series with the muscle). The model looks like this:

$$\text{Muscle contraction} \to \text{Fascia deformation} \to \text{Bone movement}$$

In this model, when the muscle wants to pull the bone, it must first stretch the surrounding fascia. The amount of stretch depends on the muscle force and the stiffness of the fascia.

The stress-strain behavior of fascia is nonlinear. At low strains (small deformations), the tissue is relatively stiff as collagen fibers engage. At medium strains, the relationship becomes more linear (the stiffness is roughly constant). At high strains, the tissue stiffens again as the collagen fibers reach their limit. The elastin provides a small amount of elasticity throughout.

Mathematically, we can write a simple model:

$$\sigma = E(\epsilon)\,\epsilon + c\,\dot{\epsilon}$$

where:

- σ is the stress (force per unit area).
- ϵ is the strain (deformation as a fraction of original length).
- $E(\epsilon)$ is the strain-dependent elastic modulus (increasing with strain).
- c is the viscous damping coefficient (the term $c\dot{\epsilon}$ is proportional to the rate of deformation).

This is a **viscoelastic** model: the tissue has both elastic and viscous properties. The elastic part stores energy (like a spring); the viscous part dissipates energy (like a damper).

The viscous part is crucial. When fascia is stretched quickly, it acts stiffer than when stretched slowly. This is called **strain-rate dependence**. In a rapid golf swing, the fascia is stretched very quickly, so it acts almost like a stiff solid. Over seconds (like in a slow stretching session), it acts more like a liquid, deforming more easily.

In Plain Language 21.2. Viscoelasticity and Stretch Speed

Imagine fascia as a material with both springs and dashpots (viscous dampers) embedded in it. When you stretch it slowly, the springs extend, and the dashpots allow the material to deform smoothly. When you stretch it quickly, the dashpots resist strongly (because they oppose the rate of deformation), and the springs don't extend as much. So the tissue acts stiffer when stretched fast.

In the golf swing, the swing duration is about 0.2 seconds, and the deformation happens in tens of milliseconds. This is very fast. Fascia, being viscous, acts almost like a stiff solid at these timescales. It does not easily absorb or store energy; it just resists the motion.

Conversely, in a slow stretching session, the deformation happens over seconds, and the viscous resistance is minimal. The tissue absorbs more energy and deforms more. This is why there is no special value in "fascial training." The mechanical properties that matter to the golf swing (high-rate stiffness and damping) are not the same as the properties that matter to slow stretching (elasticity and plasticity).

21.3.3 Fascia Stiffness Values: Quantitative Reality

Let us put numbers on the mechanical properties of fascia and compare them to tendon and muscle.

Table 21.1: Elastic Moduli of Biological Tissues (Approximate Ranges)

Tissue	Elastic Modulus (MPa)	Typical Strain at Failure
Bone (cortical)	15,000–20,000	1–3%
Tendon (collagen-rich)	200–500	5–10%
Ligament	100–300	10–20%
Fascia (deep)	0.5–2.0	20–40%
Fascia (superficial)	0.1–0.5	40–80%
Muscle (passive)	10–100	50–100%+

The key observations:

1. **Fascia is 100–1000 times softer than tendon.** For the same applied stress, fascia deforms much more. This means fascia is not a load-bearing structure; it is a deformation-accommodating structure.

2. **Superficial fascia is much softer than deep fascia.** This makes sense: superficial fascia needs to move with the skin and accommodate deformation; deep fascia needs to organize and distribute forces among muscles.

3. **Muscle (passive) is softer than tendon but stiffer than fascia.** When a muscle is not contracting, its passive elastic properties (from the filament proteins like titin) are stiffer than the surrounding fascia.

4. **The strain-at-failure numbers show that fascia is very extensible.** Fascia can stretch 20–40% before failing, whereas tendon fails at only 5–10% strain. This

extensibility is a feature: it allows large deformations without tearing, distributing stress over a larger volume.

Now, let us estimate the elastic energy stored in fascia during a golf swing. Suppose a region of fascia (say, $10\,\mathrm{cm}^2$ cross-section, $10\,\mathrm{cm}$ thick) is stretched by 5% strain during the swing.

$$E_{\text{stored}} = \frac{1}{2}E \times \epsilon^2 \times V = \frac{1}{2} \times 1\,\mathrm{MPa} \times (0.05)^2 \times (0.01\,\mathrm{m}^2 \times 0.1\,\mathrm{m}) \tag{21.1}$$

$$E_{\text{stored}} = \frac{1}{2} \times 10^6\,\mathrm{Pa} \times 0.0025 \times 10^{-3}\,\mathrm{m}^3 = 1.25\,\mathrm{J} \tag{21.2}$$

For comparison, the kinetic energy of the clubhead at impact is typically 150–300 J. So the elastic energy stored in this one region of fascia is about 0.5–1% of the clubhead's kinetic energy.

There are many regions of fascia in the arm and torso, so the total might be 5–10 J. But this is still only 3–7% of the total kinetic energy. And much of this energy is dissipated by viscous damping before impact.

The conclusion: **fascia does store some elastic energy, but the amount is small—a few percent of the total energy available. It is not a major power source.**

21.3.4 The Role of Fascia in the AffineDrift Framework

In the control-affine framework, how does fascia affect the dynamics?

The primary effect is that fascia changes the **parameters** of the dynamics, not the **structure**. Specifically:

1. **Fascia affects $G(\boldsymbol{x})$, the control authority map.** Because fascia couples muscles mechanically, the control input (muscle torque) at one joint is partially distributed to other joints. This is not a change in the structure of the equation $\dot{\boldsymbol{x}} = f(\boldsymbol{x}) + G(\boldsymbol{x})\boldsymbol{u}$, but a change in the coefficients of G.

2. **Fascia affects the damping in $f(\boldsymbol{x})$.** The viscous properties of fascia add energy dissipation to the drift field. This is a passive damping effect, part of the drift.

3. **Fascia does not change the fundamental constraint structure.** The kinematic constraints (joint connectivity, ground contact) are determined by the skeleton, not by fascia. Fascia wraps around this structure but does not change it.

The practical consequence: if you want to understand the basic structure of the golf swing and how it is controlled, you can ignore fascia. The affine structure remains. But if you want to predict the precise values of velocity and acceleration at each joint, or the energy dissipation, you need to account for fascial damping and force distribution.

Key Principle 21.1. Fascia in Drift and Control

- **In Drift:** Fascia contributes passive damping to $f(\boldsymbol{x})$. The viscous resistance of fascia dissipates energy, slowing the swing slightly and adding a term $-c\dot{\boldsymbol{x}}$ to the drift field.

- **In Control:** Fascia couples muscles mechanically, changing the effective control authority. A control input at one joint is partially transmitted to other joints through fascial pathways. The control authority $G(\boldsymbol{x})$ is reduced because some of the muscle force is lost to lateral transmission.
- **Not a Power Source:** Fascia does not generate force. It does not store and return significant energy. It is a force transmission and distribution structure, and a source of passive damping.

21.4 Where Fascia Genuinely Matters

Despite the lack of mystical power, fascia does matter in real, concrete ways.

21.4.1 Load Distribution and Injury Prevention

Fascia distributes load across a wide area, preventing stress concentrations. Without fascia, muscles would pull directly on bone, creating high-stress points at the attachments. With fascia, the load is distributed laterally, reducing the peak stress.

This has direct implications for injury. A structure with distributed, low-stress loading can sustain high forces repeatedly. A structure with concentrated, high-stress loading will fatigue and tear.

In the golf swing, the forces are enormous—the clubhead is accelerating at thousands of $\mathrm{m/s^2}$ during the swing (Chapter 4 estimates values on the order of 180 $\mathrm{m/s^2}$ at the swing phase, with much higher peak values during ball contact; [Jorgensen, 1994a] reports similar orders of magnitude). These forces are transmitted through the arm to the shoulder to the torso, and eventually to the ground. The fascia in the rotator cuff, the arm, and the torso is critical for distributing these forces so that no single tendon or muscle attachment is overloaded.

An injury (like a rotator cuff tear) often occurs when the load distribution is disrupted—for example, when a muscle is weak and cannot contribute its share of the load, forcing the adjacent tissues to compensate. Conversely, a well-conditioned golfer with balanced muscle development and healthy fascia can distribute loads evenly and resist injury.

Example 21.2. Fascia and Rotator Cuff Health

The rotator cuff (supraspinatus, infraspinatus, teres minor, subscapularis) is wrapped in fascia that is continuous with the overlying trapezius, deltoid, and latissimus dorsi fascia. When you swing a golf club, the rotator cuff muscles must control the shoulder joint against enormous forces. If one rotator cuff muscle is weak, the load shifts to its neighbors, and the fascia must transmit this asymmetric load.

If the load distribution is poor, the peak stress at the tendon attachment can exceed the tissue's strength, leading to a tear. A fascia that is healthy and well-distributed (maintained by balanced muscle development) is protective.

Conversely, foam rolling or stretching does not change the load distribution in any meaningful way. The way to protect the rotator cuff is to maintain balanced muscle strength through proper training.

21.4.2 Proprioception and Position Sensing

Fascia is rich in sensory receptors (mechanoreceptors) that sense stretch, pressure, and vibration. These receptors provide proprioceptive feedback: they tell the nervous system the position and state of the body.

In the golf swing, proprioceptive feedback is crucial. The golfer must sense the position of the arms, the tension in the muscles, and the loads on the joints. Much of this feedback comes from proprioceptors in the muscle spindles and tendons (Golgi organs), but some comes from receptors in the fascia itself.

This is a real and important function of fascia. But it is a **sensory** function, not a mechanical function. It affects the nervous system and the control inputs, not the drift field.

In Plain Language 21.3. Why Proprioceptive Feedback Matters

A golfer cannot see their arms during the swing (they are moving too fast, and they are behind the body). The golfer's knowledge of arm position comes entirely from proprioceptive feedback. This feedback comes from receptors in the muscles and fascia, sending continuous signals to the brain about position and load.

If this feedback is disrupted (for example, by a local anesthetic or by nerve damage), the golfer loses the sense of where their arms are and cannot control them effectively. This is why proprioceptive training (like balancing exercises or slow-motion practice) can improve swing control—it sharpens the proprioceptive feedback.

Fascia contributes to this feedback through its mechanoreceptors. But there is no special value in "fascial training" to improve proprioception. Any movement practice that challenges balance and control will engage the proprioceptive system and improve feedback.

21.4.3 Injury Mechanics and Recovery

When fascia is injured (torn, inflamed, or scarred), it can form adhesions—abnormal attachments to adjacent tissues. These adhesions can limit movement and cause pain.

A classic example is adhesive capsulitis (frozen shoulder), where the shoulder joint capsule (a fascial structure) becomes inflamed and develops adhesions, severely limiting shoulder motion. Another example is plantar fasciitis, where inflammation and small tears in the plantar fascia cause foot pain.

In golfers, fascia-related injuries most commonly occur in the:

- Rotator cuff fascia and shoulder capsule (shoulder injuries).
- Thoracolumbar fascia in the lower back (back pain).
- Plantar fascia in the foot (foot pain).

Recovery from fascial injuries is slow because fascia has a relatively poor blood supply. Healing typically takes weeks to months. The accepted treatments are:

1. Rest and inflammation management (reducing the inflammatory signal).
2. Gradual re-mobilization (movement that does not re-aggravate the injury, but maintains blood flow).

3. Strengthening and conditioning (ensuring balanced muscle development so loads are evenly distributed).

There is limited evidence for other specific fascial therapies. Massage and soft tissue work may provide temporary pain relief through neurological mechanisms (reducing protective muscle tension) but do not heal the tissue faster.

Key Principle 21.2. Fascial Injury and Recovery

- Fascia can be injured by overload or trauma, just like muscle or tendon.
- Recovery is slow because fascia is poorly vascularized.
- The keys to recovery are inflammation management, gradual re-mobilization, and prevention of re-injury.
- There is no evidence that specific fascial therapies (rolling, myofascial release) speed recovery beyond the effects of standard injury management.
- Prevention is through balanced muscle development, adequate hydration, and varied movement patterns.

21.5 The Honest Summary: Connective Tissue, Not Magic Tissue

Let us summarize what we have learned and what it means for the golf swing.

What fascia is: Connective tissue composed of collagen, elastin, and ground substance. It wraps around muscles, bones, and organs, providing structural support and distributing forces.

What fascia does in the golf swing:

1. Distributes loads across a wide area, preventing stress concentrations and protecting against injury.
2. Couples muscles mechanically, changing how forces are transmitted from muscle to bone.
3. Provides damping (viscous resistance) that dissipates energy and slows movements.
4. Provides proprioceptive feedback through mechanoreceptors.

What fascia does NOT do:

1. Generate force. (Muscles generate force; fascia only transmits it.)
2. Store or release significant elastic energy. (The amount is small and mostly dissipated.)
3. Create discrete “meridians” or kinetic chains. (The skeleton and muscles do that.)
4. Respond to specialized training in ways that transform performance. (Standard training works better.)

Practical implications:

1. A well-conditioned golfer with balanced muscle development and healthy fascia is better protected against injury.

2. Strength training and movement practice are more effective for improving swing performance than fascial-specific interventions.

3. If a golfer suffers a fascial injury (shoulder, back, foot), recovery requires patience, gradual re-mobilization, and careful management of loads.

4. The mysteries of the golf swing are not hidden in the fascia. They are explained by joint mechanics, muscle activation patterns, and the interplay of drift and control.

For coaches and trainers: Do not claim that fascia is a magic power source or that specialized fascial training will transform a golfer's swing. These claims are not honest. Instead, focus on developing balanced, functional strength; improving movement coordination; and protecting the athlete from injury. These are the proven, evidence-based approaches.

For golfers: Your fascia is important for your health and longevity as a golfer, but it is not the secret to a better swing. The secret is understanding how drift and control work, and practicing the movements that exploit that understanding. Train intelligently, listen to your body, and respect the connective tissue you have—not because it is magical, but because it is essential.

Key Principle 21.3. Key Takeaways

1. **Myth-busting is necessary.** The golf community has absorbed many false claims about fascia. We must separate fact from fiction to have honest conversations about training.

2. **Fascia does not generate force.** It is a force transmission structure, not a power source.

3. **Fascia stores minimal elastic energy.** The amount stored in fascia during a golf swing is small (3–7% of kinetic energy) and is mostly dissipated by viscous damping.

4. **Myofascial meridians are anatomical fiction.** Fascia is a continuous network, not a set of discrete meridians. Forces spread out in all directions, not along meridian paths.

5. **Fascia has real, important functions:**

   - Load distribution, protecting against injury.
   - Proprioceptive feedback.
   - Mechanical coupling of muscles.
   - Passive damping in the drift field.

6. **Fascia-specific training has little evidence.** Foam rolling, myofascial release, and specialized stretching may have placebo or neurological benefits, but they do not specifically improve fascial properties or swing performance.

7. **The keys to a healthy swing:** Balanced muscle development, varied movement patterns, adequate hydration, and understanding of how forces flow through the body. Not mysticism.

8. **Fascia in the framework:** In the affine dynamics $\dot{\boldsymbol{x}} = f(\boldsymbol{x}) + G(\boldsymbol{x})\boldsymbol{u}$, fascia affects the parameters of f (damping) and G (mechanical coupling), not the fundamental structure.

9. **For injury prevention:** Balanced strength, proper load distribution, and gradual training progressions are more effective than fascial-specific interventions.

10. **The honest conclusion:** Fascia is important for the mechanical function and health of the body. But it is not the source of swing power, the secret to performance, or a magic tissue waiting to be unlocked by the right training method. It is honest-to-God connective tissue doing its honest-to-God job.

Exercises 21.1. Chapter Exercises

Exercise. *In plain language, explain why the claim that "fascia generates 50% of swing power" is mechanically incoherent. What is likely being confused?*

Exercise. *Compare the elastic moduli of tendon ($E \approx 200$ MPa) and fascia ($E \approx 1$ MPa). For the same applied stress, which tissue deforms more? Why is this important for load bearing?*

Exercise. *Explain the difference between force transmission through fascia and force generation by fascia. Why is this distinction critical to the myths?*

Exercise. *A golfer has a frozen shoulder (adhesive capsulitis) involving the shoulder joint capsule (a fascial structure). Explain why the shoulder is immobilized despite the muscles being intact and capable of producing torque.*

Exercise. *Suppose a golfer foam-rolls for 10 minutes before a swing. Explain one plausible neurological mechanism by which this might improve swing performance, without invoking any special fascial properties.*

Exercise. *Calculate the elastic energy stored in a 5% strain region of fascia (10 cm by 10 cm by 5 cm) with $E = 1$ MPa. Compare this to the kinetic energy of a 200 g clubhead moving at 50 m/s. What fraction of the kinetic energy does the fascia store?*

Exercise. *Explain why fascia's viscoelastic properties (both elastic and viscous) mean that it behaves differently during a rapid golf swing (high-speed stretch) versus a slow stretching session (low-speed stretch).*

Exercise. *In the affine framework, does fascia primarily affect the drift field $f(\boldsymbol{x})$, the control authority $G(\boldsymbol{x})$, or both? Explain.*

Chapter 22

Soft Tissue and Pliable Systems: Beyond the Rigid Body

In Plain Language 22.1. Why Soft Tissue Matters in Golf

A body segment can move approximately as a rigid object while its skin, muscle and other tissues move relative to its bones. That relative motion matters in two different ways: it changes forces and stored energy, and it changes what a surface-mounted sensor reports. A marker can therefore give a repeatable measurement without being a faithful measurement of the underlying bone.

Abdominal pressure adds another connection. Pressure acts on the surrounding structures, while muscles and connective tissues help contain that pressure. The resulting support depends on the whole force balance. A large pressure reading does not, by itself, reveal the load on a spinal joint, the energy reaching the club, or the likelihood of an injury.

This chapter builds small models that make those distinctions calculable. Their purpose is to decide which effects a golf or humanoid model must resolve and what evidence would validate it. The numerical examples are declared teaching systems, not measured anatomical parameters or instructions to change breathing or muscle activation.

22.1 The Rigid Body Assumption: Where It Works and Where It Fails

Rigid-segment modeling assumes that distances between material points assigned to a segment remain fixed. It does not require all joints to be hinges, all muscles to be inactive, or the club shaft to be rigid. A model can combine rigid bones, multiaxis joints, deformable tendons and a flexible club. The choice should follow the question and the available measurements.

Stiffness and strength are different properties. A bone's failure load does not establish how accurately a rigid approximation predicts its deformation under a particular load. Likewise, a rigid segment's center of mass and inertia must include the tissue assigned to it. Adding a wobbling mass to an anthropometric segment without subtracting that same mass from the rigid part counts tissue twice.

A useful rigid approximation reproduces specified outputs over a specified operating range. It may be adequate for a gross pose while inadequate for tissue stress or an impact peak. A small pose error does not guarantee a small force error, especially when forces depend on acceleration or contact deformation. There is no universal 15-degree-of-freedom model, or universal percentage accuracy, for coaching, inverse dynamics or injury assessment.

Pain and Challis modeled two heel-first drop landings by one man using subject-specific segment estimates and a partly calibrated wobbling-mass model [Pain and Challis, 2006]. Their model and a rigid alternative produced similar gross kinematics but substantially different computed loading. Separate controlled impacts supplied tissue-motion comparisons; the trunk response lacked equivalent independent data. This is evidence that model choice can matter under those impact conditions. It does not establish a correction percentage or tissue-motion amplitude for golf.

The practical decision is therefore output-specific. Predicting where the ball flies requires a sufficiently accurate launch state and flight model; it does not automatically require every internal tissue mode. Predicting how the golfer creates that state, or estimating a particular tissue load, introduces different requirements. Validation must follow that distinction.

22.2 Wobbling Mass: A Coupled Mechanical Model

Begin with two positive masses moving along a horizontal line. Let x_r be the rigid part's position and x_w the effective tissue mass's position. A spring of stiffness $k > 0$ and a damper with coefficient $c \geq 0$ connect them. Coordinates are measured from a configuration in which the spring is unstrained. The applied force u acts on the rigid part. Gravity has no component along this particular line; that is a boundary assumption, not a property of soft tissue.

Define relative displacement $z = x_w - x_r$. The spring and damper exert equal and opposite forces, so the closed equations are

$$\begin{aligned} m_r \ddot{x}_r &= u + kz + c\dot{z}, \\ m_w \ddot{x}_w &= -kz - c\dot{z}. \end{aligned}$$

Adding them gives $(m_r + m_w)\ddot{x}_{\text{COM}} = u$. Internal coupling can redistribute motion but cannot accelerate the complete system's center of mass without an external force. An effective tissue mass represents a selected deformation mode; it is not automatically the mass of every anatomically soft structure in the segment.

The mechanical energy and its rate are

$$E = \tfrac{1}{2} m_r \dot{x}_r^2 + \tfrac{1}{2} m_w \dot{x}_w^2 + \tfrac{1}{2} k z^2, \qquad \dot{E} = u\dot{x}_r - c\dot{z}^2.$$

The spring stores energy; the damper removes mechanical energy at a nonnegative rate. Energy entering tissue motion is not necessarily already dissipated. It can return through the coupling. Over a simulation, integrated actuator work must equal the change in mechanical energy plus accumulated dissipation. This balance provides a stronger implementation check than a plausible-looking trajectory.

22.2.1 Prescribed Motion and Free Motion Have Different Modes

If the rigid part follows an externally prescribed trajectory, the tissue equation becomes

$$m_w \ddot{z} + c\dot{z} + kz = -m_w \ddot{x}_r.$$

The base acceleration drives relative motion. Maintaining that prescribed base requires whatever reaction force the model predicts; it is not the same experiment as applying a fixed force to a freely moving rigid part.

For a fixed base, the undamped natural angular frequency and damping ratio are

$$\omega_n = \sqrt{k/m_w}, \qquad \zeta = \frac{c}{2\sqrt{km_w}}.$$

Take $m_w = 1\,\text{kg}$, $k = 5000\,\text{N/m}$ and $c = 200\,\text{N s/m}$. Then $\omega_n = 70.7107\,\text{rad/s}$, equivalent to $11.2540\,\text{Hz}$, and $\zeta = 1.4142$. The homogeneous roots are $-29.2893\,\text{s}^{-1}$ and $-170.7107\,\text{s}^{-1}$. This system is overdamped. A positive displacement released from rest decays without repeated oscillation. Describing these same parameters as an underdamped system with several cycles would contradict the equation.

If both masses move freely, subtract their acceleration equations. With reduced mass $\mu = m_r m_w/(m_r + m_w)$,

$$\mu\ddot{z} + c\dot{z} + kz = -\frac{m_w}{m_r + m_w}u.$$

The unforced relative mode uses $\sqrt{k/\mu}$, not $\sqrt{k/m_w}$. For $m_r = 1.5\,\text{kg}$ and the preceding tissue parameters, $\mu = 0.6\,\text{kg}$ and the undamped relative frequency is $91.2871\,\text{rad/s}$. Boundary conditions change the mode even when the component properties stay the same.

These examples do not identify a golfer's tissue parameters. A large acceleration also does not specify high-frequency excitation: a sustained acceleration and a short pulse can have the same peak and very different spectra. Oscillation amplitude additionally depends on initial conditions, forcing history, damping and the validity of the linear approximation.

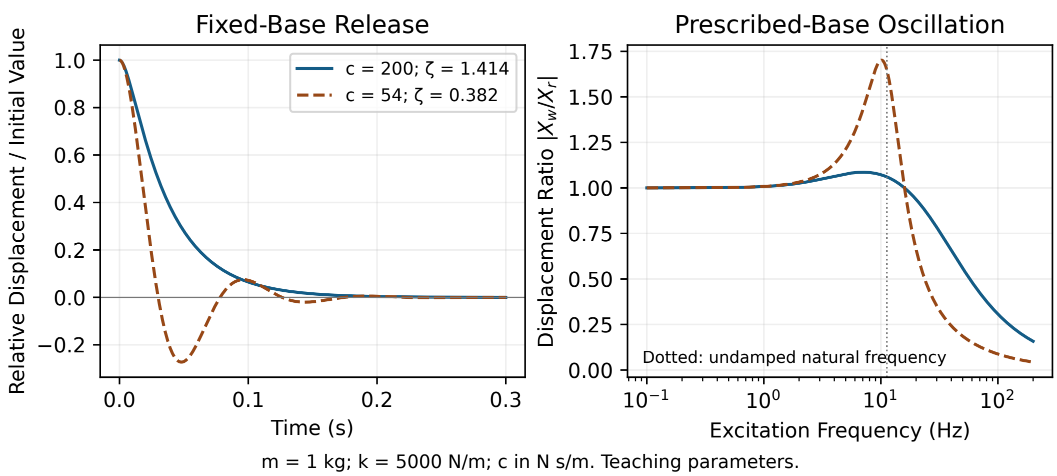


Figure 22.1: Boundary Conditions and Damping Determine Tissue Response. Both Plots Use a Fixed or Prescribed Base, a 1 kg Tissue Mass and 5000 N/m Stiffness; Parameters Illustrate a Model, Not Measured Golf Anatomy.

22.2.2 Effects of Wobbling Mass on Inverse Dynamics

Soft-tissue motion is physical, while its contribution to a bone-motion estimate is a measurement artifact. Those statements are compatible. A surface marker follows its local attachment, not necessarily the bone or the center of mass of an effective tissue mode.

In a local one-dimensional observation model,

$$y(t) = x_r(t) + z_{\text{marker}}(t) + \epsilon(t).$$

The sensor observes their sum. Without additional information, changing the proposed bone trajectory by $a(t)$ and changing marker motion by $-a(t)$ leaves the same observation. A fitted decomposition therefore needs calibration, independent observations or justified constraints. Adding a tissue model does not automatically make the components identifiable.

Differentiation magnifies rapid motion and noise. For a sinusoidal marker displacement of amplitude A and frequency f, acceleration amplitude is $A(2\pi f)^2$. At $A = 0.01\,\text{m}$ and $f = 10\,\text{Hz}$, it is $39.4784\,\text{m/s}^2$. The same displacement at another frequency gives a different acceleration; an amplitude alone is insufficient.

Filtering can reduce noise or an identifiable transient, but tissue artifact need not occupy a separate frequency band. Fuller and colleagues compared skin-mounted and bone-pin marker arrays and found overlapping spectral content [Fuller et al., 1997]. Their results oppose a universal rule that low-pass filtering recovers skeletal motion. The task, attachment, smoothing method and reference measurement all matter.

An accelerometer further measures specific force in its sensor frame. Converting it to a point's coordinate acceleration requires orientation, gravity and reference-point conventions. Rotational motion, sensor offset, attachment motion, bias and timing errors can all enter. Neither a marker nor an accelerometer alone directly measures an individual muscle force.

For inverse dynamics, distinguish errors in motion, external forces, inertial parameters and the mechanical model. A net generalized-force estimate remains a combined result of muscles, passive tissues and constraints. Estimating individual tissue loads requires additional constitutive and force-allocation assumptions. The inverse-dynamics chapters (Chapter 19) develop that distinction.

22.2.3 Effects of Wobbling Mass on the Drift Field

A useful frequency-domain calculation exposes what an effective inertia means. Prescribe a small harmonic base displacement $x_r(t) = \text{Re}(X_r e^{i\omega t})$ in the preceding linear model. Substitution into the tissue equation gives

$$H(\omega) = \frac{X_w}{X_r} = \frac{k + ic\omega}{k - m_w\omega^2 + ic\omega}, \qquad m_{\text{dyn}}(\omega) = m_r + m_w H(\omega).$$

The required external force amplitude is $U = -\omega^2 m_{\text{dyn}} X_r$. Both the spring and damper transmit base motion, so both belong in the numerator. Omitting its damping term changes the phase, reaction and energy balance.

At low frequency, H approaches one and the dynamic mass approaches $m_r + m_w$. At sufficiently high frequency, H approaches zero and the response approaches that of m_r, within this model's valid range. Around a mode, amplitude and phase depend on damping. A complex dynamic mass is a compact input-output description under harmonic forcing, not a new physical mass; its real part need not behave monotonically.

The mean input power provides an independent check:

$$\overline{P} = \tfrac{1}{2}\,\text{Re}[U(i\omega X_r)^*] = \tfrac{1}{2}c\omega^2|X_w - X_r|^2.$$

The star denotes complex conjugation. In steady harmonic motion the average stored-energy change is zero, so input power equals damper loss. This identity and direct time integration test the response without relying on the same algebraic rearrangement.

Here ω is the frequency of a small imposed oscillation. It is not simply the golfer's angular speed. A rapid rotation need not excite the same mode as a rapidly varying perturbation, and a measured swing is not a steady sinusoid. The calculation therefore supplies a testable coupling mechanism, not an explanation that elite athletes exploit tissue decoupling.

In a state-space model, adding z and $\dot{z}$ changes the state and the equations. It can also change the input map and observation map. Eliminating those states can introduce memory; a complex harmonic response should not be inserted into a time-domain equation as an ordinary constant inertia. Drift still means the vector field at the declared zero input, with the retained contact, gravity, elastic and activation mechanisms specified.

22.3 Internal Motion in the Torso

The torso contains deformable organs, muscles, connective tissues and fluids, with different attachments and material properties. It is not adequately characterized by assigning a single fraction of its mass to an unrestrained liquid. Total-body blood mass is not all located in the torso cavity, and organ masses cannot be added to an existing whole-torso estimate without checking what that estimate already includes.

For a declared rotating-frame illustration, a point at radius $r = 0.15\,\mathrm{m}$ in a frame rotating steadily at $\omega = 10.5\,\mathrm{rad/s}$ has a centripetal acceleration magnitude $\omega^2 r = 16.5375\,\mathrm{m/s^2}$ if it remains fixed in that frame. In rotating-frame force balance the corresponding centrifugal term points outward. This calculation gives an acceleration scale; it does not give an organ displacement or stress.

The complete acceleration transformation is

$$\mathbf{a} = \mathbf{a}_O + R\big[\ddot{\boldsymbol{\rho}} + 2\boldsymbol{\omega} \times \dot{\boldsymbol{\rho}} + \dot{\boldsymbol{\omega}} \times \boldsymbol{\rho} \\ + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \boldsymbol{\rho})\big].$$

Here R maps components from the moving frame to the inertial frame, $\boldsymbol{\rho}$ is position relative to its origin, and the dotted relative quantities and $\boldsymbol{\omega}$ use moving-frame components. Frame translation, angular acceleration and relative motion add terms that a centrifugal-only argument omits. Attachment forces, gravity, contact and material response must balance the resulting dynamics.

Predicting deformation requires a constitutive law and boundary conditions. In a continuum description, local momentum balance is $\rho\mathbf{a} = \nabla\cdot\boldsymbol{\sigma} + \rho\mathbf{b}$, where ρ is density, $\boldsymbol{\sigma}$ stress and $\mathbf{b}$ body force per unit mass. An approximately incompressible fluid can transmit pressure while having no static shear modulus; surrounding tissues can resist shear and tension. A lumped oscillator, a deformable solid and a fluid model make different assumptions.

22.3.1 Time-Varying Inertia Tensor

A rigid body's inertia about its center of mass is constant in a fixed body frame, while its components rotate in an inertial frame. Deformation can change even the body-frame inertia. Those are separate effects, and neither alone determines the complete angular momentum of a deforming body.

Choose an origin at the complete modeled system's center of mass and a rotating frame. With all quantities below expressed in that frame,

$$\mathbf{H} = I\boldsymbol{\omega} + \mathbf{h}_{\mathrm{rel}}, \qquad \boldsymbol{\tau}_{\mathrm{ext}} = I\dot{\boldsymbol{\omega}} + \dot{I}\boldsymbol{\omega} + \dot{\mathbf{h}}_{\mathrm{rel}} + \boldsymbol{\omega} \times (I\boldsymbol{\omega} + \mathbf{h}_{\mathrm{rel}}).$$

The relative angular momentum $\mathbf{h}_{\mathrm{rel}}$ comes from material motion relative to the chosen frame. It generally cannot be inferred from the instantaneous inertia alone. Changing the system boundary or using an accelerating origin away from its center of mass requires the appropriate additional accounting.

A special scalar model with no relative angular momentum gives $\tau = d(I\omega)/dt$. At zero external torque, angular momentum is conserved. If I changes from 3.00 to $3.18\,\mathrm{kg\,m^2}$ with initial angular speed 10 rad/s, the final speed is 9.43396 rad/s. The associated rotational energy falls from 150 to 141.5094 J. Reversing the shape change reverses that energy change in an ideal reversible process. It is not automatically heat.

If instead an actuator holds angular speed at 10 rad/s while that inertia rises uniformly over 0.2 s, it must supply a 9 N m torque in this scalar model. Work associated with the shape change must also be accounted for. The term $\dot{I}\omega$ is not a viscous damper merely because it appears next to a velocity.

22.3.2 Practical Implications

Internal motion can affect momentum, energy, sensor observations and local loading together. Its importance must be estimated from measurements or sensitivity analysis for the intended task. The chapter does not assign a universal percentage change to torso inertia or a measured displacement to organs during a golf transition.

Transition deserves investigation because velocities, activation and contact forces evolve there; a claim that visceral motion makes it the most dangerous or most powerful phase needs separate evidence. The spine chapter (Chapter 23) develops segment-level loading models. The damping chapter (Chapter 16) distinguishes elastic storage, local dissipation and effects on the complete coupled system.

22.4 Intra-Abdominal Pressure: Define the Pressure Boundary

Intra-abdominal pressure is pressure within the abdominal compartment. Its measured value depends on the measurement method, reference pressure, location, posture and task. Gastric or bladder measurements can provide useful information under defined protocols, but a pressure reading is not a direct measurement of every boundary traction.

22.4.1 Anatomy of the Pressure-Generating System

The diaphragm separates thoracic and abdominal compartments; the abdominal wall and pelvic floor complete much of the surrounding boundary. Vertebral, fascial and muscular structures provide additional attachments and constraints. These structures deform and exert forces together. Treating the abdomen as a pressure vessel can clarify force balance, but a literal hydraulic cylinder with rigid end plates is only a simplified model.

Contraction and respiratory motion can change pressure, geometry and muscular tension at the same time. Their mechanical effects must be distinguished rather than assigned to pressure alone. The pressure difference across the diaphragm involves thoracic pressure as well as abdominal pressure. Raising an abdominal pressure relative to atmosphere does not specify that difference if thoracic pressure changes too.

22.4.2 IAP as a Structural Support System

For a surface patch S with unit normal $\mathbf{n}$ pointing from the cavity into its wall, the net pressure force and moment on the wall are

$$\mathbf{F}_p = \int_S \Delta p \, \mathbf{n} \, dA, \qquad \boldsymbol{\tau}_p = \int_S (\mathbf{r} - \mathbf{r}_O) \times (\Delta p \, \mathbf{n}) \, dA.$$

The pressure difference Δp must compare the two sides of that wall. Wall tension, muscle forces and attachments remain separate terms. The force direction varies across a curved surface, so multiplying pressure by its total curved area does not usually give the vertical resultant.

For uniform Δp, the vertical force uses signed projected area:

$$F_{p,z} = \Delta p \int_S n_z \, dA = \Delta p \, A_{\text{proj}}.$$

A hemispherical cap of radius R has curved area $2\pi R^2$ but vertical projected area πR^2. Its axial resultant is therefore $\Delta p \pi R^2$, not twice that value. Numerical surface integration verifies the distinction.

Using $1\,\text{mmHg} \approx 133.3224\,\text{Pa}$, a declared pressure difference of $100\,\text{mmHg}$ over projected area $0.04\,\text{m}^2$ gives approximately $533.29\,\text{N}$. Rounding pressure to $13{,}300\,\text{Pa}$ gives $532\,\text{N}$. Neither number is automatically a reduction in spinal compression; it is one patch's resultant under the stated pressure and geometry.

For a complete closed boundary under uniform pressure,

$$\oint_S p \, \mathbf{n} \, dA = \mathbf{0}, \qquad \oint_S (\mathbf{r} - \mathbf{r}_O) \times (p \, \mathbf{n}) \, dA = \mathbf{0}.$$

Opposing regions balance globally while individual regions remain loaded. A nonuniform pressure field can have a resultant associated with fluid weight or acceleration; the corresponding fluid balance must then be included. For the complete body, internal action-reaction pairs do not create a net external supporting force. They can nevertheless redistribute loads between structures.

This is why a diaphragm force cannot simply be subtracted from a joint load. The muscles and connective tissues containing the pressure contribute their own forces and moments. Their effects depend on geometry and activation. A comparison with one supposed typical back-extensor force omits that balance.

22.4.3 IAP and Torso Stiffness

Pressure can contribute to the response of a deformable structure through geometry and wall prestress. The work calculation shows the limit of a pressure-only explanation. For uniform internal pressure relative to a constant external reference, the pressure power delivered to a moving boundary is

$$P_p = \int_S p \, \mathbf{n} \cdot \mathbf{v} \, dA = p\dot{V}, \qquad Q_{p,j} = p \frac{\partial V}{\partial q_j}.$$

Here V is cavity volume and q_j a boundary-deformation coordinate. If a proposed ideal twist preserves volume, its direct hydrostatic pressure torque is zero. A fluid does not acquire a static shear modulus merely because its pressure is high.

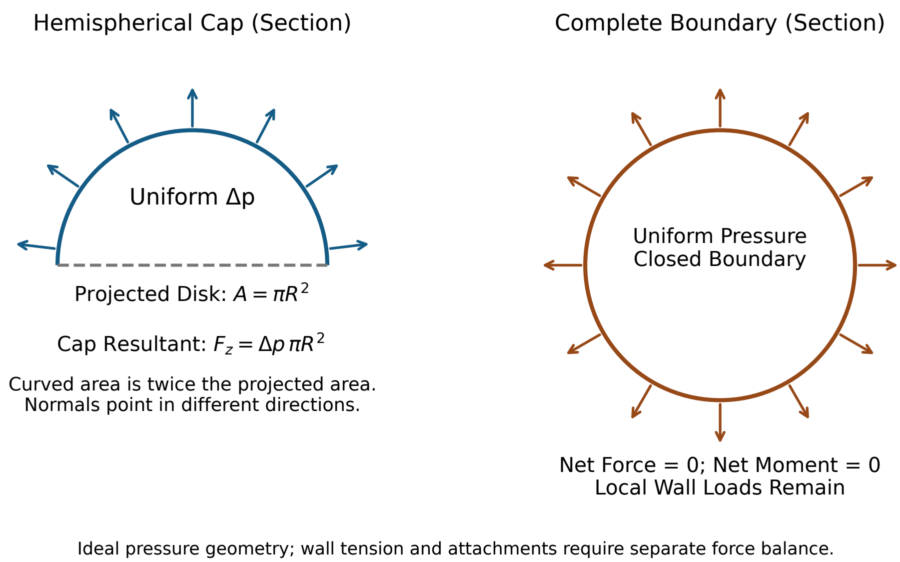


Figure 22.2: Pressure Acts Normal to a Surface. A Hemispherical Cap Has an Axial Resultant Based on Its Projected Disk; a Complete Uniform-Pressure Boundary Has Zero Net Force and Moment Despite Local Loading.

The surrounding wall can still resist that twist through its material stiffness, tension and constrained geometry. Pressure can change those quantities and the resulting tangent response. A model must include them to predict the change. It cannot replace that derivation with a universal stiffness proportional to pressure or assume that damping increases by the same rule.

Hodges and colleagues used phrenic-nerve stimulation to raise pressure in three prone subjects and measured an increase in posteroanterior lumbar indentation stiffness [Hodges et al., 2005]. This supports a mechanical contribution under their controlled conditions. It does not supply a golf-specific torsional stiffness, prove stability of a moving swing, or establish an optimum pressure profile.

22.4.4 Breathing Coordination and the Exhale Cue

Inspiration, expiration, glottal state, abdominal activation and posture interact. Inhalation does not universally reduce IAP, and exhalation does not uniquely determine it. The diaphragm's action and pressure difference depend on what the surrounding structures do.

Shirley and colleagues measured a composite posteroanterior spinal response during respiratory tasks in eight subjects [Shirley et al., 2003]. Both inspiratory and expiratory efforts could increase stiffness. Their analysis distinguished pressure, muscle activity and the measurement's composite nature. This contradicts a simple account in which inhalation necessarily removes support while exhalation guarantees it.

A breathing cue is therefore a behavioral intervention with several possible pathways. It may change timing, attention, muscle activity, pressure or comfort together. To explain its effect, measure the intended intermediate variables and the outcome. A plausible physiological account does not establish increased club speed or protection from a cardiovascular event.

22.5 Pressure, Support and Golf Performance

The central question is how pressure and muscle action jointly change a specified outcome. Local stiffness, stability, tissue load, club speed and perceived effort are different outcomes. Evidence about one can guide the next experiment but does not settle it.

22.5.1 Spine Stabilization and Load Distribution

Stokes and colleagues' static model included curved abdominal muscle paths, specified pressures and a force-allocation objective [Stokes et al., 2010]. It predicted lower spinal compression when pressure increased under its tested loads. The authors also explained that optimized activation may underestimate real coactivation and compression. The result is conditional on the complete model, not on pressure alone.

An experiment by Nachemson and colleagues measured pressures and muscle activity in four subjects performing five isometric tasks [Nachemson et al., 1986]. Four tasks showed increased rather than decreased compression during Valsalva maneuvers. The two studies ask related questions under different activation and task conditions. Together they motivate explicit force balance and experimental validation rather than a universal unloading percentage.

Stability additionally concerns how perturbations evolve. A larger measured indentation stiffness does not by itself establish stability for a multiaxis moving system with delays, muscle limits and changing contacts. Nor does lower estimated compression prove lower injury risk: the relevant tissue, loading direction, duration, exposure and capacity still matter.

22.5.2 Torso Stiffness and Energy Transmission

An elastic deformation can store recoverable energy. A dissipative process converts mechanical energy into other forms. A compliant connection can therefore transmit and return energy without being a source of loss, while a stiff connection can still dissipate energy or demand substantial muscle work.

For a linear torsional spring with fixed reference,

$$\tau_{\text{spring}} = -K\theta, \qquad V = \tfrac{1}{2}K\theta^2.$$

At fixed deformation, increasing K increases stored energy. At fixed applied torque magnitude T in static equilibrium, however, $\theta = T/K$ and $V = T^2/(2K)$, so increasing K decreases the stored energy. The comparison changes when its boundary condition changes.

Neither result establishes that a stiffer torso produces greater shoulder speed for the same hip torque. The delivered motion also depends on timing, inertia, geometry, muscle activation, damping and constraints. A finite-horizon simulation or experiment should compare the desired output under matched conditions and include the work needed to produce the changed mechanical response.

22.5.3 The X-Factor Stretch Mechanism

A measured thorax-pelvis separation depends on the segment definitions and angle convention. It is not automatically the strain of a single spring or the sum of lumbar joint rotations. A three-dimensional motion can change that reported angle through several coupled rotations.

To calculate elastic storage, identify a tissue or an effective constitutive element, its reference configuration and its force-deformation relation. For a nonlinear one-dimensional elastic element, stored energy is the integral of its restoring-force magnitude over deformation along the defined loading path. Hysteresis or active force production requires additional accounting.

If stiffness changes while the element is deformed,

$$\frac{d}{dt}\left(\tfrac{1}{2}K(t)\theta^2\right) = K\theta\dot{\theta} + \tfrac{1}{2}\dot{K}\theta^2.$$

The last term is not free energy. Raising K from 100 to 200 N m/rad at a clamped 0.1 rad deformation adds 0.5 J to this ideal storage element. The source of that energy and any losses must be modeled. Pressure or muscle activation can participate in such a change, but naming the mechanism does not remove its work cost.

22.5.4 Breathing Coordination as a Performance Cue

A useful test starts with a reproducible outcome and a tolerable intervention that the participant can perform consistently. Record delivery, ball outcome and the relevant timing or pressure measurements over comparable trials. Separate an immediate performance change from retention, adaptation and changes in reported effort.

Within-person comparisons should address order, fatigue, familiarization and strike variability. Generalization requires additional golfers and sessions. If a cue changes pressure and activation together, the result identifies the combined intervention unless the design can distinguish their contributions. A single better shot cannot establish the proposed internal mechanism.

22.6 Pressure and Health Outcomes

Mechanical models are useful for formulating loading hypotheses. They do not establish a safe pressure threshold, a disease mechanism or a rehabilitation prescription without relevant clinical evidence. The chapter's teaching examples consequently do not prescribe breath holding, maximal bracing or a core-training protocol.

22.6.1 IAP and Cardiovascular Pressures

Abdominal, thoracic, arterial and intracranial pressures are different quantities, with different reference boundaries and time courses. Blood flow depends on pressure gradients and vascular properties; a pressure reading in one compartment cannot simply be assigned to another. Changes during a strain and its release need not have the same cardiovascular effect.

Haykowsky and colleagues reported three cases of subarachnoid hemorrhage associated with weight training and discussed a possible relation to straining [Haykowsky et al., 1996].

A case report can identify a concern but cannot estimate a golfer's event probability or show that a particular exhalation cue prevents it. The physiological and clinical question requires its own measurements and study design.

22.6.2 Pelvic-Floor and Hernia Claims

Pressure loads pelvic-floor and abdominal-wall structures, but loading is not identical to injury. Tissue properties, geometry, activation, exposure and individual history affect response. A mechanical force calculation does not show that repeated golf swings cause pelvic-floor weakness, hernia or incontinence.

Measurement also matters. Bø and Sherburn's review concerns methods for assessing pelvic-floor function and strength [Bø and Sherburn, 2005]; those are distinct outcomes, measured with different instruments. It does not establish a long-term golfer injury rate or an optimum swing-pressure schedule. A study claiming prevention must measure the relevant health outcome, not substitute a pressure peak or a model's preferred activation pattern.

22.6.3 Testing Pressure Strategies

Specify what the proposed strategy should improve and how that effect will be measured. Pressure control may compete with mobility, comfort, fatigue or task performance, and a policy can shift load between structures. A numerical optimum is only an optimum for the chosen objective, constraints and model.

Clinical testing requires an appropriate population, exposure accounting, comparison and follow-up. Mechanical verification should precede that step, but it cannot replace it. The synthesis chapter (Chapter 32) distinguishes a plausible mechanism, an identified intervention effect and evidence of improved health outcomes.

22.7 Modifying the Equations of Motion for Pliable Systems

Start from a coordinate and energy definition. For independent generalized coordinates $\mathbf{q}$ and a declared input $\mathbf{u}$, a mechanical model can take the form

$$M(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{c}(\mathbf{q}, \dot{\mathbf{q}}) + \mathbf{g}(\mathbf{q}) + \mathbf{f}_{\text{passive}} = B(\mathbf{q})\mathbf{u} + J(\mathbf{q})^T\boldsymbol{\lambda} + \mathbf{Q}_{\text{ext}}.$$

The generalized mass matrix is not merely a list of segment inertia tensors. The vector $\mathbf{c}$ contains the velocity-dependent terms consistent with those coordinates. If $\mathbf{u}$ contains muscle forces, B can contain moment-arm mappings; if it contains generalized torques, the corresponding map has a different meaning. Contact reactions, external loads and passive forces must be assigned once to the intended system boundary.

22.7.1 Wobbling Mass Extension

For the earlier two-mass example, choose coordinates (x_r, z), with $x_w = x_r + z$. Substitution into the kinetic energy gives

$$M = \begin{bmatrix} m_r + m_w & m_w \\ m_w & m_w \end{bmatrix}.$$

Its determinant is $m_r m_w > 0$, and its off-diagonal terms express inertial coupling. The equations are

$$M \begin{bmatrix} \ddot{x}_r \\ \ddot{z} \end{bmatrix} + \begin{bmatrix} 0 \\ kz + c\dot{z} \end{bmatrix} = \begin{bmatrix} u \\ 0 \end{bmatrix}.$$

The zero second input does not mean that the relative mode is uncontrollable; force on the rigid part acts through the coupled dynamics. It also does not mean that living tissue is passive in general. An active contractile element would require an explicit force law and activation state.

If these same coordinates instead point vertically upward, gravitational potential is $g[(m_r + m_w)x_r + m_w z]$. The generalized gravity vector is therefore $(g(m_r + m_w), gm_w)^T$. Its soft-coordinate component is not zero. Coordinate transformations determine which equations receive gravity and other forces.

22.7.2 Time-Varying Inertia

For a Lagrangian $L(\mathbf{q}, \dot{\mathbf{q}}, t)$, derive

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = Q_i.$$

If $L = \frac{1}{2}\dot{\mathbf{q}}^T M(\mathbf{q}, t)\dot{\mathbf{q}} - V(\mathbf{q}, t)$, this gives

$$\sum_j M_{ij}\ddot{q}_j + \sum_{j,k} \Gamma_{ijk}\dot{q}_j\dot{q}_k + \sum_j \frac{\partial M_{ij}}{\partial t}\dot{q}_j + \frac{\partial V}{\partial q_i} = Q_i,$$

where $\Gamma_{ijk} = \frac{1}{2}(\partial_k M_{ij} + \partial_j M_{ik} - \partial_i M_{jk})$. The explicit partial time derivative is separate from the coordinate-dependent terms. Adding the full total $\dot{M}\dot{\mathbf{q}}$ to an already consistent Coriolis vector generally counts part of the inertia variation twice.

For a nondimensional example, $L = \frac{1}{2}(1 + q^2)\dot{q}^2$ gives $(1 + q^2)\ddot{q} + q\dot{q}^2 = u$. At $q = 1$, $\dot{q} = 2$ and $u = 0$, acceleration is -2. Appending an extra $\dot{M}\dot{q}$ incorrectly gives -6 and violates conservation of this system's kinetic energy.

When shape coordinates evolve dynamically, including them in the augmented state is often clearer than prescribing an unexplained $M(t)$. If their motion is prescribed, the actuation and work maintaining that prescription remain part of the model boundary. The general angular-momentum relation above is another way to expose omitted relative-motion terms.

22.7.3 IAP Effects

Pressure can enter generalized forces through boundary tractions and can influence wall geometry, prestress and activation-dependent material response. It is not necessarily an independent control input. A pressure state requires a closure relation involving cavity volume, wall motion and the chosen physiological approximation.

A fitted local torsional impedance may summarize small perturbations around one posture and activation condition. Its coefficients require identification, uncertainty estimates and a stated frequency range. Labeling coefficients $k_{\mathrm{IAP}}(P)$ and $d_{\mathrm{IAP}}(P)$ does not establish their signs, linearity, causal interpretation or portability to a golf swing.

22.7.4 Model Error Requires Validation

More detail can improve representation while increasing uncertainty or making parameters harder to identify. A model with many soft-tissue states can fit surface motion without correctly estimating internal forces. A simpler model can also be adequate for one output and inadequate for another.

Use a sequence of checks: conservation and units, numerical convergence, parameter sensitivity, independent measurements, and performance on held-out tasks or participants. Validation should specify the output, reference method, error measure, sampling and uncertainty. A universal error percentage attached to a model class conceals those choices.

22.8 When to Use Which Model

For gross kinematics, begin with segment and joint definitions that match the observations and assess attachment error. For inverse dynamics, add synchronized external forces, consistent inertial properties and sensitivity to filtering and tissue motion. For local loading, specify the tissues, constitutive laws, geometry and force-allocation assumptions. For impact vibration, resolve the modes and contacts relevant to the measured bandwidth.

For a pressure hypothesis, compare measured pressure and activation with the complete modeled boundary forces; test whether the predicted intermediate response occurs. For a clinical claim, add the corresponding exposure and health outcomes. For a ball-flight question, first determine whether uncertainty in the launch state already dominates the proposed internal-model refinement.

These choices can be connected without forcing every investigation into one maximum-detail model. The strongest connection is a shared record of coordinates, system boundaries, measurements, uncertainty and the outputs each model has actually reproduced.

22.8.1 Looking Ahead

Soft tissues link dynamics, estimation and load distribution. The spine chapter develops a more detailed anatomical model, while the interdisciplinary synthesis explains how to connect its predictions to control, measurement and outcomes. Neither a rigid approximation nor a soft-tissue extension earns general authority from complexity alone.

22.9 Chapter Exercises

1. For a tissue mass of 1 kg attached to a fixed base by $k = 5000\,\mathrm{N/m}$ and $c = 200\,\mathrm{N\,s/m}$, calculate ω_n, frequency in hertz, damping ratio and characteristic roots. Does release from a positive displacement at zero velocity ring? How does the unforced relative natural frequency change if a 1.5 kg base is free to move?

2. Derive the closed two-mass momentum and energy balances. Then use (x_r, z) coordinates with a vertically upward axis to derive the mass matrix and gravity vector. Explain why zero direct input to z does not mean zero response to force applied at x_r.

3. Derive $H(\omega)$ for prescribed harmonic base motion and verify that its mean input power equals damper loss. For a skin displacement of 0.01 m at 10 Hz, calculate acceleration amplitude. Explain why filtering alone need not identify bone motion.

4. A hemispherical cap has radius 0.1 m and a uniform pressure difference of 100 mmHg. Calculate its axial pressure resultant using 133.3224 Pa/mmHg. Explain why total curved area gives the wrong answer and why a closed surface under the same uniform pressure has no net force or torque.

5. In the declared zero-external-torque scalar model, inertia increases from 3.00 to 3.18 kg m^2 from an initial 10 rad/s. Calculate final speed and rotational-energy change. Contrast this with constant-speed motion over a 0.2 s inertia change, and explain why neither calculation proves viscous dissipation.

6. Compare a torsional spring with stiffness 100 and 200 N m/rad under a fixed 0.1 rad deformation and under a fixed 10 N m torque. Calculate the energy in each case. What additional energy term appears when stiffness changes while deformed?

7. Design an evidence chain for the claim that a breathing intervention reduces a specified spinal load and improves a golfer's outcome. Distinguish pressure, activation, load estimation, performance and clinical outcomes; identify a measurement ambiguity and an alternative explanation that the experiment must address.

22.10 Worked Answers

1. The fixed-base values are 70.7107 rad/s, 11.2540 Hz and $\zeta = 1.4142$. Roots are -29.2893 and $-170.7107\,\mathrm{s}^{-1}$; the stated release decays without ringing. A free base gives reduced mass 0.6 kg and relative undamped frequency 91.2871 rad/s. This comparison changes the boundary condition, not the material.

2. Adding the two force equations gives external force equal to total momentum rate. Multiplying each by its velocity and adding gives actuator power equal to mechanical-energy rate plus $c\dot{z}^2$. The transformed mass matrix is the one derived above, and the gravity vector is $(g(m_r + m_w), gm_w)^T$. Off-diagonal inertia couples the base force to the relative motion.

3. Substitution of $e^{i\omega t}$ motion gives the stated numerator $k + ic\omega$ and denominator $k - m_w\omega^2 + ic\omega$. Averaging force times velocity over a cycle gives the nonnegative loss $\frac{1}{2}c\omega^2|X_w - X_r|^2$. The marker acceleration amplitude is 39.4784 m/s^2. Bone and marker-relative motions can share a frequency and sum to the same observation, so removing high frequencies cannot uniquely separate them.

4. Projected area is $\pi(0.1)^2\,\mathrm{m}^2$, giving approximately 418.84 N. Curved area is twice as large, but its normals do not all point axially. On a complete uniform-pressure boundary, opposite contributions cancel in the vector force and moment integrals. Local loading remains even though the resultant is zero.

5. Angular momentum is 30 kg m^2/s, final speed is 9.43396 rad/s and rotational energy changes by -8.49057 J. Holding speed constant during the specified change instead requires 9 N m in the scalar angular equation. Internal shape work and actuator work complete the energy budget. Reversing an ideal shape change can restore the rotational energy; a viscous-loss explanation would require an actual dissipative mechanism.

6. At fixed angle the energies are 0.5 and 1.0 J. At fixed torque they are 0.5 and 0.25 J. The stiffness-change contribution to storage rate is $\frac{1}{2}\dot{K}\theta^2$; clamping the angle while increasing stiffness in this example adds 0.5 J. Identify its source before attributing later motion to passive recoil.

7. Calibrate the pressure reference and synchronize pressure, activation, motion and external load measurements. Validate the load estimate against independent information and vary uncertain force allocations. Compare matched intervention and control trials, including order, fatigue and strike variability. A change in pressure can accompany altered muscle coactivation, so reduced compression cannot be inferred from pressure alone. Better ball performance is another outcome, and a prevention claim additionally requires appropriate prospective health outcomes and exposure accounting.

Part VII

The Musculoskeletal Model

Chapter 23

Modeling the Spine: Motion, Load, and Evidence

In Plain Language 23.1. One Swing, Several Different Questions

A golfer can reproduce a similar shoulder turn with different distributions of motion, muscle activity and spinal loading. A video may resolve the large-scale motion well while leaving those internal differences uncertain. The purpose of a spine model is to make the connection explicit: which parts move, which forces produce that motion, how those forces are shared, and what evidence tests the calculation.

The spine participates in a coupled system that includes the pelvis, rib cage, abdominal wall, head, limbs and club. It transmits forces and moments, deforms, stores and dissipates mechanical energy, and receives active muscle forces. None of those roles alone establishes an efficient swing or predicts pain. This chapter develops the mechanics needed to investigate them together, from measured segment orientations to multibody dynamics and tissue-level questions.

23.1 Why the Spine Matters for Golf

23.1.1 The Mechanical Link

The pelvis and thorax interact through several anatomical load paths: vertebral structures, muscles spanning multiple levels, fascia, and the pressurized abdominal wall. A model that represents their interaction by one joint wrench aggregates those paths. It does not show that every joule arriving at the club passed through one disc, or that the legs generated all the energy later present in the upper body.

For a chosen body or subsystem, linear momentum changes according to its external forces and angular momentum according to its external moments about a fixed inertial point or the subsystem center of mass. An arbitrary accelerating reference point requires the corresponding transport terms. Internal muscle and joint forces redistribute momentum within the whole golfer–club system. They do not generate its total angular momentum without a compensating exchange. Ground contact can supply an external moment even when a stationary contact point does no mechanical work. Muscle work can occur throughout the body. Thus a sequence of segment-speed peaks is a useful kinematic observation, but it is not by itself a measurement of energy transferred along an anatomical chain.

At an interface point O, the instantaneous power delivered to a rigid segment is

$$P_O = \boldsymbol{F} \cdot \boldsymbol{v}_O + \boldsymbol{M}_O \cdot \boldsymbol{\omega}. \tag{23.1}$$

Force, moment, velocity and angular velocity must refer to the same frame and compatible points. Across a compliant connection, the two endpoint velocities can differ; the signed sum of powers delivered into the connection contributes to stored energy, dissipation and any active work inside it. Deformation is therefore not synonymous with lost power.

23.1.2 What the X-Factor Measures

An X-factor is a declared measure of pelvis–upper-trunk separation. A difference between projected shoulder and hip lines is not generally the same as a three-dimensional relative axial angle. A shoulder-line marker set can also move relative to the rib cage through scapular motion and soft-tissue artifact. Report the segment definitions, projections, rotation sequence and sign convention before interpreting its magnitude.

If R_{WP} and R_{WT} map pelvis and thorax coordinates into a common world frame, the relative orientation is

$$R_{PT} = R_{WP}^T R_{WT}. \tag{23.2}$$

This describes the thorax relative to the pelvis, without yet assigning motion to individual vertebrae. Cervical motion changes the head relative to the thorax; it cannot be added as a source of this pelvis–thorax relative rotation. Similarly, a global change in thorax orientation can include pelvic rotation. Subtracting loosely defined rotation magnitudes does not separate these contributions.

Increasing separation may change muscle lengths, tissue deformation, moment arms and feasible future motion. Whether it improves club delivery, changes spinal load or aggravates symptoms requires additional information. There is no universal target angle implied by the definition.

23.1.3 Pain and Injury Statistics Need Denominators

Low-back problems are a substantial concern in golf, but the proportion of injuries affecting the back is different from the proportion of golfers with pain. A retrospective survey of 1,634 Australian amateur golfers reported that 17.6% had sustained at least one golf-related injury in the preceding year; the lower back accounted for 25% of reported injuries [McHardy et al., 2007b]. A separate one-year prospective survey of 588 amateurs reported 15.8 injuries per 100 golfers, with the lower back comprising 18.3% of injury sites [McHardy et al., 2007a]. These results do not establish that half of all golfers have chronic spinal injury.

A history of pain, an imaging finding, a modeled load and a diagnosed injury are different observations. A person may change a swing because of pain, so an association between movement and pain does not by itself establish which came first. Mechanical models help formulate testable explanations; prospective observation and appropriate clinical outcomes are needed to evaluate injury-related predictions.

23.2 Anatomy and the Model Boundary

23.2.1 Vertebrae, Articulations, and Discs

The usual description has seven cervical, twelve thoracic and five lumbar vertebrae above the sacrum. The sacrum ordinarily comprises five fused elements; the coccygeal count and degree of fusion vary. The familiar developmental count of about 33 is not a count of 33 independently moving adult joints. The atlas has no vertebral body, and upper-cervical articulations differ from typical disc-bearing levels [Betts et al., 2022].

In typical anatomy there are 23 intervertebral discs, from C2–C3 through L5–S1. There is no disc between the skull and atlas or between atlas and axis. A model must list its actual connections: L1 through L5 supplies four internal lumbar intervertebral levels; including L5–S1 gives five, and adding T12–L1 gives six. Counts depend on the endpoints. An articulated chain with a fixed base and n modeled three-rotation joints has $3n$ configuration coordinates; using compliant six-coordinate connections gives a different model. A floating base adds its own coordinates and constraints.

23.2.2 Structures Carry Different Loads

A typical vertebra has a body, pedicles, laminae and processes. Intervertebral discs join neighboring bodies; paired facet articulations and their capsules provide posterior contact and restraint. Endplates, discs, facets and ligaments share loads according to geometry, deformation and contact state. Their contributions cannot be assigned from the net joint moment alone. The rib cage also alters thoracic mechanics; a model without it requires a correspondingly limited interpretation.

23.2.3 The Intervertebral Disc

The nucleus pulposus, annulus fibrosus and endplates form a heterogeneous, hydrated structure. The annulus has alternating collagen-fiber families rather than a universal single fiber angle. Microscopy documents spatial gradients in lamellar organization [Cassidy et al., 1989]. The angle must be stated relative to its reference plane or axis; an angle measured from a transverse plane is not the same angle measured from the spine's long axis.

A water-rich nucleus is not mechanically identical to a volume of pure water. Experiments show time-dependent shear behavior, including different responses in stress relaxation and dynamic loading [Iatridis et al., 1997]. Whole-disc response additionally depends on the surrounding matrix, annular restraint, fluid transport, swelling and endplate conditions. An elastic spring can approximate a specified operating range; it does not reproduce all those mechanisms. A viscoelastic or poroelastic model needs internal states or a history law if loading duration and recovery matter.

23.2.4 Ligaments and Contact

Ligaments provide passive tensile restraint, while facet contact is unilateral: a surface can push when engaged but cannot pull another surface through a gap. Different ligaments have different attachments, slack lengths and prestrain. It is inaccurate to assume that every ligament is slack in every neutral posture. The anterior longitudinal ligament resists extension; posterior structures contribute to restraint in flexion, with load sharing depending on posture and level [Betts et al., 2022].

The activation of a muscle is a different mechanism. A balanced pair of active muscles can exert little net joint moment while still producing substantial compression and stiffness. A small net moment does not mean the connection is passive or unloaded.

23.2.5 Curvature, Posture, and Hip Flexion

Lordosis and kyphosis describe anatomical sagittal curvatures. They should not be confused with the differential-geometric curvature of a configuration-space metric. Changes in posture affect lever arms, muscle lengths and contact, but a visible spinal curve does not directly determine tissue stress.

Forward inclination at address can combine hip flexion, pelvic tilt and regional spinal motion. Hip flexion is relative motion between pelvis and femur; lumbar flexion is relative motion within a specified spinal region. They are not numerically interchangeable. Report these separately rather than converting a global trunk inclination into an assumed lumbar deformation. Neither a drawing of a neutral posture nor a universal flexion cutoff supplies a clinical safety boundary.

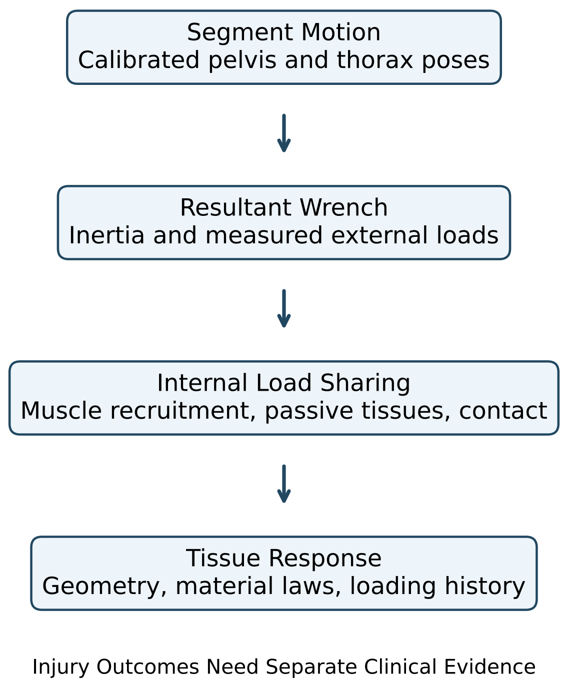


Figure 23.1: From Motion to Tissue Response: Each Stage Requires Its Own Evidence. A Single Measured Shoulder Turn Does Not Identify Every Internal Load.

23.3 Motion Segments and Measured Mobility

23.3.1 Three Rotations and Three Translations

A typical functional spinal unit contains two adjacent vertebrae and their connecting disc, ligaments and facet articulations. Their relative rigid-body pose has three rotational and three translational components. Those six components are not six unrestrained, independently actuated motions: stiffness, contact, loading and neighboring anatomy govern their response.

For this chapter use right-handed axes x anterior, y left and z superior in a declared local reference posture. Flexion–extension is associated with rotation about a mediolateral axis, lateral bending with an anteroposterior axis, and axial rotation with a superior–inferior axis. A motion occurring in a sagittal plane rotates about an axis normal to that plane; naming the plane as the rotation axis creates a convention error. Once the segment tilts, its local superior axis need not coincide with world vertical.

Anterior–posterior translation is along the local anterior direction, not along the spinal long axis. Compression and distraction concern the local axial component. Small translations can matter for contact, disc strain and load sharing even when they have little effect on a coarse reconstruction of shoulder turn. Omitting them is a task-dependent approximation, not a consequence of their being unimportant mechanically.

23.3.2 Range of Motion Is a Test Result

A range-of-motion value requires a level or region, reference posture, direction, applied load, preload and measurement method. One-sided excursion from a reference differs from total excursion between opposite endpoints. Passive cadaveric loading, voluntary in-vivo movement and a particular golf swing are different experiments.

For example, Wilke and colleagues tested 68 isolated thoracic functional units using pure moments of $\pm 7.5\,\mathrm{N\,m}$, with short rib remnants and no axial preload. Reported mean one-sided axial rotations included 5.9° at T6–T7 and 3.3° at T11–T12 [Wilke et al., 2017]. These observations expose the error in assigning 20–30° to every thoracic level. They are not individual golfer limits, and the preparation does not represent an intact loaded rib cage.

For coaxial rotations about the same fixed axis, signed angles add. For general spatial rotations, regional orientation is a product,

$$R_{0n} = R_{01}R_{12}\cdots R_{n-1,n}. \tag{23.3}$$

Adding independently measured maximum excursions may combine incompatible postures and loading conditions. Even if a scalar sum is a useful approximation for a particular test, its assumptions must be stated. A model result outside an expected range is a prompt to examine calibration, conventions and evidence, not an automatic diagnosis or a proof of anatomical impossibility.

23.4 Coordinate Coupling and Physical Coupling

23.4.1 A Tilted Axis Changes Components

Let R map local vector components into the world frame. Angular velocity components satisfy $\boldsymbol{\omega}_b = R^T\boldsymbol{\omega}_W$. For a segment tilted by ϕ about y, take $R = R_y(\phi)$. A world-vertical angular velocity $\boldsymbol{\omega}_W = \Omega\boldsymbol{e}_z$ then has local components

$$\boldsymbol{\omega}_b = \Omega(-\sin\phi,\, 0,\, \cos\phi)^T. \tag{23.4}$$

At $\phi = 30°$ and $\Omega = 4\,\text{rad/s}$ these are $(-2, 0, 2\sqrt{3})\,\text{rad/s}$. The local anteroposterior component appears because the local frame is tilted. This identity does not require a ligament, an intervertebral coupling law or an additional physical rotation imposed by anatomy. It is one vector described in two frames.

Actual anatomical coupling is a different question: for a declared applied wrench and constraints, which relative motions occur? A local linear compliance relation might be $\delta\boldsymbol{\eta} = C\,\delta\boldsymbol{m}$. Off-diagonal entries describe coupled response in those coordinates. A single projected angle cannot identify that matrix. Coupling can also arise through the whole-body inertia, other joint loads and contact changes.

23.4.2 Euler Products Must Match the Stated Sequence

Use active right-handed matrices and column vectors. As a coordinate chart, define

$$R = R_z(\alpha)R_y(\beta)R_x(\gamma). \tag{23.5}$$

This is the product for successive rotations about fixed x, fixed y and fixed z axes, applied in that order. Equivalently it can describe moving-axis rotations in the reversed axis order. It is not the product for successively rotating about moving x, moving y, then moving z axes. Here β coincides with positive forward flexion only in the corresponding reference configuration; the chart labels are not three universal anatomical hinges.

Writing $c_a = \cos\alpha$, $s_a = \sin\alpha$, and similarly for β, γ, multiplication gives

$$R = \begin{bmatrix} c_a c_b & c_a s_b s_g - s_a c_g & c_a s_b c_g + s_a s_g \\ s_a c_b & s_a s_b s_g + c_a c_g & s_a s_b c_g - c_a s_g \\ -s_b & c_b s_g & c_b c_g \end{bmatrix}. \tag{23.6}$$

All three angles are represented, although not every entry depends on every angle. Differentiating the product and using $\widehat{\boldsymbol{\omega}_b} = R^T\dot{R}$ gives

$$\boldsymbol{\omega}_b = \begin{bmatrix} 1 & 0 & -\sin\beta \\ 0 & \cos\gamma & \sin\gamma\cos\beta \\ 0 & -\sin\gamma & \cos\gamma\cos\beta \end{bmatrix} \begin{bmatrix} \dot{\gamma} \\ \dot{\beta} \\ \dot{\alpha} \end{bmatrix} = E\dot{\boldsymbol{q}}_e. \tag{23.7}$$

Thus Euler-angle rates are not generally angular-velocity components. Since $\det E = \cos\beta$, this chart becomes singular at $\beta = \pm\pi/2$. The orientation and physical angular velocity can remain regular there. A different chart, rotation matrix or unit quaternion can represent the motion; constraints and double covering must still be handled correctly.

23.4.3 Moments and Generalized Torques Transform Together

Power invariance supplies the conjugate generalized load:

$$\boldsymbol{m}_b^T\boldsymbol{\omega}_b = (E^T\boldsymbol{m}_b)^T\dot{\boldsymbol{q}}_e, \qquad \boldsymbol{Q}_e = E^T\boldsymbol{m}_b. \tag{23.8}$$

Using a physical moment component as an Euler generalized torque without this mapping changes the work. A consistent coordinate transformation does not make inverse dynamics physically ambiguous; it changes the components used to describe the same result. Muscle forces remain underdetermined for a separate reason: multiple muscles and passive structures can produce the same resultant.

If a local stiffness law is expressed in Euler coordinates, differentiating $E^T \boldsymbol{m}_b$ introduces derivatives of E as well as the tissue law. A configuration-dependent coordinate matrix can therefore create apparent coupling in the reported tangent stiffness. Comparing stiffness numbers requires the operating load, pose, coordinate definitions and whether the derivative is taken at fixed activation or with feedback operating.

23.4.4 What Motion Capture Can Identify

An isolated marker provides position, not a complete segment orientation. A noncollinear marker cluster plus calibration can estimate a rigid segment frame, with uncertainty from skin motion and model fit. Pelvis and thorax frames yield R_{PT}; recovering every intermediate vertebral pose requires additional measurements or assumptions. Many products of intervertebral rotations give the same endpoint orientation.

There is no single Euler sequence that reveals the historical order in which the anatomy moved. A declared convention makes measurements comparable within its domain. It does not remove missing observations. For a golf investigation, preserve the measured poses, document the decomposition and test conclusions against reasonable alternative calibrations. Distinguish a change of representation from a change in the physical model.

23.5 Choose a Model That Can Answer the Question

23.5.1 From Segment Motion to Tissue Deformation

A rigid thorax connected to a separate pelvis can represent an aggregate trunk rotation. A single rigid body containing both cannot: their relative pose is fixed by construction. Neither model identifies the motion of individual lumbar levels. An aggregate joint is useful for whole-body momentum and club-delivery questions when its geometry and inertial properties adequately represent the task. Its joint torque is a resultant, not the force in a disc.

A model with separate lumbar and thoracic regions allows their contributions to vary and can include bending, twist and translation. Each additional freedom needs an inertia, a constraint or force law, and information to identify its state. More segments do not automatically supply those data. If only pelvis and thorax motion are observed, adding intermediate joints may create multiple equally plausible distributions of motion and load.

A vertebral chain exposes level-specific joint reactions and muscle moment arms. Its joint count follows the included interfaces: L1 through L5 supplies four internal interfaces, adding the sacrum supplies L5–S1, and adding T12 supplies T12–L1. The chosen fixed base, floating base, rotational joints and translational constraints determine the number of generalized coordinates. An anatomical count alone does not establish the model's degrees of freedom.

A finite element model resolves deformation within discs, vertebrae, ligaments or cartilage. It can estimate spatial stress and strain given boundary conditions, contact laws and constitutive properties. It cannot repair an incorrect applied load or an unidentified material law. Mesh convergence addresses discretization error; agreement with independent measurements addresses a different question. A detailed model with uncertain loading can give precise-looking stress maps with large physical uncertainty.

The useful hierarchy is therefore a sequence of questions: Does the model reproduce measured motion? Does it satisfy the measured external wrench balance? Are muscle and contact-force estimates supported? Are local deformation and material laws validated for the loading regime? Does any proposed injury relationship have separate clinical evidence? Success at one step does not establish all later steps.

23.5.2 A Coupled Equation for the Golfer and Club

For independent joint coordinates $\boldsymbol{q}$ in a nonsingular chart, consider a fixed, ideal contact mode:

$$M(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{c}(\boldsymbol{q}, \dot{\boldsymbol{q}}) + \nabla_{\boldsymbol{q}}\big(V_g(\boldsymbol{q}) + U(\boldsymbol{q})\big) = B(\boldsymbol{q})\boldsymbol{\tau} - D(\boldsymbol{q})\dot{\boldsymbol{q}} + J_c^T\boldsymbol{\lambda}. \tag{23.9}$$

Here M is the coupled mass matrix; $\boldsymbol{c}$ contains velocity-dependent inertial terms; V_g and U are gravitational and passive elastic potentials; $B\boldsymbol{\tau}$ is the generalized active load; and D is a positive semidefinite damping matrix. The contact multiplier $\boldsymbol{\lambda}$ and acceleration must satisfy the constraint equations together. Applied club, ground or apparatus loads belong in the appropriate generalized-force terms if they are not already represented by included bodies and contacts. Do not count them twice.

The equation makes an important connection visible. Changing a hip or arm torque can change spinal acceleration through M^{-1}, even if no spine torque changes. Changing posture also changes inertia, moment arms, gravity, passive forces and contact geometry. A collection of uncoupled scalar springs cannot represent all those effects. Conversely, including a coupled mass matrix does not identify the individual muscles that generated a measured resultant.

For quaternion or other redundant orientation coordinates, write $\dot{\boldsymbol{q}} = N(\boldsymbol{q})\boldsymbol{v}$ and formulate dynamics in generalized velocity $\boldsymbol{v}$. One must then use the corresponding force and energy gradients rather than silently identifying $\boldsymbol{v}$ with $\dot{\boldsymbol{q}}$. For contact impacts, lift-off or slip, the active equations and possibly the state reset change. A smooth equation applies within its stated mode.

23.5.3 Inverse Dynamics and Internal Load Sharing

Choose a cut through the model and declare which tissues cross it. Measured motion, inertia and external loads can constrain the resultant wrench required across that cut. A simplified axial balance illustrates the remaining ambiguity:

$$N_{\text{contact}} + N_{\text{muscle}} + N_{\text{ligament}} = N_{\text{required}}, \tag{23.10}$$

with each quantity signed in the same free-body convention. Muscle forces also contribute moments through their attachment geometry. A joint reaction from one software model may include or exclude explicit muscle, disc or constraint forces differently from another. Compare definitions before comparing numbers.

For an illustrative antagonist pair with equal moment-arm magnitude r, the net moment is $\tau = r(F_+ - F_-)$. Increasing both tensions by ΔF leaves that moment unchanged. If both have compressive components at the joint, the compression increases. Thus identical motion and net moment can coexist with different internal loads. EMG, muscle geometry, recruitment criteria and physiological constraints can narrow the possibilities; none should be hidden inside an unexplained "joint force."

23.6 Spinal Loads and Injury Evidence

23.6.1 Measured Inputs and Modeled Contact Forces

Lim, Chow and Chae studied five male collegiate golfers using an EMG-assisted optimization model. Their estimated L4–L5 mean peak compression exceeded six body weights during the downswing; reported shear peaks occurred during follow-through. These are model-based internal contact estimates informed by measurements, not forces measured by an instrument inside the disc. The participant group, recruitment assumptions, modeled level and definition of peak all matter. [Lim et al., 2012]

The peak of an ensemble-average waveform need not equal the average of individual peaks: individual maxima can occur at different times. An anterior shear peak, a compression peak and a torsional-moment peak also need not coincide. Reporting only the largest absolute value discards sign, timing and concurrent loads. For transition, early downswing and follow-through, compare complete time histories with uncertainty and a common phase definition. No universal "most dangerous phase" follows from one model's largest force.

Body-weight normalization is useful for reporting scale, but it is not a tissue-strength criterion. For an illustrative 75 kg person with $g = 9.81\,\mathrm{m/s^2}$, eight body weights correspond to 5886 N. Dividing by an assumed $1800\,\mathrm{mm}^2$ area gives a mean normal traction magnitude:

$$\frac{N}{A} = \frac{5886\,\mathrm{N}}{1800\,\mathrm{mm}^2} = 3.27\,\mathrm{MPa}. \tag{23.11}$$

The area and load here are teaching inputs, not an anatomical measurement or a safe-load recommendation. This quotient cannot locate the peak annular stress, partition disc and facet load, or determine whether a golfer will develop pain.

23.6.2 Normal and Shear Stress Are Different Tensor Components

Consider a homogeneous circular beam only to clarify the mechanics. Let z run along the beam, x, y be principal centroidal cross-section axes, and $N > 0$ denote compression. With bending moments defined by the traction moment on the positive-z face,

$$\sigma_{zz}(x, y) = -\frac{N}{A} + \frac{M_x y}{I_x} - \frac{M_y x}{I_y}. \tag{23.12}$$

For Saint-Venant torsion of this circular beam,

$$\sigma_{xz} = -\frac{Ty}{J}, \qquad \sigma_{yz} = \frac{Tx}{J}, \qquad J = I_x + I_y. \tag{23.13}$$

The normal and shear components act in different directions. Their signs depend on the declared face and axis conventions. They must enter a stress tensor and the traction relation $\boldsymbol{t} = \boldsymbol{\sigma n}$; adding their scalar magnitudes does not produce a general "total disc stress." Transverse shear from a net transverse force would require additional terms, and these circular-beam torsion equations are not formulas for an arbitrary cross-section.

A two-dimensional example makes the distinction concrete. If the nonzero components are a normal stress σ and shear stress s, the matrix

$$\begin{bmatrix} \sigma & s \\ s & 0 \end{bmatrix} \tag{23.14}$$

has principal stresses

$$\sigma_{1,2} = \frac{\sigma}{2} \pm \sqrt{\left(\frac{\sigma}{2}\right)^2 + s^2}. \tag{23.15}$$

These follow from its eigenvalues, not from $\sigma + s$. Even a correctly computed isotropic equivalent stress would not establish failure of an anisotropic, hydrated, heterogeneous disc. Fiber direction, matrix behavior, fluid transport, endplates, degeneration and loading history require their own constitutive description and evidence.

23.6.3 Loading, Damage and Pain Require Separate Evidence

Cadaver experiments by Adams and Hutton produced different lesions under compression and flexion depending on the loading conditions and disc degeneration. They help investigate mechanisms under controlled conditions; they do not supply a universal human golf-swing failure threshold. [Adams and Hutton, 1982]

An inferred high contact force is not a diagnosis of herniation, and pain is not a direct readout of one stress component. An injury argument needs a specified outcome, exposure duration and repetition, participant characteristics, competing explanations, and a study design that can test the proposed relationship. Cross-sectional differences between people with and without pain can reflect causes, consequences or adaptations. A model can generate a testable hypothesis about tissue loading without claiming that its preferred swing prevents injury.

This distinction strengthens the golf investigation. Instead of declaring a particular X-factor harmful, ask whether a change in the measured movement alters a validated load estimate under comparable club delivery, whether that result survives uncertainty in muscle recruitment and tissue properties, and whether the relevant exposure predicts the stated outcome in independent observations. Each link can succeed or fail separately.

23.7 Elasticity, Active Stiffness and the Drift Field

23.7.1 A Passive Chain Does Not Create Extra Torque

Take n coaxial, linear torsional springs in series, with positive stiffnesses k_i, no intermediate applied moments and negligible intermediate acceleration. The same moment τ passes through each spring, so

$$\theta_i = \frac{\tau}{k_i}, \qquad \Theta = \sum_i \theta_i, \qquad k_{\text{eq}} = \left(\sum_i \frac{1}{k_i}\right)^{-1}. \tag{23.16}$$

Thus $\tau = k_{\text{eq}}\Theta$. Adding an identical spring in series decreases the equivalent stiffness. At a fixed total angle it decreases the moment and stored energy. At a fixed applied moment it increases the total deflection and energy. Specifying what is held fixed resolves the apparent contradiction.

Angles in an energy calculation must be in radians. Consider five springs, each with stiffness 12 N m/degree. Their combined stiffness is $k_{\text{eq}} = 2.4\,\text{N m/degree}$. At a total angle of 20°, the moment is 48 N m and

$$U_{\text{total}} = \frac{1}{2}\tau\Theta = \frac{1}{2}(48\,\text{N m})\frac{20\pi}{180} \approx 8.38\,\text{J}. \tag{23.17}$$

These are synthetic teaching values. Real spinal joints have nonlinear, coupled responses; muscles can bridge several levels, facets can engage, and intersegmental inertia makes transmitted moments differ. The series result is a limiting example, not a method for multiplying a regional range of motion into an energy reserve.

23.7.2 A Ligament Law Must Define Length and Units

One illustrative tension-only law uses current length L, slack length L_0 and k with units $\mathrm{N/m^2}$:

$$F(L) = \begin{cases} 0, & L \le L_0, \\ k(L - L_0)^2, & L > L_0. \end{cases} \tag{23.18}$$

The continuous potential is

$$U_L(L) = \frac{k}{3}\big[\max(L - L_0, 0)\big]^3, \qquad \boldsymbol{Q}_L = -F(L)\nabla_{\boldsymbol{q}} L. \tag{23.19}$$

The second relation includes the geometry that converts tension to generalized load. If strain is used instead, its dimensionless definition and the coefficient's units must change consistently. A neutral joint pose does not imply every ligament is at its slack length; prestrain can produce tension there. The quadratic law illustrates a conservative nonlinear spring and omits hysteresis, damage and rate dependence. Those effects need additional state or constitutive terms.

23.7.3 Changing Stiffness Can Require Mechanical Work

For a one-coordinate example, let

$$U(\theta, t) = \tfrac{1}{2}k(t)\big(\theta - \theta_0(t)\big)^2, \qquad \delta = \theta - \theta_0(t). \tag{23.20}$$

With constant inertia I, applied torque τ_u, and damping coefficient $d \ge 0$, suppose

$$I\ddot{\theta} = \tau_u - d\dot{\theta} - k\delta. \tag{23.21}$$

For mechanical energy $E = \frac{1}{2}I\dot{\theta}^2 + U$, differentiation gives

$$\dot{E} = \tau_u\dot{\theta} - d\dot{\theta}^2 + \tfrac{1}{2}\dot{k}\,\delta^2 - k\delta\dot{\theta}_0. \tag{23.22}$$

Changing stiffness or equilibrium angle can therefore add or remove mechanical energy. For example, holding a nonzero deformation fixed while increasing k increases stored energy; the mechanism changing stiffness supplies that energy. This is not free concentration of an unchanged energy store. Nor does simultaneous acceleration of adjacent segments necessarily dissipate energy: dissipation is associated with the relevant constitutive losses, such as $d\dot{\theta}^2$, not with acceleration itself.

Active muscle tension, short-range mechanical response, activation dynamics and feedback can all affect an estimated joint stiffness. Co-contraction can change stability, stiffness and compression together. Calling it entirely passive hides the maintained activation and its physiological cost. A mechanical energy equation also does not measure metabolic expenditure. These distinctions matter when comparing a golfer who stiffens the trunk with one who changes timing: a similar endpoint motion may arise from different force, energy and control histories.

23.7.4 Gravity and Elastic Return Have Signed Contributions

An elastic restoring moment is $-\partial U/\partial\theta$, not an unsigned amount that always helps the downswing. Gravity contributes $-\partial V_g/\partial\theta$. Depending on pose, the two can reinforce or oppose one another, and their power depends on the sign of velocity. In a planar illustration, let θ be a mass's angle from the downward vertical, with center-of-mass distance ℓ. Then

$$V_g = -mg\ell\cos\theta, \qquad \tau_g = -mg\ell\sin\theta. \tag{23.23}$$

For a spring centered at zero, $\tau_e = -k\theta$. At a positive angle both moments point toward zero; if the angle is increasing, both have negative mechanical power. Their magnitudes alone do not identify energy transfer into the club. A full model must also include coupled motions and moving attachment points.

23.7.5 Define the Zero Input Before Computing Drift

For a free, independent-coordinate model with fixed passive properties, let $\boldsymbol{x} = (\boldsymbol{q}, \boldsymbol{v})$ and $\boldsymbol{v} = \dot{\boldsymbol{q}}$. Within a smooth mode,

$$\dot{\boldsymbol{x}} = \underbrace{\begin{bmatrix} \boldsymbol{v} \\ M^{-1}(-\boldsymbol{c} - \nabla V - D\boldsymbol{v}) \end{bmatrix}}_{\boldsymbol{f}(\boldsymbol{x})} + \underbrace{\begin{bmatrix} 0 \\ M^{-1}B \end{bmatrix}}_{G(\boldsymbol{x})} \boldsymbol{\tau}, \qquad V = V_g + U. \tag{23.24}$$

The drift is a state-velocity field. A torque expression alone has the wrong units and omits both the kinematic component and the inverse inertia. In a constrained mode, eliminate or solve the contact reaction consistently for both drift and controlled dynamics; simply deleting $J_c^T\boldsymbol{\lambda}$ changes the physical problem.

If muscle excitation is the input, include activation and other force-generating states. Zero excitation need not mean zero activation or immediate disappearance of muscle force. If stiffness or reference posture is commanded, their evolution and the associated work belong in the model. The same physical behavior can be split differently into drift and input after shifting the reference control. Consequently, a "passive percentage" is not established merely by assigning co-contraction to $\boldsymbol{f}$.

For a golf counterfactual, state which inputs become zero, which contacts remain, and which initial muscle, shaft and tissue states are retained. Compare forward trajectories over a declared interval. Removing an instantaneous torque term at one recorded posture does not by itself predict the subsequent delivery or the subsequent spinal load.

23.7.6 Abdominal Pressure Requires a Free-Body Diagram

Pressure acts through distributed surface traction. On a chosen surface S with consistent normal convention, the pressure-force and moment contributions involve

$$\boldsymbol{F}_p = \int_S p\boldsymbol{n}\,dA, \qquad \boldsymbol{M}_{p,O} = \int_S (\boldsymbol{r} - \boldsymbol{r}_O) \times p\boldsymbol{n}\,dA. \tag{23.25}$$

Uniform pressure on a closed surface has zero net force and moment, although a portion of that surface can experience a nonzero resultant. The choice of subsystem is therefore essential. Multiplying pressure by one projected area computes one contribution under stated geometry; it does not automatically give the change in spinal compression. Abdominal wall tension and the forces needed to generate pressure must satisfy the same equilibrium.

Hodges and colleagues used artificial diaphragm stimulation to measure a trunk extension moment associated with pressure under controlled laboratory conditions. Stokes and colleagues modeled pressure together with curved abdominal wall muscles and recruitment optimization, predicting unloading under their specified static conditions. The latter authors also discuss limitations of optimized muscle recruitment. Neither result establishes that a particular breath during a dynamic golf swing reduces an individual's disc load. [Hodges et al., 2001, Stokes et al., 2010]

The mechanical question is useful without a breathing prescription: when pressure changes, how do muscle forces, trunk moment, contact reactions and stiffness change together under the actual task constraints? A model that includes only the apparent extension moment and omits its muscular support cannot answer that question.

23.8 A Reproducible Spine Investigation in Golf

Start with a stated comparison. For example: does redistributing relative motion between modeled lumbar and thoracic joints change estimated lumbar contact load while preserving a specified club-delivery task? Define the delivery tolerances, participant population, measured coordinates, phase events and load outputs before optimizing a movement. Otherwise a lower load may simply accompany a slower or different swing.

Measure and retain the quantities needed for the model: calibrated segment poses, external ground and club interactions, timing and, where relevant, EMG with its processing and normalization. Document sampling, filtering and differentiation because acceleration estimates and inverse dynamics depend on them. Check whole-body force and moment balance before interpreting a level-specific internal force. A successful fit to thorax markers is not validation of every vertebral motion.

Use the simplest model that resolves the proposed mechanism, then test sensitivity to added degrees of freedom and uncertain parameters. Compare plausible muscle recruitment, joint stiffness, attachment geometry, anthropometry and contact assumptions. If the sign of a proposed improvement changes across reasonable choices, report that dependence. Increasing model detail is valuable when new data constrain it or it changes a validated prediction, not because a larger model must be more accurate.

Separate calibration from evaluation. Fit parameters on one set of observations and assess predictions on independent trials or participants appropriate to the intended claim. Report errors in the actual measured outputs; do not invent a blanket accuracy percentage for a model class. Internal contact and tissue stress remain conditional estimates unless the corresponding quantities have independent validation. Any proposed clinical benefit requires outcome evidence beyond a mechanical simulation.

The resulting account can connect the swing's components without reducing them to one ideal angle. Preparation establishes posture, activation and elastic state. Ground interaction, muscle action and coupled inertia shape the subsequent motion. That motion changes force directions and moment arms; muscle recruitment and passive structures partition the required wrench. Tissue properties and history govern local deformation. The club-delivery objective constrains which alternatives are comparable. A rigorous explanation follows these links, names the assumptions at each one, and shows which conclusions survive uncertainty.

The muscle-force model in Chapter 14 and the closed-chain mechanics in Chapter 9 supply complementary details about force generation and constraints. Chapter 33 connects these mechanisms to preparation, delivery and outcome across the complete swing.

Exercises 23.1. Chapter Exercises

1. **Joint Count.** A chain contains rigid T12, L1–L5 and sacrum bodies, with the sacrum fixed. Count adjacent interfaces. Determine the configuration dimension if each interface is an ideal three-rotation joint, and if instead each allows six independent relative motions. Explain why neither is an anatomical claim that all those motions are equally free.

2. **Tilted Spin.** For $R = R_y(\phi)$ and $\boldsymbol{\omega}_W = \Omega \boldsymbol{e}_z$, derive $\boldsymbol{\omega}_b$. Evaluate it for $\phi = 30°$ and $\Omega = 4\,\text{rad/s}$. Show that its norm is unchanged, and explain why the nonzero local x component does not identify a tissue compliance coefficient.

3. **Euler Kinematics and Power.** Derive the displayed matrix E from $R_z(\alpha)R_y(\beta)R_x(\gamma)$. Verify numerically at a nonsingular nonzero pose that $\boldsymbol{m}_b^T \boldsymbol{\omega}_b = (E^T \boldsymbol{m}_b)^T \dot{\boldsymbol{q}}_e$. What fails at $\beta = \pi/2$, and what physical quantity remains well defined?

4. **Finite Rotation Composition.** Compare $R_x(30°)R_y(20°)$ with the reversed product. Compute their action on $\boldsymbol{e}_z$. Explain why summing maximal regional angles cannot generally predict a thorax orientation.

5. **Load and Mean Traction.** Reproduce the 75 kg, eight-body-weight, $1800\,\text{mm}^2$ calculation. If the assumed area decreases by 20% at fixed force, what happens to the mean traction magnitude? List the additional information needed to interpret local disc stress and clinical risk.

6. **Stress Components.** For $\sigma = -3\,\text{MPa}$ and $s = 1\,\text{MPa}$ in the displayed two-dimensional stress tensor, calculate its principal stresses. Compare them with $\sigma + s$. Explain why neither this tensor nor an isotropic equivalent stress is a validated annular failure law.

7. **Series Compliance.** Derive the equivalent stiffness and stored energy for five identical 12 N m/degree springs at a total 20° rotation. Repeat with ten identical springs at the same total rotation, then at the same applied 48 N m moment. State the mechanical assumptions behind the comparison.

8. **Ligament Geometry.** For the tension-only law in this chapter, let $k = 2 \times 10^7\,\text{N/m}^2$, $L_0 = 0.020\,\text{m}$, and $L = 0.022\,\text{m}$. Calculate tension, tangent length stiffness and stored energy. If $dL/d\theta = 0.010\,\text{m/rad}$, determine the signed generalized elastic moment. These are illustrative parameters.

9. **Controlled Stiffness Work.** Hold $\delta = 0.10\,\text{rad}$ fixed while increasing k from 100 to 200 N m/rad. Calculate the change in stored energy. Reconcile the result with zero joint velocity and identify the energy term omitted by treating stiffness changes as free.

10. **Signed Gravity and Spring Moments.** Use the planar convention above with $m = 10\,\text{kg}$, $\ell = 0.30\,\text{m}$, $g = 9.81\,\text{m/s}^2$, $k = 20\,\text{N m/rad}$ and $\theta = 30°$. Calculate gravity and elastic moments, then their total power for $\dot{\theta} = +1$ and $-1\,\text{rad/s}$. Explain the energy interpretation.

11. **Muscle Redundancy.** Two ideal antagonists have $r = 0.050\,\mathrm{m}$, tensions $F_+ = 300\,\mathrm{N}$ and $F_- = 100\,\mathrm{N}$, and equal compressive projection factors 0.8. Compute their net moment and combined compressive contribution. Repeat after adding 150 N to each tension. What could motion and net-moment measurements alone distinguish?

12. **Pressure and Subsystems.** A uniform 5 kPa pressure acts on a planar projected area $0.10\,\mathrm{m}^2$. Calculate that surface's force contribution. Explain why the net force on the complete closed surface is zero and why neither result alone gives a 500 N reduction in lumbar compression.

13. **Coupled Acceleration.** At an instant in a two-rotation model, take $M = \left[\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right]\,\mathrm{kg\,m}^2$, zero other generalized loads, and applied torque $\boldsymbol{\tau} = (1, 0)^T\,\mathrm{N\,m}$. Find both accelerations. Explain why a zero second torque does not imply zero second acceleration, and state how fixed contact constraints would alter the calculation.

14. **Peak Definitions and Injury Claims.** Two normalized compression histories at three common times are $(2, 6, 2)$ and $(2, 2, 6)$ body weights. Compare the peak of their mean history with the mean of their individual peaks. Explain what additional evidence would be required to turn either number into a claim about an injurious phase.

15. **A Testable Swing Comparison.** Design a study or simulation comparing two distributions of trunk motion under matched club delivery. Specify measurements, the input/activation convention, contact assumptions, uncertain parameters, independent validation outputs and the limit of the intended conclusion. Explain how you would report a predicted load reduction that disappears under another plausible muscle-recruitment model.

Chapter 24

Anatomy and Joint Modeling: Choosing the Right Idealization

> The art of modeling is choosing what to ignore.
>
> George E. P. Box

In Plain Language 24.1. Why Biology and Engineering See Joints Differently

When we look at a knee joint, the biologist sees cartilage surfaces, synovial fluid, menisci, cruciate ligaments, collateral ligaments, and a joint capsule. The engineer sees a hinge—a revolute joint with one axis of rotation. Both are looking at the same knee, but the biologist is describing what it *is* made of, and the engineer is describing what it *does*.

The fundamental challenge of biomechanical modeling is this: we cannot simulate the exact geometry of a knee. There are too many degrees of freedom, too many muscles, too many constraints we don't fully understand. So we must choose a simplified model that captures the essential motion. The art is choosing the right level of simplification—neither so crude that we miss the important physics, nor so detailed that the model becomes intractable and unidentifiable.

In robotics, this problem is solved by standardizing joint types. A robot arm doesn't have ligaments—it has *joint primitives*: revolute, prismatic, spherical, universal, cylindrical, and planar. Each primitive has a well-defined kinematic structure and constraint equation. When we build a model of the human body for simulation, we must map each biological joint onto one of these engineering primitives.

This chapter is about that mapping: what it captures, where it breaks down, and why those breakdowns matter (or don't) for golf biomechanics.

24.1 The Modeling Problem — Biology Is Messy, Engineering Is Clean

Consider a real human knee. The femur (thighbone) sits on top of the tibia (shinbone). Between them lie the medial and lateral menisci—two C-shaped cartilage pads that act as shock absorbers and stabilizers. The femur and tibia are not a perfect hinge; they are curved surfaces that slide and roll relative to each other. The motion is constrained by four major ligaments: the anterior cruciate ligament (ACL), the posterior cruciate ligament (PCL), the medial collateral ligament (MCL), and the lateral collateral ligament (LCL). There are also smaller ligaments, a joint capsule, and bursae (fluid-filled sacs). The quadriceps and hamstring muscles span the joint and control motion actively.

In a detailed finite element model, we might include:

- Femoral surface geometry (captured from MRI)
- Tibial surface geometry
- Meniscal shape and material properties (viscoelastic)
- Ligament attachment points and nonlinear force-length relations
- Cartilage as a poroelastic material
- Synovial fluid with realistic viscosity
- Muscle moment arms as functions of knee angle

Such a model would have thousands of degrees of freedom and would require months of work to build and calibrate. It would be accurate. It would also be useless for real-time simulation or optimization of the golf swing.

An engineer takes a different approach. We ask: *what is the primary motion of the knee?* The answer is almost entirely flexion and extension—bending and straightening in the sagittal plane. This suggests a *revolute joint*: a single hinge axis about which the tibia rotates relative to the femur.

Definition 24.1. Revolute Joint (Hinge)

A revolute joint connects two bodies by a single rotation axis. It has one degree of freedom: θ, the angle of rotation about the axis. Mathematically, if body A is in frame $\mathcal{F}_A$ and body B is in frame $\mathcal{F}_B$, the relative orientation is:

$$\boldsymbol{R}_{AB}(\theta) = \exp(\theta\,[\boldsymbol{u}]_\times)$$

where $\boldsymbol{u}$ is the unit vector along the rotation axis (in body A's frame) and $[\boldsymbol{u}]_\times$ is the skew-symmetric matrix representation of $\boldsymbol{u}$.

With a revolute joint model, the knee has one input: the flexion angle θ_{knee}. This is computationally tractable. We can identify the stiffness, damping, and range of motion from experimental data. We can simulate hundreds of golf swings in seconds.

The tradeoff is obvious: we lose information. The revolute model assumes the rotation axis is fixed, but real knee axes shift slightly with angle. It assumes the tibia doesn't translate relative to the femur, but real knees have some anterior-posterior laxity, especially when the ACL is torn. It assumes the motion is perfectly in the sagittal plane, but the knee can varus/valgus (bow in and out) to some degree.

Key Principle 24.1. The Modeling Art

Good biomechanical modeling is a balance between fidelity and tractability. Ask: *Does this simplification matter for the question I'm trying to answer?*
If you're studying knee stability after ACL reconstruction, the anterior-posterior laxity matters. Model a 6-DOF knee.
If you're studying the golf swing, the knee is essentially a hinge. Model a 1-DOF revolute.
Never add complexity you don't need. Never remove complexity the physics requires.

24.2 Joint Types in Robotics — A Taxonomy

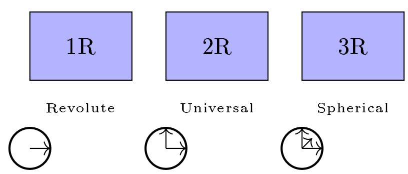


Figure 24.1: Joint Primitives: Revolute (1R) One Axis, Universal (2R) Two Axes, Spherical (3R) Three Axes.

Roboticists have standardized a small set of joint primitives. Each is characterized by:

- Number of degrees of freedom (f)
- Number of constraints imposed ($c = 6 - f$)
- Constraint Jacobian structure
- Symbolic notation

Let's enumerate them.

24.2.1 Revolute (1 DOF)

A revolute joint allows one rotation about a fixed axis. Symbol: R.

Degrees of freedom: $f = 1$. The joint variable is θ.

Constraints: $c = 5$. The joint prevents translation in all directions and rotation about the other two axes.

Constraint equation: If the axis is $\boldsymbol{u}$ in frame A, and bodies are at relative orientation $\boldsymbol{R}$, the constraints on relative position and non-axis rotations prevent unwanted motion:

$$\boldsymbol{p}_{AB} = 0 \quad \text{(position is zero at joint frame)} \tag{24.1}$$
$$\boldsymbol{R}_{AB}(:,1) \cdot \boldsymbol{u} = \boldsymbol{u} \quad \text{(rotation axis preserved)} \tag{24.2}$$
$$\boldsymbol{R}_{AB}(:,2) \cdot \boldsymbol{u} = 0 \quad \text{(perpendicular to axis)} \tag{24.3}$$
$$\boldsymbol{R}_{AB}(:,3) \cdot \boldsymbol{u} = 0 \quad \text{(perpendicular to axis)} \tag{24.4}$$

Example in golf: The knee is modeled as a revolute joint: the tibia rotates about a horizontal axis through the knee.

24.2.2 Prismatic (1 DOF)

A prismatic joint allows translation along a fixed axis. Symbol: P.

Degrees of freedom: $f = 1$. The joint variable is d, the translation distance.

Constraints: $c = 5$. All rotations are prevented, and translations perpendicular to the axis are blocked.

Constraint equation: If the axis is $\boldsymbol{u}$:

$$\boldsymbol{R}_{AB} = \boldsymbol{I} \quad \text{(no relative rotation)} \tag{24.5}$$

$$\boldsymbol{p}_{AB} \cdot \boldsymbol{u}^{\perp} = 0 \quad \text{(translation only along } \boldsymbol{u}\text{)} \tag{24.6}$$

Example in golf: Prismatic joints don't appear in the human body. They appear in golf machines—a robot that extends its arm along a rail.

24.2.3 Spherical (3 DOF)

A spherical joint (ball-and-socket) allows rotation about a point in all directions. Symbol: S or sometimes 3R.

Degrees of freedom: $f = 3$. The joint variables are three rotation angles (e.g., Euler angles ϕ, θ, ψ, or a quaternion $\boldsymbol{q}$).

Constraints: $c = 3$. The joint prevents all translation but allows free rotation.

Constraint equation:

$$\boldsymbol{p}_{AB} = 0 \quad \text{(position coincident)}$$

No constraint on $\boldsymbol{R}_{AB}$: any orientation is allowed.

Internal representation: A spherical joint is often modeled as three sequential revolute joints with orthogonal axes:

$$\boldsymbol{R}_{AB} = \mathrm{Rot}(z, \phi) \cdot \mathrm{Rot}(y, \theta) \cdot \mathrm{Rot}(x, \psi)$$

This is called an Euler angle decomposition (or ZYX convention). The order matters.

Example in golf: The hip is modeled as a spherical joint: the femoral head rotates freely in the acetabular socket.

24.2.4 Universal (2 DOF)

A universal joint (Cardan joint or Hooke's coupling) allows rotation about two perpendicular axes. Symbol: U.

Degrees of freedom: $f = 2$.

Constraints: $c = 4$. Prevents translation and rotation about one axis.

Constraint equation: If the two axes are $\boldsymbol{u}_1$ and $\boldsymbol{u}_2$ (perpendicular):

$$\boldsymbol{p}_{AB} = 0 \tag{24.7}$$

$$\boldsymbol{R}_{AB} \cdot (\boldsymbol{u}_1 \times \boldsymbol{u}_2) = \boldsymbol{u}_1 \times \boldsymbol{u}_2 \quad \text{(no rotation perpendicular to plane)} \tag{24.8}$$

Important property: Unlike a spherical joint, a universal joint can introduce *kinematic singularities*. When the two rotation axes align, the joint locks and loses controllability. This is why steering joints in cars use a universal joint with careful geometry to avoid the steering axis aligning with the input shaft axis.

Example in golf: The wrist is approximated as a universal joint: flexion/extension about one axis, radial/ulnar deviation about a perpendicular axis.

24.2.5 Cylindrical (2 DOF)

A cylindrical joint allows translation and rotation along the same axis.

Degrees of freedom: $f = 2$: translation distance d and rotation angle θ.

Constraints: $c = 4$.

Example in golf: Doesn't appear in human anatomy.

24.2.6 Planar (3 DOF)

A planar joint allows motion within a plane: two translations (in-plane) and one rotation (perpendicular to plane).

Degrees of freedom: $f = 3$.

Constraints: $c = 3$: out-of-plane translation is zero, rotation about the two in-plane axes is zero.

Example in golf: The foot-ground interface can be modeled as planar if we assume the foot stays in contact with the ground and slides.

24.2.7 6-DOF (Free, Floating)

No constraint. The two bodies move independently relative to each other.

Degrees of freedom: $f = 6$: three translations and three rotations (orientation).

Constraints: $c = 0$.

Example in golf: A club in flight between shots has a 6-DOF joint to the ground (it's not touching).

Example 24.1. Constraint Counting

A robot arm with:

- Base (1 link) fixed to ground
- Three revolute joints connecting four links

Using the full Grübler formula with $N = 5$ links (including ground) and $J = 3$ revolute joints:

$$\text{DOF} = 6(N-1) - \sum(6 - f_i) = 6 \times 4 - 3 \times 5 = 24 - 15 = 9$$

This answer (9 DOF) is incorrect for a 3-revolute chain. The discrepancy arises because the Grübler formula counts spatial DOF, but this is a **planar** mechanism with implicit constraints that eliminate out-of-plane motion.

The correct approach for a tree structure (no loops) is simply the sum of joint freedoms:

$$\text{DOF} = \sum_{i=1}^{J} f_i = 1 + 1 + 1 = 3$$

Three revolutes in series give 3 DOF. The full Grübler formula is needed only when the mechanism contains **closed loops**.

24.3 The Knee — A Revolute (Mostly)

The knee is the most obvious candidate for a revolute joint model. The primary motion is flexion and extension: the tibia bends forward and backward relative to the femur in the sagittal plane.

24.3.1 Anatomy Summary

The knee is a hinge formed between the femoral condyles (rounded knobs at the bottom of the femur) and the tibial plateau (flat-ish top of the tibia). The patella (kneecap) sits in front and slides in a groove in the femur, transmitting load from the quadriceps to the tibia.

Between the femur and tibia are two cartilage discs called menisci—the medial (inner) and lateral (outer) menisci. These crescent-shaped structures act as shock absorbers and load distributors. They are attached to the joint capsule at the periphery and can shift slightly during rotation.

Stability is provided by ligaments:

- **ACL (Anterior Cruciate Ligament)**: prevents anterior tibial translation, prevents hyperextension
- **PCL (Posterior Cruciate Ligament)**: prevents posterior tibial translation
- **MCL (Medial Collateral Ligament)**: prevents valgus (outward bowing)
- **LCL (Lateral Collateral Ligament)**: prevents varus (inward bowing)

24.3.2 Range of Motion

In a healthy knee:

- Full extension: $\theta = 0°$
- Flexion: up to $\theta \approx 140°$ (can reach $\approx 160°$ with heel-to-buttock bending)

In a golf swing, the trailing knee (right knee for a right-handed golfer) flexes significantly during the backswing (to $\approx 20-30°$) and extends during the downswing and follow-through. The leading knee (left knee) is often near full extension during the backswing and stays relatively extended during the downswing.

24.3.3 Why Revolute Works

The motion is overwhelmingly flexion-extension. If you sit on a chair and move your knee, you feel the tibia hinging forward and back. This is the revolute axis. For golf swing analysis, a 1-DOF revolute model captures the essential motion.

24.3.4 Where Revolute Breaks Down

Drift vs. Control 24.1. Revolute Approximation vs. Reality

The Revolute Model: The tibia has a fixed hinge axis. All rotation is about this axis. No translation occurs at the joint.
The Biological Reality: The knee is more complex in several ways:

1. **Screw-home mechanism:** As the knee reaches full extension, the tibia externally rotates $\approx 8 - 12°$ relative to the femur. This is an automatic locking mechanism that stabilizes the extended knee. When you extend your leg fully, your tibia rotates outward. This rotation is caused by the asymmetric shapes of the femoral condyles (the medial condyle is wider and deeper than the lateral one).

2. **Varus/valgus laxity:** In flexion (bending), the knee can adduct/abduct (twist inward/outward) by $\approx 5 - 10°$ if the ACL and collateral ligaments are intact. This is a small but real rotation perpendicular to the primary flexion axis.

3. **Anterior-posterior translation:** The tibia can slide forward/backward relative to the femur by $\approx 5 - 6$ mm when the ACL is functional. In an ACL-deficient knee, this increases dramatically.

4. **Medial-lateral translation:** Small shifts (a few mm) occur during rotation.

For golf analysis, these secondary motions are usually negligible. The screw-home mechanism introduces a small error to the revolute model (the 8–12° tibial rotation at full extension is a fraction of the total $\sim$140° flexion range). For injury risk assessment (e.g., ACL tear prediction), these details matter.

24.3.5 Modeling Recommendation

Constraint Insight 24.1. Choosing the Knee Model

For golf swing biomechanics: Use a 1-DOF revolute joint. Measure or estimate the axis orientation (approximately horizontal, slightly offset anterior-posterior from the joint center).
For injury risk analysis: Use a 6-DOF joint with nonlinear stiffness and damping constraints. Include ligament force-length relations based on experimental data. This level of detail is beyond the scope of swing analysis.
For real-time optimization: Revolute is essential. You cannot optimize over hundreds of swing variations if each variation requires a detailed 6-DOF knee model.

24.4 The Hip — A Ball-and-Socket (Almost)

The hip is a classic ball-and-socket joint. The femoral head is a nearly perfect sphere that fits into the acetabular socket of the pelvis. It allows motion in multiple directions: flexion/extension (forward/backward), abduction/adduction (sideways), and internal/external rotation.

24.4.1 Anatomy Summary

The femoral head is approximately spherical with a radius of ≈ 22 mm. The socket (acetabulum) is a hemispherical depression with a radius of ≈ 24 mm. A labrum (cartilage ring) around the rim of the socket deepens it and increases the stability. The joint is surrounded by ligaments: the iliofemoral ligament (very strong, prevents hyperextension), the pubofemoral ligament, and the ischiofemoral ligament.

24.4.2 Range of Motion

In a healthy hip:

- Flexion: $\approx 120°$
- Extension: $\approx 10 - 20°$
- Abduction: $\approx 45°$
- Adduction: $\approx 25°$
- Internal rotation: $\approx 40°$
- External rotation: $\approx 45°$

Interpreting golf measurements: Pelvis turn in the laboratory frame is not anatomical hip internal rotation. A hip rotation describes the femur relative to the pelvis; its numerical components depend on the anatomical frames, rotation sequence, sign convention and reference posture. Cheetham and colleagues computed X-factor from pelvis and upper-torso sensor orientations near transition [Cheetham et al., 2001]. That measurement does not establish a 35–50-degree lead-hip internal-rotation range or a universal requirement for professional swings.

The approximate ranges above are reference values, not fixed limits for every person or every combined hip posture. A comparison with measured swing motion requires compatible definitions, loading and assessment conditions. In a model, use subject-appropriate admissible joint configurations and check the whole motion, including pelvic orientation and foot placement. A large pelvis turn alone is insufficient to conclude that a hip is near its limit or that greater isolated rotation would improve performance.

24.4.3 Why Spherical Works

The femoral head is nearly spherical. As long as the hip motion stays within the bony geometry (no impingement), the joint behaves like a ball-and-socket with three rotational degrees of freedom. There is no fixed translation—the center of rotation is (approximately) the center of the femoral head sphere.

24.4.4 Where Spherical Breaks Down

Drift vs. Control 24.2. Spherical Approximation vs. Reality

The Spherical Model: The femoral head is a perfect sphere. It rotates about a fixed center (the geometric center of the sphere). Any orientation is achievable.
The Biological Reality:

1. **Shift of center of rotation:** The femoral head is not perfectly spherical (it's distorted by muscle attachment sites and loading). The center of rotation shifts slightly with joint angle—on the order of ≈ 5 mm, which is small but measurable.

2. **Femoroacetabular impingement (FAI):** The femur and pelvis have bony anatomy that limits motion in certain directions. When the hip is flexed and internally rotated, the femoral neck can contact the acetabular rim. This contact limits further motion and can cause pain and cartilage damage over time. FAI is one of the most common hip pathologies in athletes. Professional golfers with FAI often modify their swing to avoid impingement.

3. **Ligamentous envelope:** The hip ligaments are not equally tight in all directions. Iliofemoral ligament is very tight, limiting hyperextension. This means that not all combinations of flexion/abduction/rotation are equally accessible. The "range of motion envelope" is not a perfect ball.

4. **Muscular constraints:** The hip muscles (especially the adductors and rotators) have a preferred range. Beyond this range, they stretch and limit motion.

For the golf swing, the most important breakdown is **FAI**. A golfer with FAI cannot internally rotate the lead hip as much as a healthy golfer without impingement.

24.4.5 Modeling Recommendation

Constraint Insight 24.2. Choosing the Hip Model

For typical golfers: Use a 3-DOF spherical joint (often represented as three sequential revolutes: ZYX Euler angles). Include mechanical range-of-motion limits. Typical limits: flexion $[-20°, 120°]$, abduction $[-25°, 45°]$, internal/external rotation $[-40°, 45°]$.
For golfers with FAI: Reduce internal rotation ROM to $\approx 20-30°$ instead of $40°$. This forces the model to adopt a different swing strategy.
For competition analysis: Use 3-DOF spherical. The three angles can be recorded from motion capture and used directly in the model.

24.5 The Shoulder Complex — A Gimbal (But Much More)

The shoulder is unique: it is not a single joint. It is a *complex* of four joints working together to provide the most mobile joint system in the human body.

24.5.1 The Four Joints of the Shoulder Complex

1. **Glenohumeral (GH) joint:** The ball-and-socket joint formed by the humeral head and the glenoid fossa of the scapula. This is the primary articulation and contributes most of the motion.

2. **Acromioclavicular (AC) joint:** A small gliding joint between the acromion (top of the scapula) and the distal clavicle. It allows small rotations (a few degrees).

3. **Sternoclavicular (SC) joint:** The joint between the clavicle and the sternum (breastbone). It is the only bony attachment of the arm to the axial skeleton. It allows clavicular rotation and protraction/retraction.

4. **Scapulothoracic (ST) articulation:** This is NOT a true joint (no cartilage, no synovial membrane). The scapula glides on the surface of the ribcage. This articulation contributes significantly to shoulder motion.

24.5.2 Degrees of Freedom

- GH: 3 DOF (ball-and-socket)
- AC: ≈ 2 DOF (primarily rotation, small translation)
- SC: ≈ 2 DOF (clavicular elevation/depression, retraction/protraction)
- ST: 3 DOF (scapular upward/downward rotation, retraction/protraction, tilt)

Total: ≈ 10 DOF, but many of these are coupled. The effective DOF at the shoulder complex is ≈ 7.

24.5.3 Simple Model: GH Only

A common simplification is to model only the glenohumeral joint as a 3-DOF spherical joint and ignore the AC, SC, and ST contributions. This is valid when the scapula orientation is fixed (e.g., scapula rotates at a constant rate relative to the humerus).

24.5.4 The Problem: Scapulohumeral Rhythm

The scapula does NOT stay fixed. There is a coupling between humeral motion and scapular motion called the **scapulohumeral rhythm**. The empirical rule is:

For every 2° of glenohumeral abduction, there is approximately 1° of scapular rotation.

This means that if the humerus abducts 60° from the side of the body (raising the arm), the scapula rotates an additional 30°. The total abduction of the arm relative to the torso is 90°.

Why does this matter? The shoulder center (the location of the GH joint) is on the scapula. If the scapula rotates, the shoulder center *moves*. In a task-space control problem (e.g., "keep your hand at a fixed location"), a rotating scapula means the GH joint axis is moving in space.

For simple swing analysis where we're computing forces and moments at the joint, the scapular motion affects:

- The location of the shoulder center relative to the trunk
- The moment arms of muscles crossing the shoulder
- The available range of motion for the humerus

Example 24.2. Scapulohumeral Rhythm in the Golf Swing

During the backswing, a right-handed golfer raises the trail arm (right arm).
Without scapular contribution: The GH joint alone can abduct the humerus $\approx 90-100°$. The hand would reach a certain height and angle.
With scapulohumeral rhythm: The scapula upwardly rotates (the bottom of the scapula swings outward) as the arm is raised. This adds another $45-50°$ of combined scapular rotation, allowing the arm to reach higher and more posterior than GH alone allows.
In practice, elite golfers exploit this coupling to position their hands precisely. Ignoring scapular motion would underestimate the achievable hand positions.

24.5.5 Modeling Recommendation

Constraint Insight 24.3. Choosing the Shoulder Model

For simple swing geometry: Use a 3-DOF GH joint (spherical) at a fixed location on the trunk. The scapula rhythm is implicitly captured by the ROM limits of the GH joint (which are empirically measured and already include scapular compensation).
For detailed shoulder analysis: Use a 7-DOF shoulder complex model: 3-DOF GH (ball-and-socket), 3-DOF ST (scapular motion), and 1-DOF AC coupling. This requires measuring or estimating the scapular motion from motion capture.
For injured shoulder: If the scapular stabilizer muscles (serratus anterior, lower trapezius) are weak or damaged, the scapulohumeral rhythm is disrupted. The arm cannot achieve normal positions without compensation. Model the ST joint with reduced ROM.

24.6 The Wrist — A Universal Joint (Approximately)

The wrist is traditionally modeled as a 2-DOF universal joint: flexion/extension and radial/ulnar deviation. This is a reasonable first approximation but hides complexity.

24.6.1 Anatomy Summary

The wrist is formed by the radiocarpal joint (radius bone articulating with the carpal bones) and the midcarpal joint (carpal bones articulating with each other). There are eight carpal bones (scaphoid, lunate, triquetrum, pisiform, trapezium, trapezoid, capitate, hamate), and they don't move as a single rigid body—they slide and rotate relative to each other.

Despite this complexity, the net result is approximately a universal joint at the wrist midline.

24.6.2 Primary Motions

- Flexion (bending downward): $\approx 70 - 80°$
- Extension (bending upward): $\approx 60 - 70°$
- Radial deviation (toward thumb): $\approx 15 - 25°$
- Ulnar deviation (toward pinky): $\approx 25 - 35°$

24.6.3 The Dart-Thrower's Motion

Surprisingly, the wrist does not achieve all combinations of flexion and deviation equally. There is a preferred motion path called the **dart-thrower's motion**: a combined extension + ulnar deviation. This is the most powerful and fastest motion the wrist can perform.

From a biomechanical perspective, the dart-thrower's motion aligns with the fiber directions of the ligaments and the carpal geometry. It is the preferred movement for forceful wrist tasks.

24.6.4 Where Universal Breaks Down

Drift vs. Control 24.3. Universal Approximation vs. Reality

The Universal Model: Two perpendicular rotation axes at the wrist center. Motion is the product of rotations about these axes.
The Biological Reality:

1. **Pronation/supination confusion:** Many golfers and coaches think of "twisting the wrist" as a wrist motion. In fact, the primary twist motion is radioulnar pronation/supination: rotation of the forearm around the long axis of the ulna and radius. This is NOT a wrist joint motion—it is an elbow-region joint. The wrist cannot pronate or supinate independently (though it can assist the motion).

2. **Carpal kinematics:** The carpal bones do not rotate about fixed axes. The instantaneous axis of rotation shifts during motion, following a complex path determined by ligament constraints and bone geometry. A 2-DOF model is a simplification.

3. **Dart-thrower's motion:** The wrist does not equally prefer all (flexion, deviation) combinations. The nervous system and biomechanics prefer the dart-thrower's path. A 2-DOF planar model cannot capture this preference.

For the golf swing, these subtleties matter because the wrist cock (cocking the wrist during backswing) involves both flexion/extension and ulnar deviation. Ignoring the dart-thrower's preference means we might underestimate the achievable wrist cock angle in certain configurations.

24.6.5 Modeling Recommendation

Constraint Insight 24.4. Choosing the Wrist Model

For golf swing analysis: Use 2-DOF universal joint (flexion/extension and radial/ulnar deviation). The axes are approximately orthogonal.
For club face angle analysis: If you need to track club face rotation, note that the club shaft can rotate about its own long axis (the roll of the shaft). This is NOT a wrist motion—it is a motion of the club itself relative to the hand. Include it as a third DOF of the club, not the wrist.
For detailed wrist biomechanics: Model the radiocarpal joint separately from the midcarpal joint (both 2 DOF). Include nonlinear stiffness constraints to capture the dart-thrower's preference.

24.7 The Elbow — A Revolute (Very Good Approximation)

The elbow is one of the best candidates for a simple joint model. The primary motion—flexion and extension—is almost a pure revolute.

24.7.1 Anatomy Summary

The elbow is formed by three bones: the humerus (upper arm), the radius (forearm, thumb side), and the ulna (forearm, pinky side). There are three articulations:

1. **Humeroulnar joint:** The ulna has a U-shaped notch (trochlear notch) that wraps around the humeral trochlea (a pulley-like shape). This forms the primary hinge for flexion/extension.

2. **Humeroradial joint:** The radius sits on the humeral capitellum (a rounded knob). This articulation allows some rotation of the radius.

3. **Proximal radioulnar joint:** The head of the radius rotates within a ring formed by the radial notch of the ulna. This joint is the axis of pronation/supination (rotation of the forearm).

24.7.2 Two Separate DOF

For the purposes of the golf swing, the elbow can be modeled as **two independent revolute joints**:

- Flexion/extension: 1 DOF, centered at the humeroulnar joint
- Pronation/supination: 1 DOF, a rotation of the radius+hand about the ulnar axis

These two motions are not fully independent (they are mechanically coupled to some degree), but for golf analysis, they can be treated separately.

24.7.3 The Carrying Angle

The elbow is not a perfectly aligned hinge. The forearm makes a small angle relative to the upper arm even at full extension. This is the **carrying angle**, approximately $5-15°$ valgus (outward), larger in women than men. It means the flexion axis is slightly angled relative to the body midline.

24.7.4 Modeling Recommendation

Constraint Insight 24.5. Choosing the Elbow Model

For golf swing analysis: Use two sequential revolute joints: flexion/extension and pronation/supination. This gives 2 DOF at the elbow.
For detailed moment calculation: Account for the carrying angle when computing muscle moment arms. The axis is not perfectly sagittal.
Typical ROM limits:

- Flexion: 0° to 150°
- Extension (hyperextension): 0° to $\approx 5 - 10°$ (limited by bony geometry and ligaments)
- Pronation: $\approx 80°$
- Supination: $\approx 80°$

24.8 The Ankle/Foot — The Ground Connection

The ankle and foot are the interface between the body and the ground. For the golf swing, the feet are mostly stationary (though weight shifts during the swing), so the ankle model can be relatively simple.

24.8.1 Anatomy Summary

The ankle complex includes:

- Talocrural joint (true ankle): tibiotalar joint allowing dorsiflexion (upward) and plantarflexion (downward) in the sagittal plane
- Subtalar joint: allowing inversion (inward turn) and eversion (outward turn)
- Midtarsal joint: additional motion of the mid-foot

- Tarsometatarsal and metatarsophalangeal joints: motions of the toes

The foot has 33 joints total and can deform significantly, especially in the arch.

24.8.2 Simplified Model

For a golf swing where the feet are largely planted, a reasonable model is:

- Talocrural joint: 1 DOF (plantarflexion/dorsiflexion in the sagittal plane)
- Subtalar joint: 1 DOF (inversion/eversion in the frontal plane)

Total: 2 DOF per ankle.

24.8.3 Where This Breaks Down

The foot is not rigid. During the weight shift in the golf swing, the arch compresses and extends, storing and releasing elastic energy. For a highly detailed model that tracks power generation from the lower body, include the arch as a spring. For simpler models, assume the foot is rigid.

Constraint Insight 24.6. Choosing the Ankle/Foot Model

For weight shift analysis: Use 2-DOF ankle (plantarflexion and inversion/eversion) with the foot assumed rigid.
For ground reaction force analysis: Model the foot-ground interface as a contact problem. The foot can slide, stick, or take off from the ground. This is a constraint handled by the dynamics solver, not a joint model.
For energetics analysis: Include the arch as a spring (torsional spring about the mid-foot axis) to capture elastic energy storage.

24.9 Summary Table and Modeling Guidelines

Joint	Model	DOF	Golf Significance
Hip	Spherical (3R)	3	Lead hip internal rotation critical
Knee	Revolute	1	Flexion dominant, screw-home negligible
Ankle	Revolute + 1R	2	Weight shift, mostly fixed during swing
Shoulder	Spherical (3R)	3	Trail shoulder critical, scapular rhythm important
Elbow	1R + 1R	2	Flexion and pronation independent
Wrist	Universal (2R)	2	Cock and bow motion

Note on ROM values: The range-of-motion values cited throughout this chapter (knee flexion $\sim 140°$, hip flexion $\sim 120°$, etc.) are representative values from standard anatomy and biomechanics references [Neumann, 2017, Nordin and Frankel, 2012]. Individual variation is substantial; values for a specific subject should be obtained from clinical measurement or motion capture data.

24.9.1 Decision Flowchart

When choosing a joint model, ask:

1. **Is the motion primarily uniaxial?** (Motion about one axis dominates.)
   - **Yes:** Use revolute (1 DOF).
   - **No:** Proceed to question 2.
2. **Are there two roughly orthogonal rotation axes?**
   - **Yes:** Use universal (2 DOF).
   - **No:** Proceed to question 3.
3. **Is rotation about a fixed point with no translation?**
   - **Yes:** Use spherical (3 DOF).
   - **No:** Use 6-DOF or a more complex model.

24.9.2 Practical Guideline

The golf swing model typically uses:

- Hip: 3R (spherical)
- Knee: 1R (revolute)
- Ankle: 1R + 1R (two perpendicular revolutes)
- Shoulder: 3R (spherical at GH)
- Elbow: 1R + 1R (flexion + pronation)
- Wrist: 1R + 1R (flexion + deviation)
- Club: Rigidly attached to hand (0 DOF)

Total: $3 + 1 + 2 + 3 + 2 + 2 + 0 = 13$ DOF per arm, times 2 arms plus pelvis/spine = ≈ 30 DOF for a full-body golf swing model.

Key Principle 24.2. Key Takeaways: Choosing Joint Models

The art of biomechanical modeling is choosing the simplest joint type that captures the essential physics for your specific question.
For kinematics: Joint model choice affects the number of DOF and the constraint equations. Too simple, and you lose important motion. Too complex, and the model is unidentifiable.
For dynamics: Joint model choice affects moment arms, inertia properties, and muscle effectiveness.
For optimization: Use the simplest model that captures the essential trade-off. A golf swing optimizer with 30 DOF is already challenging; do not add DOF you don't need.

Model hierarchy: Build your model in layers. Start with revolute joints everywhere. Only add complexity (spherical, universal) where the simple model fails to capture observed motion. Only add constraint modeling (ligaments, ROM limits) if the physics requires it.
Remember: *All models are wrong, but some are useful.* — George E. P. Box.

24.10 Injury Biomechanics of the Golf Swing

The golf swing subjects the musculoskeletal system to extreme loads in a highly asymmetric pattern. Understanding injury mechanisms requires the same joint models developed throughout this chapter, now applied to failure analysis.

24.10.1 Low Back Injury

Low back pain is the most common golf injury, affecting 25–35% of amateur golfers and 22–25% of professionals [McHardy et al., 2006, Cabri et al., 2009]. The mechanism is primarily **combined loading**: the lumbar spine simultaneously experiences compression (from axial loads due to ground reaction forces transmitted through the trunk), shear (from rotational accelerations), and torsion (from the X-factor separation described in Chapter 26).

Peak lumbar compression occurs during the downswing transition, when the hips reverse direction while the torso continues rotating backward. Hosea et al. (1990) measured compressive loads of 6–8 times body weight in professional golfers at this instant [Hosea et al., 1990]—comparable to loads measured during competitive rowing or heavy deadlifting. Note that this estimate comes from a single study with a small sample size; subsequent work has generally confirmed high spinal loads in the golf swing but with varying magnitudes depending on methodology.

The injury risk increases when:

- Lumbar flexion exceeds 40–50° at address (increases disc stress by 300% [Adams et al., 2002, McGill, 2007])
- Axial rotation exceeds 5° per lumbar motion segment (facet joint overload)
- The golfer "reverse pivots," shifting weight toward the lead foot during the backswing (creates shear in the opposite direction to muscle preparation)
- Swing frequency is high (fatigue accumulation over a practice session)

Constraint Insight 24.7. Spine Loading as a Constraint Problem

The pelvis–spine–thorax connection is an articulated path, not by itself a closed kinematic loop (Chapter 9). Spinal loads depend on segment motion, inertial properties, muscle forces, passive tissues, and the surrounding contact conditions. Different muscle co-contraction patterns can produce different internal loads even with similar observed motion. Intersegment rotation alone does not determine compression, shear, stored elastic energy, or injury risk; those require a specified model and appropriate measurements. Increasing stiffness or bracing is therefore not a universal load-reduction rule.

24.10.2 Wrist and Hand Injuries

The lead wrist experiences the highest angular velocities of any joint in the swing ($\sim$50–70 rad/s at impact). Combined with the impact shock (3000–5000 N over 0.5 ms from Chapter 17), the wrist is vulnerable to:

- **Hamate fracture**: The hook of the hamate bone sits directly under the butt end of the club. Repeated impact shock can cause stress fractures. This is the most common wrist fracture in golf.
- **Extensor tendinopathy**: The wrist extensors decelerate wrist flexion after impact. At the angular velocities involved, the eccentric loads on these tendons are substantial.
- **TFCC tears**: The triangular fibrocartilage complex stabilizes the distal radioulnar joint. Forced ulnar deviation (as occurs in the lead wrist at impact) can tear this structure.

24.10.3 Medial Epicondylitis (Golfer's Elbow)

Despite its name, golfer's elbow is caused by the trail arm, not the lead arm. The trail wrist flexors and forearm pronators originate at the medial epicondyle of the humerus. During the downswing, these muscles contract eccentrically (lengthening under load) as the trail forearm supinates. The repeated eccentric loading causes micro-tears at the tendon-bone interface.

Key Principle 24.3. Injury as a Constraint Violation

From the ZTCF family perspective, injuries occur when the drift field drives the system into configurations that exceed the structural capacity of biological tissues. The golfer's muscular control authority is insufficient to prevent these configurations at high speeds (because DCR $\gg 1$ at impact). This is why injuries are more common at high swing speeds and during the downswing-to-impact phase: the system is on autopilot, and the autopilot sometimes drives through dangerous territory.

Exercises 24.1. Chapter Exercises

1. **Knee Analysis:** The screw-home mechanism causes the tibia to externally rotate 10° as the knee moves from 90° flexion to full extension. If your revolute model assumes a fixed axis, what error does this introduce in the position of the ankle (foot) as the knee extends? Assume the tibia is 40 cm long.
2. **Hip FAI:** A golfer with femoroacetabular impingement can internally rotate the lead hip only 25° instead of the normal 40°. How does this constrain the backswing? What compensatory motion might the golfer make in the spine or pelvis?
3. **Shoulder Scapular Rhythm:** During a backswing, the trail shoulder abducts 70° (humerus relative to trunk). Using scapulohumeral rhythm (2:1 ratio), what is the GH joint abduction angle, and what is the scapular upward rotation angle?

4. **Wrist Dart-Thrower's Motion:** Explain why the dart-thrower's motion (extension + ulnar deviation) is stronger than extension alone or ulnar deviation alone. What role do ligaments play in defining this preferred path?

5. **Elbow Carrying Angle:** If the carrying angle is $10°$ valgus, what does this imply for the flexion axis direction? How would you measure the carrying angle from motion capture data (marker positions)?

6. **Grübler for a Golf Body:** Construct a golf swing model with: pelvis (base), 3-DOF lumbar spine, 3-DOF thorax, 2 arms with 3-DOF shoulders, 1-DOF elbows, 2-DOF wrists, and a rigidly-attached club. Count: bodies (N), joints (J), DOF per joint, and compute the total DOF. Then subtract the loop constraint of the two-handed grip.

7. **Model Refinement:** You've built a golf swing model and found that it predicts the backswing configuration well, but the early downswing is off (wrist starts uncocking too early compared to video). What joint model refinement might explain this? (Hint: consider wrist stiffness or scapular motion.)

Chapter 25

Degrees of Freedom and Robot Models of the Human Body

How many independent numbers describe a golf swing? The answer depends on what the model retains. Joint angles describe configuration; velocities describe how that configuration is changing; muscle activation and elastic deformation may require additional states. Two hands gripping one club and two feet contacting the ground connect these descriptions through constraints and forces. Counting the entries in a motion-capture file does not settle any of these questions.

This chapter develops a consistent route from anatomy to computation. We first count admissible motions, then ask which of them change a declared club task, then derive how applied torques affect that task. Finally, we encode a physical mechanism and check the imported dynamics against independent mechanics. Each step answers a different question. Their connection is the foundation for interpreting a swing, designing an experiment and testing a proposed control explanation.

A rigid-body model uses the same mechanical laws for a robot and a golfer, but this does not make its joints, actuators or objectives biologically complete. Our upper-body teaching model has 20 internal freedoms before the two-hand loop is closed. That is a declared approximation, not a universal anatomical count. The executable example is a complete two-link mechanism whose limitations are explicit; extending it to a golfer requires the additional modeling decisions developed below.

25.1 Counting Degrees of Freedom — The Robotics Way

25.1.1 The Formula and Its Rank Assumption

For a spatial assembly containing N bodies including one fixed reference body, begin with $6(N-1)$ unconstrained freedoms. A joint permitting f_i relative freedoms nominally imposes $6 - f_i$ constraints. If those constraints are independent at a regular configuration, the mobility is

$$d = 6(N-1) - \sum_{i=1}^{J}(6 - f_i). \tag{25.1}$$

This is the spatial Grübler–Kutzbach count. The independence assumption is essential. A negative result signals an inappropriate or dependent-constraint count; it is not a physical negative mobility. Special geometric arrangements and closed loops require a rank calculation. The Modern Robotics mobility lesson illustrates precisely this limitation.

With generalized velocities $v \in \mathbb{R}^{n_v}$ before additional closures, let $A(q)v = 0$ describe independent, stationary constraints. At a regular configuration,

$$d = n_v - \operatorname{rank} A(q). \tag{25.2}$$

For holonomic constraints $\phi(q) = 0$ in local coordinates with $\dot{q} = v$, $A = D\phi$. With quaternion coordinates, $\dot{q} = E(q)v$ and $A = D\phi\, E$. Do not subtract the quaternion normalization constraint a second time from n_v: it has already been accounted for in the velocity representation.

25.1.2 Bodies, Coordinates and Reference Frames

N freely floating bodies have $6N$ absolute freedoms relative to an external world frame. Choosing one body as a reference describes only their relative configurations, with $6(N-1)$ freedoms. Physically fixing that body removes its world motion. These are different experimental assumptions even though the relative count is the same.

A free rigid body can be stored as three translation entries and a unit quaternion: seven configuration entries, six independent velocity components. Its dynamical state then uses seven plus six stored entries, subject to quaternion normalization. Likewise, three Euler angles locally parameterize orientation but their rates are not generally the Cartesian angular-velocity components. The number of coordinates, the dimension of configuration and the number of independent controls need not agree.

25.1.3 Worked Example 1: A Serial Robot Arm

A fixed base plus three moving links gives $N = 4$. Three revolute joints nominally impose 15 spatial constraints, leaving $18 - 15 = 3$ freedoms. More directly, for a tree without additional coupling constraints, sum the joint freedoms. Three revolutes give three.

Whether the endpoint can realize three independently prescribed planar task components is a separate question. A planar position-and-orientation task has three components, but its Jacobian may lose rank at particular configurations. A task dimension cannot prove the mechanism's mobility or guarantee reachability.

25.1.4 Worked Example 2: A Closed Loop

For a planar four-bar mechanism, there are four links including ground and four revolute joints. Each unconstrained planar body has three freedoms and each planar revolute removes two. The nominal mobility is $3(4-1) - 4(2) = 1$. Away from singularities, and within a chosen feasible assembly branch, one input parameter determines the others locally.

The spatial expression $6(4-1) - 4(5) = -2$ fails because the spatial constraints are dependent in this special planar arrangement. The planar count avoids that overcount, but it too needs regularity. At a toggle configuration, first-order constraint rank may change. Extra infinitesimal motions there need not extend to independent finite motions: second-order closure can restrict them. Inspect the actual constraint equations instead of applying the regular-manifold theorem at a singular point.

25.2 Worked Example — The Golf Swing Model

25.2.1 Model Definition

Use a pelvis fixed in position and orientation. Attach a lumbar body through an ideal three-DOF spherical joint, then a thorax body through another. Attach two identical seven-DOF arm reductions to the thorax. Each arm has shoulder orientation (3), elbow flexion (1), forearm pronation/supination (1), and two wrist angles (2). Fingers, scapular motion independent of the shoulder model, distributed spinal deformation and club flexibility are omitted. Pronation is a relative rotation between forearm bones in anatomy; assigning it to a compound elbow/forearm joint is a reduction, not a claim that elbow flexion and pronation share one anatomical axis.

The physical segment tree is shown schematically below. Joint names belong on the connections; they are not additional massive anatomical segments.

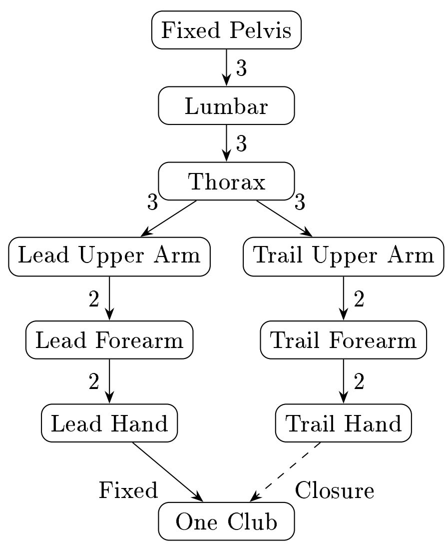


Figure 25.1: Upper-Body Tree and Two-Hand Closure. Numbers Indicate Compound-Joint Freedoms; the Dashed Connection Closes the Second Grip.

25.2.2 A Reconciled Upper-Body Count

The nine bodies are pelvis, lumbar body, thorax, and three segments in each arm. Treat the fixed pelvis as the reference body: $N = 9$. There are four three-DOF joints and four two-DOF joints. Their sum is 20 and their total nominal constraint count is $48 - 20 = 28$. Thus $6(9 - 1) - 28 = 20$. Alternatively, introduce a separate ground body and one fixed pelvis-to-ground joint: both the extra body's six freedoms and the extra joint's six constraints cancel.

This count is for both arms, not for one arm plus both legs. A serial representation of the compound joints introduces virtual intermediate links. It changes the XML link and joint counts together without creating extra independent anatomical motion. Adding an independent pelvis yaw would add one freedom; changing to a fully floating pelvis would add six. Neither change is consistent with still calling the pelvis fixed.

25.2.3 The Grip Constraint and the Club Body

First rigidly attach the club to the lead hand. A rigid attachment adds a body and a fixed joint but no relative freedom. Requiring the trail-hand frame to match its prescribed grip frame on that club then adds a loop-closure condition. A full rigid relative pose supplies up to six independent scalar constraints. For a feasible configuration with closure rank six, the mobility becomes $20 - 6 = 14$.

The same result follows by initially making the club independent. A rigid club adds six freedoms once, giving $20+6 = 26$. Two rigid hand-to-club attachments nominally add twelve constraints. If their combined Jacobian has rank twelve, the result is $26 - 12 = 14$. These are alternative formulations of the same model; subtracting both counts in one formulation would count closure twice.

Finite-stiffness grips are different. Two compliant attachments exert forces and moments but do not each create a new independent club body. If both permit finite relative displacement, the rigid-segment configuration still has $20 + 6 = 26$ freedoms before any other exact constraints. Their force laws determine how those freedoms evolve. In a very stiff limit they may approximate rigid closure, but numerical stiffness, stored energy and contact reactions require separate analysis. Internal grip states or deformable shaft modes add further states only when explicitly modeled.

A point-coincidence grip supplies up to three constraints, a full rigid grip up to six, and a slipping or rolling contact has its own permitted directions. The rank of the combined Jacobian decides the count. Neither the word "grip" nor the number of hands specifies a unique constraint model.

25.2.4 Adding Legs and Ground Contact

For each leg choose hip (3), knee (1) and ankle (2): six freedoms. Adding both legs to our fixed-pelvis upper-body tree gives $20 + 12 = 32$ internal freedoms. Releasing the pelvis to float gives 38. An earlier, different reduction using pelvis orientation (3), one spine joint (3), two legs (12) and two arms (14) also sums to 32; equal arithmetic totals do not make these the same model.

For illustration, give the floating 38-DOF body one independent rigid club, then impose two rigid feet and two rigid grips. If the combined 24 closure rows are independent and geometrically compatible, mobility is $38 + 6 - 24 = 20$. This is a local count for an ideal contact mode. It says nothing about whether the required ground wrenches lie inside friction and support limits, whether a foot should lift, or whether the golfer can supply the required joint torque.

Stack foot and hand constraints before evaluating their rank. When a heel rises or a foot slips, the contact mode and constraint matrix change. Position continuity alone does not settle velocity jumps at impact. A fixed-pelvis upper-body simulation, a floating-body simulation with ground contact, and a prescribed-pelvis simulation answer different questions about how ground interaction influences the club.

25.3 The Human Arm — A Seven-DOF Redundant Manipulator

25.3.1 DOF Count and the Selected Task

The seven-DOF arm is useful because it separates joint motion from a hand task. A hand position lies in $\mathbb{R}^3$. A full rigid hand pose lies in $SE(3)$, a six-dimensional manifold, rather than globally in a single $\mathbb{R}^6$ coordinate chart. Locally one can differentiate task coordinates or use an appropriate spatial/body twist, keeping the frame and angular/linear ordering consistent.

For a seven-DOF arm with a feasible, regular full-pose task of rank six, the instantaneous task-null space has dimension one. For a regular position-only task of rank three it has dimension four. These dimensions assume the shoulder base is fixed and no additional loop or anatomical constraints remove motion. A singular or mechanically restricted arm can have a lower task rank.

More detailed anatomy changes the chosen configuration. For example, Holzbaur et al. [2005] include 15 freedoms spanning shoulder, elbow, forearm, wrist, thumb and index finger, together with 50 muscle compartments. That model is evidence of an explicit modeling choice, not justification for assigning a universal number of freedoms to every golfer or every coaching description.

25.3.2 The Null-Space Motion: Elbow Swivel

Hold the shoulder center S and wrist center W fixed. Idealize upper arm and forearm as rigid segments with lengths a and b. Let $D = \|W - S\|$, with $|a - b| < D < a + b$. The elbow center P satisfies $\|P - S\| = a$ and $\|P - W\| = b$. The two spheres intersect in a circle. With $e = (W - S)/D$ and orthonormal e_1, e_2 perpendicular to e, define

$$\begin{aligned} c &= \frac{a^2 - b^2 + D^2}{2D}, \qquad\qquad r = \sqrt{a^2 - c^2}, \\ P(\psi) &= S + ce + r(e_1 \cos\psi + e_2 \sin\psi). \end{aligned} \tag{25.3}$$

This describes a workspace circle of possible elbow centers, not necessarily a globally closed curve in joint coordinates. If elbow flexion $\beta = 0$ denotes a straight arm, the cosine law gives $\cos\beta = (D^2 - a^2 - b^2)/(2ab)$. Fixed shoulder-to-wrist distance fixes the flexion angle in this ideal geometry. Swivel therefore requires coordinated rotations; it is not isolated elbow flexion.

Holding hand orientation as well as wrist position requires the shoulder, forearm and wrist rotations to compensate compatibly. Anatomical limits and collisions can truncate the circle to arcs or disconnected feasible pieces. At a straight or fully folded limit the circle degenerates. If $D = 0$ and $a = b$, the above formula is undefined and the sphere-intersection geometry is different. These exceptions are part of the model, not small numerical inconveniences.

25.3.3 Why Redundancy Matters for Golf

At a particular club pose, different feasible internal configurations can give different mass matrices, moment arms, joint loads, collision clearances and future reachable motions. This is a mechanical reason to study internal configuration rather than judging a swing only by the club's appearance. It does not establish that every configuration is equally useful or that every golfer can realize it.

Two arms gripping the same club cannot generally swivel independently while holding the torso and both grip frames fixed. The combined closure and task Jacobians decide which coordinated variations remain. If a model predicts alternatives, their dynamic feasibility must still be tested with torque, contact, activation and speed constraints over the relevant time interval.

25.3.4 The Pseudoinverse and Null-Space Projection

In local joint coordinates let $\dot{y} = J\dot{q}$. For a wide, full-row-rank J, its Euclidean Moore–Penrose inverse is $J^+ = J^T(JJ^T)^{-1}$. The expression $(J^TJ)^{-1}J^T$ instead requires full column rank and is invalid for this redundant case. In general use an SVD, with an explicit tolerance and physically meaningful coordinate scaling.

If $\dot{y}_d$ lies in the range of J, every exact velocity solution has the form

$$\dot{q} = J^+\dot{y}_d + (I - J^+J)w. \tag{25.4}$$

Here w has joint-velocity units. The first term has minimum Euclidean norm in the chosen coordinates, not automatically minimum energy, muscle activation or joint load. If the target is outside the range, $JJ^+\dot{y}_d$ is a projection and the remaining task residual is nonzero. A damped inverse bounds amplification near rank loss by accepting task error; its nominal "null" projector generally leaks into the task.

25.4 Kinematic Redundancy and the Manifold of Solutions

25.4.1 The Redundancy Problem

Let $\mathcal{Q}_c$ be the admissible configuration manifold after exact closures and let F map it to the declared task:

$$F : \mathcal{Q}_c \longrightarrow \mathcal{Y}, \qquad y = F(q). \tag{25.5}$$

For club pose, $\mathcal{Y} = SE(3)$; for a selected point's position, $\mathcal{Y} = \mathbb{R}^3$. A speed objective, impact event or target ball flight adds further structure. In particular, "square face with maximum speed" is not fully specified by the six club-pose coordinates. Speed belongs to velocity state and an optimization objective; impact also depends on time and collision conditions.

With $\dim \mathcal{Q}_c = d$ and task differential rank r at a regular point, local redundancy is

$$d_{\text{null}} = d - r. \tag{25.6}$$

Thus the fixed-pelvis, rigid-two-grip upper-body model has $d = 14$ under our rank assumptions. A further regular six-dimensional club-pose task would leave eight local configuration directions. Using the original open-tree count $20 - 6 = 14$ for that same closed model would omit the grip closure.

25.4.2 The Task Jacobian on Admissible Velocities

Choose a full-column-rank basis $Z(q)$ for $\ker A(q)$, so every admissible velocity can be written $v = Z\nu$. The task velocity is

$$\dot{y} = J(q)v = J(q)Z(q)\nu. \tag{25.7}$$

The relevant rank is that of JZ, not the unconstrained J. The physical task-null velocities satisfy both conditions:

$$\mathcal{N}_q = \{v : A(q)v = 0,\ J(q)v = 0\}. \tag{25.8}$$

Equivalently their dimension is $n_v - \operatorname{rank}[A^T\ J^T]^T$. The choice of basis Z changes the numerical representation but not this physical subspace. Choosing a metric is additional: it determines what counts as a small velocity, a costly torque or a statistically large deviation.

25.4.3 The Solution Manifold

For a prescribed attainable target y_0, define the level set

$$\mathcal{S}(y_0) = \{q \in \mathcal{Q}_c : F(q) = y_0\}. \tag{25.9}$$

If F has constant rank r in a neighborhood of the point, the constant-rank theorem gives a local level-set manifold of dimension $d - r$. This is a local statement. It neither proves the target is reachable nor makes disconnected inverse-kinematics branches one smooth global surface. Joint limits add boundaries and corners; singular points need separate treatment.

An instantaneous null vector preserves the task to first order. For a finite step $q + \epsilon v$, the second-order term in the Taylor expansion can change F. To maintain the task during finite motion, update the null space as the configuration changes and solve or stabilize closure throughout the integration. The curvature of the level set explains why a straight step tangent to it can leave it.

25.4.4 Different Golfers, Different Paths

Equal impact pose does not imply equal impact velocity, equal timing or equal trajectories. Even equal pose and velocity of a chosen club frame do not by themselves establish equal ball flight without matching strike location, ball state, surface interaction and the collision model. Conversely, matching a complete club trajectory can still leave internal motion unobserved.

The existence of this redundancy is a mathematical property of the chosen map. Showing that golfers organize variability along it requires data. Myers et al. [2008] measured torso/pelvis kinematics and ball velocity in 100 recreational golfers; those associations do not establish equality of club trajectories or identify null-space torques. Glazier [2011] discuss competing interpretations and methods for movement variability. Neither source supplies a universal count of freedoms "used" by novices and experts.

25.4.5 Bernstein's Degrees of Freedom Problem

The coordination question is how an organism produces reliable task behavior with many available internal variables. A mathematical redundancy creates room for coordination; it does not identify the nervous system's algorithm. Reduced variation in some joint angles can reflect a chosen strategy, physical limits, an imposed task, measurement noise, or correlations among coordinates.

Likewise, the number of principal components explaining a dataset's variance is not the number of anatomical freedoms. It depends on normalization, sampled movements, noise and the retained variance threshold. A learned low-dimensional controller can command a mechanism whose configuration remains high-dimensional. Any claim that skill "releases" a particular number of freedoms must define and measure that number rather than substitute a suggestive metaphor.

25.4.6 Self-Organization, Compensation and Evidence

Constraints and feedback can organize coordinated motion without a controller explicitly solving every equation in this chapter. Optimization is one candidate description, alongside other control models. Its success in a simulation is a prediction under assumed dynamics, information and costs, not direct evidence that the brain computes that optimization.

Restricted motion in one joint can change the feasible set and the effort needed elsewhere. Whether an alternative movement preserves a particular golf outcome is an empirical question involving the individual, task and available control. Kinematic redundancy alone does not establish that compensation succeeds, reduces tissue load or prevents injury. A useful technical account reports the predicted alternatives and the measurements that could distinguish them.

25.5 The Seven-DOF Arm — Redundancy in Miniature

25.5.1 A Concrete Elbow-Circle Calculation

Take $a = 0.30$ m, $b = 0.25$ m and $D = 0.40$ m. The circle center lies $c = 0.234375$ m from the shoulder along the shoulder-to-wrist axis and its radius is approximately 0.18727 m. Sampling the circle preserves both segment lengths; the elbow flexion is constant because $\cos\beta = 0.05$. The wrist orientation still requires compatible compensating rotations. These are dimensions of a teaching geometry, not a participant's measurements.

Adding a prescribed elbow height can remove one additional local degree of redundancy if its differential is independent on the remaining level set. For a full-pose, rank-six arm, that can leave isolated solutions locally. It need not select a unique global configuration: a horizontal plane can intersect the elbow circle at two points. At a tangent intersection or when the entire circle has that height, the rank argument changes. With a position-only task, the four-dimensional task-null space is not exhausted by describing one elbow circle.

25.5.2 Application to the Golf Swing

Suppose two feasible address configurations place both grips and the club identically. Their future behavior under the same torque history need not match: inertia and drift depend on configuration, and contact reactions may differ. If one configuration permits a better future speed/accuracy tradeoff, the reason must be demonstrated through the dynamics and the stated objective.

This also clarifies a common verbal mistake. Once a particular club velocity is prescribed, a task-null velocity cannot increase that same velocity at that instant. It can change posture while preserving the prescribed velocity, and that posture may affect subsequent acceleration or feasible control. "Preserve the present task while improving a future option" is a precise claim that can be investigated.

25.6 Redundancy and Null-Space Control

25.6.1 From Joint Velocities to Applied Torques

For the moment use unconstrained local coordinates with $v = \dot{q}$, symmetric positive-definite mass matrix $M(q)$ and generalized dynamics

$$M(q)\dot{v} + h(q, v) = B(q)u. \tag{25.10}$$

h collects the specified inertial, gravitational and passive terms on the left. B maps the chosen inputs into generalized forces. With independent joint torques, $B = I$. With muscle forces, B contains moment-arm information; with neural commands, activation dynamics may prevent the command from acting directly in this acceleration equation.

For $x = (q, v)$ the torque-driven state equation is

$$\dot{x} = f(x) + G(x)u, \qquad G = \begin{bmatrix} 0 \\ M^{-1}B \end{bmatrix}. \tag{25.11}$$

If $B = I$, G is injective: no nonzero joint torque disappears from the full-state derivative. Task redundancy therefore does not imply a nontrivial kernel of the full-state actuation map. A zero-torque counterfactual also retains the terms assigned to h and any retained activation or elastic states according to its declared intervention. Removing the specified torque commands is not synonymous with removing every biological or stored-energy contribution.

25.6.2 The Acceleration-Level Task Map

Differentiating $\dot{y} = Jv$ gives

$$\ddot{y} = JM^{-1}Bu - JM^{-1}h + \dot{J}v. \tag{25.12}$$

At the same configuration and velocity, changing the input by δu changes task acceleration by $JM^{-1}B\delta u$. The input-null condition is therefore $JM^{-1}B\delta u = 0$. It differs from the velocity-null condition $J\delta v = 0$. Dropping $\dot{J}v$ or gravity would also produce an incorrect absolute acceleration even if the input difference were computed correctly.

For an elementary counterexample, take $M = \mathrm{diag}(2, 1)$ and $J = [1\ 1]$ in consistent chosen units. The velocity $(1, -1)^T$ lies in $\ker J$. Applying the same components as a generalized torque gives $JM^{-1}(1, -1)^T = -1/2$, so the task accelerates. This calculation isolates the mass-matrix distinction without appealing to a complicated human simulation.

25.6.3 A Dynamically Consistent Projection

When J has full row rank, define

$$\begin{aligned} \Lambda &= (JM^{-1}J^T)^{-1}, \\ \bar{J} &= M^{-1}J^T\Lambda, \qquad N = I - \bar{J}J. \end{aligned} \tag{25.13}$$

Then $J\bar{J} = I$ and $JN = 0$. The velocity $\bar{J}\dot{y}$ minimizes instantaneous kinetic energy among velocities realizing $\dot{y}$, under this unconstrained model. The corresponding torque decomposition uses the transpose projector:

$$\tau = J^T F + N^T \tau_0, \qquad JM^{-1}N^T = 0. \tag{25.14}$$

Indeed $JM^{-1}N^T = JM^{-1} - JM^{-1}J^T\bar{J}^T = 0$ because $\bar{J}^T = \Lambda JM^{-1}$. This is instantaneous dynamic consistency, as developed in Khatib's operational-space analysis. F still must account for drift and the desired task acceleration. A projection does not automatically construct the complete tracking controller.

For the numerical example above, $\bar{J} = (1/3, 2/3)^T$. N maps velocities into $\ker J$, while N^T maps torques into $\ker(JM^{-1})$. Interchanging them would erase the distinction the derivation was meant to capture. Neither null torque nor null velocity implies zero internal energy change. Power remains $\tau^T v$, and a null input can alter subsequent state, limits and task sensitivity.

25.6.4 Contact, Underactuation and Internal Grip Forces

For stationary ideal constraints use

$$M\dot{v} + h = Bu + A^T\lambda, \qquad A\dot{v} + \dot{A}v = 0. \tag{25.15}$$

Assume A has independent rows. Eliminating the constraint reaction gives

$$\begin{aligned} K &= AM^{-1}A^T, \\ W &= M^{-1} - M^{-1}A^TK^{-1}AM^{-1}, \\ \dot{v} &= W(Bu - h) - M^{-1}A^TK^{-1}\dot{A}v. \end{aligned} \tag{25.16}$$

Here $AW = 0$: W maps generalized force changes into admissible acceleration changes. The task input map becomes JWB. Redundant constraints require removing dependent rows or a justified generalized inverse; unilateral/frictional contact also requires checking whether the assumed mode remains valid. A floating body generally lacks direct actuation of its base coordinates. Consequently an arbitrary generalized null torque need not be implementable by its joint actuators.

Inputs that leave all admissible accelerations unchanged may nevertheless change constraint reactions. With two hands, opposing forces can cancel in the net club wrench while changing the loads at each grip. This force redundancy is not the same as an elbow-swivel velocity. The multicontact analysis of Sentis, Park and Khatib connects support constraints, task control and internal forces under explicit actuation/contact assumptions. It does not provide evidence for a particular golfer's neural strategy.

Ideal stationary constraints do no net work on admissible motion because $(A^T\lambda)^T v = \lambda^T Av = 0$. Individual bodies can exchange power through their common constraint forces. Compliant grips can store or dissipate energy, and prescribed moving constraints can supply power. Distinguishing system boundary and constraint law is necessary before interpreting an internal force as an energy source.

25.6.5 Null-Space Optimization and Its Limits

For an unconstrained kinematic task with an Euclidean projector $P = I - J^+J$, one can choose $w = -k\nabla H(q)$ to favor a posture cost H, with $k > 0$. The null contribution to $\dot{H}$ is $-k\nabla H^T P \nabla H \leq 0$. The task-producing term $J^+\dot{y}_d$ can still increase H, so the full cost need not decrease. Joint limits, non-Euclidean metrics and closed loops require the corresponding constrained formulation.

A torque-level objective requires the acceleration dynamics, not substitution of the same w into a torque port. Force limits, muscle activation dynamics, grip feasibility and future tasks can defeat a locally attractive direction. For golf, a credible optimization specifies the horizon, impact event, observations, controls and tradeoff between speed and accuracy; then reports feasibility and sensitivity, not only the optimizer's preferred posture.

25.7 URDF — The Unified Robot Description Format

25.7.1 Basic Structure and Frames

URDF describes a tree of links connected by joints. Every non-root link has one parent joint; the root has none. Each joint specifies a parent, child, origin transform and joint type. A conventional revolute joint axis is expressed in its joint frame. The zero-position transform and rotation sequence are part of the definition: changing them while retaining the same angle values changes the physical pose. Consult the URDF model API for the underlying tree and joint representation, and the selected importer's documentation for supported features.

A root link called "world", "ground" or "pelvis" does not, by its name, specify every simulator's base behavior. A parser may fix it or add a floating joint according to its conventions and setup. Verify the imported joint types, n_q, n_v and world attachment. Likewise, a displayed mesh is not evidence that the inertia or collision model matches it.

25.7.2 A Complete Two-Link Example

The following complete tree has a base and two uniform solid cylinders of lengths $(0.30, 0.25)$ m, masses $(2, 1)$ kg and radius 0.02 m. Their long axes are local x; the joints rotate about local z. Their COMs lie halfway along each cylinder. URDF cylinders are aligned with their geometry-frame z axis, so the visual and collision frames are rotated by $\pi/2$ about y. The inertial frames remain aligned with the link frames, with the small longitudinal moment in the xx entry.

Continuous joints are used for this unrestricted teaching mechanism. These are not physiological range-of-motion or torque-limit claims. Gravity, actuation and the time integrator are configured by the simulator separately.

```
<robot name="teaching_arm">
  <link name="base" />
  <link name="link1">
    <inertial>
      <origin xyz="0.15 0 0" rpy="0 0 0" />
      <mass value="2" />
      <inertia ixx="0.0004" iyy="0.0152" izz="0.0152" ixy="0" ixz="0" iyz="0" />
    </inertial>
    <visual>
      <origin xyz="0.15 0 0" rpy="0 1.57079632679 0" />
      <geometry>
        <cylinder radius="0.02" length="0.3" />
      </geometry>
    </visual>
    <collision>
      <origin xyz="0.15 0 0" rpy="0 1.57079632679 0" />
      <geometry>
        <cylinder radius="0.02" length="0.3" />
      </geometry>
    </collision>
  </link>
```

```
  <joint name="joint1" type="continuous">
    <parent link="base" />
    <child link="link1" />
    <origin xyz="0 0 0" rpy="0 0 0" />
    <axis xyz="0 0 1" />
  </joint>
  <link name="link2">
    <inertial>
      <origin xyz="0.125 0 0" rpy="0 0 0" />
      <mass value="1" />
      <inertia ixx="0.0002" iyy="0.00530833333333" izz="0.00530833333333" ixy="0" ixz="0" iyz="0" />
    </inertial>
    <visual>
      <origin xyz="0.125 0 0" rpy="0 1.57079632679 0" />
      <geometry>
        <cylinder radius="0.02" length="0.25" />
      </geometry>
    </visual>
    <collision>
      <origin xyz="0.125 0 0" rpy="0 1.57079632679 0" />
      <geometry>
        <cylinder radius="0.02" length="0.25" />
      </geometry>
    </collision>
  </link>
  <joint name="joint2" type="continuous">
    <parent link="link1" />
    <child link="link2" />
    <origin xyz="0.3 0 0" rpy="0 0 0" />
    <axis xyz="0 0 1" />
  </joint>
</robot>
```

The downloadable `teaching_arm.urdf` and the generator `TwoLinkArm.urdf()` use the same model. A content test checks that they agree. This example is a verifiable building block; the open two-link tree does not implement a golfer's two-hand closure, foot contacts or muscle physiology.

25.7.3 Link Inertia and Joint Semantics

An inertial origin specifies the COM position and the orientation of the inertia frame relative to the link. The six independent inertia entries describe the symmetric rotational inertia about that COM in that frame, in kg m^2. They must not be copied from a tensor about a joint origin without a parallel-axis correction. For a uniform cylinder along x,

$$I_{xx} = \frac{1}{2}mr^2, \qquad I_{yy} = I_{zz} = \frac{m}{12}(3r^2 + L^2). \tag{25.17}$$

The first cylinder has moments $(0.0004, 0.0152, 0.0152)$ kg m^2. For the second cylinder they are approximately $(0.0002, 0.00530833, 0.00530833)$ kg m^2. Positive mass and symmetric positive-definite rotational inertia are necessary for nondegenerate solid bodies. Principal moments must also obey the triangle inequalities, such as $I_1 + I_2 \geq I_3$; positive eigenvalues alone do not establish physical realizability.

If R expresses the inertia frame in another frame, rotate the tensor as RI_CR^T. To move its reference point by displacement d from COM to the new origin, add $m(\|d\|^2 I - dd^T)$.

Rotation and translation perform different operations. Summing segment tensors requires expressing them about the same point and in the same frame.

25.7.4 Compound Joints and Closed Loops

Standard revolute, continuous, prismatic and fixed joints represent their specified relative motions. Floating and planar joint support varies by importer. A three-axis shoulder represented by serial ZYX revolutes needs two intermediate virtual links between the physical parent and child. The axes are successive local axes, and the Euler-angle rate map becomes singular at its chart's gimbal-lock orientations. An ideal spherical joint has three physical rotational freedoms without that particular chart singularity.

Virtual links represent coordinate frames, not new pieces of anatomical tissue. A dynamics engine may reject massless moving links or treat them specially. Adding small masses to make a parser accept them changes inertia; it requires an explicit regularization study. Where supported, native multi-DOF joints or multiple joint primitives on one physical body can avoid this artificial mass assignment.

The URDF tree cannot give a child link two parents. Choose a spanning tree and add the omitted rigid closure through the engine's constraint mechanism, or define a compliant attachment with stated force and energy laws. Do not duplicate the club as two independent bodies, silently weld away intended motion, or expect a second parent tag to create a valid two-hand model.

25.8 Building a Full-Body Golf Model

25.8.1 A Construction Sequence That Preserves the Physics

Begin with the nine-body, eight-compound-joint upper-body blueprint and document every physical segment and reference frame. Decide whether the pelvis is fixed, prescribed or floating before adding ground contact. Add legs only when the research question needs their motion and reaction forces; add a head or detailed hands when their inertia or task role matters. Assign the club once, with measured or explicitly idealized mass, COM and inertia.

Then choose the joint representation, including pronation and shoulder-girdle approximation, and verify forward kinematics at recognizable poses. Add one grip to the spanning tree and the second as an explicit closure, or use two compliant attachments. At a feasible reference pose compute the constraint rank and residual. Finally add actuators and their states, contact parameters and the controller. Each addition should have a mechanical check before interpreting a full swing.

Calling this a musculoskeletal model requires more than a file of rigid links. Muscle paths, moment arms, force–length and force–velocity behavior, activation dynamics and any tendon compliance must be defined when those mechanisms support the argument. A tendon path in a simulator is not by itself a muscle activation model. Parameter provenance and validation must match the scientific claim being made.

25.8.2 Anthropometry: Segment Definitions Come First

De Leva [1996] adjusted segment parameters derived from Zatsiorsky–Seluyanov's gamma-scanning measurements of living subjects, using additional anthropometric information to change reference landmarks. This was not a new cadaver-measurement study. Population estimates provide a starting model, not an individual's measured inertia. Using their values requires matching the segment boundaries, sex-specific table, COM endpoints and inertia-axis definitions.

For orientation, selected male segment mass fractions from Table 4 are: upper arm 2.71%, forearm 1.62%, hand 0.61%, thigh 14.16%, shank 4.33% and foot 1.37% of body mass, each for one segment. The whole-trunk value is 43.46%. A table containing whole trunk plus its subparts would double-count mass. Nor can a generic "thorax" fraction and an undefined "lumbar" body be assigned independently and presumed to preserve the original partition.

The same table gives the male forearm longitudinal radius of gyration as 12.1% of the specified segment length. That yields $I_{\text{long}} = m(0.121L)^2 = 0.014641mL^2$, not $0.121mL^2$. Alternative endpoint definitions change the normalized coefficient. The published URDF example uses derived cylinder inertias instead, so its geometry and mass properties can be checked without presenting population means as subject anatomy.

25.8.3 Scaling Mass, Length and Inertia

For an unchanged segment definition, scaling body mass from 70 to 80 kg while retaining mass fractions multiplies segment masses by 80/70. If every segment's geometry is unchanged and only density changes uniformly, its inertia scales by the same factor. If lengths also scale by s_L with geometrical similarity, inertia scales by $s_m s_L^2$ and COM distances by s_L. With unchanged density, $s_m = s_L^3$, giving inertia scaling s_L^5.

These are conditional scaling laws, not proof that two people have similar segment shapes or densities. If a lumbar segment is repartitioned, its mass, COM and inertia must be recomputed together. A subject-specific optimization must maintain physical consistency while fitting measurements; independently perturbing all tensor entries can generate nonphysical bodies even when each entered number looks plausible.

25.9 From URDF to Simulation — Drake, MuJoCo and Pinocchio

25.9.1 What the Software Supplies

Drake provides multibody modeling within a systems and optimization framework. Its Parser adds model instances to a plant; the returned model-instance identifiers are not the plant itself. A usable setup creates the plant, parses into it, specifies world attachment and actuation as needed, finalizes the plant, creates a context and supplies state/input before simulation. This distinction matters when translating a short API sketch into an executable experiment.

MuJoCo compiles URDF or its native MJCF into a model and advances a separate data/state object. MJCF exposes richer engine-specific constraints, tendons and actuators. Its modeling documentation allows multiple primitive joints on one physical body, avoiding dummy bodies for such compound representations. An RL algorithm remains a separate controller/training procedure; a physics engine does not itself establish policy learning or biological validity.

Pinocchio supplies rigid-body algorithms including inverse dynamics, generalized inertia, kinematics and Jacobians, with URDF parsing, as described in its official feature documentation. A dynamics library, numerical integrator, contact solver, derivative evaluator and trajectory optimizer are distinct components even when a larger package integrates them. Select and test the required combination rather than ranking software by an unsupported steps-per-second number.

25.9.2 An Executed Import and Dynamics Check

Run the following from the repository root with Python 3.12, NumPy and MuJoCo 3.4.0. The version is stated because signatures can change; the installed version's `mj_fullM` accepts the model, destination dense matrix and sparse inertia array in that order. The XML path is relative to the repository, and the checked root interpretation is fixed. No GUI or training run is required.

```
import mujoco
import numpy as np

model = mujoco.MjModel.from_xml_path(
    "articles/The_Physics_of_Golf/models/teaching_arm.urdf"
)
model.opt.gravity[:] = [0, -9.81, 0]
model.opt.timestep = 0.0005
model.opt.integrator = mujoco.mjtIntegrator.mjINT_RK4
data = mujoco.MjData(model)
data.qpos[:] = [0.3, 0.7]
mujoco.mj_forward(model, data)
mass = np.empty((model.nv, model.nv))
mujoco.mj_fullM(model, mass, data.qM)
print(model.nq, model.nv, model.nu)
print(mass, data.qfrc_bias.copy())
data.qfrc_applied[:] = [1.0, 0.0]
mujoco.mj_forward(model, data)
print(data.qacc.copy())
mujoco.mj_step(model, data, nstep=200)
print(data.time, data.qpos.copy(), data.qvel.copy())
```

The imported dimensions are $n_q = n_v = 2$ and $n_u = 0$: the URDF defines no actuator. Here `qfrc_applied` supplies external generalized torques directly, which is useful for a mechanics check but does not model bounded biological actuation. At $q = (0.3, 0.7)$ rad and zero velocity, the independent and imported results agree:

$$M \simeq \begin{bmatrix} 0.22849650 & 0.04961492 \\ 0.04961492 & 0.02093333 \end{bmatrix}, \qquad h \simeq \begin{bmatrix} 6.28565628 \\ 0.66254570 \end{bmatrix}. \tag{25.18}$$

The units are kg m^2 for M and N m for h. Applying $(1, 0)$ N m gives initial accelerations approximately $(-33.50095, 47.75165)$ rad/s^2. The positive distal acceleration follows from dynamic coupling and gravity; it is not evidence of a distal actuator command. After 200 RK4 steps of 0.0005 s, this model reaches $q \simeq (0.135246, 0.915810)$ rad. A numerical trajectory is a result for the specified example, not a verified golf prediction.

25.9.3 Independent Mechanics Behind the Check

Let $c_i = L_i/2$ and let I_i be each cylinder's centroidal zz moment. Expanding the Cartesian translational and rotational kinetic energies gives

$$\begin{aligned} M_{11} &= I_1 + m_1 c_1^2 + I_2 + m_2(L_1^2 + c_2^2 + 2L_1 c_2 \cos q_2), \\ M_{12} &= M_{21} = I_2 + m_2(c_2^2 + L_1 c_2 \cos q_2), \\ M_{22} &= I_2 + m_2 c_2^2. \end{aligned} \tag{25.19}$$

With downward global y gravity, potential is $V = g[(m_1c_1 + m_2L_1)\sin q_1 + m_2c_2 \sin(q_1 + q_2)]$ and the zero-velocity bias is ∇V. At nonzero velocity the bias also contains velocity-dependent terms. The tests compare $v^T M v/2$ against independently assembled Cartesian kinetic energy and compare ∇V with finite differences before checking the importer. Agreement between two calls into the same engine would be a weaker verification.

25.10 Sensitivity Analysis — What Happens When Parameters Are Wrong?

25.10.1 Sources of Uncertainty

Relevant uncertainties include segment boundaries and inertias, marker-to-bone motion, joint axes, club properties, external wrenches, initial velocities, actuator parameters and the contact model. State-estimation filtering can correlate errors across time and across coordinates. Finite differences used to estimate acceleration can magnify noise. Sampling frequency alone does not specify motion-capture accuracy or identify the correct model structure.

Uncertainty statements need units and an evidential basis. An orientation error is measured in radians or degrees; an angular-velocity error in rad/s or degrees/s. A perturbation range is an assumed scenario unless it is calibrated to measurements. There is no model-independent percentage by which a forearm mass error changes club speed, and no universal ordering that makes kinematic measurement accuracy more important than inertia accuracy for every output.

25.10.2 Fixed-Control and Feedback Sensitivities

For a smooth fixed-mode system $\dot{x} = f(x, u, p)$, differentiate with respect to a parameter vector p along a specified input history. With $S = \partial x/\partial p$,

$$\dot{S} = f_x S + f_p, \qquad S(0) = \frac{\partial x_0}{\partial p}. \tag{25.20}$$

If the controller is $u = \pi(x, p)$, the corresponding terms become $(f_x + f_u\pi_x)S + f_p + f_u\pi_p$. Holding a torque history fixed, replaying recorded commands and reoptimizing the controller for each parameter are different experiments. Report which is performed. An optimizer's improved value after reoptimization does not measure the same sensitivity as a golfer receiving an unanticipated perturbation.

For a fixed terminal time and output $z = \psi(x(T), p)$, $dz/dp = \psi_x S(T) + \psi_p$. A differentiable impact-time definition $H(x(t_*), p, t_*) = 0$ adds event-time sensitivity:

$$\frac{dt_*}{dp} = -\frac{H_x S(t_*) + H_p}{H_x f(t_*) + H_t}. \tag{25.21}$$

The denominator must be nonzero. Terminal output derivatives then include the time-shift term $\psi_x f\, dt_*/dp$ and any explicit time dependence of ψ. Grazing events, changing contact modes and discontinuous controllers may invalidate this smooth calculation and require event/reset analysis.

25.10.3 A Bounded Numerical Interpretation

If a stated model gives $\partial v_{\text{club}}/\partial m_f = -0.05$ (m/s)/kg and the only perturbation is $\delta m_f = \pm 0.2$ kg, the first-order speed change is ∓ 0.01 m/s. At 45 m/s this is about 0.0222% of the nominal speed. These are hypothetical supplied numbers, not a measured golf sensitivity. Other parameter and state errors can dominate or cancel this contribution.

For a covariance Σ of a parameter/state error vector and local output sensitivity row s, the linearized variance is $s\Sigma s^T$. Correlations matter. For interval assumptions, propagate the stated admissible set rather than labeling the result a confidence interval. Check nonlinear effects by finite perturbations and numerical convergence; do not infer a probability distribution from a convenient percentage sweep.

25.10.4 Sensitivity to Initial Conditions and Robustness

Initial-state perturbations obey the variational equation $\dot{\Phi} = f_x \Phi$ for the specified open-loop system, or the corresponding closed-loop Jacobian. Large finite-time amplification is not by itself evidence of chaos. Proving chaotic dynamics needs a more specific dynamical claim than observing that a small start error changes an impact result.

For a useful robustness assessment, define the output or conclusion, the feasible uncertainty set, the controller and the contact/event rules. Compare multiple time steps and derivative perturbation sizes. Include parameter correlations and physically realizable inertias. State which conclusions persist and which change. A successful arbitrary $\pm 20\%$ sweep establishes only the tested scenarios; it cannot certify robustness outside them or determine the best measurement investment without a sensitivity comparison.

25.11 Building Intuition: From URDF to Golf Physics

25.11.1 The Redundancy Principle

The key connection is now explicit. Constraints determine the motion the body can make; the task map determines which of those motions appear at the club; inertia and actuation determine how forces change those motions. An internal configuration can preserve the present club pose while changing future acceleration authority. Grip-force distribution can preserve a resultant club wrench while changing loads on the arms. Those are complementary mechanical possibilities, each with its own observable consequences.

Redundancy is therefore an opportunity to investigate alternative feasible solutions, not a guarantee of greater efficiency, safety or consistency. A model that omits a material wrist, grip or elastic state can hide a relevant pathway even if its joint-angle fit is excellent. A model that adds detail without identifiable parameters can create unwarranted confidence. The appropriate resolution is the one needed to answer and test a stated question.

25.11.2 The Identification Problem

Club trajectory alone generally does not recover full-body joint motion. Full-body kinematics alone generally does not recover unknown external contact wrenches. If inertial parameters, full state/acceleration and all external loads are known, inverse dynamics determines the required net generalized force in the specified model. Splitting that net force among muscles remains another inverse problem, especially with multiple muscles and co-contraction.

Likewise, a fitted inverse-optimal-control cost is not uniquely the nervous system's objective. Multiplying a cost by a positive constant preserves its minimizers, and different costs can agree along one observed trajectory while disagreeing elsewhere. Model error, unmeasured constraints and assumed feedback information can all be absorbed into fitted weights. Cross-condition predictions and controlled perturbations are more informative than fitting one successful swing.

A strong study therefore states the forward model and which quantities are measured, reports parameter/state uncertainty and residuals, and tests predictions on conditions withheld from fitting. Depending on the question, discriminating data might include force-plate wrenches, grip loads, additional segment kinematics or activation-related measurements with their own observation model. No single sensor automatically closes every level of this inverse problem.

25.11.3 What We Can Now Claim

We can count regular mobility for a declared model, compute task-null directions on its admissible velocity space, distinguish them from torque-null inputs, and verify a rigid-body file against mechanics. These tools make competing explanations concrete. They do not yet establish a unique control strategy, a complete musculoskeletal representation or a universal best swing.

The next step is to ask how control works with limited information, delay and noisy measurements. Those constraints belong alongside anatomy, mechanics and task geometry in an explanation of the golf swing. Preserving their distinctions makes their interaction more understandable, not less.

25.12 Primary Sources and Reading Guidance

1. Lynch and Park, Mobility, and Singularities: independent constraints, task Jacobians and rank; these are mechanisms lessons, not human motor-control measurements.

2. Khatib, Dynamic Consistency: operational-space inertia and torque projection. Follow the dimensions of every matrix when reading the scanned equations.

3. Sentis, Park and Khatib (2010): multicontact and internal-force control, under the paper's contact and actuation assumptions.

4. de Leva (1996): adjusted segment definitions, Table 4 and reference endpoints. Do not combine whole-trunk and trunk-subsegment fractions.

5. Holzbaur, Murray and Delp (2005): an explicit arm and hand model; compare retained joints and muscles before transferring its parameters.

6. Myers and Colleagues (2008) and Glazier (2011): torso/pelvis associations and the interpretation of movement variability, respectively; neither identifies the torque-null controller of a golfer.

7. MuJoCo 3.4 Python Bindings, the linked modeling manual, URDF API, Drake Parser and Pinocchio feature documentation: verify importer semantics and runtime version independently of scientific validation.

25.13 Chapter Exercises

1. **DOF Counting:** Reconstruct the nine-body upper-body model's 20 freedoms using both joint summation and the spatial constraint count. Add the club first as a rigid lead-hand attachment and then as an independent body. Under full-rank rigid-grip closure, obtain 14 in both formulations. Repeat with two compliant grips, then explain why nominal foot constraints must be stacked with grip constraints before making a full-body count.

2. **Null-Space Motion:** For the regular seven-DOF arm distinguish a rank-three position task from a rank-six pose task. Derive the elbow circle for fixed shoulder/wrist centers and prove flexion is constant. Add a prescribed elbow height, stating the independence assumption and the possible global multiplicity. Explain why one circle does not parameterize all four freedoms of the position-only null space.

3. **URDF Construction:** Starting from the complete two-link file, verify every COM, cylinder orientation and principal inertia. Design a kinematic spine/arm tree with two three-axis compound joints. Count the intermediate virtual links needed for six serial revolutes and identify the reference body. Describe the selected engine's treatment of massless frames or native multi-DOF joints before claiming a valid dynamic model; do not assign undocumented artificial tissue mass.

4. **Anthropometric Scaling:** Scale a 70 kg model to 80 kg with unchanged segment geometry, then repeat with an independent uniform length factor s_L. Derive the inertia factors in both cases and under fixed-density geometrical similarity. Explain why changing trunk partitions requires recomputing COM and inertia, not only multiplying masses.

5. **Sensitivity Calculation:** Use the hypothetical derivative -0.05 (m/s)/kg with mass perturbation ± 0.2 kg to calculate the local speed change and its fraction of 45 m/s. State what additional evidence is needed to call this a confidence interval. Derive how a differentiable impact-time condition modifies the output derivative and identify the grazing-event failure condition.

6. **Loop Constraints and Forces:** Contrast a point grip, rigid grip and compliant grip. For $M = \mathrm{diag}(2, 1)$ and $J = [1\ 1]$, show that $(1, -1)^T$ is velocity-null but not torque-null. Construct $\bar{J}$, N and N^T and verify their respective identities. Explain how two opposing grip forces can alter hand loads without altering the net club wrench.

7. **Redundancy in Golf:** Formulate two dynamically feasible candidate swings that agree in a declared club task but differ internally. Specify whether the agreement is at one impact pose, impact state or the entire trajectory. Identify an observable that could distinguish the candidates and a perturbation that tests their predicted future behavior. Explain why a fitted trajectory alone does not identify the brain's unique objective or prove reduced injury risk.

Part VIII

The Kinetic Chain and Performance

Chapter 26

The Kinetic Chain: Motion, Work, and Control in the Golf Swing

The central question of the kinetic chain is how a golfer's coordinated motion produces a useful club state at impact. The answer involves muscle work, contact forces, changing geometry, inertia, elastic storage, and timing. None of these can replace the others. A pelvis that slows while the club accelerates is an observation worth explaining; it is not, by itself, a measurement of energy moving from pelvis to club. This chapter develops the balances needed to make that explanation testable.

The useful intuition behind the whip analogy is that a connected system can develop high distal speed without every part moving equally fast. Its limitation is equally important: a golfer has active joints, two hands on a club, two changing foot contacts, and a three-dimensional body. A prescribed sequence of peaks is not a mechanical law for that system. We will use simple rotors and a moving-grip club to isolate mechanisms, then reconnect those mechanisms to measurements and control. The examples are deliberately specified teaching models, not fitted swings or prescriptions for an athlete.

26.1 Define the Observation Before Explaining It

26.1.1 Segments, Joints, and Comparable Rates

A segment is a body with a defined mass, center of mass, and inertia. A joint is a connection with relative motion and transmitted force or moment. A wrist is a joint region, not an additional rigid body whose energy can simply be added to forearm and hand energy. Similarly, a measured shoulder-line orientation may describe the thorax; counting both as separate energy reservoirs can count the same mass twice. A whole-body energy calculation requires disjoint segment definitions.

For a segment, the angular velocity vector describes rotation relative to a declared frame. A joint rate describes relative motion between adjacent segments. Neither is clubhead linear speed. The relation between clubhead velocity and generalized velocity is a Jacobian relation, $\mathbf{v}_C = J_C(q)\dot{q}$, including base translation when the base moves. Changing posture changes this map. Comparing a pelvic rate in degrees per second with clubhead speed in miles per hour does not establish amplification without geometry and a unit conversion.

Three-dimensional orientation also requires care. Euler-angle derivatives generally are not the Cartesian components of angular velocity. If $\boldsymbol{\omega} = E(\theta)\dot{\theta}$, a physical moment $\boldsymbol{\tau}$ has generalized force $E^T\boldsymbol{\tau}$, so that power is preserved. Angular-speed magnitude, a selected angular-velocity component, and the derivative of a projected turn angle can have different peak times.

26.1.2 A Sequence Is a Measurement Result

Define downswing onset, the instant immediately before ball contact, the measured quantity, its sign convention, smoothing, and the method for finding peaks. If t_0 and t_I are those events, normalized time is $s = (t - t_0)/(t_I - t_0)$. A peak at a percentage of downswing time is meaningful only under that event definition. Finite sampling and filtering create timing uncertainty; a broad plateau should not be represented as a precisely ordered instant.

Peak segment speed and peak acceleration are distinct events. For a smooth signed velocity, an interior velocity extremum has zero acceleration; an acceleration extremum concerns its derivative. Nothing makes the peak speed of one segment equivalent to the peak acceleration of its neighbor. A speed magnitude introduces another distinction: $d\|\boldsymbol{\omega}\|/dt = \boldsymbol{\omega} \cdot \dot{\boldsymbol{\omega}}/\|\boldsymbol{\omega}\|$ away from zero speed.

Vena and colleagues' five-subject study, using instantaneous screw-axis angular velocities, reported support for a proximal-to-distal kinematic sequence in two subjects, rather than a universal order [Vena et al., 2011]. That result does not say sequencing is useless. It says a proposed invariant sequence must survive explicit measurement choices and between-player variation. We therefore do not assign universal pelvic, thoracic, arm, or wrist peak percentages here. Ball contact also changes the dynamics: pre-impact club speed should not be inferred from a peak several milliseconds into a collision or follow-through.

26.2 Force and Power Balances for a Connected Body

26.2.1 Start at Each Segment's Center of Mass

For a rigid segment of mass m, central inertia tensor I_G, and center G, Newton–Euler equations are

$$m\mathbf{a}_G = \sum_k \mathbf{F}_k + m\mathbf{g}, \qquad \left.\frac{d\mathbf{H}_G}{dt}\right|_{\text{inertial}} = \sum_k \big[(\mathbf{r}_k - \mathbf{r}_G) \times \mathbf{F}_k + \boldsymbol{\tau}_k\big]. \tag{26.1}$$

In body-fixed principal coordinates the left angular term is $I_G\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (I_G\boldsymbol{\omega})$. All vector components must be represented consistently. A scalar $I\ddot{\theta} = \tau$ is a special fixed-axis equation, not the general balance about a moving anatomical joint. Forces at both proximal and distal boundaries contribute moments about the center of mass.

The corresponding kinetic energy and rate are

$$T = \tfrac{1}{2}m\|\mathbf{v}_G\|^2 + \tfrac{1}{2}\boldsymbol{\omega}^T I_G\boldsymbol{\omega}, \qquad \dot{T} = \sum_k (\mathbf{F}_k \cdot \mathbf{v}_k + \boldsymbol{\tau}_k \cdot \boldsymbol{\omega}) + m\mathbf{g} \cdot \mathbf{v}_G. \tag{26.2}$$

Here $\mathbf{v}_k$ is the material-point velocity at the application point. A force can supply power to a translating segment even if the joint has no relative translation. Moving the point at which a wrench is represented changes its moment and the associated velocity together; the total wrench power stays the same. This is why a club calculation must retain grip force power.

26.2.2 Transfer, Generation, Storage, and Dissipation

At a coincident ideal joint, the two segments share the same point velocity. Equal and opposite joint forces then make powers that cancel when both bodies are included. They can nevertheless transfer energy across the segment boundary. For an applied joint couple $\boldsymbol{\tau}$ acting on the distal body and $-\boldsymbol{\tau}$ on the proximal body, the combined power is

$$P_j = \boldsymbol{\tau} \cdot (\boldsymbol{\omega}_d - \boldsymbol{\omega}_p). \tag{26.3}$$

An ideal constraint couple does zero power along the allowed relative motion. An actuator couple need not. In a planar hinge $P_j = \tau_j \dot{\phi}_j$, where $\phi_j = \theta_d - \theta_p$. Calling every joint moment a "constraint torque" hides precisely the work we need to account for.

A spring and damper are another distinct case. Let $\delta = \theta_p - \theta_d - \delta_0$, and let $k\delta + c\dot{\delta}$ act positively on the distal rotor. The combined spring/damper power is

$$P_p + P_d = -(k\delta + c\dot{\delta})\dot{\delta} = -\frac{d}{dt}\left(\tfrac{1}{2}k\delta^2\right) - c\dot{\delta}^2. \tag{26.4}$$

Thus proximal power loss need not equal distal power gain at that instant. With $k = 100$ Nm/rad, $c = 2$ Nm s/rad, $\delta = 0.1$ rad, and angular speeds 5 and 3 rad/s, the two powers are -70 and 42 W. The remaining 28 W is 20 W of storage and 8 W of dissipation. This single example exposes the ambiguity in a claim that all proximal braking becomes distal acceleration.

26.2.3 Whole-System Equations Do Not Identify Individual Muscles

In local generalized coordinates with $v = \dot{q}$, a constrained mechanical model can be written

$$M(q)\dot{v} + h(q, v) = B(q)u + J(q)^T\lambda, \qquad J(q)v = 0. \tag{26.5}$$

The term h includes the chosen velocity-dependent, gravitational, elastic, and dissipative contributions. The multipliers λ are solved together with acceleration and constraint consistency. They are not an arbitrary derivative of a separate "constraint Lagrangian." Multiplying a generalized-force residual by M^{-1} yields acceleration, not torque.

Joint-coordinate mass matrices are generally coupled. A block-diagonal inertia representation is possible for separate bodies before connectivity is imposed; it does not make the assembled joint dynamics independent. An inverse-dynamics result is a model-dependent net generalized force. Many muscle-force combinations, including different amounts of co-contraction, may produce it. Segment kinematics alone do not select one biological realization.

26.3 What Coupled Oscillators Actually Predict

26.3.1 A Solvable Two-Rotor Counterexample

Consider two fixed-axis rotors with absolute angles θ_1, θ_2, inertias I_1, I_2, one torsional spring k, no gravity or damping, and a constant external torque τ on rotor 1. Set both angles and speeds to zero at $t = 0$. These are not anatomical hip and shoulder equations; they isolate elastic coupling:

$$I_1\ddot{\theta}_1 = \tau - k(\theta_1 - \theta_2), \qquad I_2\ddot{\theta}_2 = k(\theta_1 - \theta_2). \tag{26.6}$$

Define the inertia-weighted mean $\bar{\theta} = (I_1\theta_1 + I_2\theta_2)/(I_1 + I_2)$ and separation $\delta = \theta_1 - \theta_2$. Then

$$\ddot{\bar{\theta}} = \frac{\tau}{I_1 + I_2}, \qquad \ddot{\delta} + \Omega^2\delta = \frac{\tau}{I_1}, \qquad \Omega^2 = k\left(\frac{1}{I_1} + \frac{1}{I_2}\right). \tag{26.7}$$

The response combines an accelerating rigid mode and an oscillatory relative mode. It is not two independent oscillators with frequencies $\sqrt{k/I_i}$. For $I_1 = 2$, $I_2 = 0.5$ kg m^2, $k = 100$ Nm/rad, and $\tau = 50$ Nm,

$$\Omega = \sqrt{250} \text{ rad/s}, \quad \ddot{\theta}_1 = 20 + 5\cos(\Omega t), \quad \ddot{\theta}_2 = 20[1 - \cos(\Omega t)]. \tag{26.8}$$

Rotor 1 accelerates positively throughout: its acceleration never falls below 15 rad/s^2. It therefore has no interior speed peak, although rotor 2 has its first acceleration maximum at $t = \pi/\Omega \approx 0.19869$ s. The first proximal speed maximum on $[0, 1]$ s is simply the endpoint. Coupling alone has not produced the proposed universal peak order.

Integrating gives $\bar{\theta} = 10t^2$ and $\delta = 0.1[1-\cos(\Omega t)]$. The individual angles are $\theta_1 = \bar{\theta}+0.2\delta$ and $\theta_2 = \bar{\theta} - 0.8\delta$. Their total kinetic-plus-spring energy equals $50\theta_1$ joules, the input work from rest. This is a useful independent check on signs and numerical integration. A different torque history, damping, initial preload, or terminal objective can yield different ordering; that is a result to compute, not an exception to hide.

26.3.2 Normal Modes Belong to the Coupled System

For a linear chain $M\ddot{\theta} + C\dot{\theta} + K\theta = \tau(t)$, undamped modes solve $K\psi = \omega^2 M\psi$. Free overall rotation gives a zero-frequency mode when there is no spring to ground. With $M = \text{diag}(2, 1.2, 0.4)$ kg m^2 and neighboring springs 120 and 100 Nm/rad,

$$K = \begin{bmatrix} 120 & -120 & 0 \\ -120 & 220 & -100 \\ 0 & -100 & 100 \end{bmatrix}. \tag{26.9}$$

There is one rigid mode and two elastic modes. Each mode involves the chain, not a single segment. Relative-joint damping gives a similarly assembled matrix; terms $-c_i\dot{\theta}_i$ instead describe damping to a stationary reference. Neither model automatically imposes a quarter-period intersegment delay. Even a single oscillator's phase depends on whether one compares force, displacement, velocity, or acceleration and whether the forcing is transient or periodic.

For golf, this means that effective inertia, stiffness, posture, muscle activation, and boundary conditions jointly shape the response. A smaller distal inertia alone does not prove a particular timing sequence. Linear modes are local approximations; large rotations, changing contacts, and activation-dependent stiffness can make those modes vary through a swing.

26.4 X-Factor: Relative Motion Before Biological Interpretation

26.4.1 Use a Declared Angular Convention

One workable scalar convention defines $\alpha_X = \theta_T - \theta_P$, with thorax and pelvis turn angles measured from address and positive toward the backswing. Then $\dot{\alpha}_X = \dot{\theta}_T - \dot{\theta}_P$. If the pelvis turns toward the target while the thorax is still turning away, the separation increases in this convention. At a smooth interior maximum the two turn rates are equal; no particular number of degrees of pelvic motion is required.

These turn angles must come from a stated projection or orientation decomposition. The angle between two rotation axes is a different quantity. For example, thorax and pelvis could rotate about parallel vertical axes while their turn angles differ substantially. A full relative orientation $R_P^T R_T$ contains three-dimensional information that a scalar X-factor does not. Marker placement, tilt, side bending, angle unwrapping, and handedness affect interpretation.

Define the reference event t_0 and an early-downswing window $[t_0, t_1]$ before computing stretch:

$$\Delta\alpha_X = \max_{t\in[t_0,t_1]} \alpha_X(t) - \alpha_X(t_0). \tag{26.10}$$

Including t_0 makes this number nonnegative by definition; zero means no increase in that window. It does not prove that every golfer exhibits a positive stretch. A signed change between two specified events can instead be negative. Maximum pelvic and thoracic backswing positions need not occur simultaneously, and two zero angular velocities do not imply zero relative angle or zero elastic preload.

26.4.2 What the Classic Study Measured

Cheetham and colleagues compared 10 highly skilled and 9 less-skilled right-handed golfers using 5-irons and electromagnetic tracking. Their reported 30 Hz sampling and pelvis/T3 rotation measurements concern the transition region. They found a greater early-downswing X-factor increase in the higher-skill group, but no statistically significant between-group difference at the top of the backswing [Cheetham et al., 2001]. This group comparison does not establish that increasing one player's separation will increase distance.

The observed angle change also does not measure tissue strain energy, spindle response, or muscle-fiber shortening. For an assumed linear torsional spring, $U = \frac{1}{2}k(\alpha_X - \alpha_0)^2$ is a model with a specified rest angle and stiffness. In a living trunk, distributed joints, muscle routing, tendon compliance, activation, and passive tissues must justify that approximation. An increased angle can alter storage and muscle operating conditions while also changing loads and the future club path. The measured association is a reason to investigate that interaction, not a warrant to maximize separation.

26.5 Wrist Motion and Work at the Grip

26.5.1 Lag Is Geometry, Not a Muscle State

In a planar approximation, define an absolute arm angle θ_a and absolute club angle θ_c from the same reference; their signed relative angle is $\phi = \theta_c - \theta_a$. An included angle from directed forearm and shaft vectors is another convention and can differ by a supplement. State the directions and zero position. Three-dimensional wrist motion includes more than a single lag coordinate, and a flexible shaft adds deformation. A projected angle does not uniquely specify the clubface orientation.

Neither hand speed nor clubhead speed determines muscle shortening speed. A muscle-tendon length $\ell(q)$ has rate $\dot{\ell} = (\partial\ell/\partial q)\dot{q}$; fiber speed additionally depends on tendon compliance and pennation. Classical force–velocity behavior concerns the contractile element under declared activation and length conditions. It cannot be applied by converting an absolute club speed directly into a universal wrist torque limit. Active lengthening and isometric force are also different operating conditions from concentric shortening.

Sprigings and Neal's planar three-segment simulation found that suitably timed active wrist torque could increase clubhead speed by about 9 percent in its modeled conditions [Sprigings and Neal, 2000]. That is a simulation result with particular torque generators and constraints, not an experimentally established instruction to add a late wrist action. It is nevertheless enough to reject the universal claim that wrist torque cannot contribute at high club speed. Conversely, a measured small net wrist moment does not prove that all wrist muscles are inactive.

26.5.2 The Club Receives a Wrench, Not Only a Torque

For a rigid club, with grip point H, external grip force $\mathbf{F}_H$, and grip couple $\boldsymbol{\tau}_H$, its pre-impact power balance is

$$\frac{d}{dt}(T_c + U_{g,c}) = \mathbf{F}_H \cdot \mathbf{v}_H + \boldsymbol{\tau}_H \cdot \boldsymbol{\omega}_c + P_{\text{other}}. \tag{26.11}$$

The term P_{other} accounts for effects such as aerodynamic loading. For a flexible club, include shaft strain energy and dissipation and use the grip section's angular velocity for its applied couple. The power entering the club is not the same as the combined power of an anatomical wrist actuator acting between two moving bodies.

Even when $\boldsymbol{\tau}_H = 0$, a translating grip can do work through $\mathbf{F}_H \cdot \mathbf{v}_H$. The two hands may also produce a resultant couple through their separated forces. A zero net couple at one chosen port does not remove force transmission or make the club an isolated body. Nesbit and Serrano's analysis explicitly modeled both force and torque work at the club interface [Nesbit and Serrano, 2005]. The balance here follows rigid-body mechanics independently of any biological interpretation of their reconstructed loads.

26.6 A Release Model That Retains the Moving Grip

26.6.1 Derive the Unactuated Equation

Use an inertial vertical plane with horizontal x and upward y. The rigid club's center is $\mathbf{r}_G = \mathbf{r}_H + \ell\mathbf{e}$, where $\mathbf{e} = (\cos\theta, \sin\theta)$ points from grip to center and $\mathbf{e}_\perp = (-\sin\theta, \cos\theta)$. Let m be mass, I_G central inertia, and $I_H = I_G + m\ell^2$. Eliminating the grip force between translation and rotation gives

$$I_H\ddot{\theta} = \tau_H + m\ell(\mathbf{g} - \mathbf{a}_H) \cdot \mathbf{e}_\perp. \tag{26.12}$$

To see the origin of the forcing, substitute $\mathbf{a}_G = \mathbf{a}_H + \ell\ddot{\theta}\mathbf{e}_\perp - \ell\dot{\theta}^2\mathbf{e}$ into $m\mathbf{a}_G = \mathbf{F}_H + m\mathbf{g}$, then take moments about G. The radial $\dot{\theta}^2$ term has no moment about the grip; the acceleration of the grip has a tangential projection that does. Omitting $\mathbf{a}_H$ changes the problem after "release."

For an illustrative instantaneous state, set $m = 0.2$ kg, $\ell = 1$ m, $I_G = 0.02$ kg m^2, $\theta = 0$, $\dot{\theta} = 2$ rad/s, $\mathbf{v}_H = (0, -3)$ m/s, $\mathbf{a}_H = (0, -20)$ m/s^2, and $\tau_H = 0$. Then $I_H = 0.22$ kg m^2 and $\ddot{\theta} \approx 9.2636$ rad/s^2. The grip force is approximately $(-0.8, -0.18527)$ N and supplies 0.55582 W. Directly differentiating $T_c + U_{g,c}$ gives the same rate. Angular acceleration is present with no applied grip couple because the grip trajectory is still imposed and its force still acts.

Changing only grip acceleration to free fall, $\mathbf{a}_H = \mathbf{g}$, gives zero angular acceleration. A stationary grip instead gives $\ddot{\theta} \approx -8.9182$ rad/s^2 at the same horizontal orientation. These are three different boundary conditions, not three explanations of the same swing. Centrifugal or Coriolis terms in other coordinates reorganize the dynamics; they do not introduce an additional energy source.

26.6.2 Holding, Releasing, and Feasibility

Suppose an idealized holding condition imposes $\theta = \theta_a(t) + \phi_0$. The required grip couple is

$$\tau_{\text{hold}} = I_H \ddot{\theta}_a - m\ell(\mathbf{g} - \mathbf{a}_H) \cdot \mathbf{e}_\perp. \tag{26.13}$$

A bound $|\tau_{\text{hold}}| \le \tau_{\max}$ checks whether this imposed trajectory is feasible in the toy model. It does not assert that a human wrist locks perfectly until its torque limit is reached. A controller could change its command earlier; anatomical joints need not behave as binary latches. For a smooth release without an impulse, initialize the released motion with the angle and angular velocity immediately before release and continue the specified grip trajectory. If a constraint change produces an impulse, its momentum and energy effects require a separate event model.

A claim that later release improves speed must specify which intervention changes, which upstream controls or trajectories remain fixed, the admissible torque/force and joint-range limits, and the endpoint. Compare states at a common impact surface with correct clubface orientation, not just at an arbitrary common clock time. If grip motion is prescribed, its force and work generally change between alternatives. Giving both alternatives the same torque history does not make their input work equal. There is no universal percentage benefit or optimal delay that follows from this pendulum equation alone.

26.7 Ground Reaction: Momentum Channel and Contact Constraint

26.7.1 Compute the Full Moment Vector

Choose a right-handed frame: x along the target line, y horizontally across it, and z upward. About origin O, a force applied at displacement $\mathbf{r} = (r_x, r_y, r_z)$ and a free moment $\mathbf{m}$ produce

$$\boldsymbol{\tau}_O = \mathbf{r} \times \mathbf{F} + \mathbf{m} = \begin{bmatrix} r_y F_z - r_z F_y \\ r_z F_x - r_x F_z \\ r_x F_y - r_y F_x \end{bmatrix} + \mathbf{m}. \tag{26.14}$$

Vertical force at a horizontal offset produces horizontal-axis moments, not a vertical-axis yaw moment. Horizontal force at a vertical offset contributes horizontal-axis moments too. Summing those components as if all were pelvic yaw torque is invalid.

For a numerical example about a stated origin, let $\mathbf{r} = (0.4, 0.2, -1)$ m and $F_z = 1.5(80)(9.81) = 1177.2$ N. Vertical force alone gives $(235.44, -470.88, 0)$ Nm. Add $F_x = 100$ N, $F_y = 200$ N, and a free yaw moment $m_z = 10$ Nm: the complete moment, in Nm, is

$$\boldsymbol{\tau}_O = (435.44, -570.88, 70)^T. \tag{26.15}$$

If that net yaw moment stays constant for 0.05 s, its yaw angular impulse is 3.5 N m s. These are declared illustrative inputs; the calculation is not a prediction of hip angular speed.

26.7.2 Whole-Body Angular Momentum Is Not Pelvic Rotation

For the complete golfer-plus-club system about a fixed inertial origin,

$$\dot{\mathbf{H}}_O = \sum \boldsymbol{\tau}_{O,\text{external}}. \tag{26.16}$$

Joint forces and couples are internal to that system. Ground wrenches, gravity, aerodynamic loads, and ball contact where present are external. About the total center of mass, uniform gravity has zero resultant moment, and the external-moment balance also holds for angular momentum defined relative to that center. Other moving reference points require the appropriate transport terms. The force example above gives a change in whole-system angular momentum only when all other relevant external moments are accounted for.

One cannot divide that moment by an isolated pelvic inertia to obtain pelvic acceleration in a coupled swing. Nor does zero external yaw moment prevent the pelvis from rotating: internal actions can produce counter-rotation elsewhere while conserving total angular momentum. Ground contact permits exchange of momentum with the environment and changes feasible motion; it is not the sole possible cause of every observed segment rotation.

26.7.3 Center of Pressure and Ground Work

A force plate's resultant force, moment, and center of pressure are related but not interchangeable. The center of pressure describes a pressure-resultant location on the contact plane under its adopted convention. It does not by itself determine horizontal shear force or the residual free yaw moment. Near vanishing normal force, center-of-pressure estimates can become ill-conditioned. Combining two plates also requires a common origin and frame.

At a perfectly stationary, non-slipping contact, ideal reaction forces do zero mechanical work because the material contact points have zero velocity. Yet they can provide a nonzero moment and angular impulse. Muscles supply mechanical work internally from chemical energy while contact constrains the motion. Real shoes, ground compliance, slipping, and foot deformation can add storage or work terms. A peak ground force before impact does not mean zero force at impact, and it does not determine when the center-of-mass speed peaks. Timing force, impulse, work, and segment velocity answers four different questions.

26.8 Close the Energy Budget Before Naming Efficiency

26.8.1 Choose the System and the Time Interval

For a pre-impact mechanical model including the body, club, gravity, and chosen elastic elements, write

$$E = T_{\text{body}} + T_{\text{club}} + U_g + U_e, \qquad E(t_I) - E(t_0) = W_{\text{act}} + W_{\text{ext}} - D. \tag{26.17}$$

Here W_{act} is signed actuator work at the declared joint ports, W_{ext} is work from the other external interactions not already represented by potential energy, and $D \geq 0$ is the modeled dissipation. Avoid counting an elastic element both in U_e and again as an independent actuator energy input. An ideal frictionless model with ongoing positive actuator work does not conserve mechanical energy; it obeys this balance with increasing E.

Backswing history enters through initial position, velocity, and stored energy. The top of a backswing is not generally a state of complete rest. Gravity needs a scale check: raising an 80 kg whole-body center of mass by 0.1 m changes potential energy by 78.48 J. Individual segment and club height changes must also be handled consistently. Small-looking displacement is not a sufficient argument to delete the term.

Energy of a segment depends on both translation and rotation. A decrease in angular speed about one measured axis can coexist with increasing segment kinetic energy. Even a verified decrease in total segment energy does not identify its destination without boundary powers and storage. Negative net joint power also does not uniquely identify eccentric action in every spanning muscle. Tendon energy changes and simultaneous muscle actions can yield the same net value.

26.8.2 Signed Work, Positive Work, and Metabolism

For an actuator power P_j, distinguish signed work $W_j = \int P_j dt$ from positive work $W_j^+ = \int \max(P_j, 0)dt$ and absorbed-work magnitude $W_j^- = \int \max(-P_j, 0)dt$, so $W_j = W_j^+ - W_j^-$. Summing positive work over joints is not the same as taking the positive part after summing their powers. State which denominator a reported ratio uses.

Nesbit and Serrano analyzed one selected swing from each of four amateur golfers with reconstructed body and club models. Their "swing efficiency" was club work divided by total body joint work; their table reports about 20.2–26.8 percent [Nesbit and Serrano, 2005]. Those values are not a measured elite population range, a direct muscle efficiency, or a fraction of metabolic energy. A larger ratio also need not imply a faster club: numerator and denominator can change together.

For a transparent illustrative budget, suppose initial recoverable mechanical energy is 100 J, positive actuator work is 500 J, absorbed actuator work is 80 J, and other dissipation is 20 J, with zero other external work. Final mechanical energy is 500 J. It could comprise 250 J in the club and 250 J in body motion and retained potentials. Club energy divided by positive actuator work is 0.50, while club energy divided by signed actuator work is about 0.595. Neither ratio alone identifies where the rest became heat or establishes metabolic efficiency. Initial energy can even make a club-energy/work ratio exceed one without violating conservation.

Collision adds another boundary and another interval. Pre-impact club kinetic energy is not all transferred to the ball; club motion, deformation, losses, and ball spin remain relevant. A numerical impact-transfer fraction must follow an explicit collision model or measurement. It cannot be appended as a universal percentage to close an incomplete downswing budget.

26.9 How Drift and Control Connect to the Chain

26.9.1 Declare What Zero Input Removes

The full local state must at least include configuration and velocity, $x = (q, v)$. With a direct torque input and consistently eliminated regular constraints, it may admit

$$\dot{x} = f(x) + G(x)u, \qquad f(x) = \begin{bmatrix} v \\ a_0(q, v) \end{bmatrix}. \tag{26.18}$$

The first component is configuration rate, not the complete dynamics. The drift is the zero-input field for the declared input port. Its contents depend on which gravity, elastic, dissipative, actuator, and contact effects the model retains. A change of coordinates transforms both the drift and the input fields; numerical component magnitudes need a declared scaling or metric.

If input means neural excitation, zero input need not mean zero activation or zero muscle force. Activation and tendon states must then be included, and a simple affine torque model may no longer represent the same intervention. If input means a net joint torque, removing it also differs from making all muscles inactive. In a moving-grip club model, setting its grip couple to zero retains the imposed grip path and its force. These distinctions are essential to the zero-torque counterfactual framework in Chapter 6.

26.9.2 Attribution and Later Consequences

At a fixed state in a genuinely affine model, $G(x)u$ is the instantaneous difference between actual and zero-input state rates. It does not follow that the difference between two finite trajectories equals its integral along the original trajectory. Once states diverge, gravity, geometry, constraint reactions, and future control effectiveness change too. A counterfactual that crosses a contact boundary requires an event-consistent model.

The first-order endpoint sensitivity along a nominal smooth trajectory is

$$\delta x(t_I) = \Phi(t_I, t_0)\delta x(t_0) + \int_{t_0}^{t_I} \Phi(t_I, s)G(x(s))\delta u(s)\, ds, \tag{26.19}$$

where Φ is the state-transition matrix of the full linearized dynamics under the specified control policy. For a terminal output $y = h(x)$ at a fixed time, premultiply by Dh. If impact time is determined by a contact event, include its timing sensitivity and any impact map rather than silently treating t_I as fixed.

This gives the site's central argument a precise form: early action changes the state and geometry from which later action operates, and the existing state continues to evolve between inputs. The relevant question is how much admissible future action can change the required impact outputs, with what uncertainty and cost. A large instantaneous drift-control ratio does not measure neural latency, prove an optimal policy, or establish absence of useful distal control. It needs a declared model, coordinate metric, input port, and output horizon before it can inform that question.

26.10 Test the Explanation

26.10.1 Link Each Claim to an Observable

To test sequencing, collect synchronized segment kinematics with declared coordinate frames and timing uncertainty. To test power flow, add external wrenches, body/club inertial estimates, and an inverse-dynamics model whose residuals and filtering sensitivity are reported. To infer individual muscle or tissue action, additional measurements and physiological assumptions are necessary. Electromyography measures an electrical signal related to activation; it does not directly measure muscle force, joint work, or tendon energy.

Check integrated boundary powers against the change in mechanical energy, allowing for measurement uncertainty. Large unexplained residuals can arise from differentiation noise, mismatched time bases, inaccurate inertias, omitted foot or shaft deformation, or inconsistent moment references. Reproducing measured positions in a kinematically driven model is not an independent validation of all its internal forces.

26.10.2 Test Interventions Against Competing Explanations

Suppose a player gains club speed after changing transition timing. Candidate explanations include different active work, a changed hand path, different segment geometry, altered elastic storage, a different impact time, or changed clubface/contact conditions. Compare repeated trials and quantify the outputs that matter: speed, path, face orientation, strike location, ball result, and consistency. A faster isolated segment peak is not enough.

Use a model intervention to separate hypotheses only after declaring the preserved quantities. Holding the grip path fixed isolates one question; holding actuator effort fixed isolates another. Sensitivity to initial state and model parameters determines whether the predicted advantage survives measurement uncertainty. A simulation that matches an observed sequence can demonstrate compatibility with a mechanism, but competing mechanisms may match the same sequence. Successful explanation requires discriminating observations or interventions.

26.11 What the Reader Can Use in Practice

The mechanics support an investigation organized around the desired club state and the player's repeatability. They do not justify a universal instruction to maximize pelvic speed, freeze the wrists, or seek the largest possible X-factor. A change should be evaluated by its measured effect on the complete shot, including dispersion and strike, rather than by conformity to an attractive sequence diagram.

Slow practice can help a player explore positions and coordination, but it is not automatically dynamically similar to a full-speed swing. Under a time scaling, inertial terms scale differently from gravity, and activation, elastic, and contact effects need not scale together. The motion should be reassessed at the intended speed. Likewise, smoother commanded torque does not guarantee small club jerk near rapidly changing geometry or contact events.

Force-plate feedback can inform contact forces, moments, impulse, and their timing when measured correctly. It does not directly calibrate a drift-control ratio. Feeling "effortless" does not prove that muscles stopped working, and muscle activation does not identify concentric action; active muscle can shorten, remain approximately isometric, or lengthen. These distinctions allow useful coaching descriptions to coexist with technically honest explanations.

The connections are the important result. Ground contact shapes feasible momentum exchange; muscle action changes energy and state; geometry maps that state to club motion; elastic elements store and return work; and the remaining time constrains what later control can accomplish. An account that quantifies those connections is more informative than either a list of peak timings or a claim that one part of the chain supplies all the speed.

26.12 Reproduce the Mechanical Checks

The following Python functions use NumPy and SciPy to evaluate the stated teaching cases. Times are in seconds and other inputs use SI units. The rotor solution can be checked against independent numerical integration; the modal vectors are mass-normalized. The moving-grip calculation checks instantaneous force, moment, and total mechanical-energy rates, not a complete golf trajectory. The same program is printed in both editions.

```python
# Kinetic-chain mechanical checks
import numpy as np
from scipy.linalg import eigh


def rotor_case(time: np.ndarray) -> dict[str, np.ndarray]:
    """Two fixed-axis rotors, zero initial state, constant 50 Nm input."""
    frequency = np.sqrt(250.0)
    phase = frequency * time
    separation = 0.1 * (1 - np.cos(phase))
    relative_speed = 0.1 * frequency * np.sin(phase)
    mean_angle = 10 * time**2
    mean_speed = 20 * time
    angle_one = mean_angle + 0.2 * separation
    angle_two = mean_angle - 0.8 * separation
    speed_one = mean_speed + 0.2 * relative_speed
    speed_two = mean_speed - 0.8 * relative_speed
    energy = speed_one**2 + 0.25 * speed_two**2
    energy += 50 * separation**2
    return {
        "state": np.array([angle_one, angle_two, speed_one, speed_two]),
        "acceleration": np.array([20 + 5 * np.cos(phase), 20 * (1 - np.cos(phase))]),
        "energy": energy,
    }


def mode_case() -> dict[str, np.ndarray]:
    """Mass-normalized modes of the stated three-rotor model."""
    mass = np.diag([2.0, 1.2, 0.4])
    stiffness = np.array([[120, -120, 0], [-120, 220, -100], [0, -100, 100]])
    squared, modes = eigh(stiffness, mass)
    return {"squared_frequencies": squared, "modes": modes}


def grip_case(acceleration: tuple[float, float]) -> dict:
    """Horizontal rod, zero grip couple, translating/accelerating grip."""
    mass, length, central_inertia = 0.2, 1.0, 0.02
    gravity = np.array([0.0, -9.81])
    grip_velocity = np.array([0.0, -3.0])
    angular_speed = 2.0
    polar_inertia = central_inertia + mass * length**2
    alpha = mass * length * (gravity[1] - acceleration[1]) / polar_inertia
    center_acceleration = np.array(acceleration) + length * np.array([-4.0, alpha])
    force = mass * (center_acceleration - gravity)
    center_velocity = grip_velocity + np.array([0.0, length * angular_speed])
    kinetic_rate = mass * center_velocity @ center_acceleration
    kinetic_rate += central_inertia * angular_speed * alpha
    return {
        "alpha": alpha, "force": force,
        "grip_power": force @ grip_velocity,
        "energy_rate": kinetic_rate - mass * gravity @ center_velocity,
    }


```

```python
def ground_case() -> dict:
    """Moments about the declared origin; all inputs are illustrative SI values."""
    arm = np.array([0.4, 0.2, -1.0])
    vertical_force = np.array([0.0, 0.0, 1.5 * 80 * 9.81])
    force = vertical_force + np.array([100.0, 200.0, 0.0])
    moment = np.cross(arm, force) + np.array([0.0, 0.0, 10.0])
    return {
        "vertical_moment": np.cross(arm, vertical_force),
        "full_moment": moment, "yaw_impulse": moment[2] * 0.05,
    }


def joint_case() -> dict[str, float]:
    """A spring/damper stores and dissipates work while transferring power."""
    separation, relative_speed = 0.1, 2.0
    stiffness, damping = 100.0, 2.0
    torque = stiffness * separation + damping * relative_speed
    return {
        "proximal_power": -torque * 5.0,
        "distal_power": torque * 3.0,
        "storage_rate": stiffness * separation * relative_speed,
        "dissipation": damping * relative_speed**2,
    }
```

For example, evaluate the rotor function at a time grid from zero to one second and compare its energy with $50\theta_1$. For the modal case, verify $K\Psi = M\Psi\,\mathrm{diag}(\omega^2)$ and $\Psi^T M\Psi = I$; the smallest squared frequency is zero up to floating-point roundoff. Evaluate the grip case with accelerations $(0, -20)$, $(0, -9.81)$, and $(0, 0)$ to reproduce the three boundary conditions. Ground and joint functions reproduce the moment and power examples directly. Small rounding differences should not be interpreted as physical residuals.

26.13 Exercises

1. **Sequence Counterexample.** Starting from the two-rotor equations, derive the mean and relative solutions and reconstruct both angular velocities. On $[0, 1]$ s, locate rotor 1's maximum velocity and rotor 2's first acceleration maximum. Explain why their order refutes a universal proximal-speed/distal-acceleration rule without refuting every possible sequencing mechanism. Verify the input-work identity independently.

2. **X-Factor and Storage.** In the declared backswing-positive convention, let thorax and pelvis angles at the reference event be 90 and 45 degrees. At a later event let them be 88 and 38 degrees. Compute separation at each event and its change. For an assumed torsional spring with rest separation 30 degrees and stiffness 100 Nm/rad, compute the storage change in joules. State why this result is not a measurement of human trunk energy and why both turn rates becoming zero would not remove preload.

3. **A Moving-Grip Release.** Derive the club equation by eliminating grip force from Newton–Euler balances. Reproduce the three acceleration cases and check energy power. Then prescribe a smooth grip trajectory and an arm angle history of your choosing, state them explicitly, and compute the holding couple. Explain the addi-

tional endpoint, feasibility, and input-work checks needed before comparing two release times. Zero grip couple alone is not a complete boundary condition.

4. **Ground Wrench and Reference Point.** Reproduce all three moment components in the ground example. Change the origin by a known displacement and transform the moment consistently. Integrate the original yaw moment over the stated interval. Identify the additional information needed to infer pelvic acceleration from whole-system angular impulse, and explain why center of pressure alone cannot supply it.

5. **Energy and Identifiability.** Close the illustrative 100 J initial-energy budget using the stated positive work, absorbed work, and dissipation. Compute both club-energy ratios. Construct a second consistent budget with the same final club energy but different body energy and actuator work. Explain why club speed alone cannot distinguish the budgets or determine individual muscle action. Add a separately declared collision model only if estimating ball energy.

6. **Modes and Experimental Discrimination.** Solve the three-rotor generalized eigenproblem, identify its rigid mode, and verify mass normalization. Apply a finite torque pulse to one rotor and calculate the response from rest. Repeat with relative-joint damping and changed initial preload. Compare the resulting peak definitions explicitly. Propose a synchronized measurement that could distinguish a change in active work from a change in energy redistribution in a golfer; report which required quantities remain model-dependent.

Chapter 27

Induced Acceleration Analysis: Quantifying Who Moves What

> "It is not enough to know what forces are present; one must know what each force *does*."

In Plain Language 27.1. The Question Behind This Chapter

In the previous chapter, we described the kinetic chain as a sequential cascade of energy from proximal segments (hips, torso) to distal segments (arms, wrists, club). A narrower, model-bounded question is how one declared coordinate, contact, and force-partition convention divides the instantaneous acceleration ledger. That ledger does not identify a unique cause.

This chapter introduces **induced acceleration analysis** (IAA). At a frozen state, a declared force partition maps linearly through the constrained forward-dynamics operator and closes to the model acceleration when constraints, contact reactions, and residuals are handled consistently. The terms are calculations under that convention, not observations of forces literally acting alone.

Induced acceleration analysis was pioneered by Zajac and Gordon [1989] for studying muscle function in multi-joint movements and was extended into a comprehensive framework for walking by Zajac et al. [2002, 2003]. The technique has since been applied to throwing Hirashima et al. [2008, 2007], Hirashima and Ohtsuki [2008], gait pathology Riley and Kerrigan [1999], Neptune et al. [2001], sit-to-stand transfers Caruthers et al. [2016], cycling Schutte et al. [1993], and postural control Challis [2011]. This chapter brings induced acceleration analysis into the golf swing for the first time, connecting it to the control-affine framework developed in earlier chapters.

Normative Attribution Record

A publishable attribution must identify the model and revision; engine, solver, and revision; generalized coordinates and reference frame; output point; mass matrix; constraint handling; contact model and mode; complete force partition; residual treatment; and numerical tolerance with closure error. Its identifiability contract must state the measurements or assumptions supporting each term. An algebraic term does not by itself identify anatomical source, neural intent, necessity, sufficiency, or intervention effect. Missing fields make a result **unsupported or unqualified**;

an unavailable cross-engine state is reported that way rather than treated as solver parity.
Two counterexamples block unique attribution. Under $q = Tz$, the same dynamics use $M_z = T^\mathsf{T} M_q T$ and $Q_z = T^\mathsf{T} Q_q$: component values change even though $T\ddot{z} = \ddot{q}$. Likewise, if $Q = Q_A + Q_B$, then $Q = (Q_A + r) + (Q_B - r)$: a residual reassignment changes both reported terms while preserving their sum. An intervention must separately re-solve constraints and contact and declare what is held fixed.

27.1 The Mathematical Foundation

27.1.1 Starting From the Manipulator Equation

Recall from Chapter 4 that the equations of motion for an n-degree-of-freedom mechanical system take the form:

$$\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} = \boldsymbol{\tau}_{\text{muscle}} + \boldsymbol{V}(\boldsymbol{q}, \dot{\boldsymbol{q}}) + \boldsymbol{g}(\boldsymbol{q}) \tag{27.1}$$

Here $\boldsymbol{q}$ is a declared independent coordinate vector and $\boldsymbol{\tau}_{\text{muscle}}$ is only a model-assigned generalized-force term unless an actuator map and identifiability contract support the anatomical label. Contact, passive, constraint, and residual terms must also appear when retained.
The critical step is to solve for accelerations by premultiplying both sides by $\boldsymbol{M}^{-1}(\boldsymbol{q})$:

$$\boxed{\ddot{\boldsymbol{q}} = \boldsymbol{M}^{-1}(\boldsymbol{q})\left[\boldsymbol{\tau}_{\text{muscle}} + \boldsymbol{V}(\boldsymbol{q}, \dot{\boldsymbol{q}}) + \boldsymbol{g}(\boldsymbol{q})\right]} \tag{27.2}$$

For independent minimal coordinates with a regular kinetic-energy metric, $\boldsymbol{M}(\boldsymbol{q})$ is symmetric positive definite and invertible. Redundant coordinates or active constraints require a declared constrained solve. At a fixed state and active set, the resulting force-to-acceleration map is linear in the declared right-hand-side terms.

Key Principle 27.1. Instantaneous Superposition of Accelerations

At any frozen instant in time (fixed $\boldsymbol{q}$ and $\dot{\boldsymbol{q}}$), the total acceleration $\ddot{\boldsymbol{q}}$ is the sum of the accelerations induced by each individual force source:

$$\ddot{\boldsymbol{q}} = \underbrace{\boldsymbol{M}^{-1}(\boldsymbol{q})\boldsymbol{\tau}_{\text{muscle}}}_{\text{muscle-induced}} + \underbrace{\boldsymbol{M}^{-1}(\boldsymbol{q})\boldsymbol{V}(\boldsymbol{q}, \dot{\boldsymbol{q}})}_{\text{velocity-induced}} + \underbrace{\boldsymbol{M}^{-1}(\boldsymbol{q})\boldsymbol{g}(\boldsymbol{q})}_{\text{gravity-induced}} \tag{27.3}$$

Each term can be computed independently. Furthermore, the muscle torque vector can itself be decomposed into contributions from individual muscles or individual joints:

$$\boldsymbol{\tau}_{\text{muscle}} = \sum_{k=1}^{m} \boldsymbol{\tau}_k \tag{27.4}$$

where $\boldsymbol{\tau}_k$ is the torque vector produced by the k-th muscle (or the net torque at the k-th joint). Since $\boldsymbol{M}^{-1}$ is linear:

$$\ddot{\boldsymbol{q}}_{\text{muscle}} = \sum_{k=1}^{m} \boldsymbol{M}^{-1}(\boldsymbol{q})\boldsymbol{\tau}_k = \sum_{k=1}^{m} \ddot{\boldsymbol{q}}_k \tag{27.5}$$

Each $\ddot{\boldsymbol{q}}_k = \boldsymbol{M}^{-1}(\boldsymbol{q})\boldsymbol{\tau}_k$ is the term assigned to one declared generalized-force channel. Calling that channel a muscle requires a muscle-to-generalized-force map and identification evidence.

In Plain Language 27.2. What Does "Induced Acceleration" Actually Mean?

Imagine you could freeze time at one instant during the downswing. At that instant, multiple forces act on the golfer-club system: hip torque, shoulder torque, wrist torque, gravity, and the centrifugal and Coriolis forces arising from the current velocities.

IAA asks how the frozen-state operator maps one declared term while state, active constraints, and partition are fixed. This is not automatically a realizable intervention that removes other forces; such an intervention must re-solve contact and constraints.

This is not a thought experiment about what *would* happen over time if we only applied one force (that would be a different and much more complicated question, because the system's state would diverge). It is a precise statement about the *instantaneous* contribution of each force to the total acceleration at one frozen moment.

27.1.2 Connection to the Control-Affine Framework

Readers who have followed the development in Chapter 5 will recognize that Equation 27.3 is closely related to the control-affine decomposition:

$$\dot{\boldsymbol{x}} = \underbrace{\boldsymbol{f}(\boldsymbol{x})}_{\text{drift}} + \underbrace{\boldsymbol{G}(\boldsymbol{x})\boldsymbol{u}}_{\text{control}} \tag{27.6}$$

For one unconstrained partition, the acceleration block of drift may contain velocity and gravity terms while $\boldsymbol{G}(\boldsymbol{x})\boldsymbol{u}$ maps declared applied inputs. The complete drift also retains passive, contact, constraint, and internal-state terms. A control channel is not synonymous with a measured muscle input.

The connection is conditional: both frameworks may use a frozen-state linear force-to-acceleration map, but their terms are comparable only when plant, coordinates, actuator map, constraints, contact, residual treatment, and output agree. The traditions were developed in nonlinear control Murray et al. [1994], Bullo and Lewis [2004] and biomechanics Zajac and Gordon [1989], Zajac et al. [2002], respectively.

Drift vs. Control 27.1. Drift, Control, and Induced Acceleration

In the control-affine framework:

- **Drift** = the complete autonomous acceleration of the declared effective plant when applied generalized control is zero, including every retained inertial, gravitational, passive, contact, constraint, and internal-state term.

- **Control** = additional acceleration from declared applied generalized inputs

In induced acceleration analysis:

- **Velocity-dependent acceleration** = $\boldsymbol{M}^{-1}\boldsymbol{V}(\boldsymbol{q}, \dot{\boldsymbol{q}})$
- **Gravity-induced acceleration** = $\boldsymbol{M}^{-1}\boldsymbol{g}(\boldsymbol{q})$
- **Muscle-induced acceleration** = $\boldsymbol{M}^{-1}\boldsymbol{\tau}_{\text{muscle}}$

These expressions coincide only when plant, state, coordinates, actuator map, constraints, contact mode, force partition, and output match. The control term is not identified muscular effort.

27.2 Dynamic Coupling: A Declared Joint Torque Can Affect Multiple Coordinates

One of the most important and initially counterintuitive consequences of the equations of motion is **dynamic coupling**: a torque applied at one joint produces accelerations at *every* joint in the kinematic chain, not just the joint where the torque is applied.

27.2.1 Why This Happens

Consider a two-joint system (shoulder and elbow) in a horizontal plane. If we apply only a shoulder flexion torque τ_1 (with $\tau_2 = 0$), the induced accelerations are:

$$\begin{bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{bmatrix} = \boldsymbol{M}^{-1}(\boldsymbol{q}) \begin{bmatrix} \tau_1 \\ 0 \end{bmatrix} = \begin{bmatrix} A_{11}\tau_1 \\ A_{21}\tau_1 \end{bmatrix} \tag{27.7}$$

where A_{ij} denotes the (i, j) element of $\boldsymbol{M}^{-1}(\boldsymbol{q})$. The shoulder experiences an acceleration $A_{11}\tau_1$ (the *direct* effect), but the elbow also experiences an acceleration $A_{21}\tau_1$ (the *remote* or *induced* effect). The off-diagonal element A_{21} is generally nonzero because the segments are physically linked: applying a torque at the shoulder creates a reaction force at the elbow joint, which accelerates the forearm.

This phenomenon was clearly articulated by Zajac and Gordon [1989] in their foundational review: in multi-articular movement, determining the action of a muscle requires accounting for the constraint forces that propagate through the entire chain. A muscle crossing only the shoulder can accelerate the elbow, the wrist, and the club—not through direct contact, but through the mechanical coupling embedded in $\boldsymbol{M}^{-1}(\boldsymbol{q})$.

27.2.2 Posture Dependence: The Same Torque, Different Results

A second important finding, emphasized in the work of Hirashima [2011], is that the induced accelerations depend on the current posture $\boldsymbol{q}$ of the entire system, because $\boldsymbol{M}^{-1}(\boldsymbol{q})$ changes with configuration.

Consider the shoulder internal rotation torque applied to a three-segment upper limb model. When the elbow is fully extended (so the entire limb forms a straight line), the principal axes of inertia of the distal chain align with the shoulder joint axes. In this configuration, the shoulder internal rotation torque produces *only* internal rotation—there is a clean one-to-one mapping from torque to acceleration.

When the elbow is flexed at 90° in the cited model, the same declared shoulder-torque channel receives nonzero internal-rotation, abduction, and horizontal-extension components Hirashima [2011]. These are coordinate- and model-reported terms, not an identified anatomical function.

Key Principle 27.2. Configuration-Dependent Muscle Function

The acceleration induced by a given muscle force depends on:

1. The magnitude and direction of the torque vector produced by the muscle (determined by anatomy and moment arms).
2. The current posture of the *entire* kinematic chain (which determines $\boldsymbol{M}^{-1}(\boldsymbol{q})$).

Changing posture can change the declared term ledger. This supports a configuration-sensitivity hypothesis; it does not establish equal muscular effort, proper form, a unique cause, or an intervention effect.

27.2.3 The Induced Acceleration Index

Challis [2011] formalized the configuration-dependent coupling between joints by introducing the **induced acceleration index** (IAI). For a system where joint j is "active" (has an applied torque) and joint k is "inactive" (has no applied torque), the IAI quantifies the potential of the active joint to accelerate the inactive joint:

$$\mathrm{IAI}_{j\to k} = \left| M_{kk}^{-1} M_{kj} \right| \tag{27.8}$$

This dimensionless index depends only on the system's configuration and inertial properties, not on the magnitudes of the torques. It reveals the *structural* coupling between joints.

For a model of quiet standing, Challis [2011] found that the ankle has over 12 times the ability to accelerate the hip joint compared to the hip's ability to accelerate the ankle. This asymmetry helps explain why the ankle strategy is the primary postural control mechanism for small perturbations.

In the golf swing, similar asymmetries exist. During the early downswing, the large muscles of the hips and trunk have a strong ability to accelerate the distal segments (arms, wrists, club) through the off-diagonal elements of $\boldsymbol{M}^{-1}$. As the limbs extend during the late downswing, the coupling changes: the extended configuration alters the inertia matrix, potentially amplifying or diminishing the effectiveness of proximal torques on distal motion.

27.3 Instantaneous Effects Versus Cumulative Effects

One of the most important distinctions to emerge from the induced acceleration literature—and one with direct relevance to understanding the golf swing—is the difference between **instantaneous effects** and **cumulative effects**. This distinction was developed most clearly in the work of Hirashima et al. [2008] and Hirashima [2011] in the context of baseball pitching, and it maps directly onto the drift-control decomposition introduced in Chapter 5.

27.3.1 Instantaneous Effects: What the Current Torques Produce Right Now

The muscle-induced acceleration at time t is:

$$\ddot{\boldsymbol{q}}_{\text{muscle}}(t) = \boldsymbol{M}^{-1}(\boldsymbol{q}(t))\boldsymbol{\tau}_{\text{muscle}}(t) \tag{27.9}$$

This is a purely *instantaneous* quantity. It tells us what acceleration the muscles are producing at this exact moment, given the current posture. If the muscles were to vanish at time t, this contribution to the acceleration would immediately disappear. These are the instantaneous effects, and they include both the **direct effect** (a joint torque accelerating its own joint) and the **remote effect** (a joint torque accelerating distant joints through dynamic coupling).

27.3.2 Cumulative Effects: The Legacy of Past Forces

The velocity-dependent torque $\boldsymbol{V}(\boldsymbol{q}, \dot{\boldsymbol{q}})$ depends on the current angular velocities $\dot{\boldsymbol{q}}$. But these velocities are the time-integrated result of *all* previous forces—every muscle torque, every gravitational pull, every constraint reaction—from the beginning of the movement to the current instant. The acceleration induced by the velocity-dependent torque:

$$\ddot{\boldsymbol{q}}_V(t) = \boldsymbol{M}^{-1}(\boldsymbol{q}(t))\boldsymbol{V}(\boldsymbol{q}(t), \dot{\boldsymbol{q}}(t)) \tag{27.10}$$

is often called a cumulative term because it depends on the velocity state. Model-bounded language must not relabel it as a uniquely reconstructed history; no such identifiability contract is provided here.

Drift vs. Control 27.2. Instantaneous vs. Cumulative = Control vs. Drift

Instantaneous effects correspond to the **control** term in the affine decomposition: the muscle torques applied at this instant, mapped through $\boldsymbol{M}^{-1}$ to produce immediate accelerations.

Cumulative effects correspond to the **drift** term: the velocity-dependent and gravity accelerations that arise from the current state $(\boldsymbol{q}, \dot{\boldsymbol{q}})$, which is itself the integrated consequence of the entire prior force history.

The drift field in the golf swing is not an abstract mathematical construct. It is the physical manifestation of every muscle torque, gravitational pull, and elastic recoil that has occurred since the beginning of the backswing, encoded in the current velocities and configuration.

27.3.3 Why the Distinction Matters for the Golf Swing

The golf swing is a fast, ballistic movement. By the time the club is approaching impact, the angular velocities in the system are enormous—shoulder internal rotation rates of 5,000–7,000 deg/s are not unusual in elite players, with wrist uncocking rates even higher. The velocity-dependent torques (centrifugal and Coriolis) at these speeds produce accelerations that dwarf what the muscles can produce instantaneously.

A drift-control ratio is a declared magnitude ratio, not a causal certificate or induced-acceleration partition. This chapter supplies no governed human-golf attribution result satisfying the normative record.

Hirashima et al. [2008] demonstrated this phenomenon clearly in baseball pitching, where the elbow extension and wrist flexion accelerations are produced predominantly by velocity-dependent torques rather than by local joint torques. The proximal trunk and shoulder joint motions, in contrast, are produced primarily by local muscle torques (instantaneous effects). The kinetic chain, in this framework, is a mechanism by which *instantaneous* muscle efforts at proximal joints are converted into *cumulative* velocity-dependent torques that accelerate distal joints without requiring large distal muscle forces.

27.4 Lessons From Throwing: Implications for the Golf Swing

The golf swing and the overarm throw share fundamental dynamical features: both are fast, multi-joint movements performed by serial kinematic chains, and both rely on proximal-to-distal energy transfer to achieve high distal segment speeds. The induced acceleration analyses of baseball pitching performed by Hirashima et al. [2007, 2008], Hirashima and Ohtsuki [2008] provide some of the clearest demonstrations of how the kinetic chain operates at the level of individual forces and accelerations.

27.4.1 Proximal Muscles Drive the System; Distal Joints Are Driven by Cumulative Effects

Using a 13-degree-of-freedom model of the trunk and upper limb, Hirashima et al. [2008] performed induced acceleration analysis on the pitching motions of skilled baseball players. The results were striking:

- **Trunk motion** (forward translation and leftward rotation) was produced primarily by the joint torques at the trunk itself—an *instantaneous direct effect*. The large muscles of the lower body and core generate the initial kinetic energy of the system.

- **Shoulder internal rotation** was produced by a combination of the shoulder joint torque (instantaneous direct effect) and the velocity-dependent torque (cumulative effect), with the velocity-dependent contribution becoming important near ball release.

- **Elbow extension** was produced predominantly by the *velocity-dependent torque*, not by the elbow joint torque. The elbow joint torque actually *decelerated* elbow extension in the final 20 milliseconds before ball release (acting as a brake, likely to protect the joint). The elbow extension acceleration came from the angular velocities of the trunk and upper arm that had accumulated earlier in the throw.

- **Wrist flexion** was also driven primarily by velocity-dependent torques, originating from the forearm angular velocity that was itself a consequence of trunk and shoulder torques applied earlier.

27.4.2 The Route of the Kinetic Chain

Hirashima et al. [2008] decomposed the velocity-dependent torque into its kinematic sources, tracing the chain of causation:

1. Trunk and shoulder joint torques accelerate the trunk and upper arm (instantaneous effect, early phase).

2. The resulting trunk and upper arm angular velocities produce velocity-dependent torques that accelerate elbow extension (cumulative effect, mid-phase).

3. The forearm angular velocity increases, producing additional velocity-dependent torques that accelerate both wrist flexion and (during a brief window near ball release) shoulder internal rotation (cumulative effect, late phase).

This represents a "route" through which the kinetic energy flows, with each link in the chain converting the instantaneous muscle efforts of the proximal joints into the cumulative velocity-dependent torques that accelerate the distal joints. The analogy to the golf swing is direct: the ground reaction forces and hip torques initiate the sequence; the trunk and shoulder torques sustain it; and the velocity-dependent torques—the drift—deliver the final burst of acceleration to the club.

27.4.3 Torque Reversal: The Instantaneous Remote Effect

A separate mechanism, distinct from the cumulative effect, also operates in fast multi-joint movements: the **torque reversal**. Near ball release, the elbow joint torque reverses from extension to flexion (the muscles begin braking elbow extension). While this braking decelerates the elbow, it simultaneously accelerates the wrist into flexion through the instantaneous remote effect (the off-diagonal coupling in $\boldsymbol{M}^{-1}$).

Hirashima [2011] distinguishes the source model's assigned instantaneous term from its velocity-dependent term. Under the model's identifiability contract it is not a unique cause, reconstructed history, or training prescription.

An analogous wrist term in golf is an open hypothesis. It requires the complete normative record and cannot be asserted from a throwing model or drift magnitude alone.

27.5 Lessons From Walking and Clinical Biomechanics

Walking studies provide methodological examples. They do not validate a golf attribution, and cross-task or cross-engine states remain unsupported or unqualified unless the complete record is supplied.

27.5.1 The Foundational Walking Studies

The comprehensive two-part review by Zajac et al. [2002, 2003] established the modern framework for induced acceleration analysis in biomechanics. Their key contributions were:

1. **Separating quantities:** Joint power, segment energy, and acceleration terms answer different questions. A forward model computes a declared ledger but does not identify a causal relationship without additional measurement and intervention evidence.

2. **Separating individual muscle contributions:** By driving a musculoskeletal model with individual muscle forces and computing the resulting segment accelerations, it becomes possible to determine each muscle's contribution to whole-body tasks such as support (vertical center-of-mass acceleration) and forward progression (horizontal center-of-mass acceleration).

3. **Demonstrating counterintuitive muscle functions:** Muscles often accelerate segments and joints they do not cross, and their contributions to whole-body function can differ substantially from what anatomy alone would predict.

27.5.2 Soleus Versus Gastrocnemius: A Case Study in Superposition

The study by Neptune et al. [2001] applied induced acceleration analysis to a forward dynamics simulation of normal walking, focusing on the individual contributions of the soleus (a uniarticular ankle plantar flexor) and the gastrocnemius (a biarticular ankle plantar flexor that also crosses the knee).

Despite both muscles being ankle plantar flexors, their contributions to whole-body function were strikingly different:

- **Soleus**: In late stance and pre-swing, the energy generated by soleus was delivered primarily to the trunk, providing forward progression and vertical support.

- **Gastrocnemius**: In the same phase, nearly all the energy generated by gastrocnemius was delivered to the swing leg, initiating swing rather than propelling the trunk.

This dissociation—two muscles at the same joint with opposite energetic roles—was invisible to inverse dynamics, which can only compute the *net* ankle moment. Only induced acceleration analysis, which tracks each muscle's individual contribution through the constraint forces of the entire chain, could reveal this functional distinction.

For golf, this motivates actuator-map and configuration-sensitivity tests; it is not evidence that similarly located muscles have identified swing roles.

27.5.3 Clinical Applications: Stiff-Legged Gait

Riley and Kerrigan [1999] applied induced acceleration analysis to study stiff-legged gait in stroke patients, demonstrating the clinical utility of the technique. By computing the knee accelerations induced by hip, knee, and ankle moments individually, they showed that the causes of reduced knee flexion in swing were highly variable across patients—some had ankle impairments, others had hip impairments, and individual patients required individualized rehabilitation protocols.

The model can generate patient-specific hypotheses. Algebraic superposition does not identify which impairment is primarily responsible without an additional identifiability contract.

27.5.4 Sit-to-Stand Transfers

Caruthers et al. [2016] reported positive gluteus-maximus and soleus terms and an opposing quadriceps term for selected center-of-mass outputs in a 46-degree-of-freedom, 194-actuator model. These are model-reported contributions, not uniquely identified anatomical sources or intervention effects.

27.6 Implications for the Golf Swing

27.6.1 The Swing as a Superposition Problem

The results reviewed in this chapter frame the golf swing as a **superposition problem**: at each instant, the clubhead acceleration is the sum of contributions from every torque source in the body, weighted by the current configuration of the kinematic chain. This framing provides several actionable insights:

1. **Golf attribution remains open.** A future study must compare declared force partitions, coordinates, contact models, residuals, and output projections before making source or technique claims.

2. **Configuration sensitivity is testable.** The same declared generalized torque can map to different terms as posture changes. This does not establish equal human effort, proper mechanics, or an intervention effect.

3. **Late-downswing club acceleration is dominated by velocity-dependent torques.** The velocity-dependent torques (Coriolis and centrifugal) are the primary drivers of club acceleration during the critical impact-approach phase. These torques are the cumulative consequence of forces applied earlier in the swing. This is the induced acceleration community's version of the statement that drift dominates control in the late downswing.

4. **Muscles at one joint can accelerate (or decelerate) distant segments.** Through dynamic coupling ($\boldsymbol{M}^{-1}$ off-diagonal elements), a hip torque can directly affect club acceleration. This means that the kinetic chain is not simply a sequential hand-off; rather, every active muscle contributes to every joint's acceleration simultaneously, with the relative contributions determined by the current configuration.

27.6.2 A Unified Framework

The induced acceleration framework unifies the observations made throughout this book:

This Book's Framework	IAA Framework	Physical Meaning
Complete drift $\boldsymbol{f}(\boldsymbol{x})$	Declared autonomous terms	Model-defined zero-input evolution
Control $\boldsymbol{G}(\boldsymbol{x})\boldsymbol{u}$	Declared applied-input term	Not identified muscle or intent
Constrained forward operator	Force-to-acceleration map	Coordinate-, contact-, and model-dependent
Term-magnitude comparison	Declared norm and partition	Not a causal or reachability certificate
ZTCF family (Ch. 6)	Zero-control model counterfactual	What the declared effective plant produces u

Both frameworks can use the same frozen-state linear map, but they are not automatically equivalent. The plant, input definition, actuator map, constraints, contact, residual allocation, coordinates, solver qualification, and requested output must match.

27.7 Superposition and Its Limits

A critical caveat, emphasized in Chapter 3 of *The Geometry of Motion, Volume I*, is that superposition holds *instantaneously* but *not* over time. The accelerations induced by individual forces add up exactly at each frozen instant. But if we were to simulate the system forward in time under only one force, the resulting trajectory would diverge from the component of the actual trajectory "due to" that force, because the system's state evolves nonlinearly.

Hirashima and Ohtsuki [2008] showed why isolated diagonal-inertia calculations do not reproduce the coupled model. A full constrained solve is necessary for closure, but closure alone does not identify causal contributions; that stronger claim requires the normative identifiability and intervention contract.

Any golf term must be computed on the declared coupled system. Even then, it remains a model term rather than an identified statement about what each muscle does.

27.8 Summary

Key Principle 27.3. Key Takeaways from Induced Acceleration Analysis

1. **At each instant, forces superpose:** The total acceleration is the sum of individual contributions from each muscle, velocity-dependent torque, and gravity. This is exact and follows directly from the linearity of $\boldsymbol{M}^{-1}(\boldsymbol{q})$.
2. **Dynamic coupling propagates effects across joints:** A torque at one joint induces accelerations at *every* joint, with the magnitude and direction depending on the current posture. The off-diagonal elements of $\boldsymbol{M}^{-1}(\boldsymbol{q})$ encode this coupling.
3. **Instantaneous vs. cumulative:** Current muscle torques produce instantaneous accelerations; past muscle torques produce the current velocities, which generate velocity-dependent accelerations (cumulative effects). In fast movements like the golf swing, cumulative effects dominate the late phases.
4. **Configuration sensitivity is a hypothesis generator:** A modeled term can change with posture, but this does not establish proper mechanics or equal human effort.
5. **Induced acceleration analysis and the control-affine decomposition are equivalent frameworks** developed by different communities. Together, they provide a complete picture of how muscles, physics, and configuration interact to produce the golf swing.

Exercises 27.1. Chapter Exercises

1. For a two-segment planar model (shoulder and elbow in a horizontal plane), write out the 2×2 mass matrix $\boldsymbol{M}(\boldsymbol{q})$ and compute its inverse. Show that the off-diagonal elements of $\boldsymbol{M}^{-1}$ are generally nonzero, confirming that a shoulder torque induces elbow acceleration.

2. Using the double-pendulum model from Chapter 3, compute the velocity-dependent torque $\boldsymbol{V}(\boldsymbol{q}, \dot{\boldsymbol{q}})$ and the gravity torque $\boldsymbol{g}(\boldsymbol{q})$ at a representative mid-downswing posture ($\theta_1 = -45°, \theta_2 = -90°$, with angular velocities $\dot{\theta}_1 = 500$ deg/s$, \dot{\theta}_2 = 200$ deg/s). Compute the induced accelerations from each source and verify that they sum to the total acceleration.

3. The induced acceleration index (IAI) of Challis [2011] quantifies the coupling potential between joints. For a two-segment model of the golfer's lead arm and club, compute the IAI at three postures: (a) early downswing (large lag angle), (b) mid-downswing (moderate lag angle), and (c) near impact (small lag angle). Discuss how the coupling changes and what this implies for wrist release mechanics.

4. Re-express the model-reported elbow result from Hirashima et al. [2008] as a term-magnitude comparison. List the metadata needed before comparing it with a control-affine implementation, and explain why neither wording establishes a unique cause.

5. Consider the finding of Neptune et al. [2001] that soleus and gastrocnemius have opposite energetic effects on the trunk and leg during walking. Explain how this result follows from the off-diagonal structure of $\boldsymbol{M}^{-1}(\boldsymbol{q})$ and the different moment-arm vectors of uniarticular versus biarticular muscles. How might analogous reasoning apply to the pectoralis major (biarticular) versus the subscapularis (uniarticular) during the golf downswing?

Part IX

Motor Control and Learning

Chapter 28

Motor Control I: The Brain as Controller

> The brain is the ultimate controller—a device of considerable subtlety that solves the golf swing problem every time you play, without you ever consciously knowing the equations.
>
> Adapted from Richard Feynman

In Plain Language 28.1. What This Chapter Is About

We've spent the last several chapters learning the physics: the nonlinear dynamics, the drift-dominated regime, the trade-off between control and drift. But now we must face the central mystery: *the brain solves this problem.* Not perfectly—golfers miss—but with high consistency. An elite golfer produces clubhead speeds within 2% of their average, swing after swing Jorgensen [1994b], despite 30–50 millisecond neural delays, sensor noise, and a system where the downswing lasts only 300 milliseconds. This chapter reframes the nervous system as an engineering control system and asks: what kind of controller is the brain? How does it manage to work within such severe constraints? The answer is instructive: the brain does not try to control everything. Instead, it exploits the drift field—the very physics we've been studying—to do most of the work. The brain is not fighting physics; it is using physics as its primary actuator.

28.1 The Control Problem: What the Brain Must Solve

Recall from Chapters 6 and 6 that the golf swing is a control-affine system:

$$\dot{\boldsymbol{x}} = f(\boldsymbol{x}) + G(\boldsymbol{x})\boldsymbol{u}(t) \tag{28.1}$$

The state vector $\boldsymbol{x}$ contains all joint angles and angular velocities. The drift term $f(\boldsymbol{x})$ represents gravity and centrifugal effects—the physics does for free. The control term $G(\boldsymbol{x})\boldsymbol{u}(t)$ represents what the muscles can accomplish. The control input $\boldsymbol{u}(t)$ is the vector of muscle torques, subject to physical bounds: $|\boldsymbol{u}| \leq \boldsymbol{u}_{\max}$.

The brain's problem: compute $\boldsymbol{u}(t)$ as a function of time to produce a desired motion that puts the ball at the target.

This is not a simple problem. It is a *nonlinear optimal control problem in real time*, subject to harsh constraints:

Constraint Insight 28.1. Constraints on the Brain's Control Problem

1. **Neural delay:** Motor commands must travel from the motor cortex to the muscles via the spinal cord. Typical conduction time: 30–50 milliseconds. If the downswing lasts 300 milliseconds, the motor command that initiates the acceleration phase must be generated 50 milliseconds before the transition.

2. **Activation dynamics:** Once a motor neuron fires, the muscle doesn't instantly produce torque. Muscle force rises with a time constant of 50–100 milliseconds. The brain cannot produce torque changes faster than this.

3. **Sensory feedback delay:** Sensorimotor feedback is not one serial channel. Short-latency stretch responses begin at roughly 20–45 ms, long-latency responses at roughly 50–100 ms, and voluntary responses at greater than 100 ms in the upper limb; visually guided responses usually enter later and depend on the task and response being measured [Kurtzer, 2014, Pruszynski and Scott, 2012]. These are exemplar onset bands, not universal golf-swing constants.

4. **Noisy sensors:** Proprioceptive acuity varies by joint. Illustrative estimates suggest detection thresholds on the order of 1–2° at the wrist and 3–5° at the shoulder, though exact values depend on testing methodology [Proske and Gandevia, 2012]. Vision has high acuity but slow feedback. The nervous system must work with uncertain, delayed measurements.

5. **Noisy actuators:** Muscle force variability is 5–10% even when you're trying to produce exactly the same torque twice. This is called *signal-dependent noise* and increases with force level.

6. **Bounded control:** Maximum torques are limited. The shoulder can produce roughly 100–200 N·m of torque, the elbow 100–150 N·m, the wrist 30–50 N·m Enoka [2002]. The brain cannot exceed these.

Example 28.1. Delays in the Golf Swing

In an illustrative 250–300 ms downswing, several parallel pathways may contribute:

- Short-latency stretch response: approximately 20–45 ms
- Long-latency response: approximately 50–100 ms
- Voluntary response: greater than 100 ms
- Muscle activation and force development: additional, muscle- and task-dependent dynamics
- Visually guided response: typically late relative to the downswing, with timing and mechanical effect dependent on stimulus, response definition, and swing phase

The onset bands must not be added into one "round-trip" total: the pathways are partly parallel and close through different circuitry [Kurtzer, 2014, Pruszynski and Scott, 2012]. Timing alone also does not establish response authority—whether a response begins before impact is different from whether it can materially change club or body state. The defensible conclusion is narrower: late visually guided gross corrections are strongly constrained, while short- and long-latency feedback may still regulate muscle activity, limb impedance, or a local trajectory error. Which effects matter in golf remains a phase- and task-specific empirical question.

28.1.1 Feedforward and Feedback: Complementary Contributions

Feedforward and feedback are useful analytical components, but biological control can combine them rather than choosing one exclusively:

Definition 28.1. Feedback (Closed-Loop) Control

The controller measures the current state $\boldsymbol{x}(t)$, compares it to the desired state $\boldsymbol{x}_d(t)$, computes an error $\boldsymbol{e}(t) = \boldsymbol{x}_d(t) - \boldsymbol{x}(t)$, and adjusts the control input based on this error:

$$\boldsymbol{u}(t) = \boldsymbol{K}\boldsymbol{e}(t) = \boldsymbol{K}[\boldsymbol{x}_d(t) - \boldsymbol{x}(t)] \tag{28.2}$$

where $\boldsymbol{K}$ is a gain matrix.
This is reactive: if things are going wrong, the controller corrects.

Definition 28.2. Feedforward (Open-Loop) Control

The controller uses a *model* of the system to compute the required control input in advance:

$$\boldsymbol{u}(t) = \boldsymbol{u}_{\text{planned}}(t) \tag{28.3}$$

This component is prepared before the corresponding sensory consequence arrives. It can operate alongside feedback during execution.

The useful hypothesis is that learned preparation and feedforward commands carry substantial responsibility for the downswing, while feedback remains pathway- and phase-dependent.

For example, a new visual error presented at $t = 100$ ms may not produce a mechanically important correction before a $t \approx 300$ ms impact. That scenario bounds one late visually guided loop; it does not rule out faster proprioceptive pathways or responses already in progress. Long-latency responses are task-dependent and can incorporate limb mechanics and behavioral goals [Kurtzer, 2014, Pruszynski and Scott, 2012].

Myth vs. Reality 28.1. What Timing Does and Does Not Establish

Unsupported extreme: A golfer continuously sees clubface error and fully replans the downswing in real time.
Bounded interpretation: A late visual perturbation is unlikely to support a com-

plete replan before impact. Faster feedback can still modulate the ongoing action, and post-shot feedback supports learning. The response authority of each pathway depends on perturbation onset, swing phase, task, response measure, and mechanical outcome.

Short- and long-latency proprioceptive responses can begin within a downswing, but onset is not equivalent to a successful clubhead correction. Establishing their golf-specific role requires time-locked perturbations, muscle and motion measurements, and an outcome definition. Until golf-specific perturbation evidence is available, claims about elite golfers suppressing or relying on a pathway are hypotheses, not established facts.

This timing picture motivates careful study of pre-swing preparation, initial conditions, impedance, and learned commands without assuming that execution is open-loop or deterministic.

28.2 Internal Models: The Brain's Simulator

If the golf swing is feedforward, how does the brain know what command to send? It has a model. The brain maintains an *internal model*—a neural simulation of the body's dynamics.

Definition 28.3. Forward Model

A forward model predicts the sensory consequences of a motor command. Given the current state $\boldsymbol{x}(t)$ and a motor command $\boldsymbol{u}(t)$, the forward model predicts the next state:

$$\hat{\boldsymbol{x}}(t+\Delta t) = \hat{f}(\boldsymbol{x}(t)) + \hat{G}(\boldsymbol{x}(t))\boldsymbol{u}(t) \tag{28.4}$$

Mathematically, this is analogous to solving the nonlinear dynamics equation. The neural implementation is of course very different from symbolic computation, but the functional result is similar: the brain generates predictions consistent with physical dynamics.

Definition 28.4. Inverse Model

An inverse model computes the motor command needed to achieve a desired sensory outcome. Given a desired state change $\boldsymbol{x}_d(t)$, the inverse model computes:

$$\boldsymbol{u}(t) = \hat{G}(\boldsymbol{x}(t))^{-1}[\dot{\boldsymbol{x}}_d(t) - \hat{f}(\boldsymbol{x}(t))] \tag{28.5}$$

This is the inverse dynamics problem: "What muscles do I fire to move the club from here to there?"

The forward and inverse models are intimately related. The brain learns them together. A forward model is easy to differentiate numerically to obtain an approximate inverse model: $\boldsymbol{u} \approx \hat{G}^{-1}(\boldsymbol{x}_d - \hat{f})$.

28.2.1 The Cerebellum as a Candidate Substrate for Internal Models

Where in the brain do these internal models live? The leading candidate is the *cerebellum*, a structure at the base of the brain.

The cerebellum is notable for several reasons:

Key Principle 28.1. Cerebellar Architecture

- **Size:** The cerebellum is only ∼10% of the brain's volume, but contains ∼80% of all neurons—about 69 billion of the brain's 86 billion, the overwhelming majority of them granule cells Azevedo et al. [2009]. That makes it the most neuron-dense structure in the brain. It also contains roughly 15 million Purkinje cells, the large output neurons that project to the deep cerebellar nuclei Eccles [1973].
- **Computational density:** This makes the cerebellum the most densely packed neural tissue in the brain—a massively parallel neural processor.
- **Simple circuit:** The cerebellar circuit is highly stereotyped and mathematically simple. It's one of the few neural circuits we nearly understand.
- **Learning rule:** The cerebellum uses a clear, identified learning mechanism: long-term depression (LTD) of synapses. When a Purkinje cell receives simultaneous input from a climbing fiber (error signal) and parallel fibers (motor plan), the parallel fiber–Purkinje synapses are weakened. This is an error-correction learning rule.
- **Timing:** The cerebellum operates fast, with learning that can occur within a single trial for simple adaptation tasks.

Cerebellar damage produces catastrophic loss of motor coordination—a condition called *ataxia*. Patients with cerebellar ataxia cannot perform smooth, coordinated movements. They cannot hit a golf ball. They cannot reach smoothly for a cup. This is because they have lost their internal model of how their body works.

The cerebellar learning rule is thought to implement the following algorithm:

$$\Delta w_{ij} = -\eta \cdot e(t) \cdot x_i(t) \cdot y_j(t) \tag{28.6}$$

where w_{ij} is the synaptic weight between parallel fiber i and Purkinje cell j, $e(t)$ is the climbing fiber error signal, $x_i(t)$ is the firing rate of parallel fiber i, and $y_j(t)$ is the firing rate of the Purkinje cell. The negative sign indicates that errors drive depression of relevant synapses. Over time, this makes the forward model more accurate.

This is a simplified form of the Marr-Albus-Ito hypothesis. More precisely, the climbing fiber carries an error signal, and parallel fiber synapses that are co-active with climbing fiber input undergo long-term depression (LTD). The learning rule above captures the essential structure: weights decrease when error and pre-synaptic activity coincide [Wolpert et al., 1995, Flanagan et al., 2003, Kawato, 1999].

28.2.2 Why Internal Models Matter

Without an internal model, the brain cannot solve the golf swing. Here's why:

Example 28.2. Controlling Without a Model

Suppose you didn't have an internal model. You swing the club and see (after 150 ms) that the clubface was open. You generate a correction. But this correction is based on what your muscles did 150 milliseconds ago, when the club was in a different position. The dynamics are nonlinear—the effect of a torque at position $\boldsymbol{x}_1$ is different from the effect of the same torque at position $\boldsymbol{x}_2$. Applying a correction based on delayed feedback at the wrong state is futile.
With an internal model, you *predict* what will happen before it happens. You mentally simulate the swing. If the simulation predicts an open clubface, you adjust the plan *before* you swing.

The forward model enables *predictive control*. The inverse model enables *planning*. Together, they enable the feedforward computation of motor commands that reliably produce desired outcomes.

28.3 The Predictive Processing Framework

Modern neuroscience has converged on a powerful unifying idea: the brain is fundamentally a *prediction machine*. This framework, developed by Karl Friston and others, is called the *free energy principle* and the *predictive processing* framework Friston [2010], Clark [2013].

Definition 28.5. Predictive Processing

The brain doesn't passively receive sensory information. Instead, it constantly generates predictions about what it expects to sense. Perception works by comparing predictions with actual sensations. When they match, the brain's model of the world is good. When they don't match, a *prediction error* occurs, and the brain updates its model.

For the golf swing, this framework explains several phenomena:

1. **Pre-swing planning:** Before you swing, your brain generates a predicted trajectory. This is your forward model running forward: "If I fire these muscles at these times, the club will follow this path, and the ball will land there." This is an *active inference*—you're inferring which motor command will make the world match your prediction (the target).

2. **During-swing monitoring:** As you swing, your brain continuously compares predicted sensory signals with actual signals. The proprioceptive feedback is compared against the forward model's predictions. If they match, all is well. If they don't, there's a prediction error.

3. **Post-swing learning:** After the swing, a large prediction error is available: the ball lands at location $\boldsymbol{x}_{\text{actual}}$ instead of the predicted location $\boldsymbol{x}_{\text{predicted}}$. This error drives learning. The brain updates its internal model to reduce the error on the next swing.

4. **Limited time for some corrections:** A 300 ms downswing constrains late voluntary and visually guided responses, while faster parallel pathways can still contribute. Prior learning is important, but the number and effect of within-swing responses cannot be inferred by dividing duration by one nominal delay.

This framework has far-reaching implications. It says that action is not fundamentally different from perception. Both work through prediction and error correction. The goal of a motor action is to make the actual sensory consequences match your predicted sensory consequences. For the golf swing, the predicted sensory consequence is the *feeling* of a well-struck shot—the proprioceptive pattern that you've learned corresponds to good contact.

28.4 Ideomotor Theory: Think the Result, Not the Process

William James, in his 1890 *Principles of Psychology*, proposed a radical idea: **actions are represented by their effects, not by their motor commands**. When you think about throwing a ball, you think about the trajectory of the ball, not the sequence of muscle contractions. Your motor system then computes the commands needed to produce that trajectory.

Modern ideomotor theory Prinz [1997], Hommel et al. [2001] has formalized this insight:

Theorem 28.1. Ideomotor Principle (Theoretical Framework)

According to modern ideomotor theory—a framework with substantial experimental support but ongoing debate regarding its scope and mechanisms—an action is coded as a set of desired sensory consequences (the effect). When the action goal is activated, the motor system retrieves the motor commands that, in the past, produced those consequences. The action is executed by commands designed to minimize discrepancy between the predicted sensory consequences and the desired consequences.

The ideomotor principle connects to the drift field. In the ZTCF family framework, the brain doesn't need to control every detail of the trajectory. Instead, it specifies the initial conditions and lets physics (the drift field) carry the system to the outcome. In ideomotor terms, the brain specifies the desired sensory consequence (impact with the ball, ball trajectory), and the motor system retrieves the commands that, given physics, produce that consequence [Prinz, 1997, Hommel et al., 2001, Wolfensteller and Ruge, 2011].

28.5 The Nervous System Architecture: Hardware Specifications

To understand how the brain controls movement, we need to understand its hardware architecture. The nervous system is not a single, monolithic controller. It's a hierarchy of controllers, each operating at a different timescale and with different capabilities.

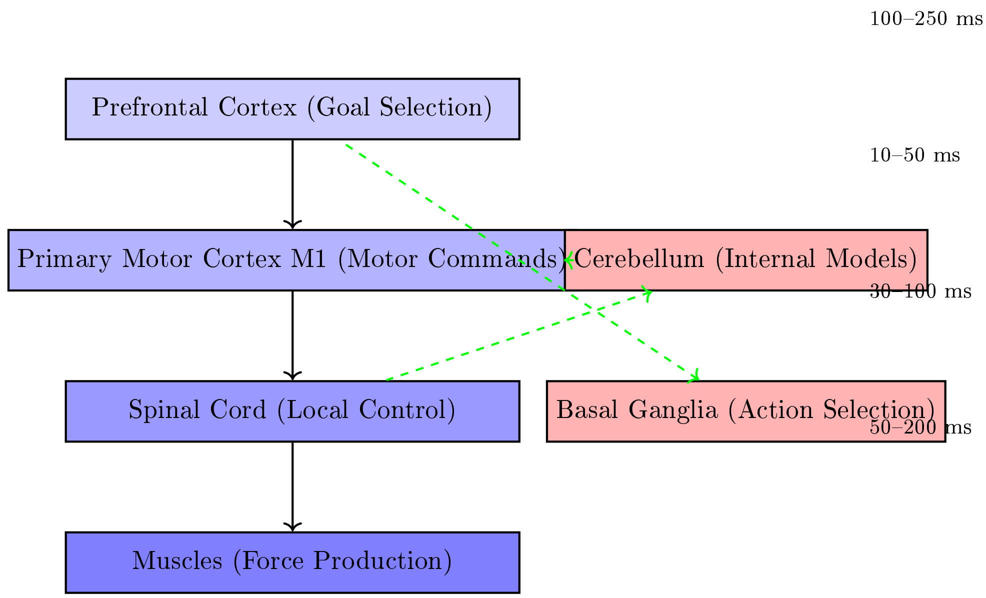


Figure 28.1: Motor Control Hierarchy in the Brain. Top-Down Commands Flow From Prefrontal Cortex (Goal Selection) Through Motor Cortex to Spinal Circuits and Muscles. The Cerebellum (Red, Right) Monitors Movement and Provides Error Signals. The Basal Ganglia Select Which Action to Execute. Each Level Operates on a Different Timescale.

28.5.1 The Motor Cortex

The primary motor cortex (M1) is located in the frontal lobe, just in front of the central sulcus. The corticospinal tract contains approximately one million fibers per hemisphere (estimates vary by source and methodology), with axons projecting down through the spinal cord to synapse onto motor neurons and interneurons.

Definition 28.6. Corticospinal Tract

The major pathway for voluntary motor control. Fibers originate in the motor cortex, travel through the brain and brainstem, cross the midline (decussate) in the medulla, and descend the spinal cord. Each M1 neuron controls the muscles on the *opposite* side of the body.

Signal transmission in these fibers is fast—roughly 120 m/s in myelinated fibers. This means a motor command takes about 30 ms to reach the hand muscles, and about 50 ms to reach the foot muscles [Kandel et al., 2013].

An important principle:

Key Principle 28.2. Motor Cortex Population Coding

The motor cortex doesn't work one neuron per muscle. Instead, it uses *population coding*. Thousands of neurons, each with slightly different directional preferences, are active simultaneously. The vector sum of their activity encodes the direction and magnitude of the desired motion. This population code is robust: loss of individual neurons has little effect because the information is distributed.

28.5.2 The Spinal Cord: Local Control Centers

The spinal cord is not just a cable carrying commands down and feedback up. It contains *local circuits* that can coordinate movement independently of the brain.

Definition 28.7. Central Pattern Generator

A neural circuit that produces rhythmic output without rhythmic input. CPGs in the spinal cord generate patterns like walking, swimming, or scratching. They can operate autonomously, though they're normally modulated by brain signals.

Definition 28.8. Stretch Reflex (Monosynaptic Reflex)

A rapid response to muscle stretch. When a muscle is stretched, spindle receptors send a signal directly to motor neurons in the spinal cord, which fire and contract the muscle. This entire loop happens in about 30 milliseconds, *without brain involvement*. The brain doesn't know about it until after it happens.

For the golf swing, short-latency spinal pathways may contribute local stabilizing responses to a perturbation. Longer-latency supraspinal and visually guided contributions would enter later. Whether any pathway materially changes the club state requires a defined perturbation, phase, and measured outcome.

28.5.3 The Basal Ganglia: Action Selection

The basal ganglia are a set of nuclei deep in the brain. Their primary function is *action selection*—deciding which motor program to execute.

Definition 28.9. Motor Program

A learned sequence of muscle activations that produces a stereotyped action (like a golf swing). Once selected and initiated, the program unfolds with minimal brain intervention.

The basal ganglia use a neurochemical signal called *dopamine* to encode reward prediction errors. When you execute a motor program and it produces a better-than-expected outcome (a good shot), dopamine neurons fire. This signal reinforces the basal ganglia circuits that led to that action. Over time, actions that produce rewards become more likely.

Example 28.3. Basal Ganglia in Golf

The pre-shot routine involves several motor programs:

1. The waggle (small oscillations of the club)
2. The step(s) to position
3. The alignment checks
4. The final address

Each of these is a learned motor program. The basal ganglia select which routine to execute and in what order. Once the routine is selected, it unfolds automatically.
When you hit a great shot, dopamine neurons fire (reward). The basal ganglia strengthen the circuits that selected that particular pre-shot routine. Over time, the routine becomes more automatic.
When you hit a poor shot, dopamine neurons are quiet (no reward). The circuits that selected that routine weaken. You're less likely to use that routine next time.
This is *reinforcement learning* implemented in neural circuits.

Parkinson's disease, which damages dopamine neurons, produces severe difficulty in action selection. Patients know what they want to do, but they cannot initiate movement. They can stand frozen, unable to take a step. This illustrates how critical the basal ganglia are for movement initiation.

28.5.4 Motor Unit Recruitment: Henneman's Size Principle

A motor unit is a motor neuron and all the muscle fibers it innervates. A single motor neuron controls 10–1000 muscle fibers, depending on the muscle. The smaller the motor neuron, the fewer fibers it controls.

Key Principle 28.3. Henneman's Size Principle

Motor neurons are recruited in order of size: small (slow, weak) units first, large (fast, strong) units last Henneman et al. [1965]. This ordering is automatic and doesn't require brain control. It's determined by the electrophysiology of the motor neuron pool.

This principle has important consequences:

1. **You can't produce maximum force instantly.** To achieve high forces, you must recruit the large motor units. But you can't recruit them without first recruiting all the small units. This takes time—typically 100–500 milliseconds.

2. **At low effort, you have precision control.** With only small units active, forces are fine-graded and controllable. This is why you can write with a pen or play piano (low forces, high precision).

3. **High effort can constrain force grading.** Recruiting larger motor units changes the available force increments and signal-dependent variability. The effect on move-

ment precision depends on task and coordination; it is not a universal proof that maximum-effort movement is ballistic.

4. **The golf swing transitions from lower to higher modeled force demand.** During setup and the backswing, forces are often lower than in the downswing. Higher effort may constrain force grading, but the control architecture and feedback contribution remain task- and phase-dependent.

28.6 Bandwidth Limitations: The Brain's Processing Speed

A fundamental constraint on any controller is its bandwidth—the rate at which it can process information and generate corrections.

Definition 28.10. Control Bandwidth

The maximum frequency at which a controller can detect errors and generate corrective responses. For the brain, the limiting factor is not neural transmission speed (which is fast—30–120 m/s) but the time taken to process information and compute responses.

The conscious brain updates motor plans at roughly 4–10 Hz (one update every 100–250 ms). These values are illustrative estimates from reaction-time studies; the exact rate depends on task complexity and attentional demands. Visual flicker fusion occurs at ∼60 Hz, but conscious decision-making and motor replanning are far slower. This mismatch is critical: sensory data arrives faster than the conscious brain can act on it.

Example 28.4. Flicker Fusion Frequency

A light flickering at 60 Hz appears steady to your eye, even though it's actually off half the time. This is because 60 Hz exceeds your conscious bandwidth. Each on–off cycle is too fast to perceive as distinct.

For the golf swing, this bandwidth limitation is catastrophic:

Constraint Insight 28.2. The Bandwidth Problem in Golf

A 300 millisecond downswing at 10 Hz bandwidth means the brain can process roughly 3 distinct time windows during the swing:

- $t = 0$–100 ms: transition phase (beginning of acceleration)
- $t = 100$–200 ms: acceleration phase (maximum force)
- $t = 200$–300 ms: deceleration phase (approaching impact)

This three-window sketch is a coarse pedagogical partition, not a count of feedback cycles. Different feedback pathways overlap these windows and have different response authority.
It is tempting to blame a bandwidth mismatch — a controller too slow for a fast plant — but the numbers do not support that reading. A driver shaft's first bending

mode is about 3–5 Hz, and the gross swing dynamics are slower still: the whole downswing is a single acceleration–deceleration cycle lasting roughly a quarter of a second. The plant is therefore *slower* than the 10–40 Hz at which the nervous system can modulate its output, not faster.

Transport delay is an important constraint, but it must be interpreted by pathway. A short-latency response can begin at roughly 20–45 ms, a long-latency response at 50–100 ms, and a voluntary response at greater than 100 ms [Kurtzer, 2014, Pruszynski and Scott, 2012]. Late visually guided gross corrections may have little response authority before impact, while earlier proprioceptive feedback may modulate impedance or local trajectory. Timing alone does not determine the mechanical outcome.

This is the classic problem of a control loop whose delay is comparable to the duration of the task itself.

One model-based response is to exploit drift and mechanical structure. When a motion is partly drift-dominated, a slow controller can perform better by preparing initial conditions, stiffness, and feedforward commands rather than relying on moment-by-moment correction. This does not mean initial conditions are the only thing that matters [Todorov, 2005, Shadmehr and Wise, 2005].

28.7 The Brain's Processing Power: Energy and Computation

How much "horsepower" does the brain have for controlling movement?

Key Principle 28.4. Brain Power Budget

- **Energy consumption:** The brain uses roughly 20 watts of power at rest, about 20% of the body's total metabolic rate. This seems like a lot, but it's actually quite efficient for the amount of computation being performed.

- **Computational capacity:** Estimates of the brain's computational capacity vary widely depending on what counts as an 'operation.' If each of the brain's $\sim 10^{14}$ synapses fires at $\sim$10 Hz, the brain performs roughly 10^{15} synaptic operations per second [Wolpert et al., 2011]. Under more liberal definitions (including subthreshold integration), estimates reach 10^{16}–10^{18}. These numbers are uncertain, but the order of magnitude is instructive.

- **Per-neuron computation:** With $\sim$86 billion neurons Azevedo et al. [2009], this yields roughly 10^4–10^5 synaptic operations per neuron per second. Direct comparison with digital processors is misleading: neurons are analog, parallel, and asynchronous, while CPUs/GPUs perform discrete, clocked operations. The brain compensates for slower individual operations with massive parallelism ($\sim$86 billion units operating simultaneously).

The brain need not solve the explicit engineering optimization written here at every instant. Learned policies, state estimation, prediction, and task-dependent feedback can all contribute online.

Example 28.5. Online Optimization vs. Offline Learning

A robotic arm might use real-time trajectory optimization: at each time step (every millisecond), solve a local optimal control problem to compute the next motor command. This requires substantial computation per time step.
Biological control can learn over repeated trials and reuse prepared policies during execution. That does not imply simple playback: prediction and task-dependent feedback can modify the ongoing command.
Robot and human adaptation rates depend on controller, task, perturbation, and learning history. This comparison is conceptual; it is not a measured energy or adaptation-time benchmark.

Practice can shape prepared commands and feedback policies. During a swing, their relative contributions remain an experimental question rather than a strict offline/online divide.

Key Principle 28.5. Key Takeaways

1. Learned preparation and feedforward commands are likely important in a short downswing, but feedback remains active through parallel pathways. The mechanical response authority of each pathway depends on latency, phase, task, and outcome.

2. The brain controls movement using internal models—forward models that predict sensory consequences, and inverse models that compute motor commands. The cerebellum is the leading candidate neural substrate for these models, supported by substantial evidence from lesion studies and neuroimaging, though the full picture likely involves multiple brain regions.

3. The brain operates as a predictive processing system. It constantly generates predictions and compares them to actual sensory input. Prediction errors drive learning.

4. Actions are represented by their effects (ideomotor theory), not by motor commands. The brain specifies a desired outcome, and the motor system retrieves the commands that produce that outcome.

5. The nervous system is a hierarchy: the motor cortex (high-level planning), the spinal cord (local control), and the basal ganglia (action selection) all contribute.

6. Delay and plant dynamics jointly constrain control. Late visually guided gross corrections may have little time before impact; faster proprioceptive pathways may still modulate the ongoing action. Golf-specific perturbation evidence is needed to quantify those contributions.

7. Practice shapes prepared commands, state estimation, and feedback policies that can all contribute during performance; this is not a strict offline-

learning/online-playback split.

Exercises 28.1. Chapter Exercises

1. The nervous system modulates output at 10–40 Hz, while the shaft's first bending mode is only 3–5 Hz — so the controller is *faster* than the plant. Why, then, can the golfer not correct the swing in flight? Frame your answer in terms of loop delay against movement duration rather than bandwidth, and state which reflex loops could close within a 250 ms downswing and what they could actually accomplish.

2. A professional golfer produces clubhead speeds with a coefficient of variation of 1–3%. Given the neural noise (5–10% muscle force variability) and sensory noise ($\sim$1–5 degree proprioceptive acuity), explain how this high consistency is possible. Hint: what does the drift field contribute?

3. Suppose you wanted to teach a robot to swing a golf club using online optimal control (computing the motor command at each millisecond). What would be the minimum computation time required? Assume the downswing lasts 300 ms and the robot must solve a 20-dimensional optimal control problem at each millisecond.

4. In Section 28.5.4, we discussed Henneman's size principle: motor units are recruited in order of size. Why is this principle important for control? What would happen if large motor units were recruited first?

5. Compare a short-latency proprioceptive response, a long-latency response, and a late visually guided response during a 300 ms downswing. What additional measurement is needed to determine whether each response has useful mechanical authority?

6. According to ideomotor theory, what should a golf coach emphasize: mechanical details of the swing, or the desired outcome and feel? Why?

7. The cerebellum contains 70% of all brain neurons but is only 10% of brain volume. What does this suggest about the cerebellum's computational strategy?

8. A golfer with cerebellar damage cannot hit a golf ball. They can move their arms, and they can understand the task. What is missing? How does this relate to the concepts of forward and inverse models?

9. Sketch a block diagram of the golf swing control system. Include the motor cortex, motor command, muscles, dynamics, sensory feedback, and feedback delays. Indicate which components operate on which timescale.

10. Compare feedforward and feedback components in a golf swing. Which pathways could respond within a 250–300 ms downswing, and what experiment would distinguish response onset from useful mechanical correction?

Chapter 29

Motor Control II: Learning the Swing

In Plain Language 29.1. What This Chapter Is About

A well-struck shot leaves several kinds of information: the ball's flight, contact sound, pressure at the hands, effort, balance and the golfer's judgment of the result. Learning involves finding which information predicts useful outcomes, then changing preparation and action so those outcomes become more reliable. The sensation of a good swing can help, but reproducing a sensation does not guarantee reproducing its mechanics or result.

This chapter connects feel to physical state, prediction, task goals and practice design. Its central distinction is between performing well now and acquiring a capability that survives a delay or a change of conditions. Neither a satisfying shot nor a compelling explanation of it is sufficient evidence of lasting learning.

29.1 Stages of Motor Learning

Fitts and Posner's cognitive, associative and autonomous stages describe changing demands on attention and skill organization [Fitts and Posner, 1967]. They are useful descriptions, not a fixed timetable, a neurological diagnosis or a guarantee that improvement proceeds monotonically. A golfer may execute a familiar putt automatically while deliberately learning an unfamiliar bunker shot.

Definition 29.1. Cognitive Stage

The learner is establishing the task, relevant cues and possible actions. Instructions and explicit strategies may occupy substantial attention. What matters is the difficulty of organizing this task for this learner, not the number of weeks since a first lesson.

Definition 29.2. Associative Stage

The learner is refining relationships among preparation, action, feedback and results. Some components become reliable while others remain uncertain. Improvement can involve a better strategy, more accurate prediction, different coordination or better use of feedback.

Definition 29.3. Autonomous Stage

Execution demands less deliberate supervision in a familiar context. Automaticity does not mean perfect repetition, immunity to pressure or absence of neural feedback. A change of equipment, lie or task can require renewed attention.

Example 29.1. The Three Stages in Golf

Consider learning distance control in putting. Initially the golfer may consciously compare stroke size and roll distance. Later, a familiar distance can evoke an effective preparation without that calculation. To assess the change, record both distance error and the attention required. A single lower error does not establish a stage transition; fatigue, warm-up and helpful instructions can temporarily alter performance.

There is no justified one-to-one sequence from prefrontal cortex to premotor cortex to cerebellum. Behavioral stages do not identify the neural structures responsible for them. Likewise, variability need not fall in every joint as skill improves. We need a task-level account before deciding which variation is beneficial or harmful.

29.1.1 Bernstein's Problem: Coordination and Task Redundancy

Definition 29.4. Bernstein's Problem

A body has many mechanically coupled degrees of freedom and many possible muscle actions. A task usually constrains only some combinations of these variables. The coordination problem is to organize feasible actions that achieve the task while respecting physical and physiological constraints [Bernstein, 1967].

The number of degrees of freedom depends on the model: its joints, constraints, flexible structures and contact assumptions. Position, velocity and acceleration are not three independently selectable commands at every joint. They are related by differentiation and equations of motion. Muscle redundancy introduces another distinction: different activation patterns may produce similar net joint torques while differing in force, stiffness and metabolic demand.

Key Principle 29.1. Three Coordination Strategies

Restricting joint motion, coupling motions and exploiting mechanical interactions are possible strategies. They are not obligatory beginner, intermediate and expert stages. Restricting motion can simplify a task but increase effort; allowing more motion can improve adaptability or make delivery less reliable. No strategy is most energy-efficient without a specified system, task and energy measure.

Example 29.2. Bernstein's Problem in Golf

Suppose two small joint deviations contribute to a scalar delivery error as $y = q_1 + q_2$. All pairs satisfying $q_2 = -q_1$ cancel in this local task model. A golfer can vary both joints while preserving delivery. If a different club or operating point changes the sensitivity to $y = q_1 + 2q_2$, the same cancellation no longer works. A coordination pattern is task- and context-dependent.

More generally, let delivery be $\boldsymbol{y} = G(\boldsymbol{x}, c)$, where c specifies club, lie and other conditions. Near a nominal state, $\delta\boldsymbol{y} \approx J\delta\boldsymbol{x}$, with $J = \partial G/\partial\boldsymbol{x}$. For zero-mean deviations with covariance Σ_x,

$$\Sigma_y \approx J\Sigma_x J^{\mathsf{T}}.$$

For a scalar output with mean error μ_y, mean squared task error is $\mu_y^2 + \mathrm{Var}(y)$. Precision and bias must both be assessed. If $\boldsymbol{x}$ mixes angles, velocities and forces, its covariance has mixed units; a raw trace is not a coordinate-independent measure of skill.

Even with two identically scaled angular coordinates, smaller joint variability can mean greater task variability. For $J = (1, 1)$, consider manufactured covariance matrices in squared degrees:

$$\Sigma_A = \begin{pmatrix} 1 & 0.9 \\ 0.9 & 1 \end{pmatrix}, \qquad \Sigma_B = \begin{pmatrix} 4 & -3.8 \\ -3.8 & 4 \end{pmatrix}.$$

Both are positive definite. Although $\operatorname{tr}\Sigma_A = 2 < 8 = \operatorname{tr}\Sigma_B$, the task variances are 3.8 and 0.4 square degrees respectively. The second pattern has greater joint variation but better cancellation. This proves a limitation of a variability metric, not that a particular golfer uses a neural synergy. Measurement error, shared mechanics and task constraints can also produce covariance.

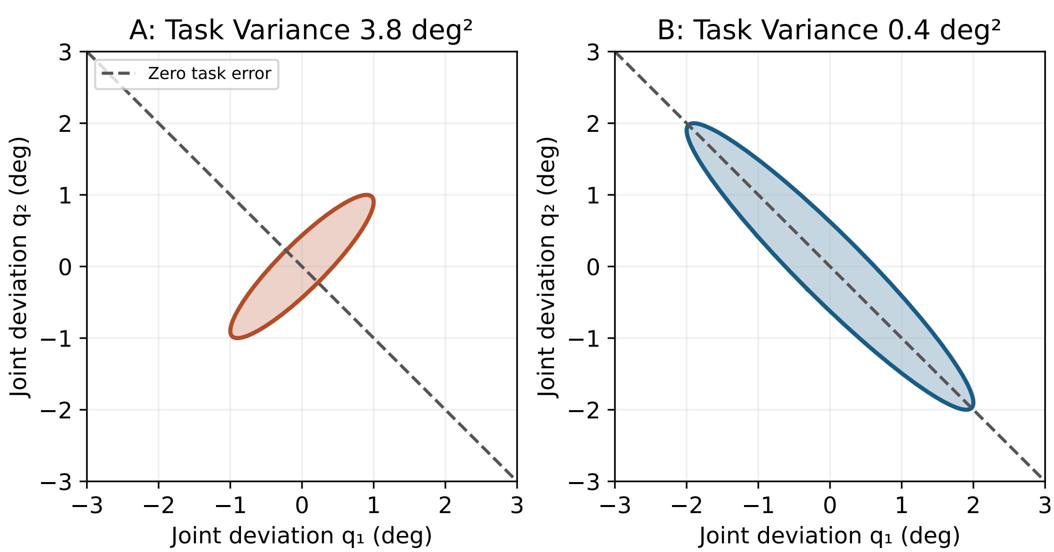


Figure 29.1: Manufactured Covariance Ellipses for the Task $y = q_1 + q_2$. Greater Joint Variation Can Coexist With Smaller Task Variance. Ellipses Have Unit Mahalanobis Radius; They Are Not Human Measurements.

29.2 The Feel of a Good Shot: Proprioception as a Control Cue

Key Principle 29.2. The Setpoint Hypothesis: Feel as a Testable Cue

A remembered sensation may serve as a cue for preparing or evaluating a swing. Calling it a reference or setpoint is a modeling choice. The hypothesis becomes useful when it predicts observable changes under altered sensory feedback or task conditions. It does not establish that the nervous system stores and replays one complete sensory trajectory.

29.2.1 Proprioception: Signals and State Estimates

Key Principle 29.3. Proprioceptive Receptors

Muscle spindle, tendon, skin and joint afferents contribute information related to movement, position and force. Their signals depend on physiological and mechanical conditions. Perception combines populations of signals with central information; there is no single receptor that directly reports a perfectly calibrated joint angle or total muscle force [Proske and Gandevia, 2012].

The golfer's report of feel also includes touch, vision, sound, effort and vestibular information. Vestibular signals concern head motion and orientation, including angular motion and gravito-inertial acceleration; they are not equivalent to eye position. Holding the gaze near the ball does not establish a stationary head or silence vestibular input. A joint's perceptual accuracy cannot be assigned a universal threshold independently of movement, load and measurement protocol.

Let the physical state include configuration $\boldsymbol{q}$, velocity $\boldsymbol{v}$, activation $\boldsymbol{a}$ and relevant elastic or contact states $\boldsymbol{z}$:

$$\boldsymbol{x} = (\boldsymbol{q}, \boldsymbol{v}, \boldsymbol{a}, \boldsymbol{z}), \qquad \dot{\boldsymbol{x}} = f(\boldsymbol{x}, \boldsymbol{u}, c).$$

An illustrative sensory channel is

$$\boldsymbol{s}_t = h(\boldsymbol{x}_{t-d}, c) + \boldsymbol{\eta}_t.$$

Here d is that channel's delay and $\boldsymbol{\eta}_t$ represents noise or unmodeled influences. An estimator combines observations, command history and a forward model to form $\hat{\boldsymbol{x}}_t$. Neither the observer nor the golfer is thereby granted direct access to the complete physical state. Different channels have different delays and reliabilities; treating them as one synchronous measurement can create apparent prediction errors.

An elementary Gaussian example clarifies the role of reliability. Suppose an unbiased scalar prediction is $20°$ with variance $9\,\mathrm{deg}^2$, and an independent unbiased observation is $22°$ with noise variance $1\,\mathrm{deg}^2$. Combining their precisions gives an estimate of $21.8°$ and variance $0.9\,\mathrm{deg}^2$. This is a calculation under stated assumptions, not an asserted neural algorithm. If the prediction and observation share an error, the independence assumption fails and confidence can be overstated.

29.2.2 Reference Trajectories, Predictions and Policies

Constraint Insight 29.1. Pedagogical Simplification

A desired trajectory, a predicted trajectory and an estimated trajectory answer different questions: what should happen, what is expected to happen, and what appears to be happening. A successful controller need not make all three identical. A reference trajectory is one possible representation; a policy can choose actions from estimated state, task and context without replaying one fixed path.

Write a policy as $\boldsymbol{u}_t = \pi_\theta(\hat{\boldsymbol{x}}_t, c, g, t)$, with goal g and adjustable parameters θ. Learning may change the estimator, the policy, the valuation of outcomes or the preparation of the initial state. A familiar feeling can cue a policy without uniquely specifying its mechanical trajectory. The same verbal description of feel can accompany different motions; similar motions can feel different when fatigue or sensory conditions change.

Example 29.3. The Feel of a Good Swing

A golfer reports a light grip and an unhurried transition after a centered strike. Record the report alongside impact location, delivery and ball flight. If the cue predicts centered contact across later trials and relevant conditions, it has practical value. If the player reports the same feel while contact moves across the face, the cue needs recalibration. Neither result requires dismissing subjective experience or treating it as an instrument reading.

Impact sensations mostly inform evaluation and later attempts. A response triggered by the collision cannot causally correct an earlier impact condition. Forces and vibrations measured at the clubhead cannot be equated to loads throughout the body without a transmission model. A prepared grip or muscle state may affect impact mechanically; that differs from a new neural correction initiated after contact. The computational-brain and impact chapters treat those timing and coupling questions separately.

29.3 Contemporary Motor Learning Frameworks

Frameworks emphasize different explanatory levels: generalization across tasks, relations between a performer and environment, coordination dynamics, or changes induced by varied practice. Their terminology does not make them mutually exclusive, and a successful drill rarely identifies one uniquely correct theory.

29.3.1 Schema Theory

Definition 29.5. Motor Schema

In schema theory, experience supports relationships among initial conditions, response specifications, sensory consequences and outcomes, allowing a skill to be parameterized for new requirements [Schmidt, 1975]. This is a theoretical account of generalization, not a claim that distance is converted into force by one universal linear rule.

Example 29.4. Golf Schema

For a projectile launched and landing at equal height in a vacuum, $R = v_0^2 \sin(2\alpha)/g$. At fixed angle α with positive $\sin(2\alpha)$, v_0 scales as $\sqrt{R}$: quadrupling range doubles launch speed. Real golf adds aerodynamic forces, spin, unequal elevations and ground interaction. Even this simplified case disproves a universal proportionality between required speed and distance. Launch speed also does not uniquely determine a muscle-force history.

Practicing several distances can supply information about a response-to-outcome relationship. Whether that information transfers depends on where practice samples the relationship, how noisy the feedback is and whether test conditions preserve it. Interpolating among familiar distances and extrapolating to a different club or lie are different problems. A theory's prediction of useful variation is not proof that every varied schedule beats every constant schedule.

29.3.2 Ecological Dynamics and Constraints-Led Approach

Definition 29.6. Ecological Dynamics

An approach that studies action in relation to the performer, task and environment, emphasizing how information and available opportunities for action shape coordination [Davids et al., 2008].

Definition 29.7. Constraints-Led Approach

Practice design that changes task, environmental or performer-related constraints to encourage exploration of useful solutions. Examples include target size, lie, club and permitted landing area. The resulting solution must be measured; the constraint does not uniquely prescribe it.

A shorter target may change backswing amplitude, speed, launch or club choice. It does not force energy transfer to occur earlier. A narrow corridor penalizes dispersion but does not mechanically require a draw. The design question is whether the drill makes the intended information and skill relevant while retaining a meaningful relationship to the eventual golf task. A drill can be difficult yet teach an adjustment that does not transfer.

29.3.3 Dynamical Systems Theory and Attractors

Definition 29.8. Movement Attractor

An attractor is an invariant set approached by trajectories from a specified neighborhood in a specified dynamical system. A finite swing that ends at impact is not automatically an attractor. One may instead study finite-time sensitivity, a time-dependent tracking system, or a return map across repeated trials, stating the system and stability definition [Kelso, 1995].

Example 29.5. Attractors in Golf

For a deliberately simplified trial model, let m_n be a learned scalar response and let r be a fixed required response. Set

$$m_{n+1} = am_n + b(r - m_n).$$

The retention factor a and error sensitivity b are dimensionless model parameters, not measured neural constants. For $a = 1$, the task error $e_n = r - m_n$ satisfies $e_{n+1} = (1-b)e_n$. With $r = 4$ and $m_0 = 0$, $b = 0.25$ gives errors $4, 3, 2.25, 1.6875, \ldots$. With $b = 1.5$, errors alternate in sign while shrinking. With $b = 2.1$, their magnitude grows. More aggressive correction can destabilize this model.

The general stability condition is $|a - b| < 1$. When stable and $1 - a + b \neq 0$, the equilibrium is $m_* = br/(1 - a + b)$, leaving error $e_* = (1 - a)r/(1 - a + b)$. For $a = 0.98$, $b = 0.25$ and $r = 4$, this error is $8/27 \approx 0.2963$. These equations separate stability from zero error and immediate change from retention. They are manufactured examples, not fitted rates of golf learning or proof of a particular practice schedule.

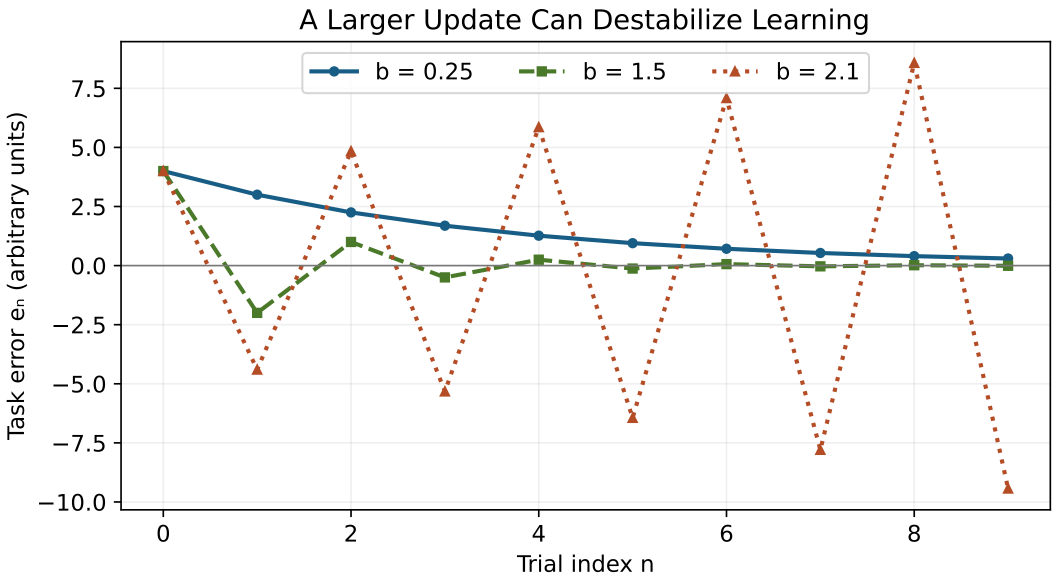


Figure 29.2: Manufactured Trial Updates With $a = 1$, $r = 4$ and $m_0 = 0$. Gains of 0.25 and 1.5 Converge; a Gain of 2.1 Diverges. These Curves Do Not Represent Measured Learning Stages.

29.3.4 Differential Learning and Noise

Definition 29.9. Differential Learning

Wolfgang Immanuel Schöllhorn's differential-learning approach deliberately varies movement conditions and emphasizes fluctuations during practice. It is conceptually distinct from merely randomizing the order of several practiced tasks [Schöllhorn, 2016].

Example 29.6. Differential Learning in Golf

Varying stance or stroke conditions introduces movement variation. Randomizing three target distances changes task order. Repeating one distance in blocks changes neither in the same way. A useful comparison specifies the actual manipulations, total attempts, feedback and later test. Calling a drill differential, variable or random does not establish its effectiveness.

Variation can reveal a useful relationship or obscure it. For example, changing stance, club, target and feedback together makes it difficult to attribute an improvement to one element. This does not forbid combined practice; it limits what that practice can establish scientifically. Added noise is not intrinsically beneficial, and its amplitude and relevance matter.

29.4 Neural Networks in the Brain: What the Evidence Establishes

29.4.1 The Neocortex: Distributed Learning

Key Principle 29.4. The Cortical Hierarchy for Motor Control

Planning, perception and action involve interacting cortical and subcortical systems. A hierarchical diagram can organize computational roles, but arrows in that diagram are not evidence for a literal serial pipeline. Behavioral adaptation, explicit strategy, reinforcement and skill acquisition need to be distinguished before assigning a mechanism [Kitago and Krakauer, 2013].

An observation that two regions are active during learning does not establish which computed the error, which changed a command or where a memory is stored. Those claims require experiments that distinguish the alternatives. Golf adds a further generalization step: a result from a laboratory pointing task may motivate a golf hypothesis without validating it.

29.4.2 The Cerebellum: Plasticity and Prediction

Key Principle 29.5. Cerebellar Learning Mechanism

One influential account involves long-term depression at parallel-fiber–Purkinje-cell synapses, with climbing-fiber activity contributing to plasticity induction. It is not a synapse between parallel and climbing fibers. Nor is that plasticity the only possible learning mechanism. Schonewille and colleagues found preserved performance on several motor-learning tasks in three mouse lines with disrupted expression of this form of LTD. The result limits an obligatory single-mechanism account; it does not establish that LTD is irrelevant in all tasks or species [Schonewille et al., 2011].

A useful mathematical teaching model is gradient learning of a prediction. Let a scalar predicted observation be $\hat{s}_n = \boldsymbol{w}_n^{\mathsf{T}}\boldsymbol{\phi}_n$ with loss $L_n = \frac{1}{2}(s_n - \hat{s}_n)^2$. Then

$$\boldsymbol{w}_{n+1} = \boldsymbol{w}_n + \eta(s_n - \hat{s}_n)\boldsymbol{\phi}_n. \tag{29.1}$$

This follows by differentiating the specified quadratic loss. The sign depends on defining error as observed minus predicted. It is not an experimentally established synaptic law for golf. Features, timing, learning rate and biological implementation are additional hypotheses.

29.4.3 The Basal Ganglia: Reinforcement Learning

Key Principle 29.6. Dopamine and Reward Prediction Error

Schultz, Dayan and Montague related phasic dopamine responses in conditioning experiments to reward prediction error. A predicted reward need not produce the phasic increase elicited by an unexpected reward; omission can produce a decrease relative to baseline. Their temporal-difference account is a computational interpretation, not a claim that every dopamine neuron reports only one scalar or that a successful golf shot stores its entire trajectory [Schultz et al., 1997].

Using time index t within an episode, a standard temporal-difference expression is

$$\delta_t = R_{t+1} + \gamma V(\chi_{t+1}) - V(\chi_t), \qquad 0 \leq \gamma \leq 1.$$

Here χ_t denotes the model's informational state; V predicts discounted future reward and R is a scalar reward in the model. A complete episode or discounting assumption is required for the return to be well-defined. At a terminal state $V(\chi_{t+1}) = 0$. A reward of 1 expected with value 1 gives $\delta = 0$; the same reward expected with value 0.25 gives 0.75. Zero error is not zero reward or absence of baseline neural firing.

Reward design matters. A practice score that rewards carry alone may favor a different behavior from one that penalizes missed fairways or poor contact. A scalar outcome also leaves a credit-assignment problem: which preparation, timing or command helped? It does not directly identify a vector of muscle corrections. Explicit reasoning, structured exploration and sensory information can all help distinguish candidate changes.

29.5 The Setpoint Hypothesis: Errors and Testable Predictions

Key Principle 29.7. The Setpoint Hypothesis: A Mechanistic Model to Test

Suppose a golfer uses a remembered sensory pattern to help prepare a policy and evaluate its execution. This model should be tested against alternatives in which task outcomes, explicit aiming or other sensory cues dominate. Successful prediction in one context does not establish a unique stored pattern, its anatomical location or a universal explanation of swing failure.

Four errors clarify what such a test must measure:

1. Tracking error compares a specified desired state with an estimate: $\boldsymbol{e}_{\text{track}} = \boldsymbol{x}_{\text{ref}} - \hat{\boldsymbol{x}}$. The reference must be defined in compatible coordinates and at a compatible time.

2. Sensory prediction error compares an observation with the predicted observation: $\boldsymbol{e}_{\text{sens}} = \boldsymbol{s} - \hat{\boldsymbol{s}}^{-}$. Predictions must account for the channel's delay. This is not desired minus actual state.

3. Task error compares the desired and observed outcome: $\boldsymbol{e}_{\text{task}} = \boldsymbol{y}_{\text{goal}} - \boldsymbol{y}_{\text{obs}}$. It depends on the chosen task, such as landing location rather than a joint angle.

4. Reward prediction error compares received and subsequently predicted value with previous expected value. It is not the task-error vector or a synonym for disappointment.

These quantities can disagree. In a manufactured pointing example, a hand at $-45°$ with a $+45°$ displayed rotation places a cursor on a $0°$ target. Task error is zero. If the predicted cursor is still $-45°$, sensory prediction error is $+45°$. Mazzoni and Krakauer's visuomotor experiment provides a related empirical dissociation: an explicit compensatory strategy initially achieved the task, yet subsequent implicit adaptation caused drift away from it [Mazzoni and Krakauer, 2006]. That result concerns the tested pointing perturbation, not an inevitability that explicit golf instruction must fail.

Example 29.7. What the Setpoint Hypothesis Can Predict

Before revealing ball-flight feedback, ask the player to rate expected contact and direction. Then compare the prediction with measured contact and delivery. Separately record the felt movement and the actual outcome. Alter one relevant sensory or mechanical condition in a controlled comparison. If feel remains constant while delivery changes, a fixed-feel target is insufficient in that condition. If calibration improves, test whether it survives a delay and transfers. Expectation, attention and feedback changes remain competing explanations unless the design separates them.

A useful protocol therefore records predictions before outcomes, avoids selecting only memorable good shots and includes unsuccessful attempts. Otherwise hindsight can make a vague sensation appear more predictive than it was. Asking about feel may itself change attention, so compare equivalent questioning across conditions.

The model also does not diagnose the yips as a corrupted reference trajectory. Involuntary movements, anxiety-related performance changes and other clinical presentations require appropriate distinctions; the observational yips literature does not support a single universal cause [Adler et al., 2011]. Likewise, a return of an old swing under pressure does not prove that two stored trajectories were activated simultaneously. Describe the observed change first, then test explanations.

29.6 Practice Design: Implications for Learning

Practice performance is measured while training. Retention assesses the learned task after a delay, with the test conditions specified. Transfer assesses performance under a changed task or context. These are related but different outcomes. Temporary guidance, warm-up, fatigue and familiarity can change immediate performance without the same change in retained capability.

For illustration, suppose practice condition A ends with mean error 2 and B with 3 in the same units. On a matched delayed test A has error 6 and B 4. A looked better during acquisition; B is better on that retention test. These invented numbers demonstrate the logic of the comparison. They are not evidence for a particular drill, neural process or schedule. A test with the guidance removed also evaluates dependence on that guidance; report the change explicitly.

29.6.1 Random vs. Blocked Practice: Contextual Interference

Definition 29.10. Blocked Practice

Repeated attempts at one task variant before switching to another, such as a block of putts at each of three distances.

Definition 29.11. Random Practice

Task variants interleaved in an unpredictable order. The task set, number of attempts and feedback can be held constant while order changes.

Constraint Insight 29.2. Contextual Interference Depends on the Task

Contextual interference concerns how mixing practiced tasks affects acquisition, retention and transfer. It is not a universal random-practice advantage. In Porter and Magill's novice putting experiment, a schedule progressing from blocked to serial to random practice produced better retention than the compared blocked and random schedules [Porter and Magill, 2010]. The result supports testing progression; it does not provide a universal dose for every golfer or full swing.

A practical design can start with enough repetition to understand the task and feedback, then introduce decisions representative of play. The transition should follow observed readiness and the purpose of practice, not a rule that difficulty is always desirable. Compare schedules with equal task exposure; otherwise increased variation, extra attempts or different feedback can explain the result.

29.6.2 External vs. Internal Focus of Attention

Definition 29.12. Internal Focus

Attention directed toward the performer's body movement, such as the action of the arms.

Definition 29.13. External Focus

Attention directed toward an effect of movement, such as the club's motion or the intended ball flight. A cue such as "smooth tempo" is not inherently external; its meaning and the player's interpretation matter.

Wulf, Lauterbach and Toole tested 22 golf novices practicing pitch shots. Attention to club motion, compared with arm motion, benefited practice and next-day retention without the instructions [Wulf et al., 1999]. This provides golf-specific evidence for that comparison. It does not show that all technical explanations impair learning, that every external cue is equivalent, or that one neurological mechanism has been uniquely established.

Technical analysis and an execution cue serve different purposes. Between attempts, a player and coach can use mechanics to diagnose delivery. During an attempt, a concise cue can direct attention toward the intended effect. Evaluate both the outcome and whether the cue is understood. The best-looking explanation is not necessarily the most useful instruction at the moment of execution.

29.6.3 The Quiet Eye: Attention Timing

Definition 29.14. Quiet Eye

A final task-relevant fixation or tracking gaze defined relative to a movement event, using explicit spatial and duration thresholds. It is an operational eye-tracking measure; the definition and relevant event must be reported for the task.

Vine, Moore and Wilson studied 22 skilled male golfers in a putting intervention combining gaze feedback and instructions. The trained group improved some laboratory and competitive measures. Retention differences were not significant for every endpoint, and gaze data from five trained participants were lost. The authors also identified possible motivational confounding and limitations of putts per round as a putting measure [Vine et al., 2011]. These results justify a bounded putting hypothesis, not a prescribed two-to-three-second fixation for every golf shot.

Gaze timing can influence information pickup, preparation and attention, but those are distinguishable candidate mechanisms. A relation between fixation duration and success does not establish how much success was caused by fixation. Extending a putting result to a driver requires a new argument and evidence. Eye stillness alone is not a measure of balance, head stability or vestibular activity.

29.6.4 Deliberate Practice

Key Principle 29.8. Characteristics of Deliberate Practice

For the practice plan here, specify the skill, a measurable criterion, feedback that can inform a change, a manageable challenge and a later test. This is an explicit design proposal, not a claim that every useful learning event requires immediate feedback or continuous conscious attention. Feedback frequency, timing and content are experimental choices.

Example 29.8. Deliberate Practice in Golf

For driver practice, choose a delivery-and-dispersion objective rather than maximum speed alone. Record club, ball, target, wind and measurement method. Obtain a baseline sample, test one defined cue or drill, and record every attempt under a stated inclusion rule. Later, repeat a matched assessment without extra coaching. Add a separate transfer test with another target or relevant condition. Report speed, strike location and dispersion together so improvement in one does not conceal deterioration in another.

Small samples deserve modest conclusions. Under an illustrative independent Bernoulli model, eight successes in ten attempts give an observed rate of 0.8, but a 95% Wilson interval is approximately $[0.490, 0.943]$. Correlated attempts or changing conditions undermine that simple model. A single excellent shot provides still less information about reliability. Report uncertainty and the practical size of a change, not just whether a preferred cue produced one memorable result.

There is no arithmetic route from a large practice total to guaranteed expertise. At 52 weeks per year, 10 hours weekly for 10 years is 5,200 hours. Reaching 10,000 hours at that rate takes about 19.23 years; at 20 hours weekly it takes about 9.62 years. These are bookkeeping examples, not thresholds or recommended workloads. Practice histories differ in content, opportunity, feedback and recovery, so hours alone cannot identify the cause of different outcomes.

29.7 Individual Differences: Why Learning Rates Differ

Key Principle 29.9. Sources of Individual Differences

Prior experience, physical capabilities, sensory information, task understanding, practice opportunity, feedback and changing circumstances are candidate contributors to different learning trajectories. Observing a faster learner does not isolate a genetic, motivational or cognitive cause. Specify the comparison and consider baseline ability, task difficulty and measurement uncertainty.

Transfer from another sport may involve perception, timing, physical capacity or useful strategies, but can also involve interference. Similarity of an action's appearance is insufficient: the relevant dynamics, equipment and success criterion may differ. Test what carries over under a defined change rather than assuming that "athleticism" explains every difference.

The affine-control viewpoint contributes a precise question: what must the golfer learn about how state, passive mechanics and commands jointly affect delivery? In a model $\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}) + B(\boldsymbol{x})\boldsymbol{u}$, passive evolution is state-dependent, and the selected definition of zero input matters. A prepared activation or elastic state can remain after subsequent commands are set to zero. Learning may improve preparation or the timing of intervention; that does not make mechanics irrelevant or prove a unique energy-optimal strategy.

To test this connection, compare specified policies or counterfactual interventions from matched initial states and report feasible actuation, contact and uncertainty. Distinguish model prediction from human validation. A reduction in simulated commanded torque is not automatically lower muscular effort, lower joint loading or better transfer to play.

Key Principle 29.10. Key Takeaways

Feel, mechanics and outcome provide related but nonidentical information. State estimates, desired trajectories, sensory predictions and reward predictions must be kept distinct. Coordination is judged against a task, including bias and covariance, rather than the smallest movement variation. Practice designs should make their proposed mechanism testable and assess retention and transfer. Neural and golf-specific claims must stay within the evidence that supports them.

Exercises 29.1. Chapter Exercises

1. Describe a golf skill you are learning. What observations distinguish increasing automaticity from a temporary reduction in task error?
2. After a poor shot, distinguish a useful report of feel from an inferred physical cause. Design one comparison that can test whether the report predicts contact quality.
3. Derive the speed ratio for vacuum ranges R and $4R$ at fixed launch angle and equal launch/landing heights. Explain why it is not a force-scaling rule for golf.
4. Explain how a schedule can look better during practice but worse in retention. Identify two design differences that would confound a comparison of blocked and random practice.
5. Separate a mechanical diagnosis between attempts from an attentional cue during execution. What evidence would justify retaining the cue?
6. Define a quiet-eye measurement and explain why a putting intervention does not establish a universal full-swing fixation duration or a vestibular mechanism.
7. Design a progression from repetition to varied decisions for a beginner. Specify a criterion for advancing and a delayed test that could reveal dependence on guidance.
8. Calculate task variance for $J = (1, 1)$ and the two covariance matrices in this chapter. Which coordination is more task-precise? Can the result rank muscular energy cost?
9. Derive the quadratic-loss prediction update. Explain the distinction between this algorithm and a demonstrated cerebellar synaptic mechanism.
10. Design a driver practice assessment with baseline, feedback, retention and transfer. Explain what eight successes in ten independent trials can and cannot establish.
11. For the scalar trial update, derive the stability condition and steady error. Relate preparation and model learning to the zero-input counterfactual without equating zero commands with zero activation.
12. Give an example of possible positive or negative transfer from another sport. Identify what must remain comparable and what must be measured before attributing the change to transfer.

29.8 Worked Answers

1. Record task error over repeated, comparable assessments and how much deliberate supervision is required. A familiar task becoming easier to execute while preserving performance is compatible with automaticity. One warm-up improvement or helpful

instruction cannot establish lasting change; add a delayed assessment.

2. Record a prediction of contact before seeing measured impact location, using the same questions in each condition. Compare all trials, not only good ones. An association supports the report's predictive use; manipulating a cue or sensory condition with appropriate controls is needed for a stronger causal claim.

3. From $v_0 = \sqrt{Rg/\sin(2\alpha)}$, $v_0(4R)/v_0(R) = 2$. The assumptions exclude golf aerodynamics and ground interaction. Mapping launch speed to force requires mass, impulse, coupling and a time history; distance alone does not provide these.

4. Temporary guidance can improve acquisition while a different condition retains more when guidance is removed. Unequal task exposure and different feedback confound an order comparison. Use matched delayed tests and state whether guidance, task order or context changed.

5. Diagnose measured face delivery, path or strike between shots, then test an understandable cue aimed at the intended effect during execution. Retain it if its practically useful benefit persists under suitable later tests, while tracking other outcomes and uncertainty. A verbal category alone is not proof of effectiveness.

6. Specify gaze location, allowable angular deviation, minimum duration and its timing relative to stroke events. The cited intervention concerns putting and bundles training elements. Neither outcome association nor eye fixation isolates head motion or vestibular processing.

7. Begin with repeated attempts that permit understanding feedback, then mix relevant distances or targets when the golfer can explain the task and respond meaningfully. Use a matched delayed test without extra prompts and a distinct transfer condition. The progression is a proposal to evaluate, not a universal optimum.

8. $J\Sigma_A J^\top = 1 + 1 + 2(0.9) = 3.8$ and $J\Sigma_B J^\top = 4 + 4 - 2(3.8) = 0.4$. B is more precise for this unbiased scalar task despite greater joint variation. Neither matrix supplies muscle work, activation or joint loading, so no energy ranking follows.

9. $\nabla_{\boldsymbol{w}} L = -(s - \boldsymbol{w}^\top \boldsymbol{\phi})\boldsymbol{\phi}$; subtracting $\eta \nabla L$ gives the stated update. Physiological interpretation additionally requires evidence for the features, teaching signal, timing, plasticity and task. A mathematical gradient is not evidence that those biological requirements hold.

10. Fix the target criterion and measurement conditions, record a baseline, compare a specified intervention, and repeat a delayed assessment plus a separate transfer test. The independent-trial Wilson interval for $8/10$ is approximately $[0.490, 0.943]$. It shows substantial uncertainty; it does not establish an 80% stable success probability or validity of independence.

11. Subtract the fixed point to get $m_{n+1} - m_* = (a-b)(m_n - m_*)$, so asymptotic stability requires $|a - b| < 1$. Solving the fixed-point equation gives $e_* = (1-a)r/(1-a+b)$. Preparation changes the initial physical state; setting later commands to zero preserves that prepared state and is not necessarily zero torque or zero muscle force.

12. A prior sport might help target selection or timing but encourage an unsuitable equipment-specific motion. Compare relevant baseline skill and practice exposure, then assess the new task under controlled conditions. Shared joint motion alone cannot identify what transferred, and a failure to transfer is informative too.

Chapter 30

Motor Control III: The Computational Brain

In Plain Language 30.1. What This Chapter Is About

A golf swing asks the nervous system to prepare a moving body, estimate what it is doing, and influence a club whose delivery depends on the whole evolving system. The interesting question is how those operations cooperate. A successful swing does not show that the brain solved a particular optimization problem, stored a complete physics engine, or stopped using feedback near impact.

This chapter connects the control and learning models to mechanics. We will count the variables in an explicitly chosen model, calculate how much time a delayed correction has to act, distinguish drift from energy supply, and examine what stiffness and muscle coordination can accomplish. We will then ask which experiments distinguish competing explanations. Every numerical controller below is an illustrative model, not a measured neural controller or a prescription for grip pressure.

The resulting picture is more useful than a contest between conscious control and passive physics. Preparation changes the state from which the swing develops; mechanics changes the consequences of commands; prediction helps interpret delayed observations; feedback can alter the remaining trajectory; learning changes future preparation and responses. Each contribution has limits that can be investigated.

30.1 The Scale of the Problem the Brain Solves

30.1.1 The Dimensionality of the System

Start by distinguishing a body from a model of that body. A coordinate count depends on which segments, joint motions, contacts and flexible modes the model retains. A shoulder represented by a ball joint has three rotational coordinates; a restricted two-axis shoulder is a different approximation. Two hands holding one club form a closed chain, and fixed-foot contact further restricts feasible motion. The club's pose is not an additional set of independent coordinates if those same coordinates have already been determined by the hands and shaft model.

Let $\boldsymbol{q}$ describe a configuration, $\boldsymbol{v}$ its independent generalized velocity, $\boldsymbol{a}$ muscle activation, and $\boldsymbol{z}$ additional tissue or actuator state. Flexible coordinates and their velocities must also be retained when their history affects delivery. A possible state is $\boldsymbol{x} = (\boldsymbol{q}, \boldsymbol{v}, \boldsymbol{a}, \boldsymbol{z})$. Its precise dimension follows from a declared model and constraint rank, not from counting named body parts. Orientation coordinates may be redundant even when the physical velocity dimension is minimal.

A count that mixes bilateral joints with aggregate trunk or lower-limb coordinates is incomplete until each coordinate and constraint is specified. A defensible report supplies the actual joints, axes, constraints and retained states so that another reader can reproduce the count.

Key Principle 30.1. Count Schedules, Dimensions and Feasible Motions Separately

Suppose, purely for a combinatorial illustration, that a controller chooses among seven command levels for each of 200 channels over 30 intervals. The number of discrete command schedules is

$$N = (7^{200})^{30} = 7^{6000}, \qquad \log_{10} N = 6000 \log_{10} 7 \approx 5070.58824.$$

There are 6000 scalar command entries, not 10^{5070} dimensions. The continuous command cube would be $[0,1]^{6000}$; at one instant it is $[0,1]^{200}$, a set with a boundary and corners. Neither number is an empirical count of neural commands or distinct successful swings. Activation dynamics, force limits, collisions and contacts make many schedules infeasible, and different schedules can produce indistinguishable observations.

Thirty 10ms intervals cover 300ms; a trajectory sampled at both endpoints has 31 sample instants. Selecting a numerical discretization does not establish a 100Hz neural clock. Evaluating every schedule at one nanosecond each would take approximately $10^{5061.58824}$ seconds. This illustrates why that particular enumeration is impractical, not why all other solution methods are impossible.

30.1.2 Constraints Reduce the Space

A smooth parameterization, a learned policy and a physical constraint simplify different parts of a problem. Five coefficients at each of 30 intervals, each with seven levels, give 7^{150} schedules, with logarithm 126.76471. Five unconstrained cubic time functions have 20 coefficients before boundary conditions, amplitude bounds and shared timing are imposed. These are alternative parameterizations, not equivalent estimates of what the brain stores.

Reducing the number of decision variables can make a numerical search easier while excluding a necessary motion. Conversely, a high-dimensional feedback policy may evaluate quickly after training, without searching over whole trajectories during execution. Learning cost, planning cost, online evaluation cost and physical response time are separate quantities. A large schedule count therefore motivates exploiting structure; it does not identify the structure or demonstrate human optimality.

For golf, the meaningful reduction is tied to the task. A family of motions that preserves clubhead speed but changes face orientation is not redundant for a directional shot. A family that preserves the full delivery state may still differ in balance, contact feasibility, tissue load or repeatability. The chosen outputs and constraints determine which differences can be ignored.

30.1.3 The Constraint of Time

Feedback uses information about a state or disturbance to change subsequent action. It need not be conscious, visual, or organized into a sequence of complete serial correction cycles. A continuous delayed loop can have many signals in transit simultaneously. Counting how often the movement duration can be divided by a latency is not a general measure of its control capacity.

Constraint Insight 30.1. Separate Delay, Activation and Remaining Time

For a perturbation at t_p, let L be the total delay before its corrective command reaches the actuator. Let T be the impact time of the nominal example. Then $T - t_p - L$ is the time available for that command's mechanical effect, if positive. Muscle activation and force development can further attenuate the response during that interval.

A first-order activation time constant is not a dead time. In $T_a\dot{a} + a = e$, a step in excitation starts changing activation immediately and reaches about 63.2% of its eventual change after T_a. Measured latency to an EMG change, force change or detectable movement are different observables. Adding published values for all three may count the same physiological processes more than once.

Primary experiments show why the distinction matters. Pruszynski and colleagues found rapid integration of shoulder and elbow information in arm perturbation tasks, with responses on a timescale of about 50ms [Pruszynski et al., 2011]. Saunders and Knill found online corrections to virtual-hand perturbations during reaching. Their mean detected latency was 163ms; their analysis estimated a positive detection bias of about 25ms [Saunders and Knill, 2003]. These are different tasks, signals and measurements. Neither result supplies a universal downswing latency, but both contradict the premise that fast movement categorically excludes feedback.

30.2 How Prediction, Feedback and Mechanics Cooperate

30.2.1 A Worked Remaining-Time Calculation

Consider the change in a single rotational coordinate caused by a corrective torque command. The illustrative actuator and rotor satisfy

$$I\delta\ddot{\theta} = \tau_m, \qquad T_a\dot{\tau}_m + \tau_m = U\,H(t - t_p - L),$$

where H is the unit step, all perturbation states initially vanish, and the command remains on until T. This is a response calculation after a correction has been chosen; it is not a complete feedback policy. Set $h = \max(T - t_p - L, 0)$. Integration gives

$$\delta\theta(T) = \frac{U}{I}\left[\frac{h^2}{2} - T_a h + T_a^2(1 - e^{-h/T_a})\right].$$

The bracket has units of time squared. For $h \ll T_a$, expansion of the exponential gives $h^3/(6T_a)$ to leading order: activation filtering makes a very late angular correction smaller than the ideal instantaneous-torque result $Uh^2/(2I)$.

Take $T = 0.300\text{s}$, $L = 0.060\text{s}$, $T_a = 0.040\text{s}$, $I = 0.050\text{kg}\,\text{m}^2$ and $U = 0.50\text{N}\,\text{m}$. Perturbations at 40, 180, 220 and 260ms give available times of 200, 60, 20 and 0ms. Their final angular changes are approximately 0.135892, 0.006430, 0.000296 and 0rad, respectively. The first is 7.786 degrees; the third is only 0.01693 degrees. Those differences arise with the same actuator and command amplitude. They are not measured golf corrections.

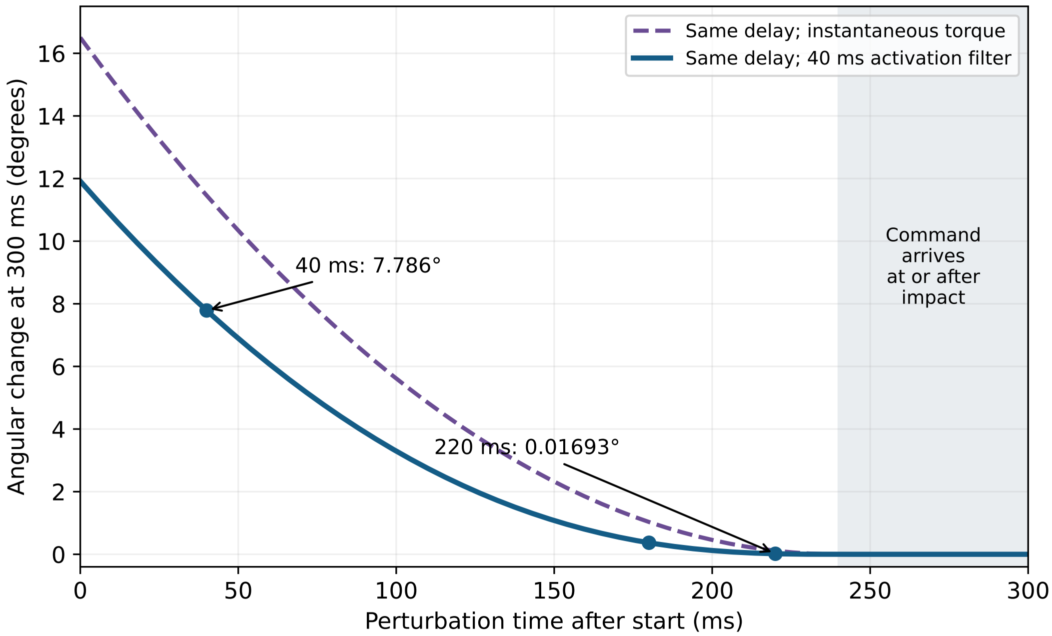


Figure 30.1: Remaining Time Changes the Same Command's Effect. The Declared Rotor Model Compares Instantaneous Torque With Activation Filtering After the Same 60ms Delay. These Are Illustrative Responses, Not Measured Golf Data.

This example explains the value of advance preparation without declaring all feedback useless. Early errors can have an appreciable correction window; late errors may have little or none. Mechanical stiffness already present at the time of a disturbance can respond before a newly generated command arrives. A correction that changes final angular velocity may have little effect on final angle, so response capacity must name the output. The real golfer also has coupled coordinates, changing leverage, force bounds, measurement uncertainty and an impact time that may itself shift.

30.2.2 Predictive Control and Anticipatory Activation

Key Principle 30.2. Prediction Uses Information Already Available

A forward model predicts a future state from a state estimate and a command. An estimator can propagate an old observation using intervening commands to improve feedback despite sensory delay. Prediction cannot reveal an unpredictable disturbance before information about it is available.

For example, a model may propagate $\hat{\boldsymbol{x}}(t-L)$ through $\dot{\boldsymbol{x}} = F(\boldsymbol{x}, \boldsymbol{u})$ using recorded commands to estimate $\hat{\boldsymbol{x}}(t)$. A discrepancy between predicted and observed signals can then update the estimate. Errors in initial state, activation, contact and the model accumulate during propagation. Replacing the unknown current state with a prediction is therefore useful only together with an account of prediction error.

Anticipatory activation changes the state before an expected event. It shifts work and information requirements earlier; it does not abolish neural conduction or muscle dynamics. A model of advance stabilization must say which muscles or actuators change, how that changes torque and impedance, and what the preparation costs. Neither address posture nor the duration of a pre-shot routine identifies those hidden quantities by itself.

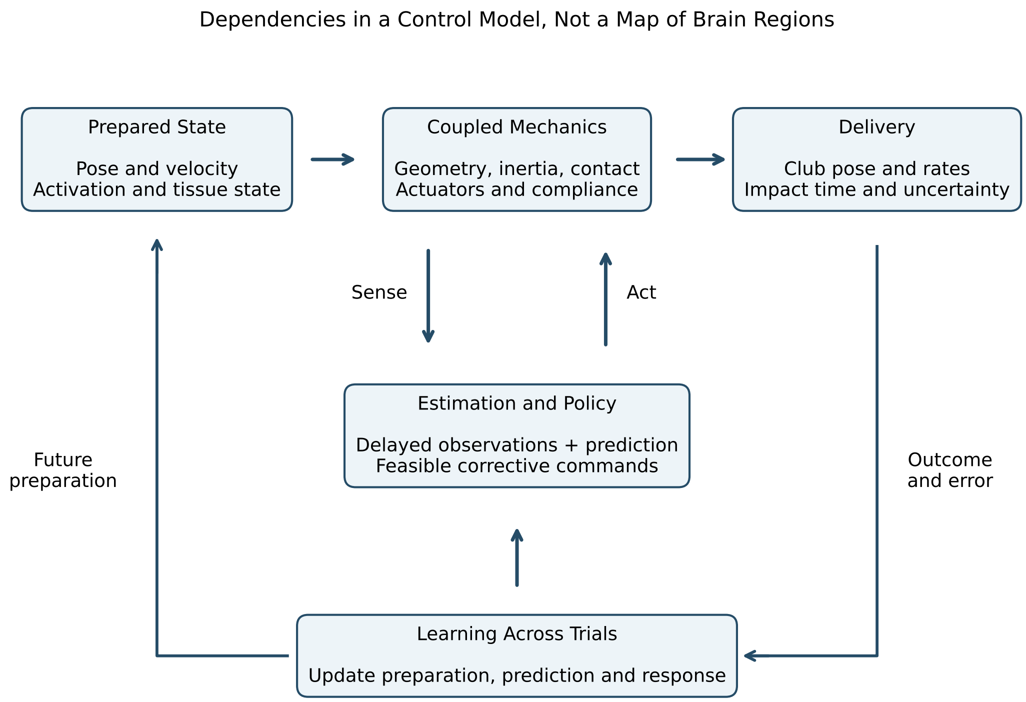


Figure 30.2: Preparation, Mechanics, Estimation, Feedback and Learning Interact. Within-Trial Responses and Changes Across Trials Must Be Distinguished Experimentally.

30.2.3 Drift Is a Model Term, Not a Free Actuator

For an input-affine model,

$$\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}) + \boldsymbol{G}(\boldsymbol{x})\boldsymbol{u},$$

$\boldsymbol{f}$ is the evolution when the declared input coordinates are zero. If the input is joint torque, this differs from a model whose input is neural excitation and whose state includes nonzero muscle activation. Zero new excitation does not instantly eliminate active force. Contact reactions and tissue forces must be treated consistently with the chosen coordinates and state.

An instantaneous ratio between drift and applied input can describe a trajectory under a declared scaling. It does not prove weak available control, low metabolic cost, or low sensitivity at impact. Linearizing a smooth, fixed-contact model around a nominal trajectory gives

$$\delta\dot{\boldsymbol{x}} = \boldsymbol{A}(t)\delta\boldsymbol{x} + \boldsymbol{B}(t)\delta\boldsymbol{u},$$

$$\delta\boldsymbol{y}(T) = \boldsymbol{C}_T\boldsymbol{\Phi}(T,0)\delta\boldsymbol{x}_0 + \int_0^T \boldsymbol{C}_T\boldsymbol{\Phi}(T,s)\boldsymbol{B}(s)\delta\boldsymbol{u}(s)\,ds.$$

Here $\boldsymbol{\Phi}$ propagates perturbations, and $\boldsymbol{C}_T$ maps state changes into the selected delivery outputs. The kernel $\boldsymbol{C}_T\boldsymbol{\Phi}(T,s)\boldsymbol{B}(s)$, together with feasible inputs and their delays, determines first-order influence on those outputs. A norm of $\boldsymbol{f}$ alone omits that propagation and output geometry. Impact-event sensitivity adds a time-shift term when T changes.

An abstract counterexample makes the logical issue explicit. Let $\dot{x} = c + ax + u$, with $x(0) = 0$, $a = 10\text{s}^{-1}$, $c = 1\text{rad/s}$ and $T = 0.3\text{s}$. The sensitivity of $x(T)$ to a constant input u is $(e^{aT} - 1)/a \approx 1.90855\text{s}$. Near the end, the nominal drift is about 20.0855rad/s. An input of 0.01rad/s is small compared with that drift but still changes the endpoint by about 0.01909rad. Large drift and appreciable input sensitivity coexist. This is a mathematical counterexample, not a proposed golf trajectory.

The zero-torque counterfactual framework can test a specified intervention in a golfer model. It must start from the same complete state, define the channels removed, evolve the resulting trajectory and verify that the assumed contacts remain feasible. Agreement with an observed phase ordering does not establish that the golfer silenced muscles or consciously relinquished control. Those are separate hypotheses requiring separate observations.

30.2.4 Mechanical Impedance During a Moving Task

Mechanical impedance describes the relation between motion and interaction force or torque. In a local linear model it can include inertia, damping and stiffness. Stiffness is a derivative of restoring force with respect to displacement under specified conditions, not simply a synonym for grip pressure. Impedance control addresses the dynamics of interaction with an environment [Hogan, 1985b].

Key Principle 30.3. A Moving Reference and Its Local Response

An illustrative joint-torque law is

$$\boldsymbol{\tau} = \boldsymbol{\tau}_{\text{ff}}(t) - \boldsymbol{K}(t)(\boldsymbol{q} - \boldsymbol{q}_d(t)) - \boldsymbol{D}(t)(\dot{\boldsymbol{q}} - \dot{\boldsymbol{q}}_d(t)).$$

The feedforward term supports the nominal motion; the other terms respond to position and velocity errors. This is a generalized-torque model, not a claim that the nervous system independently commands every matrix entry. If the displayed linear law is valid, larger errors produce larger restoring torques. Saturation or a limited range of validity must be modeled explicitly.

For a scalar spring-damper with fixed reference, positive constant stiffness K, nonnegative damping D, and no feedforward torque, write $e = q - q_d$ and $V = Ke^2/2$. Then

$$\tau\dot{q} + \dot{V} = -D\dot{q}^2 \le 0.$$

This storage balance shows passivity at that mechanical port. Positive stiffness stores energy; it does not dissipate it. A compliant element may store and return energy, whereas damping dissipates it. Nor does this passive mechanical description imply zero metabolic cost in the muscles maintaining the state.

If K varies, $\dot{V}$ contains $\dot{K}e^2/2$. At a clamped error of 0.1rad, increasing stiffness from 10 to 30N m/rad increases stored energy by 0.1J despite zero joint motion. That energy has to come from the mechanism changing stiffness. If the reference moves, its work also enters. With constant K, D, no feedforward term and $\dot{e} = \dot{q} - \dot{q}_d$,

$$\tau\dot{q} + \dot{V} = -D\dot{e}^2 + \tau\dot{q}_d.$$

The moving-reference term can be positive. A controller cannot be called passive merely because its torque equation contains a spring and a damper.

Example 30.1. What This Means for Grip and Balance

Changing grip can change local hand-club compliance, available friction, muscle activation, posture and transmission of disturbances. These changes need not have the same effect on clubhead speed, face orientation and tissue loading. A firmer grip does not necessarily lock a wrist coordinate; a lighter grip does not necessarily destabilize the club. The comparison requires a specified load, contact model, perturbation direction and outcome.

In a whole-body model, an apparently helpful local correction may alter the contact wrench needed at the feet. A controller that corrects face angle while violating friction or losing balance has not solved the original task. Joint and endpoint impedance also differ through the configuration-dependent Jacobian and, under load, geometric stiffness terms.

Burdet and colleagues investigated arm movement in an unstable robotic environment and found adaptation of impedance geometry [Burdet et al., 2001]. That supports investigating directional stabilization. It does not establish a universal optimal grip stiffness, a fixed impedance throughout a swing, or an injury-prevention rule. Those require golf-specific measurements and appropriately matched comparisons.

30.2.5 Synergies, Feasible Commands and Neural Evidence

One meaning of a muscle synergy is a low-dimensional description of recorded muscle activity. Another is a hypothesis that neural circuitry restricts commands to coordinated groups. A third is a deliberately chosen control basis in a simulation. These uses should not be interchanged.

For a data matrix $\boldsymbol{E}$ of processed EMG signals, a factorization $\boldsymbol{E} \approx \boldsymbol{W}\boldsymbol{H}$ can summarize patterns. Its result depends on the recorded muscles, filtering, normalization, task set, chosen rank and validation procedure. EMG is an electrical observation, not a direct measurement of muscle force, excitation, joint torque or mechanical work. A good reconstruction of the training recordings need not predict another posture or a novel perturbation.

Kutch and Valero-Cuevas used cadaveric measurements and computational models to show that biomechanics and task selection can produce low-dimensional patterns without a controller that groups muscles [Kutch and Valero-Cuevas, 2012]. This is a counterexample to a particular inference, not a disproof of all neural coordination. It motivates testing the mechanical alternatives alongside the neural hypothesis.

In a model, let $\boldsymbol{R}(\boldsymbol{q})$ map muscle forces to generalized torque, with moment-arm sign convention $R_{ji} = -\partial l_i/\partial q_j$ for musculotendon length l_i. Even the simplified isometric relation

$$\boldsymbol{\tau} = \boldsymbol{R}(\boldsymbol{q})\,\mathrm{diag}(\boldsymbol{F}_{\max})\boldsymbol{a}, \qquad 0 \le a_i \le 1,$$

shows why equal activation need not mean zero net torque. Antagonists with moment arms 0.04 and −0.03m, force capacities 1000 and 800N, and equal activation 0.2 produce $8 - 4.8 = 3.2$N m. Force-length-velocity effects, pennation, tendon state and history make the real relation more involved.

Example 30.2. A Compact Basis Can Remove the Needed Action

Consider two normalized opposing actuators with $\tau = a_1 - a_2$. Independent commands in $[0,1]^2$ can produce any torque in $[-1,1]$. Restricting activation to the single coactivation vector $\boldsymbol{a} = c(1,1)^\top$ gives zero torque for every c. The basis has one dimension, yet it loses both directions of net torque. The same coactivation could change stiffness in a richer muscle model; that property is absent from this simple torque map and cannot be inferred from it.

For $\boldsymbol{a} = \boldsymbol{W}\boldsymbol{c}$, the admissible coefficient set must satisfy $0 \le \boldsymbol{W}\boldsymbol{c} \le 1$ and any excitation-rate or activation constraints. Rank alone does not describe its feasible torque set. A claim that a small basis simplifies learning should therefore report both held-out prediction and the motions, forces and recovery actions the restriction excludes. No fixed number of golf synergies or universal explained-variance percentage is assumed here.

30.3 The Brain as an Internal Model: What It Learns

Predicting sensory consequences, selecting a command and evaluating an outcome are different computational functions. They can be represented by a forward model, a policy and a value or cost model, respectively. A controller may combine learned dynamics, cached actions, explicit aiming strategies and feedback corrections. Good performance does not uniquely identify which combination produced it.

Writing $\hat{\boldsymbol{f}}$ and $\hat{\boldsymbol{G}}$ is useful when building an estimator for an affine mechanical model. It does not establish that the brain separately stores those functions or that a particular brain region implements each one. Many parameter combinations can explain the same observed movement. Changes in posture, activation or contact can also look like a change in the fitted dynamics if the state is incomplete.

30.3.1 What Adaptation Experiments Can Establish

Example 30.3. Separate Task Error From Prediction Error

Mazzoni and Krakauer introduced a visuomotor rotation and supplied an explicit aiming strategy that initially corrected it. Continued implicit adaptation subsequently interfered with that strategy [Mazzoni and Krakauer, 2006]. This demonstrates that explicit and implicit processes can coexist and conflict in that task. It does not show that all rapid learning is unconscious or that successful golf requires a uniquely identified internal physics model.

A task error compares achieved and desired outcomes. A sensory prediction error compares an observation with its expected consequence. They can differ: a deliberate aiming offset may put a cursor on target while the cursor still appears away from the position predicted from the hand command. Removing a perturbation and observing an aftereffect helps distinguish adaptation from immediate compensation, but aftereffects alone do not reveal every learned variable or neural location.

Switching clubs is a useful proposed experiment, not a completed experiment with a universal trial count. The new club may change inertia, geometry, contact, visual cues and the shot objective. Comparing a driver swing with a putt changes the task as well as the tool. A stronger design perturbs a measured club property within a matched task, randomizes conditions, measures the initial response and learning curve, includes washout and retention tests, and assesses transfer to conditions withheld from training.

For a simple trial-to-trial model $z_{n+1} = az_n + be_n$, retention a and error sensitivity b are parameters to estimate under a declared observation model. Similar performance curves can arise from different hidden states, explicit strategies or changes in exploration. Fit quality is insufficient: compare predictions for error clamps, altered feedback, changed loads and delayed retention. The investigation should expose where the models disagree.

30.4 Comparison With Artificial Intelligence: What Can Machines Learn?

30.4.1 How Brain Learning Differs From AI Learning

Key Principle 30.4. Specify a Comparable Benchmark

A meaningful comparison declares the task, body and actuators, observations, prior experience, allowed feedback, training data, computation budget and success criterion. It separates real interactions from simulated interactions and separates training from execution. A trial, an episode, a sensor sample and a gradient update are not interchangeable units of experience.

An experienced adult learning a shot brings prior motor and perceptual experience. A pretrained robot can also bring demonstrations or learned representations. Counting only the final practice session on one side and all training on the other gives a misleading efficiency comparison. Conversely, a simulator can generate information unavailable to the physical agent; privileged state or perfect contact parameters must be disclosed.

There is no generic rule that reinforcement learning uses joint angles only, lacks internal models, or learns only from a final scalar reward. Dreamer, published in 2020, learned a world model from images and trained behavior using imagined latent trajectories [Hafner et al., 2020]. This is a concrete counterexample to that characterization, not evidence that it reproduces human golf learning or a claim about the best current algorithm.

30.4.2 What Human Performance Contributes to the Comparison

Key Principle 30.5. Measure Adaptation and Transfer

Useful comparisons ask how performance changes with altered club properties, uncertain ground contact, restricted sensing, fatigue or a new target. Report error distributions, recovery time, failures and retained learning. Transfer from another sport is a hypothesis to measure, not an assumed benefit for every learner. A demonstration that communicates a goal is also different from one that supplies an executable full-body policy.

Humans and machines have different embodiments, sensing and learning histories. Those differences are scientifically interesting and need not be hidden to make a comparison. A specialized machine that repeats one swing may answer a repeatability question while saying little about general adaptation. A human who adapts successfully to a changed task does not thereby establish global optimality or robustness to every disturbance.

30.4.3 What Computation Does and Does Not Guarantee

Key Principle 30.6. Deterministic Evaluation Is Not a Deterministic Shot

A deterministic policy returns the same command for an identical represented input. Sensor noise, hidden tissue or actuator state, contact variability, model error and numerical or hardware differences can still change the physical outcome. Likewise, evaluating many candidate actions does not guarantee a global optimum for a nonlinear constrained problem. Such a guarantee requires a suitable problem class, algorithm and certificate.

Frequency comparisons need equal care. A neural firing rate, a sampled command update rate, a feedback bandwidth and a shaft natural frequency describe different processes. For the delay-plus-activation part of our illustrative controller,

$$H(i\omega) = \frac{e^{-i\omega L}}{1 + i\omega T_a}, \qquad \arg H = -\omega L - \arctan(\omega T_a).$$

With $L = 0.06$s and $T_a = 0.04$s, the phase contributions sum to approximately -35.7, -101.8 and -159.5 degrees at 1, 3 and 5Hz. The mechanical plant, sensor processing and controller add their own dynamics. These numbers are not a stability certificate, but they show why comparing an alleged neural clock with a shaft mode does not determine correction capacity. Sampling faster cannot by itself remove propagation delay.

Power comparisons must use a common system boundary. An illustrative processor drawing 20W for 0.3s consumes 6J. A 0.2kg clubhead moving at 50m/s has 250J of translational kinetic energy. Neither calculation measures human brain power or whole-body metabolic expenditure, and the 6J cannot be substituted for the mechanical energy supplied to the club. A fair embodied comparison includes the actuator energy, preparation, losses, support equipment and any energy recovered, with training energy reported separately when relevant.

30.5 What Artificial Intelligence Could Learn From the Brain

30.5.1 Hierarchical Control

Key Principle 30.7. Organize Decisions by Their Role

A practical controller can separate shot selection, desired delivery, feasible motion generation and actuator feedback. Each layer passes constraints and uncertainty as well as a target. The lower layer must be able to reject an infeasible request. This is an engineering decomposition; a one-to-one map from those software layers to prefrontal, premotor and motor regions is not established by drawing the diagram.

For example, an outer planner may request a delivery distribution rather than one exact club pose. A motion planner then considers contact, balance and muscle limits; a faster estimator and feedback law respond to the current observations. If the estimator detects a changed support condition, that information may need to modify the motion goal itself. Hierarchy does not mean that task information disappears from lower-level responses.

30.5.2 Internal Models and Model-Based RL

Example 30.4. Use Prediction Where It Is Reliable

A model-based controller predicts consequences before selecting an action; a model-free policy or value estimate can select actions without an explicit dynamics rollout at decision time. Learned models, policies and value functions can be combined. The practical question is which combination performs well with the available data and computation, not whether every biological or artificial learner belongs to one category.

An inaccurate model can recommend actions that exploit its own errors. Evaluate multi-step prediction at the speeds and contacts relevant to delivery, not only one-step accuracy on a familiar trajectory. Uncertainty estimates should widen outside measured conditions, and optimization should respect actuator and contact feasibility. A cached policy can reduce online computation while still relying on substantial prior optimization or training.

30.5.3 Embodiment and Physics Exploitation

Key Principle 30.8. Mechanics Redistributes and Stores Energy

Geometry, inertia, compliant elements and contact determine how commands affect motion. They can make a task easier to execute or harder to recover from. Inertial coupling transfers influence and energy between coordinates; it is not an independent energy source. Elastic return requires earlier storage. A low-dimensional control description does not remove those energy requirements.

For the illustrative 0.2kg head above, a 1m drop supplies only $mg\Delta h = 1.962$J of gravitational work, about 0.785% of its 250J kinetic energy at 50m/s. This calculation concerns the head alone. A complete golfer-club budget must include all segments, their height changes, initial kinetic energy, elastic storage, actuator work and losses. It does not follow that gravity is negligible everywhere; it does rule out crediting the head's own 1m fall with supplying its full final energy.

Similarly, a natural frequency is a property of a specified linearization and boundary condition. A rapid, nonperiodic swing with changing supports and nonlinear muscle dynamics cannot be optimized simply by matching one resonance. A compliant design that filters high-frequency disturbances may also delay the desired action or amplify another frequency range. Mechanical design and control design must be assessed together against delivery and load constraints.

30.5.4 Prediction-Driven Learning

Example 30.5. Learn From Consequences, Then Test Transfer

A robot can record commands and subsequent observations to train a predictive model without manually labeling every desired action. That does not make the task objective unnecessary, guarantee useful exploration, or eliminate model bias. The learned representation still needs testing on the loads, viewpoints and contact transitions relevant to the task.

The analogous scientific question for human learning is which error signals alter which future behaviors. A changed sensory prediction, a missed target and an undesirable effort cost need not drive the same update. Experiments that manipulate them separately provide more information than describing every improvement as the brain learning physics. Predictive models are powerful tools for posing these questions; their success is not automatic evidence that the biological implementation uses the same variables.

30.6 The Golf Swing as a Window Into Intelligence

30.6.1 Why Golf Is a Useful Scientific Tool

Key Principle 30.9. Measure the Chain From Action to Outcome

Golf supplies a visible target and measurable delivery and ball-flight outcomes, but a stationary ball does not make the environment constant. Lie, friction, wind, club properties, contact location and the golfer's state vary. Differences between experts and novices also confound practice, age, strength, selection and other history unless the study design addresses them.

Ball landing position alone is a poor identifier of the preceding control. Different combinations of face, path, strike location, speed and spin can produce similar landing points. Synchronized club, body, force and shaft measurements can narrow the possibilities, but estimating internal muscle state and separating causes still requires assumptions. Introspective reports are observations about experience; they are not direct measurements of neural commands.

Example 30.6. A Discriminating Golf Experiment

To investigate a claimed grip-related reduction in face-angle sensitivity, measure the grip condition and a calibrated perturbation, randomize its timing and direction, and retain the same declared shot objective. Record pre-perturbation state, delivery, contact feasibility and trial-to-trial dispersion. Compare immediate mechanical response, delayed correction and adaptation across later trials. A smaller error can then be tested against alternatives such as changed initial state, increased stiffness, different input gain or a changed policy.

A feasible experiment need not apply a hazardous disturbance during a full-speed swing. Instrumented slow tasks and controlled simulator studies can establish parts of the mechanism first, with their task boundaries stated. A simulator can verify a derivation and generate competing predictions; independent physical measurements are needed to validate those predictions for people.

30.6.2 Open Questions That Golf Helps Address

Key Principle 30.10. Ask Questions That Could Change the Model

Which delivery directions remain correctable at each perturbation time? Does an apparent improvement come from a more favorable prepared state, altered passive response or delayed feedback? Does a compact activation basis predict responses to unfamiliar loads? Which parameters transfer between clubs, and which must be relearned? Do competing estimators predict different effects when visual and proprioceptive information disagree?

For small errors, a useful analysis combines the propagated initial-state covariance, disturbance covariance and the delivery Jacobian. More repeated swings do not cure systematic calibration bias. Correlated state and command errors must be retained when present; treating them as independent can misstate outcome variance. Near a grazing contact or a change of contact mode, a smooth linear covariance approximation may fail. Those limits are part of the model's domain, not inconvenient exceptions to ignore.

The same investigation applies to humanoid modeling. Contact constraints shape feasible accelerations and forces; geometry maps joint motion to an end-effector task; internal actuator state changes future force; estimation determines which corrections can be chosen; delay determines when they act. A golf swing is a demanding instance of this general coupled problem, not proof of one universal neural solution.

30.7 Why Skilled Performance Sometimes Fails

30.7.1 The Yips: Describe the Phenotype Before the Mechanism

The term yips is used for task-related disruptions such as involuntary movements or loss of control during putting and other skilled actions. It does not by itself identify one cause. A missed putt, performance anxiety and a task-specific movement disorder are not interchangeable observations.

Key Principle 30.11. Evidence Does Not Support a Single Corrupted Program

Adler and colleagues studied 50 golfers using movement recordings and surface EMG. Findings depended on whether grouping used self-report or video-defined involuntary movement; the latter groups differed in pronation-supination movement, while the co-contraction comparison was a statistical trend [Adler et al., 2011]. This supports careful phenotype measurement, not a universal diagnosis or a demonstrated corrupted cerebellar trajectory.

A model with an unstable reference, excessive gain or delayed correction can generate jerky motion. That establishes what the model can do, not what caused a particular golfer's symptoms. Several mechanisms can produce similar traces, and psychological and neurological factors need not form mutually exclusive categories. Persistent involuntary movement is a clinical assessment question; this chapter's equations do not identify its cause or treatment.

30.7.2 Choking Under Pressure

Example 30.7. A Miss Is Not Its Own Explanation

Imagine an all-square match on the last hole, with a putt that would win it. The golfer misses. That one outcome does not establish choking: a skill with a high success probability can still fail. Investigating a pressure effect requires a comparison with an appropriate performance distribution under matched conditions, preferably with pressure manipulated or measured independently.

Key Principle 30.12. Compare Attention and Pressure Hypotheses

Explicit-monitoring accounts propose that attention to execution can disrupt some well-practiced skills. Distraction, altered arousal, changed risk preferences and mechanical changes in preparation are other candidate pathways. Beilock and Carr's putting experiments investigated skill knowledge, pressure and training conditions [Beilock and Carr, 2001]. Their task-specific evidence does not establish that technical thought always harms performance or that one explanation covers every pressure-related miss.

Studying mechanics between trials, learning a new movement and monitoring a highly practiced action during execution are different activities. The fact that one attention instruction changes performance cannot justify a blanket instruction to stop analyzing the swing. Nor can an observed benefit from a routine identify a particular amygdala or cerebellar mechanism without matching neural evidence.

Key Principle 30.13. Test an Intervention Against the Intended Outcome

Compare a specific routine or attention instruction with a matched control condition, randomize order, account for skill and practice effects, and measure both performance and the intended manipulation. Distinguish immediate success from retention and transfer. A technique can be useful for a particular person or task without being a universal prevention strategy or proving the proposed neural explanation.

30.8 What the Combined Account Establishes

Key Principle 30.14. The Computation Is Distributed Across Time and Mechanisms

Preparing the state can reduce what must be corrected later. Existing mechanical impedance can shape a disturbance before a new command arrives. Prediction can improve estimates from delayed observations. Feasible feedback can change the remaining trajectory, and learning can improve future preparation and correction. The usefulness of each contribution depends on the task, state, uncertainty and time remaining.

This account does not require gravity to supply most swing energy, the brain to enumerate all muscle schedules, or feedback to switch off at a universal phase boundary. It also does not require a perfect internal model. It asks for a model detailed enough to make a discriminating prediction and measurements capable of testing it.

Key Principle 30.15. Key Takeaways

The coordinate dimension, command parameter count and number of discretized schedules are different. A finite delay limits when an observed disturbance can affect delivery; it does not forbid every correction. Drift magnitude, energy supply and terminal sensitivity require different calculations. Impedance includes dynamics and energy exchange, and its modulation can require work. A compact EMG factorization is not unique evidence of neural grouping. Adaptation, transfer and performance under pressure need controlled comparisons. Human and artificial controllers should be compared on declared tasks with common information and energy boundaries.

The purpose of understanding these connections is practical as well as scientific: to know which change a model predicts, why it predicts it, and which observation could show that the explanation is wrong. That is the footing on which a useful reference for golf and general humanoid control can be built.

30.9 Chapter Exercises

1. Compute the schedule count for 200 channels, seven levels and 30 intervals. Repeat for five channels. Explain why neither result is a dimension or a count of successful swings.

2. For $\boldsymbol{a} = \boldsymbol{W}\boldsymbol{c}$, derive the coefficient constraints induced by $0 \leq \boldsymbol{a} \leq 1$. Use $\tau = a_1 - a_2$ to explain why the basis $(1, 1)^\top$ loses torque authority despite reducing the command dimension.

3. Derive the constant-input endpoint sensitivity for $\dot{x} = c + ax + u$. Why does a large nominal drift not prove a small endpoint response? Identify the additional quantities needed for a golf delivery calculation.

4. Integrate the delayed actuator/rotor model. Reproduce the four perturbation-time examples and derive the $h^3/(6T_a)$ short-time limit. State precisely when this model predicts zero response before impact.

5. Derive the spring-damper storage balance for fixed and moving references. Calculate the energy added when stiffness increases from 10 to 30N m/rad at a clamped 0.1rad error.

6. Explain how task error can vanish while sensory prediction error remains. Propose measurements that distinguish explicit aiming, implicit adaptation and a change in mechanical impedance after a club perturbation.

7. Compare model-based rollout, a cached policy and a learned value function. Which observations would be needed before attributing one of these computational descriptions to a neural implementation?

8. Propose a hierarchy for a humanoid golf controller. Describe information flowing upward as well as downward, and specify how the controller handles an infeasible delivery request.

9. A delayed feedback model produces jerky putting. Give two competing explanations and design a measurement that could distinguish them. Explain why simulation alone does not diagnose the yips.

10. Contrast explicit-monitoring and distraction accounts of performance under pressure. State a result that would count against each account in a specified task.

11. Design an attention experiment with novice and experienced putters. Include randomization, a pressure manipulation, a manipulation check, a performance outcome, and retention or transfer testing.

12. Derive the magnitude and phase of the delay-plus-activation transfer function at 1, 3 and 5Hz. Explain why neither neural firing rate nor shaft resonance alone specifies closed-loop bandwidth or stability.

13. Specify a human/robot golf benchmark with at least five matched conditions. Include prior experience, sensing, actuation, delivery uncertainty and energy boundaries. Distinguish processor energy from actuator work.

14. Design one experiment that separates prepared-state sensitivity, immediate mechanical response, delayed feedback and adaptation across trials. State which parts require additional measurements and where the inference could remain ambiguous.

30.9.1 Worked Checks and Answer Guidance

For Exercise 1, the logarithms are 5070.58824 and 126.76471; the continuous schedule dimensions are 6000 and 150 before further constraints. Exercise 2 requires $0 \leq \boldsymbol{Wc} \leq 1$, plus any separately declared sign, rate or physiological bounds on $\boldsymbol{c}$; its example gives $\tau = 0$ for every feasible coefficient.

For Exercise 3, differentiate $x(T) = (c + u)(e^{aT} - 1)/a$ with respect to u, taking the limit T when $a = 0$. In a multi-state swing, use the propagated input kernel, the output Jacobian, feasible controls, latency and any event-time correction. For Exercise 4, integrate $\tau_m = U(1 - e^{-s/T_a})$ over the remaining interval twice. Zero response occurs when $t_p + L \geq T$ under the stated zero initial perturbation and causal command assumptions. It does not assert zero motion or zero pre-existing muscle force.

For Exercise 5, differentiate $Ke^2/2$ and retain $\dot{K}e^2/2$ and the reference-work term. The clamped stiffness change adds 0.1J. For Exercise 6, record both aiming behavior and the sensory transformation; include aftereffects and controlled loads rather than interpreting a single improved outcome. Exercises 7 and 8 require distinguishing a useful computational architecture from evidence for its neural implementation, and passing contact/actuator feasibility information back to the planner.

Exercises 9–11 need explicit competing predictions, not a presupposed corrupted motor program. Baseline variability, involuntary movement, attention and pressure must be measured separately. For Exercise 12, the illustrative actuator magnitudes are approximately 0.970, 0.798 and 0.623; the unwrapped phases are -35.7, -101.8 and -159.5 degrees. These omit the remaining loop dynamics. Exercise 13 must account for the 6J processor example and the separate 250J clubhead energy without treating their ratio as an efficiency. Exercise 14 should use perturbation timing and within-trial versus between-trial responses to distinguish mechanisms while acknowledging hidden-state and measurement ambiguity.

Chapter 31

Passive and Distributed Control: A Self-Organizing Swing Model

> The swing is not made by conscious effort but by giving the conditions where the swing can happen.
>
> After Frans Bosch

In Plain Language 31.1. What You'll Learn

Most people think the brain controls the golf swing by sending constant commands to every muscle: "tighten this, relax that, rotate here, extend there." But this view ignores a fundamental problem: the brain is slow and its bandwidth is limited. This chapter develops a more defensible modeling alternative—one that can help explain why expert golfers look effortless while novices look tense.

The body does not need to control the swing by micromanaging every degree of freedom. A more defensible hypothesis is that the nervous system sets mechanical conditions—including stiffness, damping, and baseline tension of muscles and tendons—before and during the swing. Those tissue properties can resist some perturbations and reduce feedback burden, but they do not make the swing autonomous or eliminate neural control. This is the passive-control perspective; its golf-specific claims still require empirical support from measured swings.

31.1 The Computational Problem: Why the Brain Can't Micromanage

In Chapters 28–29, we encountered the bandwidth problem: the brain has limited computational resources, slow feedback loops (100–300 ms latency), and a maximum processing rate around 10–40 bits per second. The golf swing involves roughly 20 degrees of freedom, each of which must be controlled. If the brain tried to compute independent torques for each joint at the rate of swing execution (1000 Hz or faster near impact), it would require simultaneous processing of about 20,000 decisions per second.

This exceeds the brain's capacity by orders of magnitude. Yet skilled golfers execute repeatable swings that hit the ball in the same location, with the same contact pattern, swing after swing. How?

Myth vs. Reality 31.1. The Central Control Myth

The Myth: The brain is a master computer that calculates the optimal torques for every muscle at every millisecond of the swing.
The Reality: The brain pre-computes a set of mechanical parameters—muscle stiffness, damping, co-contraction patterns—that configure the body into a self-stabilizing system. During execution, the body's tissues handle the moment-to-moment details automatically.

The alternative is more efficient: instead of computing torques in real time, the brain computes the *impedance*—the mechanical properties of the muscles and joints—*once before the swing*. Once these properties are set, the passive mechanical system takes over. The muscles and tendons become springs and dampers, and the swing executes like a mechanical oscillator responding to forces.

31.2 Passive Mechanical Systems: Springs and Dampers

Definition 31.1. Passive Control

A system exhibits *passive control* when it achieves stability and converges to a desired trajectory *without active feedback or real-time control signals*. Stability emerges from the mechanical properties of the system itself—springs, dampers, and mass distribution.

Consider a simple example: a pendulum hanging at rest. It is passively stable. If you displace it, gravity and the support structure automatically create a restoring force. The pendulum swings back to equilibrium without any external controller. The source of stability is potential energy stored in the gravitational field and kinetic energy dissipated by air resistance.

Similarly, a spring-mass-damper system is passively stable:

$$m\ddot{x} + c\dot{x} + kx = 0.$$

If displaced, the spring force kx restores the mass toward equilibrium, and the damper $c\dot{x}$ dissipates kinetic energy. With the right parameters, the system oscillates back to rest. No external controller is needed.

In Plain Language 31.2. Building Stability into Hardware

Think of a skilled golfer's body as a pre-tuned spring-mass-damper system. Before the swing, the golfer's muscles are pre-tensioned to create the right amount of stiffness (via the "spring" component) and damping. This is not voluntary muscular effort during the swing—it's the *baseline condition* set before the swing begins.
Once set, this mechanical system can be partly self-correcting. If a perturbation moves the club, muscle and tendon stiffness can resist some of that displacement. If the downswing is slightly faster than average, damping can dissipate some extra energy. This reduces the required feedback burden; it does not eliminate active neural control.

This is why expert golfers can swing with apparent ease. They are not working harder; they have tuned the mechanical system better.

31.3 Impedance Control: The Brain's Real Job

In the 1980s, the control theorist Neville Hogan proposed a radical insight: the brain does not control *position* and does not control *force*. Instead, it controls *impedance*.Hogan [1984, 1985a]

Definition 31.2. Mechanical Impedance

Mechanical impedance is the relationship between a position perturbation and the restoring force. Formally, for a joint, impedance is characterized by:

$$\boldsymbol{\tau}_{\text{resist}} = -\boldsymbol{K}(\boldsymbol{q} - \boldsymbol{q}_0) - \boldsymbol{D}\dot{\boldsymbol{q}},$$

where $\boldsymbol{K}$ is the stiffness matrix, $\boldsymbol{D}$ is the damping matrix, $\boldsymbol{q}$ is the joint position, $\boldsymbol{q}_0$ is the reference position, and $\dot{\boldsymbol{q}}$ is the joint angular velocity.
This equation assumes the desired velocity is zero ($\dot{\boldsymbol{q}}_d = 0$), appropriate for posture maintenance. During the swing, the full impedance control law includes a feedforward term and desired trajectory:

$$\boldsymbol{\tau} = \boldsymbol{\tau}_{\text{ff}}(t) - \boldsymbol{K}(\boldsymbol{q} - \boldsymbol{q}_d(t)) - \boldsymbol{D}(\dot{\boldsymbol{q}} - \dot{\boldsymbol{q}}_d(t))$$

where $\boldsymbol{\tau}_{\text{ff}}$ is the feedforward torque computed from the internal model, and $\boldsymbol{q}_d(t)$, $\dot{\boldsymbol{q}}_d(t)$ are the desired trajectory and velocity. The brain sets all three—feedforward torque, desired trajectory, and impedance gains—before the downswing begins.
High impedance: stiff, resistant to perturbation, low compliance. **Low impedance:** soft, compliant, allows motion.

The impedance parameters $\boldsymbol{K}$ and $\boldsymbol{D}$ are not commands the brain sends moment-to-moment. They are *properties set by muscle co-contraction*. When antagonist muscles (agonist and antagonist) both activate, they pull against each other, creating stiffness. More co-contraction → higher $\boldsymbol{K}$ and $\boldsymbol{D}$ → stiffer, more damped joints.

Key Principle 31.1. The Impedance Control Hypothesis

The primary job of the motor cortex is not to compute torques but to *select the impedance profile* for the task at hand. Once impedance is set, the passive mechanical system executes the motion with minimal real-time oversight.

31.3.1 The Impedance Profile of the Golf Swing

Different phases of the golf swing require different impedance profiles. Understanding this reveals why expert golfers move the way they do.

Phase	Proximal Impedance	Distal Impedance	Purpose
Address	Moderate	Moderate	Stable stance, ready position
Backswing	Low	Low	Smooth, fluid motion, low effort
Transition	**Rapid Increase**	Increasing	"Load" the system, store elastic energy
Downswing	**High**	Decreasing	Drive from the large muscles, release at the sm
Impact	Very High	Very High	Resist collision forces, prevent injury
Follow-through	Decreasing	Decreasing	Decelerate smoothly, dissipate energy

The critical distinction is the *proximal-to-distal impedance gradient*. Look at the downswing:

- **High proximal impedance:** The torso, hips, and shoulders are stiff. The large muscles can transfer power without deforming.
- **Low distal impedance:** The wrists and forearms are compliant. They release and accelerate under the inertial load of the proximal segments.

This gradient is the physical signature of the kinetic chain. It allows energy to flow from large, powerful proximal segments to smaller, faster distal segments—a cascade that produces the high speed required at impact.

The damping component of impedance—the $\boldsymbol{D}\dot{\boldsymbol{q}}$ term—is not merely a passive energy sink. As we explore in Chapter 16, damping at each joint propagates through the mass matrix coupling to affect every other joint in the chain. The brain's choice of co-contraction level at one joint therefore has system-wide consequences for the dynamics of every other joint.

In Plain Language 31.3. Why Novices Grip Too Hard

A beginning golfer, anxious about controlling the club, sets *uniformly high impedance*—they grip tightly, tense the wrists, tense the shoulders. All joints are stiff.

This prevents the distal release. The wrists cannot accelerate because they are held rigid. The proximal muscles cannot unload their energy efficiently because the distal joints will not accept it. The result: a slow, tense, inefficient swing.

An expert golfer, by contrast, maintains high proximal impedance while actively *reducing* distal impedance during the downswing. This allows the whip—the distal segments accelerate and release after the proximal segments have done their work.

31.4 The Pre-Tuned System: Setting the Springs Before Execution

Here is the key idea: *before the downswing begins*, the brain computes the *impedance parameters* and activates the muscles to set those parameters. Once set, the muscles maintain those stiffness and damping values throughout the swing.

Mathematically, the control input changes from a time-varying torque signal $\boldsymbol{u}(t)$ to a set of fixed impedance parameters:

$$\{\boldsymbol{K}, \boldsymbol{D}, \boldsymbol{q}_0\} \quad \text{(set once before the swing)}$$

During the downswing, the actual torque at each joint is then:

$$\boldsymbol{\tau}(t) = -\boldsymbol{K}(\boldsymbol{q}(t) - \boldsymbol{q}_0) - \boldsymbol{D}\dot{\boldsymbol{q}}(t) + \boldsymbol{\tau}_{\text{feedforward}}(t).$$

The three terms break down as follows:

Spring term: $-\boldsymbol{K}(\boldsymbol{q} - \boldsymbol{q}_0)$ is the restoring force when the joint deviates from the reference position $\boldsymbol{q}_0$. This term is *automatic*—it requires no neural computation during the swing.

Damping term: $-\boldsymbol{D}\dot{\boldsymbol{q}}$ is the energy dissipation when the joint moves. Again, automatic.

Feedforward term: $\boldsymbol{\tau}_{\text{feedforward}}(t)$ is a pre-computed trajectory drive based on the desired swing pattern. This is computed once before the swing, not updated in real time.

Compare this to the traditional view:

Traditional View
The brain computes $\boldsymbol{u}(t)$ at every millisecond:

Computational cost:

$$\begin{aligned} &\approx 20 \text{ DOF} \times 1000 \text{ Hz} \\ &= 20{,}000 \text{ decisions/s} \end{aligned}$$

Requires:

- Fast forward models
- Accurate state feedback
- High-bandwidth neural communication

Vulnerable to:

- Latency
- Noise
- Model error

Impedance Control View
The brain computes impedance parameters once:

Computational cost:

$$\begin{aligned} &\approx 40 \text{ parameters} \\ &(20 \text{ stiffness} + 20 \text{ damping}) \\ &+ \text{pre-computed feedforward} \end{aligned}$$

Requires:

- Coarse models (body morphology)
- Proprioceptive feedback (low bandwidth)
- Moderate bandwidth communication

Robust to:

- Small perturbations (spring restores)
- Latency (pre-tuned system)
- Noise (damping absorbs)

The impedance control view reduces the brain's computational burden by orders of magnitude. The brain sets the conditions, and the pre-tuned mechanical system handles execution.

Constraint Insight 31.1. Why Pre-Tuning Works

Pre-tuning is effective because the golf swing is a *highly stereotyped, repeatable task*. The desired swing trajectory is similar from one execution to the next. The impedance parameters that worked last time will work again.

If the task were novel or required real-time adaptation (e.g., hitting a moving target), pre-tuning alone would not suffice. But for the repeated, self-generated golf swing, pre-tuning is an excellent solution.
This is why training and practice are so important: they allow the motor system to discover and *encode* the impedance parameters that work best for each golfer's body morphology and swing style.

31.5 Passive Control in the ZTCF Family Framework

Recall from Chapter 6 the Zero Torque Counterfactual (ZTCF) decomposition:

$$\dot{\boldsymbol{x}} = f(\boldsymbol{x}) + G(\boldsymbol{x})\boldsymbol{u}.$$

The drift $f(\boldsymbol{x})$ represents the system's natural evolution without control. If you released the golfer's muscles, the arms would fall under gravity and Coriolis forces—this is the drift.

How do passive impedance forces fit into this framework?

If the golfer pre-sets muscle stiffness $\boldsymbol{K}$ and damping $\boldsymbol{D}$, these are *automatic forces*. They are not part of the feedforward control $G(\boldsymbol{x})\boldsymbol{u}$. Instead, they modify the *drift field itself*:

$$\dot{\boldsymbol{x}} = f_{\text{original}}(\boldsymbol{x}) + f_{\text{impedance}}(\boldsymbol{x}) + G(\boldsymbol{x})\boldsymbol{u}_{\text{feedforward}}.$$

The impedance-augmented drift becomes:

$$f_{\text{impedance}}(\boldsymbol{x}) = \begin{bmatrix} \dot{\boldsymbol{q}} \\ -\boldsymbol{M}^{-1}(\boldsymbol{q})[\boldsymbol{K}(\boldsymbol{q}-\boldsymbol{q}_0) + \boldsymbol{D}\dot{\boldsymbol{q}}] \end{bmatrix}.$$

This means the ZTCF—the fraction of the motion explained by gravity, inertia, and Coriolis forces *alone*—increases dramatically when impedance is included. The original drift accounts for maybe 40–50% of the swing. The impedance-augmented drift accounts for 70–80% (estimated from simulation studies comparing ZTCF trajectories with and without pre-set impedance; experimental validation with EMG-derived stiffness profiles remains an active area of research).

Drift vs. Control 31.1. Bare Drift vs. Impedance-Augmented Drift

Bare ZTCF (gravity + inertia only):

- Represents a completely passive arm: muscles off, no co-contraction
- Shows the trajectory a relaxed arm would follow
- Explains ∼40–50% of the actual swing
- Unrealistic because muscles are never completely off

Impedance-Augmented ZTCF (gravity + inertia + preset stiffness + preset damping):

- Represents an arm with pre-set muscle tension and co-contraction
- Shows the trajectory the arm follows with no additional muscular effort
- Explains ∼70–80% of the actual swing (estimated from simulation studies comparing ZTCF trajectories with and without pre-set impedance; experimental validation with EMG-derived stiffness profiles remains an active area of research)
- Realistic because muscles are always at some baseline tone
- Allows small feedforward corrections to fine-tune the motion

The impedance-augmented drift is a much better predictor of the golfer's actual swing because it accounts for the muscle tension that is actually present.

31.6 Muscle Tone and Spinal Feedback

Even a "relaxed" muscle is not inert. It maintains baseline activity called *muscle tone*. This tone serves multiple purposes, and understanding it reveals how the body distributes control away from the brain.

31.6.1 The Stretch Reflex Loop

Skeletal muscles contain specialized sensory receptors called *muscle spindles*. These spindles detect stretch—the lengthening of the muscle. When a muscle is stretched:

1. Spindles fire (increase firing rate)
2. Sensory signals travel to the spinal cord (10–20 ms)
3. Spinal circuits activate the motor neurons of the *same muscle*
4. Muscle contracts and resists the stretch

This is called the *monosynaptic stretch reflex* because it involves only one synapse in the spinal cord. The latency is remarkably short: 20–30 ms. By contrast, conscious reflexes (routing through the brain) have latencies of 100–300 ms [Kandel et al., 2013, Enoka, 2002].

The stretch reflex is a *local, automatic, distributed controller*. Each muscle has its own feedback loop built into the spinal cord. The brain does not have to manage it.

31.6.2 Gamma Drive: Configuring the Reflex

The brain does not bypass the stretch reflex. Instead, it *configures* the reflex sensitivity via *gamma motor neurons*. The motor cortex sends two types of commands:

Alpha motor neurons Innervate the main muscle fibers. They produce the primary contractile force.

Gamma motor neurons Innervate the spindle fibers themselves. They adjust the sensitivity of the stretch reflex.

High gamma drive:

- Increases spindle sensitivity
- Makes the stretch reflex stronger
- Creates a stiff, resistant muscle

Low gamma drive:

- Decreases spindle sensitivity
- Weakens the stretch reflex
- Creates a soft, compliant muscle

Before the golf swing, the motor cortex sets the gamma drive for each muscle group. This determines the baseline stiffness and the sensitivity to perturbations. Once set, the spinal circuits handle the automatic response to perturbations.

In Plain Language 31.4. The Spinal Cord as a Distributed Controller

The spinal cord is not just a cable transmitting messages from the brain to the muscles. It is a sophisticated control system in its own right.

The stretch reflex is a feedback controller operating at the spinal level. The brain configures it (via gamma drive) but does not need to manage it moment-to-moment. If an external force perturbs the arm during the swing, the spinal circuits automatically adjust muscle tension to resist the perturbation.

This is why a golfer can swing smoothly even if wind gusts occur mid-swing. The spinal reflexes handle the perturbations automatically. The brain was never aware they happened.

31.7 Self-Organization in the Kinetic Chain

A properly pre-tuned kinetic chain exhibits notable emergent behaviors. These are not taught or controlled explicitly; they arise naturally from the mechanical properties and the physics of the system.

31.7.1 Energy Flow From Proximal to Distal

In Chapter 11, we saw that power flows from proximal segments (large muscles, high inertia) to distal segments (small muscles, low inertia). The mechanism is the inertial load on the distal joints.

When the proximal joint rotates, the distal segment experiences a centripetal acceleration (and thus an inertial force) in the rotating frame. If the proximal impedance is high and the distal impedance is low, the distal segment will accelerate. The transfer of energy is a consequence of the impedance gradient and the rotational kinematics.

This is not a controlled transfer. The brain does not send a command “transfer power to the forearm.” Instead, by setting the impedance ratio (high proximal, low distal), the physics of the kinetic chain ensures that power flows naturally.

31.7.2 The Whip Effect: Emergent Distal Acceleration

One of the most striking features of an expert golf swing is the whip: the distal segments accelerate rapidly, often reaching peak velocity *after* the proximal segments have decelerated.

How does this happen? It is not because the brain sends a command to the wrists at the right moment. Instead:

1. The proximal joints (hips, torso) decelerate due to increasing impedance and muscle activation opposing the motion.

2. As the proximal joint torque decreases, the inertial load on the distal joint decreases.

3. With low distal impedance and decreasing proximal torque, the distal joint accelerates freely under the remaining inertial forces.

4. Peak distal velocity occurs when proximal velocity has nearly reached zero.

This is a pure consequence of the kinetic chain kinematics and the impedance profile. It emerges automatically.

Key Principle 31.2. Emergent Properties of Self-Organized Motion

When the body is pre-tuned with the correct impedance profile, complex behaviors like the whip effect emerge without explicit neural control. The physics of the kinetic chain, combined with the preset mechanical properties, produces the desired motion automatically.
This is the insight of the dynamical systems approach to motor control: *the brain does not compute the detailed trajectory; it computes the conditions under which the desired trajectory emerges from the physics.*

31.7.3 Impact Protection Through High Distal Impedance

At impact, the club strikes the ball with a force of hundreds of kilograms. This force propagates back up the kinetic chain through the shaft and toward the hands.

If the wrist and forearm impedance were low at impact, the force would propagate up the chain. The golfer's arm and shoulder would decelerate abruptly, creating a dangerous jerk and potential injury.

Instead, skilled golfers *increase* distal impedance at the moment of impact. The wrists become stiff, the forearms co-contract, and the hands grip firmly. This impedance barrier prevents the impact force from propagating up the chain. The energy is dissipated locally at the wrist and hand, protecting the elbow and shoulder.

A beginner who does not stiffen at impact risks tennis elbow (lateral epicondylitis) and rotator cuff injuries. An expert, by automatically stiffening, distributes the impact force and protects the soft tissues.

31.8 Stability Analysis: Is the Swing Self-Correcting?

How can we quantify whether a pre-tuned swing is self-correcting? The language of control theory offers a rigorous approach: *Lyapunov stability*.

31.8.1 The Lyapunov Function

A *Lyapunov function* is a scalar-valued function $V(\boldsymbol{x})$ that measures "distance" from a desired state. If the Lyapunov function decreases along the system's trajectories, the system is stable: perturbations shrink and the system returns to the desired state.

For a passively controlled system with stiffness $\boldsymbol{K}$, damping $\boldsymbol{D}$, and desired trajectory $\boldsymbol{q}_d(t)$, a natural Lyapunov function is the sum of kinetic and potential energy:

$$V(\boldsymbol{q}, \dot{\boldsymbol{q}}) = \frac{1}{2}\dot{\boldsymbol{q}}^T \boldsymbol{M}(\boldsymbol{q})\dot{\boldsymbol{q}} + \frac{1}{2}(\boldsymbol{q} - \boldsymbol{q}_d)^T \boldsymbol{K}(\boldsymbol{q} - \boldsymbol{q}_d).$$

The first term is the kinetic energy. The second term is the "elastic potential energy" stored in the impedance (spring-like stiffness).

Now, compute the time derivative along the trajectory:

$$\dot{V} = \dot{\boldsymbol{q}}^T \boldsymbol{M}\ddot{\boldsymbol{q}} + \frac{1}{2}\dot{\boldsymbol{q}}^T \dot{\boldsymbol{M}}\dot{\boldsymbol{q}} + (\boldsymbol{q} - \boldsymbol{q}_d)^T \boldsymbol{K}\dot{\boldsymbol{q}}.$$

Substituting the equation of motion $\boldsymbol{M}\ddot{\boldsymbol{q}} = -\boldsymbol{K}(\boldsymbol{q} - \boldsymbol{q}_d) - \boldsymbol{D}\dot{\boldsymbol{q}} + \text{other forces}$:

$$\dot{V} = -\dot{\boldsymbol{q}}^T \boldsymbol{D}\dot{\boldsymbol{q}} + \text{other force terms}.$$

If damping is sufficiently large, the term $-\dot{\boldsymbol{q}}^T \boldsymbol{D}\dot{\boldsymbol{q}}$ dominates. Since this term is negative (damping always dissipates energy), we have $\dot{V} < 0$.

Theorem 31.1. Passive Stability Condition

If a linearized impedance-controlled model is locally accurate and the damping term dominates the neglected perturbation terms in a neighborhood, then:

$$\dot{V} = -\dot{\boldsymbol{q}}^T \boldsymbol{D}\dot{\boldsymbol{q}} + (\text{small perturbation terms}) < 0.$$

This supports a local Lyapunov-stability claim for the model. It does not by itself establish exponential return of a full golf swing under large perturbations, delayed feedback, contact changes, or parameter uncertainty.

31.8.2 Conditions for Self-Correction

For a modeled golf swing to show partly self-correcting behavior, three conditions should hold:

1. **Sufficient stiffness:** The spring constant $\boldsymbol{K}$ must be large enough that restoring forces overcome small perturbations. Too soft, and the swing drifts off course.

2. **Sufficient damping:** The damping matrix $\boldsymbol{D}$ must dissipate perturbation energy. Without damping, perturbations oscillate indefinitely.

3. **Correct equilibrium:** The reference position $\boldsymbol{q}_0$ must be chosen so the spring's natural rest position aligns with the desired swing trajectory. Misalignment introduces systematic errors.

Elite golfers may approximate all three conditions more closely than novices. In that case, damping can absorb part of an over-speed perturbation and passive stiffness can resist part of a displacement. That is a robustness hypothesis, not a proof that every elite swing self-corrects.

Novices often fail at condition 1 or 2. A weak golfer may have insufficient co-contraction, leading to low stiffness and a drifting swing. A tense golfer may have excessive co-contraction, leading to high damping and a slower, more rigid swing.

In Plain Language 31.5. Why Expert Swings Look Effortless

The paradox is striking: an expert golfer swings with apparent ease, while a novice looks tense and strains.

The resolution may be that an expert has a better-matched impedance profile, so fewer large active corrections are needed once the swing starts. Passive tissues and fast local feedback then carry more of the stabilization burden.

A novice, without practice, has not yet learned the optimal impedance parameters. They compensate by increasing overall muscle tension—trying to muscle the swing under voluntary control. This looks effortful because it *is* effortful.

With training, the novice's motor system may discover a better impedance profile. The swing can then become more automatic, more robust, and apparently easier.

31.9 Training and the Discovery of Impedance

If passive control is so effective, how do golfers learn it? The answer lies in motor learning and practice.

31.9.1 The Pre-Training Problem

A beginning golfer has little practice-derived knowledge about the impedance parameters that work for their body. They face a high-dimensional search problem: what combination of muscle stiffness, damping, and co-contraction produces the desired swing?

Without knowledge, they often default to high impedance everywhere—the "grip it and rip it" approach. This is suboptimal but safe: high stiffness prevents wild motions and keeps the club on a repeatable path.

31.9.2 Practice as Impedance Search

Through repetitive practice, the motor system gradually discovers the impedance parameters that work. This is a slow process, often involving:

- **Coarse tuning:** The golfer learns rough impedance levels (low/medium/high) for each phase and body region.
- **Fine tuning:** The motor system refines the parameters based on feedback (visual, proprioceptive, kinesthetic).
- **Skill consolidation:** With sufficient repetition, the impedance parameters become fixed (automatic) in the motor program. The golfer no longer has to think about them.

The brain's role is to *search for and encode* the impedance parameters, not to compute them in real time during execution.

31.9.3 Strength and Speed Training

Understanding impedance reframes how we think about golf training.

Strength training increases the maximum stiffness and damping the muscles can produce. A stronger golfer has a broader range of impedance options. If their current impedance is too low, they can increase it. This flexibility is valuable.

Speed training teaches the motor system to reduce unnecessary co-contraction. A fast, coordinated golfer uses just enough impedance to stabilize the swing—no more. This reduces effort and increases the kinetic energy available for the club.

Key Principle 31.3. Mechanical Conditions and Self-Organization

Bosch and others have emphasized that effective motor control emerges from proper mechanical conditions rather than detailed real-time micromanagement Bosch and Cook [2015]. Mechanical conditions include grip, stance, posture, muscle co-contraction patterns, and overall body configuration.

Once these conditions are established, the coupled dynamics of the musculoskeletal system produce coordinated motion automatically, without the need for constant active correction.

31.10 Toward a Unified Picture: Control, Physics, and Self-Organization

We have now surveyed the motor control system across four chapters:

1. **Chapter 28:** The brain and spinal cord have limited bandwidth. Real-time control of 20 DOF is impossible.

2. **Chapter 29:** The motor system learns and adapts by modifying forward models and internal representations.

3. **Chapter 29:** Movement inherently exhibits variability; the motor system manages it through noise shaping and regularization.

4. **Chapter 31:** The motor system avoids micromanagement by pre-tuning mechanical properties and distributing control to spinal and local circuits.

Passive control is the *final piece*. It shows how the brain solves the fundamental control problem: not by computing fast, but by computing smart. By setting mechanical conditions that allow the physics and the body's own passive dynamics to do most of the work.

The golf swing emerges from the interaction of three elements:

Physics
Gravity, inertia, and Coriolis forces shape the motion through the drift field $f(\boldsymbol{x})$. The kinetic chain physics couples the segments; energy flows from proximal to distal naturally.

Impedance
The muscles pre-set stiffness and damping, augmenting the drift and enabling passive stability.
Spinal circuits maintain baseline control with minimal brain oversight.

Feedforward Drive
The motor cortex precomputes a trajectory and provides a feedforward command $\boldsymbol{\tau}_{\text{ff}}$ that biases the motion.
This small feedforward signal, combined with passive stability, produces the final swing.

This is a fundamentally different view from the traditional picture of the brain as an all-knowing controller. Here, the brain is a *smart configurator*. It sets conditions and allows the laws of physics and the body's mechanical properties to produce the desired motion.

Key Principle 31.4. Key Takeaways

1. **The bandwidth problem:** The brain cannot compute torques for 20 DOF at 1000 Hz. The traditional control model is unrealistic.
2. **Impedance control:** One plausible strategy is to set impedance before and during the swing so passive mechanics and fast local feedback handle part of the execution burden.
3. **Pre-tuning reduces computation:** Pre-tuning can shift some stabilization burden to mechanics and local feedback, though the exact dimensionality depends on the model.
4. **Passive stability:** With well-chosen impedance, some perturbations can be partly resisted by spring forces and damping.
5. **Distributed control:** Spinal reflexes, muscle spindles, and local feedback circuits are candidate mechanisms for part of the moment-to-moment stabilization.
6. **Emergent behaviors:** Complex behaviors like the whip effect and distal acceleration can emerge from the kinetic chain physics and the impedance profile without being scripted joint by joint.
7. **Training discovers impedance:** Through practice, the motor system discovers the impedance parameters that work best for each golfer's body. Expert performance reflects discovered, encoded impedance.
8. **Effortlessness is tuning:** Expert golfers look effortless because their passive impedance profile is well-tuned, not because they are stronger or faster.

Exercises 31.1. Chapter Exercises

Conceptual Questions

1. Explain why the impedance profile changes across the phases of the golf swing

(address, backswing, transition, downswing, impact, follow-through). For each phase, explain what impedance changes are necessary and why.

2. A Lyapunov function measures "distance from the desired state." Explain in plain language why a decreasing Lyapunov function ($\dot{V} < 0$) implies that perturbations shrink and the system returns to the desired trajectory.

3. The traditional control model requires the brain to compute ~20,000 decisions per second. The impedance control model requires the brain to compute ~40 parameters once. Explain how this 500-fold reduction in computational rate is achieved, and why it does not sacrifice precision.

4. What is the stretch reflex? Explain how it operates without conscious involvement of the brain. How does the brain modulate the stretch reflex using gamma motor neurons?

5. Why do novices grip too hard while experts grip with appropriate pressure? Relate this to the impedance control framework.

Mathematical Problems

1. Write the equation for passive joint torque in the form $\boldsymbol{\tau} = -\boldsymbol{K}(\boldsymbol{q} - \boldsymbol{q}_0) - \boldsymbol{D}\dot{\boldsymbol{q}}$. For a single joint with $K = 50$ N·m/rad, $D = 5$ N·m·s/rad, reference angle $q_0 = 0$, current angle $q = 0.1$ rad, and angular velocity $\dot{q} = 1$ rad/s, compute the restoring torque. Is the torque in the direction of decreasing q (i.e., does it resist the perturbation)?

2. For the Lyapunov function $V = \frac{1}{2}m\dot{q}^2 + \frac{1}{2}K(q - q_0)^2$, compute $\dot{V}$ along a trajectory governed by $m\ddot{q} = -K(q - q_0) - D\dot{q}$. Simplify and show that $\dot{V} = -D\dot{q}^2$. Explain why $\dot{V} < 0$ ensures Lyapunov stability.

3. Sketch the phase portrait (velocity $\dot{q}$ vs. position q) for a spring-mass-damper system with $K = 100$, $D = 20$, $m = 1$. Indicate the direction of motion and show how trajectories spiral toward equilibrium. How does the trajectory change if damping is increased to $D = 50$? Decreased to $D = 5$?

4. Suppose a golfer's wrist joint requires $(K_{\text{wrist}}, D_{\text{wrist}})$ such that the wrist releases smoothly during the downswing but is stiff enough to resist impact forces. If K is too low, the wrist collapses at impact (risk of injury). If K is too high, the wrist does not release (loss of club head speed). For a wrist with moment of inertia $I = 0.01$ kg·m^2 and impact duration $\tau = 0.005$ s, estimate the range of reasonable K values. (Hint: the system should respond to the impact without unstable resonance.)

Experimental / Applied Questions

1. Perform the waggle before your next round of golf. As you waggle, pay attention to the *feel* of the impedance (muscle tension, stiffness). If the feel is "off" (too tense or too loose), adjust your setup and waggle again. After practice, does your awareness of impedance improve? Does the correlation between "good waggle feel" and "good swing" improve?

2. Using an instrumented grip pressure sensor (or your subjective feeling), track how grip pressure changes across the phases of your swing: address, backswing, downswing, impact, follow-through. Does your grip pressure profile match the expected impedance profile? Where do you need to adjust?

3. Have a partner apply a small, unexpected perturbation (gentle push on your arm or shoulder) during different phases of your swing: backswing, downswing, impact. Can you recover and hit a good shot? What does this reveal about the passive stability of your swing? Is the stability due to impedance (spring forces) or to active corrective muscle engagement?

Part X

Synthesis

Chapter 32

Where Disciplines Collide: An Interdisciplinary Perspective

32.1 Why Disciplines Matter

A golfer changes a cue, a shaft, or a training routine and the ball flies differently. What changed between the intention and the outcome? The answer may involve the chosen movement, muscle activation, contact with the ground, elastic deformation, club delivery, ball impact, or the measurement itself. Several of these can change together. A convincing explanation must follow the connections without treating a successful shot as proof of every proposed mechanism.

The central question of this book is how a golfer prepares and regulates a coupled mechanical system so that its evolving motion delivers a useful impact. Momentum, gravity, constraints and elasticity can help carry a movement forward. Muscles prepare that motion, continue to exchange energy with it, and alter its response to disturbances. The interesting comparison is therefore between specified movement strategies under specified conditions, with their benefits and costs measured at the appropriate level.

In Plain Language 32.1. Multiple Disciplines, One Framework

A mechanical model connects forces to motion. A controller connects information and goals to actions. A measurement model connects physical events to recorded data. An injury study connects exposure and other factors to health outcomes. These models can inform one another, but their conclusions are not interchangeable. A mathematically valid simulation can motivate an experiment without establishing what a person's nervous system does or which technique prevents injury.

This chapter builds those connections explicitly. Its numerical examples are declared teaching models, not measured golf populations or fitting recommendations. The worked answers also show how to challenge the framework: specify an alternative mechanism that predicts the same observation, then identify a measurement or intervention that could distinguish them.

32.2 Interdisciplinary Connections: Overview

The useful common structure is a chain of interacting models. The golfer's policy selects commands using goals, observations and internal state. The body and club transform commands and external forces into motion. Impact converts a particular delivery and contact configuration into club and ball changes. Flight, landing and subsequent motion determine the result. Sensors observe selected parts of this process, and their reports feed into subsequent decisions.

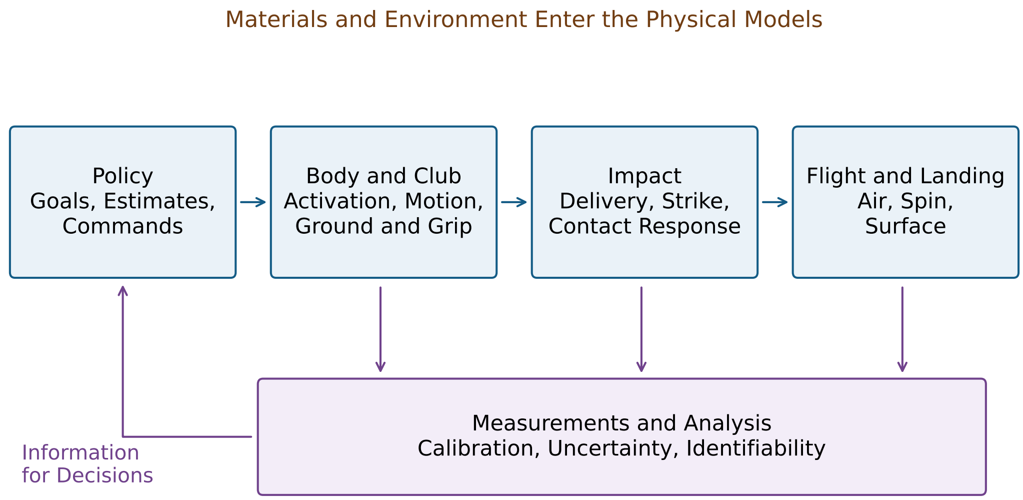


Figure 32.1: Connected Models Link Commands, Motion, Impact and Outcome. Measurements Observe Selected Stages and Inform Decisions; the Map Does Not Assign These Functions to Brain Regions.

Figure 32.1 is a map of questions, not a diagram of brain anatomy. Materials enter both the moving system and contact law. Ground and grip conditions constrain the motion. Neural and mechanical responses operate during the swing, while learning changes future policies. Uncertainty propagates through every stage; a precise final number does not imply that every upstream quantity was observed.

This chain also explains why one scalar cannot represent overall swing quality. Ball speed, directional dispersion, joint loading, effort, robustness and learning are different outcomes. Improving one may change another. A rigorous argument names the outcome, comparison and operating conditions before it calls a movement efficient, stable, safe or optimal.

32.3 Control Theory and Biomechanics: The Affine Bridge

In a local coordinate description, a constrained multibody model may be written

$$M(q)\ddot{q} + c(q,\dot{q}) + g(q) + d(q,\dot{q}) = B(q)\tau + J(q)^T\lambda + Q_{\text{ext}}.$$

Here q describes configuration, M is the inertia matrix, c collects velocity-dependent inertial terms, g is gravity, and d represents the explicitly modeled dissipative forces. Joint torques τ, contact reactions λ, and other external generalized forces must be assigned to the correct system boundary. Ideal constraints can be eliminated locally or retained with their acceleration equations. Contact creation, separation and slip may require different modes.

After choosing independent states and inputs, one possible model is

$$\dot{x} = f(x) + G(x)u.$$

This form says that the instantaneous state derivative is affine in the selected input. It does not say that motion is linear in initial conditions, that muscle force is globally linear in excitation, or that every biological model has this form. If the input is joint torque, muscle activation and tendon behavior may lie outside that simplified plant. If the input is neural excitation, activation and elastic states generally belong inside it. An input-dependent activation time constant can require a more general or piecewise description.

32.3.1 Input Coordinates and the Meaning of Zero

The drift f is the vector field obtained when the declared input is zero. It is not the set of effects that cannot be controlled. Control can change the state at which drift acts, and feedback can change the baseline used to describe the system. Under the input substitution

$$u = \alpha(x) + \beta(x)\nu, \qquad \dot{x} = \underbrace{f(x) + G(x)\alpha(x)}_{\widetilde{f}(x)} + \underbrace{G(x)\beta(x)}_{\widetilde{G}(x)}\,\nu,$$

zero residual input ν retains the baseline action $\alpha(x)$. For an invertible β, this is an equivalent input description if the admissible input set is transformed as well. If β restricts input directions, it also restricts the controller.

Consider a rotor with inertia I and torque $u = -kq + \nu$. Zero ν retains the restoring torque $-kq$; zero physical torque removes it. At $I = 0.05\,\mathrm{kg\,m^2}$, $k = 2\,\mathrm{N\,m/rad}$ and $q = 0.1\,\mathrm{rad}$, the former gives an angular acceleration of $-4\,\mathrm{rad/s^2}$. The two zero-input experiments answer different questions. A zero-excitation muscle model likewise retains whatever activation and tendon strain already exist; it is not an instantaneous zero-force intervention.

Changing the state at the top of the backswing changes the trajectory through a fixed vector field. Changing equipment or tissue parameters changes the plant itself. Changing a feedback policy changes the resulting closed-loop vector field. These are three distinct ways to obtain a different swing, even though all can change the club's eventual delivery.

32.3.2 Control Authority Requires a Task and a Time

A large nominal drift term does not prove that additional input has little effect. For the constant-inertia rotor,

$$\ddot{q} = \tau/I, \qquad \delta q(T) = \frac{\delta\tau}{2I}(T - t_0)^2$$

for a constant added torque starting at t_0, with zero initial perturbation. The torque-to-acceleration gain is $1/I$ at every angular speed. With $I = 0.05\,\mathrm{kg\,m^2}$, a $0.5\,\mathrm{N\,m}$ increment acting for $0.1\,\mathrm{s}$ changes angle by $0.05\,\mathrm{rad}$ and angular speed by $1\,\mathrm{rad/s}$, whether the initial speed is zero or $100\,\mathrm{rad/s}$.

A late intervention can nevertheless have little effect because little time remains, the command arrives after contact, activation develops slowly, torque is saturated, or available directions poorly affect the desired output. For a linearization about a specified trajectory, a fixed-time output perturbation has the form

$$\delta y(T) = C(T)\Phi(T, t_0)\delta x(t_0) + \int_{t_0}^{T} C(T)\Phi(T, s)G(x(s))\delta u(s)\,ds.$$

The state-transition matrix Φ propagates perturbations through the coupled dynamics; C maps state changes to the chosen output. A state-triggered impact also requires event-time sensitivity. Input limits, delays and units must be included before comparing authority across phases or joints. The computational-brain chapter (Chapter 30) develops a delayed-activation example that makes this distinction quantitative.

32.3.3 Minimum Effort Requires a Cost and a Horizon

Preparing a useful trajectory can reduce later correction, but proximity to the zero-input endpoint is not a universal minimum-effort criterion. For a controllable linear system with an unconstrained quadratic input cost, the minimum cost of an endpoint correction e is

$$J_{\min} = e^T W^{-1} e, \qquad W = \int_0^T e^{A(T-s)} B R^{-1} B^T e^{A^T(T-s)}\, ds,$$

where the cost is $\int_0^T u^T R u\, dt$ and R is positive definite. A singular Gramian requires a reachable endpoint and the corresponding restricted solution. Saturation, nonlinear dynamics and path constraints can invalidate this unconstrained formula.

The result follows by minimizing the cost subject to the endpoint integral constraint. A Lagrange multiplier gives

$$u_*(s) = R^{-1} B^T e^{A^T(T-s)} W^{-1} e.$$

Substitution into the endpoint integral returns $WW^{-1}e = e$, and substitution into the cost returns $e^T W^{-1} e$. The weighting therefore comes from the dynamics and input cost, not an arbitrary distance between poses.

To see why the metric matters, nondimensionalize a double integrator using a one-unit horizon, a chosen position scale and its associated velocity scale. With unit input weight,

$$W = \begin{bmatrix} 1/3 & 1/2 \\ 1/2 & 1 \end{bmatrix}, \qquad W^{-1} = \begin{bmatrix} 12 & -6 \\ -6 & 4 \end{bmatrix}.$$

The correction $(0.1, -0.1)^T$ is shorter in Euclidean distance than $(0.2, 0.3)^T$, yet costs 0.28 rather than 0.12. Position and velocity errors can reinforce or oppose what the available input can accomplish. Even this input cost is not metabolic expenditure or injury risk. Those require additional models and evidence.

32.4 Robotics: What Golf Can Learn From Machines

Robotics provides explicit ways to connect task objectives, constraints, actuator limits and feedback. A robot can be designed with rigid or compliant transmissions, delayed sensing, learned policies and adaptive parameter estimates. Human-versus-machine comparisons should specify the systems being compared rather than assign flexibility exclusively to people or fixed control exclusively to robots.

32.4.1 Principle 1: Hierarchical Control

A useful design can separate a task objective, a movement policy and an actuator controller. For example, a target distribution may constrain acceptable launch conditions; a trajectory optimizer may search for feasible club delivery; a lower-level controller may regulate torque or interaction dynamics. Information can also pass between levels during execution.

This is an engineering decomposition, not proof that the motor cortex, cerebellum and spinal cord implement those stages in that order. Nor is a hierarchy mathematically necessary for every controller. A feedback policy can map estimated state directly to actuator commands. The important questions are whether its commands are feasible, whether it handles uncertainty, and whether it produces the intended response under disturbances.

For golf, a hierarchy is useful when it keeps the instruction at the right level. A target direction is an outcome goal; an intended club path is an intermediate task; a particular muscle activation is an implementation hypothesis. A coach who changes one level should measure the outcome at the others before attributing success to a specific mechanism.

32.4.2 Principle 2: Controlling Interaction Dynamics

Mechanical impedance describes a force–velocity relation. For a scalar linear mass–damper–spring perturbation with zero initial conditions, $M > 0$, $B \geq 0$ and $K \geq 0$,

$$F(s) = (Ms + B + K/s)V(s).$$

The force–displacement relation instead has dynamic stiffness $Ms^2 + Bs + K$. These quantities depend on frequency, direction, operating state and the chosen mechanical port. A compliant shaft, an activated muscle and a feedback controller need not have the same response merely because each can resist displacement.

For an elastic rotational controller with error $e = q - q_d$, nonnegative stiffness $K(t)$ and damping D,

$$\tau = -Ke - D\dot{e}, \qquad V_{\text{stored}} = \tfrac{1}{2}Ke^2,$$

and direct differentiation gives

$$\tau\dot{q} + \dot{V}_{\text{stored}} = -D\dot{e}^2 + \tau\dot{q}_d + \tfrac{1}{2}\dot{K}e^2.$$

With fixed reference and stiffness, damping removes energy at this port. Moving the reference or increasing stiffness while displaced can supply energy. At a clamped error of 0.1 rad, raising K from 10 to 30 N m/rad increases storage by 0.1 J. Calling all such behavior passive would hide an energy source.

Impedance control therefore suggests a research question: what response to a specified perturbation is useful at each phase? It does not prescribe maximal stiffness at impact or a universal early-soft/late-stiff schedule. Resistance in one direction may help face orientation while affecting another degree of freedom or increasing internal loading. The biology and computational-brain treatments discuss the mechanical and experimental distinctions behind this question.

32.4.3 Principle 3: Task-Space vs. Joint-Space Planning

Let a chosen club task coordinate be $y = h(q)$. Locally,

$$\dot{y} = J(q)\dot{q}, \qquad \tau_{\text{task}} = J(q)^T F_{\text{task}},$$

where the task force and velocity must use compatible coordinates and units. The second relation follows from virtual work. Full orientation requires an appropriate local representation or a geometric treatment on the rotation group. A three-dimensional club pose cannot be reconstructed from face azimuth and path azimuth alone.

Redundancy means that different joint motions can satisfy the same instantaneous task velocity. Such alternatives may differ in posture, contact feasibility, energy, sensitivity and tissue loading. A kinematic null direction is not automatically dynamically harmless: moving in it can change inertia, future reachability or the next task Jacobian. This is why a good club-level result does not uniquely identify a whole-body technique.

32.5 Why Robots Teach Us About Muscles

The analogy becomes productive when actuator dynamics are made explicit. Torque is an input in one model and an output of activation, force–length–velocity behavior, tendon deformation and moment arms in another. A command can change immediately while muscle force develops over time; stored elastic energy can persist after a command changes.

Muscle recordings add evidence but need their own observation model. Pink, Jobe and Perry's shoulder EMG study reports muscle- and phase-dependent activity; its abstract does not establish that a particular muscle simply turns off once drift takes over [Pink et al., 1990]. EMG is not a direct measurement of tendon force or joint torque. Electrode placement, normalization, activation dynamics and muscle geometry matter when connecting it to a mechanical model.

The useful robot experiment is correspondingly precise. Compare a torque-controlled rotor, a motor with series elasticity, and an actuator with delayed activation under the same output task. Match initial states and input limits, then perturb each system. Similar trajectories may coexist with different internal forces and correction capacities. Differences reveal which modeling elements matter, rather than which machine resembles a golfer in appearance.

32.6 Neuroscience: Motor Control and the Cerebellum

Control theory offers computational hypotheses about prediction, estimation and action selection. Neuroscience asks how those functions are implemented and how they interact. Mechanical equations alone cannot assign an algorithm to a brain region or rank one region as more important because a swing is fast.

32.6.1 The Cerebellum and Feedforward Control

A forward model predicts consequences of a state and command. An inverse model or policy selects commands intended to achieve an outcome. Inverse kinematics solves a geometric configuration problem; it is not synonymous with either a forward dynamics model or a complete motor controller. Planning a joint trajectory also leaves the force and activation problem unresolved.

Predictive control is useful when sensory information is delayed, but feedforward action need not be independent of feedback. A policy may use a state estimate that combines prior commands with new observations. The mechanical model then helps explain what information would be useful, while physiological experiments are needed to establish where and how such computations occur. Learning a useful prediction need not mean explicitly storing the functions named f and G in this book.

32.6.2 Motor Cortex and Feedback Control

The value of feedback depends on its information, noise, latency, the task, and the time remaining. The distant location of the ball's eventual landing is not an explanation for absent visual control: vision can also concern the target, body, club or environment. Proprioception and other sensory channels provide different information and can be combined with prediction.

Controlled reaching experiments demonstrate online visual corrections, and multijoint perturbation experiments show task-relevant responses involving motor cortex [Saunders and Knill, 2003, Pruszynski et al., 2011]. These results support studying feedback in context; they do not supply a universal golf-swing latency. A command triggered too late to affect impact cannot correct that impact, while forces and feedback processes initiated earlier can still be acting. Feedback after impact can also affect follow-through and subsequent learning.

32.6.3 Learning and Adaptation

A different shot after practice may reflect an explicit aiming strategy, implicit adaptation, altered coactivation, a changed state estimate, or a different movement policy. Mazzoni and Krakauer's adaptation experiment illustrates that explicit strategy and implicit adaptation can coexist and conflict [Mazzoni and Krakauer, 2006]. One successful trial therefore does not identify what was learned.

A discriminating golf experiment records the cue or perturbation, actual delivery and outcome, then examines retention, transfer and behavior when the perturbation is removed. If the claim concerns improved mechanical robustness, apply a controlled perturbation and measure the response. If it concerns implicit learning, use appropriate behavioral comparisons rather than infer it from a fitted trajectory. If it concerns a neural location, behavioral and mechanical evidence alone is insufficient.

32.7 Material Science: Shaft Design and Ball Physics

Equipment changes both the plant and, potentially, the golfer's policy. A comparison with prescribed identical grip motion asks how two shafts respond mechanically. A comparison in which a golfer freely adapts asks how each golfer–club combination performs. Both are useful, but they estimate different effects.

32.7.1 Shaft Materials and Design

An anisotropic laminate has direction-dependent stiffness. Axially aligned fibers can strongly support axial loading; off-axis plies can contribute importantly to shear and torsional response. A statement that angled fibers are more compliant is incomplete until the loading direction, constraints and response quantity are specified. Laminate analysis combines transformed ply constitutive laws with through-thickness force and moment balances [Roylance, 2000].

For a declared plane-stress orthotropic ply, use engineering shear strain and

$$\begin{bmatrix}\sigma_1\\ \sigma_2\\ \tau_{12}\end{bmatrix} = \begin{bmatrix}Q_{11} & Q_{12} & 0\\ Q_{12} & Q_{22} & 0\\ 0 & 0 & Q_{66}\end{bmatrix}\begin{bmatrix}\epsilon_1\\ \epsilon_2\\ \gamma_{12}\end{bmatrix}.$$

Take illustrative constants $E_1 = 135$, $E_2 = 10$, $G_{12} = 5$ in GPa and $\nu_{12} = 0.3$, with $\nu_{21} = \nu_{12}E_2/E_1$. Inverting the compliance matrix gives $Q_{11} = 135.9060$, $Q_{22} = 10.0671$, $Q_{12} = 3.0201$ and $Q_{66} = 5$ in GPa.

For a 45° ply, rotation gives the constrained-strain coefficients

$$\overline{Q}_{11} = \frac{Q_{11} + Q_{22} + 2Q_{12} + 4Q_{66}}{4} = 43.0034,$$
$$\overline{Q}_{66} = \frac{Q_{11} + Q_{22} - 2Q_{12}}{4} = 34.9832$$

in GPa. Axial response decreases while the shear coefficient increases. These are stiffness-matrix entries, not freely contracting Young's and shear moduli: extension–shear coupling can matter. A balanced layup can cancel some couplings, and ply positions determine bending response. A shaft's full bending and torsional behavior additionally depends on tube geometry, layup, boundary conditions, damping and frequency.

For an ideal uniform Euler–Bernoulli beam, a mode frequency scales as $\sqrt{EI/(\mu L^4)}$, with a boundary-dependent factor, bending rigidity EI, mass per length μ, and length L. Reducing mass does not logically require reducing stiffness. Nor does a lighter shaft automatically transfer a fixed saved-energy amount to the clubhead: the golfer may change work, speed and timing. A shaft stiffness label or an undefined shaft COR cannot settle that coupled comparison.

32.7.2 Ball Compression and the Coefficient of Restitution

For a one-dimensional normal collision, restitution is the ratio of separating to approaching relative speed,

$$e = \frac{v_b' - v_c'}{v_c - v_b},$$

under the declared positive direction and an initially approaching pair. Smash factor is outgoing ball speed divided by incoming club speed, a different quantity. Restitution also is not the fraction of initial club energy transferred to the ball.

Derive the distinction in a free two-mass model: positive mass M moves at speed $U > 0$ into stationary positive mass m, external impulse is negligible, and there is no rotation or tangential contact. Take $0 \le e \le 1$ for this dissipative collision example. Momentum conservation and the restitution law give

$$v_b' = \frac{(1+e)M}{M+m}U, \qquad v_c' = \frac{M - em}{M+m}U.$$

The ball's fraction of initial kinetic energy and the total dissipated fraction are

$$\eta_b = \frac{mM(1+e)^2}{(M+m)^2}, \qquad \eta_{\text{loss}} = (1-e^2)\frac{m}{M+m}.$$

Equivalently, lost energy is $(1-e^2)\mu_r U^2/2$, where the reduced mass is $\mu_r = Mm/(M+m)$. Restitution squared governs retention of relative translational kinetic energy in this model, not the ball's share of the initial laboratory-frame energy.

With illustrative $M = 0.200\,\text{kg}$, $m = 0.04593\,\text{kg}$ and $e = 0.83$, the speed ratio is 1.48823, the ball fraction is 0.5086, and the dissipated fraction is 0.0581. The remaining approximately 43.33% stays in the moving first mass. These identities can be checked directly from both outgoing velocities.

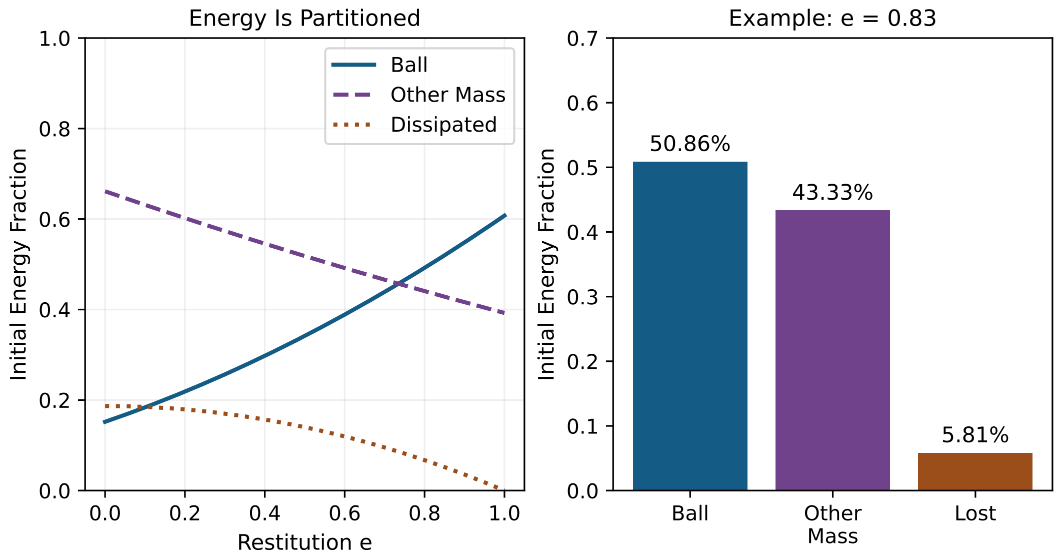


Figure 32.2: Restitution Does Not Equal Energy Transfer. The Free Two-Mass Teaching Model Uses a 0.200 kg Incoming Mass and a Stationary 0.04593 kg Ball; Fractions Sum to One.

A real club–ball impact can include rotation, off-center contact, tangential impulse, elastic waves and external coupling. Its effective normal inertia is not necessarily the head mass, much less the whole golfer's mass. The impact chapter (Chapter 17) treats those distinctions. The number 0.83 here is a model input, not a statement of a current equipment-conformance rule.

Compression describes deformation under a specified test; it does not alone determine restitution, launch, spin or dispersion. Arakawa and colleagues' normal impacts against a rigid steel target showed decreasing restitution with increasing inbound speed while normal compression increased [Arakawa et al., 2006]. That experiment contradicts a universal rule linking greater compression to greater restitution, but does not rank modern balls in a golfer's driver or wedge. Fitting requires defined impact conditions and measured outcomes, including variation across strikes.

32.8 Data Science and Launch Monitor Analysis

Data science enters through an observation equation,

$$z_k = h(x(t_k), \theta, \kappa) + \epsilon_k,$$

where θ contains physical parameters, κ calibration parameters, and ϵ_k measurement error. A launch report can combine directly observed signals with estimates and model-derived quantities. A ball-flight sensor is not automatically a full-body motion-capture system or a measurement of muscle force. The launch-monitor article explains the geometry and observability of several measurement approaches.

32.8.1 Identifying Dynamics From Data

Even complete noiseless motion need not separate a plant parameter from an unobserved input. Consider

$$I\ddot{q} + kq = u(t).$$

If one stiffness k_1 and input $u_1(t)$ reproduce an observed motion, any alternative k_2 reproduces it with

$$u_2(t) = u_1(t) + (k_2 - k_1)q(t),$$

provided that alternative input is admissible. A hundred terminal launch reports do not remove this structural ambiguity. More samples of the same confounded experiment may improve numerical precision while leaving the underlying cause unidentified.

Identification can improve with independently measured input or force, calibrated geometry, known boundary conditions, purposeful excitation, repeated conditions and external parameter measurements. Its success must be checked for the particular parameterization and experiment. Strong regularization can select one plausible model without making it uniquely supported by the data. An optimizer's convergence is not an observability result.

32.8.2 Predicting Ball Outcome From Swing Data

Prediction is a legitimate goal even when the internal mechanism is not identified. Its evaluation must match the intended use. A model trained on one golfer, club and session may interpolate well within that setting but fail after a shaft change, injury, different sensor, or new movement strategy. Separate training and evaluation by the units over which generalization is claimed, such as golfer, session or equipment condition.

The complete prediction chain is body/club state, delivery, contact, flight and landing. Each stage needs its own assumptions and uncertainty. Face angle and path do not specify attack angle, dynamic loft, strike location, angular velocity or all contact properties. Ball speed, launch angle and spin magnitude do not uniquely specify wind, aerodynamic coefficients or the full spin vector. Carry estimates require an explicit flight model; total distance additionally depends on landing and surface interaction.

A model should also be compared with simpler alternatives. If a regression from measured delivery predicts ball flight as well as a personalized full-body model, the extra body parameters have not yet demonstrated predictive value for that output. They may still be useful for other questions, but that requires separate evaluation.

32.8.3 Personalizing the Model

Personalization means estimating quantities the available experiment can support, with uncertainty and a record of calibration. Segment dimensions, inertial properties, joint constraints, actuator capacity, elastic response and policy parameters have different measurement requirements. Their estimates can be correlated.

Keep an explicit inventory of measured, estimated, assumed and unidentifiable quantities. Report held-out errors and failure conditions alongside parameter intervals. When an equipment change modifies both mechanics and behavior, distinguish a fixed-policy simulation from observed adaptation. A credible personalized model may be deliberately modest: for example, predicting a distribution of launch conditions within a calibrated range without claiming to recover every muscle's action.

32.9 Sports Medicine: From Force Estimates to Health Outcomes

32.9.1 Net Torque Is Not Tissue Load

Inverse dynamics estimates net generalized forces under its model and measurements. Multiple muscles, passive tissues and contact reactions can contribute to the same result. A low net moment therefore does not establish low tendon force or low joint compression.

Consider a planar teaching model with two parallel muscle forces that both compress a joint but have opposite 0.03 m moment arms. Forces of 200 and 100 N produce a net moment of 3 N m and a total compressive muscle contribution of 300 N. Forces of 500 and 400 N produce the same moment and a 900 N contribution. Additional external loads and joint reactions must still be included in a full balance. The example establishes non-uniqueness, not a clinical injury threshold.

Similarly, greater torque-to-acceleration gain does not imply that a larger torque is required for the same acceleration. In torque coordinates, strength limits primarily change the feasible torque set; in normalized activation coordinates, force capacity can also appear in the input map. Coordinate choice must be settled before interpreting a change in control authority as strength or tissue demand.

Injury risk concerns exposure, tissue capacity, recovery, prior injury and other factors over time. Robinson and colleagues followed 910 amateur golfers for five months and reported associations involving age, osteoarthritis and previous injury [Robinson et al., 2024]. They did not measure swing biomechanics. Self-report, population selection and incomplete questionnaire response further limit interpretation. Such surveillance helps define relevant outcomes; it does not validate a zero-torque technique as an injury-prevention intervention.

A mechanics-based hypothesis can propose that a change reduces a specified load under matched conditions. Testing prevention additionally requires suitable prospective health outcomes, exposure accounting and comparison groups. A decrease in one joint's estimated moment may shift demand elsewhere or accompany higher internal coactivation. The chapter consequently cannot derive a rehabilitation exercise, return-to-play decision or universally safer sequence from its equations. Those questions require individualized clinical assessment and relevant intervention evidence.

32.10 Coaching Science: Translating Physics Into Cues

Coaching connects a proposed mechanism to a learnable action and a measurable outcome. A cue is an intervention on behavior, not a boundary condition that guarantees a particular joint trajectory. Different golfers can interpret the same words differently, and a familiar cue can affect several mechanical variables at once.

32.10.1 Phase-Specific Coaching

An early change can alter the initial state for later motion, including momentum and elastic strain. A late change has less remaining time to propagate and may encounter activation or torque limits. Those observations motivate phase-specific experiments; they do not establish that speed problems are always drift problems or face errors are always late control problems.

For example, a changed setup might affect path, strike location and available torque directions together. An apparent improvement in face delivery might result from an earlier trajectory change rather than a conscious correction near impact. Record the relevant timing and delivery before deciding which mechanism explains the result.

32.10.2 Cue Design

Start with an observable difficulty, such as excessive variation in a specified face-to-path measure under a repeatable task. Propose one understandable cue, state its intended effect, and alternate or randomize comparable blocks with and without it. Record delivery, strike, ball outcome and perceived effort; include retention or transfer when the claim concerns learning.

Interpretation then has three levels. First, did the outcome improve beyond normal variability? Second, did the proposed intermediate variable change? Third, did that change plausibly account for the outcome under the experiment's controls? A positive answer to the first question does not settle the other two. Fatigue, order, familiarization and changed strike quality can explain part of an observed benefit.

Physics makes a cue more testable by identifying those intermediate variables and alternatives. It does not establish that a particular wording, more aggressive hip motion, grip change or release timing is universally effective. An unsuccessful cue can still teach something if the experiment reveals which assumed link failed.

32.11 Synthesis: The ZTCF Family Connects Testable Questions

A zero-torque counterfactual is most useful as a specified intervention in a specified model. Start from a declared state at a declared time, replace selected active torques with zero, retain or modify other mechanisms explicitly, and integrate to a defined horizon or event. Report what happens if balance, contact or joint constraints fail before the intended event.

Zero joint torque does not mean zero gravity, zero ground reaction, zero passive tissue force or zero previously stored energy. A branch with zero new excitation generally differs from one with zero activation or zero actuator force. These choices should be separate members of a counterfactual family, not silently conflated as the body's natural motion.

The contact mode matters for equipment comparisons too. Before any physical interaction with the ball, its compression law does not directly enter the golfer–club mechanical equations. A different ball can change the golfer's expectations, policy or prepared state, and its material response enters once contact begins. Assigning a direct pre-contact mechanical effect to ball compression would mix those pathways.

The comparison asks how the modeled future changes under that intervention. It does not uniquely apportion the actual club speed among muscles or prove that muscles did not create the motion. For a mechanically consistent model, an energy balance has the schematic form

$$\dot{E} = \dot{q}^T B\tau + P_{\text{ext}} - P_{\text{diss}},$$

with elastic storage included in E and appropriate work terms for moving constraints. Velocity-dependent inertial coupling can redistribute energy among coordinates without creating energy for the complete conservative system. Energy present at the branch point can reflect substantial earlier muscle work.

Different counterfactual interventions can interact nonlinearly. Removing two torques together need not produce the sum of their separately removed effects. A chosen attribution rule can summarize such interactions, but its result depends on the rule and baseline. This is particularly important when a zero-torque branch leaves the range in which the fitted model was validated.

The framework is therefore a common language for posing connected questions: what state was prepared, what forces remain, what alternatives are feasible, what can be observed, and what outcomes follow? Its strongest contribution is a reproducible comparison with explicit limits. Neural, coaching and clinical claims remain subject to their own experiments.

32.12 The Expanded Framework: From Rigid Bodies to Living Systems

Rigid segments are a modeling approximation, not a denial of deformation. Add degrees of freedom when they improve the questions the model can answer: spinal motion for segment-level loading, tendon or shaft states for elastic exchange, actuator states for delayed force development, or soft-tissue modes for measured relative motion. Each addition requires parameters and observations; a larger model can be less identifiable.

There is no universal anatomical degree-of-freedom count that follows by counting vertebrae or muscles. Define bodies, joints, constraints and contact modes, then determine the local independent coordinates. Redundant constraints or special configurations can change a simple counting argument. The spine and inverse-dynamics chapters (Chapters 23 and 19) develop these issues.

For a system containing the golfer and club, ground interaction and gravity are external, but so are aerodynamic loads and, during impact, the ball's action unless the ball is included in the system. A stationary ideal no-slip contact point can exert force and exchange momentum while doing zero point-contact mechanical work. That does not mean the ground force has no effect on center-of-mass kinetic energy: center-of-mass force times center-of-mass velocity is a different decomposition from the total body's external contact power. Real foot deformation, contact moments, slipping or moving supports require their own work accounting.

The same discipline applies to energy stored in tissue. Specify deformation relative to a reference, constitutive behavior, dissipation and active sources before describing recoil. Changes in mechanical work are not automatically changes in metabolic cost. A living system can maintain force while doing little external work, and the relation between those quantities is itself a physiological question.

32.13 The Future: Personalized Physics Models

A defensible development programme proceeds from calibrated measurements to identifiable models and then to tested interventions. Publish coordinate conventions, sensor timing, reference points, contact assumptions and parameter uncertainty. Verify the numerical implementation on problems with known solutions. Validate predictions against independent measurements in the intended operating range.

Next ask whether the model improves decisions. Compare a model-guided intervention with a suitable alternative, using outcomes that matter to the golfer and an evaluation long enough for the claim. Better explanation, improved launch consistency, lower estimated load and reduced injury incidence are separate milestones. Existing sensing and modeling methods make parts of this programme possible; their availability does not mean that comprehensive personalized diagnosis has already been validated.

The most useful model may change with the task. A launch predictor, a shaft bench model, a full-body inverse-dynamics model and a clinical study answer different questions. Their integration should preserve their individual validation boundaries rather than hide them inside one impressive-looking simulation.

32.14 Worked Example: What a Launch Report Can Establish

32.14.1 The Scenario

Suppose a report gives ball speed 145 mph, launch angle 14°, spin magnitude 2500 RPM, club path 3° right and face azimuth 2° right of the target. Treat these as reported quantities whose measurement definitions and uncertainty must be checked. They do not specify the complete delivery, spin vector, atmosphere or body motion.

The golfer wants more carry while maintaining directional consistency. That is a useful objective, but the report alone does not establish present carry, an achievable distance increase, a unique spin optimum, or whether a particular muscle should act earlier.

32.14.2 The Physics

If the declared two-mass collision model applies with the illustrative parameters above, the 1.48823 speed ratio implies an incoming mass speed of 97.4313 mph. This is a conditional model calculation, not an independent club-speed measurement. A different effective mass, restitution or contact configuration changes the inferred value.

In an explicitly horizontal, positive-right angle convention, the face is 1° left of the path. A local illustrative direction model

$$\ell = wf + (1 - w)p$$

with $w = 0.76$, $f = 2°$ and $p = 3°$ gives launch azimuth $\ell = 2.24°$ right. The selected weight is a local modeling assumption, not a universal law. Dynamic loft, attack angle, strike location and tangential contact remain unspecified. Face-to-path in this horizontal calculation does not by itself explain the reported backspin magnitude.

To predict carry, choose and validate a flight model with appropriate launch velocity, spin vector, aerodynamic behavior and atmosphere. To predict a zero-torque branch, additionally obtain an adequate pre-impact body/club state and a declared intervention. Neither calculation can be reconstructed uniquely from the five reported values.

32.14.3 The Feedback

A useful next session first establishes measurement reliability and repeatability, then tests a limited hypothesis. For example, compare strike-location and delivery distributions under matched conditions, determine which variation contributes most to the outcome uncertainty, and evaluate a cue or equipment change against that baseline. If the hypothesis concerns shaft response, measure that response or use an independently calibrated shaft model.

Report what improved, its variability, and what mechanism remains unresolved. A measured club-speed change can be a result; a simulation can suggest a cause; a controlled comparison can test that suggestion. Keeping these steps distinct makes the feedback actionable without pretending that a launch report uniquely diagnoses the entire swing.

32.15 Key Takeaways

1. The affine description depends on state, input and baseline choices. Drift is not synonymous with uncontrollability or absence of prior muscle work.

2. Control authority concerns feasible changes in a specified output over a specified time. High speed alone does not imply a smaller input gain.

3. Mechanical impedance, internal tissue load, input effort and metabolic cost are different quantities, with different models and measurements.

4. Equipment affects material response, system dynamics, contact and behavior. Restitution, smash factor, compression and energy transfer must be distinguished.

5. More data do not resolve a structural ambiguity without informative observations or interventions. Prediction accuracy is not unique identification.

6. A counterfactual can connect disciplines by making assumptions testable. It does not replace neural evidence, coaching evaluation or prospective clinical research.

32.16 Chapter Exercises

1. For the two antagonist force pairs in this chapter, compute net moment and compressive muscle contribution. Explain what each result does and does not say about injury.

2. Compare a torque-controlled rotor with one described by $u = -kq + \nu$. Identify the two zero-input interventions and propose a perturbation experiment that could distinguish their responses.

3. For the 145 mph worked example, calculate the conditional incoming speed and launch azimuth. List the additional quantities needed for carry prediction and for a whole-body counterfactual.

4. A golfer improves after a cue. Give three explanations that do not require learning an explicit drift field, and design observations that help distinguish them.

5. Reproduce the 45° ply coefficients and the collision energy fractions. Explain why neither a shaft stiffness label nor ball compression alone determines distance.

6. Compute the two double-integrator endpoint costs. Explain why the closer endpoint can require more effort and why neither result gives a universally effective coaching cue.

7. A new command arrives after the impact event. Under what assumptions is its effect on that impact zero? Explain why this does not imply that feedback or muscle force was absent earlier.

8. Design a study that separates a change in estimated joint loading from a claim of injury prevention. Include exposure, comparison, uncertainty and the possibility of shifted internal loads.

32.17 Worked Answers

1. The moments are $0.03(200-100) = 3$ and $0.03(500-400) = 3\,\mathrm{N\,m}$. Compressive muscle contributions are 300 and 900 N. Net moment cannot identify the individual forces; neither force sum supplies a tissue-specific injury threshold, exposure history or clinical outcome.

2. Zero physical torque gives $\ddot{q} = 0$ in the ideal rotor. Zero residual command retains $\ddot{q} = -kq/I$, which is $-4\,\mathrm{rad/s^2}$ at the declared initial displacement. Release both from matched state and measure acceleration or apply a known displacement perturbation. Distinguish a real passive spring from active feedback using power, actuator and timing measurements; matching motion alone need not distinguish them.

3. Dividing 145 mph by the conditional speed ratio 1.48822836 gives 97.4313 mph. The illustrative horizontal launch is $0.76(2) + 0.24(3) = 2.24°$ right. Carry additionally needs a sufficient three-dimensional launch state, aerodynamic model and atmosphere. A body counterfactual needs configuration, velocity, relevant activation/elastic states, physical parameters, external/contact conditions, an intervention and a stopping rule. None is uniquely supplied by the report.

4. Possible explanations include explicit re-aiming, a different movement policy, changed coactivation, altered strike location or ordinary trial variation. Measure aiming reports, delivery and strike, compare repeated conditions, and test retention, transfer or removal of the perturbation. A mechanical perturbation can probe impedance. These observations narrow hypotheses; neural localization requires additional evidence.

5. Substitution into the two rotated-ply expressions gives axial and shear numerators of 172.0134 and 139.9329 GPa. Dividing each by four gives 43.0034 and 34.9832 GPa, respectively. With the stated masses and restitution, the ball and lost-energy fractions are 0.5086 and 0.0581, leaving about 0.4333 in the other mass. Laminate/shaft boundary conditions, golfer adaptation and contact/flight conditions prevent a universal distance inference from either material label.

6. Multiplying by W^{-1} gives costs 0.28 and 0.12. The corresponding optimal dimensionless inputs are $0.8-1.8t$ and $0.6-0.6t$ for $0 \le t \le 1$; integrating each input and its remaining-time-weighted value recovers the requested velocity and position corrections. The first correction reverses part of its acceleration, making its smaller Euclidean endpoint error more expensive. These unconstrained, scaled calculations are not metabolic costs or verified human strategies.

7. If the command is causally generated after the event, initial perturbations are zero, and the model has no anticipatory dependence, it cannot alter the already completed event.

Earlier activation, stored energy, prior feedback and ongoing contact forces may still have affected that event. Post-impact feedback can affect later motion and future trials. A time constant is not automatically an additional pure delay.

8. First validate the load estimate against suitable independent measurements and vary uncertain muscle-force allocations. Compare interventions under matched tasks while recording strike, performance, exposure and relevant symptoms. A prospective prevention claim requires appropriately defined health outcomes, sufficient follow-up and a suitable comparison design, with prior injury and other confounders addressed. Randomization where feasible strengthens causal interpretation. Lower estimated net moment alone cannot substitute for that study.

Chapter 33

The Complete Golf Swing: Putting It All Together

In Plain Language 33.1. One Swing, Interacting Mechanisms

A golfer arrives at impact with more than a clubhead speed. Position, orientation, translational and angular velocity, shaft deformation, tissue state and contact conditions have developed together. Earlier muscle work changes energy and momentum; changing geometry changes how forces act; muscle activation affects both movement and response to disturbance. The collision then converts part of the delivered mechanical state into ball motion.

The purpose of a complete-swing model is to explain these connections and make comparisons testable. It should help answer why a particular change improves delivery, which assumptions that explanation needs, and what measurements could distinguish competing explanations. Calling a phase "drift-dominated" does not answer those questions by itself. This chapter connects phase descriptions to equations, worked examples and an investigation protocol.

33.1 The Full Model: Body, Arms, Club, Shaft, and Ground

Choose a system boundary before assigning forces. For the golfer and club together, hand–club interactions are internal. Gravity, ground contact, aerodynamic loads and, during impact, ball or turf contact are external. Ground reaction is therefore not the only external force besides gravity. For the club alone, the hand wrench crosses the boundary and can supply substantial power. Both descriptions are useful, but their force and energy accounts differ.

Let q contain the modeled body configuration and shaft deformation coordinates, and let v contain their corresponding generalized velocities. A floating-base orientation may require $\dot{q} = N(q)v$ rather than $\dot{q} = v$. Add internal states z when activation, tendon dynamics, viscoelastic history or other memory affects subsequent evolution. A suitable continuous state within a contact mode is

$$x = (q, v, z), \qquad \dot{q} = N(q)v. \tag{33.1}$$

Modal positions without modal velocities do not determine shaft evolution. Likewise, identical joint angles and velocities with different muscle activation need not produce identical accelerations. Ideal contact reactions are normally algebraic unknowns, not additional independent state coordinates; compliant contact may introduce deformation and memory states.

For a declared torque-input model, write the mechanical equations as

$$M(q)\dot{v} = r(q, v, z) + B(q)\tau + J(q)^T\lambda, \qquad Jv = 0. \tag{33.2}$$

Here M is positive definite on the chosen independent velocity coordinates. The vector r includes known passive, gravity, velocity-dependent and other declared loads with their signs; $B\tau$ is the modeled applied generalized load. The rows of J describe independent, stationary, active ideal constraints. A two-hand grip can close a kinematic loop; foot contact can constrain a floating base. Neither makes its reaction an independently commanded actuator.

For a fixed feasible contact mode with full-row-rank J, differentiate the velocity constraint and eliminate λ. Define

$$\begin{aligned} A_c &= JM^{-1}J^T, \\ D_c &= M^{-1} - M^{-1}J^T A_c^{-1} JM^{-1}, \\ a_0 &= D_c r - M^{-1}J^T A_c^{-1}\dot{J}v, \\ B_c &= D_c B, \qquad \dot{v} = a_0 + B_c\tau. \end{aligned} \tag{33.3}$$

The reaction follows from

$$\lambda = -A_c^{-1}\{JM^{-1}(r + B\tau) + \dot{J}v\}. \tag{33.4}$$

It depends on input. Keeping the measured reaction fixed while changing torque generally violates the same constraint. The acceleration offset contains the moving-tangent term $\dot{J}v$ even for a stationary curved constraint. These equations are a derivation of the torque-affine mechanics, not an assertion that every physiological model is globally control-affine in neural excitation.

For muscles, an excitation input can first alter z, with force subsequently determined by activation, length, shortening velocity, tendon state and history. The attainable torque set can therefore depend on both state and recent inputs. A zero-torque actuator intervention differs from zero excitation, which need not immediately remove muscle force. Use $\dot{x} = f(x) + G(x)u$ only when the selected input and constitutive model actually support that representation.

Contact feasibility remains an additional condition: normal forces cannot pull a nonadhesive foot toward the ground, tangential forces must satisfy the chosen friction law, and a contact must be removed when its gap opens. Slip, lift-off, heel/toe changes and collision can change the governing mode. A smooth equation valid in one mode is not a complete hybrid swing model.

33.2 What a Drift Ratio Measures

A norm requires compatible units and an explicit metric. Comparing meters, radians, velocities and activation entries in an unscaled state vector can make a ratio depend mainly on unit choices. For one acceleration-based diagnostic, choose a positive-definite weighting W and an admissible torque set $\mathcal{U}(x)$. Define

$$c(x) = \sup_{\tau \in \mathcal{U}(x)} \|B_c(x)\tau\|_W, \qquad D_{\text{avail}}(x) = \frac{\|a_0(x)\|_W}{c(x)}. \tag{33.5}$$

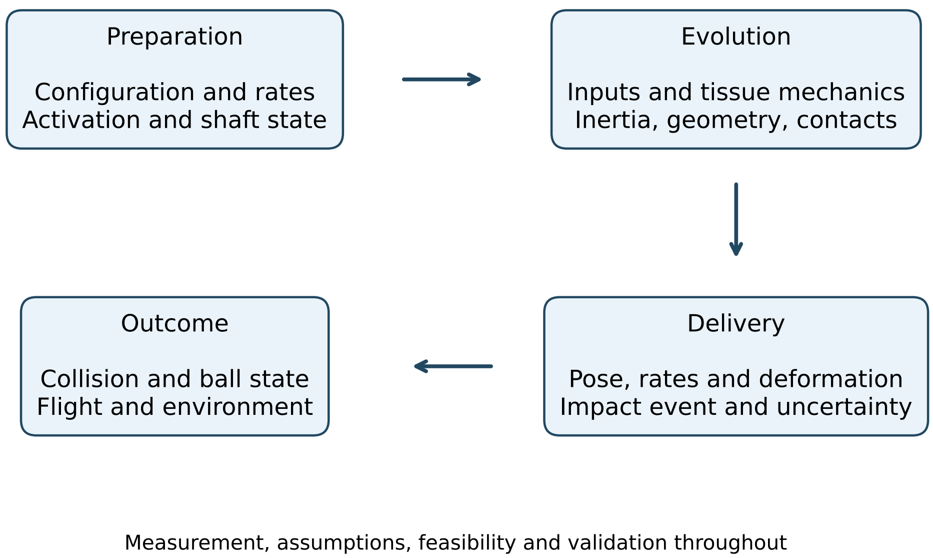


Figure 33.1: Preparation, Coupled Evolution, Delivery, and Outcome. Measurements and Model Assumptions Are Needed at Every Stage; the Arrows Describe Dependencies, Not Validated Causal Attribution.

This definition uses available input, not the torque actually applied. Using the realized torque in the denominator defines a different estimand. If $c = 0$, the ratio is infinite for nonzero numerator and undefined if both vanish. Coupled muscle limits, unilateral contacts and asymmetric torque bounds cannot generally be summarized by substituting one vector called $u_{\max}$.

The ratio loses direction. In the dimensionless system $\dot{x}_1 = 1000$, $\dot{x}_2 = u$, $|u| \leq 1$, the drift-to-available-input norm ratio is 1000. Yet the entire evolution of the output $y = x_2$ is controlled by u; drift contributes zero in that direction. A task-direction support function, or a finite-horizon reachable-output calculation, answers a different and often more relevant question.

For fixed q, fixed ideal contacts and the torque-input assumptions above, the instantaneous derivative is

$$\frac{\partial \dot{v}}{\partial \tau} = B_c(q). \tag{33.6}$$

Increasing speed can increase the velocity-dependent bias without reducing this derivative. Physiological torque limits may change with speed, and configuration can change M, J and B; those are distinct mechanisms. "The inertia becomes large because the body is moving fast" is not a general property of rigid-body mass matrices.

For an independently checkable constrained example, take

$$M = \begin{bmatrix} 2 & 0.4 \\ 0.4 & 1 \end{bmatrix}, \qquad J = \begin{bmatrix} 1 & 1 \end{bmatrix}, \qquad B = I. \tag{33.7}$$

The constraint allows only opposite coordinate accelerations. Adding the generalized input $(10, 0)^T$ changes acceleration by $(10/2.2, -10/2.2)^T$, regardless of the existing velocity-dependent bias. Equal inputs $(1, 1)^T$ produce no acceleration change and are balanced by the reaction. Thus even a large available input norm says little about a particular useful direction unless constraints are included.

There is no calculated universal DCR curve in this chapter. Earlier numerical phase bands are removed because neither a declared computation nor the cited empirical study established them. A phase trend must be computed for a stated model, metric, input set, uncertainty and contact sequence. It is not a definition of expertise, conscious effort or metabolic efficiency.

33.3 Phase 1: Address and Initial State Preparation

At an ideal steady address, generalized velocity and acceleration vanish, but supporting muscle loads need not vanish. Equilibrium requires the applied, passive and contact loads to balance, together with feasible contact forces. The reduced zero-input acceleration need not point vertically downward: geometry and support constraints shape the permitted motion. A nearly stationary person may also have small postural motion or changing activation, so a video frame alone does not establish a full-state equilibrium.

Stance, ball position, grip and alignment change geometry and the reachable set of club poses. They can also change feasible support moments and muscle moment arms. Gravitational potential changes by $\sum_i m_i g \, \Delta h_i$ for the chosen segments; a wider stance or increased forward lean does not universally raise their centers of mass. The energy claim must follow the actual geometry.

Preparation is consequently more than selecting a photograph. The initial velocities, shaft state, activation and task intention influence what follows. Repeatable address geometry can reduce one source of variation, but movement variability, fatigue, equipment and subsequent control still matter. No claim of maximal control authority at address follows merely from knowing G or standing still.

33.4 Phase 2: Backswing and Distributed Loading

The backswing changes body and club geometry while muscles perform positive and negative work at different locations. Raising a segment can store gravitational energy. Stretching a loaded elastic element can store recoverable strain energy, while hysteresis and viscosity dissipate some work. A visible torso turn does not identify the strain energy of an individual tendon, fascia sheet or muscle fiber. Activation-dependent tissue behavior also makes "elastic" different from "metabolically passive."

Angular momentum and kinetic energy develop as mass moves. They are not stored in a lag angle independently of velocity. Some segments begin decelerating or reversing while others are still moving backward. If the top is defined by reversal of a particular club-coordinate angle, that coordinate's rate crosses zero at the event; other generalized rates need not do so. The top therefore does not imply simultaneous maximal speed, maximal tissue stretch or zero kinetic energy of the entire golfer.

Keep rotation references explicit. The McTeigue study's publisher abstract reports upper-body and hip rotations relative to its measurement reference; those values are not a torso rotation of 90 degrees relative to the hips [McTeigue et al., 1994]. Subtracting two projected orientation angles also need not equal an anatomical spinal rotation in three dimensions. A useful analysis names the segments, axes, decomposition and event.

Backswing duration and amplitude are observations that vary with player, club and shot. A model can compare alternatives while constraining delivery, load and effort. The affine form alone does not establish an optimal duration, require a maximum stretch at the top, or show that the motor cortex has finished planning and the cerebellum has learned a specific drift field.

33.5 Phase 3: Transition and Coupled Sequencing

Transition is an interval of overlapping segment reversals, not necessarily one simultaneous reversal of all angular momentum. Define it with observable events, such as pelvic rotation reversal, club reversal and a specified change in hand motion. Compare the event times with measurement uncertainty; a universal 50–100ms transition is not needed to explain the mechanics.

The mass matrix couples accelerations, and velocity-dependent terms reflect the changing geometry of moving segments. Active torques, tissue loads and contact reactions act within that same system. A distal acceleration can result partly from proximal actuation, but a proximal-to-distal ordering of velocity peaks does not prove a uniquely passive cascade. Joint torques at several locations can generate similar motion, and the ordering can change with task, model, initial state and actuator limits.

For an explicit coupling example, a positive-definite two-coordinate inertia

$$M = \begin{bmatrix} a & b \\ b & c \end{bmatrix}, \qquad ac - b^2 > 0, \tag{33.8}$$

gives a second-coordinate acceleration change $-b\,\delta\tau_1/(ac-b^2)$ when only the first generalized load changes and the other loads are held fixed. Coupling can assist or oppose a chosen distal direction depending on the coordinate convention and configuration. This calculation predicts an instantaneous response, not an entire sequence of peak velocities or a causal energy-transfer story.

Ground forces and moments enter the momentum balance. For the combined golfer–club system, about its center of mass C in an inertial frame,

$$\begin{aligned} m\ddot{r}_C &= \sum F_{\text{ext}}, \\ \dot{H}_C &= \sum \big[(r_k - r_C) \times F_k + \mu_k\big]. \end{aligned} \tag{33.9}$$

Internal forces cancel from the total linear-momentum balance, but redistribute momentum among segments. A stationary foot can supply an external moment without doing contact work. For example, a vertical 800N force applied 0.2m horizontally from C has a 160N m moment magnitude even when its contact-point velocity is zero. These are illustrative values, not a prescription for a golfer's ground loading.

Ground reaction peaks need to be measured in a declared direction and reference interval. Total vertical force, each foot's vertical force, horizontal shear, center of pressure and free moment are different signals. They need not peak together or at transition. Muscles help create motion under support constraints; ground reaction does not directly identify which gluteal or other muscle was activated.

33.6 Phase 4: Early Downswing, Release Coordinates, and Shaft State

Define "lag" before measuring it. A projected arm–shaft angle describes one aspect of articulated geometry. Wrist flexion/extension, radial/ulnar deviation and forearm rotation are distinct anatomical motions. Flexible shaft lead/lag and toe-up/toe-down deflections describe deformation relative to a specified shaft reference. The shaft's angle to the horizontal is an absolute orientation, not a universal measure of wrist lag or elastic storage.

Maintaining a projected angle does not by itself prove active resistance, nor does its release prove a particular extensor contraction. Net joint moments require a dynamic model and external loads; individual muscle forces require additional information. Motion can result from active load, transmitted segment forces, inertia and passive tissue response together. Relative angles and angular velocities are informative observables, but they do not identify the control strategy on their own.

In an inertial frame, a clubhead following a curved path has an inward normal acceleration v_h^2/R_{path}, where R_{path} is the local curvature radius and v_h the speed. Tangential acceleration changes speed. Outward centrifugal terms arise in an appropriate rotating-coordinate description; Coriolis, Euler and frame-translation terms must then be included consistently. A large normal acceleration is not an additional source of outward power.

Shaft bending is driven by the coupled hand motion, grip loads, head inertia and offset geometry. Its deformation can change sign and can differ between bending directions. Earlier loading is retained through both deformation and modal velocity. High axial tension can modify transverse dynamics through geometric stiffness, but its effect depends on loading and the structural model; it does not imply a universally monotonic shaft stiffness or bend through the downswing.

The shaft-deflection simulations of MacKenzie and Sprigings show that both loading direction and previous bending matter, with radial loading capable of shifting the time of peak relative head velocity beyond a neutral shaft shape [MacKenzie and Sprigings, 2010]. This is a model result with specified boundary conditions. It does not justify assigning a fixed snapback time or the same deformation pattern to every player.

33.7 Phase 5: Mid-Downswing and the Remaining Correction Window

The time available before impact matters, but it is not interchangeable with instantaneous input gain. A fixed-inertia rotor demonstrates the distinction. With inertia $I = 0.1\text{kg m}^2$, an added constant torque of 1N m for 40ms changes angular velocity by 0.4rad/s and angle by 0.008rad, approximately 0.458 degrees:

$$\Delta\omega = \frac{\delta\tau}{I}\Delta t, \qquad \Delta\theta = \frac{\delta\tau}{2I}(\Delta t)^2. \tag{33.10}$$

These increments are independent of the initial rotor speed. They are not measured golf responses: the model omits changing geometry, muscle dynamics, contacts and flexibility. They show why limited time reduces some corrections without proving that speed has removed input authority. Whether a fraction of a degree matters depends on the delivery variable and task tolerance.

New sensory information, a planned command already being executed, and the intrinsic mechanical response of activated tissue act on different timelines. Delays restrict responses to a newly detected disturbance; they do not erase ongoing force or preplanned changes. An arm-perturbation experiment found multi-joint responses involving primary motor cortex on a roughly 50ms timescale [Pruszynski et al., 2011]. That result does not supply a universal golf delay or demonstrate that voluntary control is absent in a named swing phase.

For a first-order illustrative activation model, $\dot{a} = (e - a)/T_a$, zero excitation gives $a(t) = a(t_0)\exp[-(t-t_0)/T_a]$, not immediate zero activation. Force also depends on muscle and tendon state. A late zero-excitation experiment therefore differs fundamentally from replacing all joint torques by zero. Anticipatory impedance and feedforward timing may remain important even when little time is available for a new feedback response.

33.8 Phase 6: Late Downswing and Flexible Delivery

Delivery is the outcome of continued coupled dynamics. A useful flexible-coordinate representation is $p_h = p_g + R_g r_h(\eta)$, giving

$$\dot{p}_h = \dot{p}_g + \omega_g \times (R_g r_h) + R_g \frac{\partial r_h}{\partial \eta}\dot{\eta}. \tag{33.11}$$

All terms refer to the same physical point and inertial frame. The expression separates contributions to velocity at one state. It does not show how grip motion would change if shaft flexibility were removed. That comparison requires another dynamically consistent trajectory, because the golfer and club react to one another.

The shaft-stiffness study makes this distinction concrete: a relative kick contribution of several meters per second did not equal the speed difference between optimized rigid and flexible model swings, and the tested nonrigid stiffness changes produced little speed difference in that model [MacKenzie and Sprigings, 2009b]. Head orientation changed as well. These findings motivate separate reporting of delivered orientation, relative deformation velocity and the outcome of a declared equipment intervention; they do not imply universal irrelevance of shaft properties or guarantee a fitting benefit.

33.8.1 Force, Velocity, Power, and Stored Energy Have Different Peaks

For one unforced, undamped mode of constant modal mass m_s and stiffness k_s,

$$\eta = A\cos(\omega_s t), \quad \dot{\eta} = -A\omega_s \sin(\omega_s t), \quad \omega_s = \sqrt{k_s/m_s}. \tag{33.12}$$

Elastic force is $-k_s\eta$. At maximum $|\dot{\eta}|$, displacement and elastic force are zero. Maximum force occurs at a turning point, where velocity and instantaneous elastic power are zero. The elastic power delivered to modal motion is

$$P_e = -k_s\eta\dot{\eta} = -\frac{d}{dt}\left(\frac{1}{2}k_s\eta^2\right). \tag{33.13}$$

Its positive maximum lies between those events. A driven, damped, coupled shaft shifts them further. Maximum modal velocity at impact therefore does not imply maximum restoring force, elastic power or head acceleration at impact.

For a linear spring, stored energy is $k\eta^2/2$ at prescribed deformation, but $F^2/(2k)$ under a prescribed static force. Reducing stiffness lowers the first and raises the second. Neither boundary condition describes the entire swing. If stiffness varies with an imposed preload, the derivative of stored energy also includes $\dot{k}\eta^2/2$; the source of that parameter work must be included in the full system account.

The coupled equations and the impact objective determine useful timing. Energy transferred back toward the hands is not automatically wasted; it can alter subsequent movement and loads. A flexible mode can affect face orientation even when its contribution to speed along a chosen direction is small. Steel and graphite are material categories, not complete mechanical specifications: geometry, mass distribution, bending and torsional rigidity, damping and grip coupling also matter.

33.9 Phase 7: Impact and the Delivered State

Impact introduces contact forces on a much shorter timescale than the preceding swing. Specify the incoming ball and club states, contact location and orientation, material model, friction, and any external impulses being neglected. A rigid impulse model approximates a short collision; a compliant model resolves deformation and time. Their parameters and output meanings are different. A clubhead spring-effect test is not a universal ball COR limit, and its pendulum characteristic time is not the ball–club contact duration.

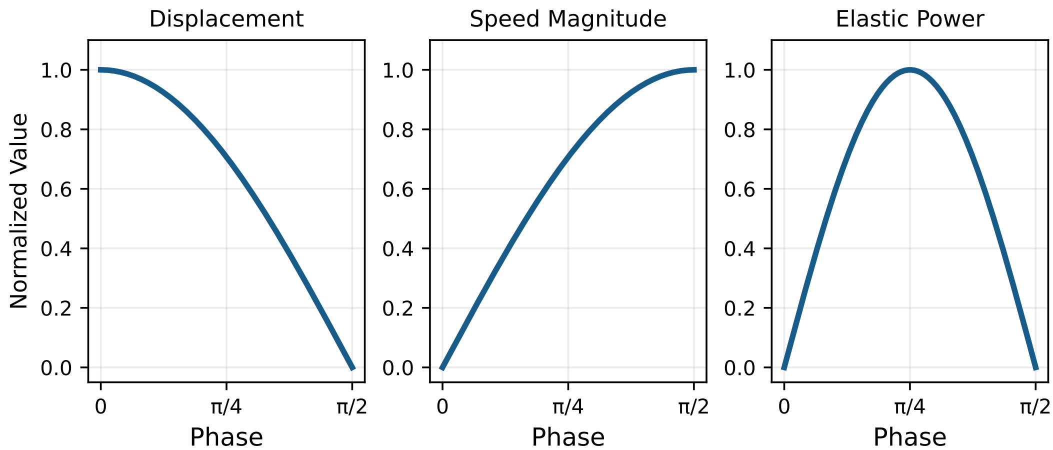


Figure 33.2: Displacement, Velocity, and Elastic Power Peak at Different Times. This Free Harmonic Mode Is an Illustrative Limit, Not a Measured Shaft Trajectory.

For a transparent reference calculation, consider only a central, one-dimensional collision: an initially stationary ball of mass m, a free translating club mass M at speed V, no spin and negligible external impulse during collision. Conservation of momentum and restitution give

$$MV = MV_c^+ + mv_b^+, \qquad v_b^+ - V_c^+ = eV, \tag{33.14}$$

where e is the relative separation-speed to approach-speed ratio. Solving,

$$v_b^+ = \frac{(1+e)M}{M+m}V, \qquad V_c^+ = \frac{M-em}{M+m}V. \tag{33.15}$$

Relative to the initial club kinetic energy, the ball fraction, retained club fraction and loss fraction are

$$\begin{aligned} \eta_b &= \frac{mM(1+e)^2}{(M+m)^2}, \\ \eta_c &= \left(\frac{M-em}{M+m}\right)^2, \\ \eta_D &= \frac{m}{M+m}(1-e^2), \qquad \eta_b + \eta_c + \eta_D = 1. \end{aligned} \tag{33.16}$$

The relative-motion kinetic energy loses the fraction $1-e^2$; the total initial laboratory-frame energy loses η_D in this particular setup. Neither is $1-e$.

For illustrative $M = 0.200\text{kg}$, $m = 0.04593\text{kg}$, $e = 0.83$ and $V = 45\text{m/s}$, the ball speed is approximately 66.970m/s and the retained club speed 29.620m/s. The club retains 65.823% of its speed, or 43.326% of its initial kinetic energy. Ball translation receives 50.863%, and 5.810% is lost from the two translational energy stores. These numbers follow from the declared two-mass model, not a population estimate or a complete driver collision.

Off-center contact couples translation and rotation. Tangential contact, frictional sticking or slipping and deformation affect ball spin; the incoming face normal, local contact velocity and contact location matter. Grip and shaft impedance, clubface flexibility, ball construction, turf contact and the collision timescale can change the effective model. A single COR and clubhead speed cannot identify every launch condition.

After separation, ball translation, spin and position initialize a flight model with atmospheric, wind and aerodynamic assumptions. The swing no longer exerts a contact force on the departed ball. A deterministic model gives a definite prediction only conditional on those complete inputs and parameters; uncertain launch and aerodynamic quantities produce uncertain flight. A ball curving right does not uniquely reveal which earlier joint or control phase was responsible.

33.10 Phase 8: Follow-Through and Energy Absorption

The ball removes only part of the preceding mechanical energy. Body and club motion continue, with active muscle work, passive losses, changing gravitational and elastic storage, and possible new contacts. Impact can abruptly alter clubhead motion while a finite-compliance shaft and grip transmit a time-dependent response to the hands. Muscles merely relaxing is not a complete explanation for the collision impulse or the subsequent deceleration.

Eccentric muscle action can absorb mechanical energy, but the sign of one net joint power does not identify all individual muscle contractions. Opposing muscles and biarticular pathways can produce different internal work distributions for the same observed net moment. Ground reactions can change total momentum without dissipating energy when the stationary contact is ideal. Slip, deformation and collision require their own power or impulse accounts.

For an ideal stationary constraint, $P_c = \lambda^T J v = 0$ in the combined system. Internal joint and grip forces can perform opposite powers on adjacent subsystems, redistributing energy while their ideal total cancels. A prescribed moving boundary can do nonzero work. This distinction explains how support can redirect motion without functioning as a brake that converts all residual energy to heat.

There is no mechanics-based reason to label every follow-through high-DCR, uncontrolled, or devoid of cortical involvement. Its duration and load distribution vary with the task and player. Load estimates can motivate investigation of tissue demand, but an injury-risk claim requires appropriate evidence beyond a computed reaction force. The physical objective is feasible deceleration and task completion, with explicit assumptions about support and tissue behavior.

33.11 The Complete Energy Budget: Work, Storage, and Transfer

The swing connects a chemical energy source to mechanical work, changing body and club energy, and a collision with the ball. Those are different accounting boundaries. A ratio measured at one boundary cannot be interpreted as efficiency at another without the intervening balances. In particular, club work divided by reconstructed body joint work does not measure the fraction of metabolic energy converted to useful motion.

Define the Mechanical System and Interval

Before contact with the ball, consider the golfer and club together. Use disjoint body segments, a declared inertial frame, and the same start and end times for all quantities. Include gravitational potential energy and whatever recoverable elastic energy the chosen model resolves. Its total mechanical energy is

$$E = T_{\text{body}} + T_{\text{club}} + U_g + U_e.$$

For ideal internal constraints, a consistent balance is

$$E_f - E_i = W_{\text{act}} + W_{\text{ext}} - D.$$

Here W_{act} is signed mechanical actuator work delivered to the modeled coordinates; W_{ext} is signed external work not already represented by the included potentials; and $D \geq 0$ is the model's separately represented passive dissipation. Do not count the same loss in both negative actuator work and D. Gravity is already in U_g, so it must not also be included in W_{ext}. A moving prescribed boundary or an inverse-dynamics residual actuator contributes work and must be recorded rather than silently omitted.

This equation is not a metabolic balance. A net joint actuator can summarize unresolved muscles and tissues, but joint motion and net moment alone do not determine individual muscle work, activation cost or heat. Such quantities require additional measurements and a physiological model.

Follow Energy Without Assuming a Universal Phase Pattern

During the backswing, positive actuator work can increase kinetic, gravitational and elastic energy while losses and negative work occur elsewhere. Raising a center of mass changes its gravitational energy by $mg\,\Delta h$; the amount must be calculated for the actual system. A turn angle alone does not identify tissue elastic energy, which depends on deformation, loading history and the constitutive model.

The top of the backswing is an event defined from a chosen observable, not simultaneous rest of every segment. Body, club and elastic energies need not attain their maxima together. Through transition and downswing, active work continues alongside conversion of previously stored energy. Whether gravity or a particular elastic element supplies substantial power is a result to calculate, not a consequence of labeling the phase as drift-dominated.

At a joint, forces can redistribute energy between segments. For the club alone, power supplied by the hands includes both the dot product of force with its application-point velocity and the dot product of couple with angular velocity. That grip power is internal to the combined golfer–club system. Ideal no-slip contact with stationary ground can transmit force and angular momentum while doing zero contact work; ground reaction force is therefore not itself a dissipative mechanism. Deforming or slipping contact requires its actual power balance.

What the Reported Work Ratio Measures

Nesbit and Serrano analyzed four amateur golfers using reconstructed body and club models. Their reported swing efficiency is club work divided by total body joint work, with values of 20.2–26.8% in Table 3 [Nesbit and Serrano, 2005]. Neither denominator nor numerator is metabolic energy or ball energy. The remaining fraction cannot be assigned to heat, and these four cases do not establish an elite-versus-amateur metabolic advantage.

When reporting a new work ratio, state whether its denominator sums signed work, positive work only, or the absolute values of work. For actuator powers P_j, define positive work and the magnitude of absorbed work by

$$W_+ = \sum_j \int \max(P_j, 0)\,dt, \qquad W_- = \sum_j \int \max(-P_j, 0)\,dt.$$

Then $W_{\text{act}} = W_+ - W_-$. This separation does not resolve individual muscle contributions when P_j represents a net joint actuator.

For an illustrative mechanical budget, let $E_i = 100$ J, $W_+ = 500$ J, $W_- = 80$ J, $W_{\text{ext}} = 0$ and $D = 20$ J. The final energy is 500 J. If 250 J is club kinetic energy and 250 J remains in the other included energy stores, club energy divided by positive actuator work is 0.50; divided by signed actuator work it is approximately 0.595. These ratios use club energy, not the cited paper's club-work numerator. Both are compatible with the same balance, and neither determines metabolic efficiency. Initial storage also prevents treating an arbitrary output-to-new-work ratio as necessarily bounded by one.

Separate Impact and Follow-Through

To include impact, enlarge the system to contain the ball and resolve an interval spanning contact. Include initial and final ball, club and body energies, work across the external boundary, and dissipative terms. Energy left in the moving club is retained energy, not collision heat. Ball translation, spin, deformation and subsequent aerodynamic flight are distinct; ball kinetic energy alone does not determine carry distance.

During follow-through, actuator absorption, passive losses and external work alter the remaining mechanical energy. Recoverable deformation and changed gravitational potential may remain even when the golfer stops. Do not assert that every joule unaccounted for at impact has become heat. Check energy closure and identify the model's residual before interpreting a loss mechanism.

Connect the Budget to Drift and Control

The drift field describes evolution under a specified zero-input convention. It is not an energy source, and a large drift contribution does not establish low metabolic cost. Stored energy may have required earlier active work; zero net joint torque can coexist with muscle activity; and a different input convention changes what is called drift. A useful comparison holds the task and system boundary explicit, reports initial storage and signed work, and measures the resulting club state. This connects preparation, timing, geometry and active effort without assuming that reducing muscle work must increase ball energy.

33.12 From Phase Inputs to Impact Sensitivity

An endpoint sensitivity connects the whole trajectory, not just the norm of a vector field at one instant. For a nominal smooth-mode trajectory of $\dot{x} = F(x, u)$, let $A(t) = F_x$ and

$B_x(t) = F_u$ evaluated along that trajectory. The first variation satisfies

$$\delta \dot{x} = A(t)\delta x + B_x(t)\delta u, \qquad \delta x(T) = \Phi(T, t_0)\delta x_0 + \int_{t_0}^{T} \Phi(T, s)B_x(s)\delta u(s)\, ds. \tag{33.17}$$

The transition matrix obeys $\partial_t \Phi(t, s) = A(t)\Phi(t, s)$ and $\Phi(s, s) = I$. For the torque-affine model, F_x includes derivatives of $G(x)u$ as well as f_x. For an output $y = h(x(T))$, left-multiply by h_x to obtain the delivered-output sensitivity. The kernel $h_x\Phi(T, s)B_x(s)$ includes later amplification, rotation, damping and task projection.

Equal input changes at different times need not have equal effects. For a torque-driven free rotor, a fixed torque impulse changes final velocity equally wherever applied, while earlier application has more time to change angle. In the stable scalar mode $\dot{x} = -10x + u$ over $0 \le t \le 1$, equal-duration unit pulses in the final versus initial 0.1s have endpoint-effect ratio e^9, approximately 8103. This counterexample does not model golf; it disproves a universal rule that early inputs must always have greater endpoint influence.

Impact is usually an event rather than a prescribed clock time. Let the contact condition be $g(x, t) = 0$ at nominal T, with a transverse crossing $g_t + g_x F^- \neq 0$. The first-order event-time shift and pre-impact output change are

$$\begin{aligned} \delta T &= -\frac{g_x \delta x(T)}{g_t + g_x F^-}, \\ \delta y^- &= h_x \delta x(T) + (h_t + h_x F^-)\delta T. \end{aligned} \tag{33.18}$$

Parameter perturbations add their direct derivatives. A post-impact comparison must additionally differentiate the reset/contact solution. Near grazing contact the denominator can become small, and a perturbation may remove impact or change the contact mode entirely; one smooth derivative then ceases to describe the comparison.

For a simple crossing $p(t) = p_0 + vt = L$, perturbing initial position and speed gives $\delta T = -(\delta p_0 + T\delta v)/v$. Position at the event remains L even though position at the original clock time changes. Thus fixed-time and event-matched errors can lead to different conclusions about the same perturbation. Report which one is being used.

Observability is separate: do the chosen measurements distinguish nearby hidden states or parameters? One measured impact error cannot generally be uniquely decomposed into address, backswing and transition errors. Many input histories map to the same small set of outputs. A sensitivity matrix can predict effects of specified perturbations; inversion needs rank, conditioning, measurements over time and additional assumptions. A regularized inverse selects one explanation according to its penalty, not necessarily the unique physical cause.

33.13 What a Forward Zero-Torque Counterfactual Can Establish

Initialize the measured/model-consistent state at a declared event, set the selected applied torque channel to zero, and evolve the model forward with its retained gravity, tissue, shaft and contact dynamics. Declare what happens to activation, prescribed boundaries, feedback, grip constraints and contact transitions. Compare actual and zero-channel trajectories in a named output, metric and interval, including whether each reaches the same impact event.

The branch tells us what this specified model predicts under that intervention. It does not recover an observed passive trajectory, identify muscle silence or demonstrate that a player should relinquish control. Existing stored energy may have been created through earlier muscle work. Changing the control channel or retaining an active support law changes the meaning of "zero." The state at which the intervention begins matters just as much as the torque being removed.

A pointwise ratio alone does not guarantee close trajectories. If the zero-input field is Lipschitz with constant L in a common valid region, and the removed input contribution has norm bounded by b there, equal initial states obey the sufficient bound

$$\|x_u(t) - x_0(t)\| \leq \begin{cases} b\,[e^{L(t-t_0)} - 1]/L, & L > 0, \\ b(t - t_0), & L = 0. \end{cases} \tag{33.19}$$

This follows by bounding the difference equation and integrating the resulting scalar inequality. It needs an absolute input bound, a growth bound and a time interval. A large ratio caused by an even larger unrelated drift component provides none of those guarantees. Contact changes require a mode-aware analysis rather than silently extending this smooth bound through a collision.

To investigate sequencing, compare the complete, zero-channel and carefully specified modified-input trajectories. Record whether a velocity-peak order persists, reverses or disappears, and examine the forces, work and state changes that accompany it. Persistence supports a statement about the chosen model and initial state. It does not prove that the measured player used the same intervention or that other muscle commands were unnecessary.

33.14 A Reproducible Complete-Swing Investigation

Begin with a concrete question: for example, whether changing a transition command improves face-orientation consistency at a specified speed, or whether a shaft change alters delivery after allowing the golfer model to respond. Define the target and tolerances before selecting the comparison. Speed, accuracy, load, robustness and effort can conflict; a claim of "better" needs an objective and feasible constraints.

1. **Specify the Model and Measurements.** Record segment geometry and inertias, joint definitions, body/club frames, shaft modes, muscle states and contact laws. Synchronize kinematics, ground wrenches and club/ball observations. Distinguish measured data, reconstructed quantities and fitted parameters.

2. **Define Events and Outputs.** State the top/transition definitions, impact detection, follow-through endpoint and whether comparisons use clock time or matched events. Report speed, face orientation, path, strike location and uncertainty without substituting one for the full delivery state.

3. **Close the Mechanical Accounts.** Check constraint residuals, contact feasibility, momentum balance and the energy budget. Identify residual actuators and moving boundaries. Reproducing motion used to drive an inverse model is not an independent forward-prediction test.

4. **Run Declared Interventions.** Change one stated parameter or input schedule from a common initial state, or explicitly reoptimize under the same objective and con-

straints. A frozen-kinematics shaft comparison and a dynamically responding golfer comparison estimate different effects.

5. **Test Sensitivity and Robustness.** Perturb plausible states, parameters and measurement errors; compare finite differences with the variational prediction where it applies. Resolve contact changes explicitly. A nominally accurate trajectory can be sensitive to timing or uncertainty, and a smaller net torque does not establish greater robustness.

6. **Separate Mechanism From Evidence.** Report whether a conclusion is an identity, an illustrative calculation, a model-specific prediction, an observational association or an independently tested intervention. Preserve failures and alternative explanations. Extending a golf model to humanoid tasks requires rechecking actuation, contacts and task constraints.

The work of Nesbit provides an instructive measurement/model example: motion reconstruction, flexible-club dynamics and work analysis reveal individual differences, while imposed motion and fitted ground support limit causal interpretations [Nesbit, 2005]. A new complete-swing investigation should state which observations were used for fitting and which were held out for validation. Evidence for one model's joint loads is not automatically evidence for individual muscle forces or a neural policy.

The resulting practical insight is specific: preparation changes the state from which later forces act; geometry determines their directions and coupling; activation and tissue dynamics constrain the available response; shaft state changes delivery; timing changes the contact event; and the collision and flight models translate delivery into ball outcome. These connections justify investigating the whole swing. They do not establish a universal passive sequence or a theorem that trying less must produce more speed.

Key Principle 33.1. Summary of Analysis

The swing is a coupled, constrained, sometimes hybrid evolution of a complete mechanical and physiological state. Drift/control labels depend on the declared input model. A drift norm does not replace directional gain, finite-time sensitivity, contact feasibility or energy accounting. Segment sequencing and shaft motion must be explained with forces and evolving state, while their causal interpretation needs explicit interventions and appropriate evidence. A useful reference connects preparation to delivery through calculations that can be checked, and leaves player-specific performance claims open to measurement.

Exercises 33.1. Chapter Exercises

1. Derive the reduced acceleration and reaction formulas from the constrained equations. For the stated two-coordinate example, calculate the response to $(10, 0)^T$ and $(1, 1)^T$. Explain why changing velocity-dependent bias alone does not change the input derivative.

2. Define an available-input DCR using a metric and admissible set. Distinguish it from an actual-input ratio. Use $\dot{x}_1 = 1000$, $\dot{x}_2 = u$ to explain why neither ratio determines authority over every output. Give conditions and measurements

needed to establish a rising phase trend.

3. For a smooth state-dependent club-velocity output $h(x)$, integrate $\dot{h} = h_x[f(x) + G(x)u]$ along the actual trajectory. Explain why these same-trajectory integrals are not the velocity difference between two independently evolved swings and are not energy fractions.

4. Derive the first variation from address through impact, including the event-time correction. Identify the extra initial states omitted if address is represented only by joint angles. Check the ballistic-crossing example.

5. Compare early and late equal-duration input pulses for a free rotor and for $\dot{x} = -10x + u$. State the output for each comparison and compute the damped-mode ratio. Explain why a desire to add 10mph does not identify the best swing phase without a model, constraints and objective.

6. Use a two-coordinate inertia matrix to show how proximal input can accelerate a distal coordinate in either sign. Specify a forward experiment that tests whether a particular peak-velocity sequence persists after an input channel is removed. Explain what that result would not establish about muscles.

7. A player has rightward ball curvature or a measured face-angle error. Form a local input-to-delivery sensitivity matrix and explain why one endpoint observation cannot uniquely allocate the error among address, backswing and transition. Identify additional measurements or interventions that would distinguish candidate explanations.

8. For the harmonic mode, find the times of maximum displacement magnitude, velocity magnitude and positive elastic power. Explain why simultaneous maximum velocity and restoring force is impossible in this example and why a driven shaft can depart from its timing.

9. Compare stored spring energy under prescribed displacement and prescribed force. Then distinguish a steel/graphite material change from a controlled change in bending stiffness. Specify fixed-input and reoptimized golfer–shaft comparisons and explain why their outcomes can differ.

10. Derive the three central-collision energy fractions and reproduce the illustrative speeds. Explain why $1-e$ is not a loss fraction, and list the additional information required for off-center impact and spin.

11. Use total linear and angular momentum balance to explain how stationary ground contact can change motion while doing zero work. Distinguish an external support moment, muscle torque and a dissipative contact force. State how to test a claim about the phase of peak ground reaction.

12. Construct a follow-through energy account with actuator absorption, passive dissipation and changing storage. Explain why ideal grip/joint reactions can redistribute subsystem energy without absorbing total energy, and why net negative joint power does not identify every muscle contraction.

13. Compare high and low late muscle activation while requiring the same delivery and feasible support. Identify measurements needed to compare mechanical work, metabolic cost and robustness. Explain why neither control magnitude nor DCR alone proves one strategy faster, cheaper or more consistent.

14. Derive the stated equal-initial-state counterfactual error bound. Explain its dependence on absolute input magnitude, state-transition growth and duration, and identify what fails when one branch misses impact or changes contact mode.

15. Formulate an inverse design problem for a desired delivery state, with initial conditions, torque/activation constraints, contact feasibility and shaft parameters. Explain why multiple solutions can exist and what additional cost and validation would be needed to choose among them.

Glossary

Reading Across Definitions

The glossary connects three questions: **what moves, what supplies or transfers energy, and what the golfer can change**. These questions share a mechanical model, but they require different evidence.

For example, a changing forearm-club angle is kinematics. A bending shaft adds deformation and possible elastic storage. A joint moment combined with the matching angular velocity gives net joint power. None of those observations alone identifies individual muscle forces or the nervous system's command. Read **Lag**, **Wrist cock**, **Shaft lag**, **Power transfer** and **Inverse dynamics** together when investigating release.

The same distinction matters for ground contact. Foot forces affect whole-body momentum and the feasible motion of a connected system. In an ideal stationary no-slip model, a contact point has zero velocity, so its contact force supplies no power there even though it changes momentum. A force can redirect motion, transmit load or enable muscle work elsewhere without itself adding mechanical energy. Read **Ground reaction force**, **Constraint forces**, **Linear momentum** and **Work** together.

Finally, a zero-input calculation is a model intervention. It retains the declared state, constraints and loads; it does not erase earlier muscle work, stored energy or activation states. Read **Affine decomposition**, **Drift**, **Control authority**, **DCR** and the **ZTCF family** together. The useful research question is which measured outcomes a declared mechanism explains, and what observation would distinguish it from an alternative.

A Collision Check

COR illustrates why definitions must be used together. Assume two translating bodies collide along one normal, external impulse is negligible, the ball starts at rest, and rotation and tangential impulse are excluded. Let m_c and m_b be club and ball masses, $V > 0$ the incident club speed, and superscript $+$ denote the post-impact value. Momentum conservation and the COR definition give

$$m_c V = m_c v_c^+ + m_b v_b^+, \qquad v_b^+ - v_c^+ = eV,$$

and therefore

$$v_b^+ = \frac{(1+e)m_c}{m_c + m_b} V, \qquad \Delta T = -\frac{1}{2}\frac{m_c m_b}{m_c + m_b}(1 - e^2)V^2.$$

Thus the ball can depart faster than the incident club even when $e < 1$. The loss ΔT is the change in the two bodies' total translational kinetic energy; it is not the fraction delivered to the ball. In an off-center three-dimensional impact, rotational inertia, contact direction and friction enter the calculation. This simple derivation explains the vocabulary, not a universal equipment test or launch prediction.

Evidence and Conventions

Unless stated otherwise, mechanical vectors use a declared ground-based frame approximated as inertial over the swing. Anatomical motion uses local segment axes. Lead and trail refer to the golfer's target-side and opposite-side limbs.

The wrist entries draw on a study that measured both wrists in local coordinates under different grips [Carson et al., 2019]. The tissue entries distinguish specimen-level mechanical evidence from whole-swing inference [Bonaldi et al., 2023]. The reaction-time example comes from an object-manipulation experiment, not a golf trial [Crevecoeur et al., 2016]. Pain terminology follows IASP's distinction between experience, nociception and tissue damage [International Association for the Study of Pain, 2020]. Equipment and stroke definitions refer to the governing USGA resources rather than treating a glossary as a complete rule book [United States Golf Association, 2026a,b].

A

Acceleration The time derivative of velocity in a declared reference frame. For a clubhead point, acceleration includes changes in both speed and direction; a large acceleration does not by itself imply large mechanical power.

Affine control system A model of the form $\dot{\mathbf{x}} = f(\mathbf{x}) + G(\mathbf{x})\mathbf{u}$. At a fixed state, changes in the declared input produce linear changes in the state derivative; the total map includes the offset f. Affinity in generalized torque does not establish affinity in neural excitation or a finite swing trajectory.

Affine decomposition The split into a zero-declared-input vector field and an input-dependent increment. It depends on the chosen actuator port and retained plant state. Elastic tension, damping, contact reactions and previously established activation may survive the intervention; the split does not identify biological passivity.

Angular acceleration The time derivative of angular velocity in a declared frame. For two points on a rigid body, $\mathbf{a}_B = \mathbf{a}_A + \boldsymbol{\alpha} \times \mathbf{r} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r})$. Here $\mathbf{r}$ points from A to B. The angular-acceleration term and the angular-speed-squared term are distinct.

Angular velocity The axial vector describing instantaneous rigid-body rotation. A body's points share angular velocity but generally have different linear velocities. For three-dimensional orientations it is not generally the vector of Euler-angle derivatives.

B

Backswing The movement taking the club away from the ball toward the transition. It establishes configuration, velocities and tissue/shaft states. Raising mass can increase gravitational potential energy, but the amount and location of elastic storage require force-deformation evidence.

Ball compression A rating derived from deformation under a specified compression test. It is not a universal material modulus, cover hardness or impact COR. Dynamic response also depends on loading rate, temperature, construction and the test protocol.

Beam theory Models relating distributed loading, deformation and internal forces in slender structures. The appropriate shaft model depends on bending, torsion, shear, rotary inertia, geometric nonlinearity and boundary motion; Euler-Bernoulli theory is one approximation.

C

Catapult effect An informal description of elastic shaft recoil contributing to clubhead motion. Recoverable energy was stored through earlier work. Its contribution depends on loading and timing; a straight shaft at impact is neither a general optimum nor proof that stored energy has all become ball energy.

Center of mass The mass-weighted position $\mathbf{r}_C = \sum_i m_i \mathbf{r}_i / \sum_i m_i$. It differs from the center of pressure under the feet and from the relative vertical loads on two force plates.

Centrifugal acceleration An apparent acceleration $-\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r})$ introduced in a rotating frame. For fixed-axis circular motion, inertial acceleration is inward, of magnitude $\omega^2 r_\perp$. Real shaft forces must be found from the complete force balance; the frame term is not an additional energy source.

Centrifugal stiffening A change in a rotating structure's transverse dynamics caused by rotation-induced axial tension. The resulting geometric stiffness depends on the configuration, loading and supports. This is not a change in the shaft material's elastic modulus, nor a universal scalar stiffness increase for every mode.

Cerebellum A brain structure involved in coordination, timing and sensorimotor adaptation. Internal-model descriptions are hypotheses about computation; they do not establish that the cerebellum explicitly learns AffineDrift's chosen mathematical drift field.

Clubhead The club's striking body. Its velocity, orientation, angular velocity, mass distribution, impact location and deformation interact with the ball and contact conditions to determine launch.

Clubhead speed The magnitude of velocity of a declared clubhead point at a declared time, commonly just before impact. Specify the point, sensor and measurement convention. Distance also depends on strike, ball launch and flight conditions; population averages are not individual targets.

Coefficient of restitution (COR) For a one-dimensional normal collision, $e = (v_b^+ - v_c^+)/(v_c^- - v_b^-)$, with signed velocities along the same contact normal and a positive approach denominator. Here c and b denote club and ball; superscripts $-$ and $+$ mean before and after impact. It compares relative separation and approach speeds, not ball speed to incident club speed. Neither a universal golf-ball COR of 0.83 nor an 83 percent energy-transfer fraction follows. Club spring-effect conformity uses governing equipment test procedures [United States Golf Association, 2026a].

Collagen A family of structural proteins important in the extracellular matrix. Tissue behavior depends on collagen type, arrangement, cross-linking and interaction with other constituents; collagen content alone does not determine stiffness or recoverable energy.

Composite material A material combining distinct constituents, such as carbon fibers in a polymer matrix. Fiber orientation, layup, resin and geometry affect a shaft's directional bending and torsional response.

Connective tissue A diverse family of tissues containing cells and extracellular matrix; examples include bone, cartilage, tendon, ligament and fascia. Their constituents and mechanical functions differ, so they cannot be assigned one stiffness, strength or elastic-storage capacity.

Constraint A modeled restriction on configuration, velocity or contact state. Joint geometry, two-hand grip closure and foot contact impose different restrictions. A real contact may slip, separate or deform, invalidating a permanently rigid constraint assumption.

Constraint forces Reactions enforcing a declared constraint model. For ideal stationary bilateral constraints they do no total work on admissible motion, although they can transfer power between bodies. Moving supports, friction and deformable contacts require their own power balances.

Constraint-force method A dynamics formulation solving accelerations and reaction multipliers together. With J the constraint Jacobian, $M\ddot{\mathbf{q}} = \mathbf{r} + B\mathbf{u} + J^{\mathsf{T}}\boldsymbol{\lambda}$ is supplemented by the differentiated constraints; $\mathbf{r}$ collects retained loads and velocity-dependent terms. Reactions generally change when applied input changes.

Control affine See affine control system.

Control authority The effect attainable with the admissible input set. Instantaneous acceleration authority is a set such as $a_0 + H\mathcal{U}$; finite-time authority additionally depends on dynamics, time and constraints. It may vary by direction and posture. Faster motion does not itself increase the mass matrix or prove loss of authority.

Control input The declared external command or effort in a model, such as joint torque, motor voltage or neural excitation. These are different ports with different internal dynamics. A net joint torque is not an individual muscle force.

Coriolis force In a rotating frame, the apparent force $-2m\boldsymbol{\omega} \times \mathbf{v}_{\mathrm{rel}}$. The relative velocity and frame rotation must be specified. Velocity-coupling terms in generalized-coordinate equations need not correspond individually to a torso-frame force.

Cortical bone The relatively dense outer bone tissue, distinguished from the porous trabecular architecture. Material modulus, apparent structural stiffness and whole-bone strength are different quantities and depend on direction and geometry.

D

DCR Abbreviation for Drift-Control Ratio. DCR compares modeled drift with bounded modeled control in the same acceleration or task-projected space. A reproducible value declares the weighting matrix and norm, admissible control set, regularizer, effective plant, and state. A ratio using realized input rather than available control capacity is a different, policy-dependent diagnostic.

Degrees of freedom The number of independent local configuration coordinates after accounting for constraints. A free spatial rigid body has six. Counts for an arm or whole golfer depend on modeled joints, flexible modes and contacts; redundant orientation coordinates are not additional physical freedoms.

Dorsiflexion At the ankle, movement bringing the dorsum of the foot toward the shin. The term is also used for wrist extension. Neither use defines the forearm-shaft angle called wrist cock; wrist deviation is a separate motion.

Downswing The movement from a declared transition event to ball contact. Segment reversals need not occur simultaneously. Duration is measured for the particular swing, club and event definition rather than fixed at 0.2 seconds.

Drift The state derivative when the declared control input is zero, with the effective plant, internal state and load protocol retained. Gravity and velocity-dependent, elastic or dissipative effects may contribute. Late-downswing dominance is a model-dependent quantitative claim requiring an explicit comparison.

Drift field The vector field $f(\mathbf{x})$ in a declared affine model. Its value depends on the state and retained loads. A sequence of pointwise zero-input evaluations along an actuated swing is generally not a trajectory generated by that field.

Drift-Control Ratio (DCR) See DCR.

Dynamics The study of how systems evolve over time under the influence of forces. The dynamics of the golf swing are governed by Newton's laws and the constraints of the body.

E

Eccentric contraction Active force production while the specified muscle or fascicle lengthens. It can absorb mechanical work, but fascicle, tendon and whole muscle-tendon length changes may differ. It is not confined to follow-through.

Effective mass Directional inertia seen at a chosen task point. For an unconstrained positive-definite M, task Jacobian J and unit direction $\mathbf{n}$, $m_{\text{eff}}^{-1} = \mathbf{n}^{\mathsf{T}} J M^{-1} J^{\mathsf{T}} \mathbf{n}$. Constraints change the mobility operator; a zero directional mobility has no finite effective mass in this sense.

Elastic energy Recoverable energy associated with deformation. A linear spring stores $U = \frac{1}{2}k\delta^2$; a bending beam stores $\frac{1}{2}\int EI\kappa^2\,ds$ in the corresponding small-strain model. Viscoelastic hysteresis and continuing external work affect how much reaches another body.

Elastic modulus A constitutive measure relating stress and strain under specified conditions. Young's modulus E describes uniaxial elastic response; anisotropic and nonlinear tissues require directional or tangent measures. It differs from a whole structure's force-displacement stiffness.

Elasticity Reversible deformation on unloading within a material's elastic regime. A material can exhibit both elastic response and irreversible plastic deformation; viscoelastic recovery may also depend on time.

Elastin An extracellular-matrix protein contributing to reversible deformation. Its abundance varies by tissue and sampling method; a single percentage cannot characterize all fascia.

Ellipsoid A surface defined by a positive-definite quadratic form, with a filled ellipsoid including its interior. Higher-dimensional ellipsoids can approximate uncertainty or reachable sets; general nonlinear reachable sets need not be ellipsoidal.

Endurance The ability to sustain effort over time. Golf endurance depends on cardiovascular fitness, muscle endurance, and mental focus.

Epimysium The connective-tissue sheath around a whole skeletal muscle, continuous with internal connective-tissue networks and attachment structures. Force can be transmitted through several paths; the sheath is not an isolated spring with a universal stiffness.

Equations of motion Mathematical equations describing how a system evolves over time, typically derived from Newton's laws or Lagrangian mechanics. The equations of motion for the golf swing are nonlinear and high-dimensional.

Equilibrium A state at which the declared state derivative vanishes for specified inputs and loads. Static force and moment balance is one mechanical case. Ongoing rotation or translation can instead be a relative equilibrium in an appropriate frame.

Euler-Bernoulli beam theory A slender-beam approximation that neglects transverse shear deformation and normally rotary inertia. Bending moment and curvature satisfy $M_b = EI\kappa$ with a consistent sign convention. Shaft taper, composite anisotropy, torsion and large motion may require extensions.

Extension An anatomical joint motion defined relative to local segment axes. At the wrist, the hand moves dorsally relative to the forearm. It is distinct from radial/ulnar deviation and cannot alone define cock or release.

F

Facet joint A synovial articulation between adjacent vertebral articular processes, also called a zygapophysial joint. It contributes to spinal guidance and load sharing.

Fascia Connective-tissue sheets and networks with region-specific architecture. Mechanical tests of fascia lata demonstrate direction- and time-dependent response; those results do not establish a universal fascia-versus-tendon ranking or whole-swing energy contribution [Bonaldi et al., 2023].

Fatigue An activity-related reduction in the capacity to produce force or power. Its measurement depends on the task and protocol; altered swing performance can have muscular, neural and other contributors.

Feedback Information about the current state or outcome that is fed back to the system for control or learning. Proprioceptive feedback informs the nervous system of body position; visual feedback informs the golfer of ball position.

Feedback control Control that uses information about measured or estimated state or outcome. The useful correction depends on sensory delay, processing, actuation and time remaining. A late swing can still be influenced by feedback to earlier events.

Feedforward control Control based on a reference or prediction without waiting for the resulting error. It can coexist with feedback throughout a movement; feedforward does not mean biologically passive.

First mode The lowest-frequency mode within the declared class and boundary conditions, such as the first elastic bending mode after excluding rigid-body modes. Whether it dominates a shaft response depends on forcing, damping, supports and the measured output.

Flexibility In a scalar linear structural model, compliance is displacement per applied force and is the reciprocal of stiffness. Anatomical flexibility instead describes range of motion under a protocol; it is not generally one material constant.

Flexion An anatomical bending motion defined for a joint. Wrist palmar flexion moves the palm toward the anterior forearm; it is a different axis from radial/ulnar deviation.

Flight Ball motion through air after separation from the club. Initial position, velocity and spin combine with gravity, aerodynamic coefficients, air properties and wind to determine the trajectory.

Follow-through The phase after impact where the golfer decelerates the club and body, coming to rest. The follow-through is important for balance and controlled deceleration.

Force A vector interaction, measured in newtons (N). For a fixed-mass body in an inertial frame, resultant external force equals mass times center-of-mass acceleration. Forces at different points can also create different moments.

Force plate An instrument measuring resultant contact forces and moments in its calibrated frame. Center of pressure is derived under declared contact-plane assumptions and becomes ill-conditioned at low normal force. It is not a direct measurement of individual muscle forces.

Forward dynamics Solving for instantaneous accelerations or the state derivative from state, input and loads, including constraint reactions when needed. Numerical integration repeatedly uses that result to predict a future state.

Forward model A model predicting motion or sensory consequences from state and command. In neuroscience it is a proposed internal computation; a useful engineering predictor is not proof of a particular neural representation.

Frame A reference origin and orientation for describing quantities. A ground-fixed frame is approximately inertial over a golf swing; Earth's rotation means that it is not exactly inertial.

Frequency Cycles per unit time, in hertz (Hz). Angular frequency is $\omega = 2\pi f$. A fixture rating of 240–280 cycles per minute corresponds to 4.00–4.67 Hz, but does not specify the modes of a moving, gripped club.

Friction Contact interaction opposing relative slip or supporting a no-slip state. Static friction can be essential at the shoes and grip even when there is little relative motion. Its magnitude cannot be dismissed by comparison with other forces without a contact analysis.

G

Generalized coordinate An independent local variable describing configuration, such as a joint angle or flexible-mode amplitude. A chosen coordinate chart and constraint reduction determine its meaning.

Generalized force A quantity dual to a generalized displacement, defined through virtual work $\delta W = \mathbf{Q}^{\mathsf{T}}\delta\mathbf{q}$. A rotational coordinate has a torque-like conjugate, but its axis and scaling must be declared.

Generalized momentum The conjugate quantity $p_i = \partial L/\partial \dot{q}_i$. Euler-Lagrange dynamics gives $\dot{p}_i = \partial L/\partial q_i + Q_i^{\mathrm{nc}}$. Conservation follows for a cyclic coordinate with zero nonconservative generalized force; zero applied force alone is insufficient.

Geometry of motion The study of how systems move in space, considering constraints and symmetries. This is the subject of a companion series to this textbook.

Glute maximus The gluteus maximus, contributing to hip extension and external rotation. Its swing contribution depends on activation, posture, loading and interactions with other muscles.

Glute medius The gluteus medius, a hip abductor with region- and posture-dependent rotational actions. Its anatomical role does not specify its activation timing in an individual swing.

Golgi organ A Golgi tendon organ, a receptor supplying sensory information related to muscle-tendon tension. Its signal is part of a neural circuit, not a direct conscious readout of net joint torque.

Gravitational potential energy For a nearly uniform field, $U_g = \sum_i m_i g h_i$ relative to chosen zero heights. Raising one segment increases its contribution, but the whole golfer-club change requires all segment heights.

Gravity The interaction between masses. Near Earth's surface, gravitational acceleration is approximately $9.81\,\mathrm{m/s^2}$ downward; weight is the associated force, not the acceleration itself.

Green The smooth, manicured grass surface on which the hole is cut. Golfers aim to land the ball on the green, from which they putt.

Grip The hand-club contact arrangement or the club's handle covering. A rigid two-hand grip creates loop constraints in a model; real grip compliance and slip may matter for force transmission.

Ground contact Interaction between shoes and the ground. Normal reactions, friction and moments support and accelerate the body subject to unilateral contact and slip limits. They do not invariably propel the golfer forward.

Ground reaction force The resultant ground force on the feet. Together with other external forces it governs whole-system momentum change. Its components, peaks and timing must be measured; weight alone does not determine them.

H

Hamstring A member of the posterior-thigh muscle group. Most cross the hip and knee and contribute to hip extension and knee flexion; the short head of biceps femoris does not cross the hip.

Hang time The interval between ball launch and first landing. It depends jointly on launch speed and direction, spin, aerodynamic properties, wind and relative landing height.

Heading An orientation or travel direction in a declared horizontal reference. Body facing direction, target direction and ball velocity direction are distinct and should not share one unlabeled angle.

Heel strike The onset of heel-ground contact, commonly used in gait analysis. A golf heel-contact event must be identified from the particular foot and swing rather than assigned to a universal phase.

Hip rotation Often used in coaching for pelvis turning, but anatomically refers to femur rotation relative to the pelvis. Pelvis orientation in the laboratory frame is a different measurement.

Holonomic constraint A restriction expressible as $\phi(\mathbf{q}, t) = 0$. It also restricts velocity through $J\dot{\mathbf{q}} + \phi_t = 0$ and acceleration through a further derivative. A joint or contact is holonomic only under the declared model.

Hook Pronounced flight curvature toward the golfer's lead side: left for a right-handed golfer. For centered contact, face-to-path geometry helps determine spin-axis tilt; off-center gear effect and aerodynamic conditions also matter.

Horizontal plane A plane perpendicular to local gravity. A sloping ground surface is not horizontal, and a golf swing generally cannot be represented by one fixed plane.

Horsepower A family of power units. Mechanical horsepower is approximately 745.7 watts; metric horsepower is approximately 735.5 watts. Use watts for an unambiguous mechanical power balance.

Hydration Water content or fluid status, with the tissue and measurement basis specified. It can influence tissue mechanics, but there is no single water fraction that characterizes all fascia.

Hypertrophy An increase in tissue or cell size; muscle hypertrophy increases muscle cross-sectional area. Strength also depends on neural drive, architecture, leverage and the testing task.

I

Inelastic collision A collision in which the bodies' modeled macroscopic kinetic energy is not conserved. Energy is redistributed into deformation, vibration, heat and other modes; it is not destroyed.

Inertia The mass-distribution properties relating momentum to velocity and force/moment to acceleration. In a rigid-body model, configuration changes can alter generalized inertia; speed alone does not increase mass.

Inertial frame A frame in which free particles move at constant velocity and Newton's equations need no frame-inertia terms. A ground-fixed frame is a useful approximation over the time and distance scales of a swing.

Infraspinatus A rotator cuff muscle contributing to external rotation and glenohumeral stabilization. Its activation must be measured rather than inferred from a named swing phase.

Injury prevention Actions intended to reduce injury risk. Mechanical analysis may identify loads and candidate mechanisms, but a claim of reduced injuries requires appropriate outcome evidence.

Input-output behavior The relationship between a declared input history and measured output, including initial state and disturbances. Torque-to-club motion and neural-excitation-to-ball outcome are different systems.

Intermuscular coordination The organization of activity across muscles. Coactivation can affect both net moments and stiffness; a small net moment does not imply small individual muscle forces.

Intervertebral disc A structure between adjacent vertebral bodies consisting of an annulus fibrosus and nucleus pulposus with associated endplates. It transmits load and permits motion; pain cannot be inferred from a rotation measurement alone.

Intramuscular coordination The organization of motor-unit recruitment and discharge within a muscle. Force also depends on muscle state and contraction history.

Inverse dynamics Estimating resultant generalized forces from measured motion, inertial properties and external loads. It does not uniquely identify individual muscles, establish neural commands or guarantee that a desired motion is dynamically feasible.

Inverse kinematics Finding configurations consistent with task positions and orientations. Solutions may be multiple, absent or sensitive to constraints. A mathematical inverse-kinematics solution does not prove that the nervous system uses that algorithm.

Isometric contraction Active force production while the specified length or joint position is held fixed. A fixed joint angle does not require every fascicle and tendon to remain the same length. Zero external mechanical work does not imply zero metabolic expenditure [OpenStax, 2022c].

J

Jacobian A derivative matrix connecting coordinate increments or velocities to a declared task representation. For point position, $\dot{\mathbf{r}} = J\dot{\mathbf{q}}$; the dual force mapping is $\mathbf{Q} = J^{\mathsf{T}}\mathbf{F}$. Orientation representations may require additional rate mappings.

Joint An articulation between bones, permitting or restricting relative motion according to its structure. Biomechanical joints may be modeled with fewer freedoms than the real anatomy.

Joint angle A relative segment-orientation measure under declared anatomical axes and rotation conventions. In three dimensions it is not generally the angle between two bone centerlines.

Joint complex A group of articulations contributing to a motion, such as glenohumeral, acromioclavicular and sternoclavicular joints with scapulothoracic motion. A modeling label does not make the complex a single hinge.

K

Kinematic chain Connected bodies and joints defining possible relative motion. A two-hand grip and ground contacts can close loops. Kinematics alone does not establish the direction or amount of energy transfer.

Kinematics The study of motion without considering forces. Kinematics describes where the body is and how fast it is moving.

Kinetic chain A description of interacting segments and the forces and moments transmitted between them. Demonstrating a particular energy-transfer pathway requires segment power balances, not merely an order of peak angular velocities.

Kinetic energy For a rigid body, $T = \frac{1}{2}m\|\mathbf{v}_C\|^2 + \frac{1}{2}\boldsymbol{\omega}^{\mathsf{T}} I_C \boldsymbol{\omega}$, with inertia and angular velocity in the same frame. Clubhead kinetic energy is not all transferred to the ball.

Knee flexion Bending at the knee. Its effect on body height and loading depends on the simultaneous ankle, hip and trunk configuration and the contact forces.

L

Lag A coaching or geometric descriptor, often the angle between a declared forearm direction and shaft direction. Specify the lines and angle convention. The same angle is possible with a rigid shaft; it does not measure shaft strain energy.

Lagrange multiplier A scalar or vector variable enforcing a constraint. With constraint Jacobian J, the generalized reaction is $J^{\mathsf{T}}\boldsymbol{\lambda}$. Multiplier magnitude and units depend on constraint normalization; a multiplier is not automatically a physical contact-force component.

Lagrangian mechanics A variational formulation using the Lagrangian, commonly $L = T - U$, together with nonconservative forces and constraints. It includes forces through generalized work and is equivalent to Newtonian mechanics under the same model assumptions.

Launch angle The vertical angle at which the ball leaves the clubface at impact, measured from horizontal. Launch angle significantly affects carry distance and ball flight.

Launch monitor An instrument estimating launch and sometimes club-delivery variables using radar, cameras or other sensors. Specify which outputs are directly measured, inferred or simulated and the calibration and uncertainty of each.

Lie angle The club's shaft-to-ground geometry under a specified sole/reference condition; dynamic lie describes delivery at impact. Lie can affect the orientation of a lofted face, but it does not by itself determine club path.

Ligament Connective tissue linking bones and contributing to joint restraint and sensory function. Its response varies with site, fiber direction, rate and loading history; it has no universal elasticity ranking relative to tendon or fascia.

Limit of motion The range reached under a specified active or passive measurement protocol. Anatomy, load, pain and muscle state influence that range; an observed endpoint is not a universal tissue-failure threshold.

Limiter Something that limits or constrains the system. In the golf swing, constraints (like the kinematic structure) act as limiters on motion.

Line of pull The local direction of muscle-tendon force at an attachment or wrapping point. It changes with posture and determines moment arms together with the joint axis.

Linear momentum For a fixed collection of bodies, $\mathbf{P} = \sum_i m_i \mathbf{v}_{C_i}$ and $\dot{\mathbf{P}} = \sum \mathbf{F}_{\text{ext}}$ in an inertial frame. Internal forces redistribute momentum but cancel in the total.

Loop closure constraint A relation ensuring that paths through a closed kinematic chain meet consistently. Two hands connected through the arms and torso while holding the same club form a loop in a rigid-grip model.

Lordosis An inward curvature of the spine viewed from the side, normally present in the cervical and lumbar regions. Its magnitude alone neither diagnoses pathology nor determines a golfer's injury risk.

Low back pain Pain in the lower-back region, with potentially multiple biological, psychological and social contributors. It is not synonymous with a particular injured structure or proof that swing rotation caused tissue damage [International Association for the Study of Pain, 2020].

Lumbar spine The lower spinal region, usually containing five vertebrae. Its motion and loading depend on posture, external forces and muscle activity; net joint moments do not uniquely identify local tissue loads.

M

Magnus effect A transverse aerodynamic force associated with rotation of a body moving through a fluid. For a golf ball, lift and side force depend on spin, relative air velocity and flow conditions; spin-axis tilt helps determine curvature.

Manipulator A mechanical device with joints and segments that move. A robot arm is a manipulator; the human arm is also a manipulator.

Mass A measure of the amount of matter in an object, measured in kilograms (kg). The mass of the clubhead and shaft affects their inertial properties.

Mass matrix The generalized inertia in $T = \frac{1}{2}\dot{\mathbf{q}}^{\mathsf{T}} M(\mathbf{q})\dot{\mathbf{q}}$. It is symmetric positive definite for independent coordinates with positive kinetic energy in every nonzero velocity direction. Dynamics also includes bias forces and constraints; redundant or massless coordinates can invalidate invertibility.

Measurement A quantitative observation with a declared method, units, reference and uncertainty. Distinguish measured signals from quantities inferred through a model.

Mechanical advantage A force or moment ratio for a specified mechanism and configuration. Ideal power balance couples force advantage to a reciprocal velocity tradeoff. Long limbs and high rotation speed do not provide unrestricted force and speed amplification.

Mechanical coupling Interaction through connections, constraints, inertia or material deformation. Coupling can transmit forces and redistribute energy, but its sign and usefulness depend on state and input.

Mechanoreceptor A sensory receptor that detects mechanical stimuli (pressure, stretch, vibration). Mechanoreceptors in fascia and tendons provide proprioceptive feedback.

Medial rotation Rotation toward the body's midline under the relevant anatomical convention, also called internal rotation. Joint-relative rotation must be distinguished from pelvis or trunk rotation in the laboratory frame.

Meridian A term with several meanings. A proposed myofascial anatomical chain and a traditional medicine energy meridian are different concepts. Anatomical continuity does not establish a special energy channel; see myofascial meridian.

Modal analysis Analysis using deformation modes of a declared linearized structure and boundary condition. For undamped vibration, $K\boldsymbol{\phi} = \omega^2 M\boldsymbol{\phi}$. Moving supports, rotation, damping and nonlinear deformation may require a time-varying or expanded model.

Modal coordinate The amplitude multiplying a chosen mode shape. In $w(s,t) = \sum_j \phi_j(s)\eta_j(t)$, rescaling ϕ_j changes η_j and the modal parameters; normalization must be stated.

Modal damping Dissipation represented in modal coordinates. A linear viscous model uses terms proportional to modal velocity; the modal damping matrix is not necessarily diagonal.

Modal decomposition Expansion of deformation in selected modes. Truncation requires checking whether omitted modes affect the forces, motion or energy relevant to the question.

Modal frequency See natural frequency.

Modal stiffness The stiffness associated with a normalized mode or modal basis. For a single mode, $k_j = \boldsymbol{\phi}_j^{\mathsf{T}} K \boldsymbol{\phi}_j$ and $\omega_j^2 = k_j/m_j$ with the matching modal mass.

Mode A characteristic deformation pattern of a specified linearized system. The lowest elastic frequency need not dominate every forced response.

Model A deliberately simplified representation with declared variables, parameters, inputs, constraints and validity limits. A rigid-body tree may need closed-loop contacts, muscle states and flexible shaft modes for a particular golf question.

Moment See torque.

Moment arm The perpendicular distance from a force's line of action to a specified axis in the corresponding planar geometry. More generally, torque is obtained from $\mathbf{r} \times \mathbf{F}$ and projected onto the axis.

Momentum See linear momentum.

Motion Change in position over time. The golf swing involves complex 3D motion of the entire body and club.

Motor control The process by which the nervous system activates muscles to produce desired motion. Motor control involves feedforward and feedback mechanisms.

Motor cortex Cortical regions contributing to movement preparation, execution and feedback within distributed neural circuits. Movement is not generated by an isolated cortical command directly specifying every muscle force.

Motor learning Changes in movement capability through practice and experience. Improved performance can reflect several neural and behavioral processes; it does not alone identify the internal dynamics model learned.

Motor unit A motor neuron and all the muscle fibers it innervates. Motor units are the basic units of muscle activation and force control.

Movement variability Variation across repetitions. Its consequences depend on task sensitivity and coordination: different joint motions can produce similar club delivery. Reducing every component of variability is not a universal objective.

Muscle Biological tissue generating active force. Skeletal muscle can shorten, hold length or lengthen while active; force depends on activation, length, velocity and history.

Muscle fiber A skeletal-muscle cell containing contractile structures. Fibers are organized into fascicles and connected through extracellular matrix; activation is coordinated through motor units.

Muscle spindle A sensory organ whose afferents provide information related to muscle length and length change, modulated by its own neural drive. Its output is not a direct measurement of whole-joint angle.

Myofascial Relating to muscle and its connective-tissue networks. Mechanical continuity permits interaction but does not quantify whole-body force transmission or recoverable energy.

Myofascial meridian A proposed chain of anatomically connected muscles and fascia. Evidence of continuity or force transmission along a particular path does not establish a dominant whole-swing energy pathway; that requires loads, deformation and power measurements.

Myofascial release A label for manual or self-applied soft-tissue techniques. A change in symptoms or range of motion does not by itself demonstrate rupture of adhesions or a specific fascial release mechanism.

Myth An informal label for a disputed belief. Technical criticism should instead identify the precise claim, its evidence and a possible falsifying observation.

N

Natural frequency An eigenfrequency of a specified linearized system. An undamped scalar oscillator has $\omega_n = \sqrt{k/m}$; its frequency in hertz is $\omega_n/(2\pi)$. Damping, shaft supports, grip and attached clubhead alter the measured response, so a fixture rating is not a universal swing frequency.

Natural motion The motion a declared effective plant exhibits after the declared control input is removed. A forward ZTCF trajectory is one precisely specified construction of this motion.

Neuroscience The scientific study of the nervous system, including the brain, spinal cord, and peripheral nerves. Neuroscience provides insight into how the golf swing is controlled.

Newton The SI unit of force, $1\,\mathrm{N} = 1\,\mathrm{kg\,m/s^2}$. A one-kilogram mass weighs approximately 9.81 N near Earth's surface.

Newton's laws In classical mechanics, free motion is uniform in an inertial frame, net force changes momentum, and interacting bodies exert equal and opposite forces. For fixed mass, $\mathbf{F}_{\text{ext}} = m\mathbf{a}_C$.

Non-holonomic constraint A velocity restriction that cannot be integrated into a configuration-only restriction on the relevant domain. Some rolling-without-slip models provide examples. Sliding contact does not automatically impose a non-holonomic constraint.

Non-muscular power A source-accounting label that must specify the boundary. Elastic recoil can deliver power while the tissue or shaft loses stored energy, but that energy may have originated in earlier muscle work. The label does not establish energy independent of muscular activity.

Nonlinear A description of a map or dynamical model that does not satisfy the relevant linear superposition properties. An affine input map can coexist with nonlinear state dynamics; a constant offset also makes a map non-linear in the strict algebraic sense.

Norm A nonnegative measure of vector magnitude satisfying definiteness, homogeneity and the triangle inequality. Comparing mixed positions, angles and velocities requires explicit scaling or weighting; a norm is not automatically an energy.

Normalized drift fraction A separate quantity written as drift / (drift + control), ranging from 0 (pure control) to 1 (pure drift) when these labels denote nonnegative, comparable magnitudes with a positive sum. Both zero makes the ratio undefined. It is not a vector sum, cancellation measure, work fraction or percentage of muscle activity.

Numerical integration Computing the future state of a system by discretizing time and integrating the equations of motion step-by-step. Numerical integration is used to simulate swings.

O

Objective A declared criterion, such as expected score, dispersion or task error. Distance and consistency can trade off; an optimization claim must state their weighting and constraints.

Oblique muscles The internal and external abdominal obliques, contributing to trunk rotation, lateral flexion and abdominal function. The resulting action depends on side, posture and coordination with other muscles.

Observation The act of measuring or perceiving something. Launch monitors provide quantitative observations of swing parameters.

Optimality Being best relative to a specified objective and admissible set. Local, global, nominal and robust optimality are different claims; no single swing is optimal for every criterion.

Optimization The process of finding the best solution to a problem given constraints. Swing optimization involves finding the control strategy that achieves the goal.

Oscillation Repeated variation around a reference or equilibrium. A moving shaft may oscillate about a changing loaded shape rather than its unloaded straight shape.

Outcome A task result such as ball position or score. Impact sets launch conditions; aerodynamics, wind, terrain, bounce and roll also affect the final result.

Overload principle Training adaptation in response to demands beyond the accustomed level. This is not an instruction to exceed tissue capacity; useful progression depends on dose, recovery and the individual.

Overstretching An informal term for stretching beyond the intended or tolerated range. Range alone does not determine tissue injury; load, rate, duration and individual state matter.

Oxidative metabolism Processes using oxygen to support ATP production. They contribute to sustained activity and recovery, while brief high-power actions also draw on other energy pathways.

P

Palmaris longus A variable forearm muscle, absent in some people, that can flex the wrist and tension the palmar aponeurosis. It is not a required late-downswing driver.

Parallel mechanism A structure in which multiple linkage paths connect bodies. A golfer holding one club with two hands can form such a closed mechanism; simply having several limbs does not make every connection parallel.

Parameter A quantity characterizing a model, such as mass or elastic modulus. A parameter is distinguished from the evolving state; uncertain or time-varying parameters require explicit treatment.

Passive stiffness Resistance to small deformation under a protocol intended to exclude active muscle contribution. It depends on posture, loading history and tissue response. Zero net joint torque does not establish zero activation or purely passive stiffness.

Performance How well a goal is achieved, measured quantitatively. Golf performance is measured by score (total strokes), distance, and consistency.

Perimysium Connective tissue surrounding and linking muscle fascicles, contributing to organization and force transmission. Its role is not described completely by a single series spring.

Periosteum A connective-tissue covering on most external bone surfaces, absent from articular cartilage-covered regions. It participates in attachment, growth and repair.

Perturbation A change or disturbance relative to a declared reference. Linear sensitivity describes sufficiently small perturbations; finite disturbances and contact changes require additional analysis.

Phase A stage defined by events, such as backswing or follow-through, or an angular position within an oscillation. Event phase and oscillator phase should not be interchanged.

Phase portrait A representation of state-space trajectories and vector fields. A low-dimensional projection can hide state variables and cannot by itself establish stability or a limit cycle.

Phase space The space of dynamical states. For an unconstrained planar point mass it has two position and two velocity dimensions; canonical mechanics often uses position and momentum instead.

Phenomenological Describing observed behavior without resolving all underlying mechanisms. A phenomenological relation may be useful within its calibration range without identifying a causal neural or tissue process.

Planar motion Motion confined to a plane under a declared model. A planar golf approximation must specify the phase, fitted plane and tolerated out-of-plane error; the full swing is three-dimensional.

Plane of motion A plane used to describe or approximate motion. An inclined fitted plane can summarize part of a club trajectory but does not impose a physical constraint or describe every segment.

Plasticity Irreversible material deformation after unloading; in neuroscience, a change in neural organization or function. These are different meanings. Permanent tissue deformation should not be equated with beneficial mobility.

Position The location of a point in a declared frame. Joint angles describe configuration; they are not Cartesian point positions.

Potential energy A scalar energy function for conservative interactions, defined up to an additive constant. Gravity and elastic deformation can contribute. Nonconservative forces do not generally have a configuration-only potential.

Power The rate of energy transfer by work. A force supplies $\mathbf{F} \cdot \mathbf{v}$ at its application point, and a moment supplies $\boldsymbol{\tau} \cdot \boldsymbol{\omega}$ in matching frames. Large force with little velocity can mean little mechanical power.

Power transfer Energy flow per unit time across a declared boundary. Interface forces can transfer power between segments while cancelling in the total balance. Net joint power, individual muscle power and shaft storage are different quantities; a proximal-to-distal peak sequence alone proves none of them.

Precision Closeness of repeated measurements or outcomes to one another. It differs from accuracy relative to a reference and from a causal explanation of the observed consistency.

Prediction A forecast conditional on a model, state estimate, input and disturbances. A successful prediction does not uniquely identify the mechanism producing the motion.

Proprioception Sensory information and perception related to body configuration, motion and force. It draws on multiple receptors and neural processing; it is not a noiseless instantaneous state measurement.

Proprioceptive feedback Sensory information about body position and movement from mechanoreceptors in muscles and connective tissue.

Proteoglycan A core protein with attached glycosaminoglycan chains, forming part of extracellular matrix. These molecules interact with water and other constituents and contribute to tissue behavior.

Putter A club designed mainly for putting, typically with low loft. A putt can initially launch and skid before rolling; it is not necessarily pure rolling from impact.

Q

Quadriceps The four anterior-thigh muscles contributing to knee extension. Rectus femoris also crosses the hip and can contribute to hip flexion; swing timing is an empirical question.

Qualitative Describing properties using words rather than numbers. Qualitative observations like "smooth" or "powerful" are important in coaching.

Quantitative Describing properties using numbers and measurements. Quantitative data from launch monitors enables objective analysis.

R

Radial deviation Movement of the wrist toward the thumb (radial) side. Radial deviation of the wrist is part of the wrist motion during the swing.

Range of motion (ROM) The angular or displacement range reached under a stated active or passive protocol. It depends on joint geometry, tissue state, loading and measurement convention.

Rate of force development The slope of a force-time curve over a declared interval. Measurement depends on the task, onset definition and filtering; a group difference does not establish that training caused it.

Reachability The set of states attainable from a specified initial state within a specified time using admissible controls and disturbances. It is model- and constraint-dependent, not a direct statement of an individual golfer's capability.

Reaction time The interval from a defined stimulus to a defined response onset. It differs from reflex latency, EMG onset and electromechanical delay. An object-manipulation experiment found goal-directed muscle responses around 60 ms; that is evidence against one universal delay, not a measured golf correction time [Crevecoeur et al., 2016].

Recoil Recovery motion of a deformed structure as loading changes. Returned elastic energy, damping and continuing work jointly determine the response; recovery need not occur only after complete unloading.

Recovery Return to normal function after fatigue or injury. Recovery time between rounds allows muscles and connective tissue to repair.

Recruitment Activation of additional motor units. Muscle force is also adjusted through motor-neuron discharge rate and depends on contraction state; fibers are not independently recruited as arbitrary continuous actuators [OpenStax, 2022c].

Reference frame See frame.

Relaxation A reduction in active muscle tension or, in material testing, stress decay under held deformation. A reduced command does not instantaneously remove all muscle-tendon force.

Repeatability Closeness of repeated results under specified conditions. It is an observable property distinct from accuracy or the mechanism producing it.

Resistance A context-dependent opposition to a specified change. Inertia opposes acceleration rather than consuming energy simply because motion occurs; friction and drag can dissipate energy.

Restitution Recovery of relative separation motion during impact, characterized in a normal collision by COR. COR is a relative speed ratio; energy loss also requires masses and the full collision model.

Restoring force A force that acts to return a deformed system to equilibrium. The spring force in Hooke's law is a restoring force.

Resultant A sum of compatible vector quantities in a common frame. Forces can be summed, but moments also depend on their application points. Clubhead velocity follows kinematic mappings, not an unqualified sum of segment speeds.

Retraction Posterior movement of a structure, such as the scapula moving toward the spine. It is an anatomical description, not a universal prescription for address posture.

Reverse engineering Inferring candidate mechanisms from observations. Swing motion alone generally does not uniquely separate drift, applied input and constraint reactions; identification needs additional measurements or assumptions and uncertainty analysis.

Rhomboid One of the rhomboid muscles, contributing to scapular retraction and downward rotation. Its activation pattern is not determined by the name of a swing phase.

Rigid body An idealized object that does not deform, maintaining its shape and size. Rigid-body models simplify golf swing analysis.

Robotics The field of engineering and computer science concerned with the design and control of robots. Robotics provides insight into human motion control.

Rotational kinetic energy For a rigid body about its center of mass, $T_{\text{rot}} = \frac{1}{2}\boldsymbol{\omega}^{\mathsf{T}} I_C \boldsymbol{\omega}$. The frame and reference point matter. A peak in angular speed need not coincide with a peak in energy when effective inertia changes.

Rotator cuff A group of four muscles (supraspinatus, infraspinatus, teres minor, subscapularis) surrounding the shoulder joint, responsible for shoulder stability and motion. The rotator cuff experiences significant loading during the golf swing.

Rough Longer grass outside closely mown playing areas. Grass or moisture between face and ball can change friction, launch and spin; spin is not invariably higher than from a clean lie.

Round Play over the holes assigned for a round, commonly 18 or 9. Duration depends on the course, format and pace of play rather than a fixed biomechanical timescale.

Run Distance traveled after first landing, including bounce and roll under the chosen convention. Landing velocity, spin, ground properties and slope determine it.

S

Saddle joint A joint with reciprocally concave and convex articular surfaces, exemplified by the thumb carpometacarpal joint. Its motion is more complex than two independent ideal hinges.

Safety Management of the chance and consequences of harm. No single swing technique or equipment choice guarantees freedom from injury.

Sagittal plane An anatomical plane separating left and right portions; the midsagittal plane passes through the midline. Its orientation follows the declared anatomical reference, not every segment throughout a swing.

Sarcomere The contractile unit between Z-discs in striated muscle. Active sarcomeres may shorten, remain approximately constant in length or lengthen; contraction means force-generating activity, not necessarily shortening [OpenStax, 2022c].

Scaphoid A carpal bone on the thumb side of the wrist, participating in wrist articulation. A diagnosis cannot be inferred from swing appearance.

Scapula The shoulder blade, a triangular bone on the back of the ribcage. The scapula moves during shoulder rotation and contributes to arm reach.

Scapular dyskinesia An observed alteration in scapular movement or position. Association with symptoms does not establish that it caused injury or that correcting the appearance will prevent injury.

Scapulohumeral rhythm Coordination of scapular and humeral motion during arm movement. It varies with task, elevation and individual; a fixed ratio is not a universal health criterion.

Second moment of inertia In beam bending, the area second moment $I = \int_A y^2 \, dA$, measured in m^4 about a declared section axis. It is distinct from mass moment of inertia in $\mathrm{kg\,m}^2$; bending rigidity is EI.

Segmentation Division of the system into parts. The golf swing can be segmented into phases or into body segments.

Sensing Detecting information about the system or environment. Proprioceptive sensing provides feedback about body position.

Sensitivity Change in an output with respect to a specified input, parameter or initial state. Derivative, finite-difference and statistical sensitivity are different measures and require units or scaling.

Sensorimotor integration The combination of sensory information with motor commands to produce coordinated movement. Sensorimotor integration is critical for movement control.

Series elastic element A compliant element in series with an actuator in a muscle-tendon model, often representing tendon and aponeurosis. Fascia includes several geometries and load paths and is not universally a series element.

Shaft The structure connecting grip and clubhead. Its bending, torsion, mass and damping interact with grip motion and head loading. Elastic energy is stored only to the extent that it deforms.

Shaft flex An industry label or measured flexibility under a stated test. Labels such as regular and stiff are not universal stiffness values across manufacturers; compare force-deflection behavior, geometry and boundary conditions.

Shaft lag A shaft-deformation descriptor requiring a reference axis and sign convention, such as head displacement relative to the grip-end tangent. It differs from the forearm-shaft angle called lag. Strain energy depends on curvature and torsion along the shaft.

Shaft plane A plane defined using the shaft and an additional independent direction or point. A shaft line alone belongs to infinitely many planes. Address geometry does not constrain the subsequent club trajectory to that plane.

Shaft release An informal term for a phase of changing shaft deformation. It should be distinguished from wrist release: one describes flexible-body motion, the other joint/club geometry.

Shank A shot struck with the club's hosel rather than the intended face region. Ball direction follows the actual contact geometry; the shot alone does not identify an arm-activation or sequencing cause.

Shape In ball-flight discussion, the trajectory's curvature, such as draw or fade. Launch velocity, spin-axis orientation and aerodynamic conditions determine the path.

Shear force A force component tangent to a declared surface or section. Its tissue effect depends on the area, direction, distribution and simultaneous loading.

Shear strain A measure of distortion involving a change of angle between material directions. In small-strain notation, engineering shear strain is twice the corresponding tensor shear component.

Simulation Running a mathematical model forward in time to predict system behavior. Swing simulation uses the equations of motion to predict ball flight.

Sinus A cavity or passage in bone or other tissue. The frontal sinus is a cavity in the forehead bone.

Sinus tarsi syndrome A clinical label for pain and related symptoms around the sinus tarsi, a space between talus and calcaneus. The label alone does not identify one lesion or establish a golf-specific cause.

Skeletal system The framework of bones and connective tissue supporting the body. The skeletal system provides the structure for motion and is involved in force transmission.

Slice Pronounced flight curvature toward the golfer's trail side: right for a right-handed golfer. Centered-strike face-to-path geometry is one contributor to spin-axis tilt; contact location and gear effect also matter.

Slop Informal mechanical terminology for play, clearance or backlash. Physiological joint motion and soft-tissue compliance should be described by their actual geometry and load response.

Slow twitch See type I fiber.

Smooth A qualitative description of motion. A technical smoothness measure may use acceleration or jerk and must specify filtering and time scaling; visual smoothness alone does not prove efficiency.

Snap An informal description of rapid motion or recoil. Replace it in a mechanical argument with the measured joint velocity, shaft deformation rate or contact event intended.

Soft tissue A broad anatomical category including muscle, tendon, ligament and other non-bony tissues. These tissues have different mechanical and physiological properties.

Sole The lower clubhead surface that interacts with turf. Bounce geometry, delivery and ground properties jointly determine the contact response.

Space The 3D region in which motion occurs. The golf swing occurs in the 3D space of the golfer's body and the area around the ball.

Span The distance between two points, like the arm span (fingertip to fingertip with arms extended).

Speed The nonnegative magnitude of velocity. Speed omits direction; equal clubhead speeds can yield different impact conditions.

Splay To spread or open apart. Foot splay at address affects balance and weight transfer.

Sprain A ligament injury, ranging from stretching or microscopic damage to partial or complete tearing. It differs from a muscle or tendon strain.

Spread The spatial distribution of a quantity. Force transmission through connective tissue can be heterogeneous; continuity does not imply uniform loading.

Spring An idealized elastic element with a force-deformation relation. A shaft can behave as a distributed elastic structure; reducing it to one spring requires a declared coordinate and loading model.

Spring constant The scalar force-displacement stiffness k in $F = k\delta$. For a uniform small-deflection cantilever with a transverse tip force, $k = 3EI/L^3$. Other supports, taper, load positions and deformation modes give different relations.

Square In a declared target reference, a square face has its horizontal normal directed along the target line. A conventional square stance has the foot-alignment line approximately parallel to that line, not perpendicular to it.

Squat A movement involving hip and knee flexion and extension with coordinated ankle and trunk motion. Its contribution to swing loading requires force and motion measurements.

Stability A property defined relative to a reference and disturbance class. Lyapunov stability keeps sufficiently small state perturbations small; attraction additionally brings them toward the reference. Balance at address and stability of a moving trajectory are different questions.

Stabilizer A muscle that provides stability to a joint or body part. The rotator cuff stabilizes the shoulder.

Stance The position of the feet at address. Stance width and position affect balance, weight transfer, and rotation.

State Variables sufficient to determine model evolution given future input and disturbances. Positions and velocities may need muscle activation, tendon or shaft states, and contact mode; measured kinematics alone may be incomplete.

Static Not moving or changing with time. Static stretching holds a stretch without movement.

Stiffness A local relation between load change and deformation change, such as $k = dF/d\delta$. It may be nonlinear, directional or history-dependent. Structural stiffness, material modulus and active joint impedance are distinct.

Stimulus A physical or sensory input that elicits a response. Proprioceptive stimuli inform the nervous system of body state.

Straightness A geometric description relative to a line or target. A straight ball flight and a planar or smooth club path are different properties.

Strain A dimensionless deformation measure relative to a reference configuration. The relation $\epsilon = \sigma/E$ applies to a uniaxial linear-elastic model, not to every tissue, loading direction or strain magnitude.

Strain rate The time derivative of strain. Material response may depend on this rate and loading history; the direction and size of a rate effect require a specified tissue and protocol.

Strength A maximum load or force capacity under a specified test. Muscle strength, structural failure load and material failure stress are different quantities.

Stress Internal force per area represented generally by a tensor. Uniform uniaxial stress is F/A; local multiaxial tissue stress cannot usually be inferred from a net joint moment alone.

Stretch An increase in a specified tissue or muscle-tendon length, or a mobility exercise. Changes in range of motion do not uniquely identify a material mechanism or establish injury prevention.

Stretch-shorten cycle Active lengthening followed by shortening. Performance effects depend on activation, tendon storage, timing and comparison conditions. A kinematic stretch alone does not quantify stored energy or guarantee enhanced power.

Striations Regular patterns of light and dark bands visible in skeletal muscle under a microscope. Striations result from the organized arrangement of contractile proteins.

Stroke In golf rules, an intentional forward movement of the club to strike the ball, subject to the governing definition and exceptions. It need not be a full swing or make contact; scoring also includes applicable penalties [United States Golf Association, 2026b].

Structural Relating to component arrangement and physical properties. These constrain possible motion but do not alone determine the motion actually selected.

Structure The arrangement of components and connections. Modeling choices about joint freedom, contact and deformation determine the mathematical representation.

Subscapularis A rotator cuff muscle contributing to internal rotation and stabilization of the humeral head. Its activity depends on the task and individual.

Substrate The material or base on which something rests. The turf is the substrate for the golfer's swing.

Supination Forearm rotation at the radioulnar joints that brings the radius and ulna toward their uncrossed anatomical relationship. Palm-up describes one arm posture, not a frame-independent definition.

Supinator A forearm muscle contributing to supination. Its mechanical contribution depends on posture and coactivation; it is not assigned a universal late-downswing activation.

Supraspinatus A rotator cuff muscle contributing to arm elevation and glenohumeral stabilization. Anatomical function alone does not determine pathology or swing-phase activation.

Surface drag The skin-friction part of fluid drag, distinct from pressure drag. Total aerodynamic loading depends on relative air velocity, shape and flow regime; neglect requires an estimate appropriate to the body or club studied.

Sway A coaching description of lateral translation, often of the pelvis or trunk. Define the segment, reference direction and time interval before evaluating its effect.

Swing The complete motion of the golfer with the club, from address through follow-through.

Swing plane A plane fitted to or defined from selected swing geometry. A full club trajectory can change plane and deviate from planarity; an on-plane label needs an explicit reference and tolerance.

Swing sequence The order of declared events, such as motion onset, peak angular speed, peak power or EMG onset. These are not interchangeable. A pelvis-to-trunk-to-arm peak sequence does not by itself prove optimality or causal energy transfer.

Swing speed See clubhead speed.

Symmetry In mechanics, a transformation leaving the relevant model or action unchanged. Continuous symmetries can imply conservation laws under suitable conditions; visual bilateral similarity alone is not such a proof.

Symptom An experience reported by a person, such as pain. A clinical sign is an observed finding. Pain may accompany tissue injury but does not establish its presence, severity or mechanism [International Association for the Study of Pain, 2020].

Synapse A junction through which a neuron communicates with another cell, chemically or electrically. Changes in synaptic function are among the processes involved in learning.

Synchronization Coordination in time. Distinguish simultaneous events from a useful relative timing pattern; optimal timing depends on the task and constraints.

Synergy Coordinated action, or a low-dimensional pattern used to describe muscle activity. A fitted synergy can summarize data without proving that the nervous system commands exactly those basis patterns.

Synthesis Combining findings into a coherent explanation while preserving their assumptions and evidential limits.

System An organized collection of components working together. The golf swing is a biomechanical system.

Systemic Affecting the system broadly. A whole-body effect or training benefit requires evidence rather than following from the adjective.

T

Tangent A local first-order direction along a smooth curve. A tangent line can cross the curve, as at an inflection point. At nonzero speed, the velocity vector lies in the trajectory's tangent direction.

Tangential acceleration For a smooth path at nonzero speed, the component along its unit tangent, equal to the time derivative of speed. It can increase or decrease speed. The normal component changes direction.

Task A specified goal and success criterion. Golf tasks can prioritize scoring, launch conditions or dispersion; minimizing immediate distance to the hole is not always the best scoring strategy.

Task-space The space of selected outputs, such as clubhead position, orientation or launch variables. Its dimension, units and metric depend on the task; it need not include every end-effector variable.

Telemetry Transmission of measured data from a remote source. Radar sensing is a measurement method, not by itself telemetry; an instrument may separately transmit its results.

Tendon Connective tissue transmitting force from muscle toward its attachment. Its recoverable energy and stiffness depend on structure, loading and history. Different tendons serve different mechanical roles.

Tensile strength The maximum tensile stress sustained under a specified test. Specimen region, direction, strain rate and area definition matter; a universal tendon-versus-fascia ranking is unsupported.

Tension A pulling force applied to a material. Tension in muscles and tendons transmits force.

Teres minor A rotator cuff muscle contributing to external rotation and glenohumeral stabilization.

Test An experiment or trial to measure performance or understand a system. Swing tests using launch monitors provide data for analysis.

Thigh The portion of the leg between the hip and knee. The thigh muscles (quadriceps and hamstrings) are large and powerful.

Thoracic spine The spinal region usually containing 12 vertebrae and articulating with the ribs. Trunk rotation is distributed across this region and other joints.

Thoracolumbar fascia A connective-tissue complex in the lower trunk, linked to several muscles and skeletal attachments. Its geometry can transmit forces; its quantitative contribution to a swing requires a model or measurement.

Threading Passing one object through another. Threading a shot between trees requires precision.

Threshold A specified boundary for an event or response. A pain threshold concerns the onset of a painful experience under a test; it is not a tissue-damage threshold [International Association for the Study of Pain, 2020].

Thrust A pushing force along a declared direction. Ground reaction has several components and should not be reduced to universal forward propulsion.

Tibia The larger, medial lower-leg bone, transmitting most axial load between knee and ankle. The fibula also contributes to the lower leg's structure and load sharing.

Timing The relative time of declared events. Wrist geometry, joint moment, shaft recoil and peak clubhead speed are separate events whose relationships can be investigated.

Tissue A collection of similar cells and their extracellular matrix. Muscle tissue, bone tissue, and connective tissue all make up the body.

Toe The clubhead region opposite the heel/hosel. An off-center strike changes impact mechanics, including head rotation and potentially gear effect; its outcome depends on the design and contact location.

Topline The upper clubhead edge visible at address. Its appearance alone does not determine launch angle.

Torque A moment, measured in newton-metres (N m). A point force contributes $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$ about a declared origin; a force couple can create a pure moment. Torque is not power.

Torso The trunk, including chest, abdomen and back. Its mechanical contribution must be evaluated with the pelvis, arms, contacts and club rather than designated the universal primary driver.

Total energy The sum of the energy forms included in a declared system. Mechanical $T+U$ is conserved only when the net power crossing the boundary and nonconservative conversion vanish and the model has no unaccounted explicit time dependence.

Total momentum See linear momentum.

Trajectory A time-parameterized evolution of position or state. A geometric path alone omits timing; ball trajectories also depend on launch speed, environment and aerodynamic behavior.

Transition The interval around the change from backswing to downswing. Different segments can reverse at different times, so a single event definition must identify the measured quantity.

Translation Rigid-body motion with unchanged orientation. Any rigid-body motion can also be decomposed into a chosen point's translation plus rotation; center-of-mass translation can coexist with rotation.

Transmitter Something that sends or conveys something else. Tendons are transmitters of muscle force to bone.

Transverse plane An anatomical plane dividing superior and inferior portions. It is horizontal in the standard upright anatomical position; a moving segment's local plane need not remain horizontal.

Trapezius A muscle with upper, middle and lower regions contributing differently to scapular elevation, retraction, depression and rotation. Activity depends on the movement and load.

Trial A single repetition of a test or swing. Learning occurs over many trials.

Triangular ligament A shape-based description that needs an anatomical name to be specific. The wrist's triangular fibrocartilage complex contains several structures and should not be treated as one generic triangular ligament.

Triceps The triceps brachii, contributing to elbow extension; its long head also crosses the shoulder. Its mechanical role is not confined to follow-through.

Triggering Initiation of an event. The first observed downswing movement does not alone identify the neural command or mechanical cause initiating the other segments.

Trunk See torso.

Trust Confidence in one's ability to execute a task. It is a psychological concept rather than a mechanical measure of reliability.

Turf Grass and soil surface on the golf course. Turf conditions affect lie and ball contact.

Turnover An informal description of changing forearm, hand or clubface orientation. Specify lead/trail side and joint axes; it is not universally forearm pronation.

Twist Rotation or deformation about a structure's longitudinal direction. Rigid rotation and material torsional strain are distinct.

Twisting moment Torque about the long axis of a structure, causing torsion. Twisting moments on the shaft cause torsional deformation.

Two-plane motion A coaching term requiring an explicit definition, often comparing planes of different segments or phases. It does not mean a two-dimensional pendulum. Spatial motion can involve multiple changing planes.

Type I fiber A relatively slow-contracting skeletal-muscle fiber type with substantial oxidative capacity and fatigue resistance. Force depends on fiber size and state; slow does not mean incapable of high tension.

Type II fiber A relatively fast-contracting skeletal-muscle fiber category with subtypes differing in metabolism and fatigue resistance. Its contribution depends on recruitment, size and contraction conditions.

U

Ulna The forearm bone on the little-finger side, prominent proximally at the elbow. It should not be described simply as the larger forearm bone; the radius is broader distally.

Ulnar deviation Wrist motion toward the little-finger side, distinct from palmar flexion. Its relationship to club release depends on hand, grip and three-dimensional motion.

Ultrasound Sound waves with frequency higher than human hearing, used to image tissues. Ultrasound can assess muscle and tendon status.

Undulation Wavelike motion or surface variation. Green undulation affects the path and speed of putts.

Unit A defined measurement standard. SI unit names include newton and watt, with symbols N and W; consistent units are required in power and energy comparisons.

Upper body The arms, shoulders, and torso, as opposed to the lower body (legs).

Upswing See backswing.

Urethane In golf equipment, commonly a polyurethane cover material. Performance depends on the full ball construction and contact conditions rather than the cover name alone.

Usage pattern The distribution of use across tasks, equipment or time. A claim about a golfer's pattern requires recorded activity.

V

Velocity The vector time derivative of position in a reference frame. Its magnitude is speed; both magnitude and direction matter at impact.

Vertebra One of the bones of the spinal column. The usual counts are seven cervical, twelve thoracic and five lumbar vertebrae, with sacral and coccygeal vertebrae below.

Vertical The local direction along gravity, with upward opposite to gravitational acceleration. It is not generally perpendicular to sloping ground.

Vibration Oscillatory motion about an equilibrium position. Shaft vibration can occur even after impact.

Vigorous An informal description of energetic effort. It does not specify maximal effort or a measurable mechanical intensity.

Viscoelastic Exhibiting time-dependent mechanical response with both recoverable storage and dissipation. Loading history, rate and relaxation matter when estimating tissue or shaft energy return.

Viscosity A constitutive relation between fluid stress and deformation rate; a Newtonian fluid uses a constant dynamic viscosity. Soft-tissue viscoelasticity is broader than a single fluid viscosity.

Visible Observable by sight. Forces and muscle activity usually require inference or instrumentation; visual appearance alone does not identify them.

Vision The sensory process of sight, distinct from a mental image or goal. Visual information can guide preparation and later responses even when a current event cannot be corrected before impact.

Visual feedback Information obtained through sight and used to update action or learning. Seeing an event and having enough sensory-motor time to change the current strike are different questions.

Vital Essential or critical. The downswing is vital to swing success.

Vitality An informal description of energy or wellbeing, not a defined mechanical or physiological measurement.

Volume The amount of space occupied by an object, measured in cubic meters or liters.

Voluntary movement Goal-directed movement involving voluntary control, often with substantial automatic and feedback processes. A late command's origin can precede the final downswing; short remaining time does not make the motion biologically passive.

Vortex A swirling motion or region of rotational flow. Air vortices around the ball affect its flight.

Vulnerable Susceptible to harm under specified conditions. Quantifying injury risk requires population, exposure and outcome evidence.

W

Walk A gait usually characterized by alternating support with periods of double support. Its energy cost and pace depend on the person, terrain and carried load.

Warmth A thermal sensation or elevated temperature. Tissue temperature can influence behavior, but feeling warm does not establish protection from injury.

Water The compound H_2O, a major tissue constituent. Tissue water fraction varies by structure and measurement basis; a universal 70 percent value is inappropriate.

Wave An oscillation or disturbance propagating through a medium. Waves in the shaft propagate along its length.

Weakness Reduced force capacity relative to a reference and task. It may affect function, but a single strength measurement does not determine injury risk or swing quality.

Weather Atmospheric conditions including wind, rain, and temperature. Weather affects ball flight and playing conditions.

Weight Gravitational force, approximately $m\mathbf{g}$ near Earth's surface. It differs from mass and from the time-varying ground reaction during acceleration.

Weight transfer A coaching term that may refer to changing foot loads, center-of-pressure motion or center-of-mass motion. These are distinct measurements; declare which is meant.

Weighted Biased or emphasized. A weighted average gives more importance to certain values.

Wellness Overall health and wellbeing. Golf can contribute to wellness through exercise and outdoor activity.

Whip An analogy for rapid motion in a linked or flexible system. A flexible shaft's phase and energy exchange must be measured or simulated rather than assumed to straighten at the ideal instant.

Wobble An informal description of oscillation or orientation variation. Oscillation can be stable; the word alone does not diagnose instability or poor performance.

Work Energy transfer through force acting along displacement, $W = \int \mathbf{F} \cdot d\mathbf{r}$, with the corresponding moment-rotation contribution for rigid bodies. Mechanical work and metabolic energy expenditure are different.

Work-energy theorem The change in kinetic energy equals the total mechanical work on the modeled particles or bodies. Clubhead energy can reflect work by interface forces, gravity and other interactions, not only an isolated muscle-work term.

Workload An exposure or effort measure defined by its protocol, such as repetition count, duration or mechanical work. It is not generally one quantity measured in joules.

Workshop A place for learning and practice. A swing workshop is where golfers improve with instruction.

World record A best verified performance under a category's rules and measurement conditions. It does not establish an absolute physiological limit.

Worst-case scenario The most adverse case within a declared uncertainty set and objective. A worst case is meaningful only relative to those bounds.

Wound-up A coaching metaphor for the configuration at the top of the backswing. Rotation or muscular tension alone does not quantify elastic energy available for recovery.

Wrap-around Something that wraps around another object. The fascia wraps around muscles.

Wrapping In composite manufacturing, laying reinforcement around a mandrel or form. In anatomy, a muscle-tendon path can wrap around a structure, changing its force direction and moment arm.

Wrench The combined moment and force acting on a rigid body about a declared origin. With moment-first ordering, $\mathcal{W} = [\boldsymbol{\tau}; \mathbf{F}]$ is dual to $\mathcal{V} = [\boldsymbol{\omega}; \mathbf{v}]$, and power is $\mathcal{W}^\mathsf{T}\mathcal{V}$. Origin, frame and ordering must match.

Wrist A joint complex permitting flexion/extension and radial/ulnar deviation with coupled carpal motion. Forearm pronation/supination is primarily radioulnar rotation, not a third interchangeable wrist hinge.

Wrist cock An informal description of forearm-club geometry, often associated with radial deviation of the lead wrist. It is not synonymous with wrist flexion or extension. Grip and three-dimensional conventions matter; the angle alone does not measure tendon or shaft energy [Carson et al., 2019].

Wrist release An informal description of changing forearm-club geometry during the downswing. It may involve wrist deviation, flexion/extension, forearm rotation and club motion. A measured angular change does not alone identify its muscular cause or elastic energy source [Carson et al., 2019].

X

X-factor A measure of relative pelvis and upper-trunk orientation using declared axes, projection and swing event. Different two- and three-dimensional definitions can yield different values. It measures geometry, not torso torque or stored energy, and a larger value is not universally better.

X-rays Electromagnetic radiation used to image bones. X-rays can reveal fractures or structural damage.

Y

Yard A unit of length exactly equal to 3 feet or 0.9144 metres.

Yardage Distance measured in yards. A golfer needs to know the yardage to the green to select the right club.

Yell A loud shout, sometimes used to psyche up for a swing. Some golfers use vocal cues to trigger execution.

Yield To give way or deform under load. Materials yield plastically (permanently deform) above the yield strength.

Young's modulus See elastic modulus.

Z

Zenith The highest point in a declared spatial direction. A segment's highest position need not coincide with its reversal event or with zero velocity of every other segment.

Zero A value defined for a particular quantity and convention. Zero declared control selects the model's drift value, which may itself be zero; it does not establish drift dominance or absence of muscle activity.

Zero-Torque Counterfactual (ZTCF) family Model interventions that set the declared applied generalized-control channel to zero while retaining the declared effective plant and protocol-specified loads. The family contains pointwise samples, stitched pointwise traces, forward trajectories, and achieved-state branched trajectories. Zero declared control does not imply zero muscle activation.

Zone A region defined by explicit boundaries. A comfort zone is subjective and should not be confused with a mechanically verified stability region.

ZVCF The instantaneous Zero-Velocity Counterfactual acceleration evaluated at the current configuration and declared internal state with both velocity and declared control set to zero. It is neither a state nor a released trajectory.

Zygapophysial joint See facet joint.

Bibliography

M.A. Adams and W.C. Hutton. The mechanics of prolapsed intervertebral disc. *International Orthopaedics*, 6(4):249–253, 1982. doi: 10.1007/bf00267146. URL https://doi.org/10.1007/bf00267146.

Michael A. Adams, Nikolai Bogduk, Kim Burton, and Patricia Dolan. *The Biomechanics of Back Pain*. Churchill Livingstone, Edinburgh, 1st edition, 2002. Comprehensive reference on spinal biomechanics. Reports that lumbar flexion beyond 40–50 degrees substantially increases intradiscal pressure and posterior annulus stress, with loading increases up to 300% compared to neutral posture in cadaveric studies.

Charles H. Adler, Debra Crews, Kanav Kahol, Marco Santello, Brie Noble, Joseph G. Hentz, and John N. Caviness. Are the yips a task-specific dystonia or "golfer's cramp"? *Movement Disorders*, 26(11):1993–1996, 2011. doi: 10.1002/mds.23824.

Kazuo Arakawa, Toshio Mada, Haruo Komatsu, Tetsuo Shimizu, Masanori Satou, Kohsei Takehara, and Goji Etoh. Dynamic deformation measurement of a golf ball during normal impact. *Transactions of the Japan Society of Mechanical Engineers Series C*, 72(724): 3826–3831, 2006. doi: 10.1299/kikaic.72.3826.

F. A. C. Azevedo, L. R. B. Carvalho, L. T. Grinberg, J. M. Farfel, R. E. L. Ferretti, R. E. P. Leite, W. F. Jacob, R. Lent, and S. Herculano-Houzel. Equal numbers of neuronal and nonneuronal cells make the human brain an isometrically scaled-up primate brain. *Journal of Comparative Neurology*, 513(5):532–541, 2009. doi: 10.1002/cne.21974. URL https://doi.org/10.1002/cne.21974.

Sian L. Beilock and Thomas H. Carr. On the fragility of skilled performance: What governs choking under pressure? *Journal of Experimental Psychology: General*, 130(4):701–725, 2001. doi: 10.1037/0096-3445.130.4.701.

Nikolai A. Bernstein. *The Co-ordination and Regulation of Movements*. Pergamon Press, Oxford, 1967.

J. Gordon Betts, Kelly A. Young, James A. Wise, Eddie Johnson, Brandon Poe, Dean H. Kruse, Oksana Korol, Jody E. Johnson, Mark Womble, and Peter DeSaix. *Anatomy and Physiology 2e*. OpenStax, Rice University, 2022. URL https://openstax.org/books/anatomy-and-physiology-2e/pages/7-3-the-vertebral-column. Section 7.3: The Vertebral Column.

Anthony M. Bloch, P. S. Krishnaprasad, Jerrold E. Marsden, and Tudor S. Ratiu. Dissipation induced instabilities. *Annales de l'Institut Henri Poincaré. C, Analyse non linéaire*, 11(1):37–90, 1994. URL https://www.numdam.org/item/AIHPC_1994__11_1_37_0/.

Kari Bø and Margaret Sherburn. Evaluation of female pelvic-floor muscle function and strength. *Physical Therapy*, 85(3):269–282, 2005. doi: 10.1093/ptj/85.3.269. Review of methods for evaluating pelvic-floor function and strength, including measurement reliability and validity. It does not establish chronic golf-pressure injury causation.

Lorenza Bonaldi, Alice Berardo, Carmelo Pirri, Carla Stecco, Emanuele Luigi Carniel, and Chiara Giulia Fontanella. Mechanical characterization of human fascia lata: Uniaxial tensile tests from fresh-frozen cadaver samples and constitutive modelling. *Bioengineering*, 10(2):226, 2023. doi: 10.3390/bioengineering10020226.

Frans Bosch. *Anatomy of Agility: movement analysis in sport*. 20/10 Publishers, 2020. ISBN 978-94-90951-59-7.

Frans Bosch and Keith Cook. *Strength Training and Coordination: An Integrative Approach*. 2010 Publishers, 2015.

Francesco Bullo and Andrew D. Lewis. *Geometric Control of Mechanical Systems*. Springer, New York, 2004. Comprehensive treatment of geometric methods in mechanical control systems, including control-affine structure of Lagrangian systems.

Etienne Burdet, Rieko Osu, David W. Franklin, Theodore E. Milner, and Mitsuo Kawato. The central nervous system stabilizes unstable dynamics by learning optimal impedance. *Nature*, 414(6862):446–449, 2001. doi: 10.1038/35106566.

Jan Cabri, Jorge P. Sousa, Miriam Kots, and João Barreiros. Golf-related injuries: A systematic review. *European Journal of Sport Science*, 9(6):353–366, 2009. doi: 10.1080/17461390903009141. Systematic review confirming the lower back as the most common injury site in golfers of all levels, with prevalence estimates of 15–34% depending on study population and methodology.

Carlos Canudas de Wit, Henrik Olsson, Karl Johan Åström, and Pablo Lischinsky. A new model for control of systems with friction. *IEEE Transactions on Automatic Control*, 40(3):419–425, 1995. doi: 10.1109/9.376053.

Howie J. Carson, Jim Richards, and Bruno Mazuquin. Examining the influence of grip type on wrist and club head kinematics during the golf swing: Benefits of a local co-ordinate system. *European Journal of Sport Science*, 19(3):327–335, 2019. doi: 10.1080/17461391.2018.1508504.

Elena J. Caruthers, Julie A. Thompson, Ajit M. W. Chaudhari, Laura C. Schmitt, Thomas M. Best, Katherine R. Saul, and Robert A. Siston. Muscle forces and their contributions to vertical and horizontal acceleration of the center of mass during sit-to-stand transfer in young, healthy adults. *Journal of Applied Biomechanics*, 32(5):487–503, 2016. Used 3D dynamic simulations with induced acceleration analysis to show that quadriceps oppose forward COM acceleration during sit-to-stand, while gluteus maximus and soleus are primary contributors to lifting and advancing the center of mass.

J. J. Cassidy, A. Hiltner, and E. Baer. Hierarchical structure of the intervertebral disc. *Connective Tissue Research*, 23(1):75–88, 1989. doi: 10.3109/03008208909103905. URL https://doi.org/10.3109/03008208909103905.

John H. Challis. An induced acceleration index for examining joint couplings. *Journal of Biomechanics*, 44(12):2320–2322, 2011. Introduced the induced acceleration index (IAI), a dimensionless quantity that captures the configuration-dependent coupling potential between joints. Applied to postural control, showing the ankle has over 12 times the ability to accelerate the hip versus the reverse.

P. J. Cheetham, P. E. Martin, R. E. Mottram, and B. F. St Laurent. The importance of stretching the X-Factor in the downswing of golf: The X-Factor Stretch. In P. R. Thomas, editor, *Optimising Performance in Golf*, pages 192–199. Australian Academic Press, Brisbane, Australia, 2001. ISBN 1875378375. URL https://www.philcheetham.com/wp-content/uploads/2011/11/Stretching-the-X-Factor-Paper.pdf.

Xavier Chiementin, Georges Kouroussis, Sébastien Murer, and Roger Serra. Experimental modal analysis of hand–arm vibration in golf: Influence of grip strength. *Applied Sciences*, 9(10):2050, 2019. doi: 10.3390/app9102050.

Andy Clark. *Whatever Next? Predictive Brains, Situated Agents, and the Future of Cognitive Science*. Oxford University Press, 2013.

Alastair J. J. Cochran and John Stobbs. *The Search for the Perfect Swing: The Man and the Science Behind the Ultimate Golf Swing*. J. B. Lippincott, Philadelphia, 1968.

Frédéric Crevecoeur, Jean-Louis Thonnard, Philippe Lefèvre, and Stephen H. Scott. Long-latency feedback coordinates upper-limb and hand muscles during object manipulation tasks. *eNeuro*, 3(1):ENEURO.0129–15.2016, 2016. doi: 10.1523/ENEURO.0129-15.2016.

Keith Davids, Chris Button, and Simon Bennett. *Dynamics of Skill Acquisition: A Constraints-Led Approach*. Human Kinetics, Champaign, IL, 2008.

Paolo De Leva. Adjustments to Zatsiorsky-Seluyanov's segment inertia parameters. *Journal of Biomechanics*, 29(9):1223–1230, 1996. doi: 10.1016/0021-9290(95)00178-6.

Scott L. Delp, J. P. Loan, M. G. Hoy, Felix E. Zajac, E. L. Topp, and J. M. Rosen. An interactive graphics-based model of the lower limb to study orthopaedic surgical procedures. *IEEE Transactions on Biomedical Engineering*, 37(8):757–767, 1990. doi: 10.1109/10.102791. Foundational musculoskeletal modeling work providing anatomical parameters and methodology for lower extremity models. Referenced for anatomical geometry and muscle moment arms.

Scott L. Delp, Frank C. Anderson, Ajay S. Arnold, Perry Loan, Ayman Habib, Chand T. John, Eran Guendelman, and Darryl G. Thelen. Opensim: Open-source software to create and analyze dynamic simulations of movement. *IEEE Transactions on Biomedical Engineering*, 54(11):1940–1950, 2007.

John C. Eccles. The cerebellum as a computer: Patterns in space and time. *The Journal of Physiology*, 229(1):1–32, 1973. doi: 10.1113/jphysiol.1973.sp010123. Seminal work establishing the cerebellum's role in motor coordination and timing. Describes how cerebellar circuitry processes spatial and temporal information to coordinate movement patterns.

Roger M. Enoka. *Neuromechanics of Human Movement*. Human Kinetics, Champaign, Illinois, 3rd edition, 2002. Comprehensive textbook integrating neuroscience and biomechanics, covering muscle physiology, neural control systems, and sensorimotor integration relevant to skilled movement tasks like the golf swing.

Roy Featherstone. *Rigid Body Dynamics Algorithms*. Springer, New York, 2008.

Paul M. Fitts and Michael I. Posner. *Human Performance*. Brooks/Cole, Belmont, CA, 1967.

J. Randall Flanagan, Philipp Vetter, Roland S. Johansson, and Daniel M. Wolpert. Prediction precedes control in motor learning. *Current Biology*, 13:146–150, 2003.

Karl Friston. The free-energy principle: A unified brain theory? *Nature Reviews Neuroscience*, 11:127–138, 2010.

J. Fuller, L.-J. Liu, M. C. Murphy, and R. W. Mann. A comparison of lower-extremity skeletal kinematics measured using skin- and pin-mounted markers. *Human Movement Science*, 16(2–3):219–242, 1997. doi: 10.1016/S0167-9457(96)00053-X.

Charles J. Gatt, Michael J. Pavol, Richard D. Parker, and Mark D. Grabiner. Three-dimensional knee joint kinetics during a golf swing. *American Journal of Sports Medicine*, 26(2):285–294, 1998.

Paul S. Glazier. Movement variability in the golf swing: Theoretical, methodological, and practical issues. *Research Quarterly for Exercise and Sport*, 82(2):157–161, 2011. doi: 10.1080/02701367.2011.10599742.

Danijar Hafner, Timothy Lillicrap, Jimmy Ba, and Mohammad Norouzi. Dream to control: Learning behaviors by latent imagination. In *International Conference on Learning Representations*, 2020. URL https://arxiv.org/abs/1912.01603.

Mark J. Haykowsky, J. Max Findlay, and Andrew P. Ignaszewski. Aneurysmal subarachnoid hemorrhage associated with weight training: Three case reports. *Clinical Journal of Sport Medicine*, 6(1):52–55, 1996. doi: 10.1097/00042752-199601000-00011. Three weight-training case reports discussing a possible relation to straining. They do not estimate golfer incidence or establish a preventive breathing intervention.

Elwood Henneman, George Somjen, and David O. Carpenter. Functional significance of cell size in spinal motoneurons. *Journal of Neurophysiology*, 28:560–580, 1965.

Erik Henrikson, Paul Wood, Chris Broadie, and Tom Nuttall. The role of friction and tangential compliance on the resultant launch angle of a golf ball. *Proceedings*, 49(1):27, 2020. doi: 10.3390/proceedings2020049027.

Archibald Vivian Hill. The heat of shortening and the dynamic constants of muscle. *Proceedings of the Royal Society B*, 126:136–195, 1938.

Masaya Hirashima. Induced acceleration analysis of three-dimensional multi-joint movements and its application to sports movements. In *Theoretical Biomechanics*, pages 303–318. IntechOpen, 2011. Comprehensive tutorial on induced acceleration analysis for 3D multi-joint movements. Demonstrates posture-dependent muscle function and the distinction between instantaneous and cumulative effects with applications to baseball pitching.

Masaya Hirashima and Tatsuyuki Ohtsuki. Exploring the mechanism of skilled overarm throwing. *Exercise and Sport Sciences Reviews*, 36(4):205–211, 2008. Review extending induced acceleration analysis to 3D multi-DOF systems. Corrected erroneous methods in previous motor control studies and established the proper forward dynamics approach for determining muscle contributions to joint angular accelerations.

Masaya Hirashima, Kazutoshi Kudo, Koji Watarai, and Tatsuyuki Ohtsuki. Control of 3D limb dynamics in unconstrained overarm throws of different speeds performed by skilled baseball players. *Journal of Neurophysiology*, 97(1):680–691, 2007. Demonstrated that skilled throwers utilize interaction torques to generate large angular velocities at distal joints, with proximal muscles creating a dynamic foundation for the entire limb motion.

Masaya Hirashima, Katsu Yamane, Yoshihiko Nakamura, and Tatsuyuki Ohtsuki. Kinetic chain of overarm throwing in terms of joint rotations revealed by induced acceleration analysis. *Journal of Biomechanics*, 41(13):2874–2883, 2008. Applied induced acceleration analysis to baseball pitching using a 13-DOF model. Showed that distal joint rotations are primarily produced by velocity-dependent torques originating from proximal muscle activity, providing a clear picture of the kinetic chain.

Paul W. Hodges, Andrew G. Cresswell, Karl Daggfeldt, and Alf Thorstensson. In vivo measurement of the effect of intra-abdominal pressure on the human spine. *Journal of Biomechanics*, 34(3):347–353, 2001. doi: 10.1016/s0021-9290(00)00206-2. URL https://doi.org/10.1016/s0021-9290(00)00206-2.

Paul W. Hodges, A. E. Martin Eriksson, Debra Shirley, and Simon C. Gandevia. Intra-abdominal pressure increases stiffness of the lumbar spine. *Journal of Biomechanics*, 38(9):1873–1880, 2005. doi: 10.1016/j.jbiomech.2004.08.016. Phrenic-nerve stimulation and posteroanterior indentation in three prone subjects; the measured stiffness is not a golf-specific torsional coefficient.

Neville Hogan. Adaptive control of mechanical impedance by coactivation of antagonist muscles. *IEEE Transactions on Automatic Control*, 29:681–690, 1984.

Neville Hogan. The mechanics of multi-joint posture and movement control. *Biological Cybernetics*, 52:315–331, 1985a.

Neville Hogan. Impedance control: An approach to manipulation: Part i—theory. *Journal of Dynamic Systems, Measurement, and Control*, 107(1):1–7, 1985b. doi: 10.1115/1.3140702.

Katherine R. S. Holzbaur, Wendy M. Murray, and Scott L. Delp. A model of the upper extremity for simulating musculoskeletal surgery and analyzing neuromuscular control. *Annals of Biomedical Engineering*, 33(6):829–840, 2005. doi: 10.1007/s10439-005-3320-7. Fifteen degrees of freedom for a single upper extremity: shoulder, elbow, forearm, wrist, thumb and index finger.

Bernhard Hommel, Jochen Müsseler, Gisa Aschersleben, and Wolfgang Prinz. The theory of event coding (TEC): A framework for perception and action planning. *Behavioral and Brain Sciences*, 24:849–878, 2001.

Timothy M. Hosea, Charles J. Gatt, Kathleen M. Galli, Noshir A. Langrana, and Joseph P. Zawadsky. Biomechanical analysis of the golfer's back. *Science and Golf: Proceedings of the First World Scientific Congress of Golf*, pages 43–48, 1990. Early instrumented study reporting peak lumbar compressive loads of approximately 6–8 times body weight during the professional golf swing, comparable to loads during competitive rowing.

Patria A. Hume, Justin Keogh, and Duncan Reid. The role of biomechanics in maximising distance and accuracy of golf shots. *Sports Medicine*, 35:429–449, 2005.

James C. Iatridis, Lori A. Setton, Mark Weidenbaum, and Van C. Mow. The viscoelastic behavior of the non-degenerate human lumbar nucleus pulposus in shear. *Journal* *of Biomechanics*, 30(10):1005–1013, 1997. doi: 10.1016/s0021-9290(97)00069-9. URL https://doi.org/10.1016/s0021-9290(97)00069-9.

International Association for the Study of Pain. Terminology, 2020. URL https://www.iasp-pain.org/resources/terminology/. Revised pain definition and terminology; accessed 7 September 2026.

Alberto Isidori. *Nonlinear Control Systems*. Springer-Verlag, London, 3rd edition, 1995.

Theodore P. Jorgensen. On the dynamics of the swing of a golf club. *American Journal of Physics*, 38(5):644–651, 1970. doi: 10.1119/1.1976419.

Theodore P. Jorgensen. *The Physics of Golf*. Springer-Verlag, New York, 2nd edition, 1994a.

Theodore P. Jorgensen. Physics of golf. *American Journal of Physics*, 62(11):991–999, 1994b. Note: This entry may be a duplicate or related to Jorgensen1994 (the full book). Verify which specific work is cited in the text. If this is the journal article version distinct from the book, it provides detailed physics analysis of golf mechanics.

Eric R. Kandel, James H. Schwartz, Thomas M. Jessell, Steven A. Siegelbaum, and A. J. Hudspeth. *Principles of Neural Science*. McGraw-Hill, New York, 5th edition, 2013.

Mitsuo Kawato. Internal models for motor control and trajectory planning. *Current Opinion in Neurobiology*, 9:718–727, 1999.

J. A. Scott Kelso. *Dynamic Patterns: The Self-Organization of Brain and Behavior*. MIT Press, Cambridge, MA, 1995.

Tomoko Kitago and John W Krakauer. Motor learning principles for neurorehabilitation. *Handbook of Clinical Neurology*, 110:93–103, 2013. doi: 10.1016/b978-0-444-52901-5.00008-3.

Isaac L. Kurtzer. Long-latency reflexes account for limb biomechanics through several supraspinal pathways. *Frontiers in Integrative Neuroscience*, 8:99, 2014. doi: 10.3389/fnint.2014.00099.

Jason J. Kutch and Francisco J. Valero-Cuevas. Challenges and new approaches to proving the existence of muscle synergies of neural origin. *PLoS Computational Biology*, 8(5): e1002434, 2012. doi: 10.1371/journal.pcbi.1002434.

Young-Tae Lim, John W. Chow, and Woen-Sik Chae. Lumbar spinal loads and muscle activity during a golf swing. *Sports Biomechanics*, 11(2):197–211, 2012. doi: 10.1080/14763141.2012.670662. URL https://doi.org/10.1080/14763141.2012.670662.

Sasho James MacKenzie and Eric J. Sprigings. A three-dimensional forward dynamics model of the golf swing. *Sports Engineering*, 11(3):165–175, 2009a. doi: 10.1007/s12283-009-0020-9. URL https://link.springer.com/article/10.1007/s12283-009-0020-9.

Sasho James MacKenzie and Eric J. Sprigings. Understanding the role of shaft stiffness in the golf swing. *Sports Engineering*, 12:13–19, 2009b. doi: 10.1007/s12283-009-0028-1.

Sasho James MacKenzie and Eric J. Sprigings. Understanding the mechanisms of shaft deflection in the golf swing. *Sports Engineering*, 12:69–75, 2010. doi: 10.1007/s12283-010-0034-3.

Pietro Mazzoni and John W. Krakauer. An implicit plan overrides an explicit strategy during visuomotor adaptation. *The Journal of Neuroscience*, 26(14):3642–3645, 2006. doi: 10.1523/jneurosci.5317-05.2006.

Stuart M. McGill. *Low Back Disorders: Evidence-Based Prevention and Rehabilitation*. Human Kinetics, Champaign, IL, 2nd edition, 2007.

Andrew McHardy, Henry Pollard, and Kai Luo. Golf injuries: A review of the literature. *Sports Medicine*, 36(2):171–187, 2006. doi: 10.2165/00007256-200636020-00006. Systematic review of golf injury epidemiology. Reports that the lower back is the most commonly injured region (25–36% of all golf injuries in amateurs; 22–24% in professionals).

Andrew McHardy, Henry Pollard, and Kehui Luo. One-year follow-up study on golf injuries in australian amateur golfers. *The American Journal of Sports Medicine*, 35(8): 1354–1360, 2007a. doi: 10.1177/0363546507300188. URL https://doi.org/10.1177/0363546507300188.

Andrew J. McHardy, Henry P. Pollard, and Kehui Luo. Golf-related lower back injuries: an epidemiological survey. *Journal of Chiropractic Medicine*, 6(1):20–26, 2007b. doi: 10.1016/j.jcme.2007.02.010. URL https://doi.org/10.1016/j.jcme.2007.02.010.

M. McTeigue, S. R. Lamb, R. Mottram, and F. Pirozzolo. Spine and hip motion analysis during the golf swing. In A. J. Cochran and M. R. Farrally, editors, *Science and Golf II: Proceedings of the World Scientific Congress of Golf*, pages 50–58. E & FN Spon, 1994. doi: 10.4324/9780203474709-9.

Matthew Millard, Thomas Uchida, Ajay Seth, and Scott L. Delp. Flexing computational muscle: Modeling and simulation of musculotendon dynamics. *Journal of Biomechanical Engineering*, 135(2):021005, 2013. doi: 10.1115/1.4023390.

Richard M. Murray, Zexiang Li, and S. Shankar Sastry. *A Mathematical Introduction to Robotic Manipulation*. CRC Press, Boca Raton, FL, 1994.

Joseph Myers, Scott Lephart, Yung-Shen Tsai, Timothy Sell, James Smoliga, and John Jolly. The role of upper torso and pelvis rotation in driving performance during the golf swing. *Journal of Sports Sciences*, 26(2):181–188, 2008. doi: 10.1080/02640410701373543.

A. L. Nachemson, B. J. Andersson, and A. B. Schultz. Valsalva maneuver biomechanics: Effects on lumbar trunk loads of elevated intraabdominal pressures. *Spine*, 11(5): 476–479, 1986. URL https://pubmed.ncbi.nlm.nih.gov/3750086/. Four subjects and five isometric tasks; intradiscal pressure, intra-abdominal pressure and muscle activity measurements.

Richard R. Neptune, Steven A. Kautz, and Felix E. Zajac. Contributions of the individual ankle plantar flexors to support, forward progression and swing initiation during walking. *Journal of Biomechanics*, 34(11):1387–1398, 2001. Applied induced acceleration and power analysis to walking simulation, revealing that soleus and gastrocnemius have distinct functional roles despite both being ankle plantar flexors.

Steven M. Nesbit. A three dimensional kinematic and kinetic study of the golf swing. *Journal of Sports Science and Medicine*, 4:499–519, 2005. URL https://www.jssm.org/volume04/iss4/cap/jssm-04-499.pdf.

Steven M. Nesbit and Monika Serrano. Work and power analysis of the golf swing. *Journal of Sports Science and Medicine*, 4:520–533, 2005. URL https://www.jssm.org/volume04/iss4/cap/jssm-04-520.pdf.

Donald A. Neumann. *Kinesiology of the Musculoskeletal System: Foundations for Rehabilitation*. Elsevier, St. Louis, MO, 3rd edition, 2017.

Antoine Nordez, Thomas Gallot, Stefan Catheline, Arnaud Guével, Christophe Cornu, and François Hug. Electromechanical delay revisited using very high frame rate ultrasound. *Journal of Applied Physiology*, 106(6):1970–1975, 2009. doi: 10.1152/japplphysiol.00221.2009.

Margareta Nordin and Victor H. Frankel, editors. *Basic Biomechanics of the Musculoskeletal System*. Wolters Kluwer, Philadelphia, PA, 4th edition, 2012.

OpenSim. First-order activation dynamics, 2020. URL https://opensimconfluence.atlassian.net/wiki/spaces/OpenSim/pages/53090590/First-Order+Activation+Dynamics. Official documentation, updated 17 April 2020; accessed 9 September 2026.

OpenStax. Anatomy and physiology 2e: Muscle fiber contraction and relaxation, 2022a. URL https://openstax.org/books/anatomy-and-physiology-2e/pages/10-3-muscle-fiber-contraction-and-relaxation. Section 10.3; accessed 9 September 2026.

OpenStax. Anatomy and physiology 2e: Types of muscle fibers, 2022b. URL https://openstax.org/books/anatomy-and-physiology-2e/pages/10-5-types-of-muscle-fibers. Section 10.5; accessed 9 September 2026.

OpenStax. Anatomy and physiology 2e: Nervous system control of muscle tension, 2022c. URL https://openstax.org/books/anatomy-and-physiology-2e/pages/10-4-nervous-system-control-of-muscle-tension. Section 10.4; accessed 7 September 2026.

Matthew T. G. Pain and John H. Challis. The influence of soft tissue movement on ground reaction forces, joint torques and joint reaction forces in drop landings. *Journal of Biomechanics*, 39(1):119–124, 2006. doi: 10.1016/j.jbiomech.2004.10.036. Two heel-first landings by one participant, with calibrated model parameters and separate controlled-impact comparisons; not a golf validation.

A. Raymond Penner. The physics of golf. *Reports on Progress in Physics*, 66(2):131–171, 2003a. doi: 10.1088/0034-4885/66/2/202.

Andrew R. Penner. Physics and golf: The interaction between science and sport. *Canadian Journal of Physics*, 81(9):899–910, 2003b.

Marilyn Pink, Frank W. Jobe, and Jacquelin Perry. Electromyographic analysis of the shoulder during the golf swing. *The American Journal of Sports Medicine*, 18(2):137–140, 1990. doi: 10.1177/036354659001800205.

Jared M Porter and Richard A Magill. Systematically increasing contextual interference is beneficial for learning sport skills. *Journal of Sports Sciences*, 28(12):1277–1285, 2010. doi: 10.1080/02640414.2010.502946.

Wolfgang Prinz. Perception and action planning. *European Journal of Cognitive Psychology*, 9(2):129–154, 1997. doi: 10.1080/713752551.

Uwe Proske and Simon C. Gandevia. The proprioceptive senses: Their roles in signaling body shape, body position and movement, and muscle force. *Physiological Reviews*, 92 (4):1651–1697, 2012. doi: 10.1152/physrev.00048.2011.

J. Andrew Pruszynski and Stephen H. Scott. Optimal feedback control and the long-latency stretch response. *Experimental Brain Research*, 218(3):341–359, 2012. doi: 10.1007/s00221-012-3041-8.

J. Andrew Pruszynski, Isaac Kurtzer, Joseph Y. Nashed, Mohsen Omrani, Brenda Brouwer, and Stephen H. Scott. Primary motor cortex underlies multi-joint integration for fast feedback control. *Nature*, 478:387–390, 2011. doi: 10.1038/nature10436.

P. M. H. Rack and D. R. Westbury. The short range stiffness of active mammalian muscle and its effect on mechanical properties. *The Journal of Physiology*, 240(2):331–350, 1974. doi: 10.1113/jphysiol.1974.sp010613.

Apoorva Rajagopal, Christopher L. Dembia, Matthew S. DeMers, Denny D. Delp, Jennifer L. Hicks, and Scott L. Delp. Full-body musculoskeletal model for muscle-driven simulation of human gait. *IEEE Transactions on Biomedical Engineering*, 63(10):2068–2079, 2016. doi: 10.1109/TBME.2016.2586891. Gait model with lower-limb muscle-tendon actuators and ideal upper-body torque actuators; not a source of forearm muscle parameters.

Patrick O. Riley and D. Casey Kerrigan. Kinetics of stiff-legged gait: Induced acceleration analysis. *IEEE Transactions on Rehabilitation Engineering*, 7(4):420–426, 1999. Applied induced acceleration analysis to stiff-legged gait in stroke patients, demonstrating that the causes of reduced knee flexion vary between patients and involve hip and ankle impairments, not just knee spasticity.

Thomas J. Roberts and Annette M. Gabaldón. Interpreting muscle function from EMG: lessons learned from direct measurements of muscle force. *Integrative and Comparative Biology*, 48(2):312–320, 2008. doi: 10.1093/icb/icn056.

Patrick Gordon Robinson, Benjamin Clarsen, Andrew Murray, Astrid Junge, Margo Lynn Mountjoy, Tomas Drobny, Lance Gill, Francois Gazzano, Mike Voight, and Jiri Dvorak. A prospective study of injuries and illnesses among 910 amateur golfers during one season. *BMJ Open Sport & Exercise Medicine*, 10:e001844, 2024. doi: 10.1136/bmjsem-2023-001844.

David Roylance. Laminated composite plates. Technical report, Massachusetts Institute of Technology, February 2000. URL `https://ocw.mit.edu/courses/3-91-mechanical-behavior-of-plastics-spring-2007/resources/15_laminates/`. Dated February 10, 2000; distributed in the Spring 2007 MIT OpenCourseWare course.

Jeffrey A. Saunders and David C. Knill. Humans use continuous visual feedback from the hand to control fast reaching movements. *Experimental Brain Research*, 152(3):341–352, 2003. doi: 10.1007/s00221-003-1525-2.

Richard A. Schmidt. A schema theory of discrete motor skill learning. *Psychological Review*, 82:225–260, 1975.

Wolfgang Immanuel Schöllhorn. Invited commentary: Differential learning is different from contextual interference learning. *Human Movement Science*, 47:240–245, 2016. doi: 10.1016/j.humov.2015.11.018.

John P. Scholz and Gregor Schöner. The uncontrolled manifold concept: Identifying control variables for a functional task. *Experimental Brain Research*, 126(3):289–306, 1999. doi: 10.1007/s002210050738.

Martijn Schonewille, Zhenyu Gao, Henk-Jan Boele, Maria F Vinueza Veloz, Wardell E Amerika, Antonia A M Simek, Marcel T De Jeu, Jordan P Steinberg, Kogo Takamiya, Freek E Hoebeek, David J Linden, Richard L Huganir, and Chris I De Zeeuw. Reevaluating the role of LTD in cerebellar motor learning. *Neuron*, 70(1):43–50, 2011. doi: 10.1016/j.neuron.2011.02.044.

Wolfram Schultz, Peter Dayan, and P. Read Montague. A neural substrate of prediction and reward. *Science*, 275:1593–1599, 1997.

Lisa M. Schutte, Mary M. Rodgers, Felix E. Zajac, and Roger M. Glaser. Improving the efficacy of electrical stimulation-induced leg cycle ergometry: An analysis based on a dynamic musculoskeletal model. *IEEE Transactions on Rehabilitation Engineering*, 1(2):109–125, 1993. Used a dynamic musculoskeletal model to analyze electrically stimulated leg cycling for spinal cord injury rehabilitation. Demonstrated how seat configuration, loading, and stimulation timing affect muscle contributions to pedaling through induced acceleration analysis.

Reza Shadmehr and Steven P. Wise. *The Computational Neurobiology of Reaching and Pointing*. MIT Press, Cambridge, MA, 2005.

D. Shirley, P. W. Hodges, A. E. M. Eriksson, and S. C. Gandevia. Spinal stiffness changes throughout the respiratory cycle. *Journal of Applied Physiology*, 95(4):1467–1475, 2003. doi: 10.1152/japplphysiol.00939.2002. Composite posteroanterior spinal response during respiratory tasks in eight subjects.

Eric J. Sprigings and Robert J. Neal. An insight into the importance of wrist torque in driving the golfball: A simulation study. *Journal of Applied Biomechanics*, 16(4):356–366, 2000. doi: 10.1123/jab.16.4.356. URL `https://doi.org/10.1123/jab.16.4.356`.

Ian A.F. Stokes, Mack G. Gardner-Morse, and Sharon M. Henry. Intra-abdominal pressure and abdominal wall muscular function: Spinal unloading mechanism. *Clinical Biomechanics*, 25(9):859–866, 2010. doi: 10.1016/j.clinbiomech.2010.06.018. URL `https://doi.org/10.1016/j.clinbiomech.2010.06.018`.

Russ Tedrake. Multi-body dynamics. Underactuated Robotics. URL `https://underactuated.mit.edu/multibody.html`. Accessed 9 September 2026; manipulator equations, double pendulum and constraint sections.

Emanuel Todorov. Stochastic optimal control and estimation methods adapted to the noise characteristics of the sensorimotor system. *Neural Computation*, 17:1084–1108, 2005.

Trackman. 40+ trackman parameters, January 2020. URL https://www.trackman.com/blog/40-trackman-parameters. Published January 14, 2020; accessed September 9, 2026. Vendor reporting conventions; point and event definitions must accompany comparisons.

Trackman. What is swing plane?, May 2024. URL https://www.trackman.com/blog/what-is-swing-plane. Published May 1, 2024; accessed September 9, 2026. Vendor definition of a finite-arc metric, not independent biomechanical validation.

United States Golf Association. Equipment standards, 2026a. URL https://www.usga.org/equipment-standards.html. Governing equipment resources; accessed 7 September 2026.

United States Golf Association. What is a stroke, 2026b. URL https://rules.usga.org/courses/rules-for-teams/lessons/playing-the-ball-2/topic/what-is-a-stroke-and-how-to-make-one/. Rules for Teams; accessed 7 September 2026.

Alessandro Vena, Dave Budney, Tom Forest, and Jason P. Carey. Three-dimensional kinematic analysis of the golf swing using instantaneous screw axis theory, part 2: golf swing kinematic sequence. *Sports Engineering*, 13(3):125–133, 2011. doi: 10.1007/s12283-010-0059-7. URL https://doi.org/10.1007/s12283-010-0059-7. Five-subject kinematic study; the abstract reports support for proximal-to-distal sequencing in two subjects.

Samuel J Vine, Lee J Moore, and Mark R Wilson. Quiet eye training facilitates competitive putting performance in elite golfers. *Frontiers in Psychology*, 2:8, 2011. doi: 10.3389/fpsyg.2011.00008.

Hans-Joachim Wilke, Andrea Herkommer, Karin Werner, and Christian Liebsch. In vitro analysis of the segmental flexibility of the thoracic spine. *PLOS ONE*, 12(5):e0177823, 2017. doi: 10.1371/journal.pone.0177823. URL https://doi.org/10.1371/journal.pone.0177823.

Uta Wolfensteller and Hannes Ruge. On the timescale of stimulus-based action–effect learning. *Quarterly Journal of Experimental Psychology*, 64(7):1273–1289, 2011. doi: 10.1080/17470218.2010.546417.

Daniel M. Wolpert, Zoubin Ghahramani, and Michael I. Jordan. An internal model for sensorimotor integration. *Science*, 269:1880–1882, 1995. doi: 10.1126/science.7569931.

Daniel M. Wolpert, Jörn Diedrichsen, and J. Randall Flanagan. Principles of sensorimotor learning. *Nature Reviews Neuroscience*, 12:739–751, 2011.

Paul Wood, Erik Henrikson, and Chris Broadie. The influence of face angle and club path on the resultant launch angle of a golf ball. *Proceedings*, 2(6):249, 2018. doi: 10.3390/proceedings2060249. 12th Conference of the International Sports Engineering Association. Horizontal ratios use filtered centered strikes; Table 1 identifies its intervals as 95% confidence intervals. The reported vertical ratios are distinct quantities.

G Wulf, B Lauterbach, and T Toole. The learning advantages of an external focus of attention in golf. *Research Quarterly for Exercise and Sport*, 70(2):120–126, 1999. doi: 10.1080/02701367.1999.10608029.

Felix E. Zajac and Michael E. Gordon. Determining muscle's force and action in multi-articular movement. *Exercise and Sport Sciences Reviews*, 16:187–230, 1989. doi: 10.1249/00003677-198900170-00009. Seminal review establishing the theoretical basis for induced acceleration analysis in multi-articular movement. Introduced the concept that a muscle produces accelerations at joints it does not cross through dynamic coupling.

Felix E. Zajac, Richard R. Neptune, and Steven A. Kautz. Biomechanics and muscle coordination of human walking—Part I: Introduction to concepts, power transfer, dynamics and simulations. *Gait and Posture*, 16(3):215–232, 2002. Part I of comprehensive two-part review on muscle-driven walking simulations. Establishes the framework for induced acceleration and power analysis of individual muscle function during gait.

Felix E. Zajac, Richard R. Neptune, and Steven A. Kautz. Biomechanics and muscle coordination of human walking—Part II: Lessons from dynamical simulations and clinical implications. *Gait and Posture*, 17(1):1–17, 2003. Part II reviews coordination principles from walking simulations, demonstrating how induced acceleration analysis reveals individual muscle contributions to support, progression, and balance.

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